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(54) **HYBRID AAV CAPSIDS**

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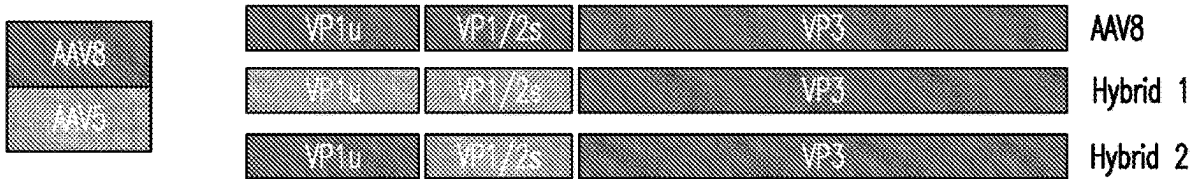
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(57) **ABSTRACT**
Provided herein are hybrid adeno-associated viruses comprising VP1u and/or VP1/2s region(s) from a first AAV and a VP3 region from a second AAV. Also provided herein are hybrid AAVs and compositions comprising hybrid AAVs that can be used for treating a subject in need thereof. Also provided herein are methods of making such hybrid AAVs.

Specification includes a Sequence Listing.



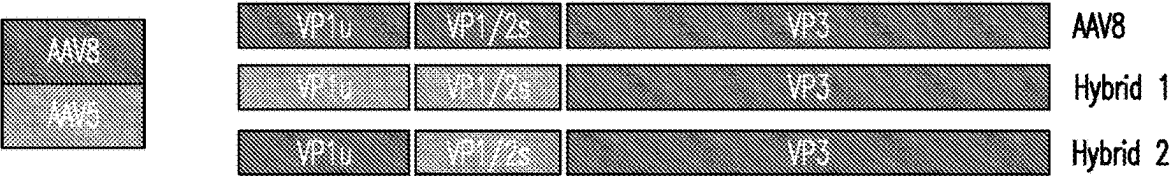


FIG. 1

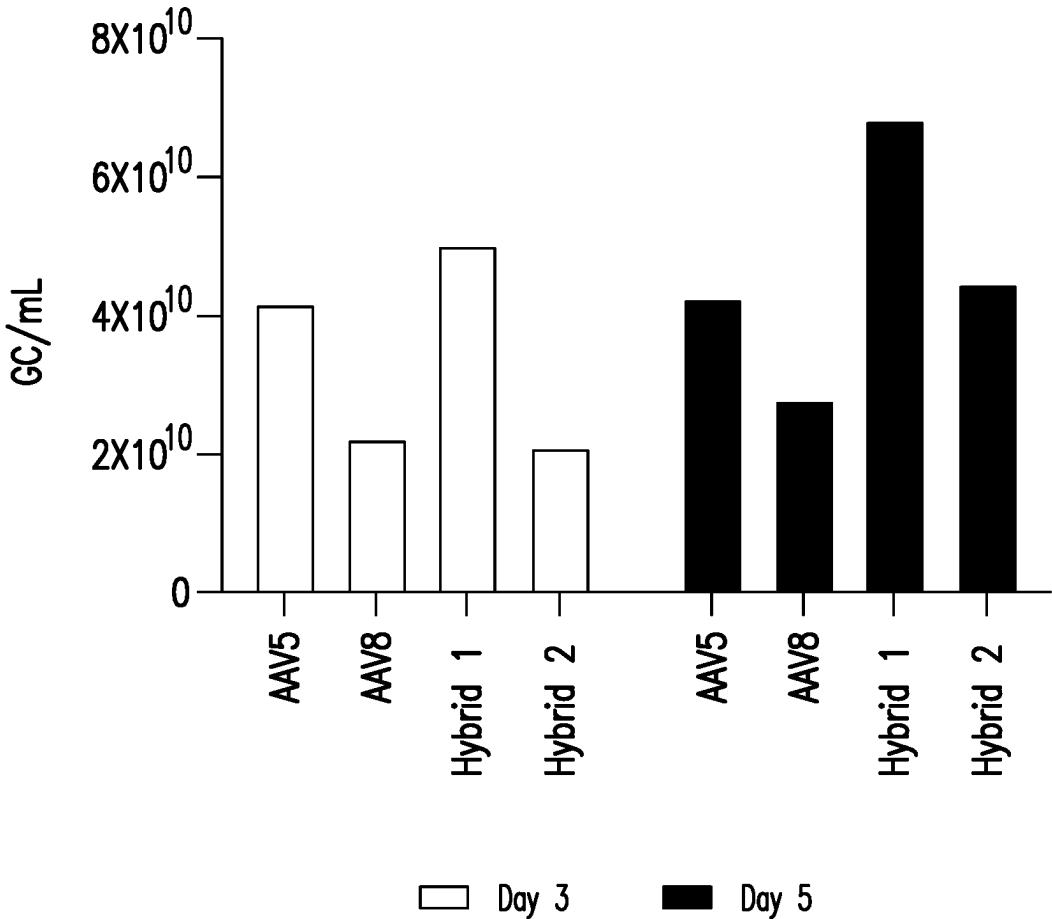
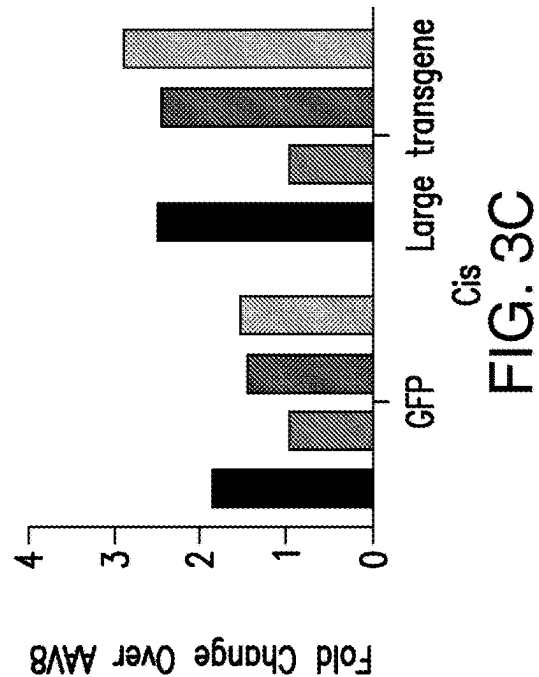
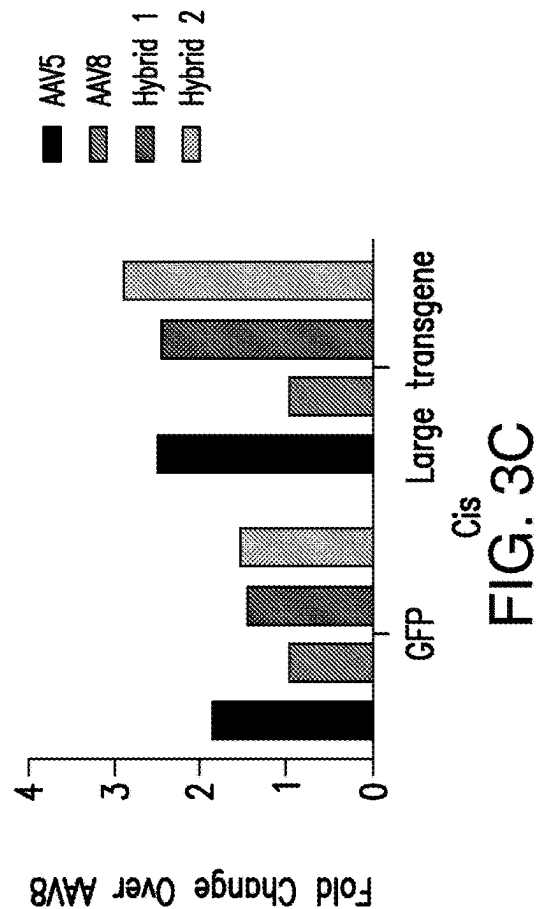
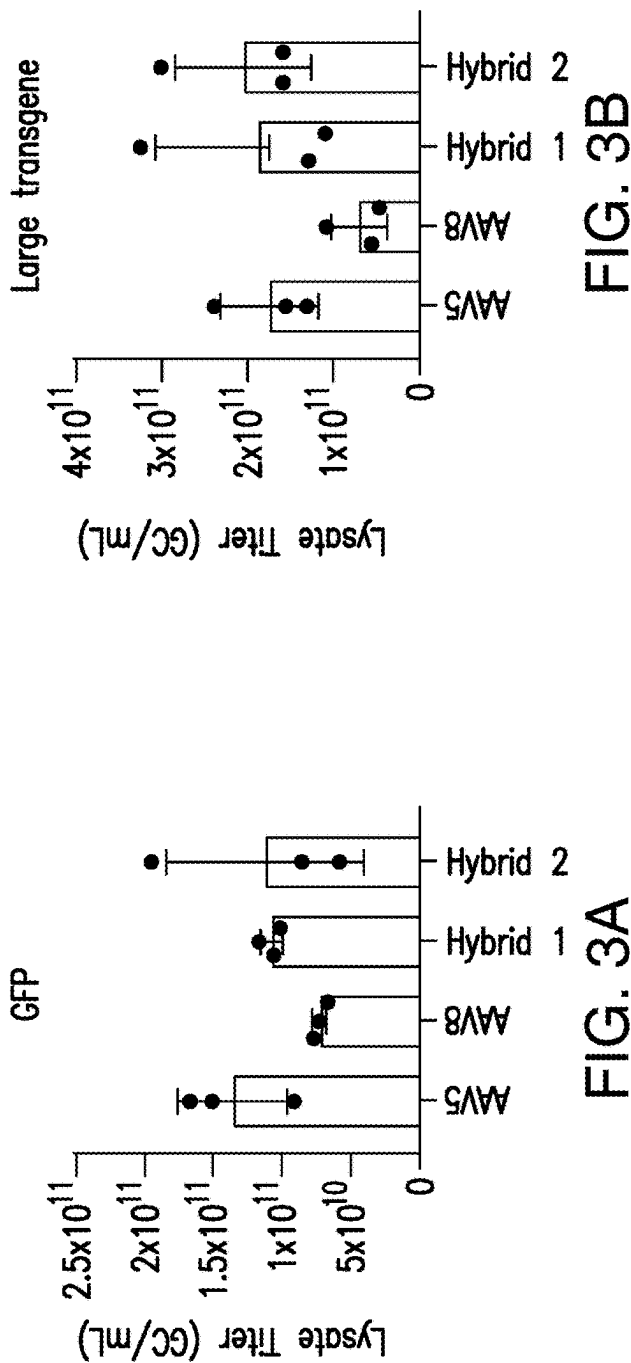


FIG. 2



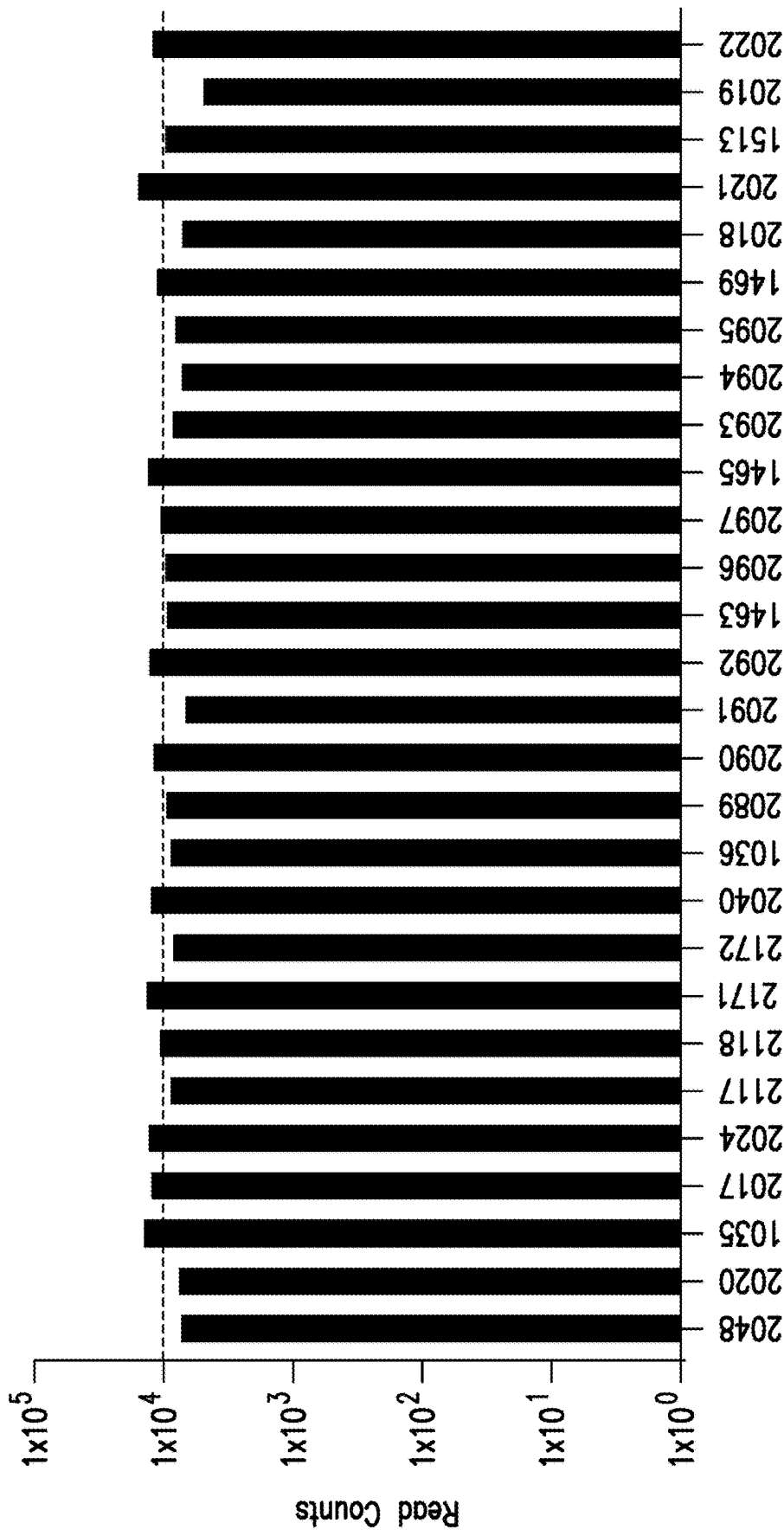
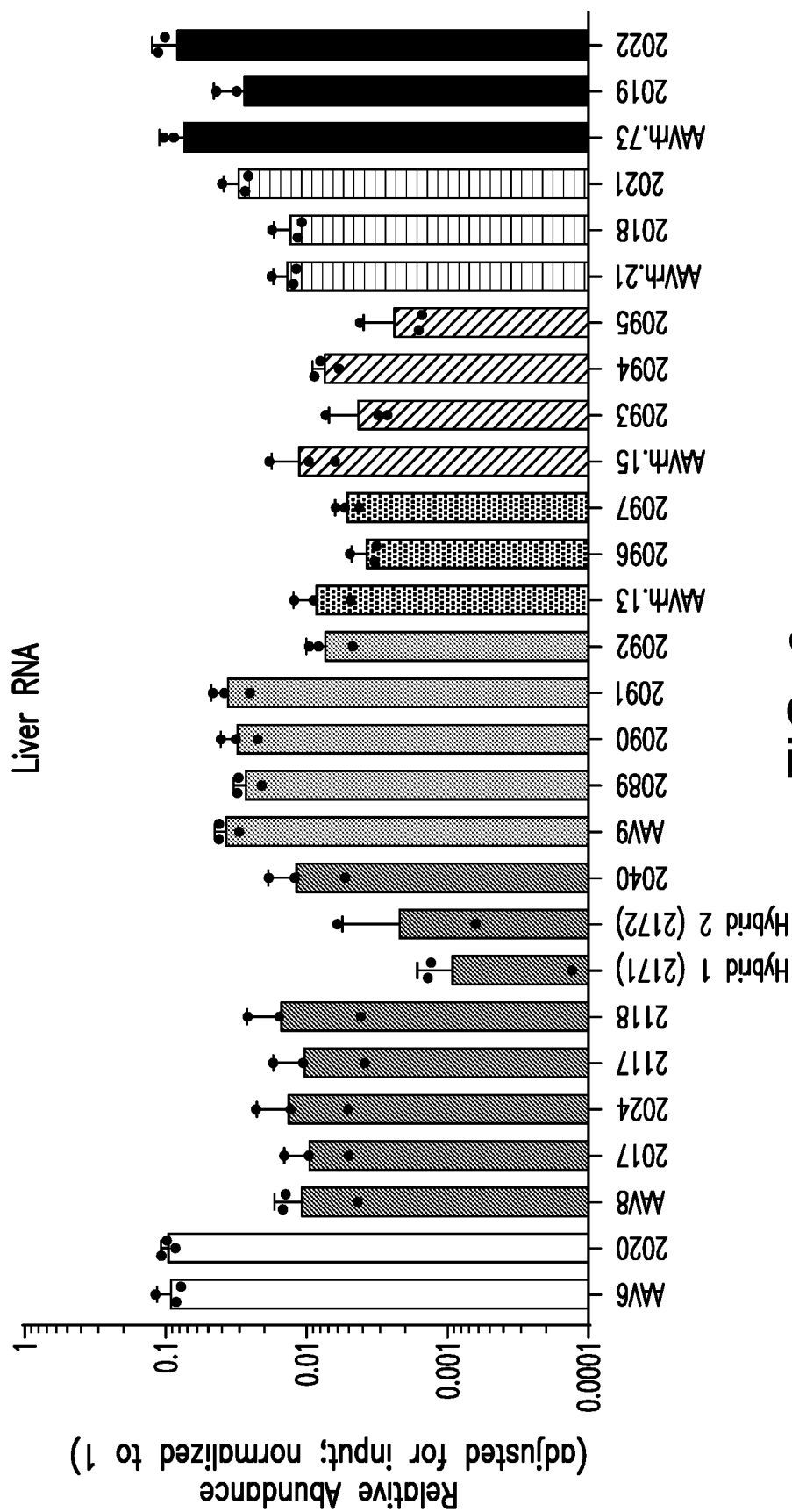


FIG. 4

5
G.
F



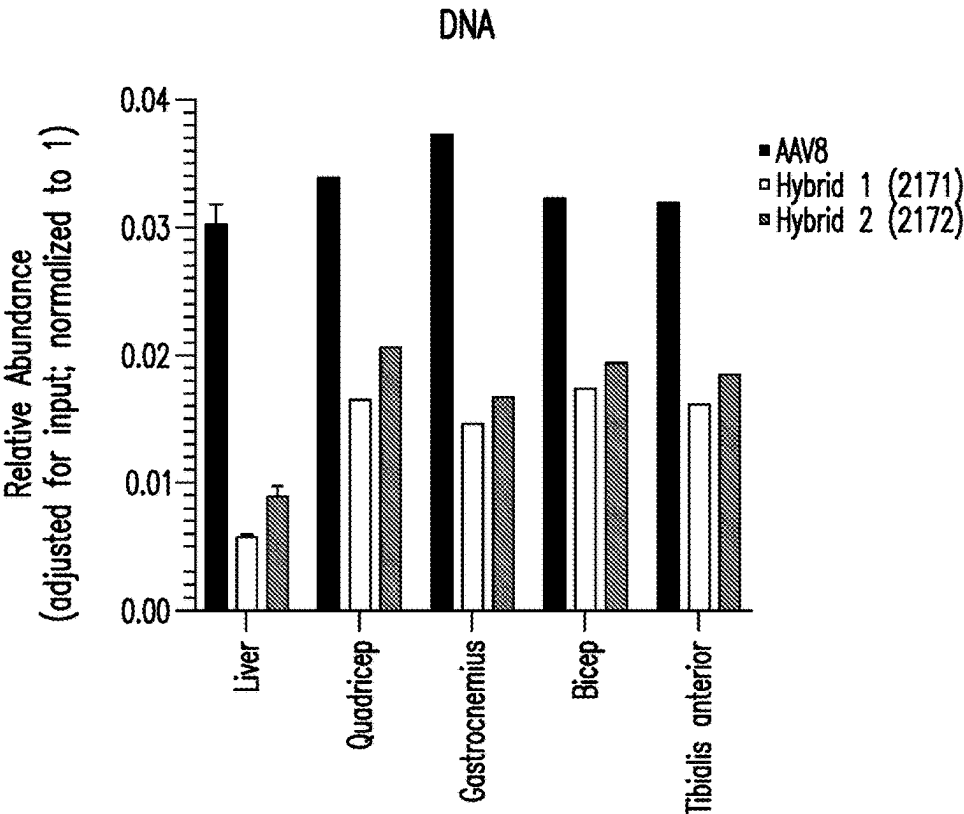


FIG. 7A

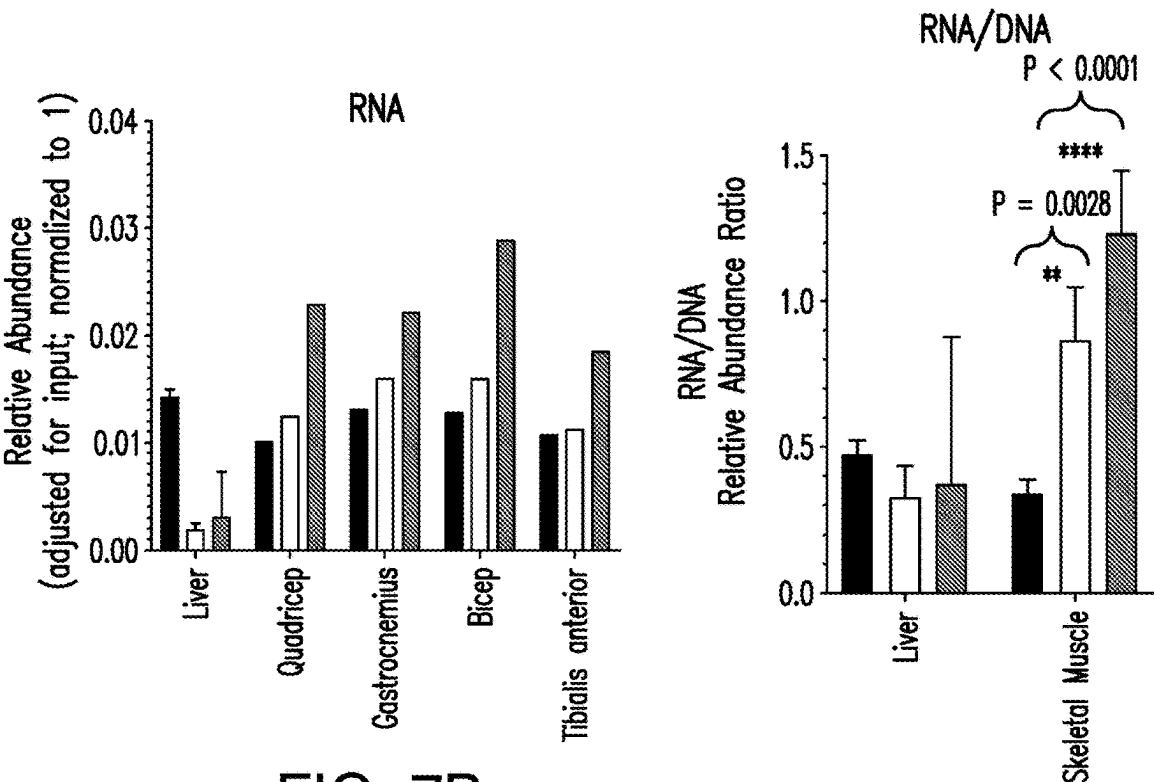


FIG. 7B

FIG. 7C

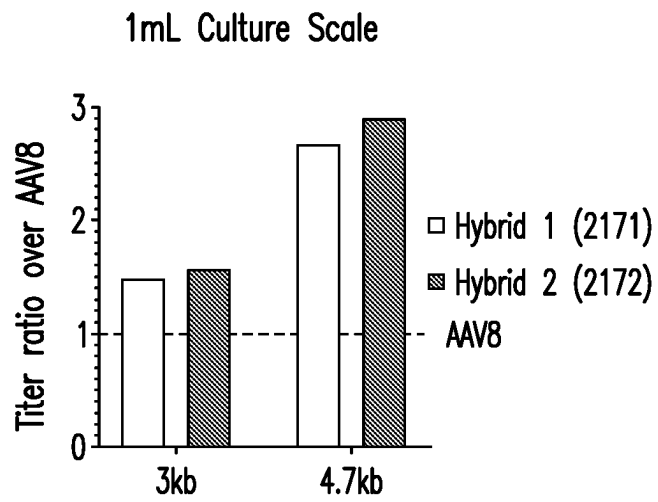


FIG. 8A

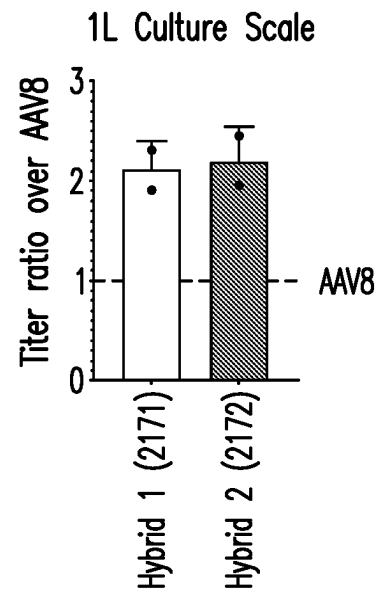


FIG. 8B

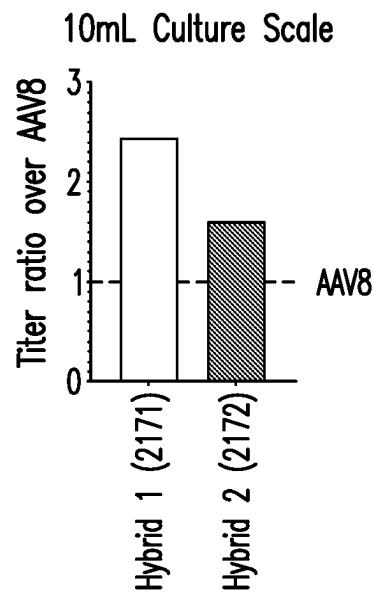


FIG. 8C

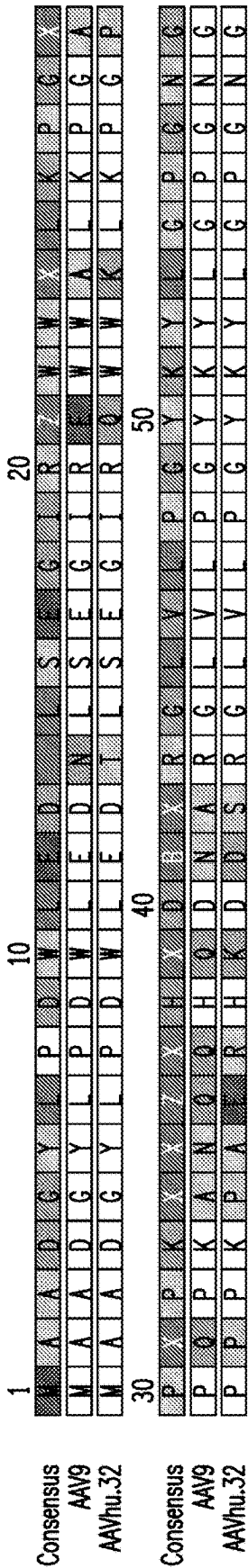


FIG. 9

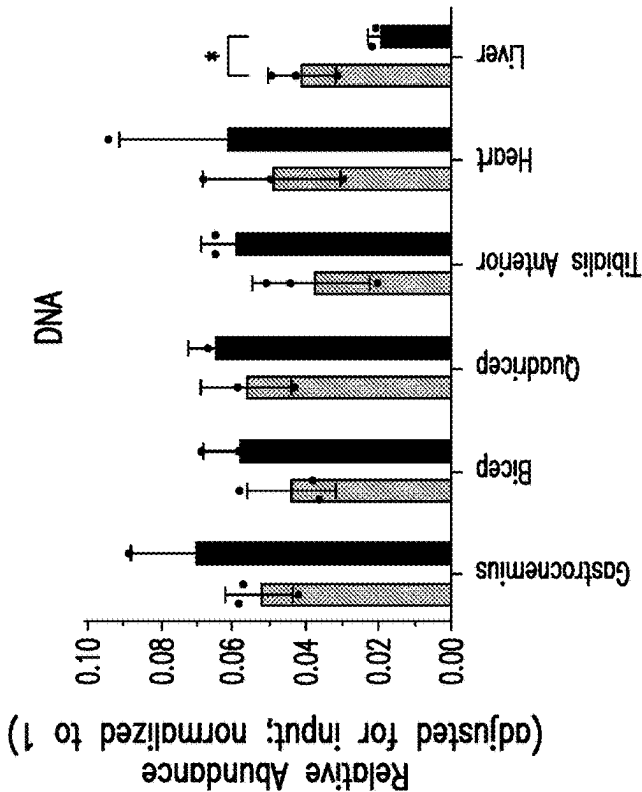


FIG. 10A

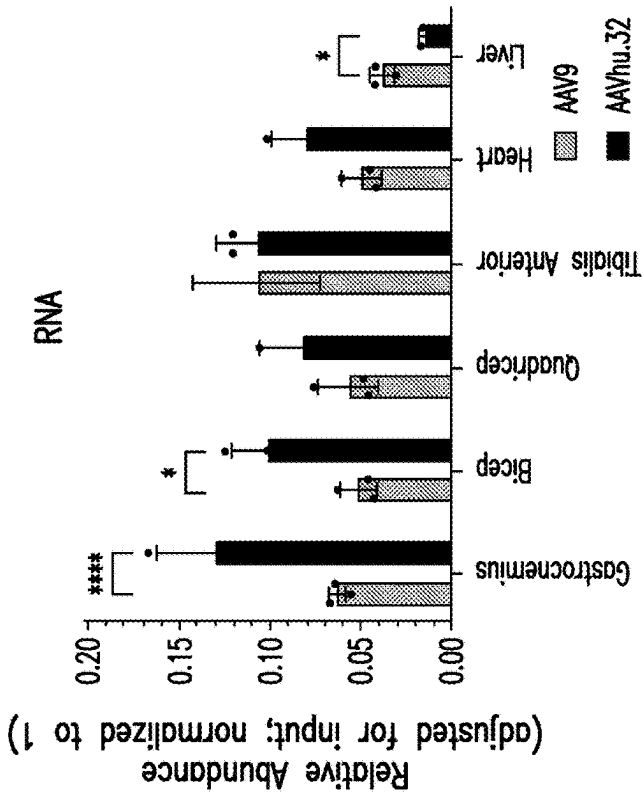
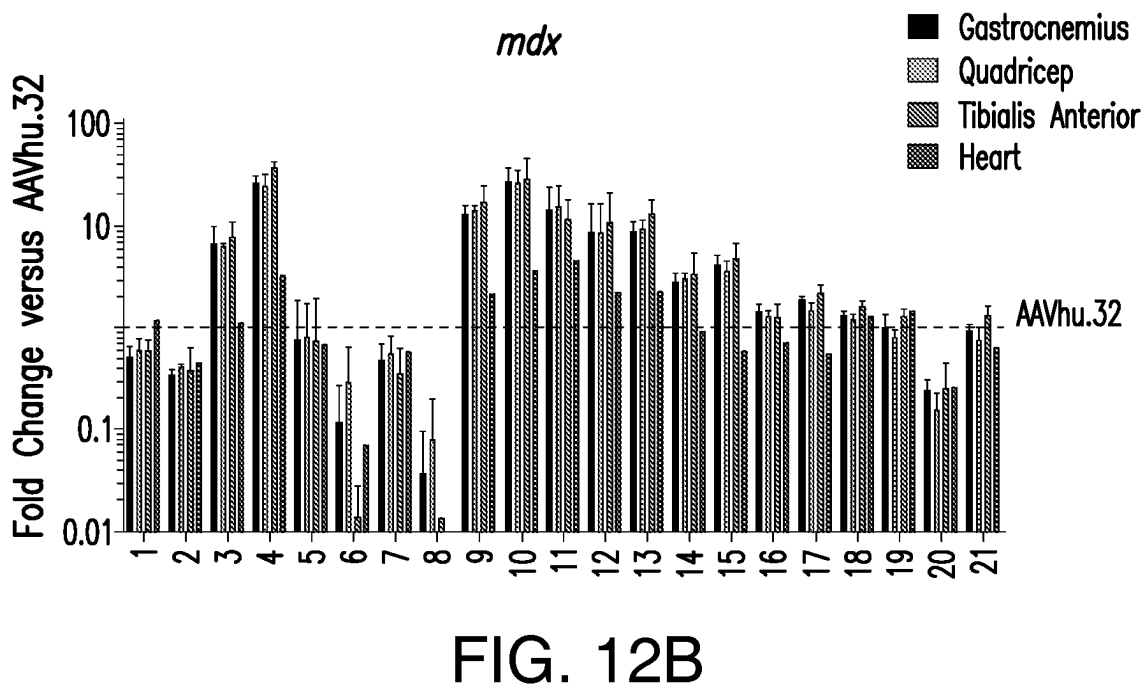
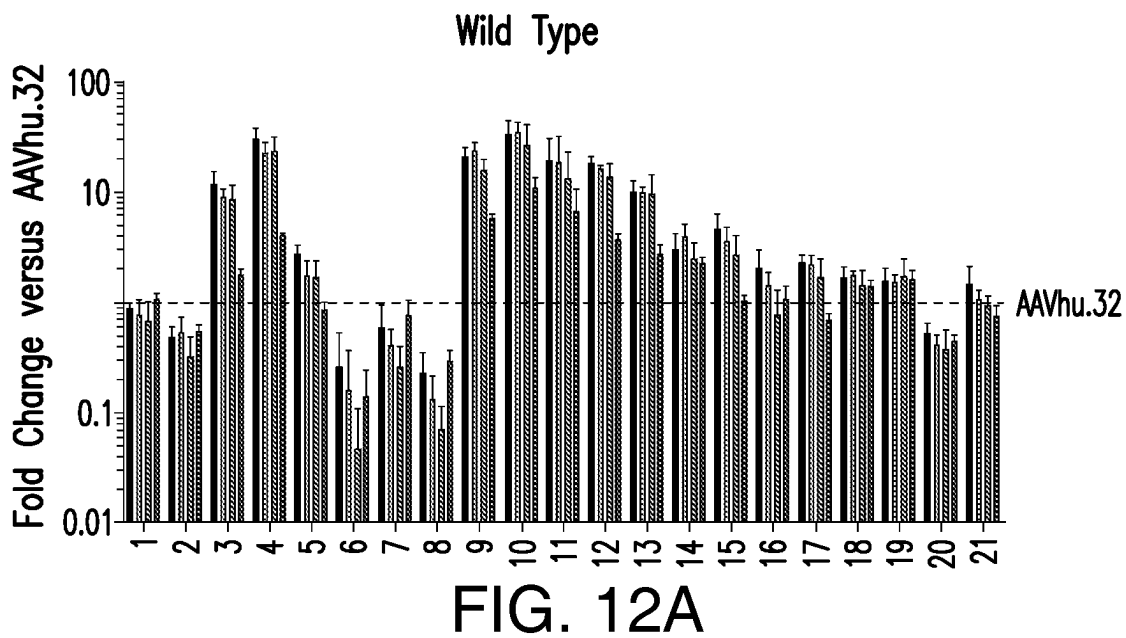


FIG. 10B



FIG. 11A

FIG. 11B



VP1₁₋₇₃₆ →

AAV1 MAADGYLPDWLEDNLSEGIREW**WDL**KPGAP**K**PKANQQ**KQDD**GRGLVLPGYKYLGP**F**NGLD 60
 AAV2 MAADGYLPDWLED**TL**SEGI**RQ****WW**KLKPG**PP**PK**PAER****H**K**DD**SRGLVLPGYKYLGP**F**NGLD 60
 AAV3-3 MAADGYLPDWLEDNLSEGIREW**W**ALKPG**V**P**Q**PKANQQ**HQDN****RR**GLVLPGYKYLGP**G**NGLD 60
 AAV4-4 **-MT**DGYLPDWLEDNLSEGV**RE****WW**AL**Q**PGAP**K**PKANQQ**HQDN**ARGLVLPGYKYLGP**G**NGLD 59
 AV5 **MSFVDH**PPDWL**EE**-**VGE**GL**REF**L**GLE**A**G**PP**K**PK**PN**QQ**HQD**QARGLVLPGY**N**YLGP**G**NGLD 59
 AAV6 MAADGYLPDWLEDNLSEGIREW**WDL**KPGAP**K**PKANQQ**KQDD**GRGLVLPGYKYLGP**F**NGLD 60
 AAV7 MAADGYLPDWLEDNLSEGIREW**WDL**KPGAP**K**PKANQQ**KQDN**GRGLVLPGYKYLGP**F**NGLD 60
 AAV8 MAADGYLPDWLEDNLSEGIREW**W**ALKPGAP**K**PKANQQ**KQDD**GRGLVLPGYKYLGP**F**NGLD 60
 hu31 MAADGYLPDWLED**TL**SEGI**RQ****WW**KLKPG**PP**PK**PAER****H**K**DD**SRGLVLPGYKYLGP**G**NGLD 60
 hu32 MAADGYLPDWLED**TL**SEGI**RQ****WW**KLKPG**PP**PK**PAER****H**K**DD**SRGLVLPGYKYLGP**G**NGLD 60
 AAV9 MAADGYLPDWLEDNLSEGIREW**W**ALKPGAP**Q**PKANQQ**HQDN**ARGLVLPGYKYLGP**G**NGLD 60
 rh10 MAADGYLPDWLEDNLSEGIREW**WDL**KPGAP**K**PKANQQ**KQDD**GRGLVLPGYKYLGP**F**NGLD 60

SUBS **-STVDHP**-----**ETVG**-**V-QFLK-QA-P-K-PAERKK-DG**-----**N-F**-----
 MF L D E V P QS
 G Q R

AAV1 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLRYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120
 AAV2 KGEPVN**E**ADAAALEHDKAYD**R**QL**DS**GDNPYLKYNHADA**E**FQERLKEDTSFGGNLGRA**V**FQ 120
 AAV3-3 KGEPVN**E**ADAAALEHDKAYDQQLKAGDNPYLKYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120
 AAV4-4 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA**E**F**Q**QRL**Q**GDT**S**FGGNLGRA**V**FQ 119
 AV5 **R**GEPVN**R**A**E****V**A**R**E**H**D**I**S**Y**N**E**Q**L**EAGDNPYLKYNHADA**E**F**Q**E**K**L**A**D**D**T**S**FGGNL**G**K**A**V**F**Q 119
 AAV6 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLRYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120
 AAV7 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLRYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120
 AAV8 KGEPVNAADAAALEHDKAYDQQL**Q**AGDNPYLRYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120
 hu31 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA**E**FQERLKEDTSFGGNLGRA**V**FQ 120
 hu32 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA**E**FQERLKEDTSFGGNLGRA**V**FQ 120
 AAV9 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA**E**FQERLKEDTSFGGNLGRA**V**FQ 120
 rh10 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLRYNHADA**E**FQERL**Q**EDTSFGGNLGRA**V**FQ 120

SUBS **R**-----**E-EV-R**-----**IS-NE-DS**-----**R**-----**QK-QD**-----**K**-----
 R R E AG
 Q

FIG. 13

VP2₁₃₈ → --HVR1--

AAV1	AKKR V LEPLGLVEE G AKTAPGKKRPVEQSPQ-EPDSS S GIGK TG Q Q PAKKRLNFGQTGD S	179
AAV2	AKKR V LEPLGLVEE PV KTAPGKKRPVE HSPV -EPDSS S G TG K AG Q Q PARKRLNFGQTGD A	179
AAV3-3	AKKR I LEPLGLVEEAAKTAPGKK GA VDQSPQ-EPDSS S G V GKSGK Q PARKRLNFGQTGD S	179
AAV4-4	AKKR V LEPLGLVE Q GE T APGKKR PLIE SPQ- Q PDSS T GIGK KG K Q PAKK KL VE DETGA	178
AV5	AKKR V LEP F GLVEE G AKTAP TG KR ID DHF----- KR KKARTEEDSKPSTSS D A	168
AAV6	AKKR V LEP F GLVEE G AKTAPGKKRPVEQSPQ-EPDSS S GIGK TG Q Q PAKKRLNFGQTGD S	179
AAV7	AKKR V LEPLGLVEE G AKTAP A KKRPVE PSPQR S SP DSS T GIGK KG Q Q PARKRLNFGQTGD S	180
AAV8	AKKR V LEPLGLVEE G AKTAPGKKRPVE PSPQR S SP DSS T GIGK KG Q Q PARKRLNFGQTGD S	180
hu31	AKKR L LEPLGLVEEAAKTAPGKKRPVEQSPQ-EPDSSAGIGKSG S QPAKK KL NFGQTGDT	179
hu32	AKKR L LEPLGLVEEAAKTAPGKKRPVEQSPQ-EPDSSAGIGKSG S QPAKK KL NFGQTGDT	179
AAV9	AKKR L LEPLGLVEEAAKTAPGKKRPVEQSPQ-EPDSSAGIGKSG A QPAKKRLNFGQTGDT	179
rh. 10	AKKR V LEPLGLVEEGAKTAPGKKRPVE PSPQR S SP DSS T GIGK KG Q Q PAKKRLNFGQTGD S	180
SUBS	— V — F — Q G G E — T G — G I D D H F — V — S — S — T — K K Q A R T R E K S V P E D E T G A	
	I PV A ALIP Q T V T K E D K STSS S	
	E A S	
	R A	

FIG. 13 Continued

-HVR2- VP2₂₀₃ →

AAV1	ESVPD-PQPLGEPPATPAAVGPTTMA	SGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	238
AAV2	DSVPD-PQPLGQPPAAPSLGTNTMATG	SGGAPMADNNEGADGVGNSSGNWHCDSTWMGDR	238
AAV3-3	ESVPD-PQPLGEPPAAPTSLGSNTMA	SGGGAPMADNNEGADGVGNSSGNWHCDSQWL GDR	238
AAV4-4	GDGP-----PEGSTSGAMS---DDSEMRAAAGGA	AVEGGQGADGVGNASGDWHCDSTWSEGH	232
AV5	EAGPSGSQQLQIPAQPASSLGADTMS	AGGGGPLGDNNGADGVGNASGDWHCDSTWMGDR	228
AAV6	ESVPD-PQPLGEPPATPAAVGPTTMA	SGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	238
AAV7	ESVPD-PQPLGEPPAAPSSVGS	GTVAAGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	239
AAV8	ESVPD-PQPLGEPPAAPSGVCPNTMA	AGGGAPMADNNEGADGVGSSSGNWHCDSTWLGDR	239
hu31	ESVPD-PQPIGEPPAAPSGVGSLTMA	SGGGAPVADNNEGADGVGSSSGNWHCDSQWL GDR	238
hu32	ESVPD-PQPIGEPPAAPSGVGSLTMA	SGGGAPVADNNEGADGVGSSSGNWHCDSQWL GDR	238
AAV9	ESVPD-PQPIGEPPAAPSGVGSLTMA	SGGGAPVADNNEGADGVGSSSGNWHCDSQWL GDR	238
rh10	ESVPD-PQPIGEPPAGPSGLGSGTMA	AGGGAPMADNNEGADGVGSSSGNWHCDSTWLGDR	239
SUBS	GDG-S-S-QLQQTSGTMASLDPNEVRAAA-GAMGEGGQ-----NA-D-----T-MEGH		
	DA	E S AQPATA AG ST S LV	S
		I --- DT A	
		TD	
		S	

HVR3

AAV1	VITTSTRTWALPTYNNHLYKQIS-SASTGASNDNH	YFGYSTPWGYDFNRFHCHFSPRDW	297
AAV2	VITTSTRTWALPTYNNHLYKQIS--SQSGASNDNH	YFGYSTPWGYDFNRFHCHFSPRDW	296
AAV3-3	VITTSTRTWALPTYNNHLYKQIS--SQSGASNDNH	YFGYSTPWGYDFNRFHCHFSPRDW	296
AAV4-4	VTTTSTRTWVLPPTYNNHLYKRLG-----ESLQ	SNTYNGFSTPWGYDFNRFHCHFSPRDW	287
AV5	VVTKSTRTWVLPSYNNHQYREIKS-GSVDGSNA	NAYFGYSTPWGYDFNRFHSHWSPRDW	287
AAV6	VITTSTRTWALPTYNNHLYKQISSAST-GASNDNH	YFGYSTPWGYDFNRFHCHFSPRDW	297
AAV7	VITTSTRTWALPTYNNHLYKQISS-ETAGSTNDNT	YFGYSTPWGYDFNRFHCHFSPRDW	298
AAV8	VITTSTRTWALPTYNNHLYKQISNGTSGCATNDNT	YFGYSTPWGYDFNRFHCHFSPRDW	299
hu31	VITTSTRTWALPTYNNHLYKQISNSTSGGSSNDN	AYFGYSTPWGYDFNRFHCHFSPRDW	298
hu32	VITTSTRTWALPTYNNHLYKQISNSTSGGSSNDN	AYFGYSTPWGYDFNRFHCHFSPRDW	298
AAV9	VITTSTRTWALPTYNNHLYKQISNSTSGGSSNDN	AYFGYSTPWGYDFNRFHCHFSPRDW	298
rh10	VITTSTRTWALPTYNNHLYKQISNGTSGGSTNDNT	YFGYSTPWGYDFNRFHCHFSPRDW	299
SUBS	-T-K-----V-S-----Q-RRLGSGSQSDATQA-T-----S-W-----		
	V	E K AATTEGL S H	
		G V	
		E A	

FIG. 13 Continued

FIG. 13 Continued

FIG. 13 Continued

HVR4

AAV1	AHQGCLPPFPADVFMIPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFTFSY	414
AAV2	AHQGCLPPFPADVFMVPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFTFSY	413
AAV3-3	AHQGCLPPFPADVFMVPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFQFSY	413
AAV4-4	GQEGSLPPFPNDVFMVPQYGYCGLVTGNTSQQQTDRNAFYCLEYFPSQMLRTGNNFEITY			407
AV5	GTEGCLPAFPQVFTLPQYGYATLNRD	NT	ENPTERRSSF	FCLEYFPSKMLRTGNNFEITY 406
AAV6	AHQGCLPPFPADVFMIPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFTFSY	414
AAV7	AHQGCLPPFPADVFMIPQYGYLTLNNG	----	SQSVGRSSFYCLEYFPSQMLRTGNNFEFSY	415
AAV8	AHQGCLPPFPADVFMIPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFQFTY	416
hu31	AHEGCLPPFPADVFMIPQYGYLTLNDG	----	GQAVGRSSFYCLEYFPSQMLRTGNNFQFSY	415
hu32	AHEGCLPPFPADVFMIPQYGYLTLNDG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFQFSY	415
AAV9	AHEGCLPPFPADVFMIPQYGYLTLNDG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFQFSY	415
rh10	AHQGCLPPFPADVFMIPQYGYLTLNNG	----	SQAVGRSSFYCLEYFPSQMLRTGNNFEFSY	416
SUBS	GQQ-S-A-PQ-TL	-----	CG-VND-CNPTD-NA-F	-----EIT-
	T		N	V
			A	T
			Q	Q
			R	E
			S	

-----HVR5-----

AAV1	TFEEVPFHSSYAHSQSLDRLMNPLIDQYLYL	NRTQ-NQSGSAQNKDLLFSRGSPAGMSV	473	
AAV2	TFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	SRTN-TPSGTTTQSRLLFSQAGASDIRD	472	
AAV3-3	TFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	NRTQGTTS GTTNQSRLLFSQAGPQMSL	473	
AAV4-4	SFEKVPFHSSMYAHSQSLDRLMNPLIDQYL	WGLQSTTTGTTLNAGTATTNFTKL RPTNFSN	467	
AV5	NFEVPFHSSFAPSQNLFLKANPLVDQYLYR	FVSTN-----NTGGVQFNKNLAGRYAN	459	
AAV6	TFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	NRTQ-NQSGSAQNKDLLFSRGSPAGMSV	473	
AAV7	SFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	ARTQSNPGGTAGNRELQFYQGGPSTMAE	475	
AAV8	TFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	SRTQT-TGGTANTQTLGFSQGGPNTMAN	475	
hu31	EFENVPFHSSYAHSQSLDRLMNPLIDQYLYL	SKTINGSG--QNQQTLKFSVAGPSNMAV	473	
hu32	EFENVPFHSSYAHSQSLDRLMNPLIDQYLYL	SKTINGSG--QNQQTLKFSVAGPSNMAV	473	
AAV9	EFENVPFHSSYAHSQSLDRLMNPLIDQYLYL	SKTINGSG--QNQQTLKFSVAGPSNMAV	473	
rh10	QFEDVPFHSSYAHSQSLDRLMNPLIDQYLYL	SRTQSTGG-TAGTQQLLFSQAGPNNMSA	475	
SUBS	T-D-MF	-----A-V-WGFNR-QTNTS-AGTKRTQ-TQGSAATFSN		
	S	E	QS NSTPT	TQNSDVN
	N	K	V	TG Q
			A	RG G
				GS E
				YRLR TRI L
				AE L

ND

FIG. 13 Continued

FIG. 13 Continued

-----HVR9-----

AAV1 DDEDKFFPMSGVMIFGKESA--GASNTALD-NVMITDEEEIKATNPVATERF~~GT~~VAVNFQ 585

AAV2 DDEKFFPQSGVLIFGKQGS--EKT~~ND~~IE-KVMITDEEEIRTTNPVATEQYGSVSTNLQ 584

AAV3-3 DDEKFFPMHG~~N~~LIFGKEGT--TASNAELD-NVMITDEEEIRTTNPVATEQYGTVANNLQ 585

AAV4-4 PADSKFS-NSQLIFAGPKQN--GNTATVPG-TLIFTSEEELAA~~T~~NATDTDMWGNLPGGDQ 583

AV5 LQGSNTYALENTMIFNSQPANPGTTATYLEGNMLITSESETQPVNRVAYNVGGQMATNNQ 574

AAV6 DDKDKFFPMSGVMIFGKESA--GASNTALD-NVMITDEEEIKATNPVATERF~~GT~~VAVNLQ 585

AAV7 DDEDRFFPSSGVLIFGKTGA--TN-KTTLE-NVLMTNEEEIRPTNPVATEEYGI~~V~~SSNLQ 586

AAV8 DDEERFFPSNGILIFGKQNA--ARDNADYS-DVMLTSEEEIKTTNPVATEEYGI~~V~~ADNLQ 587

hu31 EGEDRFFPLSGSLIFGKQGT--GRDNVDAD-KVMITNEEEIKTTNPVATESY~~G~~QVATNHQ 585

hu32 EGEDRFFPLSGSLIFGKQGT--GRDNVDAD-KVMITNEEEIKTTNPVATESY~~G~~QVATNHQ 585

AAV9 EGEDRFFPLSGSLIFGKQGT--GRDNVDAD-KVMITNEEEIKTTNPVATESY~~G~~QVATNHQ 585

rh10 DDEERFFPSSGVL~~M~~FGKQGA--GKDNVDYS-SVMLTSEEEIKTTNPVATEQYGV~~V~~ADNLQ 587

SUBS LQGSNTYAMENTMFANPKQN--TNTATVPG-TLIF-S-S-TQPV-ATDYDMW-NLPGGD-
PADEK S QHQLI SESA EASKAALE-NMLM D RA R NVF TMSN L
DDK NN V TPS AK KTY L A QG I V N
S I N EI E S S F
N Y R D

--HVR10---

AAV1 SSSTDPATGDVHAMGALPGMVWQDRDVYLQGP~~I~~WAKIPHTDGHFHPSPLMGGFGLKNPPP 645

AAV2 RGNRQAATADVNTQGVLPGMVWQDRDVYLQGP~~I~~WAKIPHTDGHFHPSPLMGGFGLKHPPP 644

AAV3-3 SSNTAPTGTGNHQAALPGMVWQDRDVYLQGP~~I~~WAKIPHTDGHFHPSPLMGGFGLKHPPP 645

AAV4-4 SNSNLPTVDRLTALGAVPGMVWQNRDIYYQGP~~I~~WAKIPHTDGHFHPSPLI~~G~~GGFGLKHPPP 643

AV5 SSTTAPATGTYNLQEI~~V~~PGSV~~M~~MERDVYLQGP~~I~~WAKIPETGAHFHPSPAMGGFGLKHPPP 634

AAV6 SSSTDPATGDVHVMGALPGMVWQDRDVYLQGP~~I~~WAKIPHTDGHFHPSPLMGGFGLKHPPP 645

AAV7 AANTAAQTQVVNNQGAALPGMVWQNRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGLKHPPP 646

AAV8 QQNTAPQIGTVNSQGAALPGMVWQNRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGLKHPPP 647

hu31 SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGMKHPPP 645

hu32 SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGMKHPPP 645

AAV9 SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGMKHPPP 645

rh10 QQNAAPIVGAVNSQGAALPGMVWQNRDVYLQGP~~I~~WAKIPHTDGNFHPSPLMGGFGLKHPPP 647

SUBS RNSNLPTVDRLTALEAV-S-ME-I-E-GAH-AI-L-N-
ASNTA AIADYHTM V N
QGTRD QT NH
Q V L
V
S

FIG. 13 Continued

---HVR11---

AAV1 QILIKNTPVPANPPAEFSATKFASFITQYSTGQVSVEIEWELQKENSkrwnPEVQYTSNY 705
 AAV2 QILIKNTPVPANPSTIFSAAKFASFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 704
 AAV3-3 QIMIKNTPVPANPPTTFSPAKFASFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 705
 AAV4-4 QIFIKNTPVPANPATTFSSTPVNSFITQYSTGQVSVDWEIQERSkrwnPEVQFTSNY 703
 AV5 MMLIKNTPVPGNI-TSFSQVPSFITQYSTGQVTVEWELKKENSkrwnPEIQYTNNY 693
 AAV6 QILIKNTPVPANPPAEFSATKFASFITQYSTGQVSVEIEWELQKENSkrwnPEVQYTSNY 705
 AAV7 QILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNF 706
 AAV8 QILIKNTPVPADPPTTFNSKLNSFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 707
 hu31 QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 705
 hu32 QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 705
 AAV9 QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 705
 rh10 QILIKNTPVPADPPTTFSSQAKLASFITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNY 707
 SUBS ~~MM~~ ~~G~~ ~~IAE~~ ~~SDVPVS~~ ~~QMD~~ ~~IK~~ ~~R~~ ~~V~~
 F SET TAA FA
 S PT
 V QS
 S

-----HVR12-----

AAV1 AKSANVDFTVDNNGLYTEPRPIGTRYLTRPL 736 (SEQ ID NO: XX)
 AAV2 NKSVNVDFTVDTNGVYSEPRPIGTRYLTRNL 735 (SEQ ID NO: XX)
 AAV3-3 NKSVNVDFTVDTNGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: XX)
 AAV4-4 GQQNSLLWAPDAAGKYTEPRAIGTRYLTHHL 734 (SEQ ID NO: XX)
 AV5 NDPQFVDFAPDSTGEYRTTRPIGTRYLTRPL 724 (SEQ ID NO: XX)
 AAV6 AKSANVDFTVDNNGLYTEPRPIGTRYLTRPL 736 (SEQ ID NO: XX)
 AAV7 EKQTGVDFAVDSQGVYSEPRPIGTRYLTRNL 737 (SEQ ID NO: XX)
 AAV8 YKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL 738 (SEQ ID NO: XX)
 hu31 YKSNNVEFAVSTEGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: XX)
 hu32 YKSNNVEFAVNTEGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: XX)
 AAV9 YKSNNVEFAVNTEGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: XX)
 rh10 YKSTNVDFAVNTDGTYSPRPIGTRYLTRNL 738 (SEQ ID NO: XX)
 SUBS ~~GQQVSL~~~~WTPDAA~~-~~K~~-~~RTT~~-~~A~~-----~~HP~~-
 NDPQF D SSN E T H
 A TG NQ L
 E A T

FIG. 13 Continued

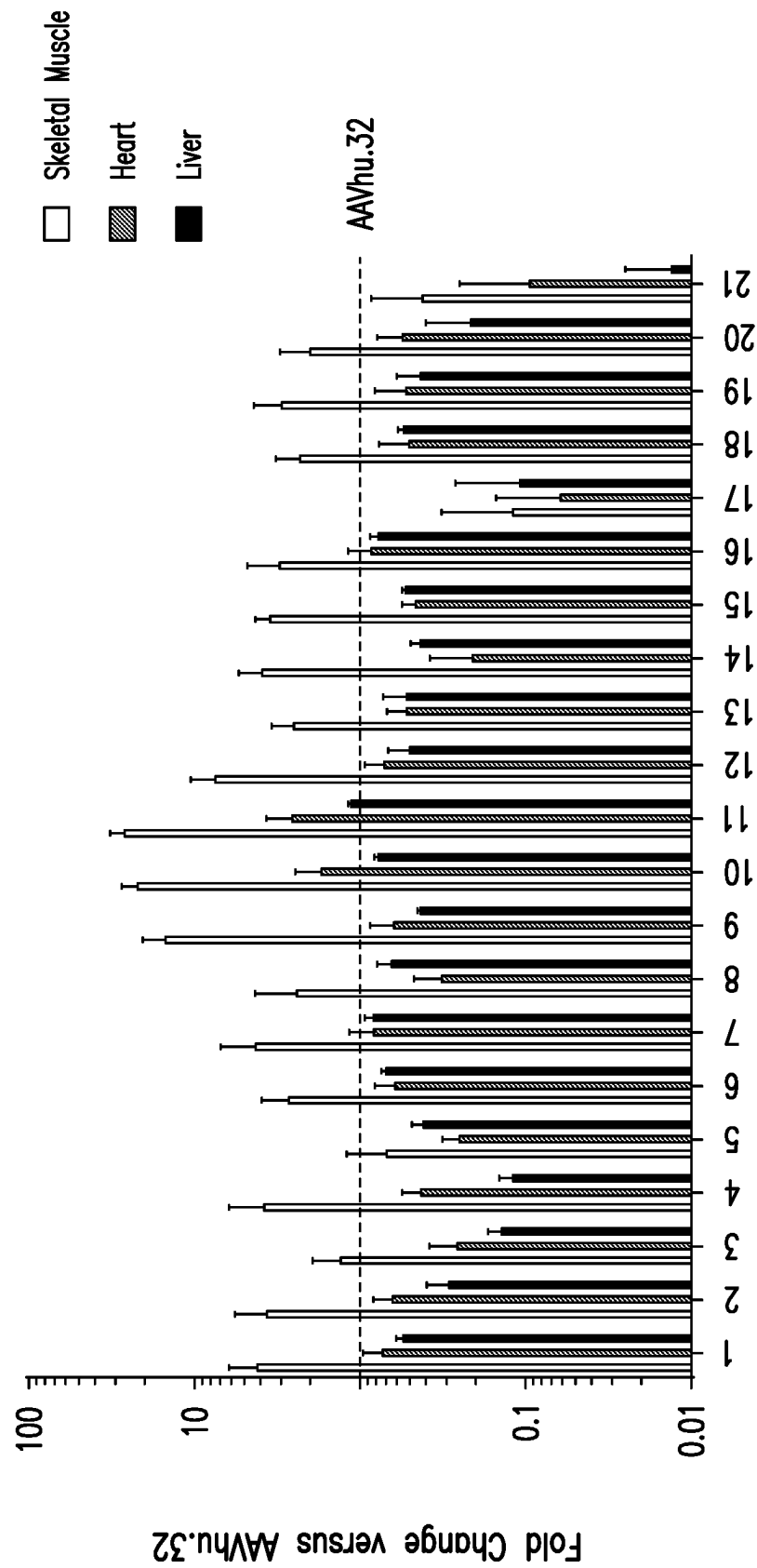


FIG. 14

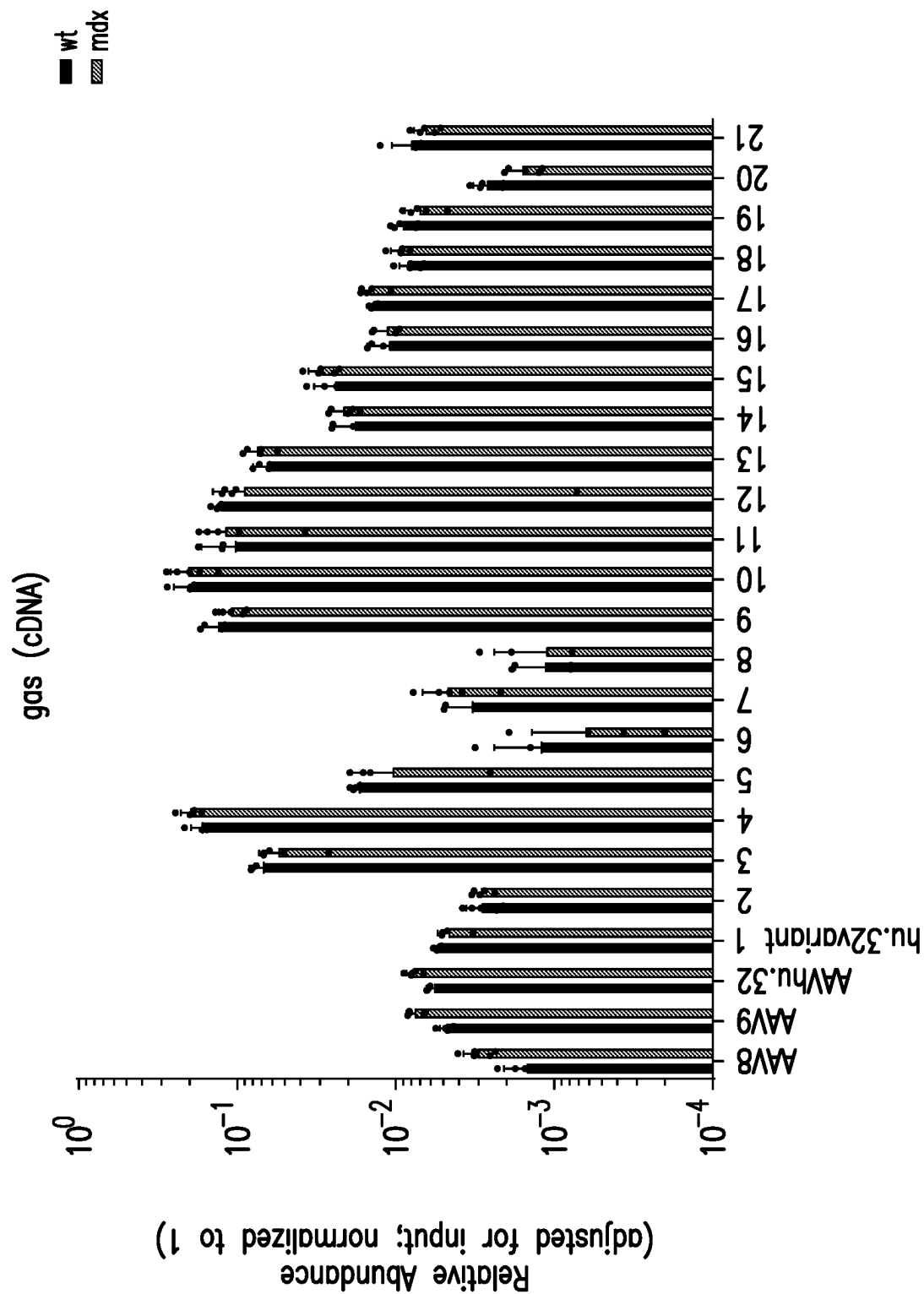


FIG. 15A

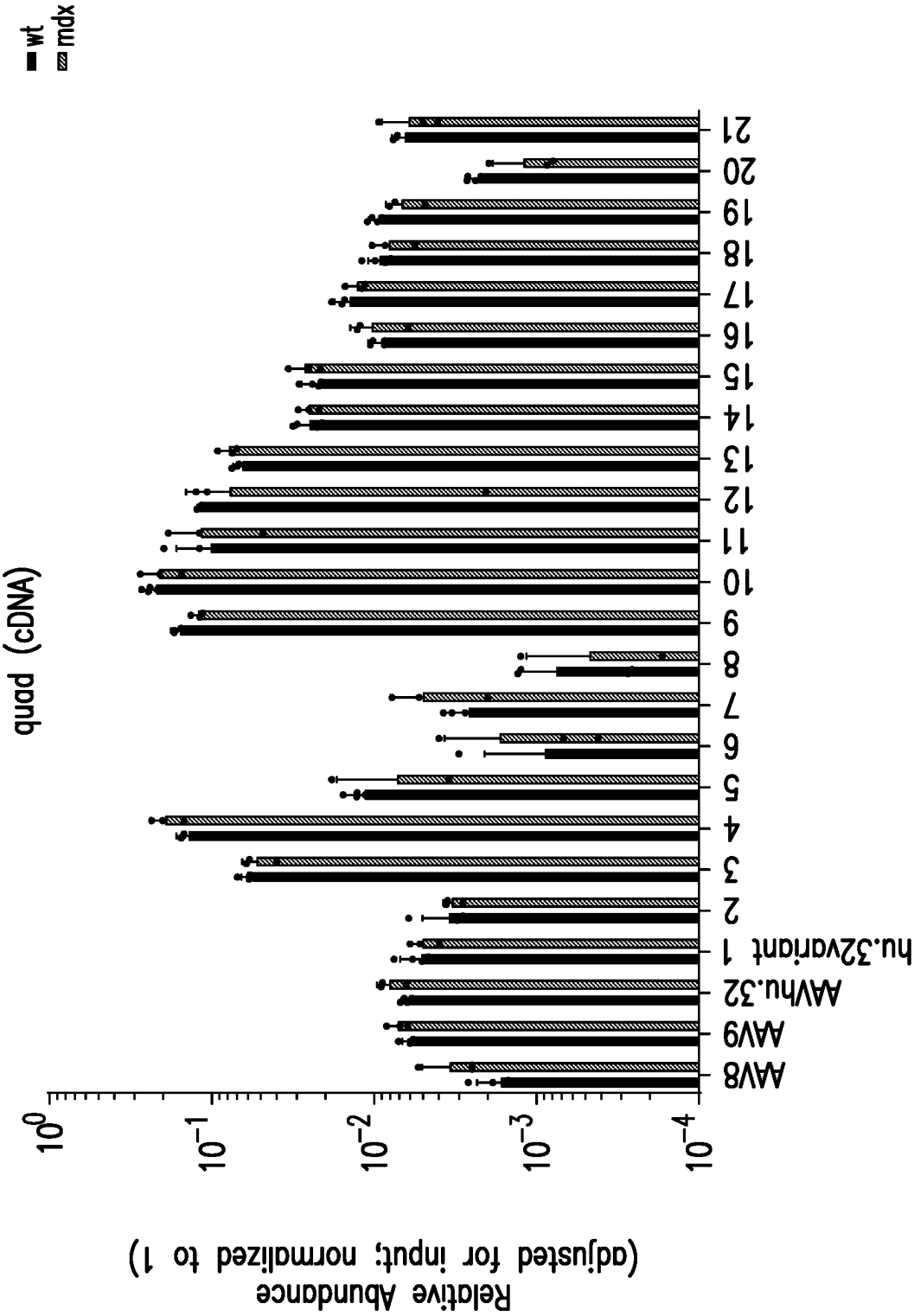


FIG. 15B

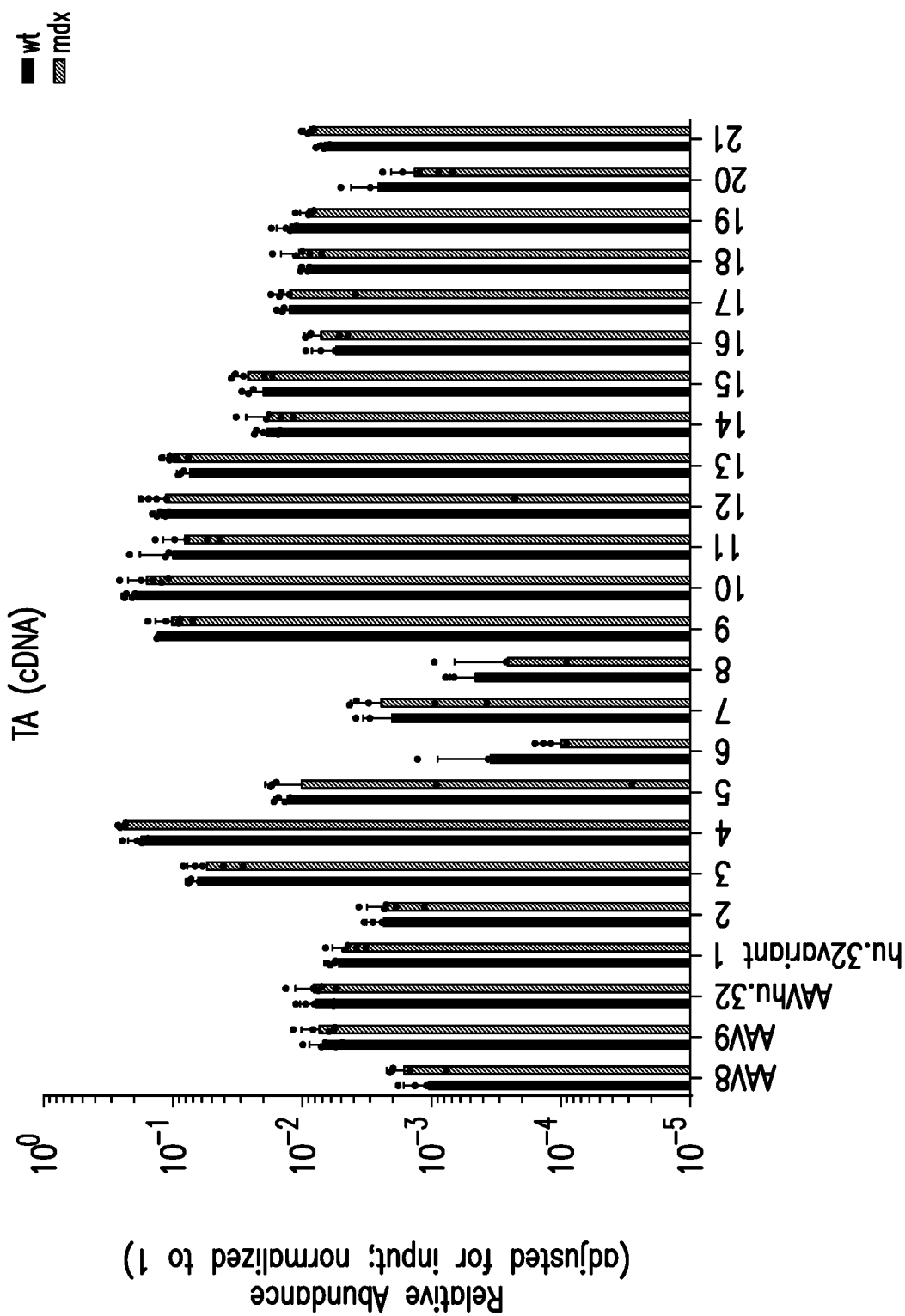


FIG. 16A

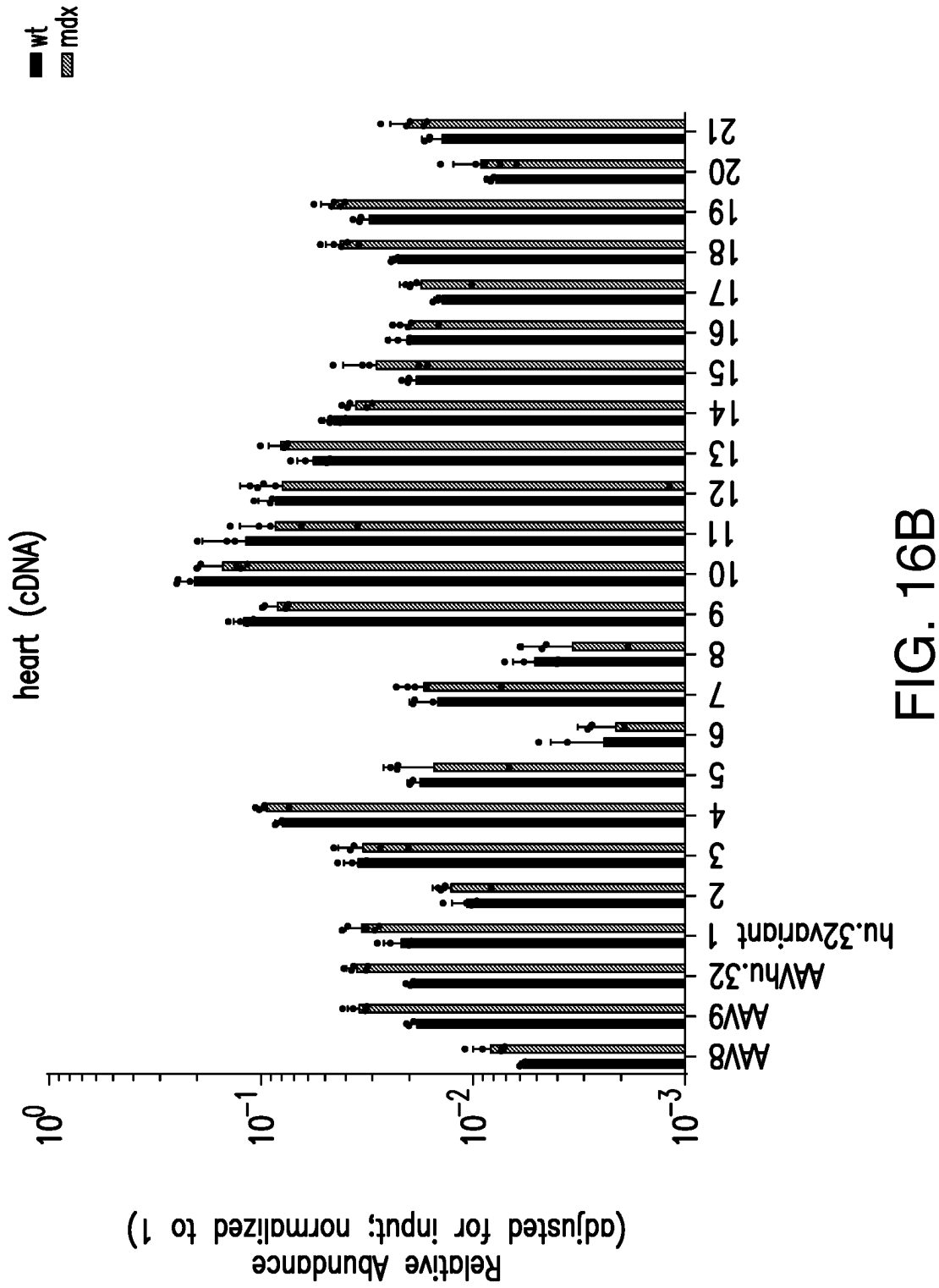


FIG. 16B

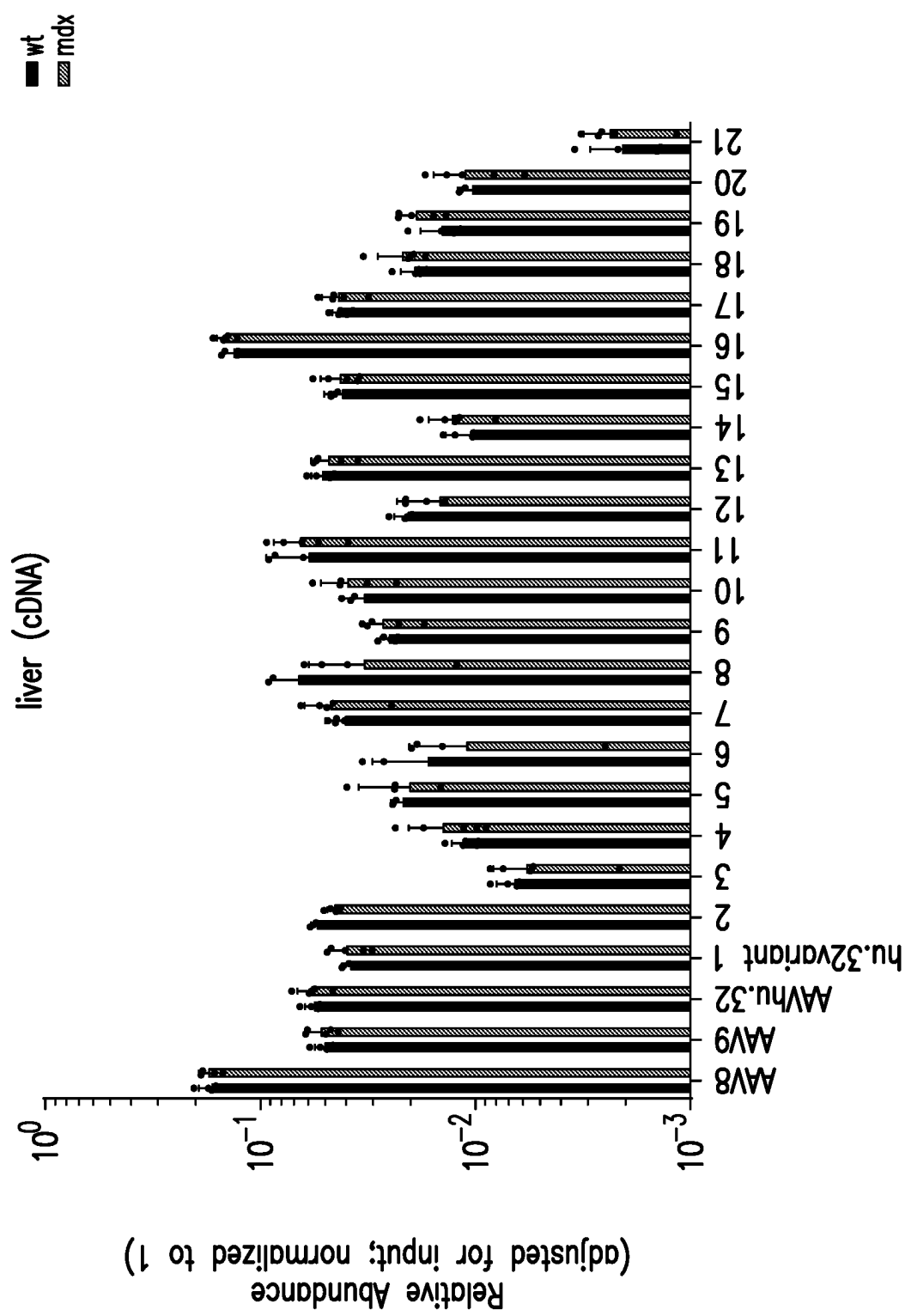


FIG. 17

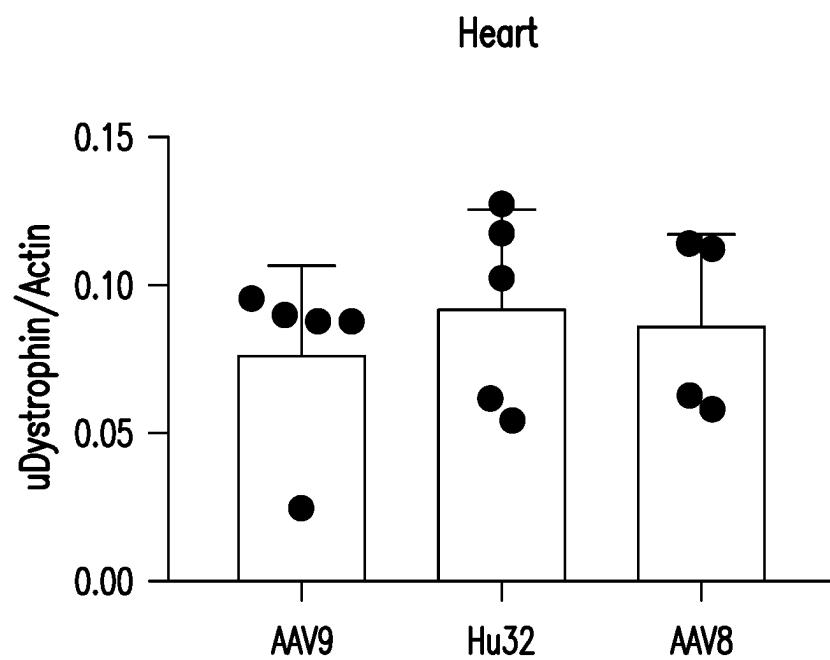


FIG. 18A

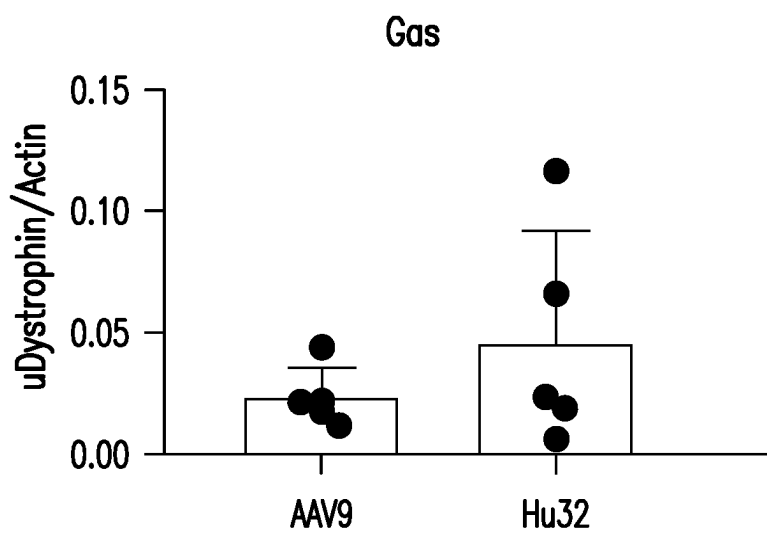


FIG. 18B

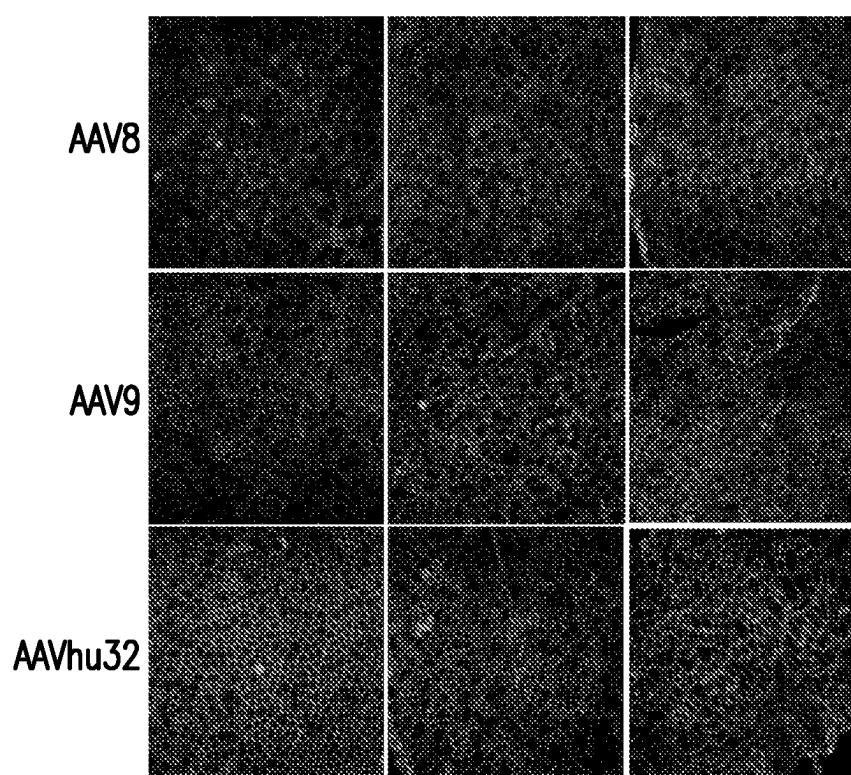


FIG. 19

AAVhu32 DNA/DAPI

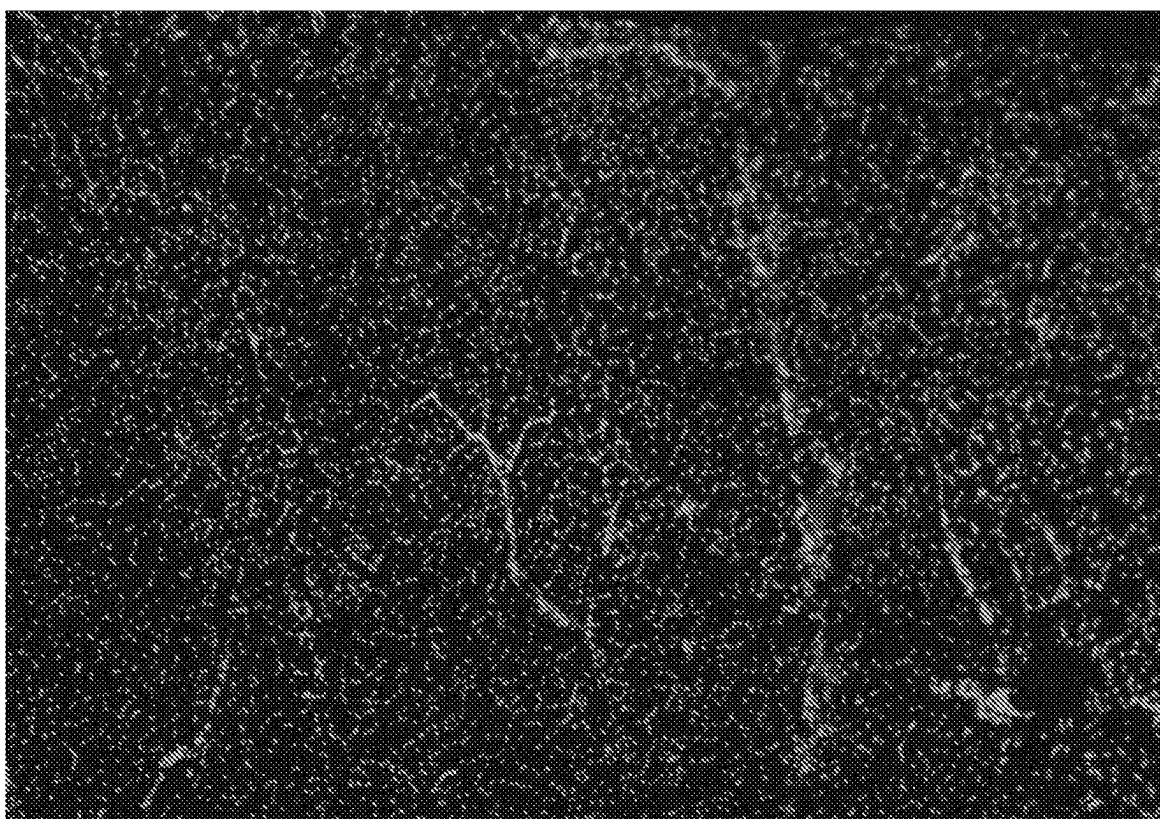


FIG. 20A

AAVhu32 RNA/DNA/DAPI

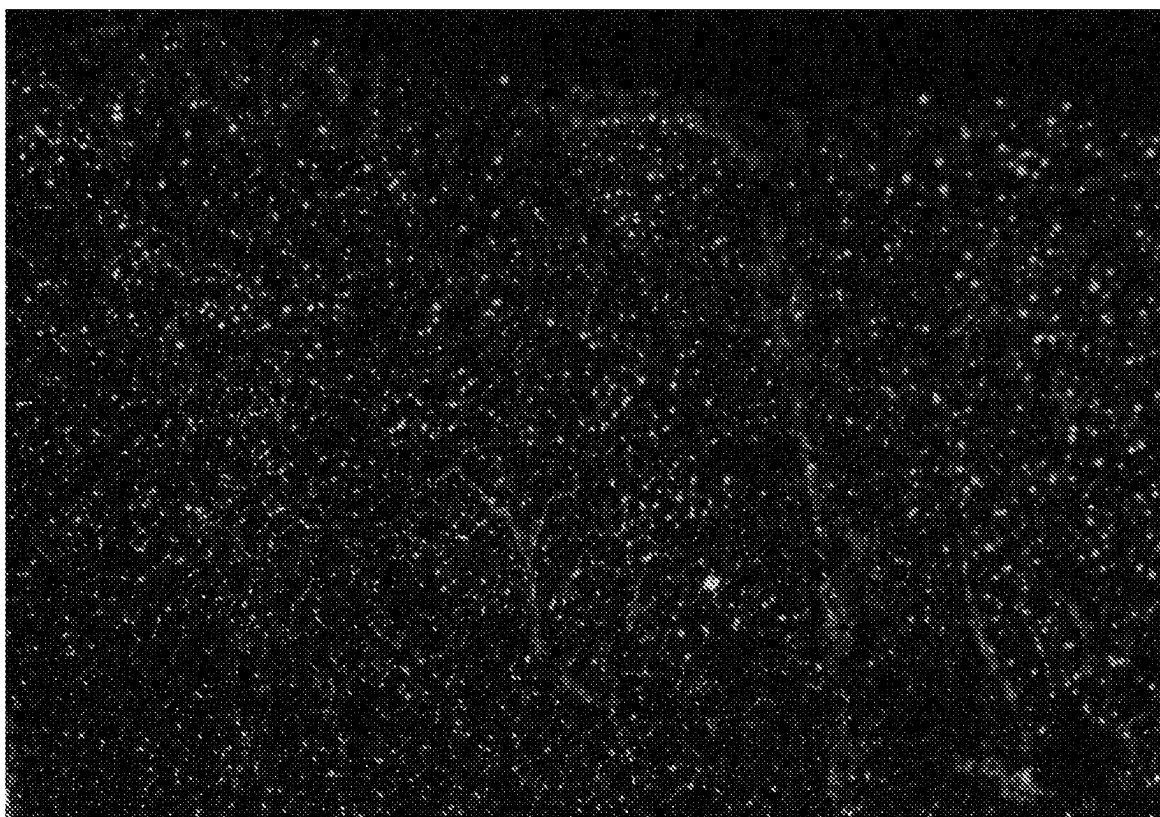


FIG. 20B

AAV8 DNA/DAPI

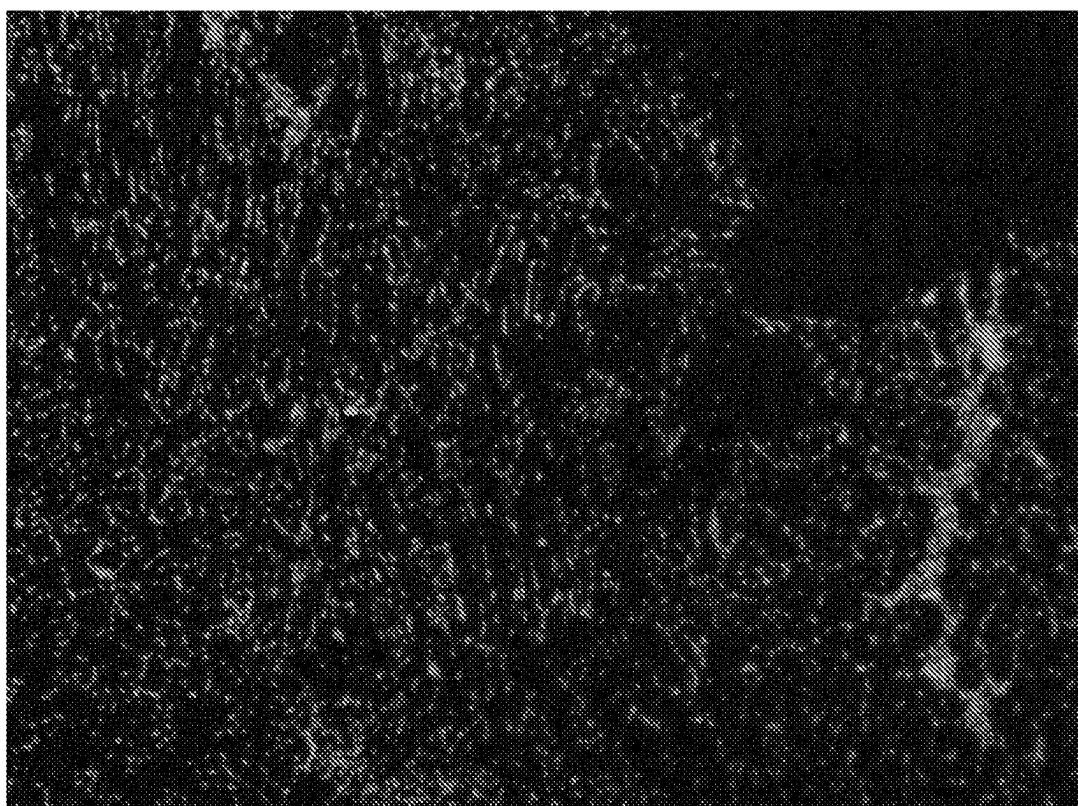


FIG. 21A

AAV9 RNA/DNA/DAPI

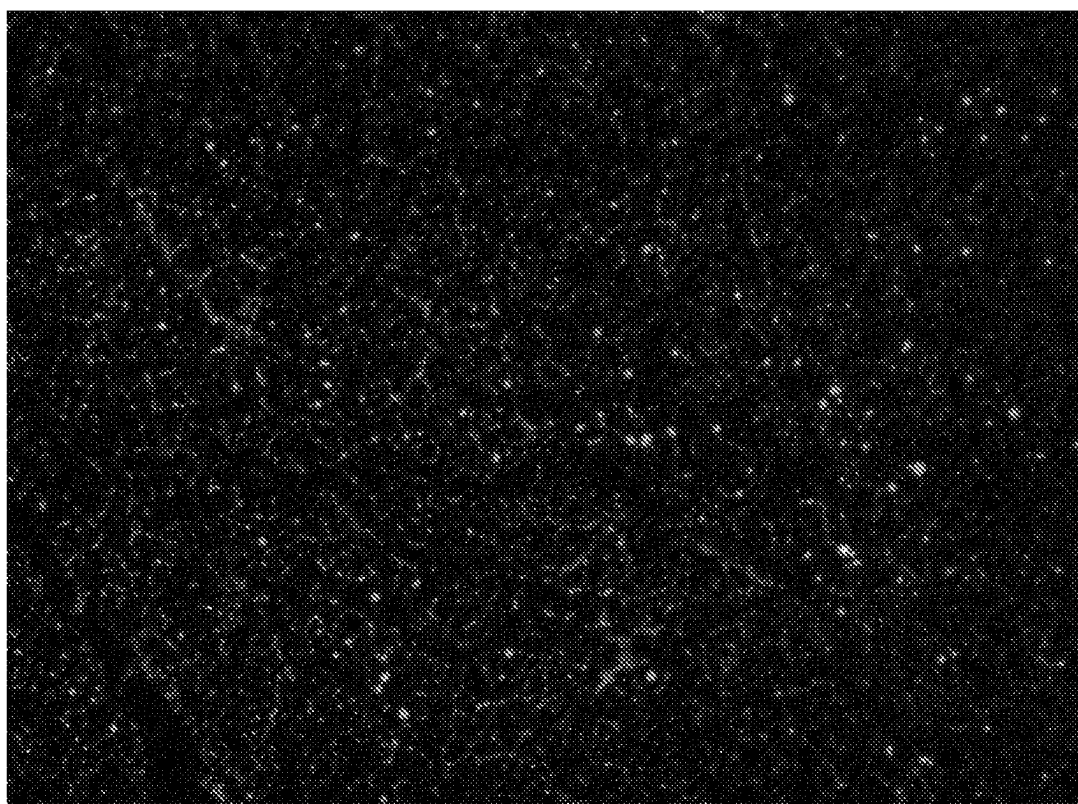


FIG. 21B

HYBRID AAV CAPSIDS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/332,132, filed Apr. 18, 2022, U.S. Provisional Patent Application No. 63/342,567, filed May 16, 2022, and U.S. Provisional Patent Application No. 63/351,796, filed Jun. 13, 2022, the content of each of which is incorporated herein by reference in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0002] This application contains a computer readable Sequence Listing which has been submitted in XML file format with this application, the entire content of which is incorporated by reference herein in its entirety. The Sequence Listing XML file submitted with this application entitled "12656-168-228_SEQ_LISTING.xml", was created on Apr. 17, 2023 and is 334,216 bytes in size.

1. FIELD

[0003] The field relates to the treatment of a disease or disorder in a subject (e.g., myopathy). Provided herein are methods and compositions involving hybrid adeno-associated viruses (hybrid AAVs) for the treatment of a disease or disorder in a subject (e.g., muscle related disease or disorder). Provided herein are hybrid adeno-associated viruses (AAVs) having hybrid capsid proteins engineered to maintain parental tropism while having improved properties (e.g., endosomal escape and nuclear trafficking properties). For example, provided herein are hybrid AAVs comprising hybrid capsid proteins comprising VP1u of AAV5, VP1/2s of AAV5, and VP3 of AAV8; VP1u of AAV8, VP1/2s of AAV5, and VP3 of AAV8, or VP1u of AAVhu32, VP1/2s of AAVhu32 and VP3 of AAVrh13. Also provided herein are hybrid capsid proteins that direct the hybrid AAVs of the disclosure to target tissues (e.g., skeletal muscle tissue) or location (e.g., targeted to a muscle related tissue or cell) in the subject, while, for example, detargeting the liver. Further provided herein are AAVhu32 capsid proteins and variant AAVhu32 capsid proteins that direct AAV vectors of the disclosure to target tissues (e.g., skeletal muscle tissue) or location (e.g., targeted to a muscle related tissue or cell) in the subject, while, for example, detargeting the liver.

2. BACKGROUND

[0004] The use of adeno-associated viruses (AAV) as gene delivery vectors is a promising avenue for the treatment of many unmet patient needs. Dozens of naturally occurring AAV capsids have been reported, and mining the natural diversity of AAV sequences in primate tissues has identified over a hundred variants, distributed in clades. AAVs belong to the parvovirus family and are single-stranded DNA viruses with relatively small genomes and simple genetic components. Without a helper virus, AAV establishes a latent infection. An AAV genome generally has a Rep gene and a Cap gene, flanked by inverted terminal repeats (ITRs), which serve as replication and packaging signals for vector production. The capsid proteins form capsids that carry genome DNA and can determine tissue tropism to deliver DNA into target cells.

[0005] Due to low pathogenicity and the promise of long-term, targeted gene expression, recombinant AAVs (rAAVs) have been used as gene transfer vectors, in which therapeutic sequences are packaged into various capsids. Such vectors have been used in preclinical gene therapy studies and several gene therapy products are currently in clinical development. However, there remains a need for hybrid AAV vectors that achieve high transduction efficiency in a specific tissue (e.g., muscle associated tissue) with minimal toxicity. There also is a need for hybrid AAV vectors with enhanced tissue-specific targeting, cell-specific tropism, and/or enhanced tissue-specific transduction to deliver therapies.

3. SUMMARY

[0006] In one aspect, provided herein is a polypeptide comprising: a) the VP1/2s region of AAV5; and b) the VP3 region of AAV8. In some aspects, the polypeptide further comprises the VP1u region of AAV5 or AAV8. In some aspects, the regions are present in the following order from amino- to carboxy-terminus of the polypeptide: a) the VP1u region of AAV5 or AAV8; b) the VP1/2s region of AAV5; and c) the VP3 region of AAV8.

[0007] In one aspect, provided herein is a polypeptide comprising: a) the VP1u region of AAVhu32; b) the VP1/2s region of AAVhu32; and c) the VP3 region of AAVrh13.

[0008] In some aspects, the VP1/2s region of AAV5 comprises SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 comprises SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 comprises SEQ ID NO: 11. In some aspects, the VP1/2s region of AAV5 comprises an amino acid sequence consisting of SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 11. In some aspects, the VP1/2s region of AAV5 consists of SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 consists of SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 consists of SEQ ID NO: 11. In some aspects, the polypeptide comprises SEQ ID NO: 23 or 25. In some aspects, the polypeptide consists of SEQ ID NO: 23 or 25. In some aspects, the polypeptide comprises and amino acid sequence consisting of SEQ ID NO: 23 or 25. In some aspects, the VP1u region of AAVhu32 comprises the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1/2s region of AAVhu32 comprises the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP3 region of AAVrh13 comprises SEQ ID NO: 27. In some aspects, the VP1u region of AAVhu32 comprises the VP1u portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1/2s region of AAVhu32 comprises the VP1/2s portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP3 region of AAVrh13 comprises an amino acid sequence consisting of SEQ ID NO: 27. In some aspects, the VP1u region of AAVhu32 consists of the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1/2s region of AAVhu32 consists of the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP3 region of AAVrh13 consists of SEQ ID NO: 27. In some aspects, the polypeptide comprises an amino acid sequence of SEQ ID NO: 29. In some aspects, the polypeptide consists

of an amino acid sequence of SEQ ID NO: 29. In some aspects, the polypeptide comprises an amino acid sequence consisting of SEQ ID NO: 29. In some aspects, provided herein is a nucleotide sequence encoding the amino acid sequence of the polypeptide. In some aspects, an expression vector comprises the nucleotide sequence. In some aspects, a host cell comprises the nucleotide sequence or the expression vector. In some aspects, a recombinant AAV comprises the polypeptide.

[0009] In one aspect, provided herein is a recombinant adeno-associated virus (AAV) comprising a capsid protein, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAV8, and wherein the rAAV comprises an expression cassette.

[0010] In some aspects, the capsid protein further comprises an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAV5 or AAV8. In some aspects, the capsid protein further comprises an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAV5 or AAV8. In some aspects, the capsid protein further comprises an amino acid sequence of the VP1u region of AAV5 or AAV8. In some aspects, the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAV8. In some aspects, the capsid protein comprises: (i) an amino acid sequence of the VP1/2s region of AAV5, and (ii) an amino acid sequence of the VP3 region of AAV8. In some aspects, the VP1/2s region of AAV5 comprises SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 comprises SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 comprises SEQ ID NO: 11. In some aspects, the VP1/2s region of AAV5 comprises an amino acid sequence consisting of SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 11. In some aspects, the VP1/2s region of AAV5 consists of SEQ ID NO: 3. In some aspects, the VP1u region of AAV5 or AAV8 consists of SEQ ID NO: 7 or 17, respectively. In some aspects, the VP3 region of AAV8 consists of SEQ ID NO: 11. In some aspects, the expression cassette comprises an open reading frame (ORF) encoding a transgene. In some aspects, the expression cassette further comprises a promoter.

[0011] In one aspect, provided herein is a recombinant adeno-associated virus (AAV) comprising a capsid protein, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAVrh13, and an expression cassette.

[0012] In some aspects, the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid

sequence that is at least about 98% sequence identical to VP3 region of AAVrh13. In some aspects, the capsid protein comprises: (i) an amino acid sequence of the VP1u region of AAVhu32, (ii) an amino acid sequence of the VP1/2s region of AAVhu32, and (iii) an amino acid sequence of the VP3 region of AAVrh13. In some aspects, the expression cassette comprises an open reading frame (ORF) encoding a transgene. In some aspects, the expression cassette further comprises a promoter.

[0013] In some aspects, the VP1u region of AAVhu32 comprises the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1u region of AAVhu32 consists of the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1u region of AAVhu32 comprises the VP1u portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 146. In some aspects, the VP1/2s region of AAVhu32 comprises the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP1/2s region of AAVhu32 consists of the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP1/2s region of AAVhu32 comprises an amino acid sequence consisting of the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148. In some aspects, the VP3 region of AAVrh13 comprises an amino acid sequence of SEQ ID NO: 27. In some aspects, the VP3 region of AAVrh13 consists of an amino acid sequence of SEQ ID NO: 27. In some aspects, the VP3 region of AAVrh13 comprises an amino acid sequence consisting of SEQ ID NO: 27. In some aspects, the capsid protein comprises SEQ ID NO: 21 or 23. In some aspects, the capsid protein consists of SEQ ID NO: 21 or 23. In some aspects, the capsid protein comprises an amino acid sequence consisting of SEQ ID NO: 21 or 23. In some aspects, the capsid protein comprises an amino acid sequence of SEQ ID NO: 29. In some aspects, the capsid protein consists of an amino acid sequence of SEQ ID NO: 29. In some aspects, the capsid protein comprises an amino acid sequence consisting of SEQ ID NO: 29. In some aspects, the rAAV further comprises an AAV inverted terminal repeat.

[0014] In some aspects, a composition comprises the rAAV and a physiologically acceptable carrier. In some aspects, the rAAV further comprises a transgene. In some aspects, provided herein is a method of delivering the transgene to a cell, comprising contacting the cell with the rAAV. In some aspects, the cell is a muscle cell. In some aspects, the transgene comprises a heterologous gene associated with a muscle related disease or disorder. In some aspects, the heterologous gene is operably linked to a regulatory sequence that controls expression of the heterologous gene in a host cell. In some aspects, the host cell is a muscle cell. In some aspects, the transgene encodes a therapeutic protein. In some aspects, the therapeutic protein is associated with a muscle related disease or disorder.

[0015] In some aspects, the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limbgirdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Ander-

sen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy. In some aspects, the muscle related disease or disorder is Duchenne muscular dystrophy. In some aspects, the transgene encodes a microdystrophin protein.

[0016] In some aspects, provided herein is a vector comprising a nucleic acid sequence encoding the capsid protein. In some aspects, the nucleic acid sequence is operably linked to a heterologous regulatory element that controls expression of the capsid protein in a host cell. In some aspects, provided herein is an in vitro cell comprising the vector.

[0017] In one aspect provided herein is a method of producing a recombinant adeno-associated virus (rAAV), the method comprising the steps of: a) culturing a cell in a cell culture to produce the rAAV, the cell comprising a nucleotide sequence encoding a capsid protein, and a nucleotide sequence comprising a transgene, and wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAV8; and b) collecting the rAAV from the cell culture.

[0018] In some aspects, the capsid protein further comprises an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAV5 or AAV8. In some aspects, the capsid protein further comprises an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAV5 or AAV8. In some aspects, the capsid protein further comprises an amino acid sequence of VP1u region of AAV5 or AAV8. In some aspects, the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAV8. In some aspects, the capsid protein comprises: (i) an amino acid sequence of VP1/2s region of AAV5, and (ii) an amino acid sequence of VP3 region of AAV8.

[0019] In one aspect provided herein is a method of producing a recombinant adeno-associated virus (rAAV), the method comprising the steps of: a) culturing a cell in a cell culture to produce the rAAV, the cell comprising a nucleotide sequence encoding a capsid protein, and a nucleotide sequence comprising a transgene, and wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 95%

sequence identical to VP3 region of AAVrh13; and b) collecting the rAAV from the cell culture.

[0020] In some aspects, the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAVrh13. In some aspects, the capsid protein comprises: (i) an amino acid sequence of VP1u region of AAVhu32, (ii) an amino acid sequence of VP1/2s region of AAVhu32, and (iii) an amino acid sequence of VP3 region of AAVrh13.

[0021] In some aspects, the cell comprises a nucleotide sequence encoding an AAV rep protein. In some aspects, the transgene is flanked by AAV inverted terminal repeats. In some aspects, the transgene comprises a heterologous gene associated with a muscle related disease or disorder. In some aspects, the heterologous gene is operably linked to a regulatory sequence that controls expression of the heterologous gene in a host cell. In some aspects, the transgene encodes a therapeutic protein. In some aspects, the therapeutic protein is associated with a muscle related disease or disorder. In some aspects, the therapeutic protein encodes microdystrophin protein. In some aspects, the transgene encodes a functional gene product. In some aspects, the cell is selected from an invertebrate cell, an insect cell, or a mammalian cell. In some aspects, the mammalian cell is selected from HEK293, HeLa, CHO, NS0, SP2/0, PER.C6, Vero, RD, BHK, HT-1080, A549, Cos-7, ARPE-19, MRC-5, or any combination thereof. In some aspects, the insect cell is selected from High Five, Sf9, Se301, SeIZD2109, SeUCR1, Sf900+, Sf21, Bti-tn-5b1-4, MG-1, Tn368, HzAm1, BM-N, Ha2302, Hz2E5, Ao38, or any combination thereof.

[0022] In some aspects, provided herein is a method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering the rAAV or the composition to the subject. In some aspects, the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG),

Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy. In some aspects, the muscle related disease or disorder is muscular dystrophy. In some aspects, the muscular dystrophy is Duchenne muscular dystrophy. In some aspects, the rAAV provides at least one improvement of packaging efficiency, yield, titer, infectivity, transduction efficiency, and transfection efficiency, as compared to a reference AAV. In some aspects, the reference AAV is AAV8. In some aspects, the reference AAV comprises VP1u of AAV8. In some aspects, the reference AAV is AAV Rh13. In some aspects, the improvement is by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold. In some aspects, the improvement is by about or at least about 2.5-fold. In some aspects, the at least one improvement is an improvement in packaging efficiency. In some aspects, the at least one improvement is an improvement in titer. In some aspects, the rAAV transduces muscle and/or liver tissues at lower levels as compared to a reference AAV. In some aspects, the rAAV results in lower RNA expression in liver as compared to a reference AAV. In some aspects, the RNA expression is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the RNA expression from a reference AAV. In some aspects, the rAAV results in higher RNA/DNA ratio in muscle tissue as compared to a reference AAV. In some aspects, the RNA/DNA ratio is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some aspects, the rAAV results in lower DNA levels in muscle tissue as compared to a reference AAV. In some aspects, the DNA levels is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the DNA levels from a reference AAV. In some aspects, the rAAV results in a lower level of liver toxicity as compared to the level of liver toxicity from a reference AAV. In some aspects, the level of liver toxicity is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the level of liver toxicity from a reference AAV. In some aspects, the rAAV results in transcription-specific liver de-targeting as compared to a reference AAV. In some aspects, the administering comprises administering a lower amount of vector genome of the rAAV as compared to the amount of vector genome of a reference AAV necessary to obtain the same therapeutic effect after the administering. In some aspects, the reference AAV is AAV8. In some aspects, the reference AAV comprises VP1u of AAV8. In some aspects, the reference AAV is AAVrh13.

[0023] In one aspect, provided herein is a polypeptide comprising one or more of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71. In one aspect, provided herein is a polypeptide consisting of one or more of any one of SEQ ID NOs: 1, 3, 5, 7,

9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71. In one aspect, provided herein is a polypeptide comprising one or more of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In one aspect, provided herein is a polypeptide consisting of one or more of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In one aspect, provided herein is a polynucleotide encoding an AAV capsid protein comprising the amino acid sequence of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71. In one aspect, provided herein is a polynucleotide encoding an AAV capsid protein consisting of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71. In one aspect, provided herein is a polynucleotide comprising any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and 72, wherein the polynucleotide encodes an AAV capsid protein. In one aspect, provided herein is a polynucleotide consisting of any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and 72, wherein the polynucleotide encodes an AAV capsid protein. In one aspect, provided herein is a polynucleotide encoding an AAV capsid protein comprising the amino acid sequence of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In one aspect, provided herein is a polynucleotide encoding an AAV capsid protein consisting of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In one aspect, provided herein is a polynucleotide comprising any one of SEQ ID NOs: 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120, wherein the polynucleotide encodes an AAV capsid protein. In one aspect, provided herein is a polynucleotide consisting of any one of SEQ ID NOs: 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120, wherein the polynucleotide encodes an AAV capsid protein.

[0024] In one aspect, provided herein is a method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering an rAAV comprising a polypeptide comprising an amino acid sequence of SEQ ID NO:31 or an rAAV encoded by SEQ ID NO:32. In some aspects, the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (sub-

acute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy. In some aspects, the muscle related disease or disorder is muscular dystrophy. In some aspects, the muscular dystrophy is Duchenne muscular dystrophy. In some aspects, the rAAV further comprises a transgene encoding a functional dystrophin, a minidystrophin, a microdystrophin, and/or dystrophin exon-skipping snRNA. In some aspects, the transgene comprises a polynucleotide sequence encoding any one of SEQ ID NOs: 73, 74, 75, 76, 77, 78, 79, or 80. In some aspects, the rAAV provides at least one improvement of packaging efficiency, yield, titer, infectivity, transduction efficiency, and transfection efficiency, as compared to a reference AAV, wherein the reference AAV is AAV9.

[0025] In some aspects, the rAAV provides a higher vector yield after the rAAV is administered to the subject, as compared to a reference AAV. In some aspects, the vector yield is higher by about or at least about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some aspects, the vector yield is higher by between about 1.5 to about 3 times. In some aspects, the RNA level is increased after the rAAV is administered to the subject, as compared to a reference AAV. In some aspects, the RNA level is increased by about or at least about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some aspects, the RNA level is increased by between about 2.5 to about 3.5 times. In some aspects, the ratio of RNA to DNA is increased after the rAAV is administered to the subject. In some aspects, the ratio of RNA to DNA is increased by about or at least about 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some aspects, the ratio of RNA to DNA is increased by between about 2.5 to about 3.5 times. In some aspects, the reference AAV is AAV5. In some aspects, the reference AAV is AAV8. In some aspects, the reference AAV comprises the expression cassette.

[0026] In one aspect, provided herein is a recombinant adeno-associated virus (rAAV) hu.32 capsid protein comprising SEQ ID NO: 31, or an rAAV capsid protein comprising an amino acid sequence that is about or at least about 90% identical to SEQ ID NO: 31, comprising a peptide insertion of at least 4 and up to 12 contiguous amino acids from a heterologous protein that is not an AAV protein, wherein the peptide insertion is inserted immediately after an amino acid residue corresponding to one of any one of amino acids 451 to 461 of AAVhu32 capsid protein of SEQ ID NO: 31.

[0027] In one aspect, provided herein is a recombinant adeno-associated virus (rAAV) hu.32 capsid protein comprising SEQ ID NO: 31, or an rAAV capsid protein com-

prising an amino acid sequence that is about or at least about 90% identical to SEQ ID NO: 31, comprising a peptide insertion of at least 4 and up to 12 contiguous amino acids from a heterologous protein that is not an AAV protein, wherein the peptide insertion is inserted immediately after an amino acid residue corresponding to one of any one of amino acids 570 to 595 of AAVhu32 capsid protein of SEQ ID NO: 31.

[0028] In some aspects, the peptide insertion is between amino acid 454 and amino acid 455 of SEQ ID NO: 31. In some aspects, the peptide insertion is between amino acid 589 and amino acid 590 of SEQ ID NO: 31. In some aspects, the peptide insertion is a muscle-homing peptide. In some aspects the muscle-homing peptide comprises an integrin receptor-binding domain or an integrin-binding domain.

[0029] In one aspect, provided herein is a recombinant adeno-associated virus (rAAV) capsid protein comprising an amino acid sequence that is about or at least about 90% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some aspects, the rAAV capsid protein comprises an amino acid sequence that is about or at least about 98% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some aspects, the rAAV capsid protein comprises an amino acid sequence of any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some aspects, provided herein is a recombinant adeno-associated virus (rAAV) vector comprising a nucleic acid sequence encoding the rAAV capsid protein.

[0030] In one aspect, provided herein is a recombinant adeno-associated virus (rAAV) vector comprising a nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some aspects, provided herein is a cell comprising the rAAV vector. In some aspects, provided herein is a cell expressing the rAAV capsid protein. In some aspects, provided herein is a recombinant AAV viral particle comprising the rAAV capsid protein. In some aspects, a first AAV viral particle comprising the rAAV capsid protein results in increased transduction of muscle cells or heart cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 19, 31, and 124. In some aspects, the first AAV viral particle has at least about 1.5 fold increase in transduction of muscle cells or heart cells relative to the second AAV viral particle. In some aspects, a first AAV viral particle comprising the rAAV capsid protein has reduced transduction of liver cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 11, 15, 19, 31, and 124. In some aspects, the first AAV viral particle has about or at least about 1 fold decrease in transduction of liver cells relative to the second AAV viral particle. In some aspects, the second AAV viral particle comprises an AAVhu32, AAV8, and/or AAV9 capsid protein.

[0031] In one aspect provided herein is a pharmaceutical composition comprising a recombinant adeno-associated

virus (rAAV) vector comprising a recombinant AAV capsid protein, wherein the recombinant AAV capsid protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

[0032] In some aspects provided herein is a method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering the rAAV viral particle, the cell, the rAAV vector, or the pharmaceutical composition to the subject. In some aspects, the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy. In some aspects, the muscle related disease or disorder is muscular dystrophy. In some aspects, the muscular dystrophy is Duchenne muscular dystrophy.

[0033] In some aspects, the rAAV further comprises a transgene encoding a functional dystrophin, a minidystrophin, a microdystrophin, or a dystrophin exon-skipping snRNA.

[0034] In one aspect, provided herein is an isolated nucleic acid molecule comprising a nucleic acid sequence encoding the rAAV capsid protein. In one aspect, provided herein is an isolated nucleic acid molecule comprising a nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some aspects, provided herein is a cultured cell comprising the nucleic acid molecule.

4. BRIEF DESCRIPTION OF THE FIGURES

[0035] The foregoing and other objects, features and advantages will be apparent from the following description of particular embodiments of the invention, as illustrated in the accompanying drawings. The drawings are not neces-

sarily to scale, emphasis instead being placed upon illustrating the principles of various embodiments of the invention.

[0036] FIG. 1 shows an illustration of the design of hybrid capsids. Hybrid 1 features a hybrid capsid protein having a VP1u and a VP1/2s of AAV5 and a VP3 of AAV8. Hybrid 2 features a hybrid capsid protein having a VP1u of AAV8, a VP1/2s of AAV5, and a VP3 of AAV8.

[0037] FIG. 2 shows packaging efficiency studies of hybrid AAVs comprising hybrid capsid proteins (Hybrid 1 and Hybrid 2). Packaging efficiency was measured by titer, expressed as genome copies per mL (GC/mL), and observed on Day 3 (left) and Day 5 (right).

[0038] FIGS. 3A-3C show packaging efficiency studies of hybrid AAVs expressing various transgenes. GFP transgene (FIG. 3A) or a large muscle protein-encoding transgene (FIG. 3B) were packaged in either wtAAV5, wtAAV8, Hybrid 1, or Hybrid 2, and transgene titers were measured from a 1 mL scale cell culture at 5 days post transfection. FIG. 3C shows the fold change over AAV8.

[0039] FIG. 4 illustrates the relative abundance of input vector pool compiled for the NHP study. The dotted line indicates target input for each vector, showing that each vector is well represented in the pool.

[0040] FIG. 5 shows biodistribution of hybrid and parental AAV vector pool assessed in NHP by obtaining DNA from various tissues harvested 21 days after dosing. FIG. 5 represents relative abundance of the vector pool in the liver by DNA vector genome copies. Bars in the graph are grouped together by shading according to same parental VP3 sequence as indicated in Table 1. Capsids containing AAV5 VP1u and/or VP1/2s region exhibit lower transduction (genome copy) in liver, e.g. Hybrid 1 (2171), Hybrid 2 (2172), 2092, and 2095, compared to their parental capsids.

[0041] FIG. 6 shows biodistribution of hybrid and parental AAV vector pool assessed in NHP by obtaining RNA quantitated from various tissues harvested 21 days after dosing. FIG. 6 represents relative abundance of the vector pool transgene in the liver by mRNA (cDNA) transcript copies. Bars in the graph are grouped together by shading according to same parental VP3 sequence as indicated in Table 1. Capsids containing AAV5 VP1u and/or VP1/2s region exhibit lower expression (RNA transcripts) in liver, e.g. Hybrid 1 (2171), Hybrid 2 (2172), 2092, and 2095, compared to their parental capsids.

[0042] FIGS. 7A-7C show biodistribution of hybrid and parental AAV vector pool assessed in NHP, by obtaining DNA and RNA quantitated in various tissues harvested 21 days after dosing. AAV5/AAV8 hybrid capsids Hybrid 1 (white bar) and Hybrid 2 (gray bar) transduce less efficiently than AAV8 (black bar; FIG. 7A) but are liver de-targeted on the transcriptional level (FIG. 7B) and express RNA more efficiently than wild type AAV8 in muscle, leading to an improved RNA/DNA ratio (2.5-3.5x increase over AAV8; FIG. 7C). Data represented is from one neutralizing antibody negative animal.

[0043] FIGS. 8A-8C show graphs depicting AAV hybrids assessed for productivity. AAV5/AAV8 hybrids, Hybrid 1 (white bar) and Hybrid 2 (gray bar), produce 1.5-3x more vector than their respective parental capsid, AAV8 (dotted line), in standard triple transfection of HEK293 suspension culture, regardless of the packaged genome size (FIG. 8A; 1

mL culture) or culture scale (FIG. 8B and FIG. 8C; 10 mL and 1 L culture, respectively). Lysate titers shown as a ratio over AAV8.

[0044] FIG. 9 illustrates an alignment of N-terminal VP1 amino acids (1-58) of AAVhu32 and AAV9 capsid proteins. The consensus sequence depicted (SEQ ID NO: 113) has the following variable amino acids (represented by X in the sequence listing and in the sequence Table at Section 7 of the disclosure): 14X=N or T; 21X=E or Q; 24X=A or K; 29X=A or P; 31X=Q or P; 33X=A or P; 34X=A or P; 35X=N or A; 36X=Q or E; 37X=Q or R; 39X=Q or K; 41X=N or D; and 42X=A or S.

[0045] FIGS. 10A-10B show the relative abundance of vector genome copies (DNA;

[0046] FIG. 10A) and transcript copies (RNA; FIG. 10B) of and AAVhu32 and AAV9 in a pooled vector NHP study.

[0047] FIGS. 11A-11B show biodistribution of AAVhu32 and AAV9 in mdx mouse tissues after intravenous administration. AAVhu32 transduction of muscle tissues are compared to AAV9, in terms of genome copy numbers (GC/cell; FIG. 11A) and RNA transcript copies (FIG. 11B).

[0048] FIGS. 12A-12B show graphs of engineered AAVhu32 capsids tested for transduction of skeletal muscle in wild type (FIG. 12A) and mdx (FIG. 12B) mice. Relative abundance of transcript copies was adjusted based on the vector input and presented as a fold change over AAVhu32.

[0049] FIG. 13 shows sequence alignment for different AAV serotypes

[0050] FIG. 14 shows AAVhu32 capsid variants following administration in the capsid pool and analyzed for the fold difference in transduction in muscle tissues versus liver compared to wild-type AAVhu32 (transcript levels). All skeletal muscle transcript data represent all bicep, gas, quad, TA, and diaphragm tissue; heart transcript data represents samples taken from aorta, apex, left and right atrium, left and right ventricle; and liver transcript data was from the left lobe sample from 2 animals.

[0051] FIGS. 15A-15B illustrate AA Vhu32 capsid variants #1 to 21 as analyzed compared to AAV8, AAV9, and wild-type AAVhu32 and the relative abundance of mRNA transcript copies (of transgene) for transduced capsid as detected in gastrocnemius (gas) (FIG. 15A) and quadriceps (quad) (FIG. 15B) tissues of wt and mdx mice.

[0052] FIGS. 16A-16B illustrate AA Vhu32 capsid variants #1 to 21 as analyzed compared to AAV8, AAV9, and wild-type AAVhu32 and the relative abundance of mRNA transcript copies (of transgene) for transduced capsid as detected in tibialis anterior (TA) (FIG. 16A) and heart (FIG. 16B) tissues of wt and mdx mice.

[0053] FIG. 17 illustrates AAVhu32 capsid variants #1 to 21 as analyzed compared to AAV8, AAV9, and wild-type AAVhu32 and the relative abundance of mRNA transcript copies (of transgene) for transduced capsid as detected in livers of wt and mdx mice.

[0054] FIGS. 18A-18B show the biodistribution of AAVhu32 and AAV9 in mdx mouse tissues after intravenous administration of AAVhu32, AAV9, or AAV8 (heart, FIG. 18A) or AAVhu32 or AAV9 (gastrocnemius/gas, FIG. 18B). Protein quantification of the transgene and a representative muscle protein, actin, is represented from a Western Blot.

[0055] FIG. 19 shows the immunofluorescent staining of gastrocnemius (gas) tissues from AAVhu32, AAV8 or AAV9-injected mdx mice expressing microdydtrophin protein.

[0056] FIGS. 20A-20B RNASCOPE experiments were performed on mdx mouse tissues following AAVhu32-microdystrophin vector systemic administration. DNA and RNA probes detected vector or transgene in gastrocnemius (gas) tissues, and scoring (semi-quantification) was based on the number of green (AAV-microdystrophin DNA), red (AAV-microdystrophin RNA/DNA), or blue (DAPI) signals (dots) per cell, for semi-quantification of the transduction events. AAVhu32-microdystrophin DNA detection was observed at about 3.73 GC/diploid cell (FIG. 20A) and RNA detection at about 82.3 microdystrophin RNA copies/TBP (TBP=TATA-box binding protein control) (FIG. 20B).

[0057] FIGS. 21A-21B illustrate RNASCOPE experiments performed analogously to that described for FIGS. 20A and 20B. AAV8-microdystrophin DNA detection was observed at about 1 copy per cell (FIG. 21A), and AAV9-microdystrophin DNA detection was observed at about 1.4 copies per cell (FIG. 21B).

5. DETAILED DESCRIPTION

[0058] Described herein are hybrid AAV capsid polypeptides (e.g., VP1u and/or VP1/2s from a first AAV and VP3 from a second AAV) as described in Section 5.1.1. Described herein are recombinant adeno-associated viruses (rAAVs) comprising hybrid AAV capsid proteins as described in Section 5.2.1. In some embodiments, such rAAVs further comprise a transgene (refer to Section 5.2.4) and have enhanced functional properties as compared to a reference AAV (e.g., an AAV that does not comprise a hybrid AAV capsid of the disclosure), as described in Section 5.2.2. In some embodiments, these rAAVs can be used to treat a disease or disorder in a subject. In some embodiments, an rAAV of the disclosure or compositions comprising the same can be used to treat or prevent a muscle related disease or disorder in a subject as described in Section 5.4. Also provided herein are rAAV vectors as described in Section 5.2.3, and pharmaceutical compositions comprising the rAAVs of the disclosure, such as described in Section 5.2.5. Also provided herein are methods of manufacturing the rAAVs of the disclosure, such as described in Section 5.3. Non-limiting illustrative examples are provided in Section 6.

5.1 Hybrid AAV Capsid Polypeptides

5.1.1 Polypeptides

[0059] Provided herein are polypeptides comprising hybrid adeno-associated virus capsid proteins. In various aspects of the disclosure, a polypeptide comprises a VP1u region and/or a VP1/2s region of a first adeno-associated virus serotype, and a VP3 region of a second adeno-associated virus serotype. In some embodiments, a polypeptide comprises a VP1u region from a first AAV serotype, a VP1/2s region of a second AAV serotype, and a VP3 region of a first AAV serotype. In some embodiments, a polypeptide comprises a VP1u region from a first AAV serotype, a VP1/2s region of a second AAV serotype, and a VP3 region of a third AAV serotype. In some embodiments, a polypeptide comprises a VP1/2s region of a second AAV serotype and a VP3 region of a first AAV serotype. In some embodiments, a polypeptide comprises a VP1u region of a second AAV serotype, a VP1/2s region of the second AAV serotype, and a VP3 region of a first AAV serotype. In some embodiments, a polypeptide comprises a VP1u region of a first AAV

serotype, a VP1/2s region of a second AAV serotype, and a VP3 region of the first AAV serotype.

[0060] In some embodiments, the first AAV is an AAV serotype 1 (AAV1), serotype 2 (AAV2), serotype 2yYF (AAV2tYF), serotype 3 (AAV3), serotype 3 (AAV3B), serotype 4 (AAV4), serotype 5 (AAV5), serotype 6 (AAV6), serotype 7 (AAV7), serotype 8 (AAV8), serotype rh8 (AAVrh8), serotype 9 (AAV9), serotype 10 (AAV10), serotype 11 (AAV11), serotype 12 (AAV12), serotype 13 (AAV13), serotype rh10 (AAVrh10), serotype rh20 (AAVrh20), serotype rh39 (AAVrh39), serotype hu.37 (AAVhu.37), serotype hu32 (AAVhu32), serotype rh13 (AAVrh13), serotype rh15 (AAVrh15), serotype rh73 (AAVrh73) and serotype rh74 (AAVrh74). In some embodiments, the second AAV is an AAV serotype 1 (AAV1), serotype 2 (AAV2), AAV2tYF, serotype 3 (AAV3), serotype 3 (AAV3B), serotype 4 (AAV4), serotype 5 (AAV5), serotype 6 (AAV6), serotype 7 (AAV7), serotype 8 (AAV8), serotype rh8 (AAVrh8), serotype 9 (AAV9), serotype 10 (AAV10), serotype 11 (AAV11), serotype 12 (AAV12), serotype 13 (AAV13), serotype rh10 (AAVrh10), serotype rh20 (AAVrh20), serotype rh39 (AAVrh39), serotype hu.37 (AAVhu.37), serotype hu32 (AAVhu32), serotype rh13 (AAVrh13), serotype rh15 (AAVrh15), serotype rh73 (AAVrh73) and serotype rh74 (AAVrh74). In some embodiments, the first AAV is an AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh10, AAV11, AAV12, AAV13, AAVrh20, AAVhu.37, AAVrh39, AAVhu32, AAVrh21, AAVrh13, AAVrh15, AAVrh73, or AAVrh74. In some embodiments, the second AAV is an AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh10, AAV11, AAV12, AAV13, AAVrh20, AAVhu.37, AAVrh39, AAV hu32, AAVrh21, AAVrh13, AAVrh15, AAVrh73, or AAVrh74.

[0061] In some embodiments, one AAV (e.g., first AAV) is AAV5 and one AAV (e.g., second AAV) is AAV8. In some embodiments, one AAV (e.g., first AAV) is AAV8 and one AAV (e.g., second AAV) is AAV5. In some embodiments, one AAV (e.g., first AAV) is AAVhu32 and one AAV (e.g., second AAV) is AAVrh13. In some embodiments, one AAV (e.g., first AAV) is AAVrh13 and one AAV (e.g., second AAV) is AAVhu32. In some embodiments, the first AAV serotype is AAV8. In some embodiments, the first AAV serotype is AAVrh13. In some embodiments, the second AAV serotype is AAV5. In some embodiments, the second AAV serotype is AAVhu32. In some embodiments, the second AAV serotype is AAV8. In some embodiments, the second AAV serotype is AAVrh13. In some embodiments, the first AAV serotype is AAV5. In some embodiments, the first AAV serotype is AAVhu32. In some embodiments, the first AAV serotype is AAV8 and the second AAV serotype is AAV5. In specific embodiments, the first AAV serotype is AAVrh13 and the second AAV serotype is AAVhu32. In some embodiments, the second AAV serotype is AAV8 and the first AAV serotype is AAV5. In some embodiments, the second AAV serotype is AAVrh13 and the first AAV serotype is AAVhu32.

[0062] In some embodiments, provided herein is a polypeptide comprising: a) the VP1/2s region of a first AAV (e.g., AAV5); and the VP3 region of a second AAV (e.g., AAV8). In some embodiments, the polypeptide further comprises the VP1u region of a first AAV (e.g., AAV5) or a second AAV (e.g., AAV8). In some embodiments, the regions are present in the following order from amino-

carboxy-terminus of the polypeptide: VP1u followed by VP1/2s followed by VP3. In some embodiments, the regions are present in the following order from amino- to carboxy-terminus of the polypeptide: the VP1u region of a first AAV (e.g., AAV5) or a second AAV (e.g., AAV8); the VP1/2s region of a first AAV (e.g., AAV5); and the VP3 region of a second AAV (e.g., AAV8). In some embodiments, a polypeptide of the disclosure comprises the VP1u region of a first AAV (e.g., AAVhu32), the VP1/2s region of a first AAV (e.g., AAVhu32), and the VP3 region of a second AAV (e.g., AAVrh13).

[0063] In certain embodiments, a polypeptide of the disclosure comprises a VP3 region of AAV5. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:1. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:1. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:1. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:1.

[0064] In certain embodiments, a polypeptide of the disclosure comprises a VP1/2s region of AAV5. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:3. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:3. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:3. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:3.

[0065] In certain embodiments, a polypeptide of the disclosure comprises a VP1/2s region and VP3 region of AAV5. In certain embodiments, a polypeptide of the disclosure comprises a VP2 of AAV5. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:3 and/or an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:1. In some embodiments, a polypeptide comprises the amino acid sequences of SEQ ID NO:3 and/or SEQ ID NO:1. In some embodiments, a polypeptide consists of SEQ ID NO:3 and/or SEQ ID NO:1. In some embodiments, a polypeptide comprises an amino acid sequence consisting of SEQ ID NO:3 and/or SEQ ID NO:1.

[0066] In some embodiments, a polypeptide comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:5. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:5. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:5.

sequence of SEQ ID NO:27. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:27.

[0079] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region of AAVhu32, a VP1/2s region of AAVhu32, and a VP3 region of AAVrh13. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:29. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:29. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:29. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:29. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAVhu32 VP1u) of SEQ ID NO:29 or SEQ ID NO: 146. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAVhu32 VP1/2) of SEQ ID NO: 29 or SEQ ID NO: 148. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVrh13 VP3) of SEQ ID NO:29. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAVhu32 VP1u) of SEQ ID NO:29 or SEQ ID NO: 146. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAVhu32 VP1/2) of SEQ ID NO:29 or SEQ ID NO: 148. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAVrh13 VP3) of SEQ ID NO:29. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAVhu32 VP1u) of an amino acid sequence consisting of SEQ ID NO:29 or SEQ ID NO: 146. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAVhu32 VP1/2) of an amino acid sequence consisting of SEQ ID NO: 29 or SEQ ID NO: 148. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVrh13 VP3) of an amino acid sequence consisting of SEQ ID NO:29. In some embodiments a nucleic acid sequence encoding VP1u portion (e.g., AAVhu32 VP1/2) of the rAAV is SEQ ID NO: 147. In some embodiments a nucleic acid sequence encoding VP1/2s portion (e.g., AAVhu32 VP1/2) of the rAAV is SEQ ID NO: 149.

[0080] In certain embodiments, a polypeptide of the disclosure comprises a VP1 region of AAVhu32. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:31. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:31. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:31. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:31.

[0081] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAV8. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:33. In some embodiments, a polypeptide comprises the amino acid

sequence of SEQ ID NO:33. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:33. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:33. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO: 33. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:33. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of an amino acid sequence consisting of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of an amino acid sequence consisting of SEQ ID NO:33. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of an amino acid sequence consisting of SEQ ID NO: 33.

[0082] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAV rh.21. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:35. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:35. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:35. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:35. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO:35. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:35. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.21 VP3) of SEQ ID NO:35. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO: 35. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:35. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV rh.21 VP3) of SEQ ID NO:35. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of an amino acid sequence consisting of SEQ ID NO: 35. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of an amino acid sequence consisting of SEQ ID NO:35. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.21 VP3) of an amino acid sequence consisting of SEQ ID NO:35.

[0083] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAV rh.73. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:37. In some embodiments, a polypeptide comprises the

amino acid sequence of SEQ ID NO:49. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:49. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV9 VP1u) of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV9 VP1/2) of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh18 VP3) of SEQ ID NO:49. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV9 VP1u) of SEQ ID NO: 49. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV9 VP1/2) of SEQ ID NO:49. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV rh18 VP3) of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV9 VP1u) of an amino acid sequence consisting of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV9 VP1/2) of an amino acid sequence consisting of SEQ ID NO:49. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh18 VP3) of an amino acid sequence consisting of SEQ ID NO:49.

[0090] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV rh.21, and a VP3 region of AAV8. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:51. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:51. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:51. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.21 VP1u) of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh.21 VP1/2) of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:51. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV rh.21 VP1u) of SEQ ID NO:51. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV rh.21 VP1/2) of SEQ ID NO:51. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.21 VP1u) of an amino acid sequence consisting of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh.21 VP1/2) of an amino acid sequence consisting of SEQ ID NO:51. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of an amino acid sequence consisting of SEQ ID NO:51.

[0091] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV rh73, and a VP3 region of AAV8. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:53. In some embodiments, a polypeptide comprises the

amino acid sequence of SEQ ID NO:53. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:53. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh73 VP1u) of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh73 VP1/2) of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:53. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV rh73 VP1u) of SEQ ID NO:53. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV rh73 VP1/2) of SEQ ID NO:53. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh 73 VP1u) of an amino acid sequence consisting of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh73 VP1/2) of an amino acid sequence consisting of SEQ ID NO:53. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of an amino acid sequence consisting of SEQ ID NO:53.

[0092] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV3B, and a VP3 region of AAV8. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:55. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:55. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:55. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:55. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV3B VP1u) of SEQ ID NO:55. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV3B VP1/2) of SEQ ID NO:55. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:55. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV3B VP1u) of SEQ ID NO:55. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV3B VP1/2) of SEQ ID NO:55. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV8 VP3) of SEQ ID NO:55. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV3B VP1u) of an amino acid sequence consisting of SEQ ID NO: 55. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV3B VP1/2) of an amino acid sequence consisting of SEQ ID NO:55. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV8 VP3) of an amino acid sequence consisting of SEQ ID NO:55.

[0093] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV rh. 15, and a VP3 region of AAVhu32. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:57. In some embodiments, a polypeptide com-

prises the amino acid sequence of SEQ ID NO:57. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:57. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.15 VP1u) of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh. 15 VP1/2) of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVhu32 VP3) of SEQ ID NO: 57. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV rh. 15 VP1u) of SEQ ID NO:57. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV rh.15 VP1/2) of SEQ ID NO:57. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAVhu32 VP3) of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.15 VP1u) of an amino acid sequence consisting of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh. 15 VP1/2) of an amino acid sequence consisting of SEQ ID NO:57. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVhu32 VP3) of an amino acid sequence consisting of SEQ ID NO:57.

[0094] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV rh.13, and a VP3 region of AAVhu32. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:59. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:59. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:59. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.13 VP1u) of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh. 13 VP1/2) of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVhu32 VP3) of SEQ ID NO: 59. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV rh.13 VP1u) of SEQ ID NO:59. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV rh.13 VP1/2) of SEQ ID NO:59. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAVhu32 VP3) of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV rh.13 VP1u) of an amino acid sequence consisting of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV rh. 13 VP1/2) of an amino acid sequence consisting of SEQ ID NO:59. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAVhu32 VP3) of an amino acid sequence consisting of SEQ ID NO:59.

[0095] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAVhu32. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID

[0097] In certain embodiments, a polypeptide of the disclosure comprises a VP1u regio and/or VP1/2s region of AAV hu.32, and a VP3 region of AAV rh.15. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to

SEQ ID NO:65. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:65. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:65. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:65. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV hu.32 VP1u) of SEQ ID NO:65. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV hu.32 VP1/2) of SEQ ID NO:65. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.15 VP3) of SEQ ID NO: 65. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV hu.32 VP1u) of SEQ ID NO:65. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV hu.32 VP1/2) of SEQ ID NO:65. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV rh. 15 VP3) of SEQ ID NO:65. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV hu.32 VP1u) of an amino acid sequence consisting of SEQ ID NO:65. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV hu.32 VP1/2) of an amino acid sequence consisting of SEQ ID NO: 65. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.15 VP3) of an amino acid sequence consisting of SEQ ID NO:65.

[0098] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAV ru.15. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:67. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:67. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:67. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:67. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO:67. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:67. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV ru. 15 VP3) of SEQ ID NO:67. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO: 67. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:67. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV ru. 15 VP3) of SEQ ID NO:67. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of an amino acid sequence consisting of SEQ ID NO: 67. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of an amino acid sequence consisting of SEQ ID NO:67. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV ru.15 VP3) of an amino acid sequence consisting of SEQ ID NO:67.

[0099] In certain embodiments, a polypeptide of the disclosure a VP1u region and/or VP1/2s region of AAV5, and a VP3 region of AAV rh.15. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%.

98%, or 99% sequence identity to SEQ ID NO:69. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:69. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:69. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:69. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV5 VP1u) of SEQ ID NO:69. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV5 VP1/2) of SEQ ID NO:69. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.15 VP3) of SEQ ID NO:69. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV5 VP1u) of SEQ ID NO: 69. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV5 VP1/2) of SEQ ID NO:69. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV rh. 15 VP3) of SEQ ID NO:69. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV5 VP1u) of an amino acid sequence consisting of SEQ ID NO: 69. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV5 VP1/2) of an amino acid sequence consisting of SEQ ID NO:69. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.15 VP3) of an amino acid sequence consisting of SEQ ID NO:69.

[0100] In certain embodiments, a polypeptide of the disclosure comprises a VP1u region and/or VP1/2s region of AAV6, and a VP3 region of AAV rh.13. In some embodiments, a polypeptide comprises an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:71. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO:71. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:71. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO:71. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO:71. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:71. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.13 VP3) of SEQ ID NO:71. In some embodiments, a polypeptide consists of the VP1u portion (e.g., AAV6 VP1u) of SEQ ID NO: 71. In some embodiments, a polypeptide consists of the VP1/2s portion (e.g., AAV6 VP1/2) of SEQ ID NO:71. In some embodiments, a polypeptide consists of the VP3 portion (e.g., AAV rh. 13 VP3) of SEQ ID NO:71. In some embodiments, a polypeptide comprises the VP1u portion (e.g., AAV6 VP1u) of an amino acid sequence consisting of SEQ ID NO: 71. In some embodiments, a polypeptide comprises the VP1/2s portion (e.g., AAV6 VP1/2) of an amino acid sequence consisting of SEQ ID NO:71. In some embodiments, a polypeptide comprises the VP3 portion (e.g., AAV rh.13 VP3) of an amino acid sequence consisting of SEQ ID NO:71.

[0101] In some embodiments, the polypeptide of the disclosure comprises about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53,

55, 57, 59, 61, 63, 65, 67, 69, and/or 71. In some embodiments, a polypeptide comprises the amino acid sequence of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and/or 71. In some embodiments, a polypeptide consists of the amino acid sequence of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and/or 71. In some embodiments, a polypeptide comprises the amino acid sequence consisting of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and/or 71.

[0102] In some embodiments, a polypeptide comprises one or more of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some embodiments, a polypeptide comprises about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some embodiments, a polypeptide consists of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

[0103] In various aspects of the polypeptide, the percent identity differences between amino acid sequences is from truncation of amino acids, addition of amino acids, or one or more amino acid substitutions of the polypeptide. In some embodiments, an amino acid substitution is a conservative substitution. Illustrative examples for conserved amino acid exchanges are amino acid substitutions that maintain structural and/or functional properties of the amino acids' side-chains, e.g., an aromatic amino acid is substituted for another aromatic amino acid, an acidic amino acid is substituted for another acidic amino acid, a basic amino acid is substituted for another basic amino acid, and an aliphatic amino acid is substituted for another aliphatic amino acid. In some embodiments, a conservative amino acid substitution is one in which the amino acid residue is replaced with an amino acid residue having a side chain with a similar charge. Families of amino acid residues having side chains with similar charges have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., asparagine, glutamine, serine, threonine, tyrosine, cysteine), non-polar side chains (e.g., glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). In contrast, examples of non-conserved amino acid exchanges are amino acid substitutions that do not maintain structural and/or functional properties of the amino acids' side-chains, e.g., an aromatic amino acid is substituted for a basic, acidic, or aliphatic amino acid, an acidic amino acid is substituted for an aromatic, basic, or aliphatic amino acid, a basic amino acid is substituted for an acidic, aromatic or aliphatic amino acid, and an aliphatic amino acid is substituted for an aromatic, acidic or basic amino acid.

[0104] In one aspect, provided herein are vectors comprising a nucleotide sequence encoding a polypeptide as described herein. In some embodiments, the vector is an expression vector. In some embodiments, a vector comprises

an expression cassette for expression of a polypeptide as described herein. In some embodiments, the vector is a viral vector.

[0105] Also provided are nucleic acids encoding the polypeptides as described herein. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising nucleotides that encode an AAV VP3 polypeptide comprising a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:2, 12, or 28. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO: 2, 12, and/or 28. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:2, 12, and/or 28. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO: 2, 12, and/or 28. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising nucleotides that encode an AAV VP1/2s comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:4 or 14. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO: 4 and/or 14. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:4 and/or 14. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO: 4 and/or 14. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising nucleotides that encode an AAV VP2 comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:6 or 16. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO: 6 and/or 16. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:6 and/or 16. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO: 6 and/or 16. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising nucleotides that encode an AAV VP1u comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:8 or 18. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO: 8 and/or 18. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO:8 and/or 18. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO: 8 and/or 18. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising nucleotides that encode an AAV VP1 and the nucleotide sequence comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, or 72. In some embodiments, a polypeptide comprises the amino acid sequence of SEQ ID NO: 10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, a polypeptide consists of the amino acid sequence of SEQ ID NO: 10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, a polypeptide comprises the amino acid sequence consisting of SEQ ID NO: 10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, the polypeptide is encoded by a nucleotide sequence comprising about or at least about 80%, 85%, 90%, 95%,

96%, 97%, 98%, 99%, 99.5%, or 100% identity to any one of SEQ ID NO: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, a nucleotide sequence comprises the amino acid sequence of any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, a nucleotide sequence consists of the amino acid sequence of any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, a nucleotide sequence comprises the amino acid sequence consisting of any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and/or 72. In some embodiments, provided herein is a host cell (e.g., in vitro cell) comprising a nucleotide sequence of the disclosure (e.g., a nucleotide sequence that encodes the polypeptide or hybrid AAV of the disclosure). In some embodiments, provided herein is a host cell (e.g., in vitro cell) comprising an expression vector of the disclosure.

[0106] As a skilled artisan would recognize, a polypeptide corresponding to a VP1 capsid protein may be truncated in vivo into a VP2 or VP3 capsid protein.

5.2 Recombinant Adeno-Associated Viruses (rAAVs) and Viral Vectors

5.2.1 Hybrid AAV Capsids

[0107] Provided herein is a recombinant adeno-associated virus (rAAV) comprising hybrid capsid proteins engineered to maintain or improve parental tropism while having improved endosomal escape and nuclear trafficking. In various aspects of the invention, an rAAV comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of a different serotype than the serotype of the VP3 capsid protein. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1/2s of a second AAV serotype and a VP3 of a first AAV serotype. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1u of a second AAV serotype, a VP1/2s of the second AAV serotype, and a VP3 of a first AAV serotype. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1u of a first AAV serotype, a VP1/2s of a second AAV serotype, and a VP3 of the first AAV serotype. In some embodiments, the first AAV serotype is AAV8 or AAV Rh13. In some embodiments, the second AAV serotype is AAV5 or AAVhu32. In specific embodiments, the first AAV serotype is AAV8 and the second AAV serotype is AAV5. In specific embodiments, the first AAV serotype is AAVrh13 and the second AAV serotype is AAVhu32. In some embodiments, a hybrid AAV of the disclosure comprises one or more than one of AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh10, AAV11, AAV12, AAV13, AAVrh20, AAVhu37, AAVrh39, AAVhu32, AAVrh21, AAVrh13, AAVrh15, AAVrh73, and/or AAVrh74.

[0108] In some embodiments, an rAAV (hybrid AAV) of the disclosure comprises one or more polypeptide of the disclosure (e.g., as described in Section 5.1.1). In some embodiments, provided herein is a recombinant adeno-associated virus (AAV) comprising a capsid protein (e.g., hybrid capsid protein) of the disclosure. In some embodiments, the rAAV comprises an expression cassette. In some

embodiments, the expression cassette comprises an open reading frame (ORF) encoding a transgene (e.g., a transgene of the disclosure). In some embodiments, the expression cassette further comprises a promoter. In some embodiments, an rAAV comprises a capsid protein comprising: (i) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), and (ii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of a second AAV (e.g., AAV8, AAVrh13, or VP3 region of any wild-type AAV or a VP3 region of any AAV of the disclosure or identified in Section 7). In some embodiments, an rAAV comprises a capsid protein comprising: (i) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1/2s region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), and (ii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP3 region of a second AAV (e.g., AAV8, AAVrh13, or VP3 region of any wild-type AAV, or a VP3 region of any AAV of the disclosure or identified in Section 7). In some embodiments, the capsid protein further comprises an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1u region of a first AAV or a second AAV (e.g., AAV5, AAVhu32, AAV8, AAVrh13, or VP1u of any wild-type AAV, or a VP1u region of any AAV of the disclosure or identified in Section 7). In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 95% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 98% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 99% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, provided herein is a recombinant adeno-associated virus (AAV) comprising a capsid protein, comprising: (i) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1u region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), (ii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1/2s region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), and (iii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP3 region of a second AAV (e.g., AAV8, AAVrh13, or VP3 region of any wild-type AAV, or a VP3 region of any AAV of the disclosure or identified in Section 7).

[0109] In certain embodiments, an rAAV of the disclosure comprises a VP3 of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP3 of AAV5. In some embodiments, an rAAV comprises a hybrid

capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:1. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:1.

[0110] In certain embodiments, an rAAV of the disclosure comprises a VP1/2s of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1/2s of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:3. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:3.

[0111] In certain embodiments, an rAAV of the disclosure comprises a VP1/2s and VP3 of AAV5. In certain embodiments, an rAAV of the disclosure comprises a VP2 of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1/2s and VP3 of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP2 of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:3 and an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:1. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequences of SEQ ID NO:3 and SEQ ID NO:1. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:5.

[0112] In certain embodiments, an rAAV of the disclosure comprises a VP1u of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1u of AAV5. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:7. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:7.

[0113] In certain embodiments, an rAAV of the disclosure comprises a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid

[0115] In certain embodiments, an rAAV of the disclosure comprises a VP1/2s and VP3 of AAV8. In certain embodiments, an rAAV of the disclosure comprises a VP2 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1/2s and VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP2 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:13 and an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:11. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:3 and SEQ ID NO:1. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:15. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:15.

[0117] In certain embodiments, an rAAV comprises a VP1u, VP1/2s, and VP3 of AAV8. In certain embodiments, an rAAV comprises a VP1 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an

[0118] In certain embodiments, an rAAV of the disclosure comprises a VP 1/2s of AAV5, and a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP 1/2s of AAV5 and a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:3, and an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:11. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequences of SEQ ID NO:3 and SEQ ID NO:11. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:21. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:21.

[0119] In certain embodiments, an rAAV of the disclosure comprises a VP1u of AAV5, a VP 1/2s of AAV5, and a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising a VP1u of AAV5, a VP 1/2s of AAV5, and a VP3 of AAV8. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:7, an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%.

[0139] In certain embodiments, an rAAV of the disclosure comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of AAV5, and a VP3 of AAVhu32. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:63. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:63.

[0140] In certain embodiments, an rAAV of the disclosure comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of AAV hu.32, and a VP3 of AAV rh.15. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:65. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:65.

[0141] In certain embodiments, an rAAV of the disclosure comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of AAV6, and a VP3 of AAV ru.15. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:67. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:67.

[0142] In certain embodiments, an rAAV of the disclosure comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of AAV5, and a VP3 of AAV rh.15. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:69. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:69.

[0143] In certain embodiments, an rAAV of the disclosure comprises a hybrid capsid protein comprising a VP1u and/or VP1/2s of AAV6, and a VP3 of AAV rh.13. In some embodiments, an rAAV comprises a hybrid capsid protein comprising an amino acid sequence having about or at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NO:71. In some embodiments, an rAAV comprises a hybrid capsid protein comprising the amino acid sequence of SEQ ID NO:71.

[0144] In some embodiments, an rAAV of the disclosure comprises a capsid protein comprising an amino acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOS: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some embodiments, an rAAV of the disclosure comprises a capsid protein

consisting of any one of SEQ ID NOS: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

[0145] In various aspects of the invention, the percent identity differences between amino acid sequences is from truncation of amino acids, addition of amino acids, or one or more amino acid substitutions. In some embodiments, an amino acid substitution is a conservative substitution. Illustrative examples for conserved amino acid exchanges are amino acid substitutions that maintain structural and/or functional properties of the amino acids' side-chains, e.g., an aromatic amino acid is substituted for another aromatic amino acid, an acidic amino acid is substituted for another acidic amino acid, a basic amino acid is substituted for another basic amino acid, and an aliphatic amino acid is substituted for another aliphatic amino acid. In some embodiments, a conservative amino acid substitution is one in which the amino acid residue is replaced with an amino acid residue having a side chain with a similar charge. Families of amino acid residues having side chains with similar charges have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., asparagine, glutamine, serine, threonine, tyrosine, cysteine), non-polar side chains (e.g., glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). In contrast, examples of non-conserved amino acid exchanges are amino acid substitutions that do not maintain structural and/or functional properties of the amino acids' side-chains, e.g., an aromatic amino acid is substituted for a basic, acidic, or aliphatic amino acid, an acidic amino acid is substituted for an aromatic, basic, or aliphatic amino acid, a basic amino acid is substituted for an acidic, aromatic or aliphatic amino acid, and an aliphatic amino acid is substituted for an aromatic, acidic or basic amino acid.

[0146] In some embodiments, provided herein are AAV vectors comprising a viral genome comprising an expression cassette for expression of a therapeutic product, under the control of regulatory elements and flanked by ITRs, and a hybrid viral capsid as described herein.

[0147] In some embodiments, an AAV of the disclosure can be an AAV derived from a naturally occurring "wild-type" virus, an AAV derived from an rAAV genome packaged into a capsid comprising capsid proteins encoded by a naturally occurring cap gene and/or from an rAAV genome packaged into a capsid comprising capsid proteins encoded by a non-naturally occurring capsid cap gene. An example of the latter includes an rAAV having a hybrid capsid protein as described herein.

[0148] In some embodiments, an AAV capsid is characterized by DNase-resistant particle which is an assembly of about 60 variable proteins (VP) which are typically expressed as alternative splice variants resulting in capsid proteins of different length of any one of SEQ ID NOS: 9, 19, or 35. The largest protein, VP1, is generally the full-length of the amino acid sequence of any one of SEQ ID NOS: 9, 19, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, or 71. In certain embodiments, an AAV hybrid VP1 capsid protein has the amino acid

sequence of 1 to 736 of SEQ ID NOS: 9, 19, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, or 71. In certain embodiments, an AAV hybrid VP2 capsid protein has the amino acid sequence of 138 to 736 of SEQ ID NOS: 9, 19, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, or 71. VP1u is the N-terminal region unique to VP1. VP1/2s is the common region shared by VP1 and VP2. In some embodiments, a VP1 region or capsid protein or fragment thereof includes VP1u, such as amino acids 1-137 of SEQ ID NO: 31, corresponding to SEQ ID NO: 146, and/or functional fragments thereof. In some embodiments, a VP1/2s includes about amino acids 138-202 of SEQ ID NO: 31, corresponding to SEQ ID NO: 148, and/or functional fragments thereof. In some embodiments, a VP1 capsid protein or fragment thereof includes VP1u region of an AAV disclosed in Section 7 of the disclosure. In some embodiments, a VP1/2s region includes VP1/2s region of an AAV disclosed in Section 7 of the disclosure. In certain embodiments, an AAV hybrid VP3 capsid protein has the amino acid sequence of 203 to 736 of SEQ ID NOS: 9, 19, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, or 71. In certain embodiments, the VP1, VP2 or VP3 proteins may have truncations (e.g., 1 or more amino acids at the N-terminus or C-terminus). An assembled AAV capsid is composed of about 60 VP proteins, in which VP1, VP2 and VP3 are present in a ratio of about one VP1, to about one VP2, to about 10 to 20 VP3 proteins. This ratio may vary depending upon the production system used. In certain embodiments, an engineered AAV hybrid capsid may be generated in which VP2 is absent.

[0149] Also provided are nucleic acids encoding an engineered capsid protein and variants thereof, packaging cells for expressing the nucleic acids to produce an rAAV vector, an rAAV vector further comprising a therapeutic product (e.g., transgene), and pharmaceutical compositions comprising an rAAV vector; as well as methods of using an rAAV vector to deliver a therapeutic protein to a cell or target tissue of a subject in need thereof. In some embodiments, the nucleotide sequence that encodes an AAV VP3 comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:2, 12, or 28. In some embodiments, the nucleotide sequence that encodes an AAV VP1/2s comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:4 or 14. In some embodiments, the nucleotide sequence that encodes an AAV VP2 comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:6 or 16. In some embodiments, the nucleotide sequence that encodes an AAV VP1u comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:8 or 18. In some embodiments, the nucleotide sequence that encodes an AAV VP1 comprises a sequence that shares about or at least about 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO:10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70 or 72.

[0150] It is within the skill in the art to design nucleic acid sequences encoding an AAV hybrid capsid protein, including DNA (genomic or cDNA), or RNA (e.g., mRNA). In certain embodiments, the nucleic acid sequence encoding

the AAV hybrid VP1 capsid protein is provided in SEQ ID NO:10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70 or 72. In other embodiments, a nucleic acid sequence of 70% to 99.9% identity to SEQ ID NO:10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70 or 72 may be selected to express the AAV hybrid VP1 capsid protein. In certain other embodiments, the nucleic acid sequence is at least about 75% identical, at least 80% identical, at least 85%, at least 90%, at least 95%, at least 97% identical, or at least 99% to 99.9% identical to SEQ ID NO:10, 20, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70 or 72.

[0151] In some embodiments, a nucleic acid sequence comprises a nucleic acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOS: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some embodiments, a nucleic acid sequence comprises the nucleic acid sequence of any one of SEQ ID NOS: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some embodiments, a nucleic acid sequence consists of the nucleic acid sequence of any one of SEQ ID NOS: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120.

[0152] In some embodiments, an rAAV comprising the hybrid capsid protein of the disclosure (e.g., an AAV capsid comprising a VP1u and/or VP1/2s of a different serotype than the serotype of the VP3 capsid protein) has enhanced cell or tissue tropism (e.g., cell or tissue associated with the musculature) as compared to an AAV comprising an unmodified capsid or a capsid without hybrid engineering of the disclosure; and/or is homogeneously expressed or uniformly distributed at a desired location (e.g., homogeneously expressed or uniformly widespread at the musculature or an area within the musculature a subject) after the rAAV is administered to a subject. In some embodiments, the rAAV comprising the hybrid capsid protein of the disclosure is detargeted from the liver.

[0153] In some embodiments, an rAAV vector comprises AAVhu32 having muscle-homing properties that are enhanced compared to other Clade F capsids, for example AAV9. In some embodiments, the muscle-specific rAAV vector, such as AAVhu32, further comprises a peptide insertion within the capsid protein, which peptide includes amino acid sequences that confer and/or enhance desired muscle-homing properties, or muscle cell tropism. In particular, the rAAV vector comprises an engineered capsid protein comprising a peptide insertion from a heterologous protein inserted within or near variable region IV (VR-IV) or, alternatively, within or near variable region VIII (VR-VIII) of the virus capsid, such that the insertion is surface exposed on the AAV particle. In some embodiments, the peptide insertion is inserted after amino acid residue 454 or 589. In some embodiments, the peptide insertion is inserted between amino acids 454 and 455. In some embodiments, the peptide insertion is inserted between amino acids 589 and 590. In some embodiments, the peptide insertion is inserted between

any two consecutive amino acids between amino acid at position 451 and amino acid at position 461. In some embodiments, the peptide insertion is inserted between any two consecutive amino acids between amino acid at position 570 and amino acid at position 589.

[0154] In some embodiments, a peptide insertion (e.g., muscle-homing peptide) is inserted before or after at least one amino acid at position: 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600 of an AAV (e.g., AAVhu32 or a corresponding position in another AAV). In some embodiments, the peptide insertion occurs immediately after one of the amino acid residues within: 450-456, 451-461, 454-456, 450-460, 440-470, 430-480, 420-500, 510-600, 520-590, 520-540, 530-540, 530-550, 530-535, 532-535, 560-590, 580-590, 570-600, 585-589, 585-590 of AAVhu32 or a corresponding position in another AAV. In some embodiments, a peptide insertion is inserted at or after position 454, particularly S454 in AAVhu32 or a corresponding position in another AAV. In some embodiments, a peptide insertion is inserted at or after position 589, particularly A589 in AAVhu32 or a corresponding position in another AAV.

[0155] In some embodiments, a peptide insertion (or an insertion of a heterologous amino acid sequence) is of about 4 to about or up to about 15 (e.g., 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15) amino acids, within or near a variable region (e.g., VR-VIII, VR-IV, and/or VR-VI of AAVhu32 or a corresponding variable region in another AAV) of a virus capsid. In some embodiments, the peptide insertion is between two amino acids without deleting any capsid amino acid(s) of an AAV capsid (e.g., AAVhu32). In some embodiments, the peptide insertion occurs within (i.e., between two amino acids without deleting any capsid amino acids) variable region VIII (VR-VIII), VR-IV, and/or VR-VI of AAVhu32 capsid, or a corresponding region in another AAV serotype.

[0156] In some embodiments, the muscle-homing peptide comprises an integrin receptor-binding domain or an integrin-binding domain, such as RGD and other recognition sequences for integrin (Ruoslahti E. *Annu Rev Cell Dev Biol.* 1996; 12:697-715. doi: 10.1146/annurev.cellbio.12.1.697. PMID: 8970741; e.g. RGD-peptide RGDLRVS (SEQ ID NO: 151), or peptide SLRSPPS (SEQ ID NO: 140), as described in Varadi, et al. 2012 *Gene Therapy*, 19, 800-809; e.g. RGD-peptide RGDGLS (SEQ ID NO: 141), as described in Michelfelder S, et al., 2009, *PLOS ONE* 4 (4): e5122. doi: 10.1371/journal.pone.0005122; e.g. 4C-RGD peptide CDCRGDCFC (SEQ ID NO: 142) or CDCRGDCFCGLS (SEQ ID NO: 143), as described in Shi and Bartlett 2003 *Mol. Ther.* 2003, 7, 515-525; and other RGD-containing peptide sequences described in Tabebordbar et al., 2021, *Cell* 184, 4919-4938, including e.g.

RGDLTTP (SEQ ID NO: 144) and RGDLSTP (SEQ ID NO: 145), each of which is incorporated herein by reference in its entirety). In certain embodiments, the peptide insertion may be a sequence of consecutive amino acids from a domain that targets muscle, or an analog, or a conformational analog designed to mimic the three-dimensional structure of said domain.

[0157] In some embodiments, a muscle-homing peptide is RRQPPRSISHP (M12; SEQ ID NO: 134) or a portion thereof. In some embodiments, a muscle-homing peptide comprises about or at least 4, 5, 6, 7, or more than 7 contiguous amino acids of RRQPPRSISHP (M12; SEQ ID NO: 134). In some embodiments, a muscle-homing peptide comprises about or at least about 80%, 85%, 90%, 95%, or 100% sequence identity to RRQPPRSISHP (M12; SEQ ID NO: 134). In some embodiments, a muscle-homing peptide is ASSLNIA (a muscle-specific peptide (MSP); SEQ ID NO: 135) or a portion thereof. In some embodiments, a muscle-homing peptide comprises about or at least 4, 5, 6, or 7 contiguous amino acids of ASSLNIA (a muscle-specific peptide (MSP); SEQ ID NO: 135). In some embodiments, a muscle-homing peptide comprises about or at least about 80%, 85%, 90%, 95%, or 100% sequence identity to ASSLNIA (a muscle-specific peptide (MSP); SEQ ID NO: 135). In some embodiments, a muscle-homing peptide is AGST-SAGSAAGSSGDRRQPPRSISHP (SEQ ID NO: 136) or a portion thereof. In some embodiments, a muscle-homing peptide comprises about or at least 4, 5, 6, 7, or more than 7 contiguous amino acids of AGST-SAGSAAGSSGDRRQPPRSISHP (SEQ ID NO: 136). In some embodiments, a muscle-homing peptide comprises about or at least about 80%, 85%, 90%, 95%, or 100% sequence identity to AGST-SAGSAAGSSGDRRQPPRSISHP (SEQ ID NO: 136). In some embodiments, a muscle-homing peptide is CYAIGSFDC (SEQ ID NO: 137) or a portion thereof. In some embodiments, a muscle-homing peptide comprises about or at least 4, 5, 6, 7, or more than 7 contiguous amino acids of CYAIGSFDC (SEQ ID NO: 137). In some embodiments, a muscle-homing peptide comprises about or at least about 80%, 85%, 90%, 95%, or 100% sequence identity to CYAIGSFDC (SEQ ID NO: 137). In some embodiments, a muscle-homing peptide is SGASAV (SEQ ID NO: 138) or a portion thereof. In some embodiments, a muscle-homing peptide comprises about or at least 4 contiguous amino acids of SGASAV (SEQ ID NO: 138). In some embodiments, a muscle-homing peptide comprises about or at least about 80%, 85%, 90%, 95%, or 100% sequence identity to SGASAV (SEQ ID NO: 138). In some embodiments, a muscle-homing peptide comprises the motif GRSGXR (SEQ ID NO: 139; wherein X can be any naturally occurring amino acid). In some embodiments, a muscle-homing peptide comprises the motif DFSGIAX (SEQ ID NO: 150; wherein X can be any naturally occurring amino acid). A muscle-homing peptide of the disclosure can be any known muscle-homing peptide or any predicted muscle-homing peptide.

[0158] The muscle-homing peptide can be inserted into an AAV capsid, for example, at sites that allow surface exposure of the peptide, such as within variable surface-exposed loops, and, in more examples, sites described herein corresponding to VR-I, VR-IV, or VR-VIII of the capsid protein or may be inserted after the first amino acid of VP2, e.g. after amino acid 137 (e.g. as in AAV4, AAV4-4, and AAV5) or at amino acid 138 (e.g. as in AAV1, AAV2, AAV3, AAV3-3,

AAV6, AAV7, AAV8, AAV9, AAVhu.31, AAVhu32, and rh.10) (FIG. 13). Recombinant AAV vectors comprising one or more muscle-homing peptides, e.g., inserted into a surface-exposed loop of an AAV capsid, are referred to herein as “rAAV muscle-homing vectors.” In some embodiments, the capsid protein is an AAVhu32 capsid protein and the muscle homing peptide insertion occurs immediately after at least one of the amino acid residues 451 to 461 or 570 to 590 of the AAVhu32 capsid (e.g., SEQ ID NO: 31). In other embodiments, the rAAV vector is an AAVhu32 variant, and the muscle-homing peptide insertion occurs immediately after an amino acid residue corresponding to at least one of the amino acid residues 451 to 461 or 570 to 590 of AAVhu32 capsid protein (e.g., SEQ ID NO: 31). The alignments of several different AAV serotypes, as shown in FIG. 13, indicates corresponding amino acid residues in these different capsid protein sequences.

[0159] Liver detargeting has also been associated with substitution of particular amino acids in capsid proteins. In some embodiments, the liver detargeting mutation and/or peptide insertion comprises any one of the capsid mutations or peptides described in PCT Publication No. WO2020206189A1 (the content of which is herein incorporated by reference in its entirety).

5.2.2 Properties of rAAV Comprising a Variant AAV Capsid

[0160] Provided herein are recombinant adeno-associated viruses (rAAVs) comprising a hybrid AAV capsid as described in Section 5.2.1 or a hybrid AAV capsid comprising a polypeptide as described in Section 5.1.1. In some embodiments, a hybrid AAV of the disclosure provides for at least one enhanced property (e.g., enhanced cell or tissue tropism, homogeneous distribution of a transgene at a specific location after such rAAVs are administered to a subject, packaging efficiency, yield, titer, infectivity, transduction efficiency, or transfection efficiency) as compared to a reference AAV, a wild-type AAV or an AAV that does not comprise a hybrid AAV capsid of the disclosure. In some embodiments, the rAAV provides at least one improvement of packaging efficiency, yield, titer, infectivity, transduction efficiency, and transfection efficiency, as compared to a reference AAV. In some embodiments, a hybrid capsid promotes transduction and/or tissue tropism as described herein (i.e., muscle tropism and/or liver detargeting). In some embodiments, an rAAV vector comprising a hybrid capsid of the disclosure enhances targeted delivery, improves transduction and/or treatment of disorders associated with the target tissue as compared to a reference AAV, a wild-type AAV vector or an AAV vector that does not comprise a hybrid capsid of the disclosure. In some embodiments, an rAAV vector comprising a hybrid capsid of the disclosure enhances packaging efficiency as compared to a reference AAV, a wild-type AAV vector or an AAV vector that does not comprise a hybrid capsid of the disclosure. In some embodiments, an rAAV vector comprising a hybrid capsid of the disclosure enhances infectivity, transduction efficiency, or transfection efficiency as compared to a reference AAV, a wild-type AAV vector or an AAV vector that does not comprise a hybrid capsid of the disclosure. In some embodiments, a reference AAV is AAV8. In some embodiments, a reference AAV is wild-type AAV. In

some embodiments, a reference AAV comprises VP1u of AAV8. In some embodiments, a reference AAV is AAVrh13. In some embodiments, a reference AAV is AAVhu32. In some embodiments, a reference AAV is AAV5. In some embodiments, reference AAV is AAV9.

[0161] In some embodiments, the rAAV transduces muscle and/or liver tissues at lower levels as compared to a reference AAV. In some embodiments, the rAAV results in lower RNA expression in liver as compared to a reference AAV. In some embodiments, an improved property as compared to a reference AAV is an improvement by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold. In some embodiments, the improvement is by about or at least about 2 fold. In some embodiments, the improvement is by about or at least about 2.5 fold. In some embodiments, the improvement is by about or at least about 3 fold. In some embodiments, the RNA expression is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the RNA expression from a reference AAV. In some embodiments, the rAAV results in higher RNA/DNA ratio in muscle tissue as compared to a reference AAV. In some embodiments, the RNA/DNA ratio is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some embodiments, the rAAV results in lower DNA levels in muscle tissue as compared to a reference AAV. In some embodiments, the DNA levels is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the DNA levels from a reference AAV.

[0162] In some embodiments, an rAAV of the disclosure results in a higher relative abundance of mRNA transcript copies (of transgene) in a muscle cell or tissue (e.g., gastrocnemius, quadriceps, and/or tibialis anterior) and/or in a heart cell or tissue compared to a reference AAV. In some embodiments, the relative abundance of mRNA transcript copies (of transgene) is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some embodiments, the relative abundance of mRNA transcript copies (of transgene) is higher by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold. In some embodiments, an rAAV of the disclosure results in a higher relative abundance of cDNA in a muscle cell or tissue (e.g., gastrocnemius, quadriceps, and/or tibialis anterior) and/or in a heart cell or tissue compared to a reference AAV. In some embodiments, the relative abundance of cDNA is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some embodiments, the relative abundance of cDNA is higher by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold.

[0163] In some embodiments, an rAAV of the disclosure results in a higher biodistribution or transgene expression (e.g., microdystrophin) in a muscle cell or tissue (e.g., gastrocnemius, quadriceps, and/or tibialis anterior) and/or in a heart cell or tissue compared to a reference AAV. In some embodiments, the biodistribution or transgene expression (e.g., microdystrophin) is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some embodiments, the biodistribution or transgene expression (e.g., microdystrophin) is higher by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold. In some embodiments, an rAAV viral particle of the disclosure has increased tropism for a particular cell or tissue (e.g., a muscle cell or tissue; and/or a heart cell or tissue) as compared to a reference AAV. In some embodiments, the increase is about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or 50% increase in tropism for a particular cell type or tissue (e.g., a muscle cell or tissue; and/or a heart cell or tissue) as compared to a reference AAV. In some embodiments, the increase is about 5% to about 20%, about 25% to about 50% increase in tropism for a particular cell type or tissue (e.g., a muscle cell or tissue; and/or a heart cell or tissue) as compared to a reference AAV. In some embodiments, the increase is about 0.1 fold, about 0.2 fold, about 0.3 fold, about 0.4 fold, about 0.5 fold, about 0.6 fold, about 0.7 fold, about 0.8 fold, about 0.9 fold, or about 1 fold increase in tropism for a particular cell type or tissue (e.g., a muscle cell or tissue; and/or a heart cell or tissue) as compared to a reference AAV. In some embodiments, the increase is about 0.1 to 1 fold increase in tropism for a particular cell type or tissue (e.g., a muscle cell or tissue; and/or a heart cell or tissue) as compared to a reference AAV. In some embodiments, an rAAV viral particle of the disclosure has increased transduction of muscle and/or heart cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 19, 31, and 124 and/or comprises AAV5, AAV8, AAV-9, AAVrh13, or AAVhu32 capsid protein. In some embodiments, the rAAV viral particle has about a 0.10 fold to about a 1 fold increase in transduction of muscle and/or heart cells relative to the second AAV viral particle. In some embodiments, the rAAV viral particle has about 0.10 fold, about 0.2 fold, about 0.3 fold, about 0.4 fold, about 0.5 fold, about 0.6 fold, about 0.7 fold, about 0.8 fold, about 0.9 fold, or about 1 fold increase in transduction of muscle and/or heart cells relative to the second AAV viral particle (e.g., a reference AAV). In some embodiments, an AAV of the disclosure has increased muscle and/or heart tropism as compared to a reference or second AAV (e.g., AAV5, AAV8, AAV-9, AAVrh13, or AAVhu32).

[0164] In some embodiments, the rAAV results in a lower level of liver toxicity as compared to the level of liver toxicity from a reference AAV. In some embodiments, the level of liver toxicity is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the level of liver toxicity from a reference AAV. In some embodiments, an rAAV viral particle of the disclosure has decreased tropism for a particular cell or tissue (e.g., a liver cell or the liver) as

compared to a reference AAV. In some embodiments, the decrease is about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or 50% decrease in tropism for a particular cell type or tissue (e.g., a liver cell or the liver) as compared to a reference AAV. In some embodiments, the decrease is about 5% to about 20%, about 25% to about 50% decrease in tropism for a particular cell type or tissue (e.g., a liver cell or the liver) as compared to a reference AAV. In some embodiments, the decrease is about 0.1 fold, about 0.2 fold, about 0.3 fold, about 0.4 fold, about 0.5 fold, about 0.6 fold, about 0.7 fold, about 0.8 fold, about 0.9 fold, or about 1 fold decrease in tropism for a particular cell type or tissue (e.g., a liver cell or the liver) as compared to a reference AAV. In some embodiments, the decrease is about 0.1 to 1 fold decrease in tropism for a particular cell type or tissue (e.g., a liver cell or the liver) as compared to a reference AAV. In some embodiments, an rAAV viral particle of the disclosure has reduced transduction of liver cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 19, 31, and 124 and/or AAV5, AAV8, AAV-9, AAVrh13, or AAVhu32 capsid protein. In some embodiments, the rAAV viral particle has about a 0.10 fold to about a 1 fold decrease in transduction of liver cells relative to the second AAV viral particle. In some embodiments, the rAAV viral particle has about 0.10 fold, about 0.2 fold, about 0.3 fold, about 0.4 fold, about 0.5 fold, about 0.6 fold, about 0.7 fold, about 0.8 fold, about 0.9 fold, or about 1 fold decrease in transduction of liver cells relative to the second AAV viral particle (e.g., a reference AAV). In some embodiments, the rAAV results in transcription-specific liver de-targeting as compared to a reference AAV. In some embodiments, an AAV of the disclosure has decreased liver tropism as compared to a reference AAV (e.g., AAV5, AAV8, AAV-9, AAVrh13, or AAVhu32). In some embodiments, a reference AAV is AAV8. In some embodiments, a reference AAV is wild-type AAV. In some embodiments, reference AAV comprises VP1u of AAV8. In some embodiments, reference AAV is AAVrh13. In some embodiments, reference AAV is AAVhu32. In some embodiments, reference AAV is AAV5. In some embodiments, reference AAV is AAV9. In some embodiments, a reference AAV is the same AAV as the AAV of the VP3 region of a hybrid AAV. In some embodiments, a reference AAV is the same AAV as the AAV of the VP1/2s region of a hybrid AAV. In some embodiments, a reference AAV is the same AAV as the AAV of the VP1u region of a hybrid AAV.

[0165] In some embodiments a lower amount of vector genome of the rAAV is sufficient to obtain a comparable therapeutic effect as compared to the amount of vector genome of a reference AAV necessary to obtain the same therapeutic effect after administration in a subject. In some embodiments, the amount is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%. In some embodiments, the amount is lower by 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold.

[0166] In some embodiments, the rAAV provides a higher vector yield after the rAAV is administered to the subject, as compared to a reference AAV. In some embodiments, the vector yield is higher by about or at least about: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5,

6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some embodiments, the vector yield is increased by about or at least about: 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than about 100%. In some embodiments, the vector yield is higher by between about 1.5 to about 3 times. In some embodiments, the vector yield is higher from about 2.5 to about 3.5, from about 1 to about 4 times, from about 1 to about 5 times, from about 1 to about 7 times, from about 1 to about 10 times, from about 2 to about 5 times, from about 2 to about 7 times, from about 2 to about 10 times, from about 3 to about 5 times, from about 3 to about 7 times, from about 3 to about 10 times, from about 4 to about 5 times, from about 4 to about 7 times, from about 4 to about 10 times, from about 5 to about 7 times, from about 5 to about 10 times, from about 6 to about 8 times, from about 6 to about 7 times, from about 6 to about 10 times, from about 7 to about 10, or more than from about 7 to about 10. In some embodiments, the RNA level is increased after the rAAV is administered to the subject, as compared to a reference AAV. In some embodiments, the RNA level is increased by about or at least about: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some embodiments, the RNA level is increased by about or at least about: 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than about 100%. In some embodiments, the RNA level is increased by between about 2.5 to about 3.5 times. In some embodiments, the RNA level is increased from about 1.5 to about 3, from about 2.5 to about 3.5, from about 1 to about 4 times, from about 1 to about 5 times, from about 1 to about 7 times, from about 1 to about 10 times, from about 2 to about 5 times, from about 2 to about 7 times, from about 2 to about 10 times, from about 3 to about 5 times, from about 3 to about 7 times, from about 3 to about 10 times, from about 4 to about 5 times, from about 4 to about 7 times, from about 4 to about 10 times, from about 5 to about 7 times, from about 5 to about 10 times, from about 6 to about 8 times, from about 6 to about 7 times, from about 6 to about 10 times, from about 7 to about 10, or more than from about 7 to about 10. In some embodiments, the ratio of RNA to DNA is increased after the rAAV is administered to the subject. In some embodiments, the ratio of RNA to DNA is increased by about or at least about: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times. In some embodiments, the ratio of RNA to DNA is increased by about or at least about: 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than about 100%. In some embodiments, the ratio of RNA to DNA is increased by between about 2.5 to about 3.5 times. In some embodiments, the ratio of RNA to DNA is increased from about 1.5 to about 3, from about 2.5 to about 3.5, from about 1 to about 4 times, from about 1 to about 5 times, from about 1 to about 7 times, from about 1 to about 10 times, from about 2 to about 5 times, from about 2 to about 7 times, from about 2 to about 10 times, from about 3 to about 5 times, from about 3 to about 7 times, from about 3 to about 10 times, from about 4 to about 5 times, from about 4 to about 7 times, from about 4 to about 10 times, from about 5 to about 7 times, from about 5 to about 10 times, from about 6 to about 8 times, from

about 6 to about 7 times, from about 6 to about 10 times, from about 7 to about 10, or more than from about 7 to about 10.

[0167] In some embodiments, a reference AAV is AAV5. In some embodiments, a reference AAV is AAV8. In some embodiments, a reference AAV is AAV9. In some embodiments, a reference AAV is AAVhu32. In some embodiments, a reference AAV comprises a capsid protein that comprises or consists of any one of SEQ ID NOs: 19, 31, and 124. In some embodiments, a reference AAV comprises an expression cassette. In some embodiments, a reference AAV comprises or encodes a transgene.

[0168] In some embodiments, an rAAV vector comprising a hybrid capsid of the disclosure targets a tissue or cell associated with the musculature. In some embodiments, the rAAV comprising a hybrid capsid of the disclosure facilitates delivery of therapeutic agents or transgenes for treating diseases or disorders of the musculature. In some embodiments, a hybrid capsid increases muscular tropism, or directs the rAAV to target the musculature or a muscle cell of the subject. The term “muscle cell” or “muscle cell” refers to one or more of the cell types of skeletal muscle cells, cardiac muscle cells, smooth muscle cells, or any other cell associated with a muscle, and the like.

[0169] In some embodiments, the rAAV vector comprises a peptide insertion for muscle cell-homing, the vector is administered by in vivo injection, such as subcutaneous injection, intradermal injection, or intramuscular injection. In some embodiments, injection is directly into a muscle. For example, the rAAV comprising a hybrid capsid as described herein has enhanced muscle tropism and is injected intramuscularly. In some embodiments, the rAAV for increasing muscle tropism is administered to deltoid, dorsogluteal, rectus femoris, vastus lateralis, or ventrogluteal muscular injection.

[0170] Provided herein are recombinant adeno-associated viruses (rAAVs) having hybrid capsid proteins engineered to confer and/or enhance desired properties (e.g., enhanced tissue or cell specific targeting, cell-specific tropism, and/or enhanced transduction efficacy). In some embodiments, the tissue or cell specific targeting or cell-specific tropism is related to a tissue or cell associated with the musculature of a subject (e.g., a myopathy or dystrophy). In some embodiments, an rAAV of the disclosure has about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99%, or 100% greater tropism for a particular muscle cell type or tissue as compared to a reference AAV (e.g., an AAV that does not comprise a hybrid capsid of the disclosure). In some embodiments, an rAAV of the disclosure has about 125%, 150%, 175%, 200%, 250%, 300%, 350%, 400%, or 500% greater tropism for a particular muscle cell type or tissue compared to a reference AAV. In some embodiments, an rAAV of the disclosure has about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99%, or 100% greater tropism for the heart or a heart cell or tissue as compared to a reference AAV (e.g., an AAV that does not comprise a hybrid capsid of the disclosure). In some embodiments, an rAAV of the disclosure has about 125%, 150%, 175%, 200%, 250%, 300%, 350%, 400%, or 500% greater tropism for the heart or a heart cell or tissue compared to a reference AAV. In some embodiments, the reference AAV is a wild-type AAV8 or another wild-type AAV. In some embodiments, the reference

AAV is a wild-type AAV9 or another wild-type AAV. In some embodiments, the reference AAV is a wild-type AAVhu32 or another wild-type AAV. In some embodiments, the reference AAV is any AAV that does not comprise a hybrid capsid of the disclosure (e.g., does not comprise a hybrid capsid comprising a polypeptide described in Section 5.1.1 or a hybrid capsid as disclosed in Section 5.2.1).

[0171] In some embodiments, immunohistochemistry analysis (e.g., myocyte immunohistochemistry) can be performed to determine the cellular tropism of an rAAV vector of the disclosure.

[0172] In some embodiments, administration of an rAAV of the disclosure comprising a hybrid capsid comprising a polypeptide described in Section 5.1.1 or a hybrid capsid as disclosed in Section 5.2.1 and a transgene (or therapeutic product) to a subject, results in detectable expression levels of the transgene in a subject (e.g., at injection site, throughout the musculature or muscle type of the subject, in a specific area of a muscle of the subject, and/or homogeneous expression of the transgene in an area of a muscle. In some embodiments, expression levels of a transgene can be monitored in the muscle of a subject to whom an rAAV of the disclosure has been administered. Transgene expression may be measured by any suitable assay known in the art, including, without limitation, Western Blotting, electrochemiluminescent (ECL) immunoassays implemented using the Meso Scale Discovery (MSD) platform, and ELISA. In some embodiments, expression levels of the transgene in the eye may vary between different areas of the musculature.

[0173] In some embodiments, expression levels of the transgene may vary between different areas of the musculature. In some embodiments, administration of an rAAV of the disclosure to a subject results in detectable (e.g., detectable using a method described herein, or a method known in the art) transgene expression levels in an area of a muscle of the subject within about 1 hour, about 3 hours, about 5 hours, about 10 hours, about 15 hours, about 20 hours, about 24 hours, about 48 hours, about 72 hours, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 2 months, about 3 months, about 6 months, about 9 months, about 12 months, about 15 months, about 18 months, about 21 months, or about 24 months after administration of the rAAV to the subject, wherein the level of the transgene expression in the area of the muscle was undetectable prior to administration of the rAAV. In some embodiments, administration of an rAAV of the disclosure to a subject results in detectable (e.g., detectable using a method described herein, or a method known in the art) transgene expression levels in an area of a muscle of the subject within about 3 months, about 6 months, about 9 months, about 12 months, about 15 months, about 18 months, about 21 months, or about 24 months of administration of the rAAV to the subject, wherein the level of the transgene expression in the area of the muscle was undetectable prior to administration of the rAAV, and wherein the transgene expression levels in serum of the subject remain undetectable (e.g., undetectable using a method described herein, or a method known in the art). In some embodiments, an increased expression level of a transgene is detectable after an rAAV of the disclosure is administered to a subject (e.g., detectable using a method described herein, or a method known in the art) in an area of a muscle of the subject as compared an expression level of a transgene after a wild-type AAV or an AAV that does not comprise a hybrid capsid of the disclo-

sure is administered to a subject. In some embodiments, a transgene is detectable or undetectable using an immunofluorescence assay. In some embodiments, a transgene is detectable or undetectable using an immunostaining assay. In some embodiments, administration of an rAAV of the disclosure to a subject results in an increase in the transgene expression levels (e.g., an increase of about or at least about 100-fold to about 500-fold, about 500-fold to about 1000-fold, about 1000-fold to about 1500-fold, about 1500-fold to about 2000-fold, about 2000-fold to about 2500-fold, about 2500-fold to about 3000-fold, about 3000-fold to about 4000-fold, about 4000-fold to about 5000-fold, about 5000-fold to about 10,000-fold, about 10,000-fold to about 15,000-fold, about 15,000-fold to about 20,000-fold, or more than about 20,000-fold) in an area of a muscle of the subject within about 1 hour, about 3 hours, about 5 hours, about 10 hours, about 15 hours, about 20 hours, about 24 hours, about 48 hours, about 72 hours, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 2 months, about 3 months, about 6 months, about 9 months, about 12 months, about 15 months, about 18 months, about 21 months, or about 24 months of administration of the rAAV to the subject compared to the transgene expression levels prior to administration of the rAAV or compared to the transgene expression levels after a reference AAV (e.g. AAV that does not comprise a hybrid capsid of the disclosure) is administered to a subject. In some embodiments, administration of an rAAV provided herein to a subject results in an increase in the transgene expression levels (e.g., an increase of about or at least about 100-fold to about 500-fold, about 500-fold to about 1000-fold, about 1000-fold to about 1500-fold, about 1500-fold to about 2000-fold, about 2000-fold to about 2500-fold, about 2500-fold to about 3000-fold, about 3000-fold to about 4000-fold, about 4000-fold to about 5000-fold, about 5000-fold to about 10,000-fold, about 10,000-fold to about 15,000-fold, about 15,000-fold to about 20,000-fold, or more than about 20,000-fold) in an area of a muscle of the eye of the subject within about 1 hour, about 3 hours, about 5 hours, about 10 hours, about 15 hours, about 20 hours, about 24 hours, about 48 hours, about 72 hours, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 2 months, about 3 months, about 6 months, about 9 months, about 12 months, about 15 months, about 18 months, about 21 months, or about 24 months of administration of the rAAV to the subject compared to the transgene expression levels prior to administration of the rAAV or compared to the transgene expression levels after a reference AAV (e.g. AAV that does not comprise a hybrid capsid of the disclosure) is administered to a subject, wherein the transgene expression levels in serum of the subject remain about constant (e.g., remain within about 10%) compared to the transgene expression levels prior to administration.

[0174] In some embodiments, administration of an rAAV provided herein to a subject results in the transgene expression in the musculature of the subject. In some embodiments, the transgene is homogeneously expressed in a muscle or throughout a muscle of the subject. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level throughout an area of the muscle. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the

level of transgene expression within $\frac{1}{4}$ radius from injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the level of transgene expression within $\frac{1}{3}$ radius from injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the level of transgene expression within $\frac{1}{2}$ radius from injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the level of transgene expression within $\frac{1}{8}$ radius from injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the level of transgene expression within $\frac{1}{16}$ radius from injection site. In some embodiments, homogeneous expression is measured by detecting transgene expression level at injection site and comparing such expression level with the level of transgene expression within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site. In some embodiments, homogeneous expression of the transgene means that the level of expression level of the transgene at injection site is about the same as the expression level of the transgene within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site. In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is about or at most about 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, or 20% of the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is about or at most about 5% of the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is about or at most about 10% of the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is about or at most about 15% of the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is about or at most about 20% of the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some

embodiments, “about the same” refers to an expression level (e.g., at injection site) that is within 1 standard deviation (SD), 2SD, 3SD, or 4 SD from the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is within 1 standard deviation (SD) from the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site). In some embodiments, “about the same” refers to an expression level (e.g., at injection site) that is within 2SD from the expression level at the comparison location (e.g., within $\frac{1}{2}$ radius from injection site, within $\frac{1}{3}$ radius from injection site, within $\frac{1}{4}$ radius from injection site, within $\frac{1}{8}$ radius from injection site, and/or within $\frac{1}{16}$ radius from injection site).

[0175] In some embodiments, the transgene is more homogeneously expressed in a muscle of a subject after an rAAV of the disclosure is administered to the subject by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to after a corresponding wild-type AAV is administered to the same subject or a different subject (e.g., an AAV that does not comprise a variant capsid comprising a peptide insertion of the disclosure). In some embodiments, homogeneity is measured using a fluorescence assay (e.g., fluorescence resonance energy transfer (FRET)). In some embodiments, transgene expression is detected and characterized by immunohistochemistry. In some embodiments, the transgene is more uniformly expressed in a muscle of the subject after an rAAV of the disclosure is administered to the subject as compared to after a reference AAV is administered to the same subject or a different subject (e.g., an AAV that does not comprise a variant capsid of the disclosure). In some embodiments, the transgene is expressed in a wider area of the muscle of the subject after an rAAV of the disclosure is administered to the subject as compared to after a corresponding wild type AAV is administered to the same subject or a different subject (e.g., an AAV that does not comprise a hybrid capsid of the disclosure). In some embodiments, the rAAV of the disclosure provides a widespread transgene expression in the musculature of the subject. Transgene product levels can be measured in patient samples of the treated muscle, or non-treated muscles.

5.2.3 rAAV Vectors

[0176] Provided herein are rAAV vectors comprising a hybrid AAV capsid as disclosed in Section 5.1.1. In some embodiments, a recombinant viral vector of the disclosure comprises a capsid region from one or more than one of: an AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAVrh10, AAV11, AAV12, AAV13, AAVrh20, AAVhu.37, AAVrh39, AAVhu32, AAVrh21, AAVrh13, AAVrh15, AAVrh73, or AAVrh74. In some embodiments, a recombinant viral vector of the disclosure comprises a capsid region from one or more than one of: AAV serotype 1 (AAV1), serotype 2 (AAV2), serotype 2tYF (AAV2tYF), serotype 3 (AAV3), serotype 3B (AAV3B), serotype 4 (AAV4), serotype 5 (AAV5), serotype 6 (AAV6), serotype 7 (AAV7), serotype 8 (AAV8), serotype rh8

(AAVrh8), serotype 9 (AAV9), serotype 10 (AAV10), serotype 11 (AAV11), serotype 12 (AAV12), serotype 13 (AAV13), serotype rh10 (AAVrh10), serotype rh20 (AAVrh20), serotype rh39 (AAVrh39), serotype hu.37 (AAVhu.37), serotype hu32 (AAVhu32), serotype rh13 (AAVrh13), rh15 (AAVrh15), serotype rh73 (AAVrh73), and serotype rh74 (AAVrh74). In some embodiments, the second AAV is an AAV serotype 1 (AAV1), serotype 2 (AAV2), serotype 2tYF (AAV2tYF), serotype 3 (AAV3), serotype 3B (AAV3B), serotype 4 (AAV4), serotype 5 (AAV5), serotype 6 (AAV6), serotype 7 (AAV7), serotype 8 (AAV8), serotype rh8 (AAVrh8), serotype 9 (AAV9), serotype 10 (AAV10), serotype 11 (AAV11), serotype 12 (AAV12), serotype 13 (AAV13), serotype rh10 (AAVrh10), serotype rh20 (AAVrh20), serotype rh39 (AAVrh39), serotype hu.37 (AAVhu.37), serotype hu32 (AAVhu32), serotype rh13 (AAVrh13), serotype rh15 (AAVrh15), serotype rh73 (AAVrh73), and serotype rh74 (AAVrh74).

[0177] In some embodiments, the AAV vector of the disclosure is a hybrid AAV. In some embodiments, the AAV vector of the disclosure is a chimeric AAV. In certain embodiments, the recombinant viral vector is a hybrid vector, e.g., an AAV vector placed into a “helpless” adeno-viral vector. In some embodiments, the rAAV of the disclosure is useful for the treatment of a subject having, suspected of having, or experiencing a symptom associate with a disease (e.g., a human subject having, suspected of having, or experiencing a symptom associated with a disease associated with muscle), which rAAV comprises a hybrid AAV capsid as disclosed in Section 5.1.1 and 5.2.1. In certain embodiments, the AAV-based viral vector (e.g., hybrid AAV) provided herein retains tropism for a specific cell or tissue (e.g., muscle cell or tissue). In certain embodiments, the AAV-based vector provided herein encodes the AAV rep gene (required for replication) and/or the AAV cap gene (required for synthesis of the capsid protein). In some embodiments, the rAAV vector of the disclosure comprises a viral genome comprising an expression cassette for expression of a transgene or therapeutic product, under the control of regulatory elements, flanked by ITRs, and a variant capsid of the disclosure (e.g., a hybrid AAV capsid). In some embodiments, the AAV-based viral vector is a targeted vector, e.g., a vector targeted to muscle cells. In some embodiments, an rAAV vector of the disclosure comprises a nucleic acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some embodiments, an rAAV vector of the disclosure comprises a nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some embodiments, an rAAV vector of the disclosure consists of a nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. In some embodiments, an rAAV vector of the disclosure comprises a nucleic acid sequence encoding an rAAV capsid protein of the disclosure (Section 5.1.1 or 5.2.1). In some embodiments, an rAAV vector of the dis-

closure comprises a nucleic acid sequence encoding a polypeptide of the disclosure (Section 5.1.1 or 5.2.1). In some embodiments, an rAAV vector of the disclosure comprises a nucleic acid sequence encoding an rAAV capsid protein comprising an amino acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some embodiments, an rAAV vector of the disclosure comprises a nucleic acid sequence encoding an rAAV capsid protein consisting of any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

[0178] Hybrid AAV-based viral vectors are used in certain embodiments of the methods described herein. Nucleic acid sequences of AAV based viral vectors and methods of making recombinant AAV and AAV capsids are taught, for example, in U.S. Pat. No. 7,282,199 B2, U.S. Pat. No. 7,790,449 B2, U.S. Pat. No. 8,318,480 B2, U.S. Pat. No. 8,962,332 B2 and International Patent Application No. PCT/EP2014/076466, each of which is incorporated herein by reference in its entirety. In one aspect, provided herein are AAV-based viral vectors encoding a therapeutic product.

[0179] Provided in some embodiments are hybrid AAV vectors comprising a viral genome comprising an expression cassette for expression of a transgene, under the control of regulatory elements, and flanked by ITRs and an engineered viral capsid as described herein (refer to Section 5.1.1 or 5.2.1) or is at least 95%, 96%, 97%, 98%, 99% or 99.9% identical to the amino acid sequence of a hybrid AAV capsid protein or a wild-type AAV capsid protein, while retaining the biological function of the engineered hybrid AAV capsid or wild-type AAV capsid.

[0180] In certain embodiments, a single-stranded AAV (ssAAV) may be used. In certain embodiments, a self-complementary vector, e.g., scAAV, may be used (see, e.g., Wu, 2007, Human Gene Therapy, 18 (2): 171-82, McCarty et al, 2001, Gene Therapy, Vol 8, Number 16, Pages 1248-1254; and U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety).

[0181] In some embodiments, a viral vector may be replication-deficient. A “replication-defective virus” or “viral vector” refers to a synthetic or recombinant viral particle in which an expression cassette containing a gene of interest is packaged in a viral capsid or envelope, where any viral genomic sequences also packaged within the viral capsid or envelope are replication-deficient; i.e., they cannot generate progeny virions but retain the ability to infect target cells. In some embodiments, the genome of the viral vector does not include genes encoding the enzymes required to replicate (the genome can be engineered to be “gutless”—containing only the transgene of interest flanked by the signals required for amplification and packaging of the artificial genome), but these genes may be supplied during production. Therefore, it is deemed safe for use in gene therapy since replication and infection by progeny virions cannot occur except in the presence of the viral enzyme required for replication.

[0182] Fragments of AAV may be readily utilized in a variety of vector systems and host cells. Among desirable

AAV fragments are the cap proteins, including the vp1, vp2, vp3 and hypervariable regions, the rep proteins, including rep 78, rep 68, rep 52, and rep 40, and the sequences encoding these proteins. Such fragments may be used alone, in combination with other AAV serotype sequences or fragments, or in combination with elements from other AAV or non-AAV viral sequences. As used herein, artificial AAV serotypes include, without limitation, AAV with a non-naturally occurring capsid protein. An artificial AAV serotype may be, without limitation, a chimeric AAV capsid, a recombinant AAV capsid, or a “humanized” AAV capsid. In one embodiment, a vector contains AAV cap and/or rep sequences (e.g., the cap and/or rep sequences from a first AAV or a second AAV of the disclosure). See, U.S. Pat. No. 7,906,111, which is incorporated by reference herein.

A. Promoters and Modifiers of Gene Expression

[0183] In certain embodiments, the recombinant vectors provided herein comprise components that modulate delivery or expression of the therapeutic product (e.g., “expression control elements”). In certain embodiments, the recombinant vectors provided herein comprise components that modulate expression of the transgene or therapeutic product. In certain embodiments, the recombinant vectors provided herein comprise components that influence binding or targeting to cells. In certain embodiments, the recombinant vectors provided herein comprise components that influence the localization of the polynucleotide encoding the therapeutic product within the cell after uptake. In certain embodiments, the recombinant vectors provided herein comprise components that can be used as detectable or selectable markers, e.g., to detect or select for cells that have taken up the polynucleotide encoding the therapeutic product.

[0184] In certain embodiments, the recombinant vectors provided herein comprise one or more promoters. In certain embodiments, the promoter is a constitutive promoter. In certain embodiments, the promoter is an inducible promoter. Inducible promoters may be preferred so that expression of the therapeutic product may be turned on and off as desired for therapeutic efficacy. Such promoters include, for example, hypoxia-induced promoters and drug inducible promoters, such as promoters induced by rapamycin and related agents. Hypoxia-inducible promoters include promoters with HIF binding sites, see, for example, Schödel, et al., 2011, *Blood* 117 (23): e207-e217 and Kenneth and Rocha, 2008, *Biochem J.* 414:19-29, each of which is incorporated by reference for teachings of hypoxia-inducible promoters. In addition, hypoxia-inducible promoters that may be used in the constructs include the erythropoietin promoter and N-WASP promoter (see, Tsuchiya, 1993, *J. Biochem.* 113:395 for disclosure of the erythropoietin promoter and Salvi, 2017, *Biochemistry and Biophysics Reports* 9:13-21 for disclosure of N-WASP promoter, both of which are incorporated by reference for the teachings of hypoxia-induced promoters). Alternatively, the recombinant vectors may contain drug inducible promoters, for example promoters inducible by administration of rapamycin and related analogs (see, for example, International Patent Application Publication Nos. WO94/18317, WO 96/20951, WO 96/41865, WO 99/10508, WO 99/10510, WO 99/36553, and WO 99/41258, and U.S. Pat. No. 7,067,526 (disclosing rapamycin analogs), which are incorporated by reference herein for their disclosure of drug inducible promoters). The

inducible promoter may also be selected from known promoters including the ecdysone promoter, the estrogen-responsive promoter, and the tetracycline-responsive promoter, or heterodimeric repressor switch. See, Sochor et al, An Autogenously Regulated Expression System for Gene Therapeutic Ocular Applications. *Scientific Reports*, 2015 Nov. 24; 5:17105 and Daber R, Lewis M., A novel molecular switch. *J Mol Biol.* 2009 Aug. 28; 391 (4): 661-70, Epub 2009 Jun. 21 which are both incorporated herein by reference in their entirety.

[0185] In certain embodiments the promoter is a hypoxia-inducible promoter. In certain embodiments, the promoter comprises a hypoxia-inducible factor (HIF) binding site. In certain embodiments, the promoter comprises a HIF-1 α binding site. In certain embodiments, the promoter comprises a HIF-2 α binding site. In certain embodiments, the HIF binding site comprises an RCGTG motif. For details regarding the location and sequence of HIF binding sites, see, e.g., Schödel, et al., *Blood*, 2011, 117 (23): e207-e217, which is incorporated by reference herein in its entirety. In certain embodiments, the promoter comprises a binding site for a hypoxia induced transcription factor other than a HIF transcription factor. In certain embodiments, the recombinant vectors provided herein comprise one or more IRES sites that is preferentially translated in hypoxia. For teachings regarding hypoxia-inducible gene expression and the factors involved therein, see, e.g., Kenneth and Rocha, *Biochem J.*, 2008, 414:19-29, which is incorporated by reference herein in its entirety.

[0186] In another embodiment, the promoter is cell-specific. The term “cell-specific” means that the particular promoter selected for the recombinant vector can direct expression of the optimized transgene coding sequence in a particular cell or tissue type. In some embodiments, the promoter is a ubiquitous or constitutive promoter.

[0187] In some embodiments, a promoter is a muscle specific promoter. In some embodiments, a muscle-specific promoter may be operably linked to a transgene and then the artificial genome delivered via any of the AAV capsids described herein. In some embodiments, the muscle-specific promoter is selected from an SPc5-12 promoter, a muscle creatine kinase myosin light chain (MLC) promoter, a myosin heavy chain (MHC) promoter, a desmin promoter, a MHCK7 promoter, a CK6 promoter, a CK8 promoter, a MCK promoter, an alpha actin promoter, an beta actin promoter, an gamma actin promoter, an E-syn promoter, a cardiac troponin C promoter, a troponin I promoter, a myoD gene family promoter, or a muscle-selective promoter residing within intron 1 of the ocular form of Pitx3, or a truncated form or variant of any of the foregoing promoters.

[0188] In one example, synthetic promoter c5-12 (Li, X. et al. *Nature Biotechnology* Vol. 17, pp. 241-245, March 1999), known as the SPc5-12 promoter, has been shown to have cell type restricted expression, specifically muscle-cell specific expression. At less than 350 bp in length, the SPc5-12 promoter is smaller in length than most endogenous promoters, which can be advantageous when the length of the nucleic acid encoding the therapeutic protein is relatively long. Muscle-specific promoters may be combined with other cell- or tissue-specific promoters, for example liver-specific promoters as described in International Publication No. WO2021021661A1, which is hereby incorporated by reference.

[0189] In certain embodiments, the promoter is a CB7 promoter (see Dinculescu et al., 2005, *Hum Gene Ther* 16:649-663, incorporated by reference herein in its entirety). In certain embodiments, the CB7 promoter includes other expression control elements that enhance expression of the therapeutic product driven by the vector, e.g. (1) a CAG promoter; (2) a CBA promoter; (3) a CMV promoter; (4) a 1.7-kb red cone opsin promoter (PR1.7 promoter); (5) a Rhodopsin Kinase (GRK1) photoreceptor-specific enhancer-promoter (Young et al., 2003, *Retinal Cell Biology*; 44:4076-4085); (6) an hCARp promoter, which is a human cone arrestin promoter; (7) an hRKp, which is a rhodopsin kinase promoter; (8) a cone photoreceptor specific human arrestin 3 (ARR3) promoter; (9) a rhodopsin promoter; and (10) a U6 promoter (in particular when the therapeutic product is a small RNA such as shRNA or siRNA).

[0190] In certain embodiments, the promoter is a hybrid chicken β -actin (CBA) promoter with cytomegalovirus (CMV) enhancer elements. In another embodiment, the promoter is the CB7 promoter. Other suitable promoters include the human β -actin promoter, the human elongation factor-1 α promoter, the cytomegalovirus (CMV) promoter, the simian virus 40 promoter, and the herpes simplex virus thymidine kinase promoter. See, e.g., Damdindorj et al, (August 2014) *A Comparative Analysis of Constitutive Promoters Located in Adeno-Associated Viral Vectors*. PLOS ONE 9 (8): e106472. Still other suitable promoters include viral promoters, constitutive promoters, regulatable promoters (see, e.g., WO 2011/126808 and WO 2013/04943). Alternatively a promoter responsive to physiologic cues may be utilized in the expression cassette, rAAV genomes, vectors, plasmids and viruses described herein. In one embodiment, the promoter is of a small size, under 1000 bp, due to the size limitations of the AAV vector. In another embodiment, the promoter is under 400 bp. Other promoters may be selected by one of skill in the art.

[0191] In a further embodiment, the promoter is selected from SV40 promoter, the dihydrofolate reductase promoter, a phage lambda (PL) promoter, a herpes simplex viral (HSV) promoter, a tetracycline-controlled trans-activator-responsive promoter (tet) system, a long terminal repeat (LTR) promoter, such as a RSV LTR, MoMLV LTR, BIV LTR or an HIV LTR, a U3 region promoter of Moloney murine sarcoma virus, a Granzyme A promoter, a regulatory sequence(s) of the metallothionein gene, a CD34 promoter, a CD8 promoter, a thymidine kinase (TK) promoter, a B19 parvovirus promoter, a PGK promoter, a glucocorticoid promoter, a heat shock protein (HSP) promoter, such as HSP65 and HSP70 promoters, an immunoglobulin promoter, an MMTV promoter, a Rous sarcoma virus (RSV) promoter, a lac promoter, a CaMV 35S promoter, a nopaline synthetase promoter, an MND promoter, or an MNC promoter. The promoter sequences thereof are known to one of skill in the art or available publically, such as in the literature or in databases, e.g., GenBank, PubMed, or the like.

[0192] In certain embodiments, the other expression control elements include chicken β -actin intron and/or rabbit β -globin polA signal. In certain embodiments, the promoter comprises a TATA box. In certain embodiments, the promoter comprises one or more elements. In certain embodiments, the one or more promoter elements may be inverted or moved relative to one another. In certain embodiments, the elements of the promoter are positioned to function

cooperatively. In certain embodiments, the elements of the promoter are positioned to function independently. In certain embodiments, the recombinant vectors provided herein comprise one or more promoters selected from the group consisting of the human CMV immediate early gene promoter, the SV40 early promoter, the Rous sarcoma virus (RS) long terminal repeat, and rat insulin promoter. In certain embodiments, the recombinant vectors provided herein comprise one or more long terminal repeat (LTR) promoters selected from the group consisting of AAV, MLV, MMTV, SV40, RSV, HIV-1, and HIV-2 LTRs. In certain embodiments, the recombinant vectors provided herein comprise one or more tissue specific promoters (e.g., a retinal pigment epithelial cell-specific promoter). In certain embodiments, the recombinant vectors provided herein comprise a RPE65 promoter. In certain embodiments, the recombinant vectors provided herein comprise a VMD2 promoter.

[0193] In certain embodiments, the recombinant vectors provided herein comprise one or more regulatory elements other than a promoter. In certain embodiments, the recombinant vectors provided herein comprise an enhancer. In certain embodiments, the recombinant vectors provided herein comprise a repressor. In certain embodiments, the recombinant vectors provided herein comprise an intron or a chimeric intron. In certain embodiments, the recombinant vectors provided herein comprise a polyadenylation sequence.

B. Untranslated Regions

[0194] In certain embodiments wherein the therapeutic product is a therapeutic protein, the recombinant vectors provided herein comprise one or more untranslated regions (UTRs), e.g., 3' and/or 5' UTRs. In certain embodiments, the UTRs are optimized for the desired level of protein expression. In certain embodiments, the UTRs are optimized for the half-life of the mRNA encoding the therapeutic protein. In certain embodiments, the UTRs are optimized for the stability of the mRNA encoding the therapeutic protein. In certain embodiments, the UTRs are optimized for the secondary structure of the mRNA encoding the therapeutic protein.

C. Inverted Terminal Repeats

[0195] In certain embodiments, the recombinant viral vectors provided herein comprise one or more inverted terminal repeat (ITR) sequences. ITR sequences may be used for packaging the recombinant therapeutic product expression cassette into the virion of the recombinant viral vector. In certain embodiments, the ITR is from an AAV, e.g., AAV9, AAV8 or AAV2 (see, e.g., Yan et al., 2005, *J. Virol.*, 79 (1): 364-379; U.S. Pat. No. 7,282,199 B2, U.S. Pat. No. 7,790, 449 B2, U.S. Pat. No. 8,318,480 B2, U.S. Pat. No. 8,962,332 B2 and International Patent Application No. PCT/EP2014/076466, each of which is incorporated herein by reference in its entirety.

[0196] The ITRs or other AAV components may be readily isolated or engineered using techniques available to those of skill in the art from an AAV. Such AAV may be isolated, engineered, or obtained from academic, commercial, or public sources (e.g., the American Type Culture Collection, Manassas, VA). Alternatively, the AAV sequences may be engineered through synthetic or other suitable means by

reference to published sequences such as are available in the literature or in databases such as, e.g., GenBank, PubMed, or the like. AAV viruses may be engineered by conventional molecular biology techniques, making it possible to optimize these particles for cell specific delivery of nucleic acid sequences, for minimizing immunogenicity, for tuning stability and particle lifetime, for efficient degradation, for accurate delivery to the nucleus, etc.

5.2.4 Transgene and Therapeutic Product

[0197] In some embodiments, provided herein is a method of delivering a therapeutic product or transgene to a subject (e.g., to treat a muscle related disease or disorder). In some embodiments, the rAAV of the disclosure mediates delivery of a transgene to a muscle tissue or cell of a subject. In some embodiments, provided herein is a method for delivering a transgene to a muscle tissue, e.g., for treating or preventing a disease or disorder associated with a muscle. In some embodiments, methods provided herein include delivering to a muscle of a subject an effective amount of rAAV, wherein the rAAV comprises (i) a hybrid capsid protein (e.g., from Section 5.1.1 or 5.2.1) and (ii) a nucleic acid comprising a promoter operably linked to a transgene or a nucleic acid encoding a transgene or therapeutic product. In some embodiments, the rAAV of the disclosure further comprises two AAV inverted terminal repeats (ITRs), wherein the ITRs flank the transgene. In some embodiments, the transgene encodes a gene or at least part of a gene associated with a muscle disease or disorder. A transgene incorporated into an rAAV of the disclosure is not limited and may be any heterologous nucleotide sequence of interest (e.g., a heterologous gene of interest). In some embodiments, a transgene is a nucleic acid sequence, heterologous to the vector genome sequences flanking the transgene, which encodes a polypeptide, protein, or other product, of interest. The nucleic acid coding sequence is operatively linked to one or more regulatory components (e.g., promoter, enhancer, poly-A, 3'UTR, Woodchuck Hepatitis Virus Posttranscriptional Regulatory Element (WPRE)) in a manner which permits transgene transcription, translation, and/or expression in a host cell. The composition of the heterologous transgene sequence will depend upon the application (e.g., the therapeutic application or indication to be treated). In some embodiments, an rAAV of the disclosure comprises two or more heterologous transgenes, for example, two, three, four or five heterologous transgenes. In other embodiments, an rAAV of the disclosure comprises one heterologous transgene incorporated into the rAAV viral particle. In some embodiments, the transgene comprises a heterologous gene associated with a disease or disorder (e.g., muscle related disease or disorder). In some embodiments, the heterologous gene is operably linked to a regulatory sequence that controls expression of the heterologous gene in a host cell. In some embodiments, a host cell is a muscle cell. In some embodiments, a transgene encodes a therapeutic protein. In some embodiments, a therapeutic protein is associated with a muscle related disease or disorder. In some embodiments, a transgene encodes a functional dystrophin, a minidystrophin, a microdystrophin, and/or a dystrophin exon-skipping snRNA. In some embodiments, a transgene comprises a polynucleotide sequence encoding one or more of any one of SEQ ID NOs: 73, 74, 75, 76, 77, 78, 79, and/or 80. In some embodiments, a transgene comprises a polynucleotide sequence encoding

at least a portion (e.g., about or at least about: 10 amino acids, 15 amino acids, 20 amino acids, 25 amino acids, 30 amino acids, 35 amino acids, 40 amino acids, 45 amino acids, 50 amino acids, 60 amino acids, 65 amino acids, 70 amino acids, 75 amino acids, 80 amino acids, 85 amino acids, 90 amino acids, 95 amino acids, 100 amino acids, or more than 100 amino acids) of one or more of any one of SEQ ID NOs: 73, 74, 75, 76, 77, 78, 79, and/or 80. In some embodiments, a transgene consists of a polynucleotide sequence encoding one or more of any one of SEQ ID NOs: 73, 74, 75, 76, 77, 78, 79, and/or 80.

[0198] The size of the nucleotide sequence of a transgene can vary. For example, the nucleotide sequence of a transgene encoding a therapeutic protein can be at least about 1.4 kb, at least about 1.5 kb, at least about 1.6 kb, at least about 1.7 kb, at least about 1.8 kb, at least about 2.0 kb, at least about 2.2 kb, at least about 2.4 kb, at least about 2.6 kb, at least about 2.8 kb, at least about 3.0 kb, at least about 3.2 kb, at least about 3.4 kb, at least about 3.5 kb in length, at least about 4.0 kb in length, at least about 5.0 kb in length, at least about 6.0 kb in length, at least about 7.0 kb in length, at least about 8.0 kb in length, at least about 9.0 kb in length, or at least about 10.0 kb in length. In some embodiments, the nucleotide sequence of a transgene encoding a therapeutic protein is at least about 1.4 kb in length. In certain embodiments, the nucleotide sequence of a transgene encoding a therapeutic protein is about 1.4 kb to 5 kb in length. In some embodiments, the nucleotide sequences of a transgene encoding a therapeutic protein is 1.4 kb to 5 kb or 5 kb to 10 kb. Alternatively, the nucleotide sequence of a transgene is at least about 30 nucleotides, at least about 40 nucleotides, at least about 50 nucleotides, at least about 75 nucleotides in length, at least about 100 nucleotides in length, at least about 150 nucleotides in length, at least about 200 nucleotides in length, at least about 250 nucleotides in length, at least about 300 nucleotides in length, at least about 350 nucleotides in length, at least about 400 nucleotides in length, at least about 500 nucleotides in length, at least about 600 nucleotides in length, at least about 700 nucleotides in length, at least about 800 nucleotides in length, at least about 900 nucleotides in length, at least about 1000 nucleotides in length, or at least about 1200 nucleotides in length. In some embodiments, the nucleotide sequence of a transgene is about 30 to 150 nucleotides in length or about 150 to 500 nucleotides in length. In certain embodiments, the nucleotide sequence of a transgene is about 100 to 500 nucleotides in length or 500 to 1000 nucleotides in length. In some embodiments, the nucleotide sequence of a transgene is 500 nucleotides to 1200 nucleotides in length.

[0199] In some embodiments, an rAAV of the disclosure comprises a therapeutic transgene. A therapeutic transgene of the disclosure can be a sequence that encodes a biomolecule (e.g., a therapeutic biomolecule) which is useful in biology and treatment of a disease, such as a protein (e.g., an enzyme), polypeptide, peptide, RNA (e.g., tRNA, dsRNA, ribosomal RNA, catalytic RNAs, siRNA, miRNA, pre-miRNA, lncRNA, snoRNA, small hairpin RNA, trans-splicing RNA, and antisense RNA), one or more components of a gene or base editing system, e.g., CRISPR gene editing system, antisense oligonucleotides (AONs), antisense oligonucleotide (AON)-mediated exon skipping, a poison exon (s) that triggers nonsense mediated decay (NMD), or a dominant negative mutant. In some embodiments, a transgene comprises a nucleic acid sequence encoding a sequence

useful for gene therapy applications. For example, certain diseases come about when one or more loss-of-function mutations within a gene reduce or abolish the amount or activity of the protein encoded by the gene. In certain embodiments, a transgene utilized herein encodes a functional version of the protein. In other embodiments, an rAAV of the disclosure comprises a transgene comprising a nucleic acid sequence encoding a sequence useful for gene therapy applications that benefit from gene silencing. For example, certain diseases come about when gain-of-function mutations within a gene result in an aberrant amount or activity of the protein encoded by the gene. In certain embodiments, a transgene utilized herein encodes an inhibitory polynucleotide, e.g., an inhibitory RNA such as an miRNA or siRNA, or one or more components of gene editing system, e.g., a CRISPR gene editing system. In some embodiments, a transgene comprises a nucleic acid encoding a CRISPR-Cas system for targeted gene disruption or correction. In other embodiments, a transgene comprising a nucleic acid sequence encodes a sequence useful for gene therapy applications that benefit from gene addition. In certain embodiments, a transgene utilized herein encodes a gene product, e.g., a protein, not present in a recipient, e.g., a human subject, of the gene therapy.

[0200] In some embodiments, a transgene comprises a nucleic acid sequence encoding an RNA sequence useful in biology and medicine, such as, e.g., tRNA, dsRNA, ribosomal RNA, catalytic RNA, siRNA, miRNA, pre-miRNA, lncRNA, snoRNA, small hairpin RNA, trans-splicing RNA, and antisense RNA. One example of a useful RNA sequence is a sequence which inhibits or extinguishes expression of a targeted nucleic acid sequence in a treated subject. Suitable target nucleic acid sequences may include oncologic sequences and viral sequences. In some embodiments, a transgene comprises a nucleic acid sequence encoding a small nuclear RNA (snRNA) construct which induces exon skipping. In certain embodiments, an RNAi agent targets a gene of interest at a location of a single-nucleotide polymorphism (SNP) or a variant within the nucleotide sequence.

[0201] In some embodiments, an RNAi agent is an siRNA duplex, wherein the siRNA duplex contains an antisense strand (guide strand) and a sense strand (passenger strand) hybridized together forming a duplex structure, wherein the antisense strand is at least partially complementary to the nucleic acid sequence of the targeted gene, and wherein the sense strand is at least partially homologous to the nucleic acid sequence of the targeted gene. In some embodiments, the 5' end of the antisense strand has a 5'phosphate group and the 3' end of the sense strand contains a 3' hydroxyl group. In some embodiments, there are none, one or 2 nucleotide overhangs at the 3' end of one or both strands. In some embodiments, one or more than one nucleotide of an antisense strand and/or a sense strand is modified. Non-limiting examples of nucleotide modifications include 2'deoxy, 2'-fluoro, 2' O-methyl, 2'deoxy-2'fluoro, a phosphorothioate, 5'-morpholino, a universal base modified nucleotide, a terminal cap molecule at the 3'-end, the 5'-end, or both 3' and 5'-ends, an inverted abasic, or an inverted abasic locked nucleic acid modification at the 5'-end and/or 3' end.

[0202] In some embodiments, each strand of an siRNA duplex targeting a gene of interest is about 19 to 25, 19 to 24 or 19 to 21 nucleotides in length. In some embodiments, an siRNA or dsRNA includes at least two sequences that are

complementary to each other. The dsRNA includes a sense strand having a first sequence and an antisense strand having a second sequence. In some embodiments, the antisense strand includes a nucleotide sequence that is substantially complementary to at least part of an mRNA encoding the target gene, and the region of complementarity is 30 nucleotides or less, and at least 15 nucleotides in length. In some embodiments, the dsRNA is 19 to 25, 19 to 24 or 19 to 21 nucleotides in length. In some embodiments, the dsRNA is from about 15 to about 25 nucleotides in length. In some embodiments, the dsRNA is from about 25 to about 30 nucleotides in length. In some embodiments, the dsRNA is about, at least about, or at most about 15 nucleotides in length, 16 nucleotides in length, 17 nucleotides in length, 18 nucleotides in length, 19 nucleotides, 20 nucleotides, 21 nucleotides, 22 nucleotides, 23 nucleotides, 24 nucleotides, 25 nucleotides in length, 26 nucleotides in length, 27 nucleotides in length, 28 nucleotides in length, 29 nucleotides in length, or 30 nucleotides in length.

[0203] In some embodiments, an rAAV of the disclosure comprises a transgene comprising a nucleic acid sequence encoding a protein, peptide or other product that corrects or ameliorates a genetic deficiency or other abnormality in a subject. Such genetic deficiencies may include deficiencies in which gene products are expressed at less than levels considered normal for a particular subject (e.g., a human subject) or deficiencies in which a functional gene product is not expressed. In some embodiments, an rAAV of the disclosure comprises multiple transgenes to, e.g., correct or ameliorate a genetic defect caused by a multi-subunit protein. In some instances, a different transgene may be used to encode each subunit of a protein, or to encode different peptides or proteins. This may be desirable when the size of the nucleic acid sequence encoding the protein subunit is large, non-limiting examples include e.g., for an immunoglobulin, the platelet-derived growth factor, or a dystrophin protein. A host cell may be infected with an rAAV of the disclosure containing transgenes, wherein each transgene comprises a nucleic acid sequence encoding a different subunit of a multi-subunit protein, in order to produce the multi-subunit protein. Alternatively, an rAAV of the disclosure may comprise a single transgene comprising nucleic acid sequences encoding different subunits of a multi-subunit protein. In this case, a single transgene comprises nucleic acid sequences encoding each of the subunits and the nucleic acid sequence encoding each subunit may be separated by an internal ribozyme entry site (IRES). This may be desirable when the size of the nucleic acid sequence encoding each of the subunits is small, e.g., the total size of the nucleic acid sequences encoding the subunits and the IRES is less than five kilobases. As an alternative to an IRES, the nucleic acid sequence may be separated by sequences encoding a peptide, such as, e.g., 2A peptide, which self-cleaves in a post-translational event. See, e.g., Donnelly et al., *J. Gen. Virol.*, 78 (Pt 1): 13-21 (January 1997); Furler, et al., *Gene Ther.*, 8 (11): 864-873 (June 2001); Klump et al., *Gene Ther.*, 8 (10): 811-817 (May 2001). A 2A peptide is significantly smaller than an IRES, making it well suited for use when space is a limiting factor. More often, when a transgene is large, consists of multi-subunits, or both, two or more AAV viral particles (including an rAAV of the disclosure) each carrying a desired transgene may be co-administered to allow them to concatamerize in vitro or in vivo to form a single vector genome. See, e.g., Yang et al., *J Virol.*

1999 November; 73 (11): 9468-9477 for information regarding the concatamerization of AAV. For example, a first AAV viral particle may comprise a single transgene and a second AAV viral particle may comprise a different transgene for co-expression in a host cell.

[0204] In some embodiments, a transgene comprises a nucleic acid sequence encoding a protein heterologous to AAV (e.g., a therapeutic protein). In certain embodiments, a transgene comprises a nucleic acid sequence encoding a therapeutic protein that is endogenously expressed in, for example, a muscle tissue of a subject.

[0205] In some embodiments, a transgene comprises a nucleic acid sequence, which upon expression produces a detectable signal. In some instances, such a nucleic acid sequence encodes an enzyme (such as, e.g., β -lactamase, β -galactosidase (LacZ), alkaline phosphatase, thymidine kinase, chloramphenicol acetyltransferase (CAT), and luciferase), a fluorescent protein (such as, e.g., green fluorescent protein (GFP), yellow fluorescent protein, and red fluorescent protein), a membrane bound protein (such as, e.g., CD2, CD4, CD8, the influenza hemagglutinin protein, and others well known in the art, to which high affinity antibodies directed thereto exist or can be produced by conventional means) or a fusion protein comprising a membrane bound protein appropriately fused to an antigen tag domain. These nucleic acid sequences, when associated with regulatory elements which drive their expression, provide signals detectable by conventional means, including enzymatic, radiographic, colorimetric, fluorescence or other spectrographic assays, fluorescent activating cell sorting assays and immunological assays, including enzyme linked immunosorbent assay (ELISA), radioimmunoassay (RIA) and immunohistochemistry (IHC). For example, where the nucleic acid sequence encodes LacZ, the presence of an AAV vector genome expressing LacZ may be detected by assays for beta-galactosidase activity. In another example, where the transgene comprises a nucleic acid sequence encoding green fluorescent protein or luciferase, an AAV vector genome expressing the green fluorescent protein or luciferase may be detected visually by color or light production in a luminometer. An AAV viral particle comprising a transgene that comprises a nucleotide sequence encoding a product with a detectable signal may be used a selectable marker or may be used to trace the virus.

[0206] A gene associated with a muscle related disease or a disease can be a protein, polypeptide, antibody or fragment thereof (e.g., ScFv), toxin, or interfering RNA (e.g., siRNA, dsRNA, miRNA, artificial miRNA (ami-RNA), antagomir). Examples of genes associated with a muscle related disease or a disease include, but are not limited to DYS1, DYS3, DYS5, human MD1 (R4-R23/ Δ CT), human microdystrophin, Dys3978, human MD3, and/or human MD4. In some embodiments, the rAAV comprises a transgene comprising at least a portion of a DYS1 sequence. In some embodiments, the DYS1 sequence comprises an amino acid sequence of SEQ ID NO: 73. In some embodiments, the DYS1 sequence consists of SEQ ID NO: 73. In some embodiments, the DYS1 sequence comprises an amino acid sequence consisting of SEQ ID NO: 73. In some embodiments, the rAAV comprises a transgene comprising at least a portion of DYS3 sequence. In some embodiments, the DYS3 sequence comprises the amino acid sequence of SEQ ID NO: 74. In some embodiments, the DYS3 sequence consists of SEQ ID NO: 74. In some embodiments, the

DYS3 sequence comprises an amino acid sequence consisting of SEQ ID NO: 74. In some embodiments, the rAAV comprises a transgene comprising at least a portion of DYS5 sequence. In some embodiments, the DYS5 sequence comprises an amino acid sequence of SEQ ID NO: 75. In some embodiments, the DYS5 sequence consists of SEQ ID NO: 75. In some embodiments, the DYS5 sequence comprises an amino acid sequence consisting of SEQ ID NO: 75. In some embodiments, the rAAV comprises a transgene comprising at least a portion of a human MD1 (R4-R23/ Δ CT) sequence. In some embodiments, the human MD1 (R4-R23/ Δ CT) sequence comprises an amino acid sequence of SEQ ID NO: 76. In some embodiments, the human MD1 (R4-R23/ Δ CT) sequence consists of SEQ ID NO: 76. In some embodiments, the human MD1 (R4-R23/ Δ CT) sequence comprises an amino acid sequence consisting of SEQ ID NO: 76. In some embodiments, the rAAV comprises a transgene comprising at least a portion of a human microdystrophin sequence. In some embodiments, the human microdystrophin sequence comprises the amino acid sequence of SEQ ID NO: 77. In some embodiments, the human microdystrophin sequence consists of SEQ ID NO: 77. In some embodiments, the human microdystrophin sequence comprises an amino acid sequence consisting of SEQ ID NO: 77. In some embodiments, the rAAV comprises a transgene comprising at least a portion of Dys3978 sequence. In some embodiments, the Dys3978 sequence comprises the amino acid sequence of SEQ ID NO: 78. In some embodiments, the Dys3978 sequence consists of SEQ ID NO: 78. In some embodiments, the Dys3978 sequence comprises an amino acid sequence consisting of SEQ ID NO: 78. In some embodiments, the rAAV comprises a transgene comprising at least a portion of human MD3 sequence. In some embodiments, the human MD3 sequence comprises the amino acid sequence of SEQ ID NO: 79. In some embodiments, the human MD3 sequence consists of SEQ ID NO: 79. In some embodiments, the human MD3 sequence comprises an amino acid sequence consisting of SEQ ID NO: 79. In some embodiments, the rAAV comprises a transgene comprising a human MD4 sequence. In some embodiments, the human MD4 sequence comprises the amino acid sequence of SEQ ID NO: 80. In some embodiments, the human MD4 sequence consists of SEQ ID NO: 80. In some embodiments, the human MD4 sequence comprises an amino acid sequence consisting of SEQ ID NO: 80. In some embodiments, the transgene is selected from SMN, mini/micro dystrophin gene, human-alpha-sarcoglycan, gamma-sarcoglycan, huFollistatin344, GALGT2, and/or hSGCB. In some embodiments, the transgene comprises or consists of SMN. In certain embodiments, the rAAV comprising a SMN transgene is for treating Spinal Muscular Atrophy (SMA) in the subject. In some embodiments, the transgene comprises or consists of a mini/micro dystrophin gene. In certain embodiments, the rAAV comprising a mini/micro dystrophin gene transgene is for treating Duchenne Muscular Dystrophy in the subject. In some embodiments, the transgene comprises or consists of a human-alpha-sarcoglycan. In certain embodiments, the rAAV comprising a human-alpha-sarcoglycan transgene is for treating Duchenne Muscular Dystrophy in the subject. In certain embodiments, the rAAV comprising a human-alpha-sarcoglycan transgene is for treating Limb Girdle Muscular Dystrophy Type 2C or Gamma-sarcoglycanopathy in the subject. In some embodiments, the transgene comprises or consists of a gamma-sarcoglycan. In certain embodiments,

the rAAV comprising a gamma-sarcoglycan transgene is for treating Limb Girdle Muscular Dystrophy Type 2C or Gamma-sarcoglycanopathy in the subject. In some embodiments, the transgene comprises or consists of a huFollistatin344. In certain embodiments, the rAAV comprising a huFollistatin344 transgene is for treating Becker Muscular Dystrophy or Sporadic Inclusion Body Myositis in the subject. In some embodiments, the transgene comprises or consists of GALGT2. In certain embodiments, the rAAV comprising a GALGT2 transgene is for treating Duchenne Muscular Dystrophy in the subject. In some embodiments, the transgene comprises or consists of hSGCB. In certain embodiments, the rAAV comprising a GALGT2 transgene is for treating Limb-Girdle Muscular Dystrophy Type 2E in the subject.

5.2.5 Pharmaceutical Compositions and Kits

[0207] In certain embodiments, provided herein are pharmaceutical compositions comprising an rAAV of the disclosure and a pharmaceutically acceptable carrier. In some embodiments, a pharmaceutical composition comprises a recombinant adeno-associated virus (rAAV) vector comprising a recombinant AAV capsid protein, wherein the recombinant AAV capsid protein comprises an amino acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119. In some embodiments, a pharmaceutical composition comprises a recombinant adeno-associated virus (rAAV) vector comprising a nucleic acid sequence that is about or at least about 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical to any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120. The pharmaceutical composition may be prepared as individual, single unit dosage forms. The pharmaceutical compositions provided herein can be formulated for, for example, parenteral, subcutaneous, intramuscular, intravenous, intraperitoneal, intranasal, intrathecal, transdermal, suprachoroidal, retinal, subretinal, juxtascleral, intravitreal, subconjunctival, and/or intraretinal administration.

[0208] Compositions are described comprising a recombinant vector encoding a therapeutic product described herein and a suitable carrier. A suitable carrier (e.g., for suprachoroidal, subretinal, juxtascleral, intravitreal, subconjunctival, and/or intraretinal administration) would be readily selected by one of skill in the art.

[0209] The disclosure provides a pharmaceutical composition comprising a recombinant adeno-associated virus (AAV), potassium phosphate monobasic, sodium chloride, sodium phosphate dibasic anhydrous, sucrose, and surfactant.

[0210] In some embodiments, the disclosure provides a pharmaceutical composition comprising a recombinant adeno-associated virus (AAV), ionic salt excipient or buffering agent, sucrose, and poloxamer 188. In some embodiments, the ionic salt excipient or buffering agent can be one or more components from the group consisting of potassium phosphate monobasic, potassium phosphate, sodium chloride, sodium phosphate dibasic anhydrous, sodium phos-

phate hexahydrate, sodium phosphate monobasic monohydrate, tromethamine, tris(hydroxymethyl)aminomethane hydrochloride (Tris-HCl), amino acid, histidine, histidine hydrochloride (histidine-HCl), sodium succinate, sodium citrate, sodium acetate, and (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid) (HEPES), sodium sulfate, magnesium sulfate, magnesium chloride 6-hydrate, calcium sulfate, potassium chloride, calcium chloride, calcium citrate.

[0211] In certain embodiments, the pharmaceutical composition has a ionic strength about 60 mM to 115 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 60 mM to 100 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 65 mM to 95 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 70 mM to 90 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 75 mM to 85 mM.

[0212] In certain embodiments, the pharmaceutical composition has a ionic strength about 30 mM to 100 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 35 mM to 95 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 40 mM to 90 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 45 mM to 85 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 50 mM to 80 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 55 mM to 75 mM. In certain embodiments, the pharmaceutical composition has a ionic strength about 60 mM to 70 mM.

[0213] In certain embodiments, the pharmaceutical composition has a ionic strength ranging from 60 mM to 115 mM. In certain embodiments, the pharmaceutical composition has a ionic strength ranging from 60 mM to 100 mM. In certain embodiments, the pharmaceutical composition has a ionic strength ranging from 65 mM to 95 mM. In certain embodiments, the pharmaceutical composition has a ionic strength ranging from 70 mM to 90 mM. In certain embodiments, the pharmaceutical composition has a ionic strength ranging from 75 mM to 85 mM.

[0214] In certain embodiments, the pharmaceutical composition has a ionic strength range from 30 mM to 100 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 35 mM to 95 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 40 mM to 90 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 45 mM to 85 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 50 mM to 80 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 55 mM to 75 mM. In certain embodiments, the pharmaceutical composition has a ionic strength range from 60 mM to 70 mM.

[0215] In certain embodiments, the pharmaceutical composition comprises potassium chloride at a concentration of 0.2 g/L.

[0216] In certain embodiments, the pharmaceutical composition comprises potassium phosphate monobasic at a concentration of 0.2 g/L.

[0217] In certain embodiments, the pharmaceutical composition comprises sodium chloride at a concentration of 5.84 g/L, and

[0218] In certain embodiments, the pharmaceutical composition comprises sodium phosphate dibasic anhydrous at a concentration of 1.15 g/L.

[0219] In certain embodiments, the pharmaceutical composition comprises sucrose at a concentration of 3% (weight/volume, 30 g/L) to 18% (weight/volume, 180 g/L). In certain embodiments, the pharmaceutical composition comprises sucrose at a concentration of 4% (weight/volume, 40 g/L).

[0220] In certain embodiments, the pharmaceutical composition comprises poloxamer 188 at a concentration of 0.001% (weight/volume, 0.01 g/L).

[0221] In certain embodiments, the pharmaceutical composition comprises poloxamer 188 at a concentration of 0.0005% (weight/volume, 0.005 g/L) to 0.05% (weight/volume, 0.5 g/L). In certain embodiments, the pharmaceutical composition comprises poloxamer 188 at a concentration of 0.001% (weight/volume, 0.01 g/L).

[0222] In some embodiments, the disclosure provides a pharmaceutical composition comprises a recombinant adeno-associated virus (AAV), ionic salt excipient or buffering agent, sucrose, and surfactant. In some embodiments, the ionic salt excipient or buffering agent can be one or more components from the group consisting of potassium phosphate monobasic, potassium phosphate, sodium chloride, sodium phosphate dibasic anhydrous, sodium phosphate hexahydrate, sodium phosphate monobasic monohydrate, tromethamine, tris(hydroxymethyl)aminomethane hydrochloride (Tris-HCl), amino acid, histidine, histidine hydrochloride (histidine-HCl), sodium succinate, sodium citrate, sodium acetate, and (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid) (HEPES), sodium sulfate, magnesium sulfate, magnesium chloride 6-hydrate, calcium sulfate, potassium chloride, calcium chloride, calcium citrate. In some embodiments, the surfactant can be one or more components from the group consisting of poloxamer 188, polysorbate 20, and polysorbate 80.

[0223] In certain embodiments, the pharmaceutical composition comprises polysorbate 20 at a concentration of 0.0005% (weight/volume, 0.05 g/L) to 0.05% (weight/volume, 0.5 g/L).

[0224] In certain embodiments, the pharmaceutical composition comprises polysorbate 80 at a concentration of 0.0005% (weight/volume, 0.05 g/L) to 0.05% (weight/volume, 0.5 g/L).

[0225] In certain embodiments, the pH of the pharmaceutical composition is about 7.4.

[0226] In certain embodiments, the pH of the pharmaceutical composition is about 6.0 to 9.0.

[0227] In certain embodiments, the pH of the pharmaceutical composition is 7.4.

[0228] In certain embodiments, the pH of the pharmaceutical composition is 6.0 to 9.0.

[0229] As used herein and unless otherwise specified, the term "about" means within plus or minus 10% of a given value or range.

[0230] In certain embodiments, the pharmaceutical composition is in a hydrophobically-coated glass vial.

[0231] In certain embodiments, the pharmaceutical composition is in a Cyclo Olefin Polymer (COP) vial.

[0232] In certain embodiments, the pharmaceutical composition is in a Daikyo Crystal Zenith® (CZ) vial.

[0233] In certain embodiments, the pharmaceutical composition is in a TopLyco coated vial.

[0234] In certain embodiments, disclosed herein is a pharmaceutical composition consists of: (a) the recombinant AAV, (b) potassium chloride at a concentration of 0.2 g/L, (c) potassium phosphate monobasic at a concentration of 0.2 g/L, (d) sodium chloride at a concentration of 5.84 g/L, (e) sodium phosphate dibasic anhydrous at a concentration of 1.15 g/L, (f) sucrose at a concentration of 4% weight/volume (40 g/L), (g) poloxamer 188 at a concentration of 0.001% weight/volume (0.01 g/L), and (h) water, and wherein the recombinant AAV is AAV9.

[0235] In certain embodiments, the vector genome concentration (VGC) of the pharmaceutical composition is about 3×10^9 GC/mL, about 1×10^{10} GC/mL, about 1.2×10^{10} GC/mL, about 1.6×10^{10} GC/mL, about 4×10^{10} GC/mL, about 6×10^{10} GC/mL, about 1×10^{11} GC/mL, about 2×10^{11} GC/mL, about 2.4×10^{11} GC/mL, about 2.5×10^{11} GC/mL, about 3×10^{11} GC/mL, about 6.2×10^{11} GC/mL, about 1×10^{12} GC/mL, about 3×10^{12} GC/mL, about 2×10^{13} GC/mL or about 3×10^{13} GC/mL. In some embodiments, 2×10^9 and 1×10^{10} of vector genome is administered to a subject.

[0236] In certain embodiments, the disclosure provides a pharmaceutical composition or formulation comprising a recombinant adeno-associated virus (AAV) of the disclosure, potassium phosphate monobasic, sodium chloride, sodium phosphate dibasic anhydrous, sucrose, and poloxamer 188. In some embodiments, a pharmaceutical composition or formulation comprises a polypeptide of the disclosure (e.g., from Section 5.1.1), a hybrid AAV of the disclosure (e.g., from Section 5.2.1), a nucleotide sequence of the disclosure, a vector of the disclosure, or any other component of the disclosure.

[0237] In some embodiments, the pharmaceutical composition consists of: (a) an AAV capsid packaging vector encoding a transgene of interest, (b) potassium chloride at a concentration of 0.2 g/L, (c) potassium phosphate monobasic at a concentration of 0.2 g/L, (d) sodium chloride at a concentration of 5.84 g/L, (e) sodium phosphate dibasic anhydrous at a concentration of 1.15 g/L, (f) sucrose at a concentration of 4% weight/volume (40 g/L), (g) poloxamer 188 at a concentration of 0.001% weight/volume (0.01 g/L), and (h) water.

[0238] In some embodiments, the pharmaceutical composition is a liquid composition. In some embodiments, the pharmaceutical composition is a frozen composition. In some embodiments, the pharmaceutical composition is a lyophilized composition from a liquid composition disclosed herein. In some embodiments, the pharmaceutical composition is a reconstituted lyophilized formulation.

[0239] In some embodiments, the pharmaceutical composition is a lyophilized composition comprising a residual moisture content between about 1% and about 7%. In some embodiments, the pharmaceutical composition is a lyophilized composition comprising a residual moisture content between about 2% and about 6%. In some embodiments, the pharmaceutical composition is a lyophilized composition comprising a residual moisture content between about 3% and about 4%. In some embodiments, the pharmaceutical composition is a lyophilized composition comprising a residual moisture content about 5%.

[0240] In certain aspects, disclosed herein is a method of treating or preventing a disease in a subject, comprising administering to the subject the pharmaceutical composition. In some embodiments, a pharmaceutical composition

provided herein is suitable for administration by one, two or more routes of administration.

[0241] In certain aspects, disclosed herein is a method of treating or preventing a disease in a subject, comprising administering to the subject the pharmaceutical composition by intravenous administration, subcutaneous administration, intramuscular injection, suprachoroidal injection (for example, via a suprachoroidal drug delivery device such as a microinjector with a microneedle), subretinal injection via transvitreal approach (a surgical procedure), subretinal administration via the suprachoroidal space (for example, a surgical procedure via a subretinal drug delivery device comprising a catheter that can be inserted and tunneled through the suprachoroidal space toward the posterior pole, where a small needle injects into the subretinal space), and/or a posterior juxtasceral depot procedure (for example, via a juxtasceral drug delivery device comprising a cannula whose tip can be inserted and kept in direct apposition to the scleral surface).

[0242] In certain embodiments, the pharmaceutical composition provided herein is suitable for intravenous administration, subcutaneous administration, intramuscular injection, suprachoroidal injection (for example, via a suprachoroidal drug delivery device such as a microinjector with a microneedle), subretinal injection via transvitreal approach (a surgical procedure), subretinal administration via the suprachoroidal space (for example, a surgical procedure via a subretinal drug delivery device comprising a catheter that can be inserted and tunneled through the suprachoroidal space toward the posterior pole, where a small needle injects into the subretinal space), and/or a posterior juxtasceral depot procedure (for example, via a juxtasceral drug delivery device comprising a cannula whose tip can be inserted and kept in direct apposition to the scleral surface)).

[0243] In certain embodiments, the pharmaceutical composition has a desired viscosity that is suitable for intravenous administration, subcutaneous administration, intramuscular injection, suprachoroidal injection (for example, via a suprachoroidal drug delivery device such as a microinjector with a microneedle), subretinal injection via transvitreal approach (a surgical procedure), subretinal administration via the suprachoroidal space (for example, a surgical procedure via a subretinal drug delivery device comprising a catheter that can be inserted and tunneled through the suprachoroidal space toward the posterior pole, where a small needle injects into the subretinal space), and/or a posterior juxtasceral depot procedure (for example, via a juxtasceral drug delivery device comprising a cannula whose tip can be inserted and kept in direct apposition to the scleral surface)).

[0244] In certain embodiments, the pharmaceutical composition has a desired density that is suitable for intravenous administration, subcutaneous administration, intramuscular injection, suprachoroidal injection (for example, via a suprachoroidal drug delivery device such as a microinjector with a microneedle), subretinal injection via transvitreal approach (a surgical procedure), subretinal administration via the suprachoroidal space (for example, a surgical procedure via a subretinal drug delivery device comprising a catheter that can be inserted and tunneled through the suprachoroidal space toward the posterior pole, where a small needle injects into the subretinal space), and/or a posterior juxtasceral depot procedure (for example, via a juxtasceral drug deliv-

ery device comprising a cannula whose tip can be inserted and kept in direct apposition to the scleral surface)).

[0245] In certain embodiments, the pharmaceutical composition has a desired osmolality that is suitable for intravenous administration, subcutaneous administration, intramuscular injection, suprachoroidal injection (for example, via a suprachoroidal drug delivery device such as a microinjector with a microneedle), subretinal injection via transvitreal approach (a surgical procedure), subretinal administration via the suprachoroidal space (for example, a surgical procedure via a subretinal drug delivery device comprising a catheter that can be inserted and tunneled through the suprachoroidal space toward the posterior pole, where a small needle injects into the subretinal space), and/or a posterior juxtasceral depot procedure (for example, via a juxtasceral drug delivery device comprising a cannula whose tip can be inserted and kept in direct apposition to the scleral surface)). In specific embodiments, the desired osmolality for subretinal administration is 160 to 430 mOsm/kg H₂O. In other specific embodiments, the desired osmolality of suprachoroidal administration is less than 600 mOsm/kg H₂O.

[0246] In certain embodiments, the pharmaceutical composition has a osmolality of about 100 to 500 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 130 to 470 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 160 to 430 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 200 to 400 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 240 to 340 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 280 to 300 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of about 295 to 395 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality of less than 600 mOsm/kg H₂O. In certain embodiments, the pharmaceutical composition has a osmolality range of 200 mOsm/L to 660 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 200 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 250 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 300 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 350 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 400 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 450 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 500 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 550 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 600 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 650 mOsm/L. In certain embodiments, the pharmaceutical composition has a osmolality of about 660 mOsm/L.

[0247] In some embodiments, the recombinant vector of the disclosure is used for delivering the transgene to a cell. Such vectors can include non-replicating recombinant adeno-associated virus vectors ("rAAV"), however, other viral vectors may be used, including but not limited to

lentiviral vectors, vaccinia viral vectors, or non-viral expression vectors referred to as “naked DNA” constructs.

[0248] In certain embodiments, gene therapy constructs are supplied as a frozen sterile, single use solution of the AAV vector active ingredient in a formulation buffer. In a specific embodiment, the pharmaceutical compositions suitable for subretinal administration comprise a suspension of the recombinant vector in a formulation buffer comprising a physiologically compatible aqueous buffer, a surfactant and optional excipients. In a specific embodiment, the construct is formulated in Dulbecco's phosphate buffered saline and 0.001% poloxamer 188, pH=7.4.

[0249] The disclosure provides a pharmaceutical composition comprising a pharmaceutically acceptable carrier and an agent of the disclosure, said agent comprising a rAAV of the disclosure. In some embodiments, the pharmaceutical composition comprises rAAV combined with a pharmaceutically acceptable carrier for administration to a subject. In some embodiment, the term “pharmaceutically acceptable” means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in animals, and more particularly in humans. The term “carrier” refers to a diluent, adjuvant (e.g., Freund's complete and incomplete adjuvant), excipient, or vehicle with which the agent is administered. Such pharmaceutical carriers can be sterile liquids, such as water and oils, including those of petroleum, animal, vegetable, or synthetic origin, including, e.g., peanut oil, soybean oil, mineral oil, sesame oil and the like. Water is a common carrier when the pharmaceutical composition is administered intravenously. Saline solutions and aqueous dextrose and glycerol solutions can also be employed as liquid carriers, particularly for injectable solutions. Suitable pharmaceutical excipients include starch, glucose, lactose, sucrose, gelatin, malt, rice, flour, chalk, silica gel, sodium stearate, glycerol monostearate, talc, sodium chloride, dried skim milk, glycerol, propylene, glycol, water, ethanol and the like. Additional examples of pharmaceutically acceptable carriers, excipients, and stabilizers include, but are not limited to, buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid; low molecular weight polypeptides; proteins, such as serum albumin and gelatin; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEENTM, polyethylene glycol (PEG), and PLURONICSTM as known in the art. The pharmaceutical composition of the present disclosure can also include a lubricant, a wetting agent, a sweetener, a flavoring agent, an emulsifier, a suspending agent, and a preservative, in addition to the above ingredients. These compositions can take the form of solutions, suspensions, emulsion, tablets, pills, capsules, powders, sustained-release formulations and the like.

[0250] In certain embodiments of the disclosure, pharmaceutical compositions are provided for use in accordance with the methods of the disclosure, said pharmaceutical compositions comprising a therapeutically and/or prophylactically effective amount of an agent of the disclosure along with a pharmaceutically acceptable carrier.

[0251] In certain embodiments, the agent of the disclosure is substantially purified (i.e., substantially free from substances that limit its effect or produce undesired side-effects). In one embodiment, the host or subject is an animal, e.g., a mammal such as non-primate (e.g., cows, pigs, horses, cats, dogs, rats etc.) and a primate (e.g., monkey such as, a cynomolgus monkey and a human). In a certain embodiment, the host is a human.

[0252] Provided herein are kits comprising a pharmaceutical composition described herein, contained in one or more containers. The containers that the pharmaceutical composition can be packaged in can include, but are not limited to, bottles, packets, ampoules, tubes, inhalers, bags, vials, and containers. In certain embodiments, the kit comprises instructions for administering the pharmaceutical administration. In certain embodiments, the kit comprises devices that can be used to administer (e.g., to musculature tissue) the pharmaceutical composition, including, but not limited to, syringes, catheters, needle-less injectors, drip bags, patches and inhalers. In another embodiment, a kit further comprises one or more other prophylactic or therapeutic agents useful for the treatment of a condition, in one or more containers.

[0253] The disclosure also provides agents of the disclosure packaged in a hermetically sealed container such as an ampoule or sachette indicating the quantity of the agent or active agent. In one embodiment, the agent is supplied as a dry sterilized lyophilized powder or water free concentrate in a hermetically sealed container and can be reconstituted, e.g., with water or saline, to the appropriate concentration for administration to a subject. Typically, the agent is supplied as a dry sterile lyophilized powder in a hermetically sealed container at a unit dosage of at least 5 mg, more often at least 10 mg, at least 15 mg, at least 25 mg, at least 35 mg, at least 45 mg, at least 50 mg, or at least 75 mg. The lyophilized agent should be stored at between 2 and 8° C. in its original container and the agent should be administered within 12 hours, usually within 6 hours, within 5 hours, within 3 hours, or within 1 hour after being reconstituted. In an alternative embodiment, an agent of the disclosure is supplied in liquid form in a hermetically sealed container indicating the quantity and concentration of agent or active agent. Typically, the liquid form of the agent is supplied in a hermetically sealed container at least 1 mg/ml, at least 2.5 mg/ml, at least 5 mg/ml, at least 8 mg/ml, at least 10 mg/ml, at least 15 mg/kg, or at least 25 mg/ml.

[0254] The compositions of the disclosure include bulk drug compositions useful in the manufacture of pharmaceutical compositions (e.g., impure or non-sterile compositions) as well as pharmaceutical compositions (i.e., compositions that are suitable for administration to a subject or patient). Bulk drug compositions can be used in the preparation of unit dosage forms, e.g., comprising a prophylactically or therapeutically effective amount of an agent disclosed herein or a combination of those agents and a pharmaceutically acceptable carrier.

[0255] The disclosure further provides a pharmaceutical pack or kit comprising one or more containers filled with one or more of the agents of the disclosure. Additionally, one or more other prophylactic or therapeutic agents useful for the treatment of the target disease or disorder can also be included in the pharmaceutical pack or kit. The disclosure also provides a pharmaceutical pack or kit comprising one or more containers filled with one or more of the ingredients of

the pharmaceutical compositions of the disclosure. Optionally associated with such container(s) can be a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, which notice reflects approval by the agency of manufacture, use, or sale for human administration.

[0256] Generally, the ingredients of compositions of the disclosure are supplied either separately or mixed together in unit dosage form, for example, as a dry lyophilized powder or water-free concentrate in a hermetically sealed container such as an ampoule or sachette indicating the quantity of agent or active agent. Where the composition is to be administered by infusion, it can be dispensed with an infusion bottle containing sterile pharmaceutical grade water or saline. Where the composition is administered by injection, an ampoule of sterile water for injection or saline can be provided so that the ingredients may be mixed prior to administration.

5.3 Manufacture of rAAVS

5.3.1 Polynucleotides, Plasmids and Cells

[0257] In some embodiments, the disclosure provides for an isolated nucleic acid comprising a nucleotide sequence encoding a variant adeno-associated virus (AAV) capsid protein of the disclosure (e.g., a hybrid AAV capsid comprising a heterologous amino acid sequence or a peptide insert). In some aspects, the disclosure provides for a nucleic acid for use, wherein the nucleic acid encodes a therapeutic product operatively linked to a promoter or enhancer-promoter described herein.

[0258] In certain embodiments, provided herein are recombinant vectors that comprise one or more nucleic acids (e.g. polynucleotides). The nucleic acids may comprise DNA, RNA, or a combination of DNA and RNA. In certain embodiments, the DNA comprises one or more of the sequences selected from the group consisting of promoter sequences, the sequence encoding the therapeutic product of interest, untranslated regions, and termination sequences. In certain embodiments, recombinant vectors provided herein comprise a promoter operably linked to the sequence encoding the therapeutic product of interest.

[0259] In certain embodiments, nucleic acids (e.g., polynucleotides) and nucleic acid sequences disclosed herein may be codon-optimized, for example, via any codon-optimization technique known to one of skill in the art (see, e.g., review by Quax et al., 2015, *Mol Cell* 59:149-161). In some embodiments, the polynucleotide is in the form of a ssDNA. In another embodiment, the polynucleotide is in the form of a dsDNA.

[0260] Also provided herein are plasmids comprising a polynucleotide provided herein (hereinafter “rAAV plasmids”). In one embodiment, the rAAV plasmid is a ssDNA plasmid. In another embodiment, the rAAV plasmid is a dsDNA plasmid. In some embodiments, the rAAV plasmid is in a circular form. In other embodiments, the rAAV plasmid is in a linear form.

[0261] Further provided herein are cells (preferably ex vivo cells) expressing (e.g., recombinantly) an rAAV provided herein. In certain embodiments, the cell (e.g., ex vivo cell) comprises a polynucleotide provided herein or an rAAV plasmid provided herein. In certain embodiments, the cell (e.g., ex vivo cell) further comprises helper polynucleotide(s) or helper plasmids providing the AAV Rep, Cap, and Ad5 functions. The cell (e.g., ex vivo cells) can be a

mammalian host cell, for example, a cell associated with a muscle, HEK293, HEK293-T, A549, WEHI, 10T1/2, BHK, MDCK, COS1, COS7, BSC 1, BSC 40, BMT 10, VERO, W138, HeLa, 293, Saos, C2C12, L, HT1080, HepG2, primary fibroblast, hepatocyte, and myoblast cells. The mammalian host cell can be derived from, for example, human, monkey, mouse, rat, rabbit, or hamster. In some embodiments, the mammalian host cell is a muscle related cell. In certain embodiments, the mammalian host cell is a human embryonic kidney 293 (HEK293) cell or HEK293-T cell. Provided are polynucleotides encoding the recombinant AAV hybrid capsid proteins, as an rAAV plasmid, and such Cap or “RepCap” constructs comprise the Cap gene encoding an hybrid capsid described herein. Also provided are host cells, such as bacterial host cells, for replication and production of these plasmid vectors. In other embodiments, the polynucleotides described herein comprise an rAAV plasmid which is replicable in a bacterial cell. In some embodiments, the host cell is a bacterial host cell comprising the rAAV plasmid.

5.3.2 Methods of Making rAAVs and Assessment of Efficacy

[0262] In some embodiments, provided herein is a method of producing a recombinant adeno-associated virus (rAAV) of the disclosure. In some embodiments, the method comprises culturing a cell in a cell culture to produce the rAAV. In some embodiments, a cell is a muscle cell. In some embodiments, a cell (e.g., muscle cell) comprises a nucleotide sequence encoding a capsid protein. In some embodiments, a cell further comprises a nucleotide sequence comprising or encoding a transgene (e.g., a transgene of the disclosure; Section 5.2.4 and 5.4). In some embodiments, the capsid protein comprises: (i) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1/2s region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), and (ii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP3 region of a second AAV (e.g., AAV8, AAVrh13, or VP3 region of any wild-type AAV, or a VP3 region of any AAV of the disclosure or identified in Section 7). In some embodiments, the method further comprises collecting the rAAV from the cell culture. In some embodiments, the capsid protein further comprises an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1u region of a first AAV or a second AAV (e.g., AAV5, AAVhu32, AAV8, AAVrh13, or VP1u of any wild-type AAV, or a VP1u region of any AAV of the disclosure or identified in Section 7). In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 95% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 98% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, the capsid protein comprises an amino acid sequence that is about or at least about 99% identical to VP1u, VP1/2s, and/or VP3 of a first or second AAV. In some embodiments, the capsid protein, comprises: (i) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to

VP1u region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), (ii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP1/2s region of a first AAV (e.g., AAV5, AAVhu32, or VP1/2s region of any wild-type AAV, or a VP1/2s region of any AAV of the disclosure or identified in Section 7), and (iii) an amino acid sequence that is about or at least about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identical to VP3 region of a second AAV (e.g., AAV8, AAVrh13, or VP3 region of any wild-type AAV, or a VP3 region of any AAV of the disclosure or identified in Section 7).

[0263] In some embodiments, the cell is selected from an invertebrate cell, an insect cell, or a mammalian cell. In some embodiments, the mammalian cell is selected from HEK293, HeLa, CHO, NS0, SP2/0, PER.C6, Vero, RD, BHK, HT-1080, A549, Cos-7, ARPE-19, MRC-5, or any combination thereof. In some embodiments, the insect cell is selected from High Five, Sf9, Se301, SeIZD2109, SeUCR1, Sf900+, Sf21, Bti-tn-5b1-4, MG-1, Tn368, HzAm1, BM-N, Ha2302, Hz2E5, A638, or any combination thereof.

[0264] Provided herein are methods of making an rAAV of the disclosure. In certain embodiments, the method comprises transfecting a cell (e.g., an ex vivo cell) with an rAAV plasmid of the disclosure and one or more helper plasmids collectively providing the AAV Rep, Cap, and Ad5 functions. In certain embodiments, the one or more helper plasmids collectively comprises the nucleotide sequences of AAV genes Rep, Cap, VA, E2a and E4.

[0265] In some embodiments, the recombinant viral vector used in the methods described herein is or is derived from a recombinant/hybrid adenovirus vector. The recombinant adenovirus can be a first generation vector, with an E1 deletion, with or without an E3 deletion, and with the expression cassette inserted into either deleted region. The recombinant adenovirus can be a second generation vector, which contains full or partial deletions of the E2 and E4 regions. A helper-dependent adenovirus retains only the adenovirus inverted terminal repeats and the packaging signal (phi). In some embodiments, the transgene or therapeutic product is inserted between the packaging signal and the 3'ITR, with or without stuffer sequences to keep the genome close to wild-type size of approx. 36 kb. An exemplary protocol for production of adenoviral vectors may be found in Alba et al., 2005, "Gutless adenovirus: last generation adenovirus for gene therapy," Gene Therapy 12: S18-S27, which is incorporated by reference herein in its entirety.

[0266] The manufacture of an rAAV provided herein for gene therapy applications can use methods known in the art, for example, as described in Clement et al., 2016, Molecular Therapy-Methods & Clinical Development, 27:16002, which is incorporated by reference herein in its entirety. In certain embodiments, transfection of the plasmid DNA is performed using calcium phosphate plasmid precipitation on human embryonic kidney 293 cells (HEK293) or HEK293-T with the rAAV plasmid and the helper plasmid(s) that provide the AAV Rep and Cap functions as well as the Ad5 genes (VA RNAs, E2a, and E4) as is described in the art. In certain embodiments, the Rep, Cap, and Ad5 genes can be on the same helper plasmid. In certain embodiments, a two-helper method (or triple transfection) is utilized where

AAV Rep, Cap, and Ad5 functions are provided from separate plasmids. In certain embodiments, the HEK293 cells can be adapted to grow in suspension in an animal component and antibiotic-free media. In some embodiments, a hybrid AAV can be produced with one or more than one plasmid or vector. In some embodiments, one or more plasmid or vector can provide or encode a capsid component from one AAV (e.g., VP1u and/or VP1/2s region) and another plasmid or vector can provide a capsid component from another AAV (e.g., VP3 region).

[0267] In certain embodiments, rAAV can be manufactured using packaging and producer cell lines. The rAAV provided herein can be manufactured using mammalian host cells, for example, A549, WEHI, 10T1/2, BHK, MDCK, COS1, COS7, BSC 1, BSC 40, BMT 10, VERO, W138, HeLa, HEK293, HEK293-T, Saos, C2C12, L, HT1080, HepG2, primary fibroblast, hepatocyte, myoblast cells, or any muscle related cell. The rAAV provided herein may be manufactured using host cells from human, monkey, mouse, rat, rabbit, or hamster. In certain embodiments, stable cell lines can be engineered by introducing the means of producing viruses in the host cells, for example, the replication and capsid genes (e.g., the rep and cap genes of AAV) and the rAAV plasmid provided herein. In some embodiments, the rAAV can be manufactured using HEK293 cells. In some embodiments, rAAV can be produced in Sf9 insect cells by coinfecting three recombinant baculovirus plasmids with genes encoding the rep gene, the cap gene, and the rAAV genome.

[0268] The cells can be cultured, transfected, and harvested according to appropriate protocols which would be readily selected by one of skill in the art. In certain embodiments, the cells can be cultured in standard Dulbecco's modified Eagle medium (DMEM), including, but not limited to, fetal calf serum, glucose, penicillin, streptomycin, and L-glutamine (McClure et al., J Vis Exp. 2011, (57): 3348; Shin et al., Methods Mol Biol. 2012, 798:267-284). Cells can be transfected in components which would be readily selected by one of skill in the art. In certain embodiments, transfection can take place in media solutions including, but not limited to, DMEM and Iscove's modified Dulbecco's medium (IMDM). In certain embodiments, the transfection time can take 46 hr, 47 hr, 48 hr, 49 hr, 50 hr, 51 hr, 52 hr, 53 hr, 54 hr, 55 hr, 56 hr, 57 hr, 58 hr, 59 hr, 60 hr, 61 hr, 62 hr, 63 hr, 64 hr, 65 hr, 66 hr, 67 hr, 68 hr, 69 hr, 70 hr, 50-55 hr, 55-60 hr, 60-65 hr, or 65-70 hr. After transfection, the cells can be harvested by scraping cells to remove them from the culture wells and washing the wells to collect all of the transfected cells.

[0269] For a method of producing rAAV, see Section IV of the Detailed Description of U.S. Pat. No. 7,282,199 B2, which is incorporated herein by reference in its entirety. Genome copy titers of said vectors may be determined, for example, by TAQMAN® analysis. Virions may be recovered, for example, by CsCl₂ sedimentation. In a specific embodiment, the rAAV described herein is an isolated or purified rAAV.

[0270] Multiple AAV serotypes have been identified. In certain embodiments, rAAVs or polynucleotides provided herein comprise one or more components derived from one or more serotypes of AAV. In certain embodiments, rAAVs or polynucleotides provided herein comprise one or more components derived from one or more of AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9,

AAVrh10, AAV11, AAV12, AAV13, AAVrh20, AAVhu.37, AAVrh39, AAVhu32, AAVrh21, AAVrh13, AAVrh15, AAVrh73, or AAVrh74. Nucleic acid sequences of AAV components and methods of making recombinant AAV and AAV capsids are described, for example, in U.S. Pat. No. 7,282,199 B2, U.S. Pat. No. 7,790,449 B2, U.S. Pat. No. 8,318,480 B2, U.S. Pat. No. 8,962,332 B2 and International Patent Application No. PCT/EP2014/076466, each of which is incorporated herein by reference in its entirety.

[0271] In vitro assays, e.g., cell culture assays, can be used to measure therapeutic product expression from a vector described herein, thus indicating, e.g., potency of the vector. Cells utilized for the assay can include, but are not limited to, A549, WEHI, 10T1/2, BHK, MDCK, COS1, COS7, BSC 1, BSC 40, BMT 10, VERO, W138, HeLa, HEK293, HEK293-T, HuH7, Saos, C2C12, L, HT1080, HepG2, primary fibroblast, hepatocyte, and myoblast cells. In some embodiments, the cells utilized in the cell culture assay comprise HuH7 cells. In some embodiments, a Cell Line can be used to assess therapeutic product expression. Once expressed, characteristics of the expressed therapeutic product can be determined, including determination of the post-translational modification patterns. In addition, benefits resulting from post-translational modification of the cell-expressed therapeutic product can be determined using assays known in the art.

5.4 Therapeutic and Prophylactic Uses

[0272] Provided herein are hybrid AAVs of the disclosure comprising a transgene for therapy use in a subject (e.g., therapeutic or prophylactic therapy). In some embodiments, the therapy of the disclosure (e.g., hybrid AAV comprising a transgene) delays, prevents, treats, manages a disease or disorder, and/or ameliorates one or more symptoms associated with a disease or disorder in a subject (e.g., a muscle associated disease or disorder). In some embodiments a therapeutic transgene is a component of an artificial genome encapsidated by an AAV capsid protein to produce the rAAV vector of the disclosure. In some embodiments, an rAAV vector comprises a hybrid AAV capsid (see Sections 5.1.1 and 5.2.1). In some embodiments, a subject in need of treatment includes a subject suffering from the disease or disorder, or a subject pre-disposed thereto, e.g., a subject at risk of developing or having a recurrence of the disease or disorder (e.g., a muscle related disease or disorder). In some embodiments, a rAAV carrying a particular transgene finds use with respect to a given disease or disorder in a subject where the subject's native gene, corresponding to the transgene, is defective in providing the correct gene product, or correct amounts of the gene product. The transgene then can provide a copy of a gene that is defective in the subject.

[0273] In some embodiments, the transgene comprises cDNA that restores protein function to a subject having a genetic mutation(s) in the corresponding native gene. In some embodiments, the cDNA comprises associated RNA for performing genomic engineering, such as genome editing via homologous recombination. In some embodiments, the transgene encodes a therapeutic RNA, such as a shRNA, artificial miRNA, or element that influences splicing.

[0274] In some embodiments, the therapeutic transgene is a microdystrophin protein (e.g., such as those disclosed herein). Microdystrophins include, but are not limited to, those described in Table 3. In some embodiments, microdystrophins comprises or consists of an amino acid sequence

of at least one of SEQ ID NO: 73-80 or a nucleotide sequence encoding one or more of any one of SEQ ID NOS: 73-80. Also provided herein are methods of treating human subjects for a muscular dystrophy disease that can be treated, for example, by providing a functional dystrophin. In some embodiments, the functional dystrophin is one or more of the microdystrophins disclosed herein (e.g., one or more of SEQ ID NOS: 73-80). In some embodiments, the disease to be treated is Duchenne Muscular Dystrophy (DMD). In some embodiments, the disease to be treated includes, but it is not limited to, Becker muscular dystrophy (BMD), myotonic muscular dystrophy (Steinert's disease), Facioscapulohumeral disease (FSHD), limb-girdle muscular dystrophy, X-linked dilated cardiomyopathy, and/or oculopharyngeal muscular dystrophy.

[0275] In some embodiments, provided herein is a method of treating a muscle related disease or disorder in a subject in need thereof. In some embodiments, the method comprises administering an rAAV comprising a polypeptide comprising an amino acid sequence of SEQ ID NO: 31 or an rAAV encoded by SEQ ID NO:32.

[0276] In some embodiments, the transgene is a muscle-specific disease therapeutic. In some embodiments, one or more of the transgene listed in Tables 1-2 is a muscle-specific transgene and can be used as a muscle specific disease therapeutics (e.g., a hybrid AAV of the disclosure comprising at least a portion of a transgene identified in Tables 1-2). A muscle-specific disease therapeutic can be a therapeutic that treats a muscle-specific disease. A muscle-specific disease can be diseases such as myopathies. In some aspects, a myopathy includes, but it is not limited to, muscular dystrophies (e.g. Duchenne Muscular Dystrophy (DMD), Becker muscular dystrophy (DMD)), human limb girdle muscular dystrophy (LGMD)), X-linked dilated cardiomyopathy, Myasthenia Gravis, Rhabdomyolysis, Amyotrophic Lateral Sclerosis (ALS), and/or Sarcopenia.

[0277] In some embodiments, the rAAVs of the disclosure is used in delivery to target tissues associated with the disorder or disease to be treated/prevented. In some embodiments, a disease or disorder associated with a particular tissue or cell type is one that largely affects the particular tissue or cell type, in comparison to other tissue of cell types of the body, or one where the effects or symptoms of the disorder appear in the particular tissue or cell type. Methods of delivering a transgene to a target tissue of a subject in need thereof involve administering to the subject an rAAV wherein the AAV capsid is an AAV hybrid serotype described herein (e.g., Section 5.1.1 and 5.2.1).

[0278] In some embodiments, the rAAV vector is administered systemically, and following transduction, the vector's production of the protein product may be further enhanced by an expression cassette employing tissue-specific promoter, for example, a muscle-specific promoter operably linked to a transgene. In some embodiments, the rAAV vector is administered by intravenous, intramuscular, and/or intra-peritoneal administration.

[0279] In some embodiments, expression of transgene from muscle tissue provides secretion of the transgene into the serum of the subject. In some embodiments, with respect to the therapeutic antibodies shown in Table 2, expression (of transgene) in muscle, or both muscle and liver may be desirable to deliver the expressed therapeutic antibody systemically.

[0280] Tables 1-2 below provide a list of transgenes that can be used in any of the rAAV vectors comprising a hybrid AAV capsid of the disclosure. In some embodiments, the disease or disorder to be treated with an rAAV of the disclosure comprising a transgene is a disease or disorder listed in Tables 1-2. The AAV hybrid serotype utilized to produce the rAAV vector is specifically engineered to optimize the tissue tropism and transduction of the vector.

TABLE 1

list of diseases associated with a transgene	
Disease	Transgene
Spinal Muscular Atrophy (SMA)	SMN
Duchenne Muscular Dystrophy	Mini-/Micro-Dystrophin Gene
Limb Girdle Muscular Dystrophy Type 2C/Gamma-sarcoglycanopathy	human-alpha-sarcoglycan

TABLE 1-continued

list of diseases associated with a transgene	
Disease	Transgene
Limb Girdle Muscular Dystrophy Type 2C/Gamma-sarcoglycanopathy	gamma-sarcoglycan
Becker Muscular Dystrophy and Sporadic Inclusion Body Myositis	huFollistatin344
Duchenne Muscular Dystrophy	GALGT2
Limb-Girdle Muscular Dystrophy, Type 2E	hSGCB

TABLE 2

list of antigens, transgene/antibodies, and indications		
ANTIGENS	TRANSGENE/ANTIBODIES	INDICATIONS
Repulsive guidance molecule-A Complement Component 5 Connective tissue growth factor (CTGF)	elezanumab ravulizumab pamrevlumab	multiple sclerosis Myasthenia Gravis fibrotic diseases, e.g. diabetic nephropathy, liver fibrosis, idiopathic pulmonary fibrosis ulcerative colitis, Crohn's disease
Integrin beta 7	etrolizumab	
Sclerostin	romosozumab (EVENTY ®)	Osteoporosis, abnormal bone loss or weakness
Interleukin receptor 6 (IL6R) or Interleukin 6 (IL6)	Satralizumab Sarilumab Tocilizumab siltuximab, clazakizumab sirukumab olokizumab gerilimumab omolizumab	Adverse immune responses (e.g. cytokine storm, CAR-T therapy)
Immunoglobulin E (IgE)		Asthma, COPD, eosinophilic asthma, chronic idiopathic urticaria
Thymic stromal lymphopoietin (TSLP)	tezelpumab	Asthma, COPD
Interleukin 5 (IL5) Interleukin 5 receptor (IL5R)	benralizumab reslizumab	Asthma, COPD Asthma, COPD, eosinophilic asthma
Interleukin 13 (IL13) Interleukin 31 receptor alpha (IL31RA) IL17A	tralokinumab nemolizumab Ixekizumab secukinumab	Atopic dermatitis Atopic dermatitis Plaque psoriasis, psoriatic arthritis, ankylosing spondylitis
Interleukins or IL-17A interleukin receptors	ixekizumab (TALTZ ®) secukinumab (COSENTYX ®) mepolizumab (NUCALA ®)	Plaque psoriasis, psoriatic arthritis, ankylosing spondylitis Asthma
IL-5	ustekinumab (STELARA ®)	Psoriasis & Crohn's disease
IL-12/IL-23	dupilumab	Atopic dermatitis
IL-4R	vedolizumab	Ulcerative colitis & Crohn's disease
Integrin	(ENTYVIO ®) Natalizumab (anti-integrin alpha 4)	Multiple sclerosis & Crohn's disease

TABLE 2-continued

list of antigens, transgene/antibodies, and indications			
ANTIGENS		TRANSGENE/ANTIBODIES	INDICATIONS
Cardiovascular Targets	PCSK9	alirocumab (PRALUENT ®)	HeFH & HoFH
		evolucumab (REPATHA ®)	
	ANGPTL3	evinacumab	HoFH & severe forms of dyslipidemia
	Proinflammatory/proatherogenic phospholipids	E06-scFv	Cardiovascular diseases such as atherosclerosis
	RANKL	denosumab (XGEVA ® and PROLIA ®)	Osteoporosis, increasing bone mass in breast and prostate cancer patients, & preventing skeletal-related events due to bone metastasis
	PD-1, or PD-L1 or PD-L2	nivolumab (OPDIVO ®)	Metastatic melanoma, lymphoma, non-small cell lung carcinoma
	BlyS (B-lymphocyte stimulator, also known as B-cell activating factor (BAFF))	pembrolizumab (KEYTRUDA ®)	Systemic lupus erythematosus
		belimumab (BENLYSTA ®)	
	TNF-alpha	adalimumab (HUMIRA ®) and infliximab (REMICADE ®)	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, plaque psoriasis, ulcerative colitis
Plasma Protein targets	C5, C5a	eculizumab (SOLIRIS ®)	Paroxysmal nocturnal hemoglobinuria, atypical hemolytic uremic syndrome, complement-mediated thrombotic microangiopathy
		ravulizumab, or crovalimuab	
	Plasma kallikrein	lanadelumab	Hereditary angioedema (HAE)

TABLE 3

amino acid sequences of microdystrophin proteins		
Structure	SEQ ID NO:	Amino Acid Sequence
DYS1	73	MLWWEVEDCYEREDVQKKTFTKWVNAQFSKFGKQHIENLFSDL QDGRLLDLLLEGLTGQKLPKEKGSTRVHALNNVNKALRVLQNNN VDLVNIGSTDIVDGNHKLTLGLIWNILHWQVKNVMKNIMAGLQ QTNSEKILLSWVRQSTRNYPQVNVINFTTSWSDGLALNALIHS RPDLFDWNSVVCQQSATQRLAHAFNIARYQLGIKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSEIAIQEVEMLPRPPKVTKEE HFQLHHQMHSQQITVSLAQGYERTSSPKPRFKSYAYTQAAYVT TSDPTRSPFPSPQHLEAPEDKSPGSSLMSEVNLDRYQTALAEVL SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTKLSEDEETEVQEQMNLNLSRWECLRVASME KQSNLHRVLMDLNQKLKELNDWLTKTEERTRKMEEEPGLPDLE DLKRQVQHKVLQEDLEQEQVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTEDRVLLQDILLKWQLRTTEQCFLS AWLSEKEDAVNKIHTTGFKDQNEMLSSLOKLAVLKADLEKKQS MGKLYSLKQDLLSTLKNKSVTQKTEAWLDNFARCDNLVQKLEK STAQISQQPDLAPGLTTIGASPTQTVTLVTQPVVTKETAISKLE MPSSLMLLEVPTLERLQELQEATDELDLKLQAQEVIKGSWQPVGD LLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLGIQL SPYNLSTLEDLNRWKLQVAVEDRVRQLHEAHRDFGPASQHFPL STSVQGPWERAI SPNKVPYYINHETQTTCDWHPKMTELYQSLAD LNNVRFSAYRTAMKLRLQKALCLDLLSLAACDALDQHNLKQN DQPM DILQI INCLTTIYDRLEQEHNNLVNVPLCVDMLNWLNV YDTGRTGRIRVLSFKTGIISLCAHLEDKYRYLKFQVASSTGFC DQRLGLLLHDSIQIPRLGEVASFGGSNIEPSVRSCFQFANNK PEIEAALFLDWMRLPQSMVWLPLVHRVAAETAHQAKCNICK

TABLE 3-continued

amino acid sequences of microdystrophin proteins		
Structure	SEQ ID NO:	Amino Acid Sequence
		ECP I I GFRYRS LKHFN YD I C Q S C F F S G R V A K G H K M H Y P M V E Y C T P T T S G E D V R D F A K V L K N K F R T K R Y F A K H P R M G Y L P V Q T V L E G D N M E T P V T L I N F W P V D S A P A S S P Q L S H D D T H S R I E H Y A S R L A E M E N S N G S Y L N D S I S P N E S I D D E H L L I Q H Y C Q S L N Q D S P L S Q P R S P A Q I L I S L E S E E R G E L E R I L A D L E E N R N L Q A E Y D R L K Q Q H E H K G L S P L S P P E M M P T S P Q S P R
DYS3	74	M L W W E E V E D C Y E R E D V Q K K T F T K W V N A Q F S K F G K Q H I E N L F S D L Q D G R R L L D L L E G L T G Q K L P K E K G S T R V H A L N N V N K A L R V L Q N N N V D L V N I G S T D I V D G N H K L T L G L I W N I I L H W Q V K N V M K N I M A G L Q Q T N S E K I L L S W V R Q S T R N Y P Q V N V I N F T T S W S D G L A L N A L I H S H R P D L F D W N S V V C Q Q S A T Q R L E H A F N I A R Y Q L G I E K L L D P E D V D T T Y P D K K S I L M Y I T S L F Q V L P Q Q V S I E A I Q E V E M L P R P P K V T K E E H F Q L H H Q M H Y S Q Q I T V S L A Q G Y E R T S S P K P R F K S Y A Y T Q A A Y V T T S D P T R S P F P S Q H L E A P E D K S F G S S L M E S E V N L D R Y Q T A L E E V L S W L L S A E D T L Q A Q G E I S N D V E V V K D Q F H T H E G Y M M D L T A H Q G R V G N I L Q L G S K L I G T G K L S E D E E T E V Q E Q M N L L N S R W E C L R V A S M E K Q S N L H R V L M D L Q N Q K L K E L N D W L T K T E E R T R K M E E P L G P D L E D L K R Q V Q Q H K V L Q E D L E Q E Q V R V N S L T H M V V V D E S S G D H A T A A L E E Q L K V L G D R W A N I C R W T E D R W V L L Q D I L L K W Q R L T E E Q C L F S A W L S E K E D A V N K I H T T G F K D Q N E M L S S L Q K L A V L K A D L E K K K Q S M G K L Y S L K Q D L L S T L K N K S V T Q K T E A W L D N F A R C W D N L V Q K L E K S T A Q I S Q Q P D L A P G L T T I G A S P T Q T V T L V T Q P V V T K E T A I S K L E M P S S L M L E V P T L E R L Q E L Q E A T D E L D L K L R Q A E V I K G S W Q P V G D L L I D S L Q D H L E K V K A L R G E I A P L K E N V S H V N D L A R Q L T T L G I Q L S P Y N L S T L E D L N T R W K L L Q V A V E D R V R Q L H E A H R D F G P A S Q H F L S T S V Q G P W E R A I S P N K V P Y Y I N H E T Q T T C W D H P K M T E L Y Q S L A D L N N V R F S A Y R T A M K L R R L Q K A L C L D L L S L S A A C D A L D Q H N L K Q N D Q P M D I L Q I I N C L T T I Y D R L E Q E H N N L V N V P L C V D M C L N W L L N V Y D T G R T G R I R V L S F K T G I I S L C A H L E D K Y R Y L F K Q V A S S T G F C D Q R R L G L L H D S I Q I P R Q L G E V A S F G G S N I E P S V R S C P Q F A N N K P E I E A A L F L D W M R L E P Q S M W V L P V L H R V A A E T A K H Q A K C N I C K E C P I I G F R Y R S L K H F N Y D I C Q S C F F S G R V A K G H K M H Y P M V E Y C T P T T S G E D V R D F A K V L K N K F R T K R Y F A K H P R M G Y L P V Q T V L E G D N M E T
DYS5	75	M L W W E E V E D C Y E R E D V Q K K T F T K W V N A Q F S K F G K Q H I E N L F S D L Q D G R R L L D L L E G L T G Q K L P K E K G S T R V H A L N N V N K A L R V L Q N N N V D L V N I G S T D I V D G N H K L T L G L I W N I I L H W Q V K N V M K N I M A G L Q Q T N S E K I L L S W V R Q S T R N Y P Q V N V I N F T T S W S D G L A L N A L I H S H R P D L F D W N S V V C Q Q S A T Q R L E H A F N I A R Y Q L G I E K L L D P E D V D T T Y P D K K S I L M Y I T S L F Q V L P Q Q V S I E A I Q E V E M L P R P P K V T K E E H F Q L H H Q M H Y S Q Q I T V S L A Q G Y E R T S S P K P R F K S Y A Y T Q A A Y V T T S D P T R S P F P S Q H L E A P E D K S F G S S L M E S E V N L D R Y Q T A L E E V L S W L L S A E D T L Q A Q G E I S N D V E V V K D Q F H T H E G Y M M D L T A H Q G R V G N I L Q L G S K L I G T G K L S E D E E T E V Q E Q M N L L N S R W E C L R V A S M E K Q S N L H R V L M D L Q N Q K L K E L N D W L T K T E E R T R K M E E P L G P D L E D L K R Q V Q Q H K V L Q E D L E Q E Q V R V N S L T H M V V V D E S S G D H A T A A L E E Q L K V L G D R W A N I C R W T E D R W V L L Q D I L L K W Q R L T E E Q C L F S A W L S E K E D A V N K I H T T G F K D Q N E M L S S L Q K L A V L K A D L E K K K Q S M G K L Y S L K Q D L L S T L K N K S V T Q K T E A W L D N F A R C W D N L V Q K L E K S T A Q I S Q Q P D L A P G L T T I G A S P T Q T V T L V T Q P V V T K E T A I S K L E M P S S L M L E V P T L E R L Q E L Q E A T D E L D L K L R Q A E V I K G S W Q P V G D L L I D S L Q D H L E K V K A L R G E I A P L K E N V S H V N D L A R Q L T T L G I Q L S P Y N L S T L E D L N T R W K L L Q V A V E D R V R Q L H E A H R D F G P A S Q H F L S T S V Q G P W E R A I S P N K V P Y Y I N H E T Q T T C W D H P K M T E L Y Q S L A D L N N V R F S A Y R T A M K L R R L Q K A L C L D L L S L S A A C D A L D Q H N L K Q N D Q P M D I L Q I I N C L T T I Y D R L E Q E H N N L V N V P L C V D M C L N W L L N V Y D T G R T G R I R V L S F K T G I I S L C A H L E D K Y R Y L F K Q V A S S T G F C D Q R R L G L L H D S I Q I P R Q L G E V A S F G G S N I E P S V R S C P Q F A N N K P E I E A A L F L D W M R L E P Q S M W V L P V L H R V A A E T A K H Q A K C N I C K E C P I I G F R Y R S L K H F N Y D I C Q S C F F S G R V A K G H K M H Y P M V E Y C T P T T S G E D V R D F A K V L K N K F R T K R Y F A K H P R M G Y L P V Q T V L E G D N M E T P V T L I N F W P V D S A P A S S P Q L S H D D T H S R I E H Y A S R L A E M E N S N G S Y L N D S I S P N E S I D D E H L L I Q H Y C Q S L N Q D S P L S Q P R S P A Q I L I S L E S
human MD1 (R4- R23/ACT)	76	M L W W E E V E D C Y E R E D V Q K K T F T K W V N A Q F S K F G K Q H I E N L F S D L Q D G R R L L D L L E G L T G Q K L P K E K G S T R V H A L N N V N K A L R V L Q N N N V D L V N I G S T D I V D G N H K L T L G L I W N I I L H W Q V K N V M K N I M A G L Q Q T N S E K I L L S W V R Q S T R N Y P Q V N V I N F T T S W S D G L A L N A L I H S H

TABLE 3-continued

amino acid sequences of microdystrophin proteins		
Structure	SEQ ID NO:	Amino Acid Sequence
		RPDLPDWN SVVCQQSATQRLEHAFNIARYQLGIEKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSI EAIQEVEMLP RPPKVTKEE HFQLHHQMHSYQQITVSLAQGYERTSSPKPRFKSYAYTQAAYVT TSDPTRSPFP SQHLEAPEDKSFSSLMSEVNLD RYQTAL EEV SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTGKLS EDETEVQE QMNLLNSRWECLRVASME KQSNLHRVLM DLQNQKLKELNDWLTKEERTRKME EEP L GPDLE DLKRQVQQHKVLQEDLEQE QVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTE DRWVLLQDILLKWQRLTEEQCLFS AWLSEKEDAVNKIHTTGFKDQNEMLS LQKLAVLKADLEKKKQS MGKLYSLKQDLLSTLKNKSVTQKTEAWLDNFARCDNLVQKLEK STAQISQAVTTTQPSLTQTVMETVTTVTTR EQLV KHAQEELP PPPPQKKRQITVDTLERLQELQEATDELDLKL RQAEVIKGSWQP VGDLLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLG IQLSPYNLSTLEDLNRWKL LQVAVEDRVRQLHEAHRDFGPASQ HELSTSVQGPWERAISP NKVPYYINHETQTT CWDHPKMT ELYQS LADLNNVRFSAYRTAMKLRRLQKALCLD LLSAACDALDQHNL KQNDQPM DILQIINCLTTIYDRLEQEHNNLVN VPLCVDMLNWL LNVYDTGRTGRIRVLSFKTGII SLCKAHLEDKYRYL FQKVASST GFCDQRRLG LLLHDSIQIPRQLGEVASFGGSNI EPSVRSCFQFA NNKPEIEAALFLDWMRLEPQSMVWLPVLHRVAAAE TAKHQAKCN ICEKCP IIGFRYRSLKHFN YD ICQSCFFSGR VAKGHKMHPMVE YCTPTTSGEDVRDFAKVLKNKFR TKRYFAKHPRMGYLPVQTVLE GDNMETDTM
Human microdystrophin	77	MLWWEVEDCYEREDVQKKTF TKWVNAQFSKFGKQHI ENLFS DL QDGRRLDLLEGLTGQKL PKEKGS TRVHALNNVN KALRVLQNNN VDLVNI GSTDIVDGNHKL TLGLIWN IILHWQVKNV MKNIMAGLQ QTNSEKILLSWVRQSTRNYPQVNVINFTTSWSDGLALNALI HSH RPDLPDWN SVVCQQSATQRLEHAFNIARYQLGIEKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSI EAIQEVEMLP RPPKVTKEE HFQLHHQMHSYQQITVSLAQGYERTSSPKPRFKSYAYTQAAYVT TSDPTRSPFP SQHLEAPEDKSFSSLMSEVNLD RYQTAL EEV SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTGKLS EDETEVQE QMNLLNSRWECLRVASME KQSNLHRVLM DLQNQKLKELNDWLTKEERTRKME EEP L GPDLE DLKRQVQQHKVLQEDLEQE QVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTE DRWVLLQDILLKWQRLTEEQCLFS AWLSEKEDAVNKIHTTGFKDQNEMLS LQKLAVLKADLEKKKQS MGKLYSLKQDLLSTLKNKSVTQKTEAWLDNFARCDNLVQKLEK STAQISQAVTTTQPSLTQTVMETVTTVTTR EQLV KHAQEELP PPPPQKKRQITVDTLERLQELQEATDELDLKL RQAEVIKGSWQP VGDLLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLG IQLSPYNLSTLEDLNRWKL LQVAVEDRVRQLHEAHRDFGPASQ HELSTSVQGPWERAISP NKVPYYINHETQTT CWDHPKMT ELYQS LADLNNVRFSAYRTAMKLRRLQKALCLD LLSAACDALDQHNL KQNDQPM DILQIINCLTTIYDRLEQEHNNLVN VPLCVDMLNWL LNVYDTGRTGRIRVLSFKTGII SLCKAHLEDKYRYL FQKVASST GFCDQRRLG LLLHDSIQIPRQLGEVASFGGSNI EPSVRSCFQFA NNKPEIEAALFLDWMRLEPQSMVWLPVLHRVAAAE TAKHQAKCN ICEKCP IIGFRYRSLKHFN YD ICQSCFFSGR VAKGHKMHPMVE YCTPTTSGEDVRDFAKVLKNKFR TKRYFAKHPRMGYLPVQTVLE GDNMETDTM
Dys3978	78	MLWWEVEDCYEREDVQKKTF TKWVNAQFSKFGKQHI ENLFS DL QDGRRLDLLEGLTGQKL PKEKGS TRVHALNNVN KALRVLQNNN VDLVNI GSTDIVDGNHKL TLGLIWN IILHWQVKNV MKNIMAGLQ QTNSEKILLSWVRQSTRNYPQVNVINFTTSWSDGLALNALI HSH RPDLPDWN SVVCQQSATQRLEHAFNIARYQLGIEKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSI EAIQEVEMLP RPPKVTKEE HFQLHHQMHSYQQITVSLAQGYERTSSPKPRFKSYAYTQAAYVT TSDPTRSPFP SQHLEAPEDKSFSSLMSEVNLD RYQTAL EEV SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTGKLS EDETEVQE QMNLLNSRWECLRVASME KQSNLHRVLM DLQNQKLKELNDWLTKEERTRKME EEP L GPDLE DLKRQVQQHKVLQEDLEQE QVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTE DRWVLLQDQPD LAPGLTTIGASPT QVTVLVTQPVVTKETAISKLEMPSSLMLEVPT HRLLQQFPLDLE KFLAWLTEAETTANVLQDATRKERLLED SKGVKELMKQWQDLQG EIEAHTDVYHNL DENSQKILRSL EGSDDAVLLQRRLDNMNFKWS ELRKKS LNIRSHLEASSDQWKRLHLSLQELLVWLQLKDD ELSRQ

TABLE 3-continued

amino acid sequences of microdystrophin proteins		
Structure	SEQ ID NO:	Amino Acid Sequence
		APIGGDFPAVQKQNDVHRAFKRELKTEKPEVIMSTLETVRIFLTE QPLEGLEKLYQEPRELPPERAQNVTRLRKQAEVNTTEWEKLN LHSADWQRKIDETLERLQELQEATDELDLKLQAEVIKGSWQPV GDLLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLGI QLSPYNLSTLEDLNRWKLQVAVEDRVRQLHEAHRDFGPASQ FLSTSVQGPWERAISPKNVPYYINHETQTTCDWHPKMTELYQSL ADLNNVRFSAYRTAMKLRLQKALCLDLLSLSAACDALDQHNK QNDQPMILQIINCLTTIYDRLEQEHNNLVNVPLCVDMLNWL NVYDTGRTGRIRVLSFKTGIIISLCKAHLEDKYRYLFKQVASSTG FCDQRRLLGLLHDSIQIPRLQGEVASFGGSNI EPSVRS CFQFAN NKPEIEAALFLDWMRLPQSMVWLPVLHRVAAAEATAKHQAKCNI CKECPIIGFRYRSLKHFNIDICQSCFFSGRVAKGHKMHPMVEY CPTTSGEDVRDFAKVLKKNKFRTRKYFAKHPRMGYLPVQTVLEG DNMET
Human MD3	79	MLWWEVEDCYEREDVQKKFTFTKWVNAQFSKFGKQHIENLFSDL QDGRRLDLLEGLTGQKLPEKKGSTRVHALNNVNKALRVLQNNN VDLVNIIGSTDIVDGNHKLTLGLIWNIIHLHWQVKNVMKNIAGLQ QTNSEKILLSWVRQSTRNYPQVNVINFTTSWSDGLALNALIHSH RPDLPDWNVVCQOSATQRLAHAFNIARYQLGIEKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSEIAIEQVEMLPRPPKVTKEE HFQLHHQMHSYQQITVSLAQGYERTSSPKPRFKSYAYTQAAVVT TSDPTRSPFPQSQHLEAPEDKSFGLSMESEVNLDRYQTALAEVL SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTGKLSDEETEVEQEMNLLNSRWECRLVASME KQSNLHRVMDLQNKQLKELNDWLTKEERTRKMEEPPLGPDLE DLKRQVQQHKVLQEDLEQEQVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTEDRWVLLQDILLKWQRLTEEQCLFS AWLSEKEDAVNKIHTTGFKDQNEMLSSQLKAVLKADLEKKKQS MGKLYSLKQDLLSTLKNKSVTQKTEAWLDFNARCWDNLVQKLEK STAQISQAVTTTQPSLTQTTVMETVTVTVTRQIILVKHAQEELP PPPPQKKRQITVDTLERLQELQEATDELDLKLQAEVIKGSWQPV VGDLLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLGI QLSPYNLSTLEDLNRWKLQVAVEDRVRQLHEAHRDFGPASQ HFLSTSVQGPWERAISPKNVPYYINHETQTTCDWHPKMTELYQS LADLNNVRFSAYRTAMKLRLQKALCLDLLSLSAACDALDQHNK QNDQPMILQIINCLTTIYDRLEQEHNNLVNVPLCVDMLNWL LNVYDTGRTGRIRVLSFKTGIIISLCKAHLEDKYRYLFKQVASST GFCDQRRLLGLLHDSIQIPRLQGEVASFGGSNI EPSVRS CFQFAN NKPEIEAALFLDWMRLPQSMVWLPVLHRVAAAEATAKHQAKCNI CKECPIIGFRYRSLKHFNIDICQSCFFSGRVAKGHKMHPMVEY YCTPTTSGEDVRDFAKVLKKNKFRTRKYFAKHPRMGYLPVQTVLE GDNMETPVTLINFWPVDSPASSQLSHDDTHSRIEHYASRLAE MENSNGSYLNDISPNESIDDEHLIHYCQSLNQDPSLSQPRS PAQILISLESEERGELERILADLEEENRNLAQAEYDRLKQQHEHK GLSPLSPPEMMPTSPQS PRDAELIAEAKLLRQHKGRLEARMQI LEDHNKQLESQHLRLRQLLEQPAEDTM
Human MD4	80	MLWWEVEDCYEREDVQKKFTFTKWVNAQFSKFGKQHIENLFSDL QDGRRLDLLEGLTGQKLPEKKGSTRVHALNNVNKALRVLQNNN VDLVNIIGSTDIVDGNHKLTLGLIWNIIHLHWQVKNVMKNIAGLQ QTNSEKILLSWVRQSTRNYPQVNVINFTTSWSDGLALNALIHSH RPDLPDWNVVCQOSATQRLAHAFNIARYQLGIEKLLDPEDVDT TYPDKKSILMYITSLFQVLPQQVSEIAIEQVEMLPRPPKVTKEE HFQLHHQMHSYQQITVSLAQGYERTSSPKPRFKSYAYTQAAVVT TSDPTRSPFPQSQHLEAPEDKSFGLSMESEVNLDRYQTALAEVL SWLLSAEDTLQAQGEISNDVEVVKDQFHTHEGYMMDLTAHQGRV GNILQLGSKLIGTGKLSDEETEVEQEMNLLNSRWECRLVASME KQSNLHRVMDLQNKQLKELNDWLTKEERTRKMEEPPLGPDLE DLKRQVQQHKVLQEDLEQEQVRVNSLTHMVVVVDESSGDHATAA LEEQLKVLGDRWANI CRWTEDRWVLLQDILLKWQRLTEEQCLFS AWLSEKEDAVNKIHTTGFKDQNEMLSSQLKAVLKADLEKKKQS MGKLYSLKQDLLSTLKNKSVTQKTEAWLDFNARCWDNLVQKLEK STAQISQAVTTTQPSLTQTTVMETVTVTVTRQIILVKHAQEELP PPPPQKKRQITVDTLERLQELQEATDELDLKLQAEVIKGSWQPV VGDLLIDSLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLGI QLSPYNLSTLEDLNRWKLQVAVEDRVRQLHEAHRDFGPASQ HFLSTSVQGPWERAISPKNVPYYINHETQTTCDWHPKMTELYQS LADLNNVRFSAYRTAMKLRLQKALCLDLLSLSAACDALDQHNK QNDQPMILQIINCLTTIYDRLEQEHNNLVNVPLCVDMLNWL LNVYDTGRTGRIRVLSFKTGIIISLCKAHLEDKYRYLFKQVASST

TABLE 3-continued

amino acid sequences of microdystrophin proteins	
Structure	SEQ ID NO: Amino Acid Sequence
	GFCDQRRRLGILLHDSIQIPRQLGEVASFGGSNIEPSVRSCFQFA NNKPEIEAALFLDWMRLPQSMVWLPVLHRVAAAEATAKHQAKCN ICKECPIIGFRYRSLKHFNQYDQCSCFFSGRVAKGHKMHYPMVE YCTPTTSGEDVRDFAKVLKNKFRTKRYFAKHPRMGYLPVQTVLE GDNMETPVTLINFWPVDSPAPSSPQLSHDDTHSRIEHYASRLAE MENSNGSYLNDISPNESIDDEHLIQHYCQSLNQDSPLSQPRS PAQILISLESEERGELERILADLEEENRNLAQYDRLKQQHEHK GLSPLPSPPEMPTSPQSPRDAELIAEAKLLRQHKGRLEARMQI LEDHNKQLESQHLRLQLLEQPQAEAKVNGTTVSSPSTSLQRSD SSQPMLLRVVGSQTSDSMGEDLLSPQDTSGLQEEVMEQLNNS FPSRGRNTPGKPMREDTM

5.4.1 Target Patient Populations

[0281] In some embodiments, a subject is a subject with a disease or disorder associated with a muscle or is at risk of having a disease or disorder associated with a muscle. In some embodiments, a subject is a subject with a muscle related disease (e.g., muscular dystrophy) or is at risk of having a muscle related disease. In some embodiments, a subject is a subject with at least one symptom associated with a muscle related disease or disorder. Non-limiting examples of muscle related disease or disorder that a subject may have been diagnosed with or is at risk of developing, is a disease or disorder including at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy. In some embodiments, the subject has been diagnosed with or is suspected of having Duchenne muscular dystrophy. In some embodiments, a subject has at least one mutation of at least one gene associated with a muscle related disease or disorder.

In some embodiments, a subject has overexpression or downexpression of a gene associated with a muscle related disease or disorder.

[0282] In certain embodiments, the subject treated in accordance with the methods described herein is female. In certain embodiments, the subject treated in accordance with the methods described herein is male. In certain embodiments, the subject treated in accordance with the methods described herein is a child. In certain embodiments, the subject treated in accordance with the methods described herein is a juvenile subject (e.g., 18 years or younger). In certain embodiments, the subject treated in accordance with the methods described herein is 1 month old, 2 months old, 3 months old, 4 months old, 5 months old, 6 months old, 7 months old, 8 months old, 9 months old, 10 months old, 11 months old, 1 year old, 1.5 years old, 2 years old, 2.5 years old, 3 years old, 3.5 years old, 4 years old, 4.5 years old, or 5 years old. In certain embodiments, the subject treated in accordance with the methods described herein is less than 1.5 months old, 2 months old, 3 months old, 4 months old, 5 months old, 6 months old, 7 months old, 8 months old, 9 months old, 10 months old, 11 months old, 1 year old, 1.5 years old, 2 years old, 2.5 years old, 3 years old, 3.5 years old, 4 years old, 4.5 years old, or less than 5 years old. In another specific embodiment, the subject treated in accordance with the methods described herein is 1-2 months old, 2-3 months old, 3-4 months old, 4-5 months old, 5-6 months old, 6-7 months old, 7-8 months old, 8-9 months old, 9-10 months old, 10-11 months old, 11 months to 1 year old, 1-1.5 years old, 1.5-2 years old, 2-2.5 years old, 2.5-3 years old, 3-3.5 years old, 3.5-4 years old, 4-4.5 years old, or 4.5-5 years old. In another specific embodiment, the subject treated in accordance with the methods described herein is 6 months to 5 years old. In another embodiment, the subject treated in accordance with the methods described herein is a human adult over 18 years old. In some embodiments, the subject treated in accordance with the methods described herein is a human child under 18 years. In some embodiments, the subject treated in accordance with the methods described herein is a human child under 84 months of age.

[0283] In a specific embodiment, the subject is an adult (at least age 16). In another specific embodiment, the subject is an adolescent (age 12-15). In another specific embodiment, the subject is a child (under age 12). In some embodiments, the subject is under age 6, under age 10, under age 15, under age 18, under age 21, under age 25, under age 30, under age

35, under age 40, under age 45, under age 50, under age 55, under age 60, under age 65, under age 70, under age 75, under age 80, under age 85, under age 90, or under age 95. In some embodiments, the subject is over age 6, over age 10, over age 15, over age 18, over age 21, over age 25, over age 30, over age 35, over age 40, over age 45, over age 50, over age 55, over age 60, over age 65, over age 70, over age 75, over age 80, over age 85, over age 90, or over age 95. In some embodiments, the subject is a subject who is not responsive to a previous treatment. In some embodiments, the subject is a subject who has not received a treatment for a disease or disorder associated with a muscle. In some embodiments, the subject is a subject who is currently undergoing treatment for a disease or disorder associated with a muscle. In some embodiments, the subject is a subject who has undergone surgery. In some embodiments, the subject is a subject who has received chemotherapy.

5.4.2 Administration and Dosage

[0284] The dosage amounts and frequencies of administration provided herein are encompassed by the terms therapeutically effective and prophylactically effective. The dosage and frequency will typically vary according to factors specific for each patient depending on the specific therapeutic or prophylactic agents administered, the severity and type of disease, the route of administration, as well as age, body weight, response, and the past medical history of the patient, and should be decided according to the judgment of the practitioner and each patient's circumstances. Suitable regimens can be selected by one skilled in the art by considering such factors and by following, for example, dosages reported in the literature and recommended in the Physician's Desk Reference (56th ed., 2002). Prophylactic and/or therapeutic agents can be administered repeatedly. Several aspects of the procedure may vary such as the temporal regimen of administering the prophylactic or therapeutic agents, and whether such agents are administered separately or as an admixture.

[0285] The amount of an agent of the disclosure that is effective can be determined by standard clinical techniques. Effective doses may be extrapolated from dose-response curves derived from in vitro or animal model test systems. For any agent used in the method of the invention, the therapeutically effective dose can be estimated initially from cell culture assays. A dose may be formulated in animal models to achieve a circulating plasma concentration range that includes the IC₅₀ (i.e., the concentration of the test compound that achieves a half-maximal inhibition of symptoms) as determined in cell culture. Such information can be used to more accurately determine useful doses in humans. Levels in plasma may be measured, for example, by high performance liquid chromatography.

[0286] Prophylactic and/or therapeutic agents, as well as combinations thereof, can be tested in suitable animal model systems prior to use in humans. Such animal model systems include, but are not limited to, rats, mice, chicken, cows, monkeys, pigs, dogs, rabbits, etc. Any animal system well-known in the art may be used. Such model systems are widely used and well known to the skilled artisan. In some embodiments, animal model systems for a CNS condition are used that are based on rats, mice, or other small mammal other than a primate.

[0287] Once the prophylactic and/or therapeutic agents of the invention have been tested in an animal model, they can

be tested in clinical trials to establish their efficacy. Establishing clinical trials can be done in accordance with common methodologies known to one skilled in the art, and the optimal dosages and routes of administration as well as toxicity profiles of agents of the disclosure can be established. For example, a clinical trial can be designed to test an rAAV molecule of the disclosure for efficacy and toxicity in human patients.

[0288] Toxicity and efficacy of the prophylactic and/or therapeutic agents of the disclosure can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD₅₀/ED₅₀. In some embodiments, prophylactic and/or therapeutic agents exhibit large therapeutic indices. While prophylactic and/or therapeutic agents that exhibit toxic side effects may be used, care should be taken to design a delivery system that targets such agents to the site of affected tissue in order to minimize potential damage to uninfected cells and, thereby, reduce side effects.

[0289] An rAAV molecule of the disclosure generally is administered for a time and in an amount effective for obtain a desired therapeutic and/or prophylactic benefit. The data obtained from the cell culture assays and animal studies can be used in formulating a range and/or schedule for dosage of the prophylactic and/or therapeutic agents for use in humans. The dosage of such agents lies within a range of circulating concentrations that include the ED₅₀ with little or no toxicity. The dosage may vary within this range depending upon the dosage form employed and the route of administration utilized.

[0290] In some embodiments, a therapeutically effective dosage of an rAAV vector is from about 0.1 ml to about 100 ml of solution containing concentrations of from about 1×10^9 to about 1×10^{16} genomes rAAV vector, or about 1×10^{10} to about 1×10^{15} , about 1×10^{12} to about 1×10^{16} , or about 1×10^{14} to about 1×10^{16} AAV genomes. Levels of expression of the transgene can be monitored to determine/adjust dosage amounts, frequency, scheduling, and the like.

[0291] Treatment of a subject with a therapeutically or prophylactically effective amount of the agents of the disclosure can include a single treatment or can include a series of treatments. For example, pharmaceutical compositions comprising an agent of the disclosure may be administered once a day, twice a day, or three times a day. In some embodiments, the agent may be administered once a day, every other day, once a week, twice a week, once every two weeks, once a month, once every six weeks, once every two months, twice a year, or once per year. In some embodiments, an rAAV of the disclosure increase or decrease over the course of treatment.

[0292] In some embodiments, ongoing treatment is indicated, e.g., on a long-term basis, such as in the ongoing treatment and/or management of a muscle related disease or disorder. For example, in particular embodiments, an rAAV of the disclosure is administered over a period of time, e.g., for at least 6 months, at least one year, at least two years, at least five years, at least ten years, at least fifteen years, at least twenty years, or for the rest of the lifetime of a subject in need thereof.

[0293] The rAAV molecules of the disclosure may be administered alone or in combination with other prophylactic and/or therapeutic agents (e.g., an agent for treating a muscle related disease or disorder, or an agent for reducing pain or inflammation or any side effect from the rAAV treatment). Each prophylactic or therapeutic agent may be administered at the same time or sequentially in any order at different points in time; however, if not administered at the same time, they should be administered sufficiently close in time so as to provide the desired therapeutic or prophylactic effect. Each therapeutic agent can be administered separately, in any appropriate form and by any suitable route.

[0294] In some embodiments, the rAAV of the disclosure, a polypeptide of the disclosure, or any pharmaceutical composition or formulation of the disclosure is administered in combination with an immune suppression therapy. Immune suppression therapies involving a regimen of tacrolimus or rapamycin (sirolimus) in combination with mycophenolic acid, or other immune suppression regimens procedures can be employed. Such immune suppression treatment may be administered during the course of therapy, and in certain embodiments, pre-treatment and/or post treatment with immune suppression therapy can be preferred. Immune suppression therapy can be continued subsequent to the therapy treatment of the disclosure, based on the judgment of the treating physician, and may thereafter be withdrawn when immune tolerance is induced; e.g., after 180 days.

[0295] In various embodiments, the different prophylactic and/or therapeutic agents are administered less than 1 hour apart, at about 1 hour apart, at about 1 hour to about 2 hours apart, at about 2 hours to about 3 hours apart, at about 3 hours to about 4 hours apart, at about 4 hours to about 5 hours apart, at about 5 hours to about 6 hours apart, at about 6 hours to about 7 hours apart, at about 7 hours to about 8 hours apart, at about 8 hours to about 9 hours apart, at about 9 hours to about 10 hours apart, at about 10 hours to about 11 hours apart, at about 11 hours to about 12 hours apart, no more than 24 hours apart, or no more than 48 hours apart. In certain embodiments, two or more agents are administered within the same patient visit.

[0296] In another specific embodiment, agents of the disclosure may be delivered in a sustained release formulation, e.g., where the formulations provide extended release and thus extended half-life of the administered agent. Controlled release systems suitable for use include, without limitation, diffusion-controlled, solvent-controlled, and chemically-controlled systems. Diffusion controlled systems include, for example reservoir devices, in which the molecules of the disclosure are enclosed within a device such that release of the molecules is controlled by permeation through a diffusion barrier. Common reservoir devices include, for example, membranes, capsules, microcapsules, liposomes, and hollow fibers. Monolithic (matrix) device are a second type of diffusion controlled system, wherein the dual antigen-binding molecules are dispersed or dissolved in an rate-controlling matrix (e.g., a polymer matrix). Agents of the disclosure can be homogeneously dispersed throughout a rate-controlling matrix and the rate of release is controlled by diffusion through the matrix. Polymers suitable for use in the monolithic matrix device include naturally occurring polymers, synthetic polymers and synthetically modified natural polymers, as well as polymer derivatives.

[0297] Any technique known to one of skill in the art can be used to produce sustained release formulations comprising one or more agents described herein. See, e.g. U.S. Pat. No. 4,526,938; PCT publication WO 91/05548; PCT publication WO 96/20698; Ning et al., "Intratumoral Radioimmunotherapy of a Human Colon Cancer Xenograft Using a Sustained-Release Gel," *Radiotherapy & Oncology*, 39:179-189, 1996; Song et al., "Antibody Mediated Lung Targeting of Long-Circulating Emulsions," *PDA Journal of Pharmaceutical Science & Technology*, 50:372-397, 1995; Cleek et al., "Biodegradable Polymeric Carriers for a bFGF Antibody for Cardiovascular Application," *Pro. Intl. Symp. Control. Rel. Bioact. Mater.*, 24:853-854, 1997; and Lam et al., "Microencapsulation of Recombinant Humanized Monoclonal Antibody for Local Delivery," *Proc. Intl. Symp. Control Rel. Bioact. Mater.*, 24:759-760, 1997, each of which is incorporated herein by reference in its entirety. In one embodiment, a pump may be used in a controlled release system (see Langer, supra; Sefton, *CRC Crit. Rev. Biomed. Eng.*, 14:20, 1987; Buchwald et al., *Surgery*, 88:507, 1980; and Saudek et al., *N. Engl. J. Med.*, 321:574, 1989). In another embodiment, polymeric materials can be used to achieve controlled release of agents comprising dual antigen-binding molecule, or antigen-binding fragments thereof (see e.g., Medical Applications of Controlled Release, Langer and Wise (eds.), CRC Pres., Boca Raton, Fla. (1974); Controlled Drug Bioavailability, Drug Product Design and Performance, Smolen and Ball (eds.), Wiley, N.Y. (1984); Ranger and Peppas, J., *Macromol. Sci. Rev. Macromol. Chem.*, 23:61, 1983; see also Levy et al., *Science*, 228:190, 1985; During et al., *Ann. Neurol.*, 25:351, 1989; Howard et al., *J. Neurosurg.*, 71:105, 1989); U.S. Pat. Nos. 5,679,377; 5,916,597; 5,912,015; 5,989,463; 5,128,326; PCT Publication No. WO 99/15154; and PCT Publication No. WO 99/20253). In yet another embodiment, a controlled release system can be placed in proximity of the therapeutic target (e.g., an affected joint), thus requiring only a fraction of the systemic dose (see, e.g., Goodson, in Medical Applications of Controlled Release, supra, vol. 2, pp. 115-138 (1984)). Other controlled release systems are discussed in the review by Langer, *Science*, 249:1527-1533, 1990.

[0298] In certain embodiments, dosages are measured by genome copies per ml, the number of genome copies, or vg/kg administered to a location in the subject. In certain embodiments, 1×10^9 genome copies per ml or vg/kg to 1×10^{15} genome copies per ml or vg/kg are administered. In a specific embodiment, 1×10^9 genome copies per ml or vg/kg to 1×10^{10} genome copies per ml or vg/kg are administered. In another specific embodiment, 1×10^{10} genome copies per ml or vg/kg to 1×10^{11} genome copies per ml or vg/kg are administered. In another specific embodiment, 1×10^{10} to 5×10^{11} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{11} genome copies per ml or vg/kg to 1×10^{12} genome copies per ml or vg/kg are administered. In another specific embodiment, 1×10^{12} genome copies per ml or vg/kg to 1×10^{13} genome copies per ml or vg/kg are administered. In another specific embodiment, 1×10^{13} genome copies per ml or vg/kg to 1×10^{14} genome copies per ml or vg/kg are administered. In another specific embodiment, 1×10^{14} genome copies per ml or vg/kg to 1×10^{15} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^9 genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least

about, or at most about 1×10^{10} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^{11} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^{12} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^{13} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^{14} genome copies per ml or vg/kg are administered. In another specific embodiment, about, at least about, or at most about 1×10^{15} genome copies per ml or vg/kg are administered. In certain embodiments, 1×10^9 to 1×10^{15} genome copies or vg/kg are administered. In a specific embodiment, 1×10^9 to 1×10^{10} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{10} to 1×10^{11} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{10} to 5×10^{11} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{11} to 1×10^{12} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{12} to 1×10^{13} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{13} to 1×10^{14} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{13} to 1×10^{14} genome copies or vg/kg are administered. In another specific embodiment, 1×10^{14} to 1×10^{15} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^9 genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{10} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{11} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{12} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{13} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{14} genome copies or vg/kg are administered. In another specific embodiment, about 1×10^{15} genome copies or vg/kg are administered. In certain embodiments, about 3.0×10^{13} genome copies or vg/kg are administered. In certain embodiments, up to 3.0×10^{13} genome copies or vg/kg are administered.

[0299] In certain embodiments, about, at least about, or at most about 2.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 6.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 1.0×10^{10} to 2.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 2.0×10^{10} to 3.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 3.0×10^{10} to 4.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 4.0×10^{10} to 5.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 5.0×10^{10} to 6.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 6.0×10^{10} to 7.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 7.0×10^{10} to 8.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 8.0×10^{10} to 9.0×10^{10} genome copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 9.0×10^{10} to 1.0×10^{11} genome

copies or vg/kg are administered. In certain embodiments, about, at least about, or at most about 1.0×10^{11} to 2.0×10^{11} genome copies or vg/kg are administered. In certain specific embodiments, about, at least about, or at most about 2.0×10^{10} genome copies or vg/kg are administered. In certain specific embodiments, about, at least about, or at most about 6.0×10^{10} genome copies or vg/kg are administered.

6. EXAMPLES

[0300] Certain embodiments provided herein are illustrated by the following non-limiting examples.

6.1 Example 1: Construction of AAV Hybrid Capsids and Assessment of Packaging Efficiency

[0301] AAV-based gene therapies have the potential to treat many severe diseases through a wide range of modalities, from gene addition to gene editing and modulation of gene expression. For successful delivery of the DNA payload, AAV vectors must efficiently bind to the intended target tissue, enter the cell, and traffic to the nucleus. Cell entry depends on endocytosis followed by endosomal escape, which requires the action of the PLA2 domain located in the VP1-unique (VP1u) region of the AAV capsid, and trafficking to the nucleus, mediated by basic residues located in VP1u and the VP1/2s shared (VP1/2s) regions. With the goal of engineering capsids that maintained parental tropism with improved endosomal escape and nuclear trafficking, hybrid capsids were designed by swapping VP1u and VP1/2s (FIG. 1 and Table 4) from different serotypes selected based on data gathered from previous study (Giles et al., ASGCT 2021). Criteria for selection included several factors such as RNA/DNA ratio and tissue-specificity of gene expression. Parental capsid sequences were aligned, and regions swapped as desired, then plasmids were synthesized and constructed at GenScript. Efforts were made to avoid interruption in the assembly-activating protein (AAP) reading frame of each capsid gene. Examples shown in Table 4 were synthesized and constructed at GenScript. Overall productivity of vectors was not significantly negatively impacted by the introduction of hybrid domains as measured by small scale production by triple transfection in suspension HEK293 cells. Without ascribing to any particular theory, it was hypothesized that VP3 is the primary driver of tropism and therefore genomic DNA biodistribution would be largely unchanged by swapping VP1u and VP1/2s regions. It was anticipated that the primary differences would be seen at the RNA expression level, indicating altered endosomal escape and trafficking properties.

TABLE 4

Capsid Number	Capsid Composition		
	VP1u	VP1/2s	VP3
2048		AAV6	
2020	AAV8		AAV6
1035		AAV8	
2017	AAV6		AAV8
2024	AAV9		AAV8
2117	AAVrh.21		AAV8
2118	AAVrh73		AAV8
2171	AAV5		AAV8
(Hybrid 1)			
2172	AAV8	AAV5	AAV8
(Hybrid 2)			

TABLE 4-continued

Capsid Number	Capsid Composition		
	VP1u	VP1/2s	VP3
2040	AAV3B	AAV9	AAV8
1036			
2089	AAVrh.15		AAV9
2090	AAVrh.13		AAV9
2091	AAV6	AAVrh.13	AAV9
2092	AAV5		AAV9
1463			
2096	AAVhu32		AAVrh.13
2097	AAV6	AAVrh.15	AAVrh.13
1465			
2093	AAVhu32		AAVrh.15
2094	AAV6		AAVrh.15
2095	AAV5	AAVrh.21	AAVrh.15
1469			
2018	AAV6		AAVrh.21
2021	AAV8		AAVrh.21
1513		AAVrh.73	
2019	AAV6		AAVrh.73
2022	AAV8		AAVrh.73

[0302] A barcoded, AAV vector pool containing these hybrids and their parental capsids was created and administered intravenously to non-human primates. Biodistribution of vector genomes was similar to their respective parental VP3 serotypes for most of the hybrid capsids, except for those containing VP1u and VP1/2s of AAV5. The AAV5 VP1u and VP1/2s-containing hybrids had lower vector genome copy numbers in all tested tissues compared to their parental, wild type VP3 serotypes (AAV8, AAV9, and AAVrh.15); however, relative abundance of RNA transcripts from each capsid showed a liver-specific transcriptional down-regulation of AAV5-containing hybrids. For example, AAV5/AAV8 hybrid capsids had lower genome copy numbers than AAV8 in all tissues tested, but at the transcriptional level were liver de-targeted and expressed RNA more efficiently than wild type AAV8 in skeletal muscle, leading to an improved RNA/DNA ratio. These results suggest increased endosomal escape or nuclear trafficking abilities of AAV5 VP1u and VP1/2s-containing capsids. Furthermore, AAV5/AAV8 hybrids produced 2.5-3× more vector than their parental capsid, AAV8, in standard triple transfection of HEK293 suspension culture. Ultimately, AAV5 hybrid capsids amongst other hybrid capsids described herein present an important new technique for liver de-targeting at the transcriptional level and offer improved production.

[0303] Several AAV hybrid capsid nucleotide sequences were designed and plasmids comprising the polynucleotides were constructed, such that each plasmid (trans plasmid) could be utilized to make recombinant AAV (rAAV) vectors. Briefly, the rAAV vectors were produced by triple transfection of three plasmids: 1) a plasmid containing a polynucleotide encoding a functional rep gene and a polynucleotide encoding a capsid gene as described herein, 2) a plasmid containing a genome to be packaged into a capsid (comprises at least one AAV inverted terminal repeat (ITR) and a non-AAV nucleic acid sequence encoding a transgene (e.g. fluorescent protein or muscle protein) operably linked to sequences which direct expression of the gene product in a target cell, such as a CAG promoter, and 3) a polynucleotide comprising sufficient helper functions to permit packaging of the genome into the AAV capsid protein under conditions

which permit packaging of the genome into the AAV capsid. Following transfection in HEK293 cells, the cells were maintained in cell culture under conditions that allowed production of the rAAV particles.

[0304] FIG. 2 depicts high packaging efficiency in terms of titer, expressed as genome copies per mL (GC/mL) of wild type AAV5 or AAV8 compared to two hybrid vectors having hybrid capsids as illustrated in FIG. 1 (e.g., Hybrid 1 and Hybrid 2), and having the sequences of SEQ ID NO: 23 (Hybrid 1 in FIG. 2) and 25 (Hybrid 2 in FIG. 2). All vectors were packaged with a muscle protein-encoding transgene and were titered following transfection and harvest from crude lysate of small flask cell cultures. Titer was determined by polyA qPCR. Assessment of titer from cells infected with viral vectors facilitated determining which insertion points did not interrupt capsid packaging and function; error bars represent standard error of the mean. As shown in FIG. 2, Hybrid 1 packages GFP with high efficiency greater than wtAAV5 or wt AAV8, and Hybrid 2 packages GFP equal to wt AAV8 at day 3, and equal to wtAAV5 but greater than wtAAV8 at day 5. It was observed that inclusion of the VP1u region of AAV5 (e.g. Hybrid 1) specifically increases the packaging efficiency of AAV8 by ~2.5× on day 5.

[0305] Different transgenes were compared, such that a GFP transgene or the muscle protein-encoding transgene were packaged in either wtAAV5, wtAAV8, Hybrid 1 or Hybrid 2, and transgene titers were measured from a 1 mL scale cell culture at 5 days post transfection. The muscle protein featured a large coding sequence (total expression cassette including ITRs is greater than 4 kb, “large transgene”) which is near the limit of AAV packaging capacity. The large transgene exhibited a larger difference in packaging efficiency between the capsids than GFP, as shown in FIGS. 3A and 3B. The AAV5/8 hybrids remain ~2.5× higher producers than AAV8 of the large transgene (FIG. 3C).

6.2 Example 2: Evaluation of AAV Hybrid Capsids in a Non-Human Primate (NHP) Study

[0306] A barcoded, AAV vector pool was created containing 44 capsids including the hybrids described herein and several parental capsids, and administered intravenously to non-human primates, as follows.

[0307] Vector Production. Vector particles were produced by triple transfection of suspension-adapted HEK293 cells. Vector packaging a TdTomato transgene with CAG promoter and 15 base pair, unique barcode were harvested via freeze-thaw and purified via an iodixanol gradient. The final product was resuspended in phosphate-buffered saline (PBS) and 0.001% Pluronic F-68 to 500 µL.

[0308] Vector Pooling. A total of 44 different iodixanol-purified, endotoxin-removed vector preparations, including vectors comprising the capsids of Table 4, were pooled in approximately equal particle numbers and quantified by ddPCR.

[0309] Dosing. Two female and one male cynomolgus macaques of approximately two to four years old were screened for AAV2, AAV8, and AAV9 neutralizing antibody status prior to injection, with any positive titers being below 1:10. Included animals were injected intravenously at 2×10¹³ GC/kg total and subsequently sacrificed and tissues harvested 21 days after dosing.

[0310] NGS and Data Analysis. DNA extracted from NHP tissue was used as a template for amplification of the

barcode region in each AAV transgene. Secondary amplification was performed to add Illumina Nextera adapters and indices to the amplicons, which were then pooled and purified. The resultant amplicon library was denatured and sequenced on a MiSeq instrument (Illumina) using the MiSeq v3 chemistry reagent kit. Barcode counts were quantified from forward reads using a custom Galaxy workflow and converted to relative abundance, adjusted for the input pool distribution, and normalized to 1. Extracted RNA was used for cDNA generation and subjected to a similar process as above to prepare for NGS analysis.

[0311] Vector Production Analysis. Vector particles were produced by triple transfection of suspension HEK293 cells packaging a small (3 kb) or large (4.7 kb) transgene at a variety of culture scales ranging from 1 mL to 1 L. Cell lysates were clarified by centrifugation and subjected to DNase digestion before ddPCR analysis.

[0312] Biodistribution of vector genomes was similar to their respective parental VP3 serotypes for most of the hybrid capsids, except for those containing VP1u and VP1/2s of AAV5. Relative abundance of vector pool in the liver by DNA vector genome copies (FIG. 5) and mRNA (cDNA) transcript copies (FIG. 6) is shown as a representative data set. Capsids containing AAV5 VP1u and/or VP1/2s region, e.g., Hybrid 1 (2171), Hybrid 2 (2172), 2092, and 2095, exhibit reduced lower transduction (genome copy, FIG. 5), as well as lower expression (RNA transcripts, FIG. 6) in liver compared to their parental capsids. It was observed that AAV5-containing capsids decrease the transduction of all NHP tissues surveyed in this study, as measured by DNA genome copy number. Transcription of AAV-containing capsids in the liver was especially decreased, leading to a liver de-targeted phenotype.

[0313] AAV5/AAV8 hybrid capsids transduce less efficiently (FIG. 7A) but are liver de-targeted on the transcriptional level (FIG. 7B) and express RNA more efficiently than wild type AAV8 in muscle, leading to an improved RNA/DNA ratio (2.5-3.5 $\times$ increase over AAV8; FIG. 7C). Despite a decrease of transduction in skeletal muscle, transcription of genomes delivered by AAV5 hybrids was increased, leading to an improved RNA/DNA ratio. Without ascribing to any particular theory, AAV5 VP1u and VP1/2s may enable increased endosomal escape or nuclear trafficking abilities. Upon further evaluation of production capabilities, AAV5/AAV8 hybrids specifically produce 1.5-3 $\times$ more vector than their parental capsid, AAV8, in standard triple transfection of HEK293 suspension culture (FIGS. 8A-8C), regardless of the packaged genome size (FIG. 8A; 1 mL culture) or culture scale (FIGS. 8B-8C; 10 mL and 1 L culture, respectively). In summary, AAV5 hybrid capsids may present a unique method of gene therapy liver de-targeting at the transcriptional level, and offer improved vector production.

6.3 Example 3: Evaluation of AAV Hybrid Capsids in a Non-Human Primate (NHP) Study

[0314] In this experiment, AAVhu32 was identified as a potentially superior clade F capsid candidate for muscle-specific disease indications. Such capsid was characterized in the context of a pool of 44 barcoded vectors administered intravenously into cynomolgus macaques (n=3) at 2×10^{13} genome copies per kg, and treated animal tissues were harvested three weeks post-injection and nucleic acids were extracted as detailed below.

[0315] Vector Production. Vector particles were produced by triple transfection of suspension-adapted HEK293 cells. Vector packaging included a TdTomato transgene with CAG promoter and 15 base pair. Uniquely barcoded AAV vectors were harvested via freeze-thaw and purified via an iodixanol gradient.

[0316] Vector Pooling. A total of 44 different iodixanol-purified, endotoxin-removed vector preparations were pooled in approximately equal particle numbers and quantified by ddPCR.

[0317] Dosing. Two female and one male cynomolgus macaques of approximately two to four years old were screened for AAV2, AAV8, and AAV9 neutralizing antibody status prior to injection, with any positive titers being below 1:10. Animals included in the study were injected intravenously at 2×10^{13} GC/kg total and subsequently sacrificed and tissues harvested 21 days after dosing.

[0318] NGS and Data Analysis. DNA extracted from NHP tissue was used as a template for amplification of the barcode region in each AAV transgene. Secondary amplification was performed to add Illumina Nextera adapters and indices to the amplicons, which were then pooled and purified. The resultant amplicon library was denatured and sequenced on a MiSeq instrument (Illumina) using the MiSeq v3 chemistry reagent kit. Barcode counts were quantified from forward reads using a custom Galaxy workflow and converted to relative abundance, adjusted for the input pool distribution, and normalized to 1. Extracted RNA was used for cDNA generation and subjected to a similar process as above to prepare for NGS analysis. Although the pool contained 44 different capsids, data analysis, as represented in FIGS. 10A and 10B, was focused on AAVhu32 DNA and RNA relative abundance compared to AAV9.

[0319] Compared to AAV9, AAVhu32 transcripts were significantly higher by double in gastrocnemius ($P < 0.0001$) and bicep ($P = 0.03$) tissue. The two capsids transduced quadriceps and tibialis anterior similarly. In heart, AAVhu32 produced two-fold more transcript copies than AAV9, but it was not statistically significant ($P = 0.36$), likely due to animal-to-animal variability. As hepatotoxicity is a concern in gene therapy, it is significant to note that AAVhu32 transduced the liver at half the level of AAV9 and produced one third of RNA transcript copies compared to AAV9 ($P = 0.03$).

[0320] The AAVhu32 capsids (variants) represented in the capsid pool were analyzed for their transduction capability in muscle tissues versus liver. All skeletal muscle transcript data (bicep, gas, quad, TA, and diaphragm), heart transcript data (samples taken from aorta, apex, left and right atrium, left and right ventricle), and liver transcript data (left lobe sample) from 2 animals is represented in FIG. 14. Most variants expressed higher levels of transcript (relative abundance of mRNA transcript copies) compared to wild-type hu32.

6.4 Example 4: Evaluation of AAV Hybrid Capsids in a Single Vector Study in Mice

[0321] Additionally, AAVhu32 compared to AAV9 was evaluated in a single vector administration study in a mouse model of Duchenne muscular dystrophy (mdx, n=5) and recapitulated results seen in the pooled vector NHP study. Most of the amino acid differences between AAVhu32 and AAV9 lie in the VP1 unique (VP1u) region (FIG. 9), potentially adjacent to the PLA2 domain; however, the

enzymatic (phospholipase) activity of recombinantly expressed AAVhu32 VP1u was found to be equivalent to that of AAV9 VP1u.

[0322] AAV9 and AAVhu32 vectors packaging a microdystrophin construct driven by a muscle-specific promoter (transgene) were produced. Five, 5 to 6-week-old mdx mice were dosed via tail vein injection with 5×10^{13} GC/kg. Tissues were harvested four weeks after vector dosing for DNA and RNA analysis using ddPCR.

[0323] Biodistribution of AAVhu32 and AAV9 was assessed in mdx mouse tissues after intravenous administration (FIGS. 11A and 11B). Compared to AAV9, AAVhu32 had increased transduction of muscle tissues in terms of genome copy numbers (GC/cell) and RNA transcript copies by approximately double, similarly to NHP. Liver “de-targeting” of AAVhu32 was observed, but less reduced in the mdx mouse model compared to the differences seen between AAVhu32 and AAV9 in the NHP study.

[0324] The biodistribution of AAVhu32 and AAV9 in mdx mouse tissues was also done by quantifying protein expression by Western Blot. After (systemic) intravenous administration, heart tissues from AAVhu32, AAV9, or AAV8-injected mice (FIG. 18A) or gastrocnemius (gas) tissues from AAVhu32 or AAV9-injected mice (FIG. 18B) were evaluated by protein quantification of the transgene (microdystrophin) and a representative muscle protein (actin) using standard protein blot transfer techniques. The protein data correlates well with the RNA transcript results for these tissues. Immunofluorescent staining of the gastrocnemius (gas) tissues from AAVhu32, AAV8 or AAV9-injected mice shows strong fluorescence detection of the microdystrophin transgene, especially in AAVhu32-injected mice (FIG. 19).

[0325] RNASCOPE experiments were performed on mdx mouse tissues following AAVhu32, AAV8 or AAV9-microdystrophin vector systemic administration. DNA and RNA probes were designed to detect vector or transgene in tissues, also markers for the specific tissue were used. RNASCOPE scoring was based on the number of green (AAV-microdystrophin DNA), red (AAV-microdystrophin RNA/DNA), or blue (DAPI) signals (dots) per cell, for semi-quantification of the transduction events. AAVhu32-microdystrophin DNA detection was observed at about 3.73 GC/diploid cell (FIG. 20A) and RNA detection at about 82.3 microdystrophin RNA copies/TBP (TBP=TATA-box binding protein control) (FIG. 20B). AAV8-microdystrophin DNA detection was observed at about 1 copy per cell (FIG. 21A), and AAV9-microdystrophin DNA detection was observed at about 1.4 copies per cell (FIG. 21B).

6.5 Example 5: Evaluation of AAV Hybrid Capsids in a Pooled, Engineered Vector Study in Mice

[0326] The ability of AAVhu32 and engineered variants of AAVhu32 to package a barcoded microdystrophin construct driven by a muscle-specific promoter (transgene) were produced by triple transfection in suspension HEK293 cells and purified by iodixanol gradient. Vector preparations were pooled in approximately equal particle numbers and quantified by ddPCR. 5 to 6-week-old, male C57BL10J and mdx mice were dosed via tail vein injection with 2×10^{13} GC/kg. Tissues were harvested three weeks post injection for nucleic acid extraction and NGS analysis as described above.

[0327] Engineered AAVhu32 capsids were capable of further enhancing transduction of skeletal muscle in wild type mice, as illustrated in FIGS. 12A and 12B. Some engineered AAVhu32-based capsids were capable of expression (relative abundance of mRNA transcript copies) up to $35 \times$ that of wild type AAVhu32 in several skeletal muscle tissues. Cardiac muscle expression was also improved, but to a lesser degree. Relative abundance of transcript copies was adjusted based on the vector input and presented as a fold change over AAVhu32. Similar trends were seen in the mdx mouse model.

[0328] AAVhu32 capsid variants #1 to 21 were also analyzed compared to AAV8, AAV9, and wild-type AAVhu32 and the relative abundance of mRNA transcript copies (of transgene) were detected in each tissue, such as: gastrocnemius (gas) (FIG. 15A), quadriceps (quad) (FIG. 15B), tibialis anterior (TA) (FIG. 16A), heart (FIG. 16B), and liver (FIG. 17) tissues of wt and mdx mice.

[0329] As shown in the Examples hereinabove, Clade F vector AAVhu32 increased skeletal and cardiac muscle in NHP by approximately two fold compared to AAV9. AAVhu32 also showed improved transduction of muscle in mdx mice. In NHP, AAVhu32 showed decreased liver transduction compared to AAV9.

[0330] Although most similar to Clade F capsids by overall VP1 sequence alignment, AAVhu32 VP1u contains a proline-rich region (aa27-34) that is also present in Clade B (AAV2-like) vectors. AAVhu32 is considered an improved clade F capsid for muscle-specific disease indications, as compared to AAV9. Engineered AAVhu32 vectors may further increase transduction and/or transcription in wild type and mdx mouse muscle, and therefore AAVhu32-based vectors find therapeutic utility in muscle disease gene therapy treatments.

7. SEQUENCES

SEQ ID NO.	DESCRIPTION	SEQUENCE
1	AAV5 VP3 capsid protein	MSAGGGGPLGDNNQGADGVGNASGDWHCDSTWMDRVRTK STRTWLPSYNNHQYREIKSGSVDGSNANAYPGYSTPWGY FDENRFHSHWSPRDWQRLINNYWGFRRSLRVKIFNIQVK EVTVDSTTTIANLNTSTVQVFTDDDYQLPYVVVNGTGC LPAPPPQVFTLPQYGYATLNRDNTENPTERSFFCLEYFP SKMLRTGNNFEFTYNFEEVFPFHSSFAPSQNLFKLANPLVD QYLYRFVSTNNTGGVQFNKNLAGRYANTYKNWPPGPMGRT QGWNLGSGVNRASVSFAFATNRMELGASYQVPPQNGMT NNLQGSNTYALENTMIFNSQPANPGTTATYLEGNMLITSE SETQPVNRVAYNVGGQMATNNQSSTTAPATGTYNLQEI GVSVMERDVYLQGPFIWAKIPETGAHFHPSAMGGFGLKHP PPMMLIKNTPVPGNITSFSDVPVSSFITQYSTGQVTVEME WELKKENSKRWNPEIQYTNNYNDPQFVDFAPDSTGEYRTT RPIGTRYLTRPL

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SEQ ID NO.	DESCRIPTION	SEQUENCE
2	AAV5 VP3 capsid DNA	ATGTCTGCGGGAGGTGGCGGCCCATTTGGGCGACAATAACC AAGGTGCCGATGGAGTGGGCAATGCCTCGGAGATTGGCA TTGCGATTCCACGTGGATGGGGACAGAGTCGTCAACCAAG TCCACCCGAACCTGGGTGCTGCCCAGCTACAACAACCACC AGTACCGAGAGATCAAAGCGGCTCCGTGACCGGAAGCAA CGCCAACGCCTACTTTGGATACAGCACCCCTGGGGGTAC TTTGACTTTAACCCTTCCACAGCCACTGGAGCCCCGAG ACTGGCAAAGACTCATCAACAATACTGGGGCTTCAGACC CCGGTCCCTCAGAGTCAAATCTTCAACATTCAAGTCAA GAGGTACGGTGCAGGACTCCACCACCACCATCGCCAACA ACCTCACCTCCACCGTCCAAGTGTTTACGGACGACGACTA CCAGCTGCCCTACGTGTCGGCAACGGGACCGAGGGATGC CTGCCGGCCTTCCCTCCGCAGGTCTTTACGCTGCCGCAGT ACGGTTACGCGACGCTGAACCGGACACACAGAAAAATCC CACCGAGAGGAGCAGCTTCTTCTGCTAGAGTACTTTCCC AGCAAGATGCTGAGAACGGGCAACAACCTTTGAGTTTACCT ACAACTTTGAGGAGGTGCCCTTCCACTCCAGCTTCGCTCC CAGTCAGAACCTGTTCAAGCTGGCCAACCCGCTGGTGGAC CAGTACTTGTACCGCTTCGTGAGCACAAATAAATACTGGCG GAGTCCAGTTCAACAAGAACCTGGCCGGGAGATACGCCAA CACCTACAAAACTGGTTCCCGGGGCCATGGGCCGAACC CAGGGCTGGAACCTGGGCTCCGGGGTCAACCGCGCCAGTG TCAGCGCCTTCGCCACGACCAATAGGATGGAGCTCGAGGG CGCGAGTTACCGGTGCCCCCGCAGCGCAACGGCATGACC AACAACTCCAGGGCAGCAACACCTATGCCCTGGAGAACA CTATGATCTTCAACAGCCAGCCGGCGAACCAGGACACAC CGCCACGTACCTCGAGGGCAACATGCTCATCACCAGCGAG AGCGAGACGCGAGCCGGTGAACCGCGTGGCGTACAACGTCG GCGGGCAGATGGCCACCAACACAGAGCTCCACCACATGC CCCCAGCAGCGGACGTACAACCTCCAGGAATCGTGCCC GGCAGCGTGTGGATGGAGAGGACGTGTACCTCCAAGGAC CCATCTGGGCCAAGATCCAGAGACGGGGGCGCACTTTCA CCCCCTCCCGCCATGGGCGGATTCGGAATCAAACACCCA CCGCCCATGATGCTCATCAAGAACACGCTGTGCCCGGAA ATATCACCAGCTTCTCGGACGTGCCCGTCAGCAGCTTCAT CACCCAGTACAGCACCGGCGAGTCAACGTGGAGATGGAG TGGGAGCTCAAGAAGGAAAATCCAAGAGGTGGAACCCAG AGATCCAGTACACAAACAATAACAGACCCCGAGTTTGT GGACTTTGCCCGGACAGCACCGGGGAATACAGAACCACC AGACCTATCGGAACCCGATACCTTACCCGACCCCTTTAA
3	AAV5 VP1/2s common region	TAPTGKRIDDHFKRKKARTEEDSKPSTSSDAEAGPSGSQ QLQIPAPASSLGADT
4	AAV5 VP1/2s common region	ACGGCCCCCTACCGAAAGCGGATAGACGACCACTTTCCAA AAAGAAAGAAGGCTCGGACCGAAGAGGACTCCAAGCCTTC CACCTCGTCAGACGCGAAGCTGGACCCAGCGGATCCAG CAGCTGCAATCCAGCCCAACAGCCTCAAGTTTGGGAG CTGATACA
5	AAV5 VP2 capsid protein	TAPTGKRIDDHFKRKKARTEEDSKPSTSSDAEAGPSGSQ QLQIPAPASSLGADTMSAGGGPLGDNNGADGVGNASG DWHCDSWMDRVVTKSTRTWLPSYNNHQYREIKSGSVD GSNANAYFGYSTPWGYFDNRFHSHWSPRDWQRLINNYWG FRPRSLRVKIFNIQVKEVTVDSTTTIANNLSTVQVFTD DDYQLPYVVGNGTEGCLPAFPQVFTLPQYGYATLNRDNT ENPTERSFFCLEYFSPKMLRTGNNEFTYNFEEVPHSS FAPSQNLFLANPLVDQYLRFVSTNNTGGVQFNKNLAGR YANTYKNWFPQPMGRGTQGNLGSVNRASVSAFATTNRME LEGASYQVPPQPNGMTNNLQGSNTYALENTMIFNSQPANP GTTATYLEGNMLITSESETQPVNRVAYNVGGQMATNNQSS TTAPATGTYNLQEIYPGSVWMERDVYLQGPWAKIPETGA HFHPSPAMGGFGLKHPPMMLIKNTVPVGNITSFSDVPVS SFITQYSTGQVTEMEWELKKENSKRWNPFIQYTNNYNDP QFVDFAPDSTGEYRTTRPIGTRYLTRPL
6	AAV5 VP2 capsid DNA	ACGGCCCCCTACCGAAAGCGGATAGACGACCACTTTCCAA AAAGAAAGAAGGCTCGGACCGAAGAGGACTCCAAGCCTTC CACCTCGTCAGACGCGAAGCTGGACCCAGCGGATCCAG CAGCTGCAATCCAGCCCAACAGCCTCAAGTTTGGGAG CTGATACAATGCTCGCGGAGGTGGCGGCCCATTTGGGCGA CAATAACCAAGGTGCCGATGGAGTGGGCAATGCTTCGGGA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GATTGGCATTGCGATTCCACGTGGATGGGGGACAGAGTCG TCACCAAGTCCACCCGAACCTGGGTGCTGCCCAGCTACAA CAACCACCAGTACCAGAGATCAAAAGCGGCTCCGTCGAC GGAAGCAACGCCAACGCCCTACTTTGGATACAGCACCCCT GGGGGTACTTTGACTTTAACCGCTTCCACAGCCACTGGAG CCCCCGAGACTGGCAAAGACTCATCAACAATACTGGGGC TTCAGACCCCGTCCCTCAGAGTCAAAATCTTCAACATTC AAGTCAAAGAGGTACGGTGCAGGACTCCACCACCACCAT CGCCAACAACCTCACCTCCACCGTCCAAGTGTTTACGGAC GACGACTACCAGTGCCTACGTCGTGGCAACGGGACCG AGGGATGCCTGCCGGCTTCCCTCCGCAAGTCTTTACGCT GCCGCAGTACGGTTACGCGACGCTGAACCGCGACAACACA GAAAATCCACCGAGAGGAGCAGCTTCTTCGCTAGAGT ACTTTCCAGCAAGATGCTGAGAACGGGCAACAACCTTGA GTTTACCTACAACCTTGAGGAGGTGCCCTTCCACTCCAGC TTCGCTCCAGTCAAGAACCTGTTCAAGTGGCCAACCGC TGGTGGACCACTTGTACCGCTTCGTGAGCACAATAA CACTGGCGGAGTCCAGTTCAACAAGAACCTGGCCGGGAGA TACGCCAACCTACAAAATGTTCCCGGGGCCATGG GCCGAACCCAGGGCTGGAACCTGGGCTCCGGGGTCAACCG CGCCAGTGTACGCGCTTCGCCACGACCAATAGGATGGAG CTCGAGGGCGCGAGTTACCAGGTGCCCGCAGCCGAACG GCATGACCAACAACCTCCAGGGCAGCAACCTATGCCCT GGAGAACACTATGATCTTCAACAGCCAGCCGGCAACCCG GGCACCAACCGCCACGTACCTCGAGGGCAACATGCTCATCA CCAGCGAGAGCGAGACGCGAGCCGGTGAACCGCGTGGCGTA CAACGTGCGCGGGCAGATGGCCACCAACAACAGAGCTCC ACCACTGCCCCGCGACCGGCACGTACAACCTCCAGGAAA TCGTGCCCGCGCAGCGTGTGGATGGAGAGGACGTGTACCT CCAAGGACCCATCTGGGCCAAGATCCAGAGACGGGGGCG CACTTTCACCCCTCTCCGGCCATGGCGGATTGCGACTCA AACACCCACCGCCCATGATGCTCATCAAGAACACGCCTGT GCCCGGAAATATCACCACTCTTCGAGCGTGCCTGTGAGC AGCTTCATCACCCAGTACAGCACCGGGCAGGTACCCGTGG AGATGGAGTGGGAGCTCAAGAAGGAAAACCTCAAGAGGTG GAACCCAGAGATCCAGTACACAAACAATAACGACCCC CAGTTTGTGACTTTGCCCGGACAGCACCGGGGAATACA GAACCACCAGACCTATCGGAACCCGATACCTTACCCGACC CCTTTAA
7	AAV5 VP1u region (protein)	MSFVDHPPDWLEEVGEGLEFLGLEAGPPKPKPNQQHQDQ ARGLVLPGYNYLPGNGLDRGEPVNRADDEVAREHDI SYNE QLEAGDNPYLKYNHADA EFQEK LADDT SFGNL GKAVFQA KKRVL EPPFLVEEGAK
8	AAV5 VP1u region (DNA)	ATGTCCTTTGTGTGATCACCTCCAGATTGGTTGGAAGAAG TTGGTGAAGTCTTCGCGAGTTTTTGGGCTTGAAGCGGG CCCACCGAAACCAAAACCAATCAGCAGCATCAAGATCAA GCCCGTGGTCTGTGCTGCCTGGTTATAACTATCTCGGAC CCGGAAACGGTCTCGATCGAGGAGAGCTGTCAACAGGGC AGACGAGTTCGCGGAGAGCAGCATCTCGTACAACGAG CAGCTTGAGGCGGAGACAACCCCTACCTCAAGTACAACC ACGCGGACGCCAGTTTCAGGAGAAGCTCGCCGACGACAC ATCCTTCGGGGGAAACCTCGGAAAGGCAGTCTTTCAGGCC AAGAAAAGGGTCTCGAACCTTTTGGCCTGGTTGAAGAGG GTGCTAAG
9	AAV5 VP1 capsid protein	MSFVDHPPDWLEEVGEGLEFLGLEAGPPKPKPNQQHQDQ ARGLVLPGYNYLPGNGLDRGEPVNRADDEVAREHDI SYNE QLEAGDNPYLKYNHADA EFQEK LADDT SFGNL GKAVFQA KKRVL EPPFLVEEGAKTPTGKRIDDFPKRKKARTEEDS KPSTSSDAEAGPSGSQQLQIPAPASSLGADTMSAGGGP LGDNNQGADGVGNASGDWHCDSTWMDRVVTKSTRTWVLP SYNHNQYREIKSGSVDSNANAYFGYSTPWGYFDNRFHS HWSPRDWQRLINNYWGFPRSLRVKIFNIQVKEVTVDST TTIANNLSTVQVFTDDYQLPYVVNGTEGCLPAFPQPQV FTL PQYGYATLNRDNTENPTERSFFCLEYFPSKMLRTGN NFETYNFEFVFPFHSFAPSQNLFLKLANPLVDQYLRFVS TNNTGGVQFNKNLAGRYANTYKNWFPGPMGRTQGWNLGSG VNRASVSATATNRMELEGASYQVPPQPNGMTNNLQGSNT YALENTMIFNSQPANPGTTATYLEGNMLITSESETQPVNR VAYNVGGQMATNNQSTTAPATGTYNLQEI VPGSVWMERD VYLQGP IWAKI PETGAHFHPS PAMGGFGLKHPPPMMLIKN

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TPVPGNITSFSDVPVSSFITQYSTGQVTMEMEWELKKENS KRWNPETIQTNNYNDPQFVDFAPDSTGEYRTRTRPIGTRYL TRPL
10	AAV5 VP1 capsid DNA	ATGTCTTTTGTGATCACCTCCAGATTGGTTGGAAGAAG TTGGTGAAGGTCTTCGCGAGTTTTTGGGCCTTGAAGCGGG CCCACCGAAACAAAACCAATCAGCAGCATCAAGATCAA GCCCGTGGTCTTGTGCTGCCTGGTTATAACTATCTCGGAC CCGGAACGGTCTCGATCGAGGAGAGCCTGTCAACAGGGC AGACGAGGTCGCGCGAGAGCACGACATCTCGTACAACGAG CAGCTTGAGGCGGGAGACAACCCCTACCTCAAGTACAACC ACGCGGACGCGAGTTTCAGGAGAAGCTCGCCGACGACAC ATCCTTCGGGGGAAACCTCGGAAAGGCAGTCTTTCAGGCC AAGAAAAGGGTTCGGAACCTTTTGGCCTGGTTGAAGAGG GTGCTAAGACGGCCCCCTACCGAAAGCGGATAGACGACCA CTTTCCAAAAAGAAAGAGGCTCGGACCGAAGAGGACTCC AAGCCTTCACCTCGTCAGACGCCGAAGCTGGACCCAGCG GATCCCAGCAGCTGCAATCCCAGCCCAACAGCCTCAAG TTTGGGAGCTGATACAAATGTCTGCGGGAGGTGGCGGCCCA TTGGGCGACAATAACCAAGGTGCCGATGGAGTGGGCAATG CCTCGGGAGATTGGCATTGCGATTCCACGTGGATGGGGGA CAGAGTCGTACCAAGTCCACCCGAACCTGGGTGCTGCCC AGCTACAACAACCCAGTACCGAGAGATCAAAAGCGGCT CCGTCGACGGAAGCAACGCCAACGCCCTACTTTGGATACAG CACCCCTGGGGGTACTTTGACTTTAACCGCTTCCACAGC CACTGGAGCCCCGAGACTGGCAAGACTCATCAACAAC ACTGGGGCTTCAGACCCCGTCCCTCAGAGTCAAAATCTT CAACATTCAAGTCAAAGAGGTACGGTGACGAGACTCCACC ACCACCATCGCCAACAACCTCACTCCACCGTCCAAGTGT TTACGGACGACGACTACAGCTGCCCTACGTCGTGCGCAA CGGGACCGAGGGATGCCGCGGCCTTCCCTCCGACAGTTC TTTACGCTGCCGAGTACGGTTACGCGACGCTGAACCGCG ACAACACAGAAAAATCCCACGAGAGGAGCAGCTTCTTCTG CCTAGAGTACTTTCCAGCAAGATGTGAGAACGGGCAAC AACTTTGAGTTTACCTACAACCTTTGAGGAGGTGCCCTTC ACTCCAGCTTCGCTCCAGTCAGAACCTGTTCAGCTGGC CAACCCGCTGGTGGACAGTACTTGTACCGCTTCGTGAGC ACAATAAACACTGGCGGAGTCCAGTTCAACAAGAACCTGG CCGGGAGATACGCCAACACCTACAAAAACTGGTTCCTGGG GCCATGGGCCGAACCCAGGGCTGGAACCTGGGCTCCGGG GTCAACCGCGCCAGTGTGAGCGCCTTCGCCACGACCAATA GGATGGAGCTCGAGGGCGCGAGTTACCAGGTGCCCCCGCA GCCGAACGGCATGACCAACAACCTCCAGGGCAGCAACACC TATGCCCTGGAGAACAATATGATCTTCAACAGCCAGCCGG CGAACCCGGGCCAACCGGCCACGTACCTCGAGGGCAACAT GCTCATCACCAGCGAGAGCGAGACGACGCGGTGAACCGC GTGGCGTACAACGTCGGCGGGCAGATGGCCACCAACAACC AGAGCTCCACCACTGCCCGCGCAGCGGACGTACAACCT CCAGGAAATCGTGCCCGGCGAGCGTGTGGATGGAGAGGGAC GTGTACTCCAAGGACCATCTGGGCCAAGATCCAGAGA CGGGGGCGCACTTTACCCCTCTCCGGCCATGGGCGGATT CGGACTCAAAACACCCACCGCCATGATGCTCATCAAGAAC ACGCCTGTGCCCGGAAATATCACCAGCTTCTCGGACGTGC CCGTCAGCAGCTTATCACCAGTACAGCACCGGGCAGGT CACCGTGGAGATGGAGTGGAGCTCAAGAAGGAAAACCTCC AAGAGGTGGAACCCAGAGATCCAGTACACAAACAACCTACA ACGACCCCGAGTTTGTGGACTTTGCCCCGACAGCACCGG GGAATACAGAACCACGACCTATCGGAACCCGATACCTT ACCCGACCCCTTAA
11	AAV8 VP3 capsid protein	MAAGGGAPMADNNEGADGVGSSSGNWHCDSTWLGDRVITT STRTWALPTYNNHLYKQISNGTSGGATNDNTYFGYSTPWG YFDNFRFHCHFSRWDQRLINNNGWFRPKRSLFKLFNIQV KEVTQNEGTKTIANNLSTIQVFTDSYQLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEYFPS QMLRTGNNFQFTYTFEDVPPHSSYAHQSGLDRMLNPLIDQ YLYLSRTQTGGTANTQTLGFSQGGPNTMANQAKNWLPG PCYRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLANPG IAMATHKDDEERFFPSNGILIFGKQNAARDNADYSDVMLT SEEEIKTTNPVATEEYGIVADNLQQQNTAPQIGTVNSQGA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		LPGMVWQNRDVYLQGPWAKIPHTDGNFHPSPLMGGFGLK HPPPIQLIKNTPVPADPPTTFNQSKINSFITQYSTGQVSV EIEWELQKENSKRWNPEIQYTSNYYKSTSVDFAVNTEGVY SEPRPIGTRYLTRNL
12	AAV8 VP3 capsid DNA	ATGGCTGCAGGCGGTGGCGCACCAATGGCAGACAATAACG AAGGCGCCGACGGAGTGGGTAGTTCTCGGAAATGGCA TTGCGATTCCACATGGCTGGGCGACAGAGTCATCACCACC AGCACCCGAACCTGGGCCCTGCCACCTACACCAACCACC TCTACAAGCAAATCTCCAACGGGACATCGGAGGAGCCAC CAACGACAACACCTACTTCGGCTACAGCACCCCTGGGGG TATTTTGACTTTAACAGATTCCACTGCCACTTTTCACCAC GTGACTGGCAGCGACTCATCAACAACACTGGGGATTCCG GCCCAAGAGACTCAGCTTCAAGCTCTTCAACATCCAGGTC AAGGAGGTCACGCAGAATGAAGGCACCAAGACCATCGCCA ATAACCTCACCAGCACCATCCAGGTGTTACGACTCGGA GTACCAGCTGCCGTACGTTCTCGGCTCTGCCACCAGGGC TGCTTGCCTCCGTTCCGGCGGACGTGTTCATGATTCCTC AGTACGGCTACCTAACACTCAACAACGGTAGTCAGGCCGT GGGACGCTCCTCTTCTACTGCCTGGAATACTTCTCTTCG CAGATGCTGAGAACCGGCAACAACCTCCAGTTTACTTACA CCTTCGAGGACGTGCTTTCCACAGCAGCTACGCCACAG CCAGAGCTTGGACCGGCTGATGAATCCTCTGATTGACCAG TACCTGTACTACTTGTCTCGGACTCAAACAACAGGAGGCA CGGCAATACGCAGACTCTGGGCTCAGCCAAGGTGGGCC TAATACAATGGCCAATCAGGCAAGAAGCTGGCTGCCAGGA CCCTGTTTACCGCCAACACCGCTCTCAACGACAACCGGGC AAAACAACAATAGCAACTTTGCCTGGACTGCTGGGACCAA ATACCATCTGAATGGAAGAAATTCATTGGCTAATCCTGGC ATCGCTATGGCAACACAAAGACGACGAGGAGCGTTTTT TTCCAGTAACGGGATCCTGATTTTGGCAACAATAATGC TGCCAGAGACAATGCGGATTACAGCGATGTCTGCTCACC AGCGAGGAAGAAATCAAACCACTAACCTGTGGCTACAG AGGAATACGGTATCGTGGCAGATAACTTGACGACGCAAAA CACGGCTCCTCAAATGGAAGTGTCAACAGCCAGGGGGCC TTACCCGGTATGGTCTGGCAGAACCGGACGTGTACCTGC AGGGTCCCATCTGGGCCAAGATTCTCACACGACGCGCAA CTTCCACCCGTCTCCGCTGATGGGCGGCTTTGGCCTGAAA CATCTCCGCTCAGATCCTGATCAAGAACCGCCTGTAC CTGCGGATCCTCCGACCACCTCAACCAGTCAAAGCTGAA CTCTTTCATCACGCAATACAGCACCGGACAGGTGAGCGTG GAAATTTGAATGGGAGCTGCAGAAGGAAACAGCAAGCGCT GGAACCCGAGATCCAGTACACCTCCAACCTACTACAATC TACAAGTGTGGACTTTGCTGTTAATACAGAAGCGGTGTAC TCTGAACCCCGCCCATTGGCACCCGTTACCTCACCCGTA ATCTGTAA
13	AAV8 VP1/2s common region (protein)	TAPGKKRPVEPSPQRSPTSSTGIGKKGQOPARKRLNFGQT GDESVPDPQPLGEPPAAPSGVGPNT
14	AAV8 VP1/2s common region (DNA)	ACGGCTCCTGGAAAGAAGAGACCGGTAGAGCCATCACCCC AGCGTTCTCCAGACTCCTCTACGGGCATCGCAAGAAAGG CCAACAGCCCGCCAGAAAAAGACTCAATTTTGGTCAGACT GGCGACTCAGAGTCAGTTCCAGACCTCAACCTCTCGGAG AACCTCCAGCAGCGCCCTCTGGTGTGGGACCTAATACA
15	AAV8 VP2 capsid protein	TAPGKKRPVEPSPQRSPTSSTGIGKKGQOPARKRLNFGQT GDESVPDPQPLGEPPAAPSGVGPNTMAAGGAPMADNNE GADGVSSSGNWHCDSTWLGDVRTTSTRTWALPTYNNHL YKQISNGTSGGATNDNTYFGYSTPWGYFDNRFCHFSR DWQRLINNNWGFPRKLSFKLFNIQVKEVTQNEGKTIAN NLSTIQVFTDSEYQLPYVLGSAHQGLPPFPADVFMIPO YGYLTLNNGSQAVGRSSPYCLEYFPSQMLRTGNFQFTYT FEDVPFHSSYAHQSQSLDRLMNLIDQYLYLSRTQTTGGT ANTQTLGFSQGGPNTMANQAKNWLPGPCYRQQRVSTTTGQ NNNSNFAWTAGTKYHLNGRNSLANPIAMATHKDDDEEFF PSNGILIFGKQNAARDNADYSDVMLTSEEEIKTTNPVATE EYGI VADNLQQNTAPQIGTVNSQGALPGMVWQNRDVYLQ GPIWAKIPHTDGNFHPSPLMGGFGLKHPPPIQLIKNTPVP ADPPTTFNQSKLNSFITQYSTGQVSV EIEWELQKENSKR NPEIQYTSNYYKSTSVDFAVNTEGVYSEPRPIGTRYLTRN L

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SEQ ID NO.	DESCRIPTION	SEQUENCE
16	AAV8 VP2 capsid DNA	ACGGCTCCTGGAAAGAAGAGACCGGTAGAGCCATCACCCC AGCGTTCTCCAGACTCCTCTACGGGCATCGGCAAGAAAGG CCAACAGCCCGCCAGAAAAGACTCAATTTTGGTCAGACT GGCGACTCAGAGTCAGTTCAGACCCCTAACCTCTCGGAG AACCTCCAGCAGCGCCCTCTGGTGTGGGACCTAATACAAT GGCTGCAGGCGGTGGCGCACCATGGCAGACAATAACGAA GGCGCCGACGGAGTGGGTAGTTCTCGGAAATTGGCATT GCGATTCCACATGGCTGGGCGACAGAGTCATCACCACCAG CACCCGAACCTGGGCCCTGCCACCTACAACAACCCCTC TACAAGCAAATCTCCAACGGGACATCGGGAGGAGCCACCA ACGACAACACCTACTTCGGCTACAGCACCCCTTGGGGTA TTTTGACTTTAACAGATTCCACTGCCACTTTTCACCACGT GACTGGCAGCGACTCATCAACAACAAGTGGGGATTCCGGC CCAAGAGACTCAGTTCAAGCTCTTCAACATCCAGGTCAA GGAGGTACGCGAGAATGAAGGCACCAAGACCATCGCCAAT AACCTCACAGCACCATCCAGGTGTTTACGGACTCGGAGT ACCAGCTGCCGTACGTTCTCGGCTCTGCCACCAAGGCTG CCTGCCTCCGTTCGGCGGAGCGTGTTCATGATTTCCCGAG TACGGCTACCTAACACTCAACAACGGTAGTCAGGCCGTGG GACGCTCCTCCTTCTACTGCCTGGAATACTTCTTTCGCA GATGCTGAGAACCGGCAACAACCTCCAGTTTACTTACACC TTCGAGGACGTGCCTTTCCACAGCAGCTACGCCACAGCC AGAGCTTGGACCGGCTGATGAATCCTCTGATTGACCAGTA CCTGTACTACTTGTCTCGGACTCAAACAACAGGAGGCACG GCAAATACGCAGACTCTGGGCTTCAGCCAAGGTGGGCCTA ATACAATGGCCAATCAGGCAAGAAGTGGCTGCCAGGACC CTGTTACCGCCAACAACGCGTCTCAACGACAACCGGGCAA AACAACAATAGCAACTTTGCCTGGACTGCTGGGACCAAAT ACCATCTGAATGGAAGAAATTCATTGGCTAATCCTGGCAT CGCTATGGCAACACACAAGACGACGAGGAGCGTTTTTTT CCCAGTAACGGGATCCTGATTTTGGCAACAAAATGCTG CCAGAGACAAATGCGGATTACAGCGATGTCTGCTCACCAG CGAGGAAGAAATCAAACCACTAACCTGTGGCTACAGAG GAATACGGTATCGTGGCAGATAACTTGCAGCAGCAAAACA CGGCTCCTCAAATTGGAACTGTCAACAGCCAGGGGGCCTT ACCGGTATGGTCTGGCAGAACCGGACGTGTACCTGCAG GGTCCCATCTGGGCCAAGATTCTCACACGGACGGCAACT TCCACCGTCTCCGCTGATGGGCGGCTTGGCCTGAAACA TCCTCCGCTCAGATCCTGATCAAGAACACGCTGTACCT GCGGATCCTCCGACCCTTCAACCAGTCAAAGCTGAACT CTTTCATCACGCAATACAGCACCGGACAGGTACGCGTGA AATTGAATGGGAGCTGCAGAAGGAAACAGCAAGCGCTGG AACCCCGAGATCCAGTACACCTCCAACCTACTACAAATCTA CAAGTGTGGACTTTGCTGTTAATACAGAAGCGGTGACTC TGAACCCCGCCCATTTGGCACCCGTTACCTCACCCGTAAT CTGTAA
17	AAV8 VP1u region (protein)	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEPVNAADAAALEHDKAYD QQLQAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAK
18	AAV8 VP1u region (DNA)	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGGAGTGGTGGGCGCTGAAACC TGGAGCCCGAAGCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTCTTCTGGCTACAAGTACCTCG GACCTTCAACGGAATCGACAAGGGGAGCCCGTCAACGC GGCGGACGACGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTGCAGGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCGGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAG
19	AAV8 VP1 capsid protein	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEPVNAADAAALEHDKAYD QQLQAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPVEPSPQRSPTSSTGI GKKGQQPARKRLNFGQTDSESVDPQPLGEPAPSPGVG PNTMAAGGAPMADNNEGADGVSSSGNWHCDSTWLGDV ITTSRTWALPTYNNHLYKQISNGTSGGATNDNTYFGYST PWGYFDNRFHCHFSRWDQRLINNNWGFPRKRLSFKLPN

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		IQVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSA HQGCLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEY FPSQMLRTGNFQFTYTFEDVPHSSYAHSQSLDRLMNPL IDQYLYLSRTQTGGTANTQTGLFSQGGPNTMANQAKNW LPGPCYRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLA NPGIAMATHKDDERFFPSNGILIFGKQNAARDNADYSDV MLTSEEEIKTTNPVATEEYGVADNLQQONTAPQIGTVNS QGALPGMVWQNRDVYLQGPWAKIPHDTGNFHPSPLMGGF GLKHPPPQILIKNTPVPADPPTTFNQSKLNSFITQYSTGQ VSVEIEWELQKENSkrwnPEIQYTSNYYKSTSVDFAVNTE GVYSEPRPIGTRYLTRNL
20	AAV8 VP1 capsid DNA	ATGGCTGCCGATGGTTATCTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGGAGTGGTGGGCGCTGAAACC TGGAGCCCCGAAGCCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTGCTTCCTGGCTACAAGTACCTCG GACCCCTTCAACGGACTCGACAAGGGGGAGCCCGTCAACGC GGCGGACGCAGCGGCCCTCGAGCACGACAAGGCCCTACGAC CAGCAGCTGCAGGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCCGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCCTGGAAGAAGAGACCGGTAGA GCCATCACCCAGCGTTCTCCAGACTCCTCTACGGGCATC GGCAAGAAGGCCAACAGCCCGCCAGAAAAAGACTCAATT TTGGTCAGACTGGCGACTCAGAGTCAGTTCAGACCCCTCA ACCTCTCGGAGAACCTCCAGCAGCGCCTCTGGTGTGGGA CCTAATACAATGGCTGCAGGCGGTGGCGCACCATTGGCAG ACAATAACGAAGGCGCCGACGGAGTGGGTAGTTCTCTCGGG AAATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTC ATCACCAAGCAGCACCCGAACTGGGCGCTGCCACCTACA ACAACCACTCTACAAGCAATCTCCAACGGGACATCGGG AGGAGCCACCAACGACAAACCTACTTCGGCTACAGCACC CCCTGGGGGTATTTGACTTTAACAGATTCCACTGCCACT TTTCAACACGTGACTGGCAGCGACTCATCAACAACAACCTG GGGATTCGGGCCCCAAGAGACTCAGTTCAGCTCTTCAAC ATCCAGGTCAAGGAGGTCAAGCAGAAATGAAGGCACCAAGA CCATCGCCAATAACCTCACCAGCACCATCCAGGTGTTTAC GGACTCGGAGTACCAGCTGCCGTACGTTCTCGGCTCTGCC CACCAGGGCTGCCCTCGCTTCCCGCGGACGTGTTC TGATTCCCCAGTACGGCTACCTAACACTCAACAACGGTAG TCAGGCGGTGGGACGCTCCTCCTTCTACTGCCTGGAATAC TTTCCTTCGAGATGCTGAGAACCAGCAACAACCTCCAGT TTACTTACACCTTCGAGGACGTGCCCTTCCACAGCAGCTA CGCCACAGCCAGAGCTTGGACCGGCTGATGAATCCTCTG ATTGACCAGTACCTGTACTACTTGTCTCGGACTCAACAA CAGGAGGCACGGCAATACGAGACTCTGGGCTTCAGCCA AGGTGGGCTTAATACAAATGGCCAATCAGGCAAGAACTGG CTGCCAGGACCCTGTACCGCCAACACGCGTCTCAACGA CAACCGGCAAAACAACAATAGCAACTTTGCCCTGGACTGC TGGGACCAAAATACCATCTGAATGGAAGAAATTCATTGGCT AATCCTGGCATCGCTATGGCAACACAAAGACGACGAGG AGCGTTTTTTTCCAGTAACGGGATCCTGATTTTGGCAA ACAAAATGCTGCCAGAGACAATGCGGATTACAGCGATGTC ATGCTCACGAGCGAGGAAGAAATCAAACCACTAACCTG TGGCTACAGAGGAATACGGTATCGTGGCAGATAACTTGCA GCAGCAAAACACGGCTCCTCAAATTGAAGTGTCAACAGC CAGGGGGCTTACCAGGTATGGTCTGGCAGAACCGGGACG TGACCTGCAGGGTCCCATCTGGGCCAAGATTCTCACAC GGACGGCAACTTCAACCGCTCTCCGCTGATGGGCGGCTT GGCCTGAAACATCTCCGCTCAGATCCTGATCAAGAACA CGCCTGTACCTGCGGATCCTCCGACCCTTCAACCACTC AAAGCTGAACCTTTCATCACGCAATACAGCACCGGACAG GTCAGCGTGGAAATTGAATGGGAGCTGCAGAAGGAAAAACA GCAAGCGCTGGAACCCGAGATCCAGTACACCTCCAACCTA CTACAAATCTACAAGTGTGGACTTTGCTGTTAATACAGAA GGCGTGTAATCTGAACCCCGCCCATTTGGCACCCGTTACC TCACCCGTAATCTGTAA
21	AAV5-AAV8 hybrid VP2 capsid protein	TAPTGKRIDDHFPKRKKARTEEDSKPSTSSDAEAGPSGSQ QLQIPAPASSLGADTMAAGGGAPMADNNEGADGVGSSSG NWHCDSTWLGDRVITTSRTRWALPTYNNHLYKQISNGTSG GATNDNTYFGYSTPWGYFDNRFCHFSRWDQRLINNW

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GFRPKRLSFKLFNIQVKEVTQNEGTKTIANNLTSTIQVFT DSEYQLPYVLGSAHQGCLPPFPADVFMIPOQYGYLTLNNGS QAVGRSSFYCLEYFPSQMLRTGNNFQFTYTFEDVPPHSSY AHSQSLDRLMNLIDQYLYLSRTQTGGTANTQTLGFSQ GGPNTMANQAKNWLPGPCYRQQRVSTTTGQNNNSNFAWTA GTKYHLNGRNSLANPGIAMATHKDDEERFFPSNGILIFGK QNAARDNADYSDVMLTSEEEIKTTNPVATEEYGIVADNLQ QQNTAPQIGTVNSQGALPGMVWQNRDVYLQGPWAKIPHT DGNFHPSPLMGGFGLKHPPQILIKNTFVPADPPTTFNQSS KLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNY YKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL
22	AAV5-AAV8 hybrid VP2 capsid DNA	ACGGCCCCCTACCGGAAAGCGGATAGACGACCACTTCCAA AAAGAAAGAGGCTCGGACCGAAGAGGACTCCAAGCCTTC CACCTCGTCAGACGCGAAGCTGGACCCAGCGGATCCAG CAGCTGCAATCCAGCCCAACGAGCTCAAGTTGGGAG CTGATACAATGGCTGCAGGCGGTGGCGCACCAATGGCAGA CAATAACGAAGGCGCGGAGGAGTGGGTAGTTCTCGGGA AATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCA TCACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAA CAACCACCTCTACAAGCAATCTCCAACGGGACATCGGGA GGAGCCACCAACGACAACACCTACTTCGGCTACAGCACC CCTGGGGGTATTTGACTTTAACAGATTCCACTGCCACTT TTCACCACGTGACTGGCAGCGACTCATCAACAACAACCTGG GGATTCCGGCCCAAGAGACTCAGCTTCAAGCTCTCAACA TCCAGGTCAAGGAGGTACGCGAGAATGAAGGCACCAAGAC CATCGCCAATAACCTCACCAGCACCATCCAGGTGTTTACG GACTCGAGTACCAAGCTGCCGTACGTTCTCGGCTCTGCC ACCAGGGCTGCCTGCCCTCCGTTCCCGGGCGGACGTGTTT GATTCCCGCAGTACGGCTACCTAACACTCAACAACGGTAGT CAGGCCGTGGGACGCTCCTCCTTCTACTGCTGGAATACT TTCCTTCGCAGATGCTGAGAACCGGCAACAACCTCCAGTT TACTTACACCTTCGAGGACGTGCCCTTCCACAGCAGCTAC GCCACAGCCAGAGCTTGAGCCGGCTGATGAATCCTCTGA TTGACCAGTACCTGTACTACTTGTCTCGGACTCAAACAAC AGGAGGCACGGCAATAACGAGACTCTGGGCTTCAGCCAA GGTGGGCTAATACAATGGCCAATCAGGCAAGAAGTGGC TGCCAGGACCTGTTACCGCCAACAACGCGTCTCAACGAC AACCAGGCAAAACAACATAGCAACTTTCCTGGAGTCTGCT GGGACCAATACCATCTGAATGGAAGAAATTCATTGGCTA ATCCTGGCATCGCTATGGCAACACACAAGACGACGAGGA GCGTTTTTTTCCAGTAACGGGATCCTGATTTTGGCAAA CAAAATGCTGCCAGAGACAATGCGGATTACAGCGATGTCA TGCTCACCAGCGAGGAAGAAATCAAAACCACTAACCCCTGT GGCTACAGAGGAATACGGTATCGTGGCAGATAACTTGACG CAGCAAAACACGCTCCTCAAATTGGAAGTGTCAACAGCC AGGGGGCCTTACCGGTATGGTCTGGCAGAACCGGACGT GTACCTCGAGGTCCTATCTGGGCCAAGATTCTCCACACG GACGGCAACTTCCACCGCTCTCCGCTGATGGGCGGCTTG GCCTGAAACATCCTCCGCTCAGATCTGATCAAGAACAC GCCTGTACTCGCGATCTCCGACCACCTTCAACCACTCA AAGCTGAACCTTTTCATCAGCAATACAGCACCGGACAGG TCAGCGTGGAATGAATGGGAGCTGCAGAAGGAAAACAG CAAGCGTGGAACCCGAGATCCAGTACACCTCCAACCTAC TACAAATCTACAAGTGTGACTTTGCTGTTAATACAGAAG GCGTGACTCTGAACCCCGCCCCATTGGCACCCGTTACCT CACCCTAATCTGTAA
23	AAV5-AAV8 hybrid VP1 capsid protein	MSFVDHPPDWLEEVGEGLEFLGLEAGPPKPKPNQQHQDQ ARGLVLPGYNYLPGNGLDRGEPVNRADDEVAREHDI SYNE QLEAGDNPYLKYNHADAEFQEKLADDTSGFNGLKAVFQA KKRVLEPPGLVEEGAKTAPTGRIDDHFPKRKKARTEEDS KPSTSSDAEAGPSGSQQLQIPAQPASSLGADTMAAGGGAP MADNNEGADGVSSSGNWHCDSTWLGDREVIITSTRTWALP TYNNHLYKQISNGTSGGATNDNTYFGYSTPWGYPDFNRFH CHFSPRDWQRLINNNWGFRPKRLSFKLFNIQVKEVTQNEG TKTIANNLTSTIQVFTDSEYQLPYVLGSAHQGCLPPFPAD VFMIPOQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNN FQFTYTFEDVPPHSSYAHQSGLDRLMNLIDQYLYLSRT QTTGGTANTQTLGFSQGGPNTMANQAKNWLPGPCYRQQRV STTTGQNNNSNFAWTAGTKYHLNGRNSLANPGIAMATHKD DEERFFPSNGILIFGKQNAARDNADYSDVMLTSEEEIKTT NPVATEEYGIVADNLQQNTAPQIGTVNSQGALPGMVWQN

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		RDVYLQGPIWAKIPHDTGNFHPSPLMGGFGLKHPPPQILI KNTVPVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQK ENSKRWNPEIQYTSNYYKSTSVDFAVNTGEGVYSEPRPIGT RYLTRNL
24	AAV5-AAV8 hybrid VP1 capsid DNA	ATGTCTTTTGTGATCACCTCCAGATTGGTTGGAAGAAG TTGGTGAAGGTCTTCGCGAGTTTTTGGGCTTGAAGCGGG CCCACCGAAACAAAACCCAATCAGCAGCATCAAGATCAA GCCCCGTGGTCTTGTGCTGCCTGGTTATAACTATCTCGGAC CCGGAACCGTCTCGATCGAGGAGACCTGTCAACAGGGC AGACGAGGTGCGCGAGAGCAGCATCTCGTACAACGAG CAGCTTGAGGCGGGAGACAACCCCTACCTCAAGTACAACC ACGCGGACGCGGAGTTTCAGGAGAAGCTCGCCGACGACAC ATCCTTCGGGGGAAACCTCGGAAAGGCAGTCTTTCAGGCC AAGAAAAGGGTTCTCGAACCTTTTGGCTGGTTGAAGAGG GTGCTAAGACGCGCCCTACCGAAAGCGGATAGACGACCA CTTCCAAAAAGAAAGGCTCGGACCGAAGAGGACTCC AAGCCTTCCACCTCGTCAGACGCCGAAGCTGGACCCAGCG GATCCCAGCAGCTGCAATCCAGCCCAACAGCCTCAAG TTTGGGAGCTGATACAATGGCTGCAGGCGGTGGCGCACCA ATGGCAGACAATAACGAAGGCGCCGACGGAGTGGGTAGTT CCTCGGGAATTGGCATTGCGATTCCACATGGCTGGCGCA CAGAGTCATCACACCAGCAGCCCCGAACCTGGGCCCTGCCC ACCTACAACAACCCCTCTACAAGCAATCTCCAACGGGA CATCGGAGGAGCCACCAACGACAACCTACTTCGGCTA CAGCACCCCTGGGGGTATTTGACTTTAACAGATTCCAC TGCCACTTTTACCACGTGACTGGCAGCGACTCATCAACA ACAACTGGGGATTCCGGCCCAAGAGACTCAGCTTCAAGCT CTTCAACATCCAGGTCAAGGAGGTACGCGAATGAAGGC ACCAAGACCATCGCCAACTACCTCACCAGCACCATCCAGG TGTTTACGGAAGTCCGAGTACCAGCTGCCGTACGTTCTCGG CTCTGCCCCACGAGGCTGCCTGCCTCCGTTCCCGGCGGAC GTGTTTCATGATTCCCGAGTACGGCTACCTAACACTCAACA ACGGTAGTCAGGCGGTGGGACGCTCCTCCTTCTACTGCCT GGAATACTTTCCTTCGCGAGATGCTGAGAACCGGCAACAAC TTCAGTTTACTTACACCTTCGAGGACGTGCCTTTCCACA GCAGCTACGCCCACAGCCAGAGCTTGGACCGGCTGATGAA TCCTCTGATTGACCAGTACCTGTACTACTTGTCTCGGACT CAAAACAACGAGGACGCGCAATACGCGAGACTCTGGGCT TCAGCCAAGGTGGGCTAATACAATGGCCAATCAGGCAAA GAACTGGCTGCGAGGACCTGTTACCGCCAACAACGCGTC TCAACGACAACCGGCAAAACAATAAGCAACTTTGCCT GGACTGCTGGGACCAATACCATCTGAATGGAAGAAATTC ATTGGCTAATCCTGGCATCGCTATGGCAACACAAAAGAC GACGAGGAGCGTTTTTTTCCAGTAACGGGATCCTGATTT TTGGCAAAACAAATGCTGCAGAGACAATGCGGATTACAG CGATGTCATGCTCACGCGAGGAAGAAATCAAAACCACT AACCCTGTGGTACAGAGGAATACGGTATCGTGGCAGATA ACTTGCAGCAGCAAAACACGGCTCCTCAAATTGGAAGTGT CAACAGCCAGGGGCGCTTACCCGGTATGGTCTGGCAGAAC CGGGACGTGTACCTGCAGGGTCCCATCTGGGCCAAGATT CTCACACGGACGGCAACTTCCACCCGCTCCTCGCTGATGGG CGGCTTTGGCCTGAAACATCCTCCGCTCAGATCCTGATC AAGAACACGCTGTACCTGCGGATCCTCCGACCACCTTCA ACCACTCAAAGCTGAACCTTTTCATCAGCAATACAGCAC CGGACAGTCAAGCTGGAAATGAATGGGAGCTGCAGAAG GAAACAGCAAGCGCTGGAACCCCGAGATCCAGTACACCT CCAATACTACAATCTACAAGTGTGACTTTGCTGTAA TACAGAAGGCGTGTACTCTGAACCCCGCCCATTTGGCACC CGTTACCTCACCCGTAATCTGTAA
25	AAV8/5-AAV8 hybrid VP1 capsid protein	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNDKGEVNVNADAAAEHDKAYD QQLQAGDNPLYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPTGRKIDDFPKRKKARTEED SKPSTSSDAEAGPSGSQQLQIPAQPASSLGADTMAAGGGA PMADNNEGADVGSSSGNWHCDSTWLGDRVITSTRTWAL PTYNNHLYKQISNGTSGGATNDNTYFGYSTPWGYFDENRF HCHFSPRDWQRLINNNWGFPRKRLSFKLFNIQVKEVTQNE GKTITANNLTSTIQVFTDSEYQLPYVLGSAHQGCLPPFPA DVFMIPQYGYLTLNNGSQAVGRSSFYCLEYPPSQMLRTGN NFQPTYTFEDVPFHSSYAHSSQLDRLMNPIDQYLYLSR TQTTGGTANTQTLGFSQGGPNTMANQAKNWLPGPCYRQQR

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		VSTTTGQNNNSNFAWTAGTKYHLNGRNSLANPGIAMATHK DDEERFFPSNGILIFGKQNAARDNADYSDVMLTSEEEIKT TNPVATEEYGI VADNLQQONTAPQIGTVNSQALPGMVWQ NRDVYLQGPWAKIPHTDGNFHPSPLMGGFGLKHPPPIIL IKNTPVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQ KENSKRWNPEIQYTSNYKSTSVDFAVNTGEGVYSEPRPIG TRYLTRNL
26	AAV8/5-AAV8 hybrid VP1 capsid DNA	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGCGCTGAAACC TGGAGCCCCGAAGCCCAAAGCCAACAGCAAAGCAGGAC GACGGCCGGGTCTGGTCTTCTGGCTACAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGACGCAGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTGCAGGCGGGTGACAATCCGTACCTGCGGTATA ACCAAGCCGACGCGAGTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGGCCCTACCGGAAGCGGATAGACGA CCACTTTCCAAAAAGAAAGAGGCTCGGACCGAAGAGGAC TCCAAGCCTTCCACCTCGTCAGACGCGAAGCTGGACCCA GCGGATCCAGCAGCTGCAATCCAGCCCAACCAAGCTC AAGTTTGGGAGCTGATACAATGGCTGCAGGCGGTGGCGCA CCAATGGCAGACAATAACGAAGGCGCGACGAGTGGGTA GTTCTCTGGGAAATTGGCATTGCGATTCCACATGGCTGGG CGACAGAGTCATCACCAACGACCCGAACCTGGGCCCTG CCCACCTACAACAACCACTCTACAAGCAAATCTCCAACG GGACATCGGGAGGAGCCACCAACGACAACCTACTTCGG CTACAGCACCCCTGGGGGTATTTGACTTTAACAGATTCT CACTGCCACTTTTCAACAGTGAAGTGGCAGCAGTCACTCA ACAACAAGTGGGATTCGCGCCCAAGAGACTCAGCTTCAA GCTCTTCAACATCCAGGTCAAGGAGTCAACGAGAATGAA GGCACCAGACCATCGCCAAATCACTCAACGACCACTCC AGGTGTTTACGGAAGTGGGAGTACAGCTGCGTACGTTCT CGGCTCTGCCACCAAGGCTGCCTGCCTCCGTTCCCGGCG GACGTGTTTCATGATTCCCAAGTACGGTACCTAACACTCA ACAACGCTAGTCAAGCGGTGGGACGCTCTCTTCTACTG CCTGGAATACTTTCTTCGAGATGCTGAGAACCGGCAAC AACTTCCAGTTTACTTACACCTTCGAGGACGTGCCTTCC ACAGCAGCTACGCCCCACAGCCAGAGCTTGGACCGGCTGAT GAATCCTCTGATTGACCACTGCTGCTACTTGTCTCGG ACTCAAAACAGGAGGACGCGCAATACGAGACTCTGG GCTTCAAGCAAGGTGGGCTTAATACAATGGCAATCAGGC AAAGAAGTGGCTGCCAGGACCTGTTACCGCCAACAACGC GTCTCAACGACAACCGGCAAAACAATAGCAACTTTG CCTGGACTGCTGGGACCAATACCATCTGAATGGAAGAAA TTCATTGGCTAATCCTGGCATCGCTATGGCAACACACAAA GACGACGAGGAGCGTTTTTTTCCAGTAACGGGATCTGTA TTTTTGGCAAAACAAATGCTGCCAGAGCAATGCGGATTA CAGCGATGTCATGCTCACCAAGGAGGAAGAAATCAAAACC ACTAACCTGTGGCTACAGAGGAATACGGTATCGTGGCAG ATAACTTGCAGCAGCAAAACACGGCTCTCAAATTGAAC TGTCACAGCCAGGGGGCTTACCGGTATGGTCTGGCAG AACCGGGACGTGTACCTGCAGGGTCCCATCTGGGCCAAGA TTCCTCACACGACGGCAACTTCCACCGTCTCCGCTGAT GGGCGGCTTTGGCCTGAAACATCTCCGCTCAGATCCTG ATCAAGAACACGCTGTACCTGCGGATCCTCCGACCACT TCAACAGTCAAGCTGAAGTCTTTCATCACGCAATACAG CACCGGACAGGTGAGCTGGAAATGAATGGGAGCTGCAG AAGGAAAACAGCAAGCGCTGGAACCCGAGATCCAGTACA CCTCCAACACTACAAATCTACAAGTGTGACTTTGTGT TAATACAGAAGGCGTGTACTCTGAACCCCGCCCATTTGGC ACCCGTTACCTCACCCGTAATCTGTAA
27	AAVrh13 VP3 capsid protein	MAAGGGAPMADNNEGADGVGNASGNWHCDSTWLGDVIT STRTWALPTYNNHLYKQISSQSGATNDNHFFGYSTPWGYF DFNRFHCHFSRWDQRLINNNWGFPRKLRFKLFNIQVKE VTNDGVTTIANNLSTIQVFSDEYQLPYVLGSAHQGCL PPFPADVFMIPQYGYLTLNNGSQSVGRSSFYCYLEYFSPQM LRTGNNEFSYTFEEVPHSSYAHSQLDRLMNPLIDQYL YYLARTQSTTGSTRELQPHQAGPNTMAEQSKNWLPGPCYR QORLSKNIDSNNSNFAWTGATKYHLNGRNSLTNPGVAMA TNKDEDDQFPFINGVLVFGETGAANKTTLENVLMTSEEEI

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		KTTNPVATEEYGVVSSNLQSSSTAGPQTQTVNSQGALPGMV WQNRDVYLGQPIWAKIPHTDGNFHPSPLMGGFGLKHPPPO ILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVEIEWE LQKENSKRWNPEIQYTSNYAKSNNVEFAVNNEGVYTEPRP IGTRYLTRNL
28	AAVrh13 VP3 capsid DNA	ATGGCTGCAGGCGGTGGCGCACCAATGGCAGACAATAACG AAGGCGCCGACGGAGTGGGTAATGCCTCCGGAAATTGGCA TTGCGATTCCACATGGCTGGGCGACAGAGTCATCACCACC AGCAGCCGACCTGGGCCCTGCCACCTACAACAACCACC TCTACAAGCAGATATCAAGTCAGAGCGGGGTACCAACGA CAACCACTTCTTCGGCTACAGCACCCCTCGGGGTATTTT GACTTCAACAGATTCCACTGCCACTTCTACCACGTGACT GGCAGCGACTCATCAACAACAACCTGGGGATTCCGGCCCAG AAAGCTGCGGTTCAAGTTGTTCAACATCCAGGTCAAGGAG GTCACGACGAACGACGGCGTTACGACCATCGCTAATAACC TTACCAGCAGATTCAAGTCTTCTCGGACTCGGAGTACCA ACTGCCGTACGTCTCTCGGCTCTGCGCACCAGGGTGCCTC CCTCCGTTCCCTGCGGACGTGTTGATGTTCTCTAGTACG GATATCTGACTCTAAACAACGGCAGTCAGTCTGTGGGACG TTCTCTCTTCTACTGCTGGAGTACTTCTCTCTCAGATG CTGAGAACGGGCAATAACTTTGAATTGAGTACACCTTTG AGGAAGTGCCTTTCCACAGCAGCTATGCGCACAGCCAGAG CCTGGACCGGCTGATGAATCCCCTCATCGACAGTACCTG TACTACCTGGCCCGGACCCAGAGCACTACGGGGTCCACAA GGGAGCTGCAGTTCATCAGGCTGGGCCCAACACCATGGC CGAGCAATCAAAGAACTGGCTGCCCGGACCTGTTATCGG CAGCAGAGACTGTCAAAAAACATAGACAGCAACAACAACA GTAACCTTGCTGGACCGGGGCCACTAAATACCATCTGAA TGGTAGAAATTCATTAAACCAACCCGGGCGTAGCCATGGCC ACCAACAGGACGACGAGGACCAAGTCTTTCCCATCAACG GAGTGTGGTTTTTGGCGAAACGGGGCTGCCAACAAGAC AACGCTGGAACAGTGTCTAATGACCAGCAGGAGGAGATC AAAACCAACCAATCCCGTGGCTACAGAAGAATACGGTGTGG TCTCCAGCAACCTGCAATCGTCTACGGCCGACCCAGAC ACAGACTGTCAACAGCCAGGGGGCTCTGCCCGGATGGTC TGGCAGAACCGGACGTGTACCTGCAGGGTCCCATCTGGG CCAAAATTCCTCACACGGACGGCAACTTTCACCCGTCTCC CCTGATGGGCGGATTTGGACTCAAACACCCGCCCTCTCAA ATTCTCATCAAAACACCCCGGTACCTGCTAATCTCCAG AGGTGTTTACTCCTGCCAAGTTTGCTCATTTATCACGCA GTACAGCACCGGCCAGGTGACGCTGGAGATCGATGGGAA CTGCAGAAAGAAACAGCAACGCTGGAATCCAGAGATT AGTACACCTCAAATTATGCCAAGTCTAATAATGTGGAATT TGCTGTCAACAACGAAGGGTTTATACTGAGCCTCGCCCC ATTGGCACCCGTTACCTCACCCGTAACCTGTAA
29	AAVhu32- AAVrh13 hybrid VP1 capsid protein	(AAVhu32 VP1u + AAVhu32 VP1/2s + AAVrh13 VP3) MAADGYLPDWLEDTLSEGIQWVKLPGPPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLYKNHADAEPQERLKEDTSFGNLRGAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKKLNFGQTGDTESVDPQPIGEPPAAPSGVGS LTMAAGGAPMADNNNEGADGVGNASGNWHCDSTWLGDRVI TTSTRTWALPTYNNHLYKQISSQSGATNDNHFFGYSTPWG YDFNRFHCHFSPRDWQLRINNNGFRPRKLRPKLFNIIQV KEVTTNDGVTTIANNLSTSIQVFSDEYQLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQSVGRSSFYCLEYFPS QMLRTGNNFESYTFEEVPHSSYAHQSLSLDRMLNPLIDQ YLYLARTQSTTGSTRELQFHQAGPNTMAEQSKNWLPGPC YRQQLRSKNIDSNNSNFAWTGATKYHLNGRNSLTNPQVA MATNKDDEDQFFPINGVLVFGETGAANKTTLENVLMTSEE EIKTTNPVATEEYGVVSSNLQSSSTAGPQTQTVNSQGALPG MVWQNRDVYLGQPIWAKIPHTDGNFHPSPLMGGFGLKHPP PQILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVEIE WELQKENSKRWNPEIQYTSNYAKSNNVEFAVNNEGVYTEP RPIGTRYLTRNL
30	AAVhu32- AAVrh13 hybrid VP1 capsid DNA	(AAVhu32 VP1u + AAVhu32 VP1/2s + AAVrh13 VP3) ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TGGCCCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCCTTTGGGGCAACCTCGGCGGAGCAGTCTCCAG GCCAAAAGAGGGCTTCTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCTTGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTGCAGGCGGTGGCGCACCAATGGCAGACA ATAACGAAGCGCCGACGGAGTGGGTAAATGCCTCCGGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCATC ACCACCAGCACCCGCACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAGATATCAAGTCAGAGCGGGGTAC CAACGACAACCACCTTCTTCGGCTACAGCACCCCTGGGGC TATTTTGACTTCAACAGATTCCACTGCCACTTCTACCAC GTGACTGGCAGCGACTCATCAACAACACTGGGATTCCG GCCCAGAAAGCTGCGGTTCAAGTTGTCAACATCCAGGTC AAGGAGGTACGACGAAACGACGGCGTTACGACCATCGCTA ATAACCTTACCAGCACGATTAGGTCTTCTCGGACTCGGA GTACCAACTGCCGTACGTCCTCGGCTCTGCGCACCGGGC TGCCCTCCCTCCGTTCCCTGCGGACGTGTTCAATGATTCTC AGTACGGATATCTGACTCTAAACAACGGCAGTCAGTCTGT GGGACGTTCCCTCTTCTACTGCCTGGAGTACTTTCCTTCT CAGATGCTGAGAACGGGCAATACTTTGAATTAGCTACA CCTTTGAGGAAGTGCCTTTCCACAGCAGCTATGCGCACAG CCAGAGCCTGGACCGGCTGATGAATCCCTCATCGACCAG TACCTGTACTACCTGGCCCGGACCCAGAGCACTACGGGGT CCACAAGGGAGCTGCAGTTCATCAGGCTGGGCCAACAC CATGGCCGAGCAATCAAAGAACTGGCTGCCCGGACCCGTGT TATCGGCAGCAGAGACTGTCAAAAAACATAGACAGCAACA ACAACAGTAACCTTTGCCGTGACCGGGGCCACTAATAACCA TCTGAATGGTAGAAATTCATTAACCAACCCGGCGTAGCC ATGGCCACCAACAAGGACGACGAGGACAGTTCCTTCCCA TCAACGGAGTGCTGGTTTTTGGCGAAACGGGGCTGCCAA CAAGACAACGCTGGAAAACGTGCTAATGACCAGCGAGGAG GAGATCAAAACCACCAATCCGTGGCTACAGAAGAAATACG GTGTGGTCTCCAGCAACCTGCAATCGTCTACGGCCGGACC CCAGACACAGACTGTCAACAGCCAGGGGGCTCTGCCCGGC ATGGTCTGGCAGAACCGGACGTGTACCTGCAGGGTCCCA TCTGGGCCAAAATTCCTCACCGGACGGCAACTTTCACCC GTCTCCCTGATGGCGGATTGGACTCAAACACCCCGCT CCTCAAATTCATCAAAAACACCCGGTACCTGCTAATC CTCCAGAGGTGTTTACTCTGCCAAGTTTGCCTCATTAT CACGCAGTACAGCACCGGCCAGGTACGCTGGAGATCGAG TGGGAACGTCAGAAAAGAAAACAGCAAACGCTGGAATCCAG AGATTAGTACACCTCAAATTATGCCAAGTCTAATAATGT GGAATTTGCTGTCAACAACGAAGGGTTTATACTGAGCCT CGCCCCATTGGCACCCGTTACCTCACCCGTAACTGTAA
31	AAVhu32 VP1 region protein	MAADGYLPDWLEDTLSEGIQWKKLKP PPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSPGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSONLGDRI TTSTRTWALPTYNNHLYKQISNSTSGSSNDNAYFGYSTP WGYFDNRFCHFSRDPWQRLINNNWGRPKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLEPPFPADVFMIPQYGYLTNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVFPFSSYHSQSLDRMLNPLI DQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVI TNEEEIKTNPVATESYGVATNHQSAQAQAQTGWVQNGG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		ILPGMVWQDRDVYLQGPWIWAKIPHTDGNFHPSPLMGGFGM KHPPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVS VEIEWELQKENSkrwnPEIQYTSNYYKSNNVEFAVNTGV YSEPRPIGTRYLTRNL
32	AAVhu32 VP1 region DNA	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAGCTCAAACC TGGCCCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTTCG GTGAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGA TTGGCATTGCGATTCCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGTCTTACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGTCGGCTCAC GAGGGCTGCCTCCCGCGGTTCACAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTGTCGCTTTTACTGCCTGGAATATTT CCGTCGCAAAATGCTAAGAACGGGTAAACAACCTCCAGTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTGAGCTGGCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACTGCTATGGCCAGCCACAAAGAAGGAGGAGACCGTT TCTTCTCTTGTCTGGATCTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATTAACACTAACCCTGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGCACAGGCGCAGACCGGCTGGGTTCAAACCAAGGA ATACTTCCGGGTATGGTTTGGCAGGACAGAGATGTGTACC TGCAAGGACCCATTGGGCCAAAATTCTCACACGGACGG CAACTTTCACCTTCTCCGCTAATGGGAGGTTTGGAAATG AAGCACCCGCTCTCAGATCTCATCAAAACACACCTG TACCTGCGGATCTTCAACGGCTTCAATAAGGACAAGCT GAACTCTTTCATCACCCAGTATTCTACTGGCCAAGTCAGC GTGGAGATTGAGTGGGAGCTGCAGAAGGAAACAGCAAGC GCTGGAACCCGAGATCCAGTACACTTCAACTATTACAA GTCTAATAATGTTGAATTTGCTGTTAATACGAAGGTGTA TATAGTGAACCCCGCCATTGGCACACAGATACCTGACTC GTAATCTGTAA
33	AAV6/AAV8 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEVPVNAADAALEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFO AKKRVLEPFLVEEGAKTAPGKKRPVEQSPQEPDSSSGIG KTGQOPAKKRLNFGQTGDSVSPDPQLGEPATPAAVGP TTMAAGGAPMADNNNEGADGVSSSGNWHCDSTWLGDRI TTSTRTWALPTYNNHLYKQISNGTSGGATNDNTYFGYSTP WGYFDNRFHCHFSRWDQRLINNNWGRPKRLSFKLFNI QVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSAH QGCLPPFPADVFMIPQYGYLTNNGSQAVGRSSFYCLEYF PSQMLRTGNNFQFTYTFEDVPFHSSYAHQSGLDRLMNPLI

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		DQYLYLSRTQTGGTANTQTGFSQGGPNTMANQAKNWL PGPCYRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLAN PGIAMATHKDDERFFPSNGILIFGKQNAARDNADYSVDM LTSEEEIKTTNPVATEEYGIIVADNLQQQNTAPQIGTVNSQ GALPGMVWQNRDVYLQGPWIWAKIPHTDGNFHPSPLMGGFG LKHPPQILIKNTVPADPPTTFNQSKLNSFITQYSTGQV SVEIEWELQKENSKRWNPEIQYTSNYKSTSVDFAVNTEG VYSEPRPIGTRYLTRNL
34	AAV6/AAV8 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTGCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGATGCAGCGGCCCTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAAGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCCGAGTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGAGGGTTCTCGAACCTTTGGTCTGGTTGAGG AAGGTGCTAAGACGGCTCCTGGAAGAAACGTCCGTTAGA GCAGTCGCCACAAGAGCCAGACTCCTCCTCGGGCATTGGC AAGACAGGCCAGCAGCCCGCTAAAGAGAGACTCAATTTTG GTCAGACTGGCGACTCAGAGTCAGTCCCCGACCCACAACC TCTCGGAGAACCTCCAGCAACCCCGCTGCTGTGGGACCT ACTACAATGGCTGCAGGCGGTGGCGCACCAATGGCAGACA ATAACGAAGGCGCCGACGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCACATGGCTGGGCGCAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAATCTCCAACGGGACATCGGGAGG AGCCACCAACGACAAACCTACTTCCGCTACAGCACCCCC TGGGGGTATTTTGACTTTAACAGATTCCACTGCCACTTTT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCCCAAGAGACTCAGCTTCAAGCTCTTCAACATC CAGGTCAAGGAGGTCACGCAGAATGAAGGCACCAAGACCA TCGCCAATAACCTCACCAGCACCATCCAGGTGTTTACGGA CTCGGAGTACCAGCTGCCGTACGTTCTCGGCTCTGCCAC CAGGGCTGCCTGCCCTCCGTTCGGCGGACGTGTTTATGA TTCCCCAGTACGGCTACCTAACACTCAACAACGGTAGTCA GGCCGTGGGACGCTCCTCCTTCTACTGCTGGAATACTTT CCTTCGCAGATGCTGAGAACCGGCAACAACCTCCAGTTTA CTTACACCTTCGAGGACGTGCCCTTCCACAGCAGCTACGC CCACAGCCAGAGCTTGGACCGGCTGATGAATCCTCTGATT GACCAGTACCTGTACTACTTGTCTCGACTCAAACAACAG GAGGCACGGCAAATACGCAGACTCTGGGCTTCAGCCAAGG TGGGCCTAATAACAATGGCCAATCAGGCAAGAACTGGCTG CCAGGACCTGTTACCGCCAACAACGCGTCTCAACGACAA CCGGGCAAAACAACAATAGCAACTTTCCTGGACTGCTGG GACCAAAATACCATCTGAATGGAAGAAATTCATTGGCTAAT CCTGGCATCGCTATGGCAACACAAAGACGACGAGGAGC GTTTTTTTCCCAGTAACGGGATCCTGATTTTGGCAAAACA AAATGCTGCCAGAGACAATGCGGATTACAGCGATGTCATG CTCACCAGCGAGGAAGAAATCAAACCACTAACCTGTGG CTACAGAGGAATACGGTATCGTGGCAGATAACTTGCAGCA GCAAAACACGGCTCCTCAAATTGGAACGTCAACAGCCAG GGGGCCTTACCGGTATGGTCTGGCAGAACCGGGACGTGT ACCTGCAGGGTCCCATCTGGGCCAAGATTCCTCACACGGA CGGCAACTTCCACCCGTCTCCGCTGATGGGCGGCTTTGGC CTGAAACATCCTCCGCTCAGATCCTGATCAAGAACACGC CTGTACCTGCGGATCCTCCGACCACCTTCAACAGTCAAA GCTGAACTCTTTCATCACGCAATACAGCACCGGACAGGTC AGCGTGGAATGGAATGGGAGCTGCAGAAGGAAAACAGCA AGCGCTGGAACCCGAGATCCAGTACACCTCCAACACTA CAAATCTACAAGTGTGGACTTGTGTGTTAATACAGAAGGC GTGTACTCTGAACCCCGCCCATTTGGCACCCGTTACCTCA CCCGTAATCTGTAA
35	AAV6/AAVrh. 21 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGFNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPVEPSPQSPDSSTGI GKKGQQPARKRLNFGQTGDSSESVDPQPLGEPPAAPSSVG SGTMAAGGGAPMADNNEGADGVGSSGNWHCDSTWLGDRV ITTSRTWALPTYNNHLYKQISNGTSGGATNDNTYFYGYST

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		PWGYFDENRFHCHFSPRDWQRLINNNWGFPRKRLSFKLFN IQVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSA HQGCLPPFPADVFMIPOYGYLTLLNNGSQAVGRSSFYCLEY FPSQMLRTGNNFQFTYTFEDVPHSSYAHQSGLDRLMNPL IDQYLYLSRTQTGGTANTQTLGFSQGGPNTMANQAKNW LPGPCYRQORVSTTTQNNNSNFAWTAGTKYHLNGRNSLA NPGIAMATHKDEERFFPSNGILIFGKQNAARDNADYSDV MLTSEEEIKTTNPVATEEYGIVADNLQQONTAPQIGTVNS QGALPGMVWQNRDVYLQGPWAKIPHDTGNFHPSPLMGGF GLKHPPQILIKNTVPVADPPTTFNQSKLNSFITQYSTGQ VSVEIEWELQKENSQRWNPEIQYTSNYYKSTSVDFAVNTE GVYSEPRPIGTRYLTRNL
36	AAV6/AAVrh. 21 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACCTGAAACC TGGAGCCCCGAAACCAAGCCAACCGACAAAAGCAGGAC GACGGCCGGGTCTGGTGCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGACGCAGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAAGCGGTGACAATCCGTACTGCGGTATA ATCAGCGCGACGCGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCCTGGAAGAAGAGACCGGTAGA GCCGTACCCACAGCGTTCCCCGACTCTCCACGGGCATC GGCAAGAAGGCCAGCAGCCCGCCAGAAAGAGACTCAATT TCGGTCAGACTGGCGACTCAGAGTCAGTCCCCGACCTCA ACCTCTGGGAGAACCTCCAGCAGCGCCCTCTAGTGTGGGA TCTGGTACAATGGCTGCAGGCGGTGGCGCACC AATGGCAG ACAATAACGAAGGCGCCGACGAGTGGGTAGTTCTCTCGG AAATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTC ATCACCCAGCAGCCGAACTGGGCGCTGCCACCTACA ACAACCACCTCTACAAGCAATCTCCAACGGGACATCGGG AGGAGCCACCAACGACACACCTACTTCGGCTACAGCACC CCCTGGGGGTATTTGACTTTAACAGATTCCTGCTGCTG TTTCAACAGTGTGCTGCGAGCTCATCAACAACAACCTG GGGATTCCGGCCCAAGAGACTCAGCTTCAAGCTCTTCAAC ATCCAGGTCAAGGAGGTACGCGAGAATGAAGGCACCAAGA CCATCGCCAATAACCTCACCAGCACCATCCAGGTGTTTAC GGACTCGGAGTACAGCTGCCGTACGTTCTCGGCTCTGCC CACCAGGGCTGCCTGCCCTCCGTTCCCGGCGGACGTGTTCA TGATTCCCCAGTACGGCTACCTAACACTCAACAACGGTAG TCAGGCGTGGGACGCTCCTCCTTCTACTGCTGGAATAC TTTCCTTCGCAGATGCTGAGAACC GGCAACAATTCAGT TTACTTACACCTTCGAGGACGTGCCTTTCACAGCAGCTA CGCCACAGCCAGAGCTTGAGCCGGCTGATGAATCCTCTG ATTGACCAGTACCTGTACTACTTGTCTCGGACTCAAACAA CAGGAGGCACGGCAAAATACGAGACTCTGGGCTTCAGCCA AGGTGGGCTAATAACAATGGCCAATCAGGCAAGAAGTGG CTGCCAGGACCTGTTACCGCCAACAACGCTCTCAACGA CAACCGGGCAAAAACAATAGCAACTTTGCTTGGACTGC TGGGACCAAATACCATCTGAATGGAAGAAATTCATTGGCT AATCCTGGCATCGCTATGGCAACACAAAGACGACGAGG AGCGTTTTTTTCCAGTAACGGGATCCTGATTTTGGCAA ACAAAATGCTGCCAGAGACAATGCGGATTACAGCGATGTC ATGCTCACCAGCGAGGAAGAAATCAAAACCACTAACCTG TGGCTACAGAGGAATACGGTATCGTGGCAGATAACTTGCA GCAGCAAAACACGGCTCCTCAAATGGAAGTGTCAACAGC CAGGGGGCTTACCCGGTATGGTCTGGCAGAACCGGGACG TGTAACCTGACAGGTCCTCTGCGGCCAAGATTCTCACAC GGACGGCAACTTCCACCGCTCTCCGCTGATGGCGGCTTT GGCCTGAAACATCCTCCGCTCAGATCCTGATCAAGAACA CGCCTGTACCTGCGGATCCTCCGACCACCTTCAACCAGTC AAAGCTGAACCTCTCATCACGCAATACAGCACC GGACAG GTACGCTGGAAATGAATGGGAGCTGCAGAAGGAAACA GCAAGCGCTGGAACCCGAGATCCAGTACACCTCCAAC TA CTACAAATCTACAAGTGTGGACTTGTGTTAATACAGAA GGCGTGACTCTGAACCCCGCCCATTTGGCACCGGTTACC TCACCCGTAATCTGTAA
37	AAV6/AAVrh. 73 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEPVNAADAAALEHDKAYD QQLKAGDNPYLRYNHDAEFQERLQEDTSFGNNGRAVFQ

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		AKKRVLEPFGLVEEGAKTAPGKKRPVEQSPQEPDSSSGIG KTGQQPAKKRLNFGQTGDSESVDPQPPLGEPATPAAVGP TTMAAGGGAPMADNNNEGADGVSSSGNWHCDSTWLGDRVI TTSTRTWALPTYNNHLYKQISNGTSGGSTNDNTYFGYSTP WGYFDNRFHCHFSRDRWQRLINNNWGFRPKRLSFKLFNI QVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSAH QGCLPPFPADVFMIPQYGYLTNNQSQAAGRSSFYCLEYF PSQMLRTGNNFSFSYTFEDVPFHSSYAHQSGLDRMLNPLI DQYLYLSRTQSTGGTAGTQQLLFSQAGPSNMSAQARNWL PGPCYRQQRVSTTLSQNNNSNFAWTGATKYHLNGRDSLNV PGVAMATNKDDEDRFFPSSGILMFGKQAGKDNVDYSNM LTSEEEIKTTNPVATEQYGVVADNLQRONTAPIVGAVNSQ GALPGMVWQNRDVLQGPWAKIPHTDGNFHPSPLMGGFG LKHPPPQILIKNTVPADPPTAFNQAKLNSFITQYSTGQV SVEIEWELQKENSKRWNPEIQYTSNYKSTNVDFAVNTEG VYSEPRPIGTRYLTRNL
38	AAV6/AAVrh.73 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTCTTCTGGCTACAGTACCTCG GACCCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGATGCAGCGGCCCTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAAGCGGGTGACAATCCGTACCTGCGGTATA ACCAAGCCGACGCGAGTTTCAAGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGAGGGTTCTCGAACCTTTGGTCTGGTTGAGG AAGGTGCTAAGACGGCTCCTGGAAGAAGACGTCGGTAGA GCAGTCGCCACAAGAGCCAGACTCCTCCTCGGGCATTGGC AAGACAGGCCAGCAGCCCGCTAAAAGAGACTCAATTTTG GTCAGACTGGCGACTCAGAGTCAGTCCCGACCCACAACC TCTCGGAGAACCTCCAGCAACCCCGCTGCTGTGGGACCT ACTACAATGGGTGACGGCGGTGGCGCACCAATGGCAGACA ATAACGAAGGTGCCGACGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAATCTCAACGGGACCTCGGGAGG CAGCACCACGACAACACCTACTTTGGCTACAGCACCCCT TGGGGGTATTTTGACTTTAAGAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGG ATTCCGGCCCAAGAGACTCAGCTTCAAGCTCTTCAACATC CAGGTCAAAGAGGTACGCGAGAATGAAGGCACCAAGACCA TCGCCAATAACCTCACCAGCACCATCCAGGTGTTACGGA CTCGGAATACCAGCTGCCGTACGTCTCTCGGCTCTGCCAC CAGGGCTGCCTGCCTCCGTTCCCGGGCGGACGCTCTCATGA TTCTCTAGTACGGCTACCTGACTCTCAACAACGGTAGTCA GGCGGTGGGACGTTCTCTCTACTGCCTGGAGTACTTC CCCTCTCAGATGCTGAGAACGGGCAACAATTTTCTCTCA GCTACACTTTTCGAGGACGTGCCTTTCCACAGCAGCTACGC GCACAGCCAGAGTTTGGACAGGCTGATGAATCCTCTCATC GACCAGTACCTGTACTACCTGTCAAGAACCAGTCTACGG GAGGCACAGCGGGAACCCAGCAGTTGCTGTTTTCTCAGGC CGGGCCTAGCAACATGTCGGCTCAGGCCAGAACTGGCTG CCTGGACCTGCTACAGACAGCAGCGCTCTCCACGACAC TGTCGCAAAACAACAACAGCAACTTTGCTGGACTGGTGC CACCAGTATCATCTGAACGGCAGAGACTCTCTGGTGAAT CCGGCGTCGCCATGGCAACCAACAAGGACGACGAGGACC GCTTCTTCCATCCAGCGCATCCTCATGTTTGGCAAGCA GGGAGCTGGAAAAGACAACGTGGACTATAGCAACGTGATG CTAACAGCGAGGAAGAAATCAAGACCACCAACCCCGTGG CCACAGAACAGTATGGCGTGGTGGCTGATAACCTACAGAG ACAAAACACCGCTCCTATTGTGGGGCGGTCAACAGCCAG GGAGCCTTACCTGGCATGGTCTGGCAGAACCGGACGTGT ACCTGCAGGGTCTATTGGGCCAAGATTCCTCACACAGA TGGCAACTTTCACCCGTCTCCTTAAATGGGCGGCTTTGGA CTTAAACATCCGCCTCTCAGATCCTCATCAAAAACACTC CTGTTCTGCGGATCCTCCAACAGCGTTCAACCAGGCCAA GCTGAATTCTTTCATACGCAGTACAGCACCGGACAAGTC AGCGTGGAGATCGAGTGGGAGCTGCAGAAGGAGAACAGCA AGCGCTGGAACCCAGAGATTAGTATACCTTCAACTACTA CAAATCTACAAATGTGACTTTGCTGTTAATACTGAGGGT GTTTACTCTGAGCCTCGCCCATGGCACTCGTTACCTCA CCCGTAATCTGTAA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
39	AAV8/AAV6 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEFVNAADAAAEHDKAYD QQLQAGDNPYLRYNHADAEFQERLQEDTSFGNLRGAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPFVEFSPQRSPTSSTGI GKKGQQPARKRLNFGQTDSESVDPQPLGEPPAAPSGVG PNTMASGGGAPMADNNEGADGVGNASGNWHCDSTWLGDRV ITSTRTWALPTYNNHLYKQISSASTGASNDNHFGYSTP WGYFDNRFHCHFSRDPWQRLINNNWGFRPKRLNFKLFNI QVKEVTNDGVTTIANNLSTVQVFSDEYQLPYVLGSAH QGCLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEYF PSQMLRTGNNFTFSYTFEDVPPHSSYAHQSOLDRLMNPLI DQYLYLNRTONQSGSAQNKDLLFSRGSFAGMSVQPKNWL PGPCYRQQRVSKTKTDNNNSNFTWTGASKYNLNGRESIIN PGTAMASHKDDKDFPMSGVMIFGKESAGASNTALDNVM ITDEEIKATNPVATERFGTVAVNLQSSSTD PATGDVHVM GALPGMVWQDRDVYLQGPWAKIPHTDGHFHPSPLMGGFG LKHPPQILIKNTPVPANPPAEFSATKFASFITQYSTGQV SVEIEWELQKENSKRWNPEVQYTSNYAKSANVDFTVDNNG LYTEPRPIGTRYLTRPL
40	AAV8/AAV6 VP1 dna	ATGGCTGCCGATGGTTATCTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGCGCTGAAACC TGGAGCCCCGAAGCCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTGCTTCCTGGCTACAAGTACCTCG GACCCCTCAACGGACTCGACAAGGGGAGCCCGCTCAACGC GGCGGACGCAGCGGCCCTCGAGCAGCACAAGCCCTACGAC CAGCAGCTGCAGGGGGTGACAATCCGTACCTGCGGTATA ACCAACGCCGACGCCGAGTTTCAGGAGCGTCTGCAAGAAGA TACGCTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCCTGGAAAGAAGAGACCGGTAGA GCCATCACCCAGCGTTCTCCAGACTCCTCTACGGGCATC GGCAAGAAGAAGCCAACAGCCCGCCAGAAAAAGACTCAATT TTGGTCAGACTGGCGACTCAGAGTCAGTTCAGACCCCTCA ACCTCTCGGAGAACCTCAGCAGCGCCCTCTGGTGTGGGA CCTAATAACAATGGCTTCAGGCGGTGGCGCACAATGGCAG ACAATAACGAAGGCCGCGACGGAGTGGGTAATGCCTCAGG AAATTGGCATTGGCATTCCACATGGCTGGGCGACAGAGTC ATCACACCAGCACCCGAACTGGGCCCTTGCCACCTATA ACAACCACCTCTACAAGCAATCTCCAGTGTCTCAACGGG GGCCAGCAACGACAACTACTTCCGCTACAGACCCCC TGGGGGTATTTTGATTTCAACAGATTCCACTGCCATTCT CACCACGTGACTGGCAGCGACTCATCAACAACAATTGGGG ATTCCGGCCCCAAGAGACTCAACTTCAAGCTCTTCAACATC CAAGTCAAGGAGGTCACGACGAATGATGGCGTCACGACCA TCGCTAATAACCTTACCAGCACGGTTCAAGTCTTCTCGGA CTCGGAGTACCAAGTTCGCGTACGTCCTCGGCTCTGCGCAC CAGGGCTGCCTCCCTCCGTTCGCGCGGACGTGTTTATGA TTCCGAGTACGGCTACCTAACGCTCAACAATGGCAGCCA GGCAGTGGGACGGTCATCTTTTACTGCCTGGAATATTTT CCATCGCAGATGCTGAGAACGGCAATAACTTTACCTTCA GCTACACCTTCGAGGACGTGCCCTTCCACAGCAGCTACGC GCACAGCCAGAGCCTGGACCGGCTGATGAATCCTCTCATC GACCAGTACCTGTATTACCTGAACAGAACTCAGAATCAGT CCGGAAGTGCCCAAAACAAGGACTTGCTGTTTAGCCGGGG GTCTCCAGCTGGCATGTCTGTTACGCCCAAAACTGGCTA CCTGGACCCTGTTACCGGCGAGCAGCGGTTTCTAAACAA AAACAGACAACAACAACAGCAACTTTACCTGGACTGGTGC TTCAAAATATAACCTTAATGGGCGTGAATCTATAATCAAC CCTGGCACTGCTATGGCCTCACACAAGACGACAAAGACA AGTTCTTTCCATGAGCGGTGTCATGATTTTGGAAAGGA GAGCGCCGGAGCTTCAAACTGCTATGGACAATGTCTATG ATCACAGACGAAGAGGAACTCAAAGCCACTAACCCCGTGG CCACCGAAAGATTTGGGACTGTGGCAGTCAATCTCCAGAG CAGCAGCACAGACCTGCGACCGGAGATGTGCTATGTTATG GGAGCCTTACCTGGAATGGTGTGGCAAGACAGAGACGTAT ACCTGCAGGGTCTTATTTGGGCAAAATCTCTCACACGGA TGGACACTTTCACCCGTCTCTCTCATGGGCGGCTTTGGA CTTAAGCACCCGCTCTCAGATCTCATCAAAACACGC CTGTTCTTGCAGTCTCTCCGGCAGAGTTTTCGGCTACAAA GTTTGTTCATTATCACCAGTATTCACAGGACAAGTG AGCGTGGAGATTGAATGGGAGCTGCAGAAAGAAACAGCA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		AACGCTGGAATCCCGAAGTGCAATACATCTAACTATGC AAAATCTGCCAACGTTGATTTCACTGTGGACAACAATGGA CTTTATACTGAGCCTCGCCCCATTGGCACCCGTTACCTCA CCCGTCCCCGTAA
41	AAV8/AAVrh. 21 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFNGLDKGEPVNAADAALEHDKAYD QQLQAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFO AKKRVLEPLGLVEEGAKTAPGKKRPVEFSPQSPDSSTGI GKKGQQPARKRLNFGQTDSESVDPQPLGEPAPASGVG PNTMAAGGGAPMADNNEGADGVGNASGNWHCDSTWLGDRV ITSTRTWALPTYNNHLYKQISSQSGATNDNHFFGYSTPW GYFDENRFHCHFSRDPWQRLINNNWGFPRKLRPKLFNIQ VKEVTNDGVTTIANNLSTIQVFSDEYQLPYVLGSAHQ GCLPPFPADVPMIPQYGYLTLNNGSQSVGRSSFYCLEYFP SQMLRTGNNFEFSYTFEEVPHFSSYAHSSQLDRLMNLID QYLYLARTQSTTGSTRELQPHQAGPNTMAEQSKNWLPGP CYRQORLSKNIDSNNNSNFAWTGATKYHLNGRNSLTNPGV AMATNKDDEDQFFPINGVLVFGKTGAANKTLENVMTSE EEIKTTNPVATEEYGVVSSNLQSSSTAGPQTQTVNSQGALP GMVWQNRDVYLQGPWAKI PHTDGNFHPSPLMGGFGLKHP PPQILIKNTVPANPPEVFTPAKFASFIQYSTGQVSVEI EWELQKENSKRWNPEIQYTSNYAKSNNVEFAVNNEGVTYE PRPIGTRYLTRNL
42	AAV8/AAVrh. 21 VP1 dna	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGCGCTGAAACC TGGAGCCCCGAAGCCAAAGCCAACCGACAAAAGCAGGAC GACGGCCGGGTCTGGTGCTTCCTGGCTACAAGTACCTCG GACCTCTCAACGGACTCGACAAGGGGAGCCCGCTCAACGC GGCGGACGCAGCGGCCCTCGAGCAGACAAGGCTACGAC CAGCAGCTGCAGGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCCGAGTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGCGAGCAGTCTTCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCTTGGAAAGAGAGACCGGTAGA GCCATCACCCAGCGTTCTCCAGACTCCTCTACGGGCATC GGCAAGAAGGCCAACAGCCCGCCAGAAAAAGACTCAATT TTGGTCAGACTGGCGGACTCAGAGTCAGTTCAGACCCCTCA ACCTCTCGGAGAACCTCCAGCAGCGCCCTCTGGTGTGGGA CCTAATAAATGGCTGCAGGCGGTGGCGCTCCAATGGCAG ACAATAACGAAGGCGCCGACGAGTGGGTAATGCCTCCGG AAATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTC ATCACCAACAGCACCCGACCTGGGCGCTGCCACCTACA ACAACCACTCTACAAGCAGATATCAAGTCAGAGCGGGGC TACCAACGACAACCACTTCTTCGGCTACAGCACCCCTGG GGCTATTTGACTTCAACAGATTCCACTGCCACTTCTCAC CACGTGACTGGCAGCGACTCATCAACAACACTGGGGATT CCGGCCAGAAAGCTGCGGTTCAAGTTGTTCAACATCCAG GTCAAGGAGGTACGACGAACGACGGCGTTACGACCATCG CCAATAACCTTACAGCAGGATTAGGTCTTCTCGGACTC GGAGTACCAACTGCCGTACGTCTCGGCTCTGCGCACCAG GGCTGCTCCCTCCGTTCCCTGCGGAGCTGTTCTATGATTC CTCAGTACGGATATCTGACTCTAAACAACGGCAGTCAGTC TGTGGGACGTTCTCTCTTACTGCTTGGAGTACTTTCCT TCTCAGATGCTGAGAACGGGCAATAACTTTGAATTCAGCT ACACCTTTGAGGAAGTGCTTTCCACAGCAGCTATGCGCA CAGCCAGAGCCTGGACCGGCTGATGAATCCCTCATCGAC CAGTACCTGTACTACCTGGCCCGGACCCAGAGCACTACGG GGTCCACAAGGGAGCTGCAGTTCATCAGGCTGGGCCCAA CACCATGGCCGAGCAATCAAAGAACTGGCTGCCCGGACCC TGTTATCGGCAGCAGAGACTGTCAAAAAACATAGACAGCA ACAACAACAGTAACCTTGCCTGGACCGGGCCACTAAATA CCATCTGAATGGTAGAATTCATTAACCAACCCGGGCGTA GCCATGGCCACCAACAAGGACGACGAGGACAGTCTTTTC CCATCAACGGAGTGCTGGTTTTTGGCAAAACGGGGGCTGC CAACAAGCAACGCTGGAAAACGTGCTAATGACAGCGAG GAGGAGATCAAAACCAACATCCCGTGGCTACAGAAGAAT ACGGTGTGGTCTCCAGCAACCTGCAATCGTCTACGGCCGG ACCCAGACACAGACTGTCAACAGCCAGGGGGCTCTGCCC GGCATGGTCTGGCAGAACCGGGAGCTGTACTGCAGGGTC CCATCTGGGCCAAAATTCCTCACAGGACGGCAACTTTCA CCCGTCTCCCCGTATGGGCGGATTTGGACTCAAACACCCG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		CCTCCTCAAATTCATCAAAACACCCCGTACCTGCTA ATCCTCCAGAGGTGTTTACTCCTGCCAAGTTTGCTCATT TATCACGCAGTACAGCACCGGCCAGGTACGCTGGAGATC GAGTGGGAAGTGCAGAAAGAAAACAGCAACGCTGGAATC CAGAGATTAGTACACCTCAAATTATGCCAAGTCTAATAA TGTGGAATTTGCTGTCAACAACGAAGGGTTTATACTGAG CCTCGCCCATTTGGCACCCGTACCTCACCCGTAACTGT AA
43	AAV8/AAVrh.73 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEVNAADAAAEHDKAYD QQLQAGDNPYLRYNHADAEPQERLQEDTSFGNNGRAVFO AKKRVLEPLGLVEEAKTAPGKKRPVEFSPQSPDSSGTI GKKGQQPARKRLNFGQTGDSSEVDPQPLGEPPAAPSGVG PNTMAAGGGAPMADNNEGADGVGSSGNWHCDSTWLGDV ITTSTRTWALPTYNNHLYKQISNGTSGGSTNDNTYFGYST PWGYFDNRFHCHFSRWDQRLINNNWGFPRKLSFKLFN IQVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSA HQGCLPPFPADVFMIPQYGYLTNNNGSQAVGRSSFYCLEY FPSQMLRTGNFNSFSYTFEDVPPHSSYAHQSGLDRLMNPL IDQYLYLSRTQSTGGTAGTQQLLFSQAGPSNMSAQARNW LPPGCPYRQORVSTTLSQNNNSNFAWTGATKYHLNGRDSL NPGVAMATNKDDEDRFPSSGILMFGKQAGKDNVDYSNV MLTSEEEIKTTNPVATEQYGVVADNLQRONTAPIVGAVNS QGALPGMVWQNRDVYLQGPWAKIPTHGDNFHPSPLMGGF GLKHPPPQILIKNTPVPADPPTAFNQAKLNSFITQYSTGQ VSVEIEWELQKENSQRWNPEIQYTSNYYSKTNVDFAVNTE GYSEPRPIGTRYLTRNL
44	AAV8/AAVrh.73 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGCGCTGAAACC TGGAGCCCCGAAGCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGCTCTGGTCTTCTGGCTACAGTACCTCG GACCTTCAACGAGCTCGACAAGGGGAGCCCGTCAACGC GCGGACGCGAGCGGCTCGAGCAGCACAAGGCCTACGAC CAGCAGCTGCAGGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCGCAGCGGAGTTTCAAGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCTTGAAGAAGAGACCGGTAGA GCCATCACCCAGCGTTCTCCAGACTCCTCTACGGGCATC GGCAAGAAAGGCCAACAGCCCGCAGAAAAAGACTCAATT TTGGTCAGACTGGCGACTCAGAGTCAGTTCCAGACCTCA ACCTCTCGGAGAACCTCCAGCAGCGCCTCTGGTGTGGGA CCTAATACAATGGCTGCAGGCGGTGGCGCACCAATGGCAG ACAATAACGAAGGTGCCGACGGAGTGGTAGTTCTCGGG AAATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTC ATCACCACAGCACCCGAACCTGGGCCTGCCCACCTACA ACAACCACCTCTACAAGCAAATCTCAACGGGACCTCGGG AGGCAGCACCAACGACAACCTACTTTGGCTACAGCACC CCTGGGGGTATTTGACTTTAACAGATTCCTGCTGCACT TCTCACACGTGACTGGCAGCGACTCATCAACAACAATG GGGATTCCGGCCCAAGAGACTCAGCTTCAAGCTCTTCAAC ATCCAGGTCAAAGAGGTACGCGAGAATGAAGGCACCAAGA CCATCGCCAATAACCTCACAGCACCATCCAGGTGTTTAC GGACTCGGAATACAGCTGCCGTACGCTCTCGGCTCTGCC CACCAGGCTGCTGCTCCGTTCCCGGCGGACGTCTTCA TGATTCTCAGTACGGCTACCTGACTCTCAACAACGGTAG TCAGGCGGTGGGACGTTCTCTCTTACTGCTTGGAGTAC TTCCCTCTCAGATGCTGAGAACGGGCAACAACCTTTCT TCAGCTACACTTTGAGGACGTGCCTTTCCACAGCAGCTA CGCGCACAGCCAGAGTTTGGACAGGCTGATGAATCCTCTC ATCGACCAGTACCTGTACTACCTGTCAAGAACCAGTCTA CGGAGGCACAGCGGGAACCCAGCAGTTGCTGTTTTCTCA GGCGGGGCTAGCAACATGTCGGCTCAGGCCAGAACTGG CTGCTGGACCTGCTACAGACAGCAGCGCTCTCCACGA CACTGTGCAAAAACAACAGCAACTTTGCTTGGACTGG TGCCACCAAGTATCATCTGAACGGCAGAGACTCTCTGGTG AATCCGGGCGTGGCCATGGCAACCAAGGACGACGAGG ACCGCTTCTTCCATCCAGCGGCATCTCATGTTTGCA GCAGGAGCTGGAAAGACAACGTGGACTATAGCAACGTG ATGCTAACCGAGGAGGAAGAAATCAAGACCACCAACCCG TGGCCACAGAACAGTATGGCGTGGTGGCTGATAACCTACA

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45	AAV9/AAV4 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQD NARGLVLPGYKYLPGFNGLDKGEVNAADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTGDTESVDPQPIGEPPAAPSGVGS LTMRAAGGAAVEGGQGDGVGNASGDWHCDSTWSEGHVT TTSTRTWLPTYNHLYKRLGESLQSNLYNGFSTPWGYFD FNRPHCHFSRDPWQRLINNNWGMRPKAMRVKIFNIQVKEV TTSNGETTVANNLTSTVQIFADSSYELPYVMDAGQEGSLP PFPNDVFMVPQYGYCGLVTGNTSQQQTDNRNIFYCLEYFPS QMLRTGNNFETYSFEKVPFHSYAHQSGLDRLMNLIDQ YLWGLQSTTTGTTLNAGTATTNFTKLRPNTFSNFKKNWLP GPSIKQQGFSKTANQNYKIPATGSDSLIKYETHSTLDGRW SALTPGPPMATAGPADSKFSNSQLIFAGPKQNGNTATVPG TLIPTSEEEAATNATDTDMWGNLPGGDQSNLPTVDRL TALGAVPGMVWQNRDIYYQGPWAKIPHTDGHFHPSPILIG GFGHKPPQPIFINKTPVPANPATFSTPVSFITQYST GOVSVQIDWEIQKERSKRWNPEVQFTSNYGQNSLLWAPD AAGKYTEPRAIGTRYLTHHL
46	AAV9/AAV4 VP1 dna	ATGGCTGCCGATGGTTATCTCCAGATTGGCTCGAGGACA ACCTTAGTGAAGGAATTCGCGAGTGGTGGCTTTGAAACC TGGAGCCCTCAACCAAGGCAATCAACAACATCAAGAC AACGCTCGAGGTCTTGTGCTTCCGGGTACAAATACCTTG GACCCGGCAACGGACTCGACAAGGGGGAGCCGCTCAACGC AGCAGACGCGGCGGCCCTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCGGAGACAACCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTGACAGCCCGCTAAAAGAGACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCAGACCCCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGCGTGACAGAGTGGCGGAGCTGCAGTCGAGG GCGGACAAGGTGCCGATGGAGTGGGTAATGCCTCGGGTGA TTGGCATTGCGATTCCACCTGGTCTGAGGGCCACGTACG ACCACCAGCACCAGAACCTGGGTCTTGCCCACTACAACA ACCACCTCTACAAGCGACTCGGAGAGAGCCTGCAGTCCAA CACCTACAACGGATTCTCCACCCCTGGGGATACTTTGAC TTCAACCGCTTCCACTGCCACTTCTCACACGTGACTGGC AGCGACTCATCAACAACAACCTGGGGCATGCGACCCAAAGC CATGCGGGTCAAAATCTTCAACATCCAGGTCAAGGAGGTCT ACGACGTGAAACGGCGAGACAACGGTGGCTAATAACCTTA CCAGCACGGTTCAGATCTTTGCGGACTCGTCGTACGAACT GCCGTACGTGATGGATGCGGGTCAAGAGGGCAGCCTGCCT CCTTTTCCCAACGAGCTTTATGGTGCCCCAGTACGGCT ACTGTGGACTGGTGACCGGCAACACTTCGCAGCAACAGAC TGACAGAAATGCCCTTCTACTGCCTGGAGTACTTTCTTCG CAGATGCTGCGGACTGGCAACAACCTTTGAAATTACGTACA GTTTTGAGAAGGTGCCCTTCCACTCGATGTACGCGCACAG CCAGAGCCTGGACCGGCTGATGAACCTCTCATCGACCAG TACCTGTGGGACTGCAATCGACACCACCGGAACCAACC TGAATGCCGGACTGCCACCACCAACTTACCAAGCTGCG GCCTACCAACTTTTCCAACCTTAAAAAGAACTGGCTGCCC GGGCCTTCAATCAAGCAGCAGGGCTTCTCAAGACTGCCA ATCAAAATACAAGATCCCTGCCACCGGGTCAGACAGTCT CATCAAAATACGAGACGCACAGCACTTGGACGGAAGATGG AGTGCCCTGACCCCGGACCTCCAATGGCCACGGCTGGAC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
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47	AAV9/AAV8 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQD NARGLVLPGYKYLGPNGLDKGEVPVNAADAALEHDKAYD QQLKAGDNPYLYKNHADAERQERLKEDTSFGGNLGRAVFQ AKRRLLEPLGLVEEAATAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTGDTESVDPQPIGEPPAAPSGVGS LTMAAGGAPMADNNEGADGVGSSSGNWHCDSTWLGDRI TTSTRTWALPTYNNHLYKQISNGTSGGATNDNTYFGYSTP WGYFDNRFHCHFSRDPQRLINNNGWFRPKRLSFKLENI QVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSAH QGCLPPFPADVFMIPQYGYLTNNNGSQAVERSSFYCLEYF PSQMLRTGNFQFTYTFEDVPFHSSYAHSQSLDRLMNLPI DQYLYLSRTQTTGGTANTQTLGFSQGGPNTMANQAKNWL PGPCYRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLAN PGIAMATHKDDERFFPSNGLIFGKQNAARDNADYSDVM LTSEEEIKTTNPVATEEYGIADNLQQQNTAPQIGTVNSQ GALPGMVWQNRDVLQGGPIWAKIPHDTGNFHPSPLMGGFG LKHPPQILIKNTVPVADPPTTFNQSKLNSFITQYSTGQV SVEIEWELQKENSKRWNPEIQYTSNYYKSTSVDFAVNTEG VYSEPRPIGTRYLTRNL
48	AAV9/AAV8 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTTAGTGAAGGAATTCGCGAGTGGTGGGCTTTGAAACC TGGAGCCCTCAACCAAGGCAAAATCAACAACATCAAGAC AACGCTCGAGGTCTTGTGCTTCCGGGTTACAAATACCTTG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACTCAAGTACA ACCACGCCGACGCCGAGTTCAGAGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGCGCTGTAGA GCAGTCTCTCAGGAACCGACTCCTCCGCGGTATTGGC AAATCGGTGCACAGCCCGTAAAAAGAGACTCAATTTCTG GTCAGACTGGCGACACAGAGTCAGTCCAGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTGCAGGCGGTGGCGCACCAATGGCAGACA ATAACGAAGGCGCGACGGAGTGGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAATCTCCAACGGGACATCGGGAGG AGCCACCAACGACAACACCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTTAACAGATTCCACTGCCACTTTT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGGG ATTCGGCCCCAAGAGACTCAGCTTCAAGCTCTTCAACATC CAGGTCAAGGAGGTACGCGAGAATGAAGGCACCAAGACCA TCGCCAATAACCTCACAGCACCATCCAGGTGTTTACGGA CTCGGAGTACCAGCTGCCGTACGTTCTCGGCTCTGCCAC CAGGGCTGCCTGCCTCCGTTCGGCGGACGTGTTATGA TTCCCAGTACGGCTACCTAACACTCAACAACGGTAGTCA GGCGGTGGGACGCTCCTCCTTCTACTGCCTGGAATACTTT CCTTCGCAGATGCTGAGAACCGGCAACAACCTCCAGTTTA CTTACACCTTCGAGGACGTGCCCTTTCACAGCAGCTACGC CCACAGCCAGAGCTTGGACCGGCTGATGAATCCTCTGATT GACCAGTACCTGTACTACTTGTCTCGGACTCAACAACAG GAGGCACGGCAAAATACGAGACTCTGGGCTTCAGCCAAGG TGGGCTAATACAATGGCCAATCAGGCAAGAACTGGCTG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
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49	AAV9/AAVrh18 VP1 aa	MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQD NARGLVLPGYKYLPGNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKRLLLEPLGLVEEAAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTGDTESVPDPQPIGEPPAAPSGVGS LTMAAGGAPMADNNEGADGVSSSGNWHCDSTWLGDRV I TTSTRTWALPTYNNHLYKQISNGTSGGSTNDNTYFGYSTP WGYPDFNRFHCHFSRWDQRLINNNWGFRPKRLNFKLFNI QVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSAH QGCLPPFPADVFMIPQYGYLTNNNGSQAVGRSSPYCLEYF PSQMLRTGNNFEFSYQFEDVPFHSSYAHQSGLDRLMNPLI DQYLYYLSRTQSTGGTAGTQQLLFSQAGPNNMSAQAKNWL PGPCYRQQRVSTTLSQNNNSNFAWTGATKYHLNGRDSL VN PGVAMATHKDDERFFPSSGVLMPGKQAGKDNVDYSVM LTSEEEIKTTNPVATEQYGVVADNLQQQNAAPIVGAVNSQ GALPGMVWQNRDVYLQGPWAKIPHDTGDNFHPSPLMGGFG LKHPPPPQILIKNTPVPADPPTTFSQAKLASFITQYSTGQV SVEIEWELQKENSKRWNPEIQYTSNYKSTNVDFAVNTEG TYSEPRPIGTRYLTRNL
50	AAV9/AAVrh18 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTTAGTGAAGGAATTGCGAGTGGTGGGCTTTGAAACC TGGAGCCCCCTCAACCAAGGCAAAACAACAATCAAGAC AACGCTCGAGGTCTTGTGCTTCCGGGTACAAAACCTTG GACCCGGCAACGGAATCGACAAGGGGAGCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAGAAGA TACGTCTTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTGACAGCCCGCTAAAAGAGACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCAGACCCCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGGTGACAGCGGTGGCGCTCCAATGGCAGACA ATAACGAAGGCGCCGACGAGTGGGTAGTTCTCAGGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCATC ACCACGACACCCGAACCTGGGCCCTCCCCACCTACAACA ACCACCTCTACAAGCAAATCTCAACGGGACATCGGGAGG AAGACCAACGACAACACCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTTAAGAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCGGGCCAAGAGACTCAACTTCAAGCTCTTCAACATC CAGGTCAAGGAGGTACGCGAGAATGAAGGCACCAAGACCA TCGCCAATAACCTTACAGCAGGATTAGGTCTTTACGGA CTCGGAATACAGCTCCCGTACGTCTCGGCTCTGCGCAC CAGGGCTGCCTGCCTCCGTTCCCGGGGACGTCTTCATGA TTCTCAGTACGGGTACCTGACTCTGAACAACGGCAGTCA GGCCGTGGGCGTCTCTCTTCTACTGCCTGGAGTACTTT CCTTCTCAAATGCTGAGAACGGGCAACAACCTTGAGTTCA GCTACCAGTTTGAGGACGTGCCTTTTCACAGCAGCTACGC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
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51	AAVrh21/AAV8 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEVNEADAALHDKAYD KQLEQGDNPYLKYNHADAEPQERLQEDTSFGNLRGAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPIESPDSTGIGRKGQ QPAKKLNFQGTGDSSEVPDPQPIGEPFAGPSGLGSGTMA AGGGAPMADNNEGADGVSSGNWHCDSTWLGDVITTS RTWALPTYNNHLYKQISNGTSGGATNDNTYFGYSTPWGYF DFNRFHCHFSPRDWQRLINNNGWFRPKRLSKFLFNIVKE VTQNEGTKTIANNLSTSIQVFTDSEYQLPYVLGSAHQGCL PPFPADVFMIPQGYLTLNNGSQAVGRSSFYCLEYFPSQM LRTGNNFQFTYTFEDVPFHSSYAHSSQLDRLMNPLIDQYL YYLSRTQTGGTANTQTLGFSQGGPNTMANQAKNWLPGPC YRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLANPGIA MATHKDDERFFPSNGILIFGKQNAARDNADYSDVMLTSE EEIKTTNPVATEEYGVADNLQQQNTAPQIGTVNSQGALP GMVWQNRDVYLGQPIWAKIPHTDGNFHPSPLMGGFGLKHP PPQILIKNTPVPADPPTTFNQSKLNSFITQYSTGQVSVEI EWELQKENSkrwnPEIQYTSNYYKSTSVDFAVNTEGVYSE PRPIGTRYLTRNL
52	AAVrh21/AAV8 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGTCTGGTGTCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGAGAGCCGGTCAACGA GGCAGACGCCCGGCCCTCGAGCAGCACAAGGCTACGAC AAGCAGCTCGAGCAGGGGGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTTCAGGAGCGTCTTCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCCTGGAAAGAAGAGACCCATAGA ATCCCCGACTCCTCCACGGGCATCGGCAGGAAAGGCCAG CAGCCCCGCTAAAAAGAAGCTCAACTTTGGGCAGACTGGCG ACTCAGAGTCAGTGCCCGACCTCAACCAATCGGAGAACCC CCCCGAGGCCCTCTGGTCTGGGATCTGGTACAATGGCT GCAGGCGGTGGCGCACCATGGCAGACAATAACGAAGGCG CCGACGGAGTGGGTAGTTCCTCGGGAAATTGGCATTGCGA TTCCACATGGCTGGGCGACAGAGTCATCACCACAGCACC CGAACCTGGGCCCTGCCACCTACAACAACCACTCTACA AGCAAATCTCCAACGGGACATCGGGAGGAGCCACCAACGA CAACACCTACTTCGGCTACAGCACCCCTGGGGGTATTTT GACTTTAACAGATTCCACTGCCACTTTTACCACGTGACT GGCAGCGACTCATCAACAACAACTGGGGATTCCGGCCCAA GAGACTCAGCTTCAAGCTCTTCAACATCCAGGTCAAGGAG GTCACGCAGAATGAAGGCACCAAGCATCGCCAATAACC TCACCAGCACCATCCAGGTGTTTACGGACTCGGAGTACCA GCTGCCGTACGTTCTCGGCTCTGCCACCAGGGCTGCCTG CCTCCGTTCCCGGCGGACGTGTTATGATTTCCCACTAGC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GCTACCTAACACTCAACAACGGTAGTCAGGCCGTGGGACG CTCTCTCTTCTACTGCCTGGAATACTTCTCTTCGAGATG CTGAGAACCGGCAACAACCTCCAGTTTACTTACACCTTCG AGGACGTGCCTTCCACAGCAGCTACGCCACAGCCAGAG CTTGGACCGGCTGATGAATCCTCTGATTGACCACTACCTG TACTACTTGTCTCGGACTCAACAACAGGAGGCACGGCAA ATACGCAGACTCTGGGCTTCAGCCAAGGTGGGCCTAATAC AATGGCCAATCAGGCAAGAAGTGGTGCCAGGACCTGT TACCGCCAACAACGCGTCTCAACGACAACCGGGCAAAACA ACAATAGCAACTTTGCCTGGAAGTGTGGGACCAATACCA TCTGAATGGAAGAAATTCATTGGCTAATCCTGGCATCGCT ATGGCAACACACAAGACGACGAGGAGCGTTTTTTTCCCA GTAACGGGATCCTGATTTTTGGCAACAATAAGTGTCCAG AGACAATGCGGATTACAGCGATGTCATGCTCACCAGCGAG GAAGAAATCAAAACCACTAACCTGTGGCTACAGAGGAAT ACGGTATCGTGGCAGATAACTTGCAGCAGCAAAACAGGC TCCTCAAATTGGAAGTGTCAACAGCCAGGGGCTTACCC GGTATGGTCTGGCAGAACCGGGACGTGTACCTGCAGGCTC CCATCTGGGCCAAGATTCTCACAGCGACGCAACTTCCA CCCGTCTCCGCTGATGGGCGGCTTTGGCCTGAAACATCCT CCGCCTCAGATCCTGATCAAGAACACGCTGTACCTGCGG ATCCTCCGACCACCTTCAACCACTCAAGCTGAAGTCTTT CATCAGCAATACAGCACCGGACAGGTCAGCGTGGAAAT GAATGGGAGCTGCAGAGGAAACAGCAAGCGCTGGAACC CCGAGATCCAGTACACCTCCAAGTACTACAATCTACAAG TGTGGACTTTGCTGTTAATACAGAAGCGGTACTCTGAA CCCCGCCCATTTGGCACCCGTACCTCACCCTAATCTGT AA
53	AAVrh73/AAV8 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPYKYLGPFGNLDKGEVNAADAALAEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPVEFSPQRSDDSTGI GKKGQQPARKRLNFGQTDSESVDPQPLGEPPAAPSSVG SGTMAAGGAPMADNNEGADGVGSSSGNWHCDSTWLGDRV ITTSTRTWALPTYNNHLYKQISNGTSGGATNDNTYFGYST PWGYFDNRFHCHFSPRDQRLINNNWGFPRKLSFKLFN IQVKEVTQNEGKTIANNLTSTIQVFTDSEYQLPYVLGSA HQGCLPPFPADVFMI PQYGYLTLNNGSQAVGRSSFYCLEY FPSQMLRTGNFQFTYTFEDVPHSSYAHSSQLDRLMNP IDQYLYLSRTQTGTANTQTGLFSGGGPNTMANQAKNW LPPGCPYRQVRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLA NPGIAMATHKDEERFFPSNGILIFGKQNAARDNADYSDV MLTSEEEIKTTNPVATEEYGVADNLQQNTAPQIGTVNS QGALPGMVQNRDVYLQGPWAKIPHTDGNFHPSPLMGGF GLKHPPQILIKNTFPVPADPPTTFNQSKLNSFITQYSTGQ VSVIEWELQKENSQRWNPFIQYTSNYYKSTSVDFAVNTE GVYSEPRPIGTRYLTRNL
54	AAVrh73/AAV8 VP1 dna	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACCTGAAACC TGGAGCCCCGAAACCAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGTCTGGTGCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GCGGACGCAGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAAGCGGGTGACAATCCGTACCTGCGGTATA ATCAGCCGACGCGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGCGAGCAGTCTTCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCTGGAAGAAGAGACCGGTAGA GCCGTACCAACAGCGTTCCCCGACTCTCCACGGGCATC GGCAAGAAAGGCCAGCAGCCCGCCAGAAAGAGACTCAATT TCGGTACAGCTGGGACTCAGAGTCAGTCCCGACCCCTCA ACCTCTGGGAGAACCTCCAGCAGCGCCCTCTAGTGTGGGA TCTGGTACAATGGCTGAGGCGGTGGCGCACCAATGGCAG ACAATAACGAAGGCCCGACGAGTGGGTAGTTCCTCGGG AAATTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTC ATCACCACGACACCGAACCTGGGCCCTGCCACCTACA ACAACCACCTCTACAAGCAAATCTCCAACGGGACATCGGG AGGAGCCACCAACGACACACCTACTTCGGCTACAGCACC CCTGGGGGTATTTTGAAGTTTAAACAGATTCACCTGCCACT TTTACACCGTGAAGTGGCAGCGACTCATCAACAACAACG GGGATTCGGGCCAAGAGACTCAGCTTCAAGCTTCAAC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
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55	AAV3B/AAV8 VP1 aa	MAAGGGAPMADNNEGADGVSSSNWCHDSTWLGDVRVIT STRTWALPTYNNHLYKQISNGTSGGATNDNTYFYSTPWG YDFNRFHCHFSRWDQRLINNNWGFRRKLSFKLFNIQV KEVTQNEGKTIANNLTSTIQVFTDSYQLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEYFPS QMLRTGNNFQFTYTFEDVPPHSSYAHQSQSLDRLMNLIDQ YLYLSRTQTTGGTANTQTLGFSQGGPNTMANQAKNWLPG PCYRQQRVSTTTGQNNNSNFAWTAGTKYHLNGRNSLANPG IAMATHKDDEERFFPSNGILIFGKQNAARDNADYSDVMLT SEEEIKTTNPVATEEYGVADNLQQNTAPQIGTVNSQGA LPGMVWQNRDVYLGPIWAKIPHDTGNFHPSPLMGGFGLK HPPPQILIKNTPVPADPPTTFNQSKLNSFIQYSTGQVSV EIEWELQKENSKRWNPEIQYTSNYYKSTSVDFAVNTEGVY SEPRPIGTRYLTRNL
56	AAV3B/AAV8 VP1 dna	ATGGCTGCAGGCGGTGGCGCACCAATGGCAGACAATAACG AAGGCGCCGACGGAGTGGGTAGTTCCTCGGAAATTGGCA TTGCGATTCCACATGGCTGGGCGACAGAGTCATCACCACC AGCACCCGAACCTGGGCCCTGCCACCTACAACAACCACC TCTACAAGCAAATCTCCAACGGGACATCGGAGGAGCCAC CAACGACAACACCTACTTCGGCTACAGCACCCCTGGGGG TATTTTGACTTTAACAGATTCCACTGCCACTTTTACCAC GTGACTGGCAGCGACTCATCAACAACAACCTGGGATTCCG GCCCAAGAGACTCAGCTTCAAGCTCTCAACATCCAGGTC AAGGAGGTCACGCGAATGAAGGCACCAAGACCATCGCCA ATAACCTCACCAGCACCATCCAGGTGTTTACGGAATCGGA GTACCAGCTGCCGTACGTTCTCGGCTCTGCCACCAGGGC TGCCTGCCCTCCGTTCCCGGCGGACGTGTCATGATTCCCC AGTACGGCTACCTAACACTCAACAACGGTAGTCAGGCCGT GGGACGCTCCTCCTTCTACTGCCTGGAATACTTTCCTTCG CAGATGCTGAGAACCGGCAACAACCTCCAGTTTACTTACA CCTTCGAGGACGTGCCTTTCACAGCAGCTACGCCACAG CCAGAGCTTGGACCGGCTGATGAATCCTCTGATTGACCAG TACCTGTACTACTTGTCTCGGACTCAACAACAGGAGGCA CGGCAATACGAGACTCTGGGCTTCAGCCAAGGTGGGCC TAATACAATGGCCAATCAGGCAAGAACTGGCTGCCAGGA CCCTGTTACCGCCAACAACGCGTCTCAACGACAACCGGGC AAAACAACAATAGCAACTTTGCCCTGGACTGCTGGACCAA ATACCATCTGAATGGAAGAAATTCATTGGCTAATCCTGGC ATCGCTATGGCAACACAAAGACGACGAGGAGCGTTTTT

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57	AAVrh15/ AAVhu32 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFGNLDKGEPVNEADAALEHDKAYD KQLEQGDNPYLKYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPIESPDSSGTIGKKGQ QPAKKKLNFGQTGDSSEVPDQPIGEPAPGPSGLSGTMA SGGGAPVADNNEGADGVSSSGNWHCDSQWLGRVITTS RTWALPTYNNHLYKQISNSTSGSSNDNAYFGYSTPWGYF DFNRFHCHFSPRDWQRLINNNGWFRPKRLNFKLPNIQVKE VTDNNGVKTIANNLTSTVQVFTDSYQLPYVLGSAHEGCL PFPFADVFMI PQYGYLTLDGSAVGRSSFYCLEYFSPQM LRTGNNFQFSYEFENVPHSSYAHSQSLDRLMNPIDQYL YYLSKTINGSGQNQQT LKFSVAGPSNMAVQGRNYIPGPSY RQQRVSTVTQNNNSEFAWPASSWALNGRNSLMNPGPAM ASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEE EIKTTNPVATESYGVATNHQSAQAQATGWVQNQGI LPG MVWQDRDVLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPP PQILIKNTVPVADPPTAFNKDKLNSFITQYSTGQVSEIE WELQKENSKRWNPEIQYTSNYYKSNNVFAVNTBGVYSEP RPIGTRYLTRNL
58	AAVrh15/ AAVhu32 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCAAGCAACCAAGCAAGCAAGCAAGCA GACGGCCGGGTCTGGTCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGAGAGCCGGTCAACGA GGCAGACGCCGCGGCCCTCGAGCACGACAAGGCTACGAC AAGCAGCTCGAGCAGGGGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTT CAGGAGCGTCTTCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCAGCAGTCTTCCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGGCGCTAAGACGGCTCTTGGAAAGAGAGCCATAGA ATCCCCGACTCCTCCACGGGCATCGGCAAGAAAGGCAG CAGCCCGCTAAAAAGAAGCTCAACTTTGGGCAGACTGGCG ACTCAGAGTCAGTGCCCGACCTCAACCAATCGGAGAACC CCCCAGGCCCCCTCTGGTCTGGGATCTGGTACAATGGCT TCAGGTGGTGGCGCACCAGTGGCAGACAATAACGAAGGTG CCGATGGAGTGGGTAGTTCCTCGGGAATTTGGCATTGCGA TTCCAATGGCTGGGGGACAGAGTCATCACCACGACACC CGAACCTGGGCCCTGCCACCTACAACAATCACCTCTACA AGCAAATCTCCAACAGCACATCTGGAGGATCTTCAAATGA CAACGCCTACTTCCGCTACAGCACCCCTGGGGGTATTT GACTTCAACAGATTCCACTGCCACTTCTCACCACGTGACT GGCAGCGACTCATCAACAACAACCTGGGGATTCCGGCCTAA GCGACTCAACTTCAAGCTCTTCAACATTGAGTCAAGAG GTTACGGACAACAATGGAGTCAAGACCATCGCCAATAACC TTACCAGCACGGTCCAGGTCTTCACGGACTCAGACTATCA GCTCCCGTACGTGCTCGGGTCGGCTCACGAGGGCTGCCTC CCGCCGTTCCAGCGGACGTTTTCATGATTCTCAGTACG GGTATCTGACGCTTAATGATGGGAGCCAGGCGGTGGGTG TTCGTCCTTTTACTGCTGGAATATTTCCTCGCAAATG CTAAGAACGGGTAACAACCTCCAGTTGAGTACGAGTTTG AGAACGTACCTTTCCATAGCAGCTACGCTCACAGCCAAAG CCTGGACCGACTAATGAATCCACTCATCGACCAATACTTG TACTATCTCTCAAAGACTATTAACGGTTCTGGACAGAATC AACAAACGCTAAAATTGAGCGTGGCCGGACCCAGCAACAT GGCTGTCCAGGAAGAACTACATACCTGGACCCAGCTAC

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59	AAVrh13/ AAVhu32 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPYKYLGFNFGLDKGEPVNEADAALHDKAYD KQLEQGDNPYLKYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPLGLVEEGAKTAPGKKRPIESPDSSTGIGKKGO QPAKKLNFQGTGDSSEVPDPQPLGEPAPAPSGLSGTMA SGGGAPVADNNEGADGVSSSGNWHCDSQLGDRVITST RTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTPWGYF DFNRHFCHFSPRDWQRLINNHWGFRPKRLNFKLFNIQVKE VTDNNGVKTIANNLTSTVQVFTDSYQLPYVLGSAHEGCL PPFPADVFMIPQYGYLTNDGSQAVGRSSFYCLEYFSPQM LRTGNNFQFSYEFENVPHSSYAHSQLDRLMNP LIDQYL YYLSKTINGSGQNQQT LKFSVAGPSNMAVQGRNYIPGPSY RQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGPAM ASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEE EIKTTPNVATESYGVATNHQSAQAQATGWVQNGILPG MVWQDRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPP PQILIKNTVPVADPPTAFNKDKLNSFITQYSTGVSVIEIE WELQKENSKRWNPEIQYTSNYYKSNNVFAVNTBGVYSEP RPIGTRYLTRNL
60	AAVrh13/ AAVhu32 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCAAGCCAACCAAGCAAGGACGAGGAC GACGGCCGGGGTCTGGTGCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGAGAGCCGGTCAACGA GGCAGACGCGCGGCCCTCGAGCACGACAAGGCTACGAC AAGCAGCTCGAGCAGGGGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTTCAGGAGCGTCTTCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCAG GCCAAGAAGCGGGTTCTCGAACCTCTCGGTCTGGTTGAGG AAGCGCTAAGACGGCTCTTGAAAGAAGAGACCCATAGA ATCCCCGACTCCTCCACGGGCATCGGCAAGAAAGGCCAG CAGCCCGCTAAAAAGAAGCTCAACTTTGGGCAGACTGGCG ACTCAGAGTCAGTGCCCGACCCCAACCTCTCGGAGAACC TCCCGCGCGCCCTCAGGTCTGGGATCTGGTACAATGGCT TCAGGTGGTGGCGCACCAAGTGGCAGACAATAACGAAGGTG CCGATGGAGTGGGTAGTTCTCGGGAAATTGGCATTGCGA TTCCCAATGGCTGGGGACAGAGTCATCACCACCGACACC CGAACCTGGGCCCTGCCACCTACAACAATCACCTCTACA AGCAAAATCTCAACAGCACATCTGGAGGATCTTCAAATGA CAACGCCTACTTCGGCTACAGCACCCCTGGGGGTATTTT GACTTCAACAGATTCCACTGCCACTTCTCACCACGTGACT GGCAGCGACTCATCAACAACACTGGGGATTCCGCGCTAA GCGACTCAACTTCAAGCTCTTCAACATTCAAGTCAAAGAG GTTACGGACAACAATGGAGTCAAGACCATCGCCAATAACC TTACCAGCACGGTCCAGGTCTTACGGACTCAGACTATCA GCTCCCGTACGTGCTCGGGTCGGCTCAGGAGGCTGCCTC CCGCGTTCCAGCGGACGTTTTCATGATTCTCAGTACG GGTATCTGACGCTTAATGATGGGAGCCAGGCCGTGGGTG TTCGTCTTTTACTGCTTGAATATTTCCCGTCGCAATG CTAAGAACGGGTAACTTCCAGTTCAGCTACGAGTTTG AGAACGTACCTTCCATAGCAGCTACGCTCAGGCCAAG CCTGGACCGACTAATGAATCACTCATCGCCAATCTTG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
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61	AAV6/AAVhu32 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGFNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFQ AKKRVLEPFLVEEGAKTAPGKKRPVEQSPQEPDSSSGIG KTQQQPAKKRLNFGQTGDSSEVPDPQLGEPPTPAAVGP TTMASSGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI TTSTRTWALPTYNHLYKQISNSTSGSSNDNAYFGYSTP WGYFDNRFHCHFSRDPWQRLINNNWGFRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIQYGYLTLNDGSAVGRSSFYCLEYF PSQMLRTGNFQFSYEFENVPHSSYAHSQSLDRLMNLPI DQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNDADKVM TNEEEIKTNPVATESYQVATNHQSAQAQAQTGWVQNG ILPGMVWQDRDVLQGPWAKIPHDTGNFHPSPLMGGFGM KHPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVS VEIEWELQKENSKRWNPEIQYTSNYKSNNEVFAVNTGCV YSEPRPIGTRYLTRNL
62	AAV6/AAVhu32 VP1 dna	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCAAGCCAACCAAGCAAAAGCAGGAC GACGGCCGGGCTCTGGTGCTTCTGGCTACAAGTACCTCG GACCTTCAACGGACTCGACAAGGGGGAGCCCGTCAACGC GGCGGATGCAGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAAGCGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCCGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCAG GCCAAGAAGAGGGTTCTCGAACCTTTTGGTCTGGTTGAGG AAGGTGCTAAGACGGCTCCTGGAAGAAACGTCCGGTAGA GCAGTCGCCACAAGAGCCAGACTCCTCCTCGGGCATTGGC AAGACAGGCCAGCAGCCCGCTAAAAGAGACTCAATTTTG GTCAGACTGGCGACTCAGAGTCAGTCCCGACCCACAACC TCTCGGAGAACCCTCAGCAACCCCGCTGCTGTGGGACCT ACTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGAATCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCCGCGGTTCACAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTGCTCTTTTACTGCCTGGAATATTTT

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		CCGTCGCAAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAAACGCTAAAATTAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAAATACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTTCCTTTGTCTGGATCTTTAATTTTGGCAAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAACTACTAACCCTGGTAGCAA CGGAGTCCTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGCAGAGCGCAGACCGGCTGGGTTCAAAACCAAGGA ATACTTCCGGGTATGGTTTGGCAGGACAGAGATGTGTACC TGCAAGGACCCATTTGGGCCAAAATTCCTCACACGGACGG CAACCTTTCACCTTCTCCGCTAATGGGAGGGTTTGGAAATG AAGCACCCGCTCTCAGATCCTCATCAAAACACACCTG TACCTGCGGATCCTCCAACGGCTTTCAATAAGGACAAGCT GAACCTTTCATCACCCAGTATTCTACTGGCCAAGTCAGC GTGGAGATTGAGTGGGAGCTGCAGAAGGAAACAGCAAGC GCTGGAACCCGGAGATCCAGTACACTTCCAATTATTACAA GTCTAATAATGTTGAATTTGCTGTTAATACTGAAGGTGTA TATAGTGAACCCGCCCCATTGGCACCATACCTGACTC GTAATCTGTAA
63	AAV5/AAVhu32 VP1 aa	MSFVDHPPDWLEEVGEGLEFLGLEAGPPKPKPNQHQDQ ARGLVLPGYNYLPGNGLDRGEPVNRADDEVAREHDISYNE QLEAGDNPYLKYNHADADEFQEKLADDTSGFNLGKAVFQA KKRVLPPGLVEEGAKTAPTGKRIDDHFPKRKKARTEEDS KPSTSSDAEAGPSGSQQLQIPAPASSLGADTMASGGGAP VADNNEGADGVSSSGNWHCDSSQWLGDRVITSTRTWALP TYNHLYKQISNSTSGGSSNDNAYFGYSTPWGYPDFNRFH CHFSPRDWQRLINNNWGFPRKRLNFKLFNIQVKEVTDNMG VKTIANNLTSTVQVFTDSYQLPYVLGSAHEGCLPPFPAD VFMIPOYGYLTLNDGSQAVGRSSFYCLEYFPPSQMLRTGNN PQFSYEFENVPFHSSYAHQSGLDRLMNLIDQYLYYLSKT INSGQNQOTLKFVSVAGPSNMAVQGRNYIPGPSYRQQRVS TTVTONNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEG EDRFPPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTN PVATESYQVATNHQSAQAQATGWVQNQGLPGMVWQDR DVYLQGPWIWAKIPHTDGNFHPSPLMGGFGMKHPPPILIK NTPVPADPPTAFNKDKLNSFITQYSTGVSVIEIWEWQKE NSKRWNPEIQYTSNYYKSNNVEFVNTEGVYSEPRPIGTR YLTRNL
64	AAV5/AAVhu32 VP1 dna	ATGTCTTTTGTGATCACCTCCAGATTGGTTGGAAGAAG TTGGTGAAGGTCTTCGCGAGTTTGGGCTTGAAGCGGG CCCACCGAAACCAAACCCAATCAGCAGCATCAAGATCAA GCCCGTGGTCTTGTGCTGCCTGGTTATAACTATCTCGGAC CCGGAAACGGTCTCGATCAGGAGAGCCTGTCAACAGGGC AGACGAGTTCGCGCAGAGCAGCATCTCGTACAACGAG CAGCTTGAGGCGGGAGACAACCCCTACCTCAAGTACAACC ACGCGGACGCCGAGTTTCAGGAGAAGCTCGCCGACGACAC ATCTTTCGGGGGAAACCTCGGAAAGGCAGTCTTTCAGGCC AAGAAAAGGGTTCTCGAACCTTTGGCCTGGTTGAAGAGG GTGCTAAGACGCCCTTACCGAAAGCGGATAGACGACCA CTTTCCAAAAAGAAAGAGGCTCGGACCGAAGAGGACTCC AAGCTTTCACCTCGTCAGACGCCGAAGCTGGACCCAGCG GATCCCAGCAGCTGCAATCCAGCCCAACGAGCTCAAG TTTGGGAGCTGATACAATGGCTTCAGGTGGTGGCGCACCA GTGGCAGACAATAACGAAGGTGCCGATGGAGTGGGTAGTT CCTCGGAAATTTGGCATTGCGATTCCCAATGGCTGGGGGA CAGAGTCATCACCCACGACACCCGAACCTGGGCCCTGCCC ACCTACAACAATCACCTCTACAAGCAAATCTCCAACAGCA CATCTGGAGGATCTTCAAATGACAACGCCTACTTCGGCTA CAGCACCCCTGGGGGTATTTGACTTCAACAGATTCCAC TGCCACTTCTCACACGTGACTGGCAGCGACTCATCAACA ACAACTGGGGATTCCGGCTAAGCGACTCAACTTCAAGCT CTTCAACATTCAAGTCAAAGAGGTTACGGACAACAATGGA GTCAAGACCATCGCCAATAACCTTACCAGCACGGTCCAGG TCTTCACGGACTCAGACTATCAGTCCCCGTACGTGCTCGG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GTCGGCTCACGAGGGCTGCCTCCCGCGTTCCCGAGCGGAC GTTTTTCATGATTCCTCAGTACGGGTATCTGACGCTTAATG ATGGGAGCCAGGCCGTGGGTCGTTCTGCTCTTTACTGCCT GGAATATTTCCTCGTCAATGCTAAGAACGGTAACAAC TTCCAGTTCAGCTACGAGTTTGAAGACGTACCTTTCCATA GCAGCTACGCTCACAGCCAAAGCCTGGACCGACTAATGAA TCCACTCATCGACCAATACTTGTAATCTCTCTCAAAGACT ATTAACGGTTCCTGGACAGAATCAACAAACGTAAAAATTCA GCGTGGCCGGACCCAGCAACATGGCTGTCCAGGGAAGAAA CTACATACCTGGACCCAGCTACCGACAACAACGTGTCTCA ACCACTGTGACTCAAAACAACAACAGCGAATTTGCTTGGC CTGGAGCTTCTCTTGGGCTCTCAATGGACGTAATAGCTT GATGAATCCTGGACCTGCTATGGCCAGCCAAAGAAGGA GAGGACCGTTTCTTCTTCTTGTCTGGATCTTAAATTTTGG GCAACAAGGAAGTGAAGAGACAACGTGGATGCGGACAA AGTCATGATAACCAAGCAAGAGAAATTAAGTACTAATAC CCGGTAGCAACGGAGTCTATGGACAAGTGGCCACAAACC ACCAGAGTGCCCAAGCACAGGCGCAGACCGGCTGGGTTC AAACCAAGGAATACTTCCGGGTATGGTTGGCAGGACAGA GATGTGTACCTGCAAGGACCCATTGGGCCAAAATTCCTC ACACGGACGGCACTTTTACCCCTTCTCCGCTAATGGGAGG GTTTGGAATGAAGCACCCGCTCTCAGATCTCATCAAA AACACACCTGTACCTGCGGATCCTCCAACGGCTTTCAATA AGGACAAGCTGAATCTTCTCATCCAGTATTCTACTGG CCAAGTCAGCGTGGAGATTGAGTGGGAGCTGCAGAAGGAA AACAGCAAGCGCTGGAACCCGGAGATCCAGTACACTTCCA ACTATTACAAGTCTAATAATGTTGAATTTGCTGTTAATAC TGAAGGTGTATATAGTGAACCCCGCCCATTTGGCACCAGA TACCTGACTCGTAATCTGTAA
65	AAVhu32/ AAVrh15 VP1 aa	MAADGYLPDWLEDTLSEGIQWKKLKPGRPPPKPAERHKD DSRGLVLPQYKYLPGNGLDKGEVNVNADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKRLLLEPLGLVEEAAKTPGKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQQTGDTESVPDPQPIGEPPAAPSGVGS LTMAAGGAPMADNNNEGADGVGNASGNWHCDSTWLGDRVI TTSTRTWALPTYNNHLYKQISSQSGATNDNHFFGYSTPWG YFDNRFHCHFSRWDQRLINNNWGFRPRKLRFLFNQIV KEVTNDGVTTIANNLSTIQVFSDSYQLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQSVGRSSFYCLEYFPS QMLRTGNNFEFSYTFEEVPPHSSYAHQSGLDRMLNPLIDQ YLYYLARTQSTTGSTRELQPHQAGPNTMAEQSKNWLPGPC YRQQLSKNIDSNNSTSNFAWTGATKYHLNGRNSLTNPQVA MATNKDDEDQFFPINGVLVFGKTAANKTTLENVLMTSEE EIKTNTNPVATEQYGVVSNLQSSSTAGPQTQTVNSQGALPG MVWQNRDVYLQGIWAKIPHTDGNFHPSPLMGGFGLKHPP PQILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVIE WELQKENSKRWNPEIQYTSNYAKSNNVEFAVNNEGVYTEP RPIGTRYLTRNL
66	AAVhu32/ AAVrh15 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCCAACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCTCAGGAACCGGACTCTCCGCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAGAAATCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCTCAACC AATCGGAGAACCCTCCGCGAGCCCTCAGGTGTGGGATCT CTTACAATGGCTGCAGGCGGTGGCGCTCCAATGGCAGACA ATAACGAAGGCGCGCAGGAGTGGGTAAATGCTCCGGA TTGGCATTGCGATTCCACATGGCTGGGCGACAGATCATC ACCACCAGCACCCGCACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAGATATCAAGTCAGAGCGGGGTAC CAACGACAACCACTTCTTCGGCTACAGACCCCTTGGGGC TATTTTGACTTCAACAGATTCCACTGCCACTTCTCACCAC GTGACTGGCAGCGACTCATCAACAACACTGGGATTCCG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GCCCAGAAAGCTGCGGTTCAAGTTGTTCAACATCCAGGTC AAGGAGGTCACGACGAAACGACGGCGTTACGACCATCGCTA ATAACCTTACCAGCAGATTTCAGGTCTTCTCGGACTCGGA GTACCAACTGCGGTACGTCTCGGCTCTGCGCACCAGGGC TGCTTCCCTCCGTTCCCTGCGGACGTGTTTCATGATTCCTC AGTACGGATATCTGACTCTAAACAACGGCAGTCAGTCTGT GGGACGTTCTCTTCTACTGCTGGAGTACTTTCCTTCT CAGATGCTGAGAACGGGCAATAACTTTGAATTTCAGCTACA CCTTTGAGGAAGTGCTTTCCACAGCAGCTATGCGCACAG CCAGAGCCTGGACCGGCTGATGAATCCCTCATCGACCAG TACCTGTACTACCTGGCCCGGACCCAGAGCACTACGGGGT CCACAAGGGAGCTGCAGTTCATCAGGCTGGGCCCAACAC CATGGCCGAGCAATCAAAGAACTGGCTGCCCGGACCTGT TATCGGCAGCAGAGACTGTCAAAAAACATAGACAGCAACA ACACCAGTAACCTTTGCTGGACCGGGGCCACTAAATACCA TCTGAATGGTAGAAATTCATTAACCAACCGGGCGTAGCC ATGGCCACCAACAAGGACGACGAGGACCAGTTCTTTCCCA TCAACGGAGTGCTGGTTTGGCAAAACGGGGCTGCCAA CAAGACAACGCTGGAACAGTCTAATGACCAGCAGGAG GAGATCAAAACCAACATCCGTTGGCTACAGAACAGTACG GTGTGGTCTCCAGCAACCTGCAATCGTCTACGGCCGAC CCAGACACAGACTGTCAACAGCCAGGGGCTCTGCCCGGC ATGGTCTGGCAGAACCGGACGTGTACCTGCAGGGTCCCA TCTGGGCCAAAATTCCTCACACGGACGGCAACTTACCC GTCTCCCTGATGGCGGATTGGACTCAAAACCCCGCT CCTCAAATTCATCAAAACACCCCGTACCTGCTAATC CTCCAGAGGTGTTTACTCTGCCAAGTTTGCTCATTAT CACGCAGTACAGCACCGGCAGGTCAGCGTGGAGATCGAG TGGGAACCTGCAGAAAGAAAACAGCAACGCTGGAATCCAG AGATTTCAGTACACCTCAAATATGCCAAGTCTAATAATGT GGAATTTGCTGTCAACACGAGGGGTTTATACTGAGCCT CGCCCCATTGGCACCCGTTACCTCACCCGTAACCTGTAA
67	AAV6/AAVrh15 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPGYKYLGPFNGLDKGEPVNAADAAALEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGNLRGAVFQ AKKRVLEPFGLVEEGAKTAPGKKRPVEQSPQEPDSSGIG KTGQQPAKKRLNFGQTGDSESVDPQPLGEPPTPAAVGP TTMAAGGAPMADNNEGADGVGNASGNWHDSTWLGDRI TTSTRTWALPTYNNHLYKQISSQSGATNDNHFFGYSTPWG YFDNRFHCHFSRWDQRLINNNWGFPRKRLRFKLFNIQV KEVTNDGVTTIANNLSTIQVFSDESQYLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQSVGRSSFYCLEYFPS QMLRTGNNFEFSYTFEEVPFHSSYAHQSLSLDRLMNPLIDQ YLYLARTQSTTGSTRELQFHQAGPNTMAEQSKNWLPGPC YRQQLSKNIDSNNTSNFAWTGATKYHLNGRNSLTNPQVA MATNKDDEDQFFPINGVLVFGKTGAANKTTLENVMTSEE EIKTNTNPVATEQYGVVSNLQSSSTAGPQTQTVNSQGALPG MVWQNRDVYLQGIWAKIPHDTGNFHPSPLMGGFGLKHPP PQILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVIE WELQKENSKRWNPEIQYTSNYAKSNNVEFAVNNEGVYTEP RPIGTRYLTRNL
68	AAV6/AAVrh15 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGGACTTGAAACC TGGAGCCCCGAAACCCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGTCTGGTGCTTCCTGGCTACAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGATGCAGCGGCCCTCGAGCACGACAAGCCTACGAC CAGCAGCTCAAAGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCCGAGTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCAGCAGTCTTCCAG GCCAAGAAGAGGGTTCTCGAACCTTTTGGTCTGGTTGAGG AAGGTGCTAAGACGGCTCCTGGAAGAAACGTCCGGTAGA GCAGTCGCCACAAGAGCCAGACTCCTCTCGGGCATTGGC AAGACAGGCCAGCAGCCGCTAAAAGAGACTCAATTTTG GTGAGACTGGCGACTCAGAGTCAGTCCCGACCCACAACC TCTCGGAGAACCTCCAGCAACCCCGCTGCTGTGGGACCT ACTACAATGGCTGCAGGCGGTGGCGCTCCATGGCAGACA ATAACGAAGGCGCCGACGGAGTGGGTAATGCCCTCCGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGAGTCATC ACCACAGCACCCGACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAGATATCAAGTCAGAGCGGGCTAC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		CAACGACAACCACTTCTTCGGCTACAGCACCCCTGGGGC TATTTTGACTTCAACAGATTCCACTGCCACTTCTCACCAC GTGACTGGCAGCGACTCATCAACAACACTGGGGATTCCG GCCAGAAAGCTGCGGTTCAAGTTGTTCAACATCCAGGTC AAGGAGGTCACGACGAAACGACGGCGTTACGACCATCGCTA ATAACTTACCAGCAGATTTCAGGTCTTCTCGGACTCGGA GTACCACTGCGGTACGTCTCGGCTCTGCCACCGAGGC TGCCTCCCTCCGTTCCTGCGGACGTGTTATGATTCTCTC AGTACGGATATCTGACTCTAAACAACGGCAGTCAGTCTGT GGGACGTTCTCTTCTACTGCCTGGAGTACTTTCCTTCT CAGATGCTGAGAACGGGCAATAACTTTGAATTTCAGCTACA CCTTTGAGGAAGTGCCTTCCACAGCAGCTATGCGCACAG CCAGAGCCTGGACCGGCTGATGAATCCCTCATCGACCAG TACCTGTACTACCTGGCCCGGACCCAGAGCACTACGGGGT CCACAAGGGAGCTGCAGTTCATCAGGCTGGGCCCAACAC CATGGCCGAGCAATCAAAGAACTGGCTGCCCGGACCTGT TATCGGCAGCAGAGACTGTCAAAAAACATAGACAGCAACA ACACCAGTAACCTTTCCTGGACCGGGGCCACTAAATACCA TCTGAATGGTAGAAATTCATTAACCAACCGGCGTAGCC ATGGCCACCAACAAGGACGACGAGGACCGTTCCTTCCCA TCAACGGAGTGTGGTTTGGGCAACGGGGCTGCCAA CAAGACAACGCTGGAACGCTGCTAATGACGAGCAGGAG GAGATCAAAACCAACATCCCGTGGCTACAGAACGATACG GTGTGGTCTCCAGCAACCTGCAATCGTCTACGGCCGAC CCAGACACAGACTGTCAACAGCCAGGGGCTCTGCCCGGC ATGGTCTGGCAGAACCGGACGTGTACCTGCAGGGTCCCA TCTGGGCAAAATTCCTCACAGCAGCGCAACTTTCACCC GTCTCCCTGATGGCGGATTGGACTCAACACCCCGCT CCTCAATTCATCAAAACACCCCGGTACCTGCTAATC CTCCAGAGGTGTTTACTCCTGCCAAGTTTGCTCATTAT CACGAGTACAGCACCGGCCAGGTCAGCGTGGAGATCGAG TGGGAACCTGCAGAAAGAAAACAGCAACGCTGGAATCCAG AGATTAGTACACCTCAAATATGCCAAGTCTAATAATGT GGAATTTGCTGTCAACAACGAAGGGTTTATACTGAGCCT CGCCCCATTGGCACCCGTTACCTCACCCGTAACCTGTAA
69	AAV5/AAVrh15 VP1 aa	MSFVDHPPDWLEEVGEGLEFLGLEAGPPKPKPNQHQDQ ARGLVLPGYNYLPGNGLDRGEPVNRADDEVAREHDISYNE QLEAGDNPYLKYNHADADEFQEKLAADTSFGGNLKGAVFQA KKRVLPPGLVEEGAKTAPTGKRIDDHFPKRKKARTEEDS KPSTSSDAEAGPSGSQQLQIPAPASSLGADTMAAGGGAP MADNNEGADGVGNASGNWCHDSTWLGRDVIITSTRTWALP TYNNHLYKQISSQSGATNDNHFFGYSTFWGYDFNRFHCH FSPRDWQRLINNNWGRPRKLRFLFNIVKEVTNDGVT TIANNLTSTIQVSDSEYQLPYVLGSAHQGLPPFPADVF MIPQYGYLTLNNGSQSVGRSSFYCLEYFSPQMLRTGNNE FSYTFEEVPPHSSYAHSQSLDRLMNPIDQVLYYLARTQS TTGSTRELQPHQAGPNTMAEQSKNWLPGPCYRQQLRSKNI DSNNTSNFAWTGATKYHLNGRNSLTNPGVAMATNKDDQ FFPINGVLVFGKTGAANKTLENVLMTEEEIKTTNPVAT EQYGVSSNLQSSTAGPQTQTVNSQGALPGMVWQNRDVYL QGPIWAKIPHDTGNFHPSPLMGGFGLKHPPQILIKNTPV PANPPEVFTPAKFASFITQYSTGQVSVIEWELQKENSKR WNPEIQYTSNYAKSNNVEFVNNNEGVYTEPRPIGTRYLTR NL
70	AAV5/AAVrh15 VP1 dna	ATGTCTTTTGTGATCACCTCCAGATTGGTTGGAAGAAG TTGGTGAAGGTCTTCGCGAGTTTTCGGCCCTTGAAGCGGG CCCACCGAAACCAAAACCAATCAGCAGCATCAAGATCAA GCCCGTGGTCTGTGCTGCCTGGTTATAACTATCTCGGAC CCGGAAACGGTCTCGATCGAGGAGAGCTGTCAACAGGGC AGACGAGGTCGCGCGAGAGCACGACATCTCGTACAACGAG CAGCTTGAGGCGGGAGACAACCCCTACCTCAAGTACAACC ACGCGGACGCCGAGTTTCAGGAGAAGCTCGCCGACGACAC ATCCTTCGGGGGAAACCTCGGAAAGGCAGTCTTTCAGGCC AAGAAAAGGGTTCTCGAACCTTTTGGCCTGGTTGAAGAGG GTGCTAAGACGGCCCTTACCGGAAAGCGGATAGACGACCA CTTTCCAAAAAGAAAGAGGCTCGGACCGAAGAGGACTCC AAGCCTTCCACCTCGTCAGACGCCGAAGCTGGACCCAGCG GATCCAGCAGCTGCAATCCAGCCCAACAGCCTCAAG TTTGGGAGCTGATACAAATGGCTGCAGGCGGTGGCGCTCCA ATGGCAGACAATAACGAAGGCGCCGACGAGTGGGTAATG CCTCCGAAATTGGCATTGCGATTCCACATGGCTGGGCGA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		CAGAGTCATCACCACCAGCACCCGCACCTGGGCCCTGCCC ACCTACAACAACCACCTCTACAAGCAGATATCAAGTCAGA GCGGGGCTACCAACGACAACCACCTCTTCGGCTACAGCAC CCCCTGGGGCTATTTGACTTCAACAGATTCCACTGCCAC TTCTCACCACGTGACTGGCAGCGACTCATCAACAACAAC GGGGATTCCGGCCAGAAAGCTGCGGTTCAAGTTGTTCAA CATCCAGGTCAAGGAGGTACGACGAAACGACGCGTTACG ACCATCGCTAATAACCTTACCAGCACGATTCAAGTCTTCT CGGACTCGGAGTACCAACTGCCGTACGTCCTCGGCTCTGC GCACCAGGGCTGCCCTCCCTCCGTTCCCTGCGGACGTGTT ATGATTCTCAGTACGGATATCTGACTCTAAACAACGGCA GTCAGTCTGTGGGACGTTCCTCTTCTACTGCTGGAGTA CTTTCTTCTCAGATGCTGAGAACGGGCAATACTTTGAA TTCAGCTACACCTTTGAGGAAGTGCCTTTCACAGCAGCT ATGCGCACAGCCAGAGCCTGGACCGGCTGATGAATCCCT CATCGACCAGTACCTGTACTACCTGGCCCGACCCAGAGC ACTACGGGGTCCACAAGGAGCTGCAGTTCATCAGGCTG GGCCCAACACCATGGCCGAGCAATCAAAGAAGTGGCTGCC CGGACCTGTTATCGGCAGCAGAGACTGTCAAAAAACATA GACAGCAACAACCCAGTAACCTTGGCTGGACCGGGCCA CTAATATCATCTGAATGGTAGAAATTCATTAACCAACCC GGGCGTAGCCATGGCCACCAACAAGGACGACGAGGACAG TTCTTCCCATCAACGGAGTGTGGTTTTTGGCAAAACGG GGGCTGCCAACAAGACACGCTGGAACACGCTGCTAATGAC CAGCGAGGAGGAGATCAAAACCAACATCCCGTGGCTACA GAACAGTACGGTGTGGTCTCCAGCAACCTGCAATCGTCTA CGGCCGGACCCAGACACAGACTGTCAACAGCCAGGGGGC TCTGCCCGGCATGGTCTGGCAGAACCAGGACGTGTACCTG CAGGGTCCCATCTGGGCCAAAATTCCTCACACGGACGGCA ACTTTCACCCGCTCTCCCTGATGGGGCGGATTGGAGCTCAA ACACCCGCTCTCTCAAATTCATCAAAAAACACCCCGTA CCTGCTAATCTCCAGAGGTGTTTACTCCTGCCAAGTTTG CCTCATTTATCAGCAGTACAGCACCGGCCAGGTGAGCGT GGAGATCGAGTGGGAACGACAGAAAGAAAACAGCAAAACGC TGGAATCCAGAGATTCAATACACCTCAAATATGCCAAGT CTAATAATGTGGAATTTGCTGTCAACACGAAGGGGTTTA TACTGAGCCTCGCCCCATTGGCACCCGTTACCTACCCGT AACCTGTAA
71	AAV6/AAVrh13 VP1 aa	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQKQD DGRGLVLPQYKYLGPFGNLDKGEVNVNADAAALEHDKAYD QQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNLGRAVFG AKKRVLEPFGLVEEGAKTAPGKKRPVEQSPQEPDSSSGIG KTQQPAKKRLNFGQTGDSESVDPQPLGEPATPAAVGP TTMAAGGAPMADNNEGADGVGNASGNWHDCTWLGDRI TTSRTWALPTYNNHLYKQISSQSGATNDNHFGYSTPWG YFDNRFHCHFSRQWRLINNNWGFRPRKLRFLFNQV KEVTNDGVTTIANNLSTIQVFSDEYQLPYVLGSAHQG CLPPFPADVFMIPQYGYLTLNNGSQSVGRSSFYCLEYFPS QMLRTGNNFEFSYTFEEVPHSSYAHQSGLDRMLNPLIDQ YLYLARTQSTTGSTRELQPHQAGPNTMAEQSKNWLPGPC YRQQLSKNIDSNNSNFAWTGATKYHLNGRNSLTNPGVA MATNKDDEDQFFPINGVLVFGGETGAANKTTLENVMTSEE EIKTNPVATEEYGVSSNLQSSSTAGPQTQTVNSQGALPG MVWQNRDVYLQGPWAKIPHDTGNFHPSPLMGGFGLKHPP PQILIKNTPVPANPPEVFTPAKFASFITQYSTGQVSVIE WELQKENSKRWNPEIQYTSNYAKSNNVEFVNNNEGVYTEP RPIGTRYLTRNL
72	AAV6/AAVrh13 VP1 dna	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA ACCTCTCTGAGGGCATTGCGAGTGGTGGACTTGAAACC TGGAGCCCCGAAACCAAAGCCAACAGCAAAAGCAGGAC GACGGCCGGGGTCTGGTGCTTCTGGCTACAAAGTACCTCG GACCTTCAACGGACTCGACAAGGGGAGCCCGTCAACGC GGCGGATGCAGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAAGCGGGTGACAATCCGTACCTGCGGTATA ACCACGCCGACGCGAGTTTTCAGGAGCGTCTGCAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCAGCAGTCTTCAG GCCAAGAAGAGGGTTCTCGAACCTTTTGGTCTGGTTGAGG AAGGTGCTAAGACGGCTCTTGGAAAGAAACGTCGGTAGA GCAGTCGCCACAAGAGCCAGACTCCTCCTCGGGCATTGGC AAGACAGCCAGCAGCCCGCTAAAAAGAGACTCAATTTTG GTCAGACTGGCGACTCAGAGTCAGTCCCGACCCACAACC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TCTCGGAGAACCTCCAGCAACCCCGCTGCTGTGGGACCT ACTACAATGGCTGCAGGCGGTGGCGCACCAATGGCAGACA ATAACGAAGGCGCCGACGGAGTGGGTAATGCCCTCCGAAA TTGGCATTGCGATTCCACATGGCTGGGCGACAGATCATC ACCACCAGCACCCGCACCTGGGCCCTGCCACCTACAACA ACCACCTCTACAAGCAGATATCAAGTCAGAGCGGGGTAC CAACGACAACCACTTCTTCGGCTACAGCACCCCTGGGGC TATTTTGACTTCAACAGATTCCACTGCCACTTCTACCAC GTGACTGGCAGCGACTCATCAACAACACTGGGATTCCG GCCAGAAAGCTGCGGTTCAAGTTGTTCAACATCCAGGTC AAGGAGGTCACGACGAACGACGGCGTTACGACCATCGCTA ATAACCTTACCAGCACGATTGAGTCTTCTCGGACTCGGA GTACCAACTGCGGTACGTCCTCGGCTCTGCGCACCGGGC TGCCTCCCTCCGTTCCCTGCGGACGTGTTTCATGATTCCTC AGTACGGATATCTGACTCTAACAACCGGCAGTCAGTCTGT GGGACGTTCTCTCTTCTACTGCTGGAGTACTTTCCTTCT CAGATGCTGAGAACGGGCAATAACTTTGAATTGAGCTACA CCTTTGAGGAAGTGCCTTTCACAGCAGCTATGCGCACAG CCAGAGCCTGGACCGGCTGATGAATCCCTCATCGACCAG TACCTGTACTACCTGGCCCGGACCCAGAGCACTACGGGGT CCACAAGGGAGCTGCAGTTCATCAGGCTGGGCCCAACAC CATGGCCGAGCAATCAAAGAACTGGCTGCCCGGACCTGT TATCGGCAGCAGAGACTGTCAAAAAACATAGACAGCAACA ACAACAGTAACCTTTGCCGAGCCGGGGCCACTAAATACCA TCTGAATGGTAGAAATTCAATACCAACCCGGGCGTAGCC ATGGCCACCAACAAGGACGACGAGGACAGTTCTTTCCCA TCAACGGAGTGTGGTTTTTGGCGAAACGGGGCTGCCAA CAAGACAACGCTGGAACGCTGCTAATGACCAGCGAGGAG GAGATCAAAACCCCAATCCCGTGGCTACAGAAGAATACG GTGTGGTCTCCAGCAACCTGCAATCGTCTACGGCCGGACC CCAGACACAGACTGTCAACAGCCAGGGGCTCTGCCCGGC ATGGTCTGGCAGAACCGGACGTGTACCTGCAGGGTCCCA TCTGGGCAAAATCTCTCACAGCAGCGCAACTTCAACCC GTCTCCCTGATGGCGGATTGGACTCAAAACCCCGCT CCTCAAATCTCATCAAAACACCCCGTACCTGCTAATC CTCCAGAGGTGTTTACTCTGCCAAGTTGCTCATTAT CACGCAGTACAGCACCGGCCAGGTACGCTGGAGATCGAG TGGGAACCTGCAGAAAGAAAACAGCAACGCTGGAATCCAG AGATTAGTACACCTCAAATATGCCAAGTCTAATAATGT GGAAATTTGCTGTCAACACGAAGGGTTTATACTGAGCCT CGCCCCATTGGCACCCGTTACCTCACCCGTAACCTGTAA
81	Hu32 Variant 1	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEPVNAADAAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNGLGRAVFQ AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKKLNFGQTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSQWLGDRI TTSRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRDPWQRLINNNWGFPRKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPHSSYAHQSGLDRLMNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYQVATNHQSAQSTTLYSPAQAQT GWVQNQILPGMVWQDRDVLQGPWIWAKIPHDTGNFHPSP LMGGFGMKHPPQILIKNTPVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYKSNNVEF AVNTEGVYSEPRPIGTYLTRLNL
82	Hu32 Variant 1	ATGGCTGCCGATGGTTATCTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCACCACCACCACCAAGCCGACAGCGGCATAAGGAC GACAGCAGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGAGCTCGACAAGGGGGAGCCGCTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAGAAGA TACGTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GCAGTCTCCTCAGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTTACAGCCCGCTAAAAAGAACTCAATTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCCCTCAACC AATCGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCTACTTCGGCTACAGCACCCCT TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCTTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCGTTCACAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGTTTCTGTCCTTTTACTGCTGGAATATTTC CCGTCGCAAAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAAGCTACCTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACCGCTAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCCTGTGA CTCAAAAACAACAACAGCGAATTTGCTTGGCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGGAGACCGTT TCTTCTCTTTGCTCTGGATCTTTAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATTAAGCTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACACCACAGAGTGC CCAATCAACGACGCTCTATTCTCCAGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCATTTGGGG CAAAATCTCTCACACGGACGGCAACTTCAACCCTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCTCTCTCAGA TCCTCATCAAAAACACCTGTACCTGCGGATCTCTCAAC GGCTTCAATAAGGACAAGCTGAACCTTTTATCAACCCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAATATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCCGCCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
83	Hu32 Variant 2	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPYKYLPGPNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKKRLLLEPLGLVEEAAKTAPGKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVSSSGNWHCDSQLGDRV I TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFCHFSRDPWQRLINNNWGFPRKLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSQYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHSQLDRLMNP LI DQLYYLSKTINGSGQNQTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM I TNEEEIKTNPVATESYGVATNHQSAQFVVQGSYAQAQT GWVQNQGILPGMVWQDRDVYLQGP IAWAKI PHTDGNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
84	Hu32 Variant 2	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCCAACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAGTACCTCG GACCCGGCAACGGAAGTGCAGCAAGGGGAGCGGTCACACG AGCAGACGCGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCCTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTACAGCCCGCTAAAAAGAACTCAATTTCTG GTCAGACTGGCGACACAGAGTCAGTCCCGACCCCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATACGAAAGGTGCCGATGGAGTGGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCAACAGCACATCTGGAGG ATCTTCAAATGACAAACGCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCAGCGGACGCTTTTATGA TTCTCTAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTCGTCTCTTTTACTGCCTGGAATATTT CCGTCGCAAAATGCTAAGAACGGGTAAACACTTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACACAACGCTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAAGAAGGAGGAGACCGTT TCTTCTCTTTGTCTGGATCTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATAAAACTACTAACCCGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAATTCTGTTGGGACAGTCTTATGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTGGGC CAAAATTCCTCACACGGACGGCAACTTTACCCCTTCTCCG CTAATGGGAGGTTTGGAAATGAAGCACCCGCTCTCTCAGA TCCTCATCAAAAACACACCTGTACCTGCGGATCCTCCAAC GGCTTTCAATAAGGACAAGCTGAACCTTTTATCATACCCAG TATTCTACTGGCCAAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAACATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCCGCCCA TTGCAACAGATACCTGACTCGTAATCTGTAA
85	Hu32 Variant 3	MAADGYLPDWLEDTLSEGIQWVKLKPGRPPPKPAERHKD DSRGLVLPGYKYLGFNGLDKGEVNAADAAALEHDKAYD QQLKAGDNPYLYNHADAERFQERLKEDTSFGGNLGRAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKKLNFGQTGDTESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDFNRFHCHFSRDPWQRLINNNGWFRPKRLNFKLFNI QVKEVTDNNGVKTIANNLTSVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNPQFSYEFENVPFHSSYHSQSLDRMLNPLI DQYLYYLSKTINGSGQSGRDLGLGNQQTLLKFSVAGPSN MAVQGRNYPGPSYRQQRVSTTVTQNNNSEFAWPAGSSWA LNGRNSLMNPGPAMASHKEGEDRFFPLSGSLIFGKQGTGR DNVDADKVMITNEEEIKTTNPVATESYQVATNHQSAQAQ AQTGWVQNGILPGMVWQDRDVYLQGPWAKIPHTDGNFH PSLPMGGFGMKHPPQILIKNTPVPADPPTAFNKDKLNSF ITQYSTGQVSVEIEWELQKENSkrwnPEIQYTSNYKSN VEFVNTGEGVSEPRPIGTRYLTRNL
86	Hu32 Variant 3	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTGAAGGAATAAGACAGTGGTGGAGCTCAAACC TGGCCACCACCACCAAGCCGAGAGCGGCATAAGGAC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGCTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAAGAGGCCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAAACTCAATTTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGACTTCAACAGATTCCTGCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCCGCCGTTCACAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTCGTTCTGCTCTTTACTGCCTGGAATATTT CCGTCGCAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAAGCTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATCTCTCAAGACTATTAAACGGTT CTGGCCAGAGTGGCCGCGGATCTGGGCCCTGGGACAGAA TCAACAAACGCTAAAATTCAGCGTGGCCGGACCCAGCAAC ATGGCTGTCCAGGGAAGAACTACATACCTGGACCCAGCT ACCGACAACAACGTGTCTCAACCACTGTGACTCAAAACAA CAACAGCGAATTTGCTTGGCCTGGAGCTTCTTCTGGGCT CTCAATGGACGTAATAGCTTGATGAATCCTGGACCTGCTA TGGCCAGCCACAAAGAGGAGAGGACCGTTTCTTCTCTTT GTCTGGATCTTTAATTTTGGCAAAACAGGAACCTGGAAGA GACAACGTGGATGCGGACAAAGTCATGATAACCAACGAAG AAGAAATTAAACTACTAACCCGGTAGCAACGGAGTCCTA TGGACAAGTGGCCACAACACAGAGTGCCCAAGCACAG GCGCAGACCGGCTGGGTTCAAAACCAAGGAATACTTCCGG GTATGGTTTGGCAGGACAGAGATGTGTACCTGCAAGGACC CATTTGGGCCAAAATCTCTCACACGGACGGCAACTTTTAC CCTTCTCCGCTAATGGGAGGGTTTGAATGAAGCACCGGC CTCCTCAGATCCTCATAAAAACACCTGTACCTGCGGA TCCTCCAACGGCTTTCAATAAGGACAAGCTGAACCTTTT ATCACCCAGTATTCTACTGGCCAAGTCAGCGTGGAGATTG AGTGGGAGCTGCAGAAGGAAAACAGCAAGCGCTGGAACCC GGAGATCCAGTACACTTCCAATATTACAAGTCTAATAAT GTTGAATTTGCTGTTAATACTGAAGGTGTATAGTGAAC CCCGCCCATTTGGCACCAGATACCTGACTCGTAATCTGTA A
87	Hu32 variant 4	MAADGYLPDWLEDTLSEGIQWKKLPGPPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEPVNAADAAALEHDKAYD QQLKAGDNPYLYNHADAEPQERLKEDTSFGNLRGAVFQ AKKRLLLEPLGLVEEAATAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRDPWQRLINNNGWFRPKRLNFKLFNI QVKEVTDNNGVKTIANLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSPYCLEYF PSQMLRTGNNPQFSYEFENVFPFHSSYAHSQSLDRMLNPLI DQLYYLSKTINGSGQNQOTLKFVSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNS EFAWP GASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM I

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TNEEEIKTNPVATESYGVQVATNHQGSGRGDLGLSAQAA QTGWVQNQGILPGMWQDRDVYLQGP IWAKIPHTDGNFHP SPLMGGFQGMKHPPPQILIKNTPVPADPPTAFNKDKLNSFI TQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNV EFAVNTEGVYSEPRPIGTRYLTRNL
88	Hu32 variant 4	ATGGCTGCCGATGGTTATCTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCACGAC CAGCAGCTCAAGGCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAGAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCACGCGGACGTTTTCATGA TTCCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTTCGCTCTTTTACTGCCTGGAATATTTT CCGTGCAAAATGCTAAGAACGGTAAACAATTCAGATTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATACTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTACGCTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAAATACATACTT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTGCTTGGCTGGAGCTTC TCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTTCCTTTGTCTGGATCTTTAATTTTGGCAAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAGTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCACAGGGCCA GAGTGGCCGCGCGATCTGGGCCTGAGCGCCAGGCGGCC CAGACCCGCTGGGTTCAAAACCAAGGAATACTTCCGGTA TGGTTTGGCAGGACAGAGATGTGTACCTGCAAGGACCCAT TTGGGCCAAAATTCCTCACACGGACGGCACTTTCACCT TCTCCGCTAATGGGAGGGTTTGAATGAAGCACCCGCTC CTCAGATCCTCATCAAAACACACCTGTACCTGCGGATCC TCCAACGGCTTCAATAAGGACAAGCTGAATCTTTTCATC ACCCAGTATTTACTGGCCAAGTCAGCGTGGAGATTGAGT GGGAGCTGCAGAAGGAAAACAGCAAGCGCTGGAACCCGGA GATCCAGTACACTTCCAATATTACAAGTCTAATAATGTT GAATTTGCTGTATAACTGAAGGTGATATAGTGAACCCC GCCCCATTGGCACCAGATACCTGACTCGTAATCTGTAA
89	Hu32 variant 5	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPQYKYLPGNGLDKGEPVNAADAAAEHDKAYD QQLKAGDNPYLYNHADAEPQERLKEDTSFGGNLGRAVFQ AKKRLLEPLGLVEEAATAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHDCSQWLGDRVI TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTP WGYFDNRFHCHFSRDPWQRLINNNWGRPKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSPYCLEYF

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		PSQMLRTGNNFQFSYEFENVFPHSSYAHSQSLDRLMNPLI DQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGQVATNHQSAQGRDGLQAQQT GWVQNQGILPGMVWQDRDVLQGPWAKIPIHTDGNFHPSP LMGGFGMKHPPQPILIKNTPVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
90	Hu32 variant 5	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCAGCAGTCTTCAG GCCAAAAGAGGGCTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCTCAGGAACCGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCCTCGGGA TTGGCATTGCGATTCCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCAGGTCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCCGCCGTTCACAGCGGACGTTTTCATGA TTCTCTAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGTTGTCGTCCTTTTACTGCTGGAATATTT CCGTCGCAATGCTAAGAACGGGTAAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAAACAACAAGCAATTTGCTTGGCTCGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTCTCTTTGCTCGGATCTTAATTTTGGCAAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAATACTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGGCCGCGCGCATCTGGGCCTGGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCATTTGGGC CAAAATCTCTCACACGGACGGCAACTTCACCTCTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCCCTCTCAGA TCCTCATCAAAAACACCTGTACTGCGGATCTTCAAC GGCTTTCAATAAGGACAAGCTGAACCTTTTATCACCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAACATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCGCCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
91	Hu32 variant 6	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPYKYLPGNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKRLLLEPLGLVEEAATAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVGSSSGNWHCDSQLGDRVI

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRDLINNNWGFRLNFKLFNI QVKEVTDNNGVKTIANNLTSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDGSAQVGRSSPYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHQSGLDRMLNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGVATNHQSAQQRGDYNQAQQT GWVQNGILPLGMVWQDRDVYLQGPWAKIPHTDGNFHPSP LMGGFGMKHPPQILIKNTPVPADPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
92	Hu32 variant 6	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGCTTCTCGGTACAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGCTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCCTGTAGA GCAGTCTCCTCAGGAACCGACTCCTCGCGGGTATTGGC AAATCGGTTTACAGCCCGCTAAAAGAAATCAATTTCTG GTGAGTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAAGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGAAA TTGGCATTGCGATTCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACGACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACGACACCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAAGTGGG ATTCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACAGCAGCGTCCAGGTCTTCAAGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCTGGCTCAC GAGGGCTGCCTCCCGCCGTTCAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTCTGCTCCTTTTACTGCCTGGAATATTTCT CCGTCGCAATGCTAAGAACGGGTAAACAATTCCAGTTCA GCTACGAGTTTGAAGACGTACCTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCCTGTGA CTCAAAACAACAACAGCGAATTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAAGAAGGAGAGGACCGTT TCTTTCTTTGTCTGGATCTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATAAAAGTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCACAGAGTGC CCAACAGAGAGGCGACTACAACCTGACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTGGGC CAAAATCTCTCACAGGACGGCAACTTCACCCCTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCTCCTCAGA TCCTCATCAAAACACACCTGTACTGCGGATCCTCCAAC GGCTTCAATAAGGACAAGCTGAACCTTTTATCACCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAACAGCAAGCGCTGGAACCCGGAGATCCA GTACACTTCCAACATTTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
93	Hu32 variant 7	MAADGYLPDWLEDTLSEGIQWVKLKPGRPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEVNAADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRVAFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRDRWQRLINNNWGFRPKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSQVLPVVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHSQSLDRMLNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGVATNHQSAQNSRGDYNAQAQT GWVQNQGILPGMVWQDRDVYLQGPWAKIPHDTGDNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVIEWELQKENSKRWNPEIQYTSNYKSNNEVF AVNTEGVYSEPRPIGTRYLTRNL
94	Hu32 variant 7	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCGGTCAACGC AGCAGACGCGCGCGCTTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCGGAGACAACCGTACCTCAAGTACA ACCAACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGCTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAAGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGTGCCGATGGAGTGGGTAGTCTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCTTACTTCGGCTACAGCACCCTC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGGG ATTCCGGCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCAGGTCAGGTCTTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCGTTCACAGCGACGTTTTCATGA TTCCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCGGTGGGTGCTGCTCCTTTTACTGCTGGAATATTTT CCGTCGCAATGCTAAGAACGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTACGCTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCTGGAGCTTC TCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGGACCGTT TCTTCTCTTGTCTGGATCTTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATAAAACCTAACCCTGGTAGCAA CGAGTCTTATGGACAAGTGGCCACAACCACAGAGTGC CCAAAATTCTAGAGGCGACTACAACGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCATTTGGGC CAAAATTCCTCACACGGACGGCAACTTTCACCTTCTCCG CTAATGGGAGGGTTTGGAAATGAAGCACCCTCTCAGA TCCTCATCAAAACACACCTGTACCTGCGGATCCTCAAC GGCTTCAATAAGGACAAGCTGAACCTTTCATCACCCAG TATTCTACTGCCAAGTCAGCTGGAGATTGAGTGGGAGC

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		TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGGAGATCCA GTACACTTCCAAC TATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
95	Hu32 variant 8	MAADGYLPDWLEDTLSEGIQWKKLKP PPPPKPAERHKD DSRGLVLPGYKYLGPNGLDKGEPVNAADAAAEHDKAYD QQLKAGDNPYLYKNHADAEPQERLKEDTSFGGNLGRAVFO AKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQQTGDTESVDPDQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTP WGYPDFNRFCHFSRDPDQRLINNNWGFRLNFKLENI QVKEVTDNNGVKTIANNLSTVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDGSGQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPHSSYAHSQSLDRLMNPLI DQYLYYLSKTINGSGQNQQLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTVTQNNNS EFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDKVM I TNEEEIKTNPVATESYQVATNHQSAQENRRGDFQAQQT GWVQNQGILPGMVWQDRDVLQGP I WAKI PHTDGNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
96	Hu32 variant 8	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCCAACCACCAACAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGAGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGCGAGCAGTCTTCCAG GCCAAAAGAGGGCTTTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCTTGGAAAGAAGAGCCCTGTAGA GCAGTCTCTCAGGAACCGACTCTCCGCGGTATTGGC AAATCGGTTTACAGCCCGCTAAAAGAACTCAATTTCTG GTGAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCCTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCC TACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCTGCTGCTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGGTGGCTCAC GAGGGCTGCCCTCCCGCCGTTCACAGCGGACGTTTTCATGA TTCTCTAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTCTGTTCTGTCCTTTTACTGCTGGAATATTT CCGTCGCAATGCTAAGAACGGGTAACTTCCAGTTCA GCTACGAGTTTGAGAACGTACTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAAACGGTT CTGGACAGAATCAACAACCGCTAAAATTGAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACTT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTTCTTTGTCTGGATCTTAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAACCTACTAACCCTGAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGAAAACCGGAGAGGCGACTTCGACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCAATTTGGGC CAAAATCTCTACACGAGCGCAACTTTCACCTTCTCCG

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97	Hu32 variant 9	MAADGYLPDWLEDTLSEGIQWKKLKP PPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEVNAADAAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGGNLGRAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKKLNFGQTDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSQWLGDRV I TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYPDFNRFHCHFSRDPWQRLINNNWGFRPKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDGSGQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHQS LDRLMNPLI DQYLYYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNEFAWP GASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM I TNEEEIKTTNPVATESYQVATNHQSAQRGDYVGLAQQT GWVQNQGILPGMVWQDRDVLQGP IWAKI PHTDGNFHPSP LMGGFGMKHPPPIILIKNTVPVADPPTAFNKD K LNSFITQ YSTGQVSVEIEWELQKENS KRWNPEIQYTSNYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
98	Hu32 variant 9	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAGCTCAAACC TGGCCCAACCAACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGA CTGACAAGGGGAGCCGCTCAACGC AGCAGACGCGGCGGCTTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCGCAGCGGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAAATCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACGCACATCTGGAGG ATCTTCAATGACAACGCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCCAGCGGACGCTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTCTGTCCTTTTACTGCCTGGAATATTTT CCGTCGCAATGCTAAGAACGGGTAAACAATTCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTACGCTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGTATGGCCAGCCACAAGAAGGAGGAGGACCGTT TCTTTCCTTTGTCTGGATCTTAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATCGCGACAAGTCATGATA ACCAACGAAGAAGAAATAAACTACTAACCCGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCAACCAAGAGTGC

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		CCAAAGAGGCGACTACGTGGGCTGGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTGGGC CAAAATTCTTCACACGGACGCAACTTTCACCTTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCTCCTCAGA TCCTCATCAAAACACACCTGTACCTGCGGATCCTCAAC GGCTTCAATAAGGACAAGCTGAACCTTTCATCACCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGGAGATCCA GTACACTTCCAATATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
99	Hu32 variant 10	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPGYKYLPGFNGLDKGEVNAADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVSSSGNWHDCSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDFNRFCHFSRDPQRLINNNGFRPKRLNFKLFNI QVKEVTDNNGVKTIANLSTVQVFTSDYQLPYVLGSAH EGCLPPPPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNPQFSYEFENVPFHSSYAHQSGLDRLMNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGVATNHQSNRGRDYNLSLAQAQT GWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSP LMGGFGMKHPPPIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNVVEF AVNTEGVYSEPRPIGTRYLTRNL
100	Hu32 variant 10	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCAACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCGGTCACGCGC AGCAGACGCGGCGGCTTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCGGAGACAACCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTCTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCTTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCTGCTGCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGGG ATTCCGGCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTACCGGA CTGAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCTCCCGCGTTCCAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCGGTGGGTGCTCGTCTCTTTTACTGCTGGAATATTT CCGTCGCAATGCTAAGAACGGGTAAACAACCTCCAGTTCA GCTACGAGTTTGAGAAGTACCTTTCATAGCAGCTACGC TCACAGCCAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTGAGCTGGCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTAAAACAACAACAGCGAATTGTCTGGCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT

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101	Hu32 variant 11	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHK DDSRGLVLPGYKYLPGPNGLDKGEPVNAADAAALEHDKAY DQQLKAGDNPYLKYNHADAEPQERLKEDTSFGNGLGRAVF QAKKRLLEPLGLVEEAAPTAPGKKRPVEQSPQEPDSSAGI GKSGSQPAKKKLNFGQTDTESVDPDQPIGEPPAAPSGVG SLTMASGGGAPVADNNEGADVGSSSGNWHCDSSQWLGDRV ITTSRTRWALPTYNNHLYKQISNSTSGSSNDNAYFGYST PWGYFDNRFHCHFSRWDQRLINNNWGRPKRLNFKLFN IQVKEVTDNNGVKTIANNLTSTVQVFTDSYQLPYVLGSA HEGCLPPFPADVFMIPQYGYLTLDGSAVGRSSFYCLEY FPSQMLRTGNNFQFSYEFENVPFHSSYAHSSQLDRLMNPL IDQYLYLSKTINGSGQNQQTLLKFSVAGPSNMAVQGRNYI PGPSYRQQRVSTTVTQNNNSEFAWPAGSSWALNGRNSLMN PGPAMASHKEGEDRFPLSLIFGKQGTGRDNVDADKVM ITNEEEIKTTNPVATESYGQVATNHQENRRGDFNNTAQAO TGWVQNQILPGMVWQDRDVYLQGPWAKIPIHTDGNFHPS PLMGGFGMKHPPQILIKNTVPVADPPTAFNKDKLNSFIT QYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNMVE FAVNTEGVYSEPRPIGTRYLTRNL
102	Hu32 variant 11	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCCAACCACCACAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGAGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGGTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCTTGAAGAAGAGCGCTGTAGA GCAGTCTCTCAGGAACCGACTCTCCGCGGTATTGGC AAATCGGTTTCACAGCCGCTAAAAGAACTCAATTTCTG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCTAAGCGACTCAACTCAAGCTCTTCAACATT CAGGTCAAAGAGGTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGTCGGCTCAC GAGGGCTGCCTCCCGCGTTCCAGCGGACGTTTTCATGA TTCCTCAGTACGGGTATCTGACGCTTAATGATGGAGCCA GGCCGTGGGTCTGTCGCTCTTTACTGCCTGGAATATTT CCGTGCAAAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTACGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGAAGAACTACATACCT

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103	Hu32 variant 12	MAADGYLPDWLEDTLSEGIQWKKLKP PPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRVAVFQ AKRRLLEPLGLVEEAAKTPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSGVGS LTMASSGGAPVADNNEGADGVSSSGNWHCDSQWLGDRV I TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYPDFNRFHCHFSRDPWQRLINNNWGFRPKRLNFKLFNI QVKEVTDNNGVKTIANNLTSVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDNGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHQSGLDRMLNPLI DQYLYYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM I TNEEEIKTNPVATESYGVATNHQSAQRGDYVGVQAQT GWVQNQILPGMVWQDRDVLQGP I WAKI PHTDGNFHPSP LMGGFGMKHPPQILIKNTPVPADPPTAFNKDKLNSFITQ YSTGQVSVIEIWEWELQKENSKRWNPEIQYTSNYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
104	Hu32 variant 12	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCCAACCAACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGTCACGCG AGCAGACGCGGCGGCCCTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGTAAAAGAAATCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGACAGAGTCATC ACCACCAGACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACGACATCTGGAGG ATCTTCAAATGACAACGCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCGGGCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCAGGTCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCCGCGGTTCCAGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGTTCTGCTCTTTACTGCCTGGAATATTT CCGTCGCAATGCTAAGAACGGGTAACAATTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTCCATAGCAGCTACGC

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105	Hu32 variant 13	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPYKYLPGPNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPQPIGEPPAAPSGVGS LTMASSGGAPVADNNEGADGVSSSNWCHDSQWLGDREVI TTSTRWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFCHFSRPRDQRLINNNWGRFPKRLNFKLFI QVKEVTDNNGVKTIANNLSTVQVFTDSQVLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNPQFSYEFENVPFHSYAHQSGLDRMLNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYP GPSYRQQRVSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEIKTNPVATESYGVATNHQSAQRGDLVLSAQAT GWVQNQGILPGMVWQDRDVYLGPIWAKIPHTDGNFHPSP LMGGFGMKHPPQILIKNTVPVADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYKSNVVEF AVNTEGVYSEPRPIGTRYLTRNL
106	Hu32 variant 13	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCCAACCAACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGCGAACGGACTCGACAAGGGGGAGCGGTCAACGC AGCAGACGCGCGCGCTTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCCTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTACGGA CTCAGACTATCAGCTCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCGCCGTTCCAGCGGACGTTTTCATGA

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107	Hu32 variant 14	MAADGYLPDWLEDTLSEGIQWKLKPGPPPKPAERHKD DSRGLVLPGYKYLPGFNGLDKGEVNAADAALHDKAYD QQLKAGDNPYLYNHADAEPQERLKEDTSFGGNLGRAVFQ AKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSQWLGDRI TTSRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSPRDQRLINNNWGFPRKRLNFKLFI QVKEVTDNNGVKTIANNLSTVQVFTSDYQLPVVLGSAH EGCLPPFPADVFMIPIQYGYLTLDGSGQAVGRSSFYCLEYF PSQMLRTGNFQFSYEFENVFPFHSSYAHSQSLDRLMNLPI DQYLYYLSKTINGSGQNQQLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNSFEPAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVI TNEEEIKTNPVATESYQVATNHQSAQRGDFVGLAQQT GWVQNQGILPGLVWQDRDVLQGPWAKIPIHPTDGNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSFRWNPEIQYTSNYYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
108	Hu32 variant 14	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCCAACCAACCAAGCCGCGAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGCTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGCGGAGCAGTCTTCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCTTGGAAAGAGAGCCCTGTAGA GCAGTCTCTCAGGAACCGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTTCTG GTACAGCTGGCGACACAGAGTCAGTCCCGACCCCTCAACC AATCGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGTGCCGATGGAGTGGGTAGTTCCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCACCTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGG ATTCGGCCTAAGCGACTCACTTCAAGCTCTTCAACATT

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCGTTCCAGCGGACGTTTTCATGA TTCTCTAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTTCGTCTTTTACTGCTGGAATATTTTC CCGTCGCAATGCTAAGAACGGGTAACTTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAAACGCTAAAATTAGCGTGGCCGG ACCCAGCAACATGGCTGTCAGGGAAGAAATACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTCTCTTTGTCTGGATCTTAAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAGTACTAACCCTGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGAGGGGACTTCGTGGGCGTGGCAGGCGCAGACC GGCTGGGTCAAAACCAAGGAATACTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTTGGGC CAAAATTCCTCACACGGACGGCACTTTCACCTTCTCCG CTAATGGGAGGTTTGGAAATGAAGCACCCTCTCTCAGA TCCTCATCAAAACACACCTGTACCTGCGGATCTCCAAC GGCTTTCAATAAGGACAGCTGAAGTCTTTCATACCCAG TATTCTACTGGCCAGTCAAGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAATATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGATATAGTGAACCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
109	Hu32 variant 15	MAADGYLPDWLEDTLSEGIHQWKLKPGPPPKPAERHKD DSRGLVLPQYKYLPGNGLDKGEPVNAADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRVAVFQ AKRLLLEPLGLVEEAAKTAPGKRPVEQSPQEPDSSAGIG KSGSQPAKKKLNFGQTGDTESVPDPQPIGEPPAAPSGVGS LTMASSGGAPVADNNEGADGVSSSGNWHCDSQLGDRV I TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRDPQRLINNNWGFPRKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHSQLDRLMNPLI DQYLYYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNS EFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM I TNEEEIKTINPVATESYQVATNHQSAQRGDYTLQAQQT GWVQNQILPGMWQDRDVLQGP IWKIPHTDGNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKKLNSFITQ YSTGQVSVEIEWELQKENS KRWNP EIQYTSNYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
110	Hu32 variant 15	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCCAACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCACGACAAGCCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCCTGTAGA GCAGTCTCTCAGGAACCGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAAATCAATTTCTG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCCTCCCGACGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGACAGAGTCATC ACCACAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCACAGCACATCTGGAGG

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		ATCTTCAAATGACAACGCCTACTTCGGCTACAGCACCCC TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCCAGCGGACGTTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTCGTTTCGTCTTTTACTGCCTGGAATATTTC CCGTCGCAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACCGTAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGGAGGACCGTT TCTTTCCTTTGTCTGGATCTTTAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATTAACCTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCAACAGAGTGC CCAAAGAGGCGACTACACCCTGTCTGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCAATTTGGGC CAAAATCTCTCACACGGACGGCAACTTACCCTTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCTCCTCAGA TCCTCATCAAAAACACCTGTACCTGCCGATCCTCCAAC GGCTTCAATAAGGACAAGCTGAACCTTTTATCACCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAATATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCCGCCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
111	Hu32 variant 16	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPYKYLGPNGLDKGEVNAADAAAEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRVAFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPDQPIGEPPAAPSQVGS LTMASGGGAPVADNNNEGADGVSSSGNWHDCSQWLGDRVI TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFCHFSRPRDQRLINNWFGRPKRLNFKLFNI QVKEVTDNNGVKTIANNLTSTVQVFTSDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNPQFSYEFENVPFHSSYAHQSGLDRMLNPLI DQYLYLSKTINGSGQNQQLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGVQVATNHQSAQSRGDFNLAQAQT GWVQNGILPGMVWQDRDVYLQGPWAKIPHDTGDNFHPSP LMGGFGMKHPPQILIKNTVPVADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYKSNVVEF AVNTEGVSEPRPIGTRYLTRNL
112	Hu32 variant 16	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCAACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCGGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGACGCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCCACCAAGTGGCAGACA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAAACGCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCACGCGGACGTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTCGTCCCTTTTACTGCCTGGAATATTTT CCGTCGCAAAATGCTAAGAACGGGTAAACACTTCCAGTTCA GCTACGAGTTTGAGAAGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATCTCTCAAAGACTATTAAACGGTT CTGGACAGAATCAACAAACGCTAAATTCAGCGTGGCGG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCAAAGAAGGAGGAGACCGTT TCTTTCCTTTGTCTGGATCTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATAAACTACTAACCCGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAATCTAGAGGCGACTTCAACCTGGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGTTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTGGGC CAAAATTCCTCACACGGACGGCAACTTTCACCTTCTCCG CTAATGGGAGGGTTTGGAAATGAAGCACCCGCTCCTCAGA TCCTCATCAAAAACACCTGTACCTGCGGATCCTCCAAC GGCTTTCATAAGGACAAGCTGAACCTTTTCATCACCCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAACATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCGCCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
113	Hu32 variant 17	MAADGYLPDWLEDTLSEGIQWQWKLKPGPPPKPAERHKD DSRGLVLPYKYLGPGNGLDKGEPVNAADAALEHDKAYD QQLKAGDNPYLYNHADAEPQERLKEDTSFGGNLGRAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVSSSGNWHCDSQWLGDRVI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNFRFHCHFSRPDQRLINNWFGRPKRLNFKLFNI QVKEVTDNNGVKTIANNLTSTVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDGSAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVFPHSSYAHQSQSLDRLMNPLI DQYLYYLSKTINGSGQNQQLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNS EFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYQVATNHQSAQRGDYVAQAQTGW VQNQGI LPMVWQDRDYLQGP I WAKIPHTDGNFHPSPLM GGFGMKHPPQILIKNTVPVADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYKSNNEFAV NTEGVYSEPRPIGTRYLTRNL
114	Hu32 variant 17	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCAAACC TGGCCCAACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCAGCACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCCAGGAGCGGCTCAAAGAAGA TACGTCTTTTGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCTTGAAAGAAGAGGCGCTGTAGA GCAGTCTCTCTAGGAACCGGACTCTCCGCGGGTATTGGC

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		AAATCGGGTTCACAGCCCGCTAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCCTCAACC AATCGGAGAACCCTCCCGAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCT TGGGGGTATTTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCGTTCACAGCGACGTTTTCATGA TTCCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTTCGTCTTTTACTGCCTGGAATATTTT CCGTGCAAAATGCTAAGAACGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAAACGCTAAAATTGAGCGTGGCCGG ACCCAGCAACATGGCTGTCAGGGAAGAAATACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTTGGCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTTCCTTTGCTCTGGATCTTTAATTTTGGCAAAACAAGG AACTGGAAGAGACAACCTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAGTACTAACCCTGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAAGAGGGGACTACGTGGCACAGGCGCAGACCGGCTGG GTTCAAAACCAAGGAATACTTCCGGGTATGGTTTGGCAGG ACAGAGATGTGTACCTGCAAGGACCCATTTGGGCCAAAAT TCCTCACCGGACGGCAACTTTCACCTTCTCGCTAATG GGAGGGTTTGAATGAAGCACCCGCTCTCAGATCCTCA TCAAAACACACCTGTACCTGCGGATCCTCAACGGCTTT CAATAAGGACAAGCTGAACCTTTTCATCACCAGTATTCT ACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGCTGCAGA AGGAAAACAGCAAGCGCTGGAACCCGAGATCCAGTACAC TTCCAACTATTACAAGCTAATAATGTTGAATTTGCTGTT AATACTGAAGGTGTATATAGTGAACCCGCCCATTTGGCA CCAGATACCTGACTCGTAATCTGTAA
115	Hu32 variant 18	MAADGYLPDWLEDTLSEGIROWWKLKPGPPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEVPVNAADAAALEHDKAYD QQLKAGDNPYLKYNHADAEPQERLKEDTSFGNLRGAVFQ AKRLLLEPLGLVEEAAKTAPGKRPEVQSPQEPDSSAGIG KSGSQPAKKKLNFGQTGDTESVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVGSSSGNWHCDSQWLGDRI TTSRTWALPTYNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFHCHFSRWDQRLINNNWGRPKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLDGSGQAVGRSSFYCLEYF PSQMLRTGNNQFSYEFENVPFHSSYAHQSGLDRLMNPLI DQYLYLSKTINGSGQNQOTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTONNNSFARPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTINPVATESYQVATNHQSAQAQATGWVQNGG ILPGMVWQDRDVYLQGPWAKIPHDTGNFHPSPLMGGFGM KHPPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVS VEIEWELQKENSkrwnPEIQYTSNYYKSNNVEFAVNTGEGV YSEPRPIGTRYLTRL
116	Hu32 variant 18	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCCAACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCACGACAAGGCCTACGAC CAGCAGCTCAAGGCCGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAGAAGA

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		TACGTCCTTTGGGGGCAACCTCGGGCGAGCAGTCTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCCTCAGGAACCGACTCCTCCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTCG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCCTTACAAGCAAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGTCTCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCCTCCCGCCGTTCACAGCGGACGTTTTCATGA TTCTCTAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGTTCTGCTCTTTTACTGCTGGAATATTTC CCGTCGCAATGCTAAGAACGGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACCGCTAAATTCAGCGTGGCCGG ACCCAGCAACATGGCTGTCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAAACAACAACAGCGAATTTGCTCGACCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGGAGACCGTT TCTTCTCTTTGCTCTGGAATCTTAATTTTGGCAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAGTCATGATA ACCAACGAAGAAGAAATTAAGCTACTAACCCGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCAACAGAGTGC CCAAGCACAGGCGCAGACCGGCTGGGTTCAAAACCAAGGA ATACTTCCGGGTATGGTTTGGCAGGACAGAGATGTGTACC TGCAAGGACCCATTTGGGCCAAAATTCCTCACACGGACGG CAACTTTCACCTTCTCCGCTAATGGGAGGGTTTGAATG AAGCACCCGCCTCCTCAGATCCTCATCAAAACACACCTG TACCTGCGGATCCTCAACGGCTTCAATAAGGACAAGCT GAACCTTTTCATCACCCAGTATTCTACTGGCCAAGTCAGC GTGGAGATTGAGTGGGAGCTGCAGAAGGAAAACAGCAAGC GCTGGAACCCCGGAGATCCAGTACACTTCCAATATTACAA GTCTAATAATGTTGAATTTGCTGTTAATACTGAAGGTGTA TATAGTGAACCCCGCCCATTTGGCACCAGATACCTGACTC GTAATCTGTAA
117	Hu32 variant 19	MAADGYLPDWLEDTLSEGIQWKLKPGPPPPKPAERHKD DSRGLVLPGYKYLPGNGLDKGEFVNADAAALEHDKAYD QQLKAGDNPYLKYNHADAQERLKEDTSFSGNLRGAVFQ AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFHCHFSRDPWQRLINNNWGFPRKRLNFKLFNI QVKEVTDNNGVKTIANNLSTVQVFTDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVFPHSSYHSQSLDRMLNPLI DQLYYLSKTINGSGQNQTLKFSVAGPSNMAVQGRNYP GPSYRQQRVSTTVTQNNSEFARPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYQVATNHQSAQSTTLSPAQQT GWVQNQILPGMVWQDRDVLQGPWAKIPIHTDGNFHPSP LMGGFGMKHPPQILIKNTPVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSKRWNPEIQYTSNYKSNNVEF AVNTEGVYSEPRPIGTRYLTRNL
118	Hu32 variant 19	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAGTACCTCG

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGGCGGCCCTCGAGCAGACAAGGCCTACGAC CAGCAGCTCAAGGCCGGAGACAACCCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCCTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGCTAAGACGGCTCCTGGAAAGAAGAGGCCTGTAGA GCAGTCTCCTCAGGAACCGGACTCCTCCGCGGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCGACCTCAACC AATCGGAGAACCTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGCACCAGTGGCAGACA ATAACGAAGGTGCCGATGGAGTGGGTAGTTCTCGGGAAA TTGGCATTGCGATTCCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCACAGCACATCTGGAGG ATCTTCAATGACAACGCCCTACTTCGGCTACAGCACCCCC TGGGGGTATTTGACTTCAACAGATTCACCTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACCTGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTCAGACTATCAGCTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCCGCCGTTCACAGCGGACGTTTTTCATGA TTCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTCGTCCTTTTACTGCCTGGAATATTTT CCGTCGCAAAATGCTAAGAACGGGTAAACAATTCAGTTCA GCTACGAGTTTGAGAAGTACCTTTCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTACTATCTCTCAAAGACTATTAACGGTT CTGGACAGAATCAACAACGCTAAAATTACGCTGGCGCG ACCCAGCAACATGGCTGTCCAGGGAAGAACTACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTCGACCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAAGAAGGAGGAGACCGTT TCTTTCCTTTGTCTGGATCTTAATTTTGGCAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAGACTTAACCCGGTAGCAA CGGAGTCTATGGACAAGTGGCCACAACCACAGAGTGC CCAATCAACGACGCTCTATTCTCCAGCACAGGCGCAGACC GGCTGGGTTCAAAACCAAGGAATACTTCCGGGTATGGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTGGGC CAAAATTCCTCACACGGACGGCAACTTTCACCTTCTCCG CTAATGGGAGGGTTTGAATGAAGCACCCGCTCCTCAGA TCCTCATCAAAAACACACCTGTACCTGCGGATCCTCAAC GGCTTCAATAAGGACAAGCTGAACTCTTTCATCACCAG TATTCTACTGGCCAAAGTCAAGCTGGAGATTGAGTGGGAGC TGCAGAAGGAAAAACAGCAAGCGCTGGAACCCGAGATCCA GTACACTTCCAACATTACAAGTCTAATAATGTTGAATTT GCTGTTAATACTGAAGGTGTATATAGTGAACCCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
119	Hu32 variant 20	MAADGYLPDWLEDTLSEGIQWKKPKGPPPPKPAERHKD DSRGLVLPGYKYLGPNGLDKGEVNAADAAALEHDKAYD QQLKAGDNPYLYKNHADAEEFQERLKEDTSFGGNLGRAVFO AKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGSQPAKKLNFQGTGDTESVDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNNEGADGVSSSGNWHCDSQWLGRDVI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYFDNRFCHFSRDPQRLINNNGWFRPKRLNFKLFNI QVKEVTDNNGVKTIANLSTVQVFTSDYQLPYVLGSAH EGCLPPFPADVFMIPOYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVFPHSSYAHQSGLDRMLNPLI DQYLYYLSKTINGSGQNQOTLKFSVAGFSNMAVQGRNYIP GPSYRQQRVSTTVTQNNSEFARPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEIKTNPVATESYQVATNHQSAQFVVGQSYAQAT GWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSP LMGGFGMKHPPQILIKNTVPADPPTAFNKDKLNSFITQ YSTGQVSVEIEWELQKENSkrwnPEIQYTSNYKSNVVEF AVNTEGVYSEPRPIGTRYLTRNL

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SEQ ID NO.	DESCRIPTION	SEQUENCE
120	Hu32 variant 20	ATGGCTGCCGATGGTTATCTTCCAGATTGGCTCGAGGACA CTCTCTCTGAAGGAATAAGACAGTGGTGAAGCTCAAACC TGGCCACCACCACCAAGCCCGCAGAGCGGCATAAGGAC GACAGCAGGGGTCTTGTGCTTCTGGGTACAAGTACCTCG GACCCGGCAACGGACTCGACAAGGGGGAGCCGGTCAACGC AGCAGACGCGCGGCCCTCGAGCACGACAAGGCTACGAC CAGCAGCTCAAGGCCGAGACAACCGTACCTCAAGTACA ACCACGCCGACGCCGAGTTCAGGAGCGGCTCAAAGAAGA TACGTCTTTGGGGGCAACCTCGGGCGAGCAGTCTTCCAG GCCAAAAAGAGGCTTCTTGAACCTCTTGGTCTGGTTGAGG AAGCGGCTAAGACGGCTCCTGGAAAGAAGAGGCTGTAGA GCAGTCTCTCAGGAACCGGACTCCTCCGCGGTATTGGC AAATCGGGTTCACAGCCCGCTAAAAAGAACTCAATTTCG GTCAGACTGGCGACACAGAGTCAGTCCCCGACCCTCAACC AATCGGAGAACTCCCGCAGCCCCCTCAGGTGTGGGATCT CTTACAATGGCTTCAGGTGGTGGCGACCAAGTGCAGACA ATAACGAAGTGCCGATGGAGTGGGTAGTTCCTCGGGAAA TTGGCATTGCGATTCCAATGGCTGGGGGACAGAGTCATC ACCACCAGCACCCGAACCTGGGCCCTGCCACCTACAACA ATCACCTCTACAAGCAATCTCCAACAGCACATCTGGAGG ATCTTCAAATGACAACGCCCTACTTCGGCTACAGACCCCC TGGGGGTATTTGACTTCAACAGATTCCACTGCCACTTCT CACCACGTGACTGGCAGCGACTCATCAACAACAACTGGGG ATTCCGGCCTAAGCGACTCAACTTCAAGCTCTTCAACATT CAGGTCAAAGAGGTTACGGACAACAATGGAGTCAAGACCA TCGCCAATAACCTTACCAGCACGGTCCAGGTCTTCACGGA CTGAGACTATCAGTCCCGTACGTGCTCGGGTCGGCTCAC GAGGGCTGCCTCCGCGCTTCCAGCGGACGTTTTCATGA TTCCTCAGTACGGGTATCTGACGCTTAATGATGGGAGCCA GGCCGTGGGTGCTGCTCCTTTTACTGCCTGGAATATTTT CCGTCGCAAAATGCTAAGAACGGTAACAACCTCCAGTTCA GCTACGAGTTTGAGAACGTACCTTTCCATAGCAGCTACGC TCACAGCCAAAGCCTGGACCGACTAATGAATCCACTCATC GACCAATACTTGTAATACTCTCAAAGACTATTAACGTTT CTGGACAGAATCAACAACGCTAAAATTACGCTGGCCGG ACCCAGCAACATGGCTGTCAGGGAAGAAATACATACCT GGACCCAGCTACCGACAACAACGTGTCTCAACCACTGTGA CTCAAAACAACAACAGCGAATTTGCTCGACCTGGAGCTTC TTCTTGGGCTCTCAATGGACGTAATAGCTTGATGAATCCT GGACCTGCTATGGCCAGCCACAAGAAGGAGAGGACCGTT TCTTTCCTTTGCTGGATCTTTAATTTTGGCAAAACAAGG AACTGGAAGAGACAACGTGGATGCGGACAAAGTCATGATA ACCAACGAAGAAGAAATTAAGTACTAACCCTGGTAGCAA CGGAGTCTTATGGACAAGTGGCCACAACCACAGAGTGC CCAATTGCTTGTGGGACAGTCTATGACAGGCGCAGACC GGCTGGGTCAAACCAAGGAATACTTCGGGTATGTTT GGCAGGACAGAGATGTGTACCTGCAAGGACCCATTTGGGC CAAAATTCCTCACACGGACGGCAACTTACCCCTTCTCCG CTAATGGGAGGTTTGAATGAAGCACCCTCTCTCAGA TCCTCATCAAAACACACCTGTACCTGCGGATCTTCCAAC GGCTTCAATAAGGACAAGCTGAAGTCTTTCATCACCCAG TATTCTACTGGCCAAGTCAGCGTGGAGATTGAGTGGGAGC TGCAGAAGGAAAACAGCAAGCGCTGGAACCCGGAGATCCA GTACACTTCCAACATATTACAAGTCTAATAATGTTGAATTT GCTGTAAATACTGAAGGTGTATATAGTGAACCCCGCCCA TTGGCACCAGATACCTGACTCGTAATCTGTAA
121	AAV9/hu32 consensus N- terminal sequence	MAADGYLPDWLEDXLSEGIKXWWXKPGXPXPKXXXXHXD XXRGLVLPGYKYLGPNG [14X = N or T; 21X = E or Q; 24X = A or K; 29X = A or P; 31X = Q or P; 33X = A or P; 34X = A or P; 35X = N or A; 36X = Q or E; 37X = Q or R; 39X = Q or K; 41X = N or D; and 42X = A or S]
122	AAV9 N-terminal sequence	MAADGYLPDWLEDNLSEGIKXWWXKPGAPQPKANQQHQD NARGLVLPGYKYLGPNG
123	Hu32 N-terminal sequence	MAADGYLPDWLEDTLSEGIKXWWXKPGPPPPKPAERHKD DSRGLVLPGYKYLGPNG
124	AAV9	MAADGYLPDWLEDNLSEGIKXWWXKPGAPQPKANQQHQD NARGLVLPGYKYLGPNGLDKGEVNAADAALHDKAYD

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		QQLKAGDNPYLKYNHADAEFQERLKEDTSFGGNLGRAVFQ AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTGDTESVPDPQPIGEPPAAPSGVGS LTMASSGGAPVADNNNEGADGVSSSGNWHCDSQWLGDRI TTSTRTWALPTYNNHLYKQISNSTSGGSNDNAYFGYSTP WGYPDFNRFHCHFSRDPWQRLINNNGWFRPKRLNFKLFNI QVKEVTDNNGVKTIANNLTSTVQVFTSDSYQLPYVLGSAH EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYF PSQMLRTGNNFQFSYEFENVPFHSSYAHSQSLDRMLNPLI DQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP GPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVM TNEEEIKTNPVATESYGVATNHQSAQAQATGWVQNOG ILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSPLMGGFGM KHPPFQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVS VEIEWELQKENSQRWNPEIQYTSNYYKSNNVFAVNTBVG YSEPRPIGTRYLTRNL
125	AAV9	atggctgccg atggttatct tccagattgg ctcgaggaca accttagtga aggaattcgc gagtggtggg ctttgaacc tggagccct caacccaagg caaatcaaca acatcaagac aacgctcgag gtcttgtgct tccgggttac aaataccttg gacccggcaa cggactcgac aagggggagc cggtaacgc agcagacgcg gcggccctcg agcacgacaa ggcctacgac cagcagctca aggcgggaga caaccctac ctcaagtaca accacgccga cgcgagttc caggagcggc taaaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag gccaaaaaga ggcttcttga acctcttggt ctggttgagg aagcggctaa gacggctcct ggaaagaaga ggcctgtaga gcagtctcct caggaaccgg actcctccgc gggatttggc aaatcgggtg cacagccgc taaaaagaga ctcaatttcg gtcagactgg cgacacagag tcagtccag accctcaacc aatcggagaa cctcccgag cccctcagg tgtgggatct cttacaatgg ctccaggtgg tggcgaccca gtggcagaca ataacgaagg tgccgatgga gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac accaccagca cccgaacctg ggcctgcc acctacaaca atcacctcta caagcaaatc tccaacagca catctggagg atottcaaat gacaacgcct acttcggcta cagcaccccc tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc acggtccagg tottcacgga ctcagactat cagctcccg acgtgctcgg gtccgctcac gagggctgcc tcccgccgtt cccagcggac gtttcatga ttctcagta cgggtatctg acgcttaatg atggaagcca ggcgtgggt cgttcgtcct ttactgcct ggaatatctc ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagtt tgagaacgta ctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcac gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg ctaaaattca gtgtggccgg acccagcaac atggctgtcc agggaagaaa ctacatacct ggacccagct accgacaaca acgtgtctca accactgtga ctcaaaacaa caacagcgaa tttgcttggc ctggagcttc ttcttgggct ctcaatggac gtaatagctt gatgaatcct ggacctgcta tggccagcca caaagaagga gaggaccgtt totttccttt gtctggatct ttaatttttg gcaacaagg aactggaaga

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		gacaacgtgg atgcggacaa agtcatgata accaacgaag aagaaattaa aactactaac ccggtagcaa cggagtccta tggacaagtg gccacaaacc accagagtgc ccaagcacag gcgcagaccg gctgggttca aaaccaagga atacttccgg gtatggtttg gcaggacaga gatgtgtacc tgcaaggacc catttgggcc aaaattcctc acacggacgg caactttcac ccttctccgc tgatgggagg gtttgggaatg aagcacccgc ctctcagat cctcatcaaa aacacacctg tacctgcgga tectccaacg gccttcaaca aggacaagct gaactctttc atcaccagat attctactgg ccaagtcagc gtggagatcg agtgggagct gcagaaggaa aacagcaagc gctggaaccc ggagatccag tacacttcca actattacaa gtctaataat gttgaatttg ctgttaatac tgaagggtga tatagtgaac cccgccccat tggcaccaga tacctgactc gtaactctga a
126	AAV1	MAADGYLPDWLEDNLSEGIREWDLKPGAPKPKANQQK QDDGRGLVLPGYKYLGPFGLDKGEPVNAADAAALEHD KAYDQQLKAGDNPYLRYNHADAEPQERLQEDTSFGGNL GRAVFQAKKRVLEPLGLVEEGAKTAPGKKRPVEQSPQE PDSSSGIGKTGQQPAKKRLNFGQTGDSSEVPDPQPLGE PPATPAAVGPTTMASGGGAPMADNNEGADGVGNASGNW HCDSTWLGDRIITSTRTALPTYNHLYKQISSASTG ASNDNHYFGYSTPWGYFDNRFHCHFSRDRWQRLINNN WGRPKRLNFKLFNIQVKEVTNDGVTTIANNLSTVQ VFSDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLT LNNGSQAVGRSSFYCLEYFPSQMLRTGNNTFSYTFEE VPFHSSYAHSQSLDRLMNPLIDQYLYLNRTONQSGSA QNKDLLFSRGSFAGMSVQPKNLWPGPCYRQQRVSKTKT DNNNSNFTWTGASKYNLNGRESIINPGTAMASHKDDDED KFFPMSGVMIFGKESAGASNTALDNVMTDEEEIKATN PVATERFGTVAVNFQSSSTD PATGDVHAMGALPGMVWQ DRDVYLQGPWAKIPHTDGHFHPSPLMGGFGLKNPPQ ILIKNTVPANPPAEFSATKFASFITQYSTGQVSVEIE WELQKENSKRWNPEVQYTSNYAKSANVDFTVDNNGLYT EPRPIGTRYLTRPL
127	AAV2	MAADGYLPDWLEDTLSEGIQWVKLKPGLPPPKPAERH KDDSRGLVLPGYKYLGPFGLDKGEPVNEADAAALEHD KAYDRQLDSGDNPYLYKNHADAEPQERLKEDTSFGGNL GRAVFQAKKRVLEPLGLVEEPVKTAPGKKRPVEHSPVE PDSSSGTGKAGQQPARKRLNFGQTGDADSVDPDPQLGQ PPAAPSGLGNTMATGSGAPMADNNEGADGVGNSSGNW HCDSTWMDRIITSTRTALPTYNHLYKQISSQSGA SNDNHYFGYSTPWGYFDNRFHCHFSRDRWQRLINNNW GFRPKRLNFKLFNIQVKEVTQNDGTTTTIANNLSTVQV FTDSEYQLPYVLGSAHQGCLPPFPADVFMVFPQYGYLTL NNGSQAVGRSSFYCLEYFPSQMLRTGNNTFSYTFEDV PPFHSSYAHSQSLDRLMNPLIDQYLYLSRTNTPSGTTT QSRLQFSQAGASDIRDQSRNLPGPCYRQQRVSKTSAD NNNSEYSWTGATKYHLNGRDSLVPNGPAMASHKDDEEK FFPQSGVLIFGKQSGSEKTNVDIEKVMITDEEEIRTTNP VATEQYGSVSTNLQRGNRQAATADVNTQGVLPGMVWQD RDVYLQGPWAKIPHTDGHFHPSPLMGGFGLKHPPQI LIKNTVPANPSTTFSAAKFASFITQYSTGQVSVEIEW ELQKENSKRWNPEIQYTSNYKSNVNDFTVDINGVYSE PRPIGTRYLTRNL
128	AAV3	MAADGYLPDWLEDNLSEGIREWALKPGVPQPKANQQH QDNRRLVLPGYKYLGPFGLDKGEPVNEADAAALEHD KAYDQQLKAGDNPYLKYNHADAEPQERLQEDTSFGGNL GRAVFQAKKRILEPLGLVEEAAKTAPGKKGAVDQSPQE PDSSSGVGKSGKQPAKKRLNFGQTGDSSEVPDPQPLGE PPAAPTSLGSNTMASGGGAPMADNNEGADGVGNSSGNW HCDSQWLGDRIITSTRTALPTYNHLYKQISSQSGA SNDNHYFGYSTPWGYFDNRFHCHFSRDRWQRLINNNW GFRPKKLSFKLFNIQVRGVTONDGTITIANNLSTVQV FTDSEYQLPYVLGSAHQGCLPPFPADVFMVFPQYGYLTL NNGSQAVGRSSFYCLEYFPSQMLRTGNNFQFSYTFEDV

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SEQ ID NO.	DESCRIPTION	SEQUENCE
		PFHSSYAHQSQSLDRLMNPLIDQYLYYLNRTQGTTSGTT NQSRLLFSQAGPQMSLSLQARNWLPGPCYRQQRLSKTAN DNNNSNFPWTAASKYHLNGRDSLVPNGPAMASHKDDEE KFFPMHGNLIPGKEGTTASNAELDNVMI TDEEEIRTTN PVATEQYGTVANNLQSSNTAPTGTGNHQQGALPGMVWQ DRDVYLQGGPIWAKIPHTDGHFHPSPLMGGFGLKHPPQ IMIKNTPVPANPPTTFSPAKFASFITQYSTGQVSVEIE WELQKENSKRWNPEIQYTSNYNKSNNVDFTVDINGVYS EPRPIGTRYLTRNL
129	AAV4	MTDGYLPDWLEDNLSSEGVREWWALQPGAPKPKANQQHQ DNARGLVLPGYKYLGPNGGLDKGEPVNAADAAALEHDK AYDQQLKAGDNPYLYKNHADAEPQORLQGDTSFGNGLG RAVFQAKKRVLEPLGLVEQAGETAPGKKRPLIESPQQP DSSTGIGKKGQPAKKKLVEFEDTGAGDGPPEGSTSGA MSDDSEMRAAAGGAAVEGGQGADGVGNASGDWHCDSTW SEGHVTTTSTRTWLPTYNHLYKRLGESLQSNNTYNGF STPWGYFDNRFHCHFSPRDWQRLINNNWGMRPKAMRV KIFNIQVKEVTSNGETTVANNLTSTVQIFADSSYELP YVMDAGQEGSLPPFPNDVFMVPPQYGYCGLVTGNTSQQQ TDRNAFYCLEYFPPSOMLRTGNNFEITYSEKVPFHSMY AHSQSLDRLMNPLIDQYLWGLQSTTTGTTLNAGTATTN FTKLRPTNFSNFKKNWLPGPSIKQQGFSKTANQNYKIP ATGSDSLIKYETHSTLDGRWSALTGPPMATAGPADSK FSNSQLIFAGPKQNGNTATVPGLTFTSEELAAATNAT DTDWGNLPGGDQSNLPTVDRLTALGAVPGMVWQNR DIYYQGPWAKIPHTDGHFHPSPLIGGFGLKHPPQIF IKNTPVPANPATTFSSSTPVNSFITQYSTGQVSVDWE IQKERSKRWNPEVQFTSNYQQNSLLWAPDAAGKYTEP RAIGTRYLTHHL
130	AAV6	MAADGYLPDWLEDNLSSEGIREWDLKPGAPKPKANQQK QDDGRGLVLPGYKYLGPFGNGLD KGEVNAADAAALEHDKAYDQQLKAGDNPYLYKNHADA EFQERLQEDTSFGNGLGRAVFQAKKRVLEPPGLVEEGA KTAPGKKRPVEQSPQEPDSSGIGKTQQQPAKKRLNFG QTGDSSESVPDPQPLGEPATPAAVGPTTMASGGGAPMA DNNEGADGVGNASGNWHCDSTWLGDRVITTTSTRTWALP TYNHLYKQISSASTGASNDNHYPGYSTPWGYFDNRF HCHFSPRDWQRLINNNWGRPKRLNFKLFNIQVKEVTT NDGVTTIANNLTSVQVPSDSEYQLPYVLGSAHQGCLP PFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEYFPSQ MLRTGNNFTFSYTFEDVPPHSSYAHQSQSLDRLMNPLID QYLYYLNRTQNSGSAQNKDLLFSRGSPAGMSVQPKNW LPGPCYRQQRVSKTKTDNNNSNFTWTGASKYNLNGRES IINPGTAMASHKDDKDKFFPMGVMIFGKESAGASNTA LDNVMI TDEEEIKATNPVATERFGTVAVNLQSSSTDPA TGDVHVMGALPGMVWQDRDVYLQGGPIWAKIPHTDGHF PSPLMGGFGLKHPPQILIKNTPVPANPPAEFSATKFA SFITQYSTGQVSVEIEWELQKENSKRWNPEVQYTSNY AKSANVDFTVDDNNGLYTEPRPIGTRYLTRPL
131	AAV7	MAADGYLPDWLEDNLSSEGIREWDLKPGAPKPKANQQK QDNGRGLVLPGYKYLGPFGNGLDKGEPVNAADAAALEHD KAYDQQLKAGDNPYLYKNHADAEPQERLQEDTSFGNGL GRAVFQAKKRVLEPLGLVEEAKTAPAKKRPVEPSPQR SPDSTGIGKKGQPAKKRLNFGQTGDSSESVPDPQPLG EPPAAPSSVSGSTVAAGGGAPMADNNEGADGVGNASGN WHCDSTWLGDRVITTTSTRTWALPTYNHLYKQISSETA GSTNDNTYFGYSTPWGYFDNRFHCHFSPRDWQRLINN NWGFRPKKLRFKLFNIQVKEVTNDGVTTIANNLSTI QVFSDESLEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYL TLNNGSQSVGRSSFYCLEYFPSOMLRTGNNFEFSYSFE DVPPHSSYAHQSQSLDRLMNPLIDQYLYLARTQSNPGG TAGNRELQFYGGPSTMAEQAKNWLPGPCFRQQRVSKT LDQNNNSNFAWTGATKYHLNGRNSLVNPGVAMATHKDD EDRFFPSSGVLIFGKTGATNKTTLENVMTNEEIRPT NPVATEEYGIVSSNLQAANTAAQTQVNNQGALPGMVW QNRDVYLQGGPIWAKIPHTDGNFHPSPLMGGFGLKHPPP QILIKNTPVPANPPEVTPAKFASFITQYSTGQVSVEI EWELQKENSKRWNPEIQYTSNFEKQTVDFAVDSQGVY SEPRPIGTRYLTRNL

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SEQ ID NO.	DESCRIPTION	SEQUENCE
132	AAVhu31	MAADGYLPDWLEDTLSEGI RQWWK LKPGPPPPKPAERH KDDSRGLVLPGYKYLGPNGLDKGEPVNAADAAALEHD KAYDQQLKAGDNPYLKYNHADA EFQERLKEDTSFGGNL GRAVFQAKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQE PDSSAGIGKSGSQPAKKKLNFGQTDGTEVPDPQPIGE PPAAPSGVGS LTMASGGGAPVADNNEGADGVGSSSGNW HCDSQWL GDRVITTS TRTWALPTYNNHLYKQISNSTSG GSSNDNAYFGYSTPWGYFDNRFHCHFSRDPWQRLINN NWGFRPKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTV QVFTDS DYQLPYVLGSAHEGCLPPFPADV FMI PQYGYL TLNDGGQAVGRSSFYCLEYFP SQMLRTGN NFQFSYEF E NVFP HSSYAH SQSLDR LMNPLIDQYL YLSKT INGSGQ NQQT LKFSVAGPSNMAVQGRNYIPGPSYRQQRVSTTVT QNNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGED RFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTN PVATESYGOVATNHQSAQAQAGTGWVQNGILPGMVWQ DRDVYLQGP I WAKIPHTDGNFHPSPLMGGFGMKHPP PQ ILIKNT PVPADPPTAFNKDKLNSFITQYSTGQVSVEIE WELQKENS KRWNPEIQYTSNYKSNNV EFAVSTEGVYS EPRPIGTRYLTRNL
133	AAVrh10	MAADGYLPDWLEDNLSEGI REWDLKPGAPKPKANQQK QDDGRGLVLPGYKYLGPFNGLDKGEPVNAADAAALEHD KAYDQQLKAGDNPYLRYNHADA EFQERLQEDTSFGGNL GRAVFQAKKRVLLEPLGLVEEAGKTAPGKKRPVEPSPQR SPDSSTGIGKKGQQA KKR LNFNGQTDGSEVPDPQPIG EPPAGPSGLSGGTMAAGGAPMADNNEGADGVGSSSGN WHCDSTWL GDRVITTS TRTWALPTYNNHLYKQISNGTS GGSTNDNTYFGYSTPWGYFDNRFHCHFSRDPWQRLIN NNWGFRPKRLNFKLFNIQVKEVTQNEGKTIANNLTST IQVFTDS EYQLPYVLGSAHQCLPPFPADV FMI PQYGY LTLNNGSQAVGRSSFYCLEYFP SQMLRTGN NFESYQF EDVP HSSYAH SQSLDR LMNPLIDQYL YLSRTQSTGG TAGTQQLLFSQAGPNMSAQAKNWLPGPCYRQQRVSTT LSQNNNSNFAWTGATKYHLNGRDSLVPGVAMATHKDD EERFPSSGVL MFGKQAGKDNVDYSSVMLTSEEEIKT TNPVATEQYGVVADNLQQQNAAPIVGAVNSQGALPGMV WQNRDVYLQGP I WAKIPHTDGNFHPSPLMGGFG LKHPP PQILIKNT PVPADPPTTFSQAKLASFITQYSTGQVSVE IEWELQKENS KRWNPEIQYTSNYKSTNVDFAVNTDGT YSEPRPIGTRYLTRNL
146	AAVhu32 VP1u (amino acids 1-137)	MAADGYLPDWLEDTLSEGI RQWWK LKPGPPPPKPAERH KDDSRGLVLPGYKYLGPNGLDKGEPVNAADAAALEHD KAYDQQLKAGDNPYLKYNHADA EFQERLKEDTSFGGNL GRAVFQAKKRLLLEPLGLVEEAAK
147	AAVhu32 VP1u	ATGGCTGCCGATGGTTATCTTCAGATTGGCTCGAGGA CACTCTCTCTGAAGGAATAAGACAGTGGTGGAAGCTCA AACCTGGCCCAACCACCAAGCCCGCAGAGCGGCAT AAGGACGACAGCAGGGGTCTTGTGCTTCCTGGGTACAA GTACCTCGGACCCGGCAACGGACTCGACAAGGGGAGC CGGTCAACGCAGCAGACGCGCGGCCCTCGAGCACGAC AAGGCC TACGACCAGCAGCTCAAGGCCGAGACAAACC GTACCTCAAGTACAACACGCCGACGCCGAGTTCAGG AGCGGCTCAAAGAAGATACGTCTTTTGGGGCAACCTC GGCGAGCAGTCTTCAGGCCAAAAGAGGCTCTTGA ACCTCTTGGTCTGGTTGAGGAAGCGGCTAAG
148	AAVhu32 VP1/2s (amino acids 138-202)	TAPGKKRPVEQSPQEPDSSAGIGKSGSQPAKKKLNFGQ TGDTEVPDPQPIGEPPAAPSGVGS LT
149	AAVhu32 VP1/2s	ACGGCTCCTGGAAAGAAGAGGCCTGTAGAGCAGTCTCC TCAGGAACCGGACTCCTCCGCGGTATTTGGCAAATCGG GTTACAGCCCGCTAAAAGAACTCAATTTCCGGTCAG ACTGGCGACACAGAGTCAGTCCCGACCTCAACCAAT CGGAGAACCTCCCGCAGCCCTCAGGTGTGGGATCTC TTACA

8. EQUIVALENTS AND INCORPORATIONS BY REFERENCE

[0331] Although the invention is described in detail with reference to specific embodiments thereof, it will be understood that variations which are functionally equivalent are within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description and accompanying drawings. Such modifications are intended to fall within the scope

of the appended claims. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following claims.

[0332] All publications, patents and patent applications mentioned in this specification are herein incorporated by reference into the specification to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference in their entireties.

SEQUENCE LISTING

Sequence total quantity: 151

SEQ ID NO: 1 moltype = AA length = 532
 FEATURE Location/Qualifiers
 source 1..532
 mol_type = protein
 note = AAV5 VP3 capsid protein
 organism = Adeno-associated virus

SEQUENCE: 1
 MSAGGGGGLG DNNQADGVG NASGDWHCDS TWMGDRVVK STRTWVLPSP NNHQYREIKS 60
 GSVDDGSNANA YFGYSTPWGY FDFNRFHSHW SPRDQWRLIN NYWGFRRPSL RVKIFNIQVK 120
 EVTVQDSTTT IANNLTSTVQ VFTDDDYQLP YVVGNGTEGC LPAPFPQVFT LPQYGYATLN 180
 RDNTENPTER SSFFCLEYFP SKMLRTGNMF EFTYNFEEVP FHSSFAPSQN LFKLANPLVD 240
 QYLYRFVSTN NTGGVQFNKN LAGRYANTYK NWFPGPMGRT QGWNLGSGVN RASVSATFAT 300
 NRMELEGASY QVPPQPNGMT NNLQGSNTYA LENTMIFNSQ PANPGTTATY LEGNMLITSE 360
 SETQPVNRVA YNVGGQMATN NQSSTTAPAT GTYNLQEIIV GSVWMERDVY LQGPIWAKIP 420
 ETGAHFHPS AMGGFGLKHP PPMLIKNTP VPGNITSFSD VPVSSFITQY STGQVTVE ME 480
 WELKKENSKR WNPEIQYTNN YNDPQFVDFA PDSTGEYRTT RPIGTRYLTR PL 532

SEQ ID NO: 2 moltype = DNA length = 1599
 FEATURE Location/Qualifiers
 source 1..1599
 mol_type = genomic DNA
 note = AAV5 VP3 capsid DNA
 organism = Adeno-associated virus

SEQUENCE: 2
 atgtctgcgg gaggtggcgg cccattgggc gacaataacc aaggtgccga tggagtgggc 60
 aatgcctcgg gagattggca ttgcgattcc acgtggatgg gggacagagt cgtcaccag 120
 tccaccgaa cctgggtgct gccagctac aacaaccacc agtaccgaga gatcaaaagc 180
 ggctccgctg acggaagcaa cgccaacgcc tactttggat acagcaccgc ctgggggtac 240
 ttgtgactta accgcttcca cagccactgg agccccgag actggcaaaag actcatcaac 300
 aactactggg gcttcagacc ccggtccctc agagtaaaaa tcttcaacat tcaagtcaaa 360
 gaggtcacgg tgcaggactc caccaccacc atcgccaaca acctcacctc caccgtccaa 420
 gtgtttacgg acgacgacta ccagctgccc tacgtcgtcg gcaacgggac cgagggtatc 480
 ctgcccgcct tcctccgca ggtctttacg ctgcccagtg acggttacgc gacgtgaac 540
 cgcgacaaca cagaaaatcc caccgagagg agcagcttct tctgcctaga gtactttccc 600
 agcaagaatgc tgagaacggg caacaacttt gagtttacct acaactttga ggaggtgcc 660
 ttccactcca gcttcgctcc cagtcagaac ctgttcaagc tggccaaccc gctggtggac 720
 cagtacttgt accgcttctg gagcacaat aactctggcg gactccagtt caacaagaac 780
 ctggccggga gatacgccaa cacctacaaa aactggttcc cggggcccat gggccgaacc 840
 cagggtcgga acctgggtcc cggggtcaac cgcccagtg tcagcgctt cgccacgacc 900
 aataggatgg agctcgaggg cgcgagtta cagggtgccc cgagccgaa cgcatgacc 960
 aacaacctcc agggcagcaa cacctatgcc ctggagaaca ctatgatctt caacagccag 1020
 ccggcgaacc cgggcaccac cgccacgtac ctcgagggca acatgctcat caccagcgag 1080
 agcgagacgc agccggtgaa ccgctgtggc tacaacgtcg gcgggcagat ggccaccaac 1140
 aaccagagct ccaccactgc ccccgcgacc ggcaactaca acctccagga aatcgtgcc 1200
 ggcagcgtgt gtagtgagag ggacgtgtac ctccaaggac ccatctgggc caagatccca 1260
 gagacggggg cgcaacttca cccctctccg gccatgggag gattcggact caaacacca 1320
 ccgcccata gaactcatca gaacacgcct gtgcccggaa atatcaccag cttctcggag 1380
 gtgcccgtca gcagcttcat caccagtagc agcaccgggc aggtcaccgt ggagatggag 1440
 tgggagctca agaagaaaa ctccaagagg tggaaaccag agatccagta cacaacaac 1500
 tacaacgacc ccagtttgtt ggactttgcc ccggacagca ccgggaata cagaaccacc 1560
 agacctatcg gaaccgata cttaccgga cccctttaa 1599

SEQ ID NO: 3 moltype = AA length = 56
 FEATURE Location/Qualifiers
 source 1..56
 mol_type = protein
 note = AAV5 VP1/2s common region
 organism = Adeno-associated virus

SEQUENCE: 3

-continued

TAPTGKRIDD HFPKRKKART EEDSKPSTSS DAEAGPSGSQ QLQIPAPAS SLGADT 56

SEQ ID NO: 4 moltype = DNA length = 168
 FEATURE Location/Qualifiers
 source 1..168
 mol_type = genomic DNA
 note = AAV5 VP1/2s common region
 organism = Adeno-associated virus

SEQUENCE: 4

acggccccta	ccggaagcg	gatagacgac	cactttccaa	aaagaaagaa	ggctcggacc	60
gaagaggact	ccaagccttc	cacctcgta	gacgcgaag	ctggaccag	cggatcccag	120
cagctgcaaa	tcccagccca	accagcctca	agtttgggag	ctgataca		168

SEQ ID NO: 5 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 note = AAV5 VP2 capsid protein
 organism = Adeno-associated virus

SEQUENCE: 5

TAPTGKRIDD	HFPKRKKART	EEDSKPSTSS	DAEAGPSGSQ	QLQIPAPAS	SLGADTMSAG	60
GGGPLGDNQ	GADGVGNASG	DWHCDSTWVG	DRVVTKSTRT	WVLPSTYNNHQ	YREIKSGSVD	120
GSNANAYFGY	STPWGYDFDN	RFHSHWSPRD	WQRLINNYWG	FRPRSLRVKI	FNIQVKEVTV	180
QDSTTTIANN	LTSTVQVFTD	DDYQLPYVVG	NGTEGCLPAF	PPQVFTLPQY	GYATLNRDNT	240
ENPTERRSFF	CLEYFPKML	RTGNNFEFTY	NFEEVPPHSS	FAPSONLFKL	ANPLVDQYLY	300
RFVSTNNTGG	VQPNKNLAGR	YANTYKNWPP	GPMGRTQGW	LGSGVNRASV	SAFATTNRME	360
LEGASYQVPP	QPNGMTNNLQ	GSNTYALENT	MIFNSQPANP	GTTATYLEGN	MLITSESETQ	420
PVNRVAYNVG	GQMATNNQSS	TTAPATGTYN	LQEIVPGSVW	MERDVYLQGP	IWAKIPETGA	480
HFHPSPAMGG	FGLKHPPMM	LIKNTVPVGN	ITSFSDVPVS	SFITQYSTGQ	VTVEMEWELK	540
KENSKRWNE	IQYTNNYNDP	QVDFAPDST	GEYRTRPIG	TRYLTRPL		588

SEQ ID NO: 6 moltype = DNA length = 1767
 FEATURE Location/Qualifiers
 source 1..1767
 mol_type = genomic DNA
 note = AAV5 VP2 capsid DNA
 organism = Adeno-associated virus

SEQUENCE: 6

acggccccta	ccggaagcg	gatagacgac	cactttccaa	aaagaaagaa	ggctcggacc	60
gaagaggact	ccaagccttc	cacctcgta	gacgcgaag	ctggaccag	cggatcccag	120
cagctgcaaa	tcccagccca	accagcctca	agtttgggag	ctgatacaat	gtctgcggga	180
ggtggcggcc	cattgggcga	caataaccaa	ggtgccgatg	gagtgggcaa	tgctcggga	240
gattggcatt	gcgattccac	gtggatgggg	gacagagtcg	tcaccaagtc	caccggaacc	300
tgggtgctgc	ccagctacaa	caaccaccag	taccgagaga	tcaaaagcgg	ctccgtcgac	360
ggaagcaacg	ccaacgccta	ctttggatac	agcaccacct	gggggtactt	tgactttaac	420
cgcttcaca	ggcactggag	cccccgagac	tggcaagac	tcatacaaa	ctactggggc	480
ttcagacccc	ggtccctcag	agtcacaaatc	ttcaacattc	aagtcacaa	ggtcacgggtg	540
caggactcca	ccaccaccat	cgcacaacaac	ctcacctcca	ccgtccaagt	gtttacggac	600
gacgaactcc	agctgcccta	cgctgcggc	aacgggacgg	agggatgcct	gccggccttc	660
cctccgcagg	tctttacgct	gccgcagtag	ggttacgcga	cgctgaacgg	cgacaacaca	720
gaaaatccca	ccgagaggag	cagcttcttc	tgccatagag	actttccag	caagatgctg	780
agaaacggga	acaactttga	gtttacctac	aactttgagg	aggtgccctt	ccactccagc	840
ttcgctccca	gtcagaacct	gttcaagctg	gccacccgc	tggtggacca	gtacttgtac	900
cgcttcgtga	gcacaaataa	cactggcgga	gtccagttca	acaagaacct	ggccgggaga	960
tacgccaaca	cctacaaaaa	ctggttcccg	gggcccattg	gccgaaccca	gggtgggaac	1020
ctgggctccg	gggtcaacgg	cgcagtggtc	agcgccttcg	ccacgaccaa	taggatggag	1080
ctcgagggcg	cgagttacca	ggtgcccccg	cagccgaacg	gcatgaccaa	caacctccag	1140
ggcagcaaca	cctatgccct	ggagaacact	atgatcttca	acagccagcc	ggcgaacccg	1200
ggcaccaccg	ccacgtacct	cgagggcaac	atgctcatca	ccagcgagag	cgagacgcag	1260
ccggtgaacc	gcgtggcgta	caacgtcggc	gggcagatgg	ccaccaacaa	ccagagctcc	1320
accactgccc	ccgcgaccgg	cacgtacaac	ctccaggaaa	tcgtgcccg	cagcgtgtgg	1380
atggagaggg	acgtgtacct	ccaaggaccc	atctgggcca	agatcccaga	gacggggggc	1440
cactttcacc	cctctccggc	catggcgcca	ttcggactca	aacccccacc	gcccagatg	1500
ctcatcaaga	acacgcctgt	gcccggaaat	atcaccagct	tctcggacgt	gcccgtcagc	1560
agcttcata	cccagtagag	caccgggcag	gtcacctggg	agatggagtg	ggagctcaag	1620
aaggaaaact	ccaagagggtg	gaaccagag	atccagtaca	caaacaccta	caacgacccc	1680
cagtttgtgg	actttgcccc	ggcagcacc	ggggaataca	gaaccaccag	acctatcgga	1740
accgataacc	ttacccgacc	cctttaa				1767

SEQ ID NO: 7 moltype = AA length = 136
 FEATURE Location/Qualifiers
 source 1..136
 mol_type = protein
 note = AAV5 VP1u region (protein)
 organism = Adeno-associated virus

SEQUENCE: 7

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MSFVDHPPDW	LEEVEGLRE	FLGLEAGPPK	PKPNQQHQDQ	ARGLVLPGYN	YLGPGNGLDR	60
GEPVNRADVE	AREHDISYNE	QLEAGDNPYL	KYNHADAEFQ	EKLADDTSGF	GNLGKAVFQA	120
KKRVLEPFGL	VEEGAK					136

SEQ ID NO: 8 moltype = DNA length = 408
 FEATURE Location/Qualifiers
 source 1..408
 mol_type = genomic DNA
 note = AAV5 VP1u region (DNA)
 organism = Adeno-associated virus

SEQUENCE: 8

atgtcttttg	ttgatcacc	tccagattgg	ttggaagaag	ttggtgaagg	tcttcgcgag	60
tttttggggc	ttgaagcg	cccaccgaaa	ccaaaaccca	atcagcagca	tcaagatcaa	120
gcccgtgggc	ttgtgctgcc	tggttataac	tatctcggac	ccggaaacgg	tctcgatcga	180
ggagagcctg	tcaacagggc	agacgaggtc	gcgcgagagc	acgacatctc	gtacaacgag	240
cagcttgagg	cgggagacaa	cccctacctc	aagtacaacc	acgcggacgc	cgagtttcag	300
gagaagctcg	ccgacgacac	atccttcggg	ggaaacctcg	gaaaggcagt	ctttcaggcc	360
aagaaaagg	ttctcgaacc	ttttggcctg	gttgaagagg	gtgctaag		408

SEQ ID NO: 9 moltype = AA length = 724
 FEATURE Location/Qualifiers
 source 1..724
 mol_type = protein
 note = AAV5 VP1 capsid protein
 organism = Adeno-associated virus

SEQUENCE: 9

MSFVDHPPDW	LEEVEGLRE	FLGLEAGPPK	PKPNQQHQDQ	ARGLVLPGYN	YLGPGNGLDR	60
GEPVNRADVE	AREHDISYNE	QLEAGDNPYL	KYNHADAEFQ	EKLADDTSGF	GNLGKAVFQA	120
KKRVLEPFGL	VEEGAKTAPT	GKRIDDHFPK	RKKARTEEDS	KPSTSSDAEA	GPSGSQQLQI	180
PAQPASSLGA	DTMSAGGGP	LGDNNQGADG	VGNASGDWHC	DSTWMGDRVV	TKSTRTWVLP	240
SYNNHQYREI	KSGSVDGSNA	NAYFGYSTPW	GYPDFNRRHS	HWSPRDWQRL	INNYWGFRRP	300
SLRVKIFNIQ	VKEVTQDST	TTIANNLST	VQVFTDDDYQ	LPHYVVGNGTE	GCLPAFPQV	360
FTLPQYGYAT	LNRDNTENPT	ERSSFFCLEY	FPSKMLRTGN	NFEFTYNFEE	VPFHSSFPAS	420
QNLFLANPL	VDQYLYRFVS	TNNTGGVQFN	KNLAGRYANT	YKNWFPGPMG	RTQGNLGS	480
VNRASVSAPA	TTNRMELEGA	SYQVPPQPNG	MTNNLQGSNT	YALENTMIFN	SQPANPGTTA	540
TYLEGNMLIT	SESETQPVNR	VAYNVGGQMA	TNNQSSTTAP	ATGTYNLQEI	VPGSVMERD	600
VYLQGIWAK	IPETGAHFH	SPAMGGFGLK	HPPPMMLIKN	TPVPGNITSF	SDVPVSSFIT	660
QYSTGQVTVE	MEWELKKENS	KRWNPETQYT	NNYNDPQFVD	FAPDSTGEYR	TTRPIGTRYL	720
TRPL						724

SEQ ID NO: 10 moltype = DNA length = 2175
 FEATURE Location/Qualifiers
 source 1..2175
 mol_type = genomic DNA
 note = AAV5 VP1 capsid DNA
 organism = Adeno-associated virus

SEQUENCE: 10

atgtcttttg	ttgatcacc	tccagattgg	ttggaagaag	ttggtgaagg	tcttcgcgag	60
tttttggggc	ttgaagcg	cccaccgaaa	ccaaaaccca	atcagcagca	tcaagatcaa	120
gcccgtgggc	ttgtgctgcc	tggttataac	tatctcggac	ccggaaacgg	tctcgatcga	180
ggagagcctg	tcaacagggc	agacgaggtc	gcgcgagagc	acgacatctc	gtacaacgag	240
cagcttgagg	cgggagacaa	cccctacctc	aagtacaacc	acgcggacgc	cgagtttcag	300
gagaagctcg	ccgacgacac	atccttcggg	ggaaacctcg	gaaaggcagt	ctttcaggcc	360
aagaaaagg	ttctcgaacc	ttttggcctg	gttgaagagg	gtgctaagac	ggccccatcc	420
ggaaagcgga	tagacgacca	ctttccaaaa	agaaagaagg	ctcggaccga	agaggactcc	480
aagccttcca	cctcgtcaga	cgccgaagct	ggaccacagc	gatcccagca	gctgcaaatc	540
ccagcccaac	cagcctcaag	tttgggagct	gatacaatgt	ctgcgggagg	tggcggccca	600
ttgggcgaca	ataaccaagg	tgccgatgga	gtgggcaatg	cctcgggaga	ttggcatatg	660
gattccacgt	ggatggggga	cagagtcgtc	accaagtcca	cccgaacctg	ggtgctgccc	720
agctacaaca	accaccagta	ccgagagatc	aaaagcggct	ccgtcgacgg	aagcaacgcc	780
aacgcctact	ttggatacac	cacccctcgg	gggtactttg	actttaaccg	cttccacagc	840
cactggagcc	cccagactg	gcaaagactc	atcaacaact	actggggcct	cagaccccg	900
tccctcagag	tcaaaatctt	caacattcaa	gtcaaaagag	tcacgggtga	ggactccacc	960
accaccatcg	ccaacaacct	cacctccacc	gtccaagtgt	ttacggacga	cgactaccag	1020
ctgcctacg	tcgtcgccaa	cgggaccgag	ggatgcctgc	cggccttccc	tccgaggtc	1080
tttacgctgc	cgagtagcgg	ttacgcgacg	ctgaaccg	acaacacaga	aaatcccacc	1140
gagaggagca	gcttctctcg	cctagagtac	tttcccagca	agatgctgag	aacgggcaac	1200
aactttgagt	ttacctacaa	ctttgaggag	gtgcccttcc	actccagctt	cgctcccagt	1260
cagaacctgt	tcaagctggc	caaccgcgtg	gtggaccagt	acttgtaccg	cttcgtgagc	1320
acaaataaca	ctggcgaggt	ccagttcaac	aagaacctgg	ccgggagata	cgccaacacc	1380
tacaaaaact	gggtccgggc	gcccattggc	cgaacccagg	gctggaacct	gggctccggg	1440
gtcaaccg	ccagtgctcag	gccttcggcc	acgaccaata	ggatggagct	cgaggcgccg	1500
agttaccagg	tgcccccgca	gccgaacg	atgaccaaca	acctccagg	cagcaacacc	1560
tatgccctg	agaacactat	gatcttcaac	agccagccgg	cgaacccggg	caccaccgcc	1620
acgtacctcg	agggcaacat	gtcctacacc	agcgagagcg	agacgcagcc	ggtgaaccgc	1680
gtggcgtaga	acgtcgccgg	gcagatggcc	accaacaacc	agagctccac	cactgcccc	1740

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gcgaccggca cgtacaacct ccaggaatac gtgcccggca gcgtgtggat ggagagggac 1800
gtgtacctcc aaggaccceat ctggggccaag atcccagaga cggggggcgca ctttcacccc 1860
tctccggcca tggggcggtt cggactcaaa caccaccgc ccatgatgct catcaagaac 1920
acgcctgtgc ccggaaatat caccagcttc tcggacgtgc ccgtcagcag cttcatcacc 1980
cagtacagca ccgggcaggt caccgtggag atggagtggt agctcaagaa ggaaaactcc 2040
aagaggttga acccagagat ccagtacaca aacaactaca acgaccccca gtttgtggac 2100
tttgcgccgg acagcaccgg ggaatacaga accaccagac ctatcggaac ccgatacctt 2160
acccgacccc ttttaa

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SEQ ID NO: 11      moltype = AA length = 535
FEATURE           Location/Qualifiers
source            1..535
                  mol_type = protein
                  note = AAV8 VP3 capsid protein
                  organism = Adeno-associated virus

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SEQUENCE: 11
MAAGGGAPMA DNNEGADGVG SSSGNWHCDS TWLGDRVITT STRTVALPTY NNHLYKQISN 60
GTSGGATNDN TYFGYSTPWG YFDENRFHCH FSPRDWQRLI NNNWGFPRKR LSFKLFNIQV 120
KEVTQNEGTK TIANNLTSTI QVPTDSEYQL PYVLGSAHQG CLPPFPADV MIPQYGYLTL 180
NNGSQAVGRS SFYCLEYFPS QMLRTGNNFQ FTYTFEDVPF HSSYAHSQSL DRLMNPLIDQ 240
YLYLSRTQT TGGTANTQTL GFSQGGPNTM ANQAKNWLPG PCYRQQRVST TTGQNNNSNF 300
AWTAGTKYHL NGRNSLANPG IAMATHKDDE ERFFPSNGIL IFGKQNAARD NADYSDVMLT 360
SEEEIKTTNP VATEEYGIVA DNLQQQNTAP QIGTVNSQGA LPGMVWQNRD VYLQGPWAK 420
IPHTDGNFHP SPLMGGFGLK HPPPPQILIKN TPVPADPPTT FNQSKLNSFI TQYSTGQVSV 480
EIEWELQKEN SKRWNPETIQ TSNYYKSTSV DPAVNTEGVY SEPRPIGTRY LTRNL 535

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SEQ ID NO: 12      moltype = DNA length = 1608
FEATURE           Location/Qualifiers
source            1..1608
                  mol_type = genomic DNA
                  note = AAV8 VP3 capsid DNA
                  organism = Adeno-associated virus

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SEQUENCE: 12
atggctgcag gcggtggcgc accaatggca gacaataacg aaggcgccga cggagtgggt 60
agttcctcgg gaaattggca ttgcgattcc acatggctgg gcgacagagt catcaccacc 120
agcaccggaa cctgggcccct gccccacctac aacaaccacc tctacaagca aatctccaac 180
gggacatcgg gagagccac caacgacac acctacttcg gctacagcac cccctggggg 240
tattttgact ttaacagatt ccactgccac ttttcaccac gtgactggca gcgactcacc 300
aacaacaact ggggattccg gcccaagaga ctcagcttca agctcttcaa catccaggtc 360
aaggaggtca cgcagaatga aggcaccaag accatcgcca ataacctcac cagcaccatc 420
cagggtgtta cggactcgga gtaccagctg ccgtacgttc tcggctctgc ccaccagggc 480
tgctgcctc cggtcccggc ggacgtgttc atgattcccc agtacggcta cctaaccactc 540
aacaacggta gtcaggccgt gggacgtccc tccttctact gcctggaata ctttccttcg 600
cagatgctga gaaccggcaa caactccag tttacttaca ctttcgagga cgtgcctttc 660
cacagcagct acccccacag ccagagcttg gaccggctga tgaatcctct gattgaccag 720
tacctgtact actgtctcgc gactcaaaac acaggaggca cggcaaatat gcagactctg 780
ggcttcagcc aaggtgggccc taatacaatg gccaatcagg caaagaactg gctgccaggga 840
ccctgttacc gccacaacag cgtctcaacg acaaccgggc aaaacaacaa tagcaacttt 900
gcctggagtc ctgggaccaa ataccatctg aatggaagaa attcattggc taatcctggc 960
atcgctatgg caacacacaa agacgacgag gagcgttttt ttcccagtaa cgggactcctg 1020
atttttggca aacaaaatgc tgccagagac aatgcggatt acagcgatgt catgctcacc 1080
agcgagggaag aaatcaaaac cactaacctc gtggctacag aggaatacgg tatcgtggca 1140
gataacttgc agcagcaaaa cacggctcct caaattggaa ctgtcaacag ccagggggcc 1200
ttaccgggta tggctcggca gaaccgggac gtgtacctgc aggttcccat ctgggccaag 1260
attcctcaca cggacggcaa cttccaccgc tctccgtga tggcggtgtt tggcctgaaa 1320
catcctccgc ctcagatcct gatcaagaac acgcctgtac ctgcggtacc tccgaccacc 1380
ttcaaccagt caaagctgaa ctctttcatc acgcaataca gcaccggaca ggtcagcgtg 1440
gaaattgaaat gggagctgca gaaggaaaac agcaagcgct ggaaccccca gatccagtac 1500
acctccaact actacaatc tacaagtgtg gactttgtgt ttaatacaga aggcgtgtac 1560
tctgaacccc gccccattgg caccggttac ctcaccgcta atctgtaa 1608

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SEQ ID NO: 13      moltype = AA length = 66
FEATURE           Location/Qualifiers
source            1..66
                  mol_type = protein
                  note = AAV8 VP1/2s common region (protein)
                  organism = Adeno-associated virus

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SEQUENCE: 13
TAPGKKRPVE PSPQRSPDSS TGIGKKGQQP ARKRLNFGQT GDSSEVPDPQ PLGEPPAAPS 60
GVGPNT

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SEQ ID NO: 14      moltype = DNA length = 198
FEATURE           Location/Qualifiers
source            1..198
                  mol_type = genomic DNA
                  note = AAV8 VP1/2s common region (DNA)

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                                organism = Adeno-associated virus
SEQUENCE: 14
acggctcctg gaaagaagag accggttagag ccatcacccc agcgttctcc agactcctct 60
acgggcatcg gcaagaaagg ccaacagccc gccagaaaaa gactcaattt tggtcagact 120
ggcgactcag agtcagttcc agaccctcaa cctctcggag aacctccagc agcgccctct 180
ggtgtgggac ctaataca 198

SEQ ID NO: 15                moltype = AA  length = 601
FEATURE                      Location/Qualifiers
source                      1..601
                             mol_type = protein
                             note = AAV8 VP2 capsid protein
                             organism = Adeno-associated virus

SEQUENCE: 15
TAPGKKRPVE PSPQRSPDSS TGIGKKQQQP ARKRLNFGQT GDSESVDPDQ PLGEPPAAPS 60
GVGPNTMAAG GGAPMADNNE GADGVGSSSG NWHCDSTWLG DRVITTSTRT WALPTYNNHL 120
YKQISNGTSG GATNDNTYFG YSTPWGYDFD NRFHCHFSPR DWQRLINNHW GFRPKRLSFK 180
LFNIQVKEVT QNEGKTIAN NLTSTIQVFT DSEYQLPYVL GSAHQGCLPP FPADVFMIPQ 240
YGYLTLNNGS QAVGRSSFYC LEYFPSQMLR TGNNFQFTYT FEDVPFHSSY AHSQSLDRML 300
NPLIDQYLYY LSRQTQTGGT ANTQTLGFSQ GGPNTMANQA KNWLPGPCYR QQRVSTTTGQ 360
NNNSNFAWTA GTKYHLNGRN SLANPGIAMA THKDDDEERFF PSNGILIFGK QNAARDNADY 420
SDVMLTSEEE IKTTNPVATE EYGIVADNLQ QONTAPQIGT VNSQGALPGM VWQNRDVYLQ 480
GPIWAKIPHT DGNFHPSPLM GGFGLKHPPP QILIKNTVPV ADPPTTFNQS KLNSFITQYS 540
TGQVSVEIEW ELQKENSKRW NPEIQYTSNY YKSTSVDFAV NTEGVYSEPR PIGTRYLTRN 600
L 601

SEQ ID NO: 16                moltype = DNA  length = 1806
FEATURE                      Location/Qualifiers
source                      1..1806
                             mol_type = genomic DNA
                             note = AAV8 VP2 capsid DNA
                             organism = Adeno-associated virus

SEQUENCE: 16
acggctcctg gaaagaagag accggttagag ccatcacccc agcgttctcc agactcctct 60
acgggcatcg gcaagaaagg ccaacagccc gccagaaaaa gactcaattt tggtcagact 120
ggcgactcag agtcagttcc agaccctcaa cctctcggag aacctccagc agcgccctct 180
ggtgtgggac ctaatacaat ggctgcaggc ggtggcgcac caatggcaga caataacgaa 240
ggcgccgacg gagtgggtag ttctcgggga aattggcatt gcgattccac atggctggggc 300
gacagagtcg tcaccaccag caccggaacc tggggccctgc ccacctacaa caaccacctc 360
tacaagcaaa tctccaacgg gacatcggga ggagccacca acgacaacac ctacttcggc 420
tacagcaccg cctgggggta ttttgacttt aacagattcc actgccactt ttcaccacgt 480
gactggcagc gactcatcaa caacaactgg ggattccggc ccaagagact cagcttcaag 540
ctcttcaaca tccagggtcaa ggaaggtcac cagaatgaag gcaccaagac catcgccaat 600
aacctcacca gcaccatcca ggtgtttacg gactcggagt accagctgcc gtacgttttc 660
ggctctgccc accagggctg cctgcctccg ttcccggcgg acgtgttcat gattccccag 720
tacggctacc taacactcaa caacggtagt caggccgtgg gacgtctctc cttctactgc 780
ctggaatact ttctctcgca gatgctgaga accggcaaca acttcagtt tacttacacc 840
ttcgaggacg tgcctttcca cagcagctac gccacagccg agagcttgga ccggctgatg 900
aatcctctga ttgacctgac ctgtactac ttgtctcgga ctcaaaacaac aggaggcacg 960
gcaaatacgc agactctggg cttcagccaa ggtgggccta atacaatggc caatcaggca 1020
aagaactggc tgccaggacc ctgttaccgc caacaacgcg tctcaacgac aaccgggcaa 1080
aacaacaata gcaactttgc ctggactgct gggaccaaaat accatctgaa tggaagaaat 1140
tcattggcta atctctggc cgtctatggc acacacaaag acgacgagga gcgttttttt 1200
cccagtaacg ggatctcgat ttttgcaaaa caaaatgctg ccagagacaa tgcggattac 1260
agcgatgtca tgcctaccag cgaggaaagaa atcaaaaacca ctaacctgtg ggctacagag 1320
gaatacggta tcgtggcaga taacttgacg cagcaaaaaca cggctcctca aattggaaat 1380
gtcaacagcc agggggcctt acccggtatg gtctggcaga accgggacgt gtacctgcag 1440
gggtccatct gggccaagat tcctcacacg gacggcaact tccaccgctc tccgctgatg 1500
ggcggtcttg gcctgaaaca tcctccgcct cagatcctga tcaagaacac gcctgtacct 1560
gcggatcctc cgaccacctt caaccagtca aagctgaact ctttcatcac gcaatacagc 1620
accggacagg tcagcgttga aattgaatgg gagctgcaga aggaaaacag caagcgctgg 1680
aaccgccgag tccagtacac ctccaactac tacaatacta caagtgtgga ctttgcgtgt 1740
aatacagaag gcgtgtactc tgaaccccg cccattggca cccgttacct caccgtaat 1800
ctgtaa 1806

SEQ ID NO: 17                moltype = AA  length = 137
FEATURE                      Location/Qualifiers
source                      1..137
                             mol_type = protein
                             note = AAV8 VP1u region (protein)
                             organism = Adeno-associated virus

SEQUENCE: 17
MAADGYLPDW LEDNLSEGIR EWWALKPGAP KPKANQQKQD DGRGLVLPGY KYLGPFNGLD 60
KGEPVNAADA AALEHDKAYD QQLQAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEGAK 137

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SEQ ID NO: 18 moltype = DNA length = 411
 FEATURE Location/Qualifiers
 source 1..411
 mol_type = genomic DNA
 note = AAV8 VPlu region (DNA)
 organism = Adeno-associated virus

SEQUENCE: 18

atggctgcgcg	atgggttatct	tccagattgg	ctcgaggaca	acctctctga	gggcattcgc	60
gagtgggtggg	cgctgaaacc	tggagccccg	aagcccaaag	ccaaccagca	aaagcaggac	120
gacggccggg	gtctgggtgct	tcctggctac	aagtacctcg	gaccttcaa	cggactcgac	180
aagggggagc	ccgtcaacgc	ggcggacgca	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctgc	aggcgggtga	caatccgtac	ctgcgggtata	accacgccga	cgccgagttt	300
caggagcgctc	tgcaagaaga	tacgtctttt	gggggcaacc	tcggggcgagc	agtcttcacg	360
gccaagaagc	gggttctcga	acctctcggt	ctgggtgagg	aaggcgctaa	g	411

SEQ ID NO: 19 moltype = AA length = 738
 FEATURE Location/Qualifiers
 source 1..738
 mol_type = protein
 note = AAV8 VP1 capsid protein
 organism = Adeno-associated virus

SEQUENCE: 19

MAADGYLPDW	LEDNLSEGIR	EWALKKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALEHDKAYD	QQLQAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFO	120
AKKRVLLEPLG	LVEEGAKTAP	GKKRPVEPSP	QRSPDSSTGI	GKKGQQPARK	RLNFGQTGDS	180
ESVPDPQPLG	EPPAAPSGVG	PNTMAAGGGA	PMADNNEGAD	GVGSSSGNWH	CDSTWLGDV	240
ITSTRTWAL	PTYNNHLYKQ	ISNGTSGGAT	NDNTYFGYST	PWGYFDFNRF	HCHFSPRDWQ	300
RLINNWNWGR	PKRLSFKLFN	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	360
HQGCCLPPPPA	DVFMIPQYGY	LTLNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFQFTYTFED	420
VFPFSSYAHS	QSLDRLMNPL	IDQYLYLSR	TQTGGGTANT	QTLGFSQGGP	NTMANQAKNW	480
LPGPCYRQQR	VSTTTGQNNN	SNFAWTAGTK	YHLNGRNSLA	NPGIAMATHK	DDEERFFPSN	540
GILIFGKQNA	ARDNADYSV	MLTSEEEIKT	TNPVATEEYG	IVADNLQQQN	TAPQIGTVNS	600
QGALPGMWQ	NRDVYLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPPIIL	IKNTPVPADP	660
PTTFNQSKLN	SFITQYSTGQ	VSVIEIWELO	KENSKRWNPE	IQYTSNYYS	TSVDFAVNTE	720
GVYSEPRPIG	TRYLRNL					738

SEQ ID NO: 20 moltype = DNA length = 2217
 FEATURE Location/Qualifiers
 source 1..2217
 mol_type = genomic DNA
 note = AAV8 VP1 capsid DNA
 organism = Adeno-associated virus

SEQUENCE: 20

atggctgcgcg	atgggttatct	tccagattgg	ctcgaggaca	acctctctga	gggcattcgc	60
gagtgggtggg	cgctgaaacc	tggagccccg	aagcccaaag	ccaaccagca	aaagcaggac	120
gacggccggg	gtctgggtgct	tcctggctac	aagtacctcg	gaccttcaa	cggactcgac	180
aagggggagc	ccgtcaacgc	ggcggacgca	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctgc	aggcgggtga	caatccgtac	ctgcgggtata	accacgccga	cgccgagttt	300
caggagcgctc	tgcaagaaga	tacgtctttt	gggggcaacc	tcggggcgagc	agtcttcacg	360
gccaagaagc	gggttctcga	acctctcggt	ctgggtgagg	aaggcgctaa	gacggctcct	420
ggaaagaaga	gaccggtaga	gccatcacc	cagcgttctc	cagactcctc	tacgggcatc	480
ggcaagaaag	cgaacagcc	cgccagaaaa	agactcaatt	ttggtcagac	tggcgactca	540
gagtcagttc	gacacctca	acctctcgga	gaacctccag	cagcgccctc	tggtgtggga	600
cctaatacaa	tggtgcagg	cggtggcgca	ccaatggcag	acaataacga	aggcgccgac	660
ggagtggtga	gttcctcggg	aaattggcat	tgcgattcca	catggctggg	cgacagagtc	720
atcaccacca	gcacccgaac	ctggggccctg	cccacctaca	acaaccacct	ctacaagcaa	780
atctccaacg	ggacatcggg	aggagccacc	aacgacaaca	cctactctcg	ctacagcacc	840
ccctgggggt	atcttgactt	taacagattc	cactgccact	tttcaccacg	tgactggcag	900
cgactcatca	acaacaactg	gggattccgg	cccaagagac	tcagcttcaa	gctcttcaac	960
atccagggtc	aggaggtcac	gcagaatgaa	ggcaccaaga	ccatcgccaa	taacctcacc	1020
agcaccatcc	agggtgttac	ggaactcggg	taccagctgc	cgtacgttct	cggtctctgc	1080
caccagggtc	gcctgcctcc	gttcccggcg	gacgtgttca	tgattcccca	gtacggctac	1140
ctaactca	acaacggtag	tcaggccgtg	ggacgtccct	ccttctactg	cctggaatac	1200
tttctctcgc	agatgtctgag	aaccggcaac	aacttccagt	ttacttacac	cttcgaggac	1260
gtgcctttcc	acagcagcta	cgccacacgc	cagagcttgg	accggctgag	gaatcctctg	1320
attgaccagt	acctgtacta	cttgtctcgg	actcaaaaca	caggaggcac	ggcaataacg	1380
cagactctgg	gcttcagcca	aggtgggcct	aatacaatgg	ccaatcagcg	aaagaactgg	1440
ctgccaggac	cctgttacgc	ccaacaacgc	gtctcaacga	caaccgggca	aaacaacaat	1500
agcaactttg	cctggactgc	tgggaccaaa	taccatctga	atggaagaaa	ttcattggct	1560
aatctcggca	tcgctatggc	aacacacaaa	gacgacgagg	agcgtttttt	tcccagtaac	1620
gggactctga	tttttgccaa	acaaaatgct	gccagagaca	atgcggatta	cagcgatgtc	1680
atgctcacca	gcgagggaaga	aatcaaaaacc	actaaccctg	tggttacaga	ggaatacggg	1740
atcgtggcag	ataacttgca	gcagcaaaac	acggctcctc	aaattggaac	tgtcaacagc	1800
cagggggcct	taccgggtat	ggtctggcag	aaccgggacg	tgtacctgca	gggtcccatc	1860
tgggccaaga	ttcctcacac	ggacggcaac	ttccaccctg	ctccgctgat	ggcgggcttt	1920
ggcctgaaac	atcctccgcc	tcagatcctg	atcaagaaca	cgctgtgacc	tgccgatcct	1980

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ccgaccacct tcaaccagtc aaagctgaac tctttcatca cgcaatacag caccggacag 2040
gtcagcgctgg aaattgaatg ggagctgcag aagggaaaaca gcaagcgctg gaaccccgag 2100
atccagtaca cctccaacta ctacaatct acaagtgtgg actttgctgt taatacagaa 2160
ggcgtgtact ctgaaccccg ccccatggc acccgttacc tcaccgtaa tctgtaa 2217

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SEQ ID NO: 21      moltype = AA length = 591
FEATURE           Location/Qualifiers
source            1..591
                  mol_type = protein
                  note = AAV5-AAV8 hybrid VP2 capsid protein
                  organism = Synthetic construct

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SEQUENCE: 21
TAPTGKRIDD HFPKRKKART EEDSKPSTSS DAEAGPSGSQ QLQIPAQPAS SLGADTMAAG 60
GGAPMADNNE GADGVGSSSG NWHCDSTWLG DRVITTSTRT WALPTYNNHL YKQISNGTSG 120
GATNDNTYFG YSTPWGYFDF NRFCHFSRPR DWQRLINNNW GFRPKRLSFK LFNIQVKEVT 180
QNEGKTIAN NLTSTIQVFT DSEYQLPYVL GSAHQGLPP FPADVFMIPQ YGYLTLNNGS 240
QAVGRSSFYC LEYFSPQMLR TGNFQFTYT FEDVPFHSSY AHSQSLDRML NPLIDQYLYY 300
LSRTQTGTGT ANTQTLGFSQ GGPNTMANQA KNWLPGPCYR QQRVSTTTGQ NNNSNFAWTA 360
GTKYHLNGRN SLANPGIAMA THKDDERFF PSNGILIFGK QNAARDNADY SDVMLTSEEE 420
IKTTNPVATE EYGIVADNLQ QQNTAPQIGT VNSQGALPGM VWQNRDVYLQ GPIWAKIPT 480
DGNFHPSPLM GGFGLKHPPP QILIKNTPVP ADPPTTFNQS KLNSFITQYS TGQVSVEIEW 540
ELQKENSRRW NPEIQYTSNY YKSTSVDFAV NTEGVYSEPR PIGTRYLTRN L 591

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SEQ ID NO: 22      moltype = DNA length = 1776
FEATURE           Location/Qualifiers
source            1..1776
                  mol_type = other DNA
                  note = AAV5-AAV8 hybrid VP2 capsid DNA
                  organism = Synthetic construct

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SEQUENCE: 22
acggccccta ccggaagcgc gatagacgac cactttccaa aaagaaagaa ggctcggacc 60
gaagaggact ccaagccttc cacctcgtca gacgcccgaag ctggaccag cggatcccag 120
cagctgcaaa tcccagccca accagcctca agtttgggag ctgatacaat ggctgcaggc 180
ggtggcgcac caatggcaga caataacgaa ggcgccgacg gactgggtag ttctcggga 240
aattggcatt gcgattccac atggctgggc gacagagtca tcaccaccag caccggaacc 300
tgggccctgc ccacctacaa caaccacctc tacaagcaaa tctccaacgg gacatcggga 360
ggagccacca acgacaacac ctacttcggc tacagcacc cctgggggta tttgacttt 420
aacagattcc actgccactt ttcaccacgt gactggcagc gactcatcaa caacaactgg 480
ggattccggc ccaagagact cagcttcaag ctcttcaaca tccaggtcaa ggaggtcacg 540
cagaatgaag gcaccaagac catcgccaat aacctcacca gcaccatcca ggtgtttacg 600
gactcggagt accagctgcc gtacgttttc ggctctgccc accagggtcg cctgcctccg 660
ttcccgcgcg acgtgttcat gattccccag tacggctacc taacactcaa caacggtagt 720
caggccgtgg gacgtcctc ctctactgc ctggaatact ttccttcgca gatgctgaga 780
accggcaaca acttcagatt tacttacacc ttcgaggacg tgcccttcca cagcagctac 840
gcccacagcc agagcttgga ccggctgatg aatcctctga ttgaccagta cctgtactac 900
ttgtctcgga ctcaaaccaac agagggcacg gcaataacgc agactctggg cttagcaca 960
ggtgggccta atacaatggc caatcaggca aagaactggc tgccaggacc ctgttaccgc 1020
caacaacgcg tctcaacgac aaccgggcaa aacaacaata gcaactttgc ctggactgct 1080
gggacccaat accatctgaa ttggaagaaat tcattggcta atcctggcat cgctatggca 1140
acacacaaag acgacgagga gctgtttttt cccagtaacg ggatcctgat ttttgcaaaa 1200
caaaatgctg ccagagacaa tgcggtattac agcgtatgta tgctcaccag cgaggaagaa 1260
atcaaaaaca ctaacctgtg ggtctacagag gaatacggta tcgtgcagaa taactccgac 1320
cagcaaaaaca cggctcctca aattggaaat gtcaacagcc aggggggcctt acccggtatg 1380
gtctggcaga accgggacgt gtacctgcag ggtcccatct gggccaagat tcctcacacg 1440
gacggcaact tccaccgctc tccgctgatg ggcggtttg gctgaaaca tcctccgct 1500
cagatcctga tcaagaacac gccgtacct gcggatcctc cgaccacctt caaccagtca 1560
aagctgaact cttcatcac gcaatacagc accggacagg tcagcgtgga aattgaatgg 1620
gagctgcaga aggaacacg caagcgtctg aaccccgaga tccagtacac ctccaactac 1680
tacaaatcta caagtgtgga ctttctgttt aatacagaag gcgtgtactc tgaaccccg 1740
cccatggca cccgttacct caccgtaat ctgtaa 1776

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SEQ ID NO: 23      moltype = AA length = 727
FEATURE           Location/Qualifiers
source            1..727
                  mol_type = protein
                  note = AAV5-AAV8 hybrid VP1 capsid protein
                  organism = Synthetic construct

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SEQUENCE: 23
MSFVDHPPDW LEEVGEGLRE FLGLEAGPPK PKPNQQHQDQ ARGVLVLPYGN YLPGNGGLDR 60
GEPVNRADVE AREHDI SYNE QLEAGDNPYL KYNHADAEFQ EKLADDTSGF GNLGKAVFQA 120
KKRVLEPFGL VEEGAKTAPT KRIDDHFPK RKKARTEEDS KPSTSSDAEA GPSGSQQLQI 180
PAQPASSLGA DTMAAGGGAP MADNNEGADG VGSSSGNWHC DSTWLGDRVI TTSTRTWALP 240
TYNNHLYKQI SNGTSGGATN DNTYFGYSTP WGYFDNRFH CHFSPRDWQR LINNNWGFRP 300
KRLSFKLFNI QVKEVTQNEG TKTIANNLTS TIQVFTDSEY QLPYVLGSAH QGCLPPFPAD 360
VFMIPQGYLV TLNNGSQAVG RSSFYCLEYF PSQMLRTGNN FQFTYTFEDV PFHSSYAHSQ 420
SLDRMLNPLI DQYLYLSRT QTTGGTANTQ TLGFSQGGFN TMANQAKNWL PGPCYRQQRV 480

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STTTGQNNNS	NFAWTAGTKY	HLNGRNSLAN	PGIAMATHKD	DEERFFPSNG	ILIFGKQNA	540
RDNADYSDVM	LTSEEEIKTT	NPVATEEYGI	VADNLQQQNT	APQIGTVNSQ	GALPGMVWQN	600
RDVYLQGPW	AKIPHTDGNF	HPSPLMGGFG	LKHPPPPQILI	KNTVPVPADPP	TTFNQSKLNS	660
FITQYSTGQV	SVEIEWELQK	ENSKRWNPFI	QYTSNYYKST	SVDFAVNTEG	VYSEPRPIGT	720
RYLTRNL						727

SEQ ID NO: 24 moltype = DNA length = 2184
 FEATURE Location/Qualifiers
 source 1..2184
 mol_type = other DNA
 note = AAV5-AAV8 hybrid VP1 capsid DNA
 organism = Synthetic construct

SEQUENCE: 24

atgtcttttg	ttgatcaccc	tccagattgg	ttggaagaag	ttggtgaagg	tcttcgcgag	60
tttttggggc	ttgaagcggg	cccaccgaaa	ccaaaaccca	atcagcagca	tcaagatcaa	120
gccccgtgtc	ttgtgtgcgc	tggttataac	tatctcggac	ccggaacggg	tctcgatcga	180
ggagagcctg	tcaaacgggc	agacgaggtc	gcgcgagagc	acgacatctc	gtacaacggg	240
cagcttgagc	cgggagacaa	cccctacctc	aagtacaacc	acgcggacgc	cgagtttcag	300
gagaagctcg	ccgacgacac	atccttcggg	ggaaacctcg	gaaaggcagt	cttcagggcc	360
aagaaaaagg	tctcgaacc	ttttggcctg	gttgaagagg	gtgctaagac	ggcccctacc	420
ggaaagcggg	tagacgacca	ctttccaaaa	agaaagaagg	ctcggaccga	agaggactcc	480
aagccttcca	cctcgtcaga	gcgcgaagct	ggaccacgag	gatcccagca	gctgcaaatc	540
ccagcccaac	cagcctcaag	tttgggagct	gatacaatgg	ctgcaggcgg	tggcgcacca	600
atggcagaca	ataacgaagg	gcgcgacgga	gtgggttagt	cctcgggaaa	ttggcattgc	660
gattccacat	ggctggggcg	cagagtcac	accaccagca	cccgaacctg	ggccctgccc	720
acctacaaca	accacctcta	caagcaaatc	tccaacggga	catcgggagg	agccaccaac	780
gacacacact	acttcggcta	cagcaccccc	tgggggtatt	ttgactttaa	cagattccac	840
tgccactttt	caccacgtga	ctggcagcga	ctcatcaaca	acaactgggg	attccggccc	900
aagagactca	gcttcaagct	cttcaacatc	cagggtcaagg	aggtcacgca	gaatgaaggc	960
accaagacca	tgcgcaataa	cctcaccagc	accatccagg	tggttacgga	ctcggagtag	1020
cagctgcctg	acgttctcgg	ctctgcccac	cagggtctgc	tgcctccgtt	cccggcggac	1080
gtgttcacat	ttccccagta	cggtacaccta	acactcaaca	acggtagtca	ggccgtggga	1140
cgctcctcct	tctactgcct	ggaatacttt	ccttcgcaga	tgctgagaac	cggaacaac	1200
ttccagttta	cttacacctg	cgaggacgtg	cctttccaca	gcagctacgc	ccacagccag	1260
agcttggacc	ggctgatgaa	tcctctgatt	gaccagtacc	tgtagtactt	gtctcggact	1320
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acaatggcca	atcaggcaaa	gaactggctg	ccaggaccct	gttaccgcca	acaacgcgtc	1440
tcaacgcgaa	ccgggcaaaa	caacaatagc	aactttgcct	ggactgctgg	gaccaaatat	1500
catctgaatg	gaagaaatcc	attggccta	cctggcatcg	ctatggcaac	acacaagac	1560
gcagaggagc	ggttttttcc	cagttaacggg	atcctgattt	ttggcaaaaca	aaatgctgcc	1620
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aacctctgtg	ctacagagga	atacggtatc	gtggcagata	acttgacgca	gcaaaacacg	1740
gctcctcaaa	cttgaactgt	caacagccag	ggggccttac	ccggtatggt	ctggcagaa	1800
cgggacgtgt	acctgcaggg	tcccatctgg	gccaaagattc	ctcacacgga	cggaacattc	1860
caccgctctc	cgctgatggg	cggtcttggc	ctgaacacatc	ctccgcctca	gatcctgatc	1920
aagaacacgc	ctgtacctgt	ggatcctccg	accaccttca	accagtcaaa	gctgaactct	1980
ttcatcacgc	aatacagcac	cggacaggtc	agcgtggaaa	ttgaatggga	gctgcagaag	2040
gaaaacagca	agcgtgggaa	ccccgagatc	cagtacacct	ccaactacta	caaatctaca	2100
agtgtggact	ttgctgttaa	tacagaaggc	gtgtactctg	aaccccgccc	cattggcacc	2160
cgttacctca	cccgaatact	gtaa				2184

SEQ ID NO: 25 moltype = AA length = 728
 FEATURE Location/Qualifiers
 source 1..728
 mol_type = protein
 note = AAV8/5-AAV8 hybrid VP1 capsid protein
 organism = Synthetic construct

SEQUENCE: 25

MAADGYLPDW	LEDNLSEGIR	EWALKKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALHDKAYD	QQLQAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	TGKRIDDHFP	KRKKARTEED	SKPSTSSDAE	AGPSGSQLQ	180
IPAQPASSLG	ADTMAAGGGA	PMADNNEGAD	GVGSSSGNWH	CDSTWLGDVR	ITTSTRTWAL	240
PTYNNHLYKQ	ISNGTSGGAT	NDNTYFGYST	PWGYPDFNRF	HCHFSPRDWQ	RLINNNGWFR	300
PKRLSPKLFN	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	HQGCCLPPFP	360
DVFMIPQYGY	LTLNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFQFTYTFED	VPFHSSYAH	420
QSLDRLMNP	IDQLYLYLSR	TQTGGGTANT	QTLGFSQGGP	NTMANQAKNW	LPGPCYRQQR	480
VSTTTGQNNM	NFAWTAGTK	YHLNGRNSLA	NPGIAMATHK	DEERFFPSN	GILIFGKQNA	540
ARDNADYSDV	MLTSEEEIKT	TNPVATEEY	IVADNLQQQN	TAPQIGTVNS	QALPGMVWQ	600
NRDVYLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPPPQIL	IKNTVPVPADP	PTTFNQSKLN	660
SFITQYSTGQ	VSVEIEWELQ	KENSKRWNP	IQYTSNYYKS	TSVDFAVNTE	GVYSEPRPIG	720
TRYLTRNL						728

SEQ ID NO: 26 moltype = DNA length = 2187
 FEATURE Location/Qualifiers
 source 1..2187
 mol_type = other DNA

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note = AAV8/5-AAV8 hybrid VP1 capsid DNA
organism = Synthetic construct

SEQUENCE: 26

atggctgccc	atggttatct	tccagattgg	ctcgaggaca	acctctctga	gggcattcgc	60
gagtggtggg	cgctgaaacc	tggagccccg	aagcccaaa	ccaaccagca	aaagcaggac	120
gacggccggg	gtctggtgct	tccctggctac	aagtacctcg	gaccttcaa	cggactcgac	180
aagggggagc	ccgtcaacgc	ggcggacgca	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctgc	aggggggtga	caatccgtac	ctgcgggtata	accacgcoga	cgcgagttt	300
caggagcgto	tgcaagaaga	tacgtctttt	gggggcaacc	toggcgagc	agtcttcacg	360
gccaagaagc	gggttctcga	acctctcggg	ctggttgagg	aaggcgctaa	gacggccctc	420
accgaaaagc	ggatagacga	ccactttcca	aaaagaaa	aggctcggac	cgaagaggac	480
tccaagcctt	ccactcgctc	agacgccgaa	gctggaccga	gcggatccca	gcagctgcaa	540
atcccagccc	aaccagcctc	aagtttgagg	gctgatacaa	tggtcgcagg	cggtagggca	600
ccaatggcag	acaataacga	aggcgccgac	ggagtgggta	gttcctcggg	aaattggcat	660
tgcgattcca	catggctggg	cgacagagtc	atcaccacca	gcacccgaac	ctgggcccctg	720
cccacctaca	acaaccacct	ctacaagcaa	atctccaacg	ggacatcggg	aggagccacc	780
aacgacaaca	cctactctcg	ctacagcacc	cctgggggtg	atcttgactt	taacagattc	840
cactgccact	tttcaccacg	tgactggcag	cgactcatca	acaacaactg	gggattcccg	900
ccaagagac	tcagcttcaa	gctcttcaac	atccagggtc	aggaggtcac	gcagaatgaa	960
ggcaccaga	ccatcgccaa	taacctcacc	agcaccatcc	agggtgttac	ggactcggag	1020
taccagctgc	cgtacgttct	cggctctgcc	caccagggtc	gcctgcctcc	gttcccggcg	1080
gacgtgttca	tgattcccca	gtacggctac	ctaactca	acaacgtag	tcaggccgtg	1140
ggacgtctct	cctctactcg	cctggaatac	tttctctcgc	agatgctgag	aacgggcaac	1200
aaactccagt	ttaacttacac	cttcgaggac	gtgcctttcc	acagcagcta	cgcacacagc	1260
cagagcttgg	accggctgat	gaatcctctg	attgaccagt	acctgtacta	cttgtctcgg	1320
actcaaacaa	caggaggcac	ggcaaatatc	cagactctgg	gcttcagcca	aggtagggcct	1380
aatacaatgg	ccaatcaggc	aaagaactgg	ctgcacaggac	cctgttaccg	ccaacaacgc	1440
gtctcaacga	caaccgggca	aaacaacaat	agcaactttg	cctggaactg	tgggaccaaa	1500
taaccatctga	atggaagaaa	ttcatgtggc	aatcctggca	tcgctatggc	aacacacaaa	1560
gacgacgag	agcgtttttt	tcccagtaac	gggatcccta	tttttgga	acaaaatgct	1620
gcccagagaca	atgcggctga	cagcgatgtc	atgtccacca	gcgaggaaga	aatcaaaacc	1680
actaacctcg	tggctacaga	ggaatacggg	atcgtggcag	ataacttgca	gcagcaaaac	1740
acggctcctc	aaattggaac	tgtcaacagc	caggggggcct	tacccggtat	ggtctggcag	1800
aaacgggacg	tgtacctgca	gggtcccatc	tgggccaaga	ttcctcacac	ggacggcaac	1860
ttccaccctg	ctccgctgat	ggcgcgcttt	ggcctgaaac	atcctccgcc	tcagatcctg	1920
atcaagaaca	cgctgtacc	tgcggatcct	ccgaccacct	tcaaccagtc	aaagctgaac	1980
tttttcatca	gcgaatacag	caccggacag	gtcagcgtgg	aaattgaatg	ggagctgcag	2040
aaggaaaaaca	gcaagcgctg	gaaccccgag	atccagtaga	cctccaacta	ctacaaatct	2100
acaagtgtgg	actttgctgt	taatacagaa	ggcgtgtact	ctgaaccccg	ccccattggc	2160
accggttacc	tcacccgtaa	tctgtaa				2187

SEQ ID NO: 27 moltype = AA length = 530
FEATURE Location/Qualifiers
source 1..530
 mol_type = protein
 note = AAVrh13 VP3 capsid protein
 organism = Adeno-associated virus

SEQUENCE: 27

MAAGGAPMA	DNNEGADGVG	NASGNWHCDS	TWLGDREVIT	STRTWALPTY	NNHLYKQISS	60
QSGATNDNH	FGYSTPWGYF	DFNRFHCHFS	PRDWORLIM	NWGFPRPKLR	FKLFNIQVKE	120
VTTNDGVTT	ANNLTSTIQV	FSDSEYQLPY	VLGSAHQGCL	PPFPADVFMI	PQYGYLTINN	180
GSQSVGRSF	YCLEYFSPQM	LRTGNNFERS	YTFEEVPHFS	SYAHSQSLDR	LMNPLIDQYL	240
YYLARTQST	GSTRELQFHQ	AGPNTMAEQS	KNWLPGPCYR	QQRLSKNIDS	NNNSNFAWTG	300
ATKYHLNGR	SLTNPGVAMA	TNKDDDEDQF	PINGVLVFG	TGAANKTTLE	NVLMTSEEEI	360
KTTNPVATE	YGVVSSNLQS	STAGPQTQTV	NSQALPGMV	WQNRDVYLQ	PIWAKIPHTD	420
GNFHPSPLM	GFGLKHPPPQ	ILIKNTPVPA	NPPEVFTPAK	FASFITQYST	GQVSVEIEWE	480
LQKENSKRWN	PEIQYTSNYA	KSNNEFAVN	NEGVYTEPRP	IGTRYLTRNL		530

SEQ ID NO: 28 moltype = DNA length = 1593
FEATURE Location/Qualifiers
source 1..1593
 mol_type = genomic DNA
 note = AAVrh13 VP3 capsid DNA
 organism = Adeno-associated virus

SEQUENCE: 28

atggctgcag	gcggtggcgc	accaatggca	gacaataacg	aaggcgccga	cggagtgggt	60
aatgcctccg	gaaattggca	ttgcgattcc	acatggctgg	gcgacagagt	catcaccacc	120
agcaccgcga	cctggggcct	gccacacctc	aacaaccacc	tctacaagca	gatatcaagt	180
cagagcgggg	ctaccaacga	caaccacttc	ttcggtctaca	gcacccctcg	gggctatatt	240
gactcaaca	gattccactg	ccacttctca	ccacgtgact	ggcagcgact	catcaacaac	300
aaactgggat	tccggcccg	aaagctgcgg	ttcaagttgt	tcaacatcca	ggtcaaggag	360
gtcacgacga	acgacggcgt	tacgaccatc	gctaataaac	ttaccagcac	gattcaggtc	420
ttctcggact	cggagtacca	actgccgtac	gtcctcggct	ctgcgcacca	gggctgcctc	480
cctccgttcc	ctgcggacgt	gttcatgatt	cctcagtagc	gatatctgac	tctaaacaac	540
ggcagtcagt	ctgtgggacg	ttcctccttc	tactgcctgg	agtactttcc	ttctcagatg	600
ctgagacagg	gcaataactt	tgaattcagc	tacacctttg	agggaagtgc	tttccacagc	660

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agctatgcgc acagccagag cctggaccgg ctgatgaatc cctcctcgga ccagtaacctg 720
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gctggggcca acaccatggc cgagcaatca aagaactggc tgcccggacc ctgttatcgg 840
cagcagagac tgtcaaaaaa catagacagc aacaacaaca gtaactttgc ctggaccggg 900
gccactaaat accatctgaa tggtagaaat tcattaacca acccgggctg agccatggcc 960
accaacaagg acgacgagga ccagttcttt cccatcaacg gagtgtcgtt ttttggcgaa 1020
acggggggctg ccaacaagac aacgttgtaa aacgtgttaa tgaccagcga ggaggagatc 1080
aaaaccacca atcccgctgg tacagaagaa tacggtgtgg tctccagcaa cctgcaatcg 1140
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tggcagaacc gggacgtgta cctgcagggt cccatctggg ccaaaattcc tcacacggac 1260
ggcaactttc acccgctctc cctgatgggc ggatttgga ccaaacaccc gcctcctcaa 1320
atttctcatc aaaacacccc ggtacctgct aatcctccag aggtgtttac tccctgccaag 1380
tttgccctcat ttatcacgca gtacagcacc ggcaggtgca gcgtggagat cgagtgaggaa 1440
ctgcagaaag aaaacagcaa acgctggaat ccagagattc agtacacctc aaattatgcc 1500
aagtctaata atgtggaatt tgcgttcaac aacgaagggg tttatactga gcctcgcccc 1560
attggcaccg gttacctcac ccgtaacctg taa 1593

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SEQ ID NO: 29      moltype = AA length = 732
FEATURE
source            Location/Qualifiers
                  1..732
                  mol_type = protein
                  note = AAVhu32 / AAVrh13 hybrid VP1 capsid protein (AAVhu32
                        VP1u + AAVhu32 VP1/2s + AAVrh13 VP3)
                  organism = Synthetic construct

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SEQUENCE: 29
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGGLD 60
KGEPVNADA AALEHDKAYD QQLKAGDNPY LKYNHDAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFGQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMAAGGGAP MADNNEGADG VGNASGNWHC DSTWLGRDVI 240
TTSTRTWALP TYNNHLYKQI SSQSGATNDN HFFGYSTPWG YFDNFRFHCH FSPRDWQRLI 300
NNNWGFRPRK LRFKLFINIQV KEVTTNDGVT TIANNLTSTI QVFSDEYQL PYVLGSAHQG 360
CLPPFPADVF MIPQGYLTL NNGSQSVGRS SFYCLEYFPPS QMLRTGNNFE FSYTFEEVPF 420
HSSYAHSQSL DRLMNPLIDQ YLYYLARTQS TTGSTRELQF HQAGPNTMAE QSKNWLPGPC 480
YRQQRLSKNI DSNNNSNFAW TGATKYHLNG RNSLTNPQVA MATNKDDEDQ FFPINGVLVF 540
GETGAANKTT LENVMTSEE EIKTTNPVAT EEYGVVSSNL QSSTAGPQTQ TVNSQGALPG 600
MVWQNRDVYL QGPIWAKIPH TDGNFHPSP L MGGFGLKHPP PQILIKNTPV PANPPEVFTF 660
AKFASFITQY STGQVSVEIE WELQKENSKR WNPEIQYTSN YAKSNNVFEFA VNNNEGVYTEP 720
RPIGTRYLTR NL 732

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SEQ ID NO: 30      moltype = DNA length = 2199
FEATURE
source            Location/Qualifiers
                  1..2199
                  mol_type = other DNA
                  note = AAVhu32 / AAVrh13 hybrid VP1 capsid DNA (AAVhu32
                        VP1u + AAVhu32 VP1/2s + AAVrh13 VP3)
                  organism = Synthetic construct

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SEQUENCE: 30
atggctgcgc atggttatct tccagattgg ctcgaggaca ctctctctga aggaataaga 60
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gacacgaggg gtctttgtgct tcctgggtac aagtacctcg gaccgcgcaa cggactcgac 180
aagggggagc cggctcaacgc agcagacgcg gcggccctcg agcagcagaa ggccctacgac 240
cagcagctca tggccggaga caaccggtac ctcaagtaca accacgcca gcccaggttc 300
caggagcgcc tcaagaaga tacgtctttt gggggcaacc tcggggcgagc agtctctccg 360
gccaaaaaga ggcttcttga acctcttggt ctgggttagg aagcggctaa gacggctcct 420
ggaaagaaga ggctctgtag caggtctcct caggaaccgg actcctccgc gggatttgge 480
aaatcggttt cacagcccg ctaaaaagaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctcg accctcaacc aatcggagaa cctcccgag cccctcagg tgtgggattc 600
cttacaatgg ctgacgagcg tggcgacca atggcagaca ataacgaagg cgcgcagcga 660
gtgggtaatg cctccgaaa ttggcattgc gattccacat ggctgggcca cagagtcac 720
accaccagca ccgcacactg ggccctgccc acctacaaca accactcta caagcagata 780
tcaagtca ga cggggctac caacgacaac cacttcttcg gctacagcac cccctggggc 840
tattttgact tcaacagatt ccactgccac ttctcaccac gtgactggca gcgactcatc 900
aacaacaact ggggattccg gcccgaaaag ctgcgggtca agttgttcaa catccaggtc 960
aaggagggtc cgacgaacga cggcggttac accatcgcta ataaccttac cagcacgatt 1020
caggtcttct cggactcgga gtaccaactg ccgtacgtcc tcggctctgc gcaccagggc 1080
tgccctccctc cgttccctgc ggacgtgttc atgattcctc agtacggata tctgactcta 1140
aacaacggca gtcagctctg gggacgttcc tccttctact gcctggagta ctttctcttc 1200
cagatgctga gaacgggcaa taactttgaa ttcagctaca cctttgagga agtgcccttc 1260
cacagcagct atgcgcacag ccagagcctg gaccggctga tgaatccctc catcgaccag 1320
tacctgtact acctggcccc gacccagagc actacggggg ccacaaggga gctgcagttc 1380
catcaggctg ggcccaaac catggccgag caatcaaa ga actggctgcc cggaccttgt 1440
tatcggcagc agagactgtc aaaaaacata gacagcaaca acaacagtaa ctttgctctg 1500
accggggcca ctaataacca tctgaatggt agaaattcat taaccaaccc gggcgtagcc 1560
atggccacca acaaggacga cgaggaccag ttctttccca tcaacggagt gctggttttt 1620
ggcgaacggg gggctgccaa caagacaacg ctggaaaaac tgctaagac cagcgaggag 1680
gagatcaaaa ccaccaatcc cgtggctaca gaagaatacg gtgtggtctc cagcaacctg 1740

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caatcgtcta	cggccgggacc	ccagacacag	actgtcaaca	gccagggggc	tctgcccggc	1800
atgggtctggc	agaaccggga	cgtgtacctg	caggggtccca	tctggggcaa	aattcctcac	1860
acggacggca	actttcacc	gtctcccctg	atggggcgat	ttggactcaa	acaccgcct	1920
cctcaaatc	tcatacaaaa	caccccggtg	cctgctaata	ctccagaggt	gtttactcct	1980
gccaaagt	cctcatttat	cacgcagtac	agcacgggcc	aggtcagcgt	ggagatcgag	2040
tgggaactgc	agaagaaaa	cagcaaacgc	tgggaatccg	agattcagta	cacctcaaat	2100
tatgccaaagt	ctaataatgt	ggaatttgct	gtcaacaacg	aaggggttta	tactgagcct	2160
cgccccattg	gcacccggtg	ctccacccgt	aacctgtaa			2199

SEQ ID NO: 31 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 note = AAVhu32 VP1 region protein
 organism = Adeno-associated virus

SEQUENCE: 31

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNRFH	CHFSPRDWQR	300
LINNWNWGRFP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSYAHQS	SLDRMLNPLI	DQYLYLSKT	INGSGQNOQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGQV	ATNHQSAQAQ	AQTGWVQNQG	600
ILPGMWQDR	DVYLQGGPIA	KIPHTDGNFH	PSPLMGGFGM	KHPPQILIK	NTVPVADPPT	660
AFNKDKLNSF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEIQ	YTSNYYKSNN	VEFAVNTEGV	720
YSEPRPIGTR	YLTRNL					736

SEQ ID NO: 32 moltype = DNA length = 2211
 FEATURE Location/Qualifiers
 source 1..2211
 mol_type = genomic DNA
 note = AAVhu32 VP1 region DNA
 organism = Adeno-associated virus

SEQUENCE: 32

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cagtggttga	agctcaaaacc	tggcccacca	ccaccaaaagc	ccgcagagcg	gcataaggac	120
gacagcaggg	gtcttgtgtg	tcttggtgtg	aagtacctcg	gacccggcaa	cggactcgac	180
aagggggagc	cggtcaaacgc	agcagacgcg	gcgccctcgc	agcacgacaa	ggcctacgac	240
cagcagctca	agcccgagga	caaccgcgtg	ctcaagtaca	accacgcgca	cgccgagttc	300
caggagcggc	tcaaaagaaga	tacgtctttt	gggggcaacc	tcgggagcgc	agtcctccag	360
gccccaaaaga	ggcttcttga	acctcttggt	ctgggtgagg	aagcgggtaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaccgg	actcctccgc	gggtatttgc	480
aaatcggtt	ccagacccgc	taaaaagaaa	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcctcc	accctcaacc	aatcggagaa	cctcccgagc	ccccctcagg	tgtgggatct	600
cttacaatgg	cttcaggtgg	tggcgaccca	gtggcagaca	ataacgaagg	tgccgatgga	660
gtgggtagtt	cctcgggaaa	ttggcattgc	gattcccaat	ggctggggga	cagagtcatt	720
accaccagca	cccgaaacct	ggccctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tccaacagca	catctggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
tgggggtatt	ttgacttcaa	cagattccac	tgccacttct	caccacgtga	ctggcagcga	900
ctcatcaaca	acaactgggg	attccggcct	aagcgactca	acttcaagct	cttcaacatt	960
caggtcaaaag	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acggtccagg	tcttcacgga	ctcagactat	cagctcccgt	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tcccgcggtt	cccagcggac	gttttcctga	ttcctcagta	cgggtatctg	1140
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ctaaaattca	gctgggccc	accagcaaac	atggctgtcc	agggaaagaa	ctacatacct	1440
ggaccacgct	accgacaaca	acgtgtctca	accactgtga	ctcaaaacaa	caacagcgaa	1500
tttgcttggc	ctggagcttc	ttcttgggct	ctcaatggac	gtaatatgct	gatgaatcct	1560
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tacacttcca	actattacaa	gtctaataat	gttgaatttg	ctgttaatac	tgaaggtgta	2160
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SEQ ID NO: 33 moltype = AA length = 737

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FEATURE                               Location/Qualifiers
source                               1..737
                                     mol_type = protein
                                     note = AAV6/AAV8 VP1 aa
                                     organism = Synthetic construct

SEQUENCE: 33
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD DGRGLVLPGY KYLGPFNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPFG LVEEGAKTAP GKRPVPEQSP QEPDSSSSGIG KTGQQPAKKR LNFQGTGDSE 180
SVPDPQLPGE PPATPAAVGP TTMAAGGGAP MADNNEGADG VGSSSGNWHC DSTWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNGTSGGATN DNTYFGYSTP WGYFDNRFH CHFSPRDWQR 300
LINNNWGFRR KRLSFKLFNI QVKEVTQNEG TKTIANNLTS TIQVFTDSEY QLPYVLGSAH 360
QGCLPPFPAD VFMIPQYGYL TLNNGSQAVG RSSFYCLEYF PSQMLRTGNN FQPTYTFEDV 420
PFHSSYASHQ SLDRLMNPLI DQYLYLSRT QTTGGTANTQ TLGFSQGGPN TMANQAKNWL 480
PGPCYRQQRV STTTGQNNS NFAWTAGTKY HLNGRNSLAN PGIAMATHKD DEERFFPSNG 540
ILIFGKQNA RDNADYSVM LTSEEEIKTT NPVATEEYGI VADNLQQNT APQIGTVNSQ 600
GALPGMVWQN RDVYLQGPW AKIPHTDGNF HPSPLMGGFG LKHPPQILI KNTPVVPADPP 660
TTFNQSKLNS FITQYSTGQV SVEIEWELQK ENSKRWNPEI QYTSNYYKST SVDFAVNTEG 720
VYSEPRPIGT RYLTRNL 737

SEQ ID NO: 34                       moltype = DNA length = 2214
FEATURE                               Location/Qualifiers
source                               1..2214
                                     mol_type = other DNA
                                     note = AAV6/AAV8 VP1 dna
                                     organism = Synthetic construct

SEQUENCE: 34
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gacggccggg gtctgggtct tcttggttac aagtacctcg gaccttcaa cggactcgac 180
aaggggggagc ccgtcaacgc ggcgggatgca gcggccctcg agcacgaca ggctacgac 240
cagcagctca aagcgggtga caatccgtac ctgcgggtata accacgcga cgcgagttt 300
caggagcgtc tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcacg 360
gccaaagaaga gggttctcga accttttggg ctgggtgagg aaggtgctaa gacggctcct 420
ggaaagaaac tcccggtaga gcagtcgcca caagagccag actcctcctc gggcattggc 480
aagacaggcc agcagccgcg taaaaagaga ctcaattttg gtcagactgg cgactcagag 540
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SEQ ID NO: 35                       moltype = AA length = 738
FEATURE                               Location/Qualifiers
source                               1..738
                                     mol_type = protein
                                     note = AAV6/AAVrh.21 VP1 aa
                                     organism = Synthetic construct

SEQUENCE: 35
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KGEPVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEGAKTAP GKRPVPEPSP QRSPPSSSTGI GKKGQQPARK RLNFQGTGDS 180
ESVPDPQLPG EPPAAPSSVG SGTMAAGGGA PMADNNEGAD GVGSSSGNWH CDSTWLGDRV 240

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ITTSTRWTAL	PTYNNHLYKQ	ISNGTSGGAT	NDNTYFGYST	PWGYFDFNRF	HCHFSPRDWQ	300
RLINNWNWGR	PKRLSFKLFN	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	360
HQGCLPPFFA	DVFMIPQYGY	LTNNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFQFTYTFED	420
VPFHSSYAHS	QSLDRLMNP	IDQLYLYLSR	TQTTGGTANT	QTLGFSQGGP	NTMANQAKNW	480
LPGPCYRQQR	VSTTTGQNNN	SNFAWTAGTK	YHLNGRNSLA	NPGIAMATHK	DDEERFFPSN	540
GLLIFGKQNA	ARDNADYSDV	MLTSEEEIKT	TNPVATEEYG	IVADNLQQQN	TAPQIGTVNS	600
QGALPGMVWQ	NRDYYLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPQIL	IKNTPVPADP	660
PTTFNQSKLN	SFITQYSTGQ	VSVEIEWELQ	KENSKRWNP	IQYTSNYYKS	TSVDFAVNTE	720
GVYSEPRPIG	TRYLTRNL					738

SEQ ID NO: 36 moltype = DNA length = 2217
 FEATURE Location/Qualifiers
 source 1..2217
 mol_type = other DNA
 note = AAV6/AAVrh.21 VP1 dna
 organism = Synthetic construct

SEQUENCE: 36

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cagcagctca	aagcgggtga	caatccgtac	ctgcggtata	atcacgccga	cgcgagttt	300
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ccgaccacct	tcaaccagtc	aaagctgaac	tcttctcatc	cgcaatacac	caccggacag	2040
gtcagcgtgg	aaattgaatg	ggagctgcag	aaggaaaaca	gcaagcgctg	gaaccccgag	2100
atccagtaca	cctccaacta	ctacaaatct	acaagtgtgg	actttgctgt	taatacagaa	2160
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SEQ ID NO: 37 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein
 note = AAV6/AAVrh.73 VP1 aa
 organism = Synthetic construct

SEQUENCE: 37

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KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLPPFG	LVEEGAKTAP	GKKRPVEQSP	QEPDSSSGIG	KTGQQPAKKR	LNFGQGTGDS	180
SVPDPQPLGE	PPATPAVAGP	TMAAGGGAP	MADNNEGADG	VGSSSGNWHC	DSTWLGDRI	240
TTSTRWTALP	TYNNHLYKQI	SNGTSGGSTN	DNTYFGYSTP	WGYFDFNRFH	CHFSPRDWQR	300
LINNWNWGRF	KRLSFKLFNI	QVKEVTQNEG	TKTIANNLTS	TIQVFTDSEY	QLPYVLGSAH	360
QGCLPPFPAD	VFMIPQYGYL	TLNNGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FSFSYTFEDV	420
PFHSSYAHSQ	SLDRLMNP	DQLYLYLSRT	QSTGGTAGTQ	QLLFSQAGPS	NMSAQARNWL	480
PGPCYRQQRV	STTSLQNNNS	NFAWTGATKY	HLNGRDSLNV	PGVAMATNKG	DEDREFFPSSG	540
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GALPGMVWQN	RDVYLQGP	AKIPHTDGNF	HPSPLMGGFG	LKHPPQIL	KNTPVPADPP	660
TAFNQAKLMS	FITQYSTGQV	SVEIEWELQK	ENSKRWNP	QYTSNYYKST	NVDFAVNTEG	720
VYSEPRPIGT	RYLTRNL					737

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SEQ ID NO: 38 moltype = DNA length = 2214
 FEATURE Location/Qualifiers
 source 1..2214
 mol_type = other DNA
 note = AAV6/AAVrh.73 VP1 dna
 organism = Synthetic construct

SEQUENCE: 38

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aagggggagc	ccgtcaacgc	ggcggatgca	gcgccctcg	agcacgacaa	ggcctacgac	240
cagcagctca	aagcgggtga	caatccgtac	ctgcggtata	accacgccga	cgcgagttt	300
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SEQ ID NO: 39 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein
 note = AAV8/AAV6 VP1 aa
 organism = Synthetic construct

SEQUENCE: 39

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KGEPVNAADA	AALEHDKAYD	QQQAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFO	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPVEPSP	QRSPDSSTGI	GKKGQQPARK	RLNFGQTGDS	180
ESVPDPQPLG	EPPAAPSGVG	PNTMASGGGA	PMADNNEGAD	GVGNASGNWH	CDSTWLGDV	240
ITSTRTWAL	PTYNNHLYKQ	ISSASTGASN	DNHYFGYSTP	WGYFDNFRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTNDG	VTIANNLTS	TVQVFSDEY	QLPYVLGSAH	360
QGCLPPFPAD	VFMIPQYGYL	TLNNGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FTFSYTFEDV	420
PFHSYAHSQ	SLDRLMNPFI	DQYLYLNR	QNQSGSAQNK	DLLFSRGSPA	GMSVQPKNWL	480
PGPCYRQQRV	SKTKTDNNNS	NFTWTGASKY	NLNGRESIIN	PGTAMASHKD	DKDKFFPMSG	540
VMIFGKESAG	ASNTALDNVM	ITDEEEKAT	NPVATERFGT	VAVNLQSSST	DPATGDVHVM	600
GALPGMVWQD	RDVYLQGPW	AKIPHDTGHF	HPSPLMGGFG	LKHPPQILI	KNTPVNPANP	660
AEFSATKFAS	FYLYSTGQV	SVEIEWELQK	ENSKRWNPV	QYTSNYAKSA	NVDFTVDNNG	720
LYTEPRPIGT	RILTRPL					737

SEQ ID NO: 40 moltype = DNA length = 2214
 FEATURE Location/Qualifiers
 source 1..2214
 mol_type = other DNA
 note = AAV8/AAV6 VP1 dna
 organism = Synthetic construct

SEQUENCE: 40

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cagtatatat ctaactatgc aaaatctgcc aacgttgatt tcaactgtga caacaatgga 2160
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SEQ ID NO: 41      moltype = AA length = 733
FEATURE
source             Location/Qualifiers
                   1..733
                   mol_type = protein
                   note = AAV8/AAVrh.21 VP1 aa
                   organism = Synthetic construct

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SEQUENCE: 41
MAADGYLPDW LEDNLSEGIR EWWALKPGAP KPKANQKQD DGRGLVLPY KYLGPFGNGLD 60
KGEPVNAADA AALEHDKAYD QQLQAGDNPY LRYNHADA EF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEGAKTAP GKRRPVEPSP QRSPTSSTGI GKKGQQPARK RLNFQGTGDS 180
ESVDPDPQLG EPPAAPSGLV PNTMAAGGGA PMADNNEGAD GVGNASGNWH CDSTWLGDV 240
ITTSTRTWAL PTYNNHLYKQ ISSQSGATND NHFFGYSTPW GYFDFNRFHC HFSPRDWQRL 300
INNNGWFRPR KLRFKLFNIQ VKEVTTNDGV TTIANNLTST IQVFSDS EYQ LPYVLGSAHQ 360
GCLPPFPADV FMIPOYGYLT LNNGSQSVGR SSFYCLEYFP SQMLRTGNMF EFSYTFEEVP 420
FHSSYAHQSQ LDRMLNPLID QYLYYLARTQ STTGSTRELQ FHQAGPNTMA EQSKNWLPGP 480
CYRQQRLSKN IDSNNNSNFA WTGATKYHLN GRNSLTNPGV AMATNKDDED QFFPINGVLV 540
FGKTGAANKT TLENVMTSE EEIKTTNPVA TEEYGVVSSN LQSSTAGPQT QTVNSQCALP 600
GMVWQNRD VY LQGPWIWAKIP HTDGNFHPSP LMGGFGLKHP PPQILIKNTP VPANPPEVFT 660
PAKFASFTIQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYAKSNNVEF AVNNEG VYTE 720
PRPIGTRYLT RNL 733

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SEQ ID NO: 42      moltype = DNA length = 2202
FEATURE
source             Location/Qualifiers
                   1..2202
                   mol_type = other DNA
                   note = AAV8/AAVrh.21 VP1 dna
                   organism = Synthetic construct

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SEQUENCE: 42
atggctgccc atggttatct tccagattgg ctgcaggaca acctctctga gggcattcgc 60
gagtggtggg cgctgaaacc tggagccccg aagcccaaa ccaaccagca aaagcaggac 120
gacggccggg gtctggtgct tcttggtcac aagtacctcg gaccttcaa cggactcgac 180
aaggggggagc ccgtaaacgc ggcggacgca gcggccctcg agcagcagaa ggcctacgac 240
cagcagctgc aggcgggtga caatccgtac ctgcggtata accacgcga cgcgagttt 300
caggagcgtc tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
gccaagaagc gggttctcga acctctcggg ctgggttagg aaggcgctaa gacggctcct 420
ggaaagaaga gaccgtaga gccatcaccc cagcgttctc cagactcctc tacgggcate 480
ggcaagaagc gccaacagcc gccagaaaa agactcaatt ttggtcagac tggcgactca 540
gagtcagttc cagaccctca acctctcggg gaacctccag cagcgcctc tgggtgtgga 600
cctaatacaa tggctcaggc cgggtggcgt ccaatggcag acaataacga aggcgcgcac 660
ggagtgggta atgcctcggg aaattggcat tgcgattcca catggctggg cgacagagtc 720
atcaccacca gcacccgcac ctgggcccctg cccacctaca acaaccacct ctacaagcag 780

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atatcaagtc agagcggggc taccaacgac aaccacttct tcggetacag caccctctgg 840
ggctattttg acttcaacag attccactgc cacttctcac cagtgactg gcagcgactc 900
atcaacaaca actggggatt ccggcccaga aagctgctgt tcaagttgtt caacatccag 960
gtcaaggagg tcacgacgaa cgacggcggt acgaccatcg ccaataacct taccagcacg 1020
attcagggtc tctcggactc ggagtaccaa ctgccgtacg tctcggctc tgcgcaccag 1080
ggctgcctcc ctccgttccc tgcggacgtg ttcagtattc ctacgtacgg atatctgact 1140
ctaaacaacg gcagtcagtc tgtgggacgt tcctccttct actgcctgga gtactttcct 1200
tctcagatgc tgagaacggg caataacttt gaattcagct acaccttga ggaagtgcct 1260
ttccacagca gctatgcgca cagccagagc ctggaccggc tgatgaatcc cctcatcgac 1320
cagtaacctg actacctggc ccggacccag agcactacgg ggtccacaag ggagctgcag 1380
ttccatcagg ctggggccaa caccatggcc gagcaatcaa agaactggct gcccggaacc 1440
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tggaccgggg ccactaataa ccatctgaat ggtagaaatt cattaaccaa cccggggcga 1560
gccatggcca ccaacaagga cgacgaggac cagttcttcc ccatcaacgg agtgctgggt 1620
tttgccaaaa cgggggctgc caacaagaca acgtggtgaa acgtgtctat gaccagcgag 1680
gaggagatca aaaccaccaa tcccggtggc acagaagaat acggtgtggt ctccagcaac 1740
ctgcaatcgt ctacggccgg accccagaca cagactgtca acagccaggg ggctctgcc 1800
ggcatggtct ggcagaaccg ggcagtgtac ctgcagggtc ccatctgggc caaaattcct 1860
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cctcctcaaa ttctcatcaa aaacaccccg gtacctgcta atcctccaga ggtgtttact 1980
cctgccaaag ttgctcatt tatcacgcag tacagcaccg gccaggtcag cgtggagatc 2040
gagtgggaac tgcagaaga aaacagcaaa cgtctggaatc cagagattca gtacacctca 2100
aattatgcca agtctaataa tgtggaattt gctgtcaaca acgaaggggt ttatactgag 2160
cctcgcccca ttggcacccg ttacctcacc cgtaacctgt aa 2202

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SEQ ID NO: 43      moltype = AA  length = 738
FEATURE           Location/Qualifiers
source            1..738
                  mol_type = protein
                  note = AAV8/AAVrh.73 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 43
MAADGYLPDW LEDNLSEGIR EWWALKPGAP KPKANQQKQD DGRGLVLPFY KYLGPFNGLD 60
KGEFVNAADA AALEHDKAYD QQLQAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLPLEG LVEEGAKTAP GKRPVEPSP QRSPTSSTGI GKKGQQPARK RLNFGQTGDS 180
ESVDPDPQLG EPPAAPSGVG PNTMAAGGGA PMADNNEGAD GVGSSSGNWH CDSWLGLDRV 240
ITSTRTWAL PTYNNHLYKQ ISNGTSGGST NDNTYFGYST PWGYFDNRF HCHFSPRDWQ 300
RLINNWWGFR PKRLSFKLFN IQVKEVTQNE GTKTIANNLT STIQVFTDSE YQLPYVLGSA 360
HQGCLPPFFA DVFMIPQYGY LTLNNGSQAV GRSSFYCLEY FPSQMLRTGN NFSFSYTFED 420
VPFHSSYAHS QSLDRLMNPL IDQYLYLSR TQSTGGTAGT QQLLFSQAGP SNMSAQARNW 480
LPGPCYRQOR VSTTSLQNNN SNFAWTGATK YHLNGRDSLIV NPGVAMATNK DDEDRFFPSS 540
GILMPGKQGA GKDNVDYSNV MLTSEEEIKT TNPVATEQYG VVADNLQRQN TAPIVGAVNS 600
QGALPGMWQO NRDVYVLQGI WAKIPHTDGN FHPSPLMGGF GLKHPPPIIL IKNTVPVADP 660
PTAFNQAKLN SFITQYSTGQ VSVEIEWELQ KENSKRWNP IQYTSNYYS TNVDFAVNTE 720
GVYSEPRPIG TRYLTRNL 738

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SEQ ID NO: 44      moltype = DNA  length = 2217
FEATURE           Location/Qualifiers
source            1..2217
                  mol_type = other DNA
                  note = AAV8/AAVrh.73 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 44
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gagtggtggg cgctgaaaacc tggagccccg aagcccaaaag ccaaccagca aaagcaggac 120
gacggccggg gctctggtgct tcttggttac aagtacctcg gacctctcaa cggactcgac 180
aagggggagc ccgtcaaacgc ggcggacgca cgggccctcg agcacgacaa ggcctacgac 240
cagcagctgc agcggggtga caatccgtac ctgcgggtata accacgccga cgcagagttt 300
caggagcgtc tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
gccaaagaag gggttctcga acctctcggt ctggttgagg aaggcgctaa gacggctcct 420
ggaaagaaga gaccggtaga gccatcacc cagcgttctc cagactcctc tacgggcattc 480
ggcaagaaga gccaacagcc cgcagaaaa agactcaatt ttggtcagac tggcgactca 540
gagtcagttc cagacctca acctctcgga gaacctcag cagcgccctc tggtgtggga 600
cctaatacaa tggctgcagg cgttgccgca ccaatggcag acaataacga aggtgccgac 660
ggagtgggta gttcctcggg aaattggcat tgcgattcca catggctggg cgacagagtc 720
atcaccacca gcacccgaac ctgggccctg cccacctaca acaaccacct ctacaagcaa 780
atctccaacg ggacctcggg aggcagcacc aacgacaaca cctactttgg ctacagcacc 840
ccctgggggt attttgactt taacagattc cactgccact tctcaccacg tgactggcag 900
cgactcatca acaacaactg gggattccgg cccaagagac tcagcttcaa gctcttcaac 960
atccaggcca aagaggtcac gcagaatgaa ggcaccaaga ccatcgccaa taacctcacc 1020
agcaccatca aggtgtttac ggaactcgaa taccagctgc cgtactctct cggtctgccc 1080
caccagggtc gctgcctcct tctcccggcg gacgtcttca tgattcctca gtacggctac 1140
ctgactctca acaacggtag tcaggccgtg ggaagttcct ccttctactg cctggagtag 1200
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gtgcctttcc acagcagcta cgcgcacagc cagagtttgg acaggctgat gaatcctctc 1320
atcgaccagt acctgtacta cctgtcaaga acccagctca cgggaggcac agcgggaacc 1380

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cagcagttgc	tgtttttctca	ggcggggcct	agcaacatgt	cggtctcaggc	cagaaaactgg	1440
ctgcctggac	cctgtctacag	acagcagcgc	gtctccacga	cactgtcgca	aaacaacaac	1500
agcaactttg	cctggactgg	tgccaccaag	tatcatctga	acggcagaga	ctctctgggtg	1560
aatccggggc	tgccatggc	aaccaacaag	gacgacgagg	accgctctct	cccatccagc	1620
ggcatcctca	tgtttggcaa	gcagggagct	ggaaaagaca	acgtggacta	tagcaacgtg	1680
atgctaacca	gcgaggga	aatcaagacc	accaaccccg	tggccacaga	acagtatggc	1740
gtggtggctg	ataacctaca	gagacaaaac	accgctccta	ttgtgggggc	cgtcaacagc	1800
cagggagcct	tacctggcat	ggtctggcag	aaccgggacg	tgtacctgca	gggtcctatt	1860
tgggccaaga	ttcctcacac	agatggcaac	tttcacccgt	ctcctttaat	gggcggcttt	1920
ggacttaaac	atccgcctcc	tcagatcctc	atcaaaaaca	ctcctgttcc	tgcggtatcct	1980
ccaacagcgt	tcaaccaggc	caagctgaat	tctttcatca	cgcagtacag	caccggacaa	2040
gtcagcgtgg	agatcgagtg	ggagctgcag	aaggagaaca	gcaagcgtcg	gaaccagag	2100
attcagtata	cttccaacta	ctacaatct	acaaatgtgg	actttgctgt	taatactgag	2160
ggtgtttact	cgtgcctcgc	ccccattggc	actcgttacc	tcacccgtaa	tctgttaa	2217

SEQ ID NO: 45 moltype = AA length = 740
 FEATURE Location/Qualifiers
 source 1..740
 mol_type = protein
 note = AAV9/AAV4 VP1 aa
 organism = Synthetic construct

SEQUENCE: 45

MAADGYLPDW	LEDNLSEGIR	EWALKKPGAP	QPKANQQHQD	NARGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFO	120
AKKRLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGAQPAKKR	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMRAAAGGA	AVEGGQGADG	VGNASGDWHC	DSTWSEGHVT	240
TTSTRTWVLP	TYNHLYKRL	GESLQSNTRY	GFSTPWGYFD	FNRFHCHFSF	RDWQRLINNN	300
WGMRPKAMRV	KIFNIQVKEV	TTSNGETTVA	NNLTSTVQIF	ADSSYELPYV	MDAGQEGSLP	360
PPFNDVFMVP	QYGYCGLVTG	NTSQQQTDRN	AFYCLEYFPS	QMLRTGNNFE	ITYSFEKVPF	420
HSMYAHSQLS	DRLMNLIDQ	YLWGLQSTTT	GTTLNAGTAT	TNFTKLRLPTN	FSNFKKNWLP	480
GPSIKQQGFS	KTANQNYKIP	ATGSDSLIKY	ETHSTLDGRW	SALTPGPMPA	TAGPADSKFS	540
NSQLIFAGPK	QNGNTATVPG	TLIFTSEEL	AATNATDTDM	WGNLPGGDQS	NSNLPTVDRL	600
TALGAVPMGV	WQNRDIYQG	PIWAKIPHTD	GHFHPSPLIG	GFGLKHPPPO	IFIKNTVPVA	660
NPATTFSSSTP	VNSFITQYST	GQVSVQIDWE	IQKERSKRW	PEVQFTSNYG	QQNSLLWAPD	720
AAGKYTEPRA	IGTRYLTHHL					740

SEQ ID NO: 46 moltype = DNA length = 2223
 FEATURE Location/Qualifiers
 source 1..2223
 mol_type = other DNA
 note = AAV9/AAV4 VP1 dna
 organism = Synthetic construct

SEQUENCE: 46

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gagtggtggg	ctttgaaacc	tggagccct	caaccacaag	caaatcaaca	acatcaagac	120
aacgctcgag	gtcttgtgct	tccgggttac	aaataacctg	gaccgggcaa	cggactcgac	180
aagggggagc	cggteaaccg	agcagacgcg	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctca	agcccgagga	caaccctgac	ctcaagtaca	accacgcga	cgccgagttc	300
caggagcggc	tcaagaaga	tacgtctttt	gggggcaacc	tcgggcgagc	agtcctccag	360
gccccaaaaga	ggcttcttga	acctcttggt	ctgggtgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctcct	caggaaccgg	actcctccgc	gggtattggc	480
aaatcgggtg	ccagcccgcc	taaaaagaga	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcaccg	accctcaacc	aatcggagaa	cctcccgagc	ccccctcagg	tgtgggatct	600
cttacaatgc	gtgcagcagc	tggcggagct	gcagtcgagg	gcggacaagg	tgccgatgga	660
gtgggtaatg	cctcgggtga	ttggcattgc	gattccacct	ggtctgaggg	ccacgtaacg	720
accaccagca	ccagaacctg	ggtcttgccc	acctacaaca	accacctcta	caagcgactc	780
ggagagagcc	tgcagtccaa	caacctacaac	ggattctcca	ccccctgggg	atactttgac	840
ttcaaccgct	tccactgcc	cttctcacca	cgtgactggc	agcgactcat	caacaacaac	900
tggggcatgc	gacccaaagc	catgcgggtc	aaaatcttca	acatccaggt	caaggaggtc	960
acgagctcga	acggcgagac	aacggtggct	aataacctta	ccagcagcgt	tcagatcttt	1020
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ccttttccca	acgacgtctt	tatggtgccc	cagtacggct	actgtggact	ggtgaccggc	1140
aacacttcgc	agcaacagac	tgacagaaat	gccttctact	gcctggagta	ctttccttcg	1200
cagatgctgc	ggactggcaa	caactttgaa	attacgtaca	gttttgagaa	ggtgcctttc	1260
cactcgatgt	acgcgcacag	ccagagcctg	gaccggctga	tgaacctctc	catcgaccag	1320
tacctgtggg	gactgcaatc	gaccaccacc	ggaaccaccc	tgaatgccgg	gactgccacc	1380
accaaactta	ccaagctggc	gcctaccaac	ttttccaact	ttaaaaagaa	ctggctgccc	1440
gggccttcaa	tcaagcagca	gggcttctca	aagactgcc	atcaaaaact	caagatccct	1500
gccaccgggt	cagacagtct	catcaaatat	gagacgcaca	gcactctgga	cggaagatgg	1560
agtgccttga	cccccgagcc	tccaatggcc	acggctggac	ctgcggacag	caagttcagc	1620
aacagccagc	tcactcttgc	ggggcctaaa	cagaacggca	acacggccac	cgtaccctgg	1680
actctgatct	tcacctctga	ggaggagctg	gcagccacca	acgcccacga	tacggacatg	1740
tggggcaacc	tacctggcgg	tgaccagagc	aacagcaacc	tgccgaccgt	ggacagactg	1800
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ccactttggg	ccaagattcc	tcataccgat	ggacacttcc	accctccacc	gctgattggt	1920
gggtttgggc	tgaaacaccc	gcctcctcaa	atttttatca	agaacacccc	ggtacctgcg	1980

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aatcctgcaa	cgaccttcag	ctctactccg	gtaaaactcct	tcattactca	gtacagcact	2040
ggccaggtgt	cgggtgcagat	tgactgggag	atccagaagg	agcgggtccaa	acgctggaac	2100
cccagaggtcc	agttttacctc	caactacgga	cagcaaaact	ctctgtgtgtg	ggctcccgat	2160
gcggctggga	aatacactga	gcctagggct	atcggtatccc	gctacctcac	ccaccactg	2220
taa						2223

SEQ ID NO: 47 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein
 note = AAV9/AAV8 VP1 aa
 organism = Synthetic construct

SEQUENCE: 47

MAADGYLPDW	LEDNLSEGIR	EWALKKPGAP	QPKANQQHQD	NARGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLEPLG	LVERAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGAQPAKKR	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMAGGGAP	MADNNEGADG	VGSSSGNWHC	DSTWLGDRI	240
TTSTRTWALP	TYNNHLYKQI	SNGTSGGATN	DNTYFGYSTP	WGYFDFNRFH	CHFSRPDWQR	300
LINNNGWFRP	KRLSFKLFNI	QVKEVTQNEG	TKTIANNLTS	TIQVFTDSEY	QLPYVLGSAH	360
QGCLPPFPAD	VFMIPQYGYL	TLNNGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQPTYTFEDV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYLSRT	QTTGGTANTQ	TLGFSQGGPN	TMANQAKNWL	480
PGPCYRQQRV	STTTQNNNS	NFAWTAGTKY	HLNGRNSLAN	PGIAMATHKD	DEERFFPSNG	540
ILIFGKQNA	RDNADYSVDM	LTSEEBIKTT	NPVATEEYGI	VADNLQQQNT	APQIGTVNSQ	600
GALPGMVWQN	RDVYLQGPW	AKIPHTDGNF	HPSPLMGFGF	LKHPPQILI	KNTPVPADPP	660
TTFNQSKLNS	FITYYSTGVQ	SVEIEWELQK	ENSKRWNPFI	QYTSNYYKST	SVDFAVNTEG	720
VYSEPRPIGT	RYLTRNL					737

SEQ ID NO: 48 moltype = DNA length = 2214
 FEATURE Location/Qualifiers
 source 1..2214
 mol_type = other DNA
 note = AAV9/AAV8 VP1 dna
 organism = Synthetic construct

SEQUENCE: 48

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aacgctcgag	gtcttgtgtc	tccgggttac	aaataccttg	gaccggcaca	cggactcgac	180
aagggggagc	cggtcaacgc	agcagacgcg	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctca	agggcggaga	caaccgcgtac	ctcaagtaca	accacgcgca	cgccgagttc	300
caggagcggc	tcaagaagaa	tacgtctttt	gggggcaacc	tcgggcgagc	agtccttcag	360
gccaaaaaga	ggcttcttga	acctcttggt	ctgggtgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagtcctct	caggaaccgg	actcctccgc	gggtatttgc	480
aaatcgggtg	cacagcccg	taaaaagaga	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcaccg	accctcaacc	aatcggagaa	cctcccgagc	ccccctcagg	tgtgggatct	600
cttacaatgg	ctgcaggcgg	ttggcgacca	atggcgagaca	ataacgaagg	cgccgacgga	660
gtgggtagtt	cctcgggaaa	tggcatttgc	gattccacat	ggctgggcca	cagagtcatt	720
accaccagca	cccgaacctg	ggccctgccc	acctacaaca	accacctcta	caagcaaatc	780
tccaacggga	catcgggagg	agccaccaac	gacaacacct	acttcggcta	cagcaccccc	840
tgggggtatt	ttgactttaa	cagattccac	tgccactttt	caccacgtga	ctggcagcga	900
ctcatcaaca	acaactgggg	attccggccc	aagagactca	gcttcaagct	cttcaacatc	960
caggtcaagg	aggtcacgca	gaatgaaggc	accaagacca	tcgccaataa	cctcaccagc	1020
accatccagg	tgtttacgaa	ctcggagtag	cagctgcccgt	acgttctcgg	ctctgcccac	1080
cagggctgcc	tgctctcggt	cccgcgagac	gtgttcctga	ttccccagta	cggctacctc	1140
acactcaaca	acggtagtag	ggcctgaggc	cgctcctcct	tctactgcct	ggaatacttt	1200
ccttcgcaga	tgtcgagaaa	cggcaacaac	ttccagttta	cttacacctt	cgaggacgtg	1260
cctttccaca	gcagctacgc	ccacagccag	agcttggacc	ggctgatgaa	tcctctgatt	1320
gaccagtacc	tgtactactt	gtctcgagac	caaacacaag	gaggcacggc	aaatacgcag	1380
actctgggct	tcagcccaag	tgggcctaatt	acaatggcca	atcaggcaaa	gaactggctg	1440
ccaggacctt	gttaccgcca	acaaacgcgt	tcaacgacaa	ccgggcaaaa	caacaatagc	1500
aactttgcct	ggactgctgg	gaccaaatac	catctgaatg	gaagaaattc	attggctaatt	1560
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ctcaccagcg	aggaagaat	caaaaccact	aacctgtggg	ctacagagga	atcgggtatc	1740
gtggcagata	acttcgagca	gcaaaacacg	gctcctcaaa	ttggaactgt	caacagccag	1800
ggggccttac	ccggtatggt	ctggcagaac	cgggacgtgt	acctgcaggg	tccatctggg	1860
gccaagattc	ctcacacgga	cggcaacttc	caccctgtct	cgctgatggg	cggctttggc	1920
ctgaaacatc	ctccgctcca	gacccctgat	aagaacacgc	ctgtacctgc	ggatcctccg	1980
accaccttca	accagtcaaa	gctgaactct	ttcatcacgc	aatacagcac	cggacaggtc	2040
agcgtggaaa	ttgaatggga	gctgcagaag	gaaaacagca	agcgtgggaa	ccccgagatc	2100
cagtacacct	ccaactacta	caaatctaca	agtgtggact	ttgctgttaa	tacagaaggc	2160
gtgtactctg	aaccccgccc	cattggcacc	cgttacctca	cccgtaatct	gtaa	2214

SEQ ID NO: 49 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein

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note = AAV9/AAVrh18 VP1 aa
organism = Synthetic construct

SEQUENCE: 49
MAADGYLPDW LEDNLSEGIR EWWALKPGAP QPKANQQHQD NARGLVLPGY KYLGPNGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVPQ 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGAQPAKKR LNFGQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMAAGGGAP MADNNEGADG VGSSSGNWHC DSTWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNGTSGGSTN DNTYFGYSTP WGYFDNRFH CHFSPRDWQR 300
LINNNWGFRR KRLNFKLFNI QVKEVTQNEG TKTIANNLTS TIQVFTDSEY QLPYVLGSAH 360
QGCLPPFPAD VFMIPQYGYL TLNNGSQAVG RSSFYCLEYF PSQMLRTGNN FEFSYQFEDV 420
PFHSSYAHSQ SLDRLMNP LI DQYLYLSRT QSTGGTAGTQ QLLFSQAGPN NMSAQAKNWL 480
PGPCYRQQRV STTSLQNNNS NFAWTGATKY HLNGRDSL VN PGVAMATHKD DEERFFPSSG 540
VLMFGKQGAG KDNVDYSSVM LTSEEEIKTT NPVATEQYGV VADNLQQQNA APIVGAVNSQ 600
GALPGMVWQN RDVYLQGPW AKIPHTDGNF HPSPLMGGGF LKHPPQILI KNTPVVPADPP 660
TTFSSQAKLAS FITQYSTGQV SVEIEWELQK ENSKRWNPEI QYTSNYYKST NVDFAVNTEG 720
TYSEPRPIGT RYLTRNL 737

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SEQ ID NO: 50      moltype = DNA length = 2214
FEATURE           Location/Qualifiers
source            1..2214
                  mol_type = other DNA
                  note = AAV9/AAVrh18 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 50
atggctgccc atggttatct tccagattgg ctcgaggaca accttagtga aggaattcgc 60
gagtggtggg ctttgaacc tggagccct caacccaagg caaatcaaca acatcaagac 120
aacgctcgag gtcttggtct tccgggttac aaataccttg gaccgcgcaa cggactcgac 180
aagggggagc cgggtcaagc agcagacgag cgggcccctc agcagcagaa ggcctacgac 240
cagcagctca aggcgcgaga caaccctgac ctcaagtaca accacgcga cgcgcagttc 300
caggagcggc tcaagaaga tacgtctttt gggggcaacc tcggggcagc agtcttcacg 360
gccaaaaaga ggcttcttga acctcttggt ctgggtgagg aagcggcgtaa gacggctcct 420
ggaaagaaga ggctctgtag gcagctctct caggaaccgg actcctccgc gggattattgc 480
aaatcggttg cacagccgc taaaaagaga ctcaatttcg gtcagactgg cgacacagag 540
tcagtccag acctcaacc aatcggagaa cctccgcgag cccctcagg tgtgggactc 600
cttacaatgg ctgcaggcgg tggcgctcca atggcagaca ataacgaagg cgccgacgga 660
gtgggtagtt cctcaggaaa ttggcattgc gattccacat ggctggcgca cagagtcac 720
accaccagca cccgaaccgt ggcctcccc acctacaaca accacctcta caagcaaatc 780
tccaacggga catcgggagg aagcaccaac gacaacacct acttcggcta cagcaccccc 840
tggggggatt ttgacttta cagattccac tgcacctct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggccc aagagactca actcaagct cttcaacatc 960
caggtcaagg aggtcacgca gaatgaaggc accaagacca tcgccaataa cttaccagc 1020
acgattcagg tctttacgga ctcggaatac cagctcccg acgtcctcgg ctctgcgcac 1080
cagggtctgc tgcctccgtg cccggcggag ctcttcagta ttcctcagta cgggtacctg 1140
actctgaaca acggcagtc ggcctggggc cgttctcct tctactgct ggagacttt 1200
ccttctcaaa tgctggaac gggcaacaac tttgagtcca gctaccagtt tgaggacgtg 1260
ccttttcaca gtcactacgc gcacagccaa agcctggacc ggctgatgaa cccctcactc 1320
gaccagtacc gtactacct gtctcggact cagtcacgg gaggtaccgc aggaactcag 1380
cagttgctat tttctcagc cgggcctaata acatgtcgg ctcaggccaa aaactggcta 1440
cccgggccct gctacgggca gcaacgcgtc tccacgacac tgtcgcaaaa taacaacagc 1500
aactttgctt ggaccggtgc caccagatct catctgaatg gcagagactc tctggtaaat 1560
cccggtgtcg ctatggcaac gcacaaggac gacgaagagc gattttttcc atccagcgga 1620
gtcttgatgt ttgggagctga gggagctgga aaagacaacg tggactatag cagcgttatg 1680
ctaaccagtg aggaagaaat caaaaccacc aaccagtggt ccacagaaca gtacggcgtg 1740
gtggccgata acctgcaaca gcaaaacgcc gctcctattg taggggcccgt caacagtcaa 1800
ggagccttac ctggcattgt ctggcagaac cgggacgtgt acctgcaagg tcctatctgg 1860
gccaaagatt ctcacacgga cggcaacttt catccttcgc cgctgatggg aggcctttgga 1920
ctgaaacacc cgctcctca gatcctgatt aagaatacac ctgttcccgc ggatcctcca 1980
actaccttca gtcaagccaa gctggcgctg ttcacacgc agtacagcac cggacaggtc 2040
agcgtggaaa ttgaatggga gctgcagaaa gagaacagca agcgtggaa cccagagatt 2100
cagtatactt ccaactacta caaatctaca aatgtggact ttgctgtcaa tactgagggt 2160
acttattcag agcctcggcc cattggcacc cgttacctca cccgtaacct gtaa 2214

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SEQ ID NO: 51      moltype = AA length = 733
FEATURE           Location/Qualifiers
source            1..733
                  mol_type = protein
                  note = AAVrh21/AAV8 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 51
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQQKD DGRGLVLPGY KYLGPFGNGLD 60
KGEPVNEADA AALEHDKAYD KQLEQGDNPY LKYNHADA EF QERLQEDTSF GGNLGRAVPQ 120
AKKRVLLEPLG LVEEAGKTAP GKRRPIESPD SSTGIGRKQ QPAKKKLNFQ QTGDSSEVPD 180
PQIGEPFAG PSGLSGTMA AGGAPMADN NEGADGVGSS SGNWHCDSTW LGDRVITTTST 240
RTWALPTYNN HLYKQISNGT SGGATNDNTY FGYSTPWGYF DFNRFHCHFS PRDWQRLINN 300
NWGFRPKRLS FKLFNIQVKE VTQNEGKTKI ANNLTSTIQV FTDSEYQLPY VLGSAHQGCL 360
PPFPADVMI PQYGYLTLNN GSQAVGRSSF YCLEYFPSQM LRTGNNFQFT YTFEDVPFHS 420

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SYAHSQSLDR	LMNPLIDQYL	YYLSRTQTTG	GTANTQTLGF	SQGGPNTMAN	QAKNWLPGPC	480
YRQQRVSTTT	GQNNNSNFAW	TAGTKYHLNG	RNSLANPGIA	MATHKDDEER	FFPSNGILIF	540
GKQNAARDNA	DYSDVMTLSE	EEIKTTNPVA	TEEYGIVADN	LQQQNTAPQI	GTVNSQGALP	600
GMVWQNRDVY	LQGGPIWAKIP	HTDGNFHPSP	LMGGFGLKHP	PPQILIKNTP	VPADPPTTFN	660
QSKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSTSVDF	AVNTEGVYSE	720
PRPIGTRYLT	RNL					733

SEQ ID NO: 52 moltype = DNA length = 2202
 FEATURE Location/Qualifiers
 source 1..2202
 mol_type = other DNA
 note = AAVrh21/AAV8 VP1 dna
 organism = Synthetic construct

SEQUENCE: 52

atggctgccg	atggttatct	tccagattgg	ctcgaggaca	acctctctga	gggcattcgc	60
gagtggtggg	acttgaaac	tggagccccg	aaacccaaag	ccaaccagca	aaagcaggac	120
gacggccggg	gtctgggtct	tcttggttac	aagtacctcg	gaccttcaa	cggactcgac	180
aaggagagcg	cggtcaacga	ggcagacgcc	gcggccctcg	agcacgacaa	ggcctacgac	240
aagcagctcg	agcaggggga	caaccgtac	ctcaagtaca	accacgccga	cgcagagttt	300
caggagcgctc	ttcaagaaga	tacgtctttt	gggggcaacc	tcggggcgagc	agtccttcag	360
gccaagaagc	gggttctcga	acctctcggt	ctgggttgagg	aaggcgctaa	gacggctcct	420
ggaaagaaga	gaccatata	atccccgac	tctccacgg	gcacggcgag	gaaagccag	480
cagcccgcta	aaaagaagct	caactttggg	cagactggcg	actcagagtc	agtgcccgac	540
cctcaaccac	tcggagaacc	ccccgcaggc	ccctctggtc	tgggatctgg	tacaatggct	600
gcaggcggtg	gcgacccaat	ggcagacaat	aacgaagcgc	cgcacggagt	gggtagttcc	660
tcgggaaatt	gcatttgaga	ttccacatgg	ctggggcgaca	gagtcacac	caccagcacc	720
cgaacctggg	ccctgcccac	ctacaacaac	cacctctaca	agcaaatctc	caacggggaca	780
tcgggaggag	ccaccaacga	caacacctac	ttcggtctaca	gcacccctcg	ggggtatttt	840
gactttaaca	gattccactg	ccacttttca	ccacgtgact	ggcagcgact	catcaacaac	900
aactggggat	tccggcccaa	gagactcagc	ttcaagctct	tcaacatcca	ggtcaaggag	960
gtcacgcaga	atgaaggcac	caagaccatc	gccaataacc	tcaccagcac	catccagggtg	1020
tttacggact	cggagatcca	gctgcggtac	gtttctcggt	ctgcccacca	gggctgcttg	1080
cctccgttcc	cggcgagcgt	gttcatgatt	ccccagtagc	gctacctaac	actcaacaac	1140
ggtagtcagg	cgtgtgggag	ctctctcttc	tactgctctg	aatactttcc	ttcgagatg	1200
ctgagaaccg	gcaacaactt	ccagtttact	tacaccttcg	aggacgtgcc	tttccacagc	1260
agctacgccc	acagccagag	cttggaccgg	ctgatgaatc	ctctgattga	ccagtacctg	1320
tactacttgt	ctcggaactca	aacaacagga	ggcacggcaa	atacgagac	cttgggcttc	1380
agccaagggtg	ggcctaatac	aatggccaat	caggcaaaaga	actggctgcc	aggacctgt	1440
taccgccaac	aacgcgtctc	aacgacaacc	gggcaaaaaca	acaatagcaa	ctttgctctg	1500
actgctggga	ccaaatgaga	tctgaatgga	agaaattcat	tggctaattc	tggcatcgct	1560
atggcaaacac	acaagacga	caggagcgct	ttttttccca	gtaacgggat	cctgattttt	1620
ggcaaaaaca	atgctgccag	agacaatgcg	gattacagcg	atgtcatgct	caccagcgag	1680
gaagaaatca	aaacactaa	cctctggtct	acagaggaaat	acggtatcgt	ggcagataac	1740
ttgcagcagc	aaaacacggc	tcctcaaat	ggaactgtca	acagccaggg	ggccttacc	1800
ggtagtggtct	ggcagaaccg	ggacgtgtac	ctgcagggtc	ccatctgggc	caagattcct	1860
cacacggagc	gcaacttcca	ccgctctccg	ctgatgggag	gctttggcct	gaaacatcct	1920
ccgctcaga	tctctgatcaa	gaacacgcct	gtacctgcgg	atcctccgac	caccttcaac	1980
cagtcaaaagc	tgaactcttt	catcacgcaa	tacagcaccg	gacaggtcag	cgtggaaatt	2040
gaatgggagc	tcagagaagga	aaacagcaag	cgttggaaac	ccgagatcca	gtacacctcc	2100
aactactaca	aactctacaag	tgtggacttt	gctgttaata	cagaaggcgt	gtactctgaa	2160
ccccgcccca	ttggcacccg	ttacctcacc	cgtaatctgt	aa		2202

SEQ ID NO: 53 moltype = AA length = 738
 FEATURE Location/Qualifiers
 source 1..738
 mol_type = protein
 note = AAVrh73/AAV8 VP1 aa
 organism = Synthetic construct

SEQUENCE: 53

MAADGYLPDW	LEDNLSEGIR	EWWDLPKGP	KPKANQQQKD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPVEPSP	QRSPDSSTGI	GKKGQQPARK	RLNFGQTGDS	180
ESVPDPQLG	EPAPSSVG	SGTMAAGGGA	PMADNNEGAD	GVGSSSNWNH	CDSTWLGDV	240
ITTSTRTWAL	PTNNHLYKQ	ISNGTSGGAT	NDNTYFGYST	PWGYFDNRF	HCHFSPRDWQ	300
RLINNNGWGR	PKRLSFKLFN	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	360
HQCLPFPFA	DVFMPQYGY	LTLNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFQFTYTFED	420
VPFHSSYAH	QSLDRLMPL	IDQYLYLSR	TQTTGGTANT	QTLGFSQGGP	NTMANQAKNW	480
LPGCYRQQR	VSTTTGQNN	SNFAWTAGTK	YHLNGRNSLA	NPGLAMATHK	DDEERFFPS	540
GILIFGKQNA	ARDNADYSDV	MLTSEEEIKT	TNPVATEEYG	IVADNLQQQN	TAPQIGTVNS	600
QALPGMVWQ	NRDVLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPPQIL	IKNTPVPADP	660
PTTFNQSKLN	SFITQYSTGQ	VSVIEWELQ	KENSKRWNPE	IQYTSNYYSK	TSVDFAVNTE	720
GVYSEPRPIG	TRYLTRNL					738

SEQ ID NO: 54 moltype = DNA length = 2217
 FEATURE Location/Qualifiers
 source 1..2217

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mol_type = other DNA
note = AAVrh73/AAV8 VP1 dna
organism = Synthetic construct

SEQUENCE: 54
atggctgccc atgggttatct tccagattgg ctcgaggaca acctctctga gggcattcgc 60
gagtggtggg acctgaacc tggagccccg aaacccaaag ccaaccagca aaagcaggac 120
gacggccggg gtctggtgct tcttggtctac aagtacctcg gaccttcaa cggactcgac 180
aagggggagc ccgtcaacgc ggcggacgca gcggccctcg agcacgacaa ggctacgac 240
cagcagctca aagcgggtga caatccgtac ctgcggtata atcacgccga cgcgagttt 300
caggagcgct tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcag 360
gccagaagc gggttctcga acctctcggt ctgggtgagg aaggcgctaa gacggctcct 420
ggaaagaaga gaccggtaga gccgtcacca cagcgttccc cgcactctc caccggcatc 480
ggcaagaag gccagcagcc cgccagaaag agactcaatt tcggtcagac tggcgactca 540
gagtcagtc ccgacctca acctctggga gaacctccag cagcgccctc tagtgaggga 600
tctgttaca tggctgcagg cggtggcgca ccaatggcag acaataacga agcgccgac 660
ggagtggtga gttcctcggg aaattggcat tgcgattcca catggctggg cgacagagtc 720
atcacccaca gcacccgaac ctgggcccctg cccacctaca acaaccacct ctacaagcaa 780
atctccaacg ggacatcggt agggagccacc aacgacaaca cctactctcg ctacagcacc 840
ccctgggggt atttgactt taacagattc cactgccact ttccaccag tgactggcag 900
cgactcatca acaacaactg gggattccgg cccaagagac tcagcttcaa gctcttcaac 960
atccaggtca agggagtcac gcagaatgaa ggcaccaaga ccacgcccaa taacctcacc 1020
agcaccatcc aggtgtttac ggactcggag taccagctgc cgtacgttct cggtctgccc 1080
caccagggtc gctgcctcc gtctccggcg gacgtgttca tgattcccca gtacggctac 1140
ctaactca acaacggtag tcaggccgtg ggaagctcct ccttctactg cctggaatac 1200
ttctcttcgc agatgtgag aaccggcaac aacttccagt ttacttacac cttcgaggac 1260
gtgcctttcc acagcagcta cggccacagc cagagcttgg accggctgat gaatcctctg 1320
attgaccagt acctgtacta cttgtctcgg actcaaaaca caggaggcac ggcaataacg 1380
cagactctgg ccttcagcca aggtgggcct aatacaatgg ccaatcaggc aaagaactgg 1440
ctgccaggac gctgtttacc ccaacaacgc gtctcaacga caaccgggca aaacaacaat 1500
agcaactttg cctggactgc tgggacccaaa taccatctga atggaagaaa ttcatgtgct 1560
aatcttgcca ctgctatggc aacacacaaa gacgacgagg agcgtttttt tcccagtaac 1620
gggaccccta tttttgcaa acaaaatgct gccagagaca atgcccatta cagcgatgtc 1680
atgctcacca gcgagggaaga aatcaaaacc actaaccctg tggctacaga ggaatacgtt 1740
atcgtggcag ataaccttga gcagcaaaac acggctcctc aaattggaac tgtcaacagc 1800
cagggggccc tacccggtat ggtctggcag aaccgggacg tgtacctgca gggteccatc 1860
tgggccaaga ttctctcac gcagcgcaac ttccaccctg ctccgctgat gggcggtttt 1920
ggcctgaaac atcctccgct tcagatcctg atcaagaaca cgctgtacc tgcggatcct 1980
ccgaccacct tcaaccagtc aaagctgaac tctttcatca cgcaatacag caccggacag 2040
gtcagcgtgg aaattgaat ggagctgcag aaggaataca gcaagcgctg gaaccccgag 2100
atccagtaca cctccaacta ctacaaatct acaagtgtgg actttgctgt taatacagaa 2160
ggcgtgtact ctgaaccccg ccccatgtgc acccgttacc tcaccgtaaa tctgtaa 2217

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SEQ ID NO: 55      moltype = AA length = 535
FEATURE           Location/Qualifiers
source            1..535
                  mol_type = protein
                  note = AAV3B/AAV8 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 55
MAAGGGAPMA DNNEGADGVG SSSGNWHCDS TWLGDRVITT STRTWALPTY NNHLYKQISN 60
GTSGGATNDN TYFGYSTPFWG YPDFNRFRHCH FSPRDWQRLI NNNWGFPRPKR LSFKLFIQV 120
KEVTQNEGTGK TIANNLSTI QVFTDSEYQL PYVLGSAHQG CLPPFPADVF MIPQYGYLTL 180
NNGSQAVGRS SFYCLEYFPS QMLRTGNNFQ FTYTFEDVPF HSSYAHSQSL DRLMNPLIDQ 240
YLYLSRTQT TGGTANTQTL GFSQGGPNTM ANQAKNWLPG PCYRQQRVST TTGQNNNSNF 300
AWTAGTKYHL NGRNSLANPG IAMATHKDDE ERFPPSNGIL IFGKQNAARD NADYSVMLT 360
SEEEIKTTNP VATEEYGIVA DNLQQQNTAP QIGTVNSQGA LPGMVWQNRD VYLQGPWAK 420
IPHTDGNFHP SPLMGGFGLK HPPPPILIKN TPVPADPPTT FNQSKLNSFI TQYSTGQVSV 480
EIEWELQKEN SKRWNPETQY TSNYYKSTSV DPAVNTEGVY SEPRPIGTRY LTRNL 535

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SEQ ID NO: 56      moltype = DNA length = 1608
FEATURE           Location/Qualifiers
source            1..1608
                  mol_type = other DNA
                  note = AAV3B/AAV8 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 56
atggctgcag gcggtggcgc accaatggca gacaataacg aaggcgccga cggagtggtt 60
agttcctcgg gaaattggca ttgcgattcc acatggctgg gcgacagagt catcaccacc 120
agcaccggaa cctgggccct gccccacctac aacaaccacc tctacaagca aatctccaac 180
gggacatcgg gaggagccac caacgacaac acctactctg gctacagcac cccctggggg 240
tattttgact ttaacagatt ccactgccac ttttcaccac gtgactggca gcgactcatc 300
aacaacaact ggggattccg gcccaagaga ctcagcttca agctcttcaa catccaggtc 360
aaggaggtca cgcagaatga aggcaccaag accatcgcca ataacctcac cagcaccatc 420
cagggtgtta cggactcgga gtaccagctg ccgtacgttc tcggctctgc ccaccagggc 480
tgctgcctc cgttcccggc ggacgtgttc atgattcccc agtacggcta cctaaccatc 540
aacaacggta gtcaggccgt gggacgctcc tccttctact gcctggaata ctttctctgc 600

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cagatgctga gaaccggcaa caacttcag ttactttaca ccttcgagga cgtgccttcc 660
cacagcagct acgcccacag ccagagcttg gaccggctga tgaatcctct gattgaccag 720
tacctgtact acttgtctcg gactcaaaaca acaggaggca cggcaaatat gcagactctg 780
ggcttcagcc aaggtggggc taatacaatg gccaatcagg caaagaactg gctgcccagga 840
ccctgttacc gccacaacac cgtctcaacg acaaccgggc aaaacaacaa tagcaacttt 900
gcctggactg ctggggaccaa ataccatctg aatggaagaa attcattggc taatcctggc 960
atcgctatgg caacacacaa agacgacgag gagcgttttt tcccagtaa cgggactctg 1020
atttttggca aacaaaaatgc tgccagagac aatgcccatt acagcgatgt catgctcacc 1080
agcagaggaag aaatcaaaac cactaaccct gtggctacag aggaatacgg tatcgtagga 1140
gataacttgc agcagcaaaa caccggctcct caaatggaa ctgtcaacag ccagggggcc 1200
ttaccgggta tgggtctggca gaaccgggac gtgtacctgc aggggtccat ctggggccaa 1260
attcctcaca cggacggcaa ctccacccg tctccgctga tggcgcggtt tggcctgaaa 1320
catctccgc ctcagatcct gatcaagaac acgcctgtac ctgcccgatcc tccgaccacc 1380
ttcaaccagt caaagctgaa ctctttcatc acgcaataca gcaccggaa ggtcagcgtg 1440
gaaattgaat gggagctgca gaaggaaaac agcaagcgtc ggaaccccca gatccagtag 1500
acctccaact actacaaatc tacaagtgtg gactttgctg ttaatacaga aggcgtgtac 1560
tctgaacccc gccccattgg caccggttac ctcacccgta atctgtaa 1608

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SEQ ID NO: 57      moltype = AA length = 732
FEATURE           Location/Qualifiers
source            1..732
                  mol_type = protein
                  note = AAVrh15/AAVhu32 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 57
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD DGRGLVLPGY KYLGPFNGLD 60
KGEPVNEADA AALEHDKAYD KQLEQGDNPY LKYNHADA EF QERLQEDTSF GGNLGRAVFO 120
AKKRVLPLGL LVEEGAKTAP GKRPPIESP SGTGIGKKGQ QPAKKKLNFG QTGDSSEVPD 180
PQPIGEPPAG PSGLSGTMA SGGGAPVADN NEGADGVGSS SGNWHCDSQW LGDRVITTST 240
RTWALPTYNN HLYKQISNST SGGSSNDNAY FGYSTPWYF DFNRFHCHFS PRDWQRLINN 300
NWGFRPKRLN FKLFPNIQVKE VTDNNGVKTI ANNLTSTVQV FTDSDYQLPY VLGSAHEGCL 360
PPFPADVPMI PQYGYLTLDN GSQAVGRSSF YCLEYFPSQM LRTGNNQFQS YEFENVPFHS 420
SYAHSQSLDR LMNPLIDQYL YYLSKTINGS GQOQQLKFS VAGPSNMAVQ GRNYIPGPSY 480
RQQRVSTTGT QNNNSEFAWP GASSWALNGR NSLMNPGPAM ASHKEGEDRF FPLSGSLIFG 540
KQGTGRDNVD ADKVMITNEE EIKTTNPVAT ESYGQVATNH QSAQAQAQTG WVQNQGILPG 600
MVWQDRDVLV QGPIWAKIPH TDGNFHPSPL MGGFGMKHPP PQILIKNTPV PADPPTAFNK 660
DKLNSFITQY STGQVSVEIE WELQKENSKR WNPEIQYTSN YYKSNNVEFA VNTGQVYSEP 720
RPIGTRYLTR NL

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SEQ ID NO: 58      moltype = DNA length = 2199
FEATURE           Location/Qualifiers
source            1..2199
                  mol_type = other DNA
                  note = AAVrh15/AAVhu32 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 58
atggctgccg atggttatct tccagattgg ctgaggaca acctctctga gggcattcgc 60
gagtgggtgg acttgaaaac tggagccccg aaacccaaag ccaaccagca aaagcaggac 120
gacggccggg gtctgtgtct tcttggtctac aagtacctcg gaccttcaa cggactcgac 180
aaggagagag cggtcacaga ggcagacgcc gggccctcgc agcacgacaa ggctacgac 240
aagcagctcg agcaggggga caaccctgac ctcaagtaca accacgcga cgcagagttt 300
caggagcctc ttcaagaaga tacgtctttt gggggcaacc tcggcgagc agtcttcacg 360
gccaaagaag cgggttctcg accctcgggt ctgggtgagg aaggcgctaa gacggctcct 420
ggaaagaaga gacctataga atcccccgac tctccacgg gcacggcaa gaaagggcag 480
cagcccgcta aaaagaagct caactttggg cagactggcg actcagatc agtgcggac 540
cctcaaccaa tcggagaacc ccccgaggc cctctggtc tgggatctgg tacaatggct 600
tcaggtgggt gcgcaccagt ggcagacaat aacgaagggt ccgatggagt gggtagttcc 660
tcgggaaatt cggcattgca ttccaatgg ctgggggaca gagtcacac caccagcacc 720
cgaacctggg cctgcccac ctacaacaat cacctctaca agcaaatctc caacagcaca 780
tctggaggat cttcaaatga caacgcctac ttcggtctaca gcacccctg ggggtatttt 840
gacttcaaca gattccactg ccacttctca ccactgtact ggcagcgact catcaacaac 900
aactggggat tccggcctaa gcgactcaac ttcaagctct tcaacattca ggtcaagag 960
gttacggaca acaatggagt caagaccatc gccataaacc ttaccagcac ggtccaggct 1020
ttcacggact cagactatca gctcccgtac gtgctcgggt cggtcacga gggctgcctc 1080
cgcgcgttcc cagcggacgt ttctatgatt cctcagtagc ggtatctgac gcttaatgat 1140
gggagccagg ccgtgggtcg ttctgctctt tactgcctgg aatatttccc gtcgcaaatg 1200
ctaagaacgg gtaacaactt ccagttcagc tacgagtttg agaacgtacc ttcccatagc 1260
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tactatctct caaagactat taacggttct ggacagaatc acaaaacgct aaaattcagc 1380
gtggccggac ccagcaacat ggtgttcagg ggaagaacct acatactgg acccagctac 1440
cgacaacaac gtgtctcaac cactgtgact caaaacaaca acagcgaatt tgcttggcct 1500
ggagcttctt cttgggctct caatggacgt aatagcttga tgaatcctgg acctgctatg 1560
gccagccaca aagaaggaga ggaccgtttc ttctctttgt ctggatcttt aatttttggc 1620
aaacaaggaa ctggaagaga caacgtggat gcggacaaa tcatgataac caacgaagaa 1680
gaaattaaaa ctactaaccc ggtagcaacg gactcctatg gacaagtgcc cacaaccac 1740
cagagtgtccc aagcacaggc gcagaccggc tgggttcaaa accaaggaat acttccgggt 1800

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atggttttgc	aggacagaga	tgtgtacctg	caaggaccca	tttgggcca	aattcctcac	1860
acggacggca	actttcaccc	ttctccgcta	atgggagggg	ttggaatgaa	gcacccgcct	1920
cctcagatcc	tcatacaaaa	cacacctgta	cctgcgggac	ctccaacggc	tttcaataag	1980
gacaagctga	actctttcat	cacccagtat	tctactggcc	aagtcagcgt	ggagattgag	2040
tgggagctgc	agaaggaaaa	cagcaagcgc	tggaacccgg	agatccagta	cacttccaac	2100
tattacaagt	ctaataatgt	tgaatttgct	gttaataactg	aaggtgtata	tagtgaaccc	2160
cgccccattg	gcaccagata	cctgactcgt	aatctgtaa			2199

SEQ ID NO: 59 moltype = AA length = 732
 FEATURE Location/Qualifiers
 source 1..732
 mol_type = protein
 note = AAVrh13/AAVhu32 VP1 aa
 organism = Synthetic construct

SEQUENCE: 59

MAADGYLPDW	LEDNLSEGIR	EWWDLKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNEADA	AALHDKAYD	KQLEQGDNPY	LKYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPIESPD	SSTGIGKKGQ	QPAKKKLNFQ	QTGDSSEVPD	180
PQPLGEPPAA	PSGLSGGTMA	SGGGAPVADN	NEGADGVGSS	SGNWHCDSQW	LGDRVITTST	240
RTWALPTYNN	HLKQISNST	SGSSNDNAY	FGYSTPWGYF	DFNRFHCHFS	PRDWQRLINN	300
NWGFRPKRLN	FKLFNIQVKE	VTNNNGVVTI	ANNLTSTVQV	FTSDYQLPY	VLGSAHEGCL	360
PPFPADVPMI	PQGYLTLND	GSQAVGRSSF	YCLEYFPSQM	LRTGNNFQFS	YEFENVPFHS	420
SYAHSQSLDR	LMNPLIDQYL	YYLSKTINGS	GQNQQTLKFS	VAGPSNMAVQ	GRNYIPGPSY	480
RQQRVSTTVT	QNNNEFAFPW	GASSWALNGR	NSLMNPGPAM	ASHKEGEDRF	FPLSGSLIFG	540
KQGTGRDNVD	ADKVMITNEE	EIKTTNPVAT	ESYGVQVATNH	QSAQAQQTG	WVQNGILPG	600
MVWQDRDVYL	QGPIWAKIPH	TDGNFHPSP	MGGFGMKHPP	PQILIKNTPV	PADPPTAFNK	660
DKLNSFITQY	STGQVSVEIE	WELQKENSKR	WNPEIQYTSN	YYKSNNVEFA	VNTEGVYSEP	720
RPIGTRYLTR	NL					732

SEQ ID NO: 60 moltype = DNA length = 2199
 FEATURE Location/Qualifiers
 source 1..2199
 mol_type = other DNA
 note = AAVrh13/AAVhu32 VP1 dna
 organism = Synthetic construct

SEQUENCE: 60

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gagtggtggg	acttgaacc	tggagccccg	aaacccaaag	ccaaccagca	aaagcaggac	120
gacggccggg	gtctggtgct	tcttggtctac	aagtacctcg	gaccttcaa	cggactcgac	180
aagggagagc	cggtcacaag	ggcagacgccc	gcggccctcg	agcacgacaa	ggcctacgac	240
aagcagctgc	agcaggggga	caacccgtac	ctcaagtaca	accacgcccga	cggcagagtt	300
caggagcgtc	ttcaagaaga	tacgtctttt	gggggcaacc	tcggggcagc	agtcttcacg	360
gccaaagaag	gggttctcga	acctctcggt	ctggttgagg	aaggcgctaa	gacggctcct	420
ggaaagaaga	gacccataga	atcccccgac	tctctccacg	gcacggcga	gaaaggccag	480
cagcccgcta	aaaagaagct	caactttggg	cagactggcg	actcagagtc	agtgcgccac	540
ccccaacctc	tccgagaacc	tcccgccgcg	ccctcaggtc	tgggactctg	tacaatggct	600
tcaggtggtg	gcgcaccagt	ggcagacaat	aacgaagggt	ccgatggagt	gggtagttcc	660
tcgggaaatt	ggcattcgca	ttcccaatgg	ctgggggaca	gagtcacac	caccagcacc	720
cgaacctggg	ccctgcccac	ctacaacaat	cacctctaca	agcaaatctc	caacagcaca	780
tctggaggat	cttcaaatga	caacgcctac	tctggctaca	gcacccctcg	ggggatattt	840
gactcaaca	gattccactg	ccactctcca	ccacgtgact	ggcagcgact	catcaacaac	900
aaactgggat	tccggcctaa	gcgactcaac	ttcaagctct	tcaacattca	ggtcaaaag	960
gttacggaca	acaatggagt	caagaccatc	gccaaataac	ttaccagcac	gggtccaggtc	1020
ttcaccgact	cagactatca	gtctccgtac	gtgctcgggt	cggctcacga	gggctgcctc	1080
ccgcgcttcc	cagcggacgt	ttctcatgatt	cctcagtagc	ggtatctgac	gcttaatgat	1140
gggagccagg	ccgtgggtcg	ttctgctctt	tactgcctgg	aatatttccc	gtcgcaaatg	1200
ctaagaacgg	gtaacaactt	ccagttcagc	tacgagtttg	agaacgtacc	tttccatagc	1260
agctacgctc	acagccaaag	cctggaccga	ctaataatga	cactcactga	ccaatacttg	1320
tactatctct	caaagactat	taacggttct	ggacagaaac	aacaaacgct	aaaattcagc	1380
gtggccggac	ccagcaacat	ggctgtccag	ggaagaaact	acatacctgg	accagctcac	1440
cgacaacaac	gtgtctcaac	cactgtgact	caaaacaaca	acagcgaatt	tgcttggcct	1500
ggagcttctt	cttgggctct	caatggacgt	aatagcttga	tgaatcctgg	acctgctatg	1560
gccagccaca	aagaaggaga	ggaccgtttc	ttctctttgt	ctggatcttt	aatttttggc	1620
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gaaattaaaa	ctactaaccc	ggtagcaacg	gagtcctatg	gacaagtggc	cacaaaccac	1740
cagagtgccc	aagcacaggc	gcagaccggc	tgggttcaaa	accaagggaat	acttccgggt	1800
atggttttgc	aggacagaga	tgtgtacctg	caaggaccca	tttgggcca	aattcctcac	1860
acggacggca	actttcaccc	ttctccgcta	atgggagggg	ttggaatgaa	gcacccgcct	1920
cctcagatcc	tcatacaaaa	cacacctgta	cctgcgggac	ctccaacggc	tttcaataag	1980
gacaagctga	actctttcat	cacccagtat	tctactggcc	aagtcagcgt	ggagattgag	2040
tgggagctgc	agaaggaaaa	cagcaagcgc	tggaacccgg	agatccagta	cacttccaac	2100
tattacaagt	ctaataatgt	tgaatttgct	gttaataactg	aaggtgtata	tagtgaaccc	2160
cgccccattg	gcaccagata	cctgactcgt	aatctgtaa			2199

SEQ ID NO: 61 moltype = AA length = 736
 FEATURE Location/Qualifiers

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source          1..736
                 mol_type = protein
                 note = AAV6/AAVhu32 VP1 aa
                 organism = Synthetic construct

SEQUENCE: 61
MAADGYLDPDW LEDNLSEGIR EWWDLKPGAP KPKANQQQKD DGRGLVLPFY KYLGPFNGLD 60
KGEFVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPFG LVEEGAKTAP GKRRPVEQSP QEPDSSSGIG KTGQQPAKKR LNFQQTGDSE 180
SVPDPQPLGE PPATPAAVGP TTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMNLPI DQYLYLSKT INSGGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDAKVKMI TNEEEIKTTN PVATESYGGV ATNHQSAQAQ AQTGWVQNG 600
ILPGMWQDR DVYLQGPWA KIPHTDGNFH PSPLMGGFGM KHPPQILIK NTPVPADPPT 660
AFNKDKLNSF ITQYSTQVVS VEIEWELQKE NSKRWNPEIQ YTSNYYKSNN VEFVNTGEGV 720
YSEPRPIGTR YLTRNL 736

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SEQ ID NO: 62      moltype = DNA length = 2211
FEATURE           Location/Qualifiers
source            1..2211
                  mol_type = other DNA
                  note = AAV6/AAVhu32 VP1 dna
                  organism = Synthetic construct

SEQUENCE: 62
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gaaggccggg gtcctggtct tcttggtctc aagtaacctc gaccttcaa cggaactcga 180
aagggggagc ccgtcaacgc ggcggatgca cgcccccctc agcacgacaa ggcctacgac 240
cagcagctca cagcgggtga caatccgtac ctgcggtata accacgccga cgccgagttt 300
caggagcgct tgcagaaga tacgtctttt gggggcaacc tcgggagcgc agtctccag 360
gccaagaaga gggttctcga accttttggg ctggttgagg aagtgctaa gacggctcct 420
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aagacaggcc agcagccgcg taaaagaga ctcaattttg gtcagactgg cgactcagag 540
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gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
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ctcatcaaca acaactgggg attcgggctc aagcgactca actcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
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cgttcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagtt tgagaacgta 1260
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ctaaaattca gcgtggccgg acccagcaac atggctgtcc aggggaagaa ctacatacct 1440
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gccacaaacc accagagtgc ccaagcacag gcgcagaccg gctgggttca aaaccaagga 1800
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aagcaccgct ctcctcagat cctcatcaaa aacacacctg tacctgcgga tcttccaacg 1980
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tacacttcca actattacaa gtctaataat gttgaatttg ctgttaatac tgaaggtgta 2160
tatagtgaac cccgccccat tggcaccaga tacctgactc gtaactctga a 2211

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SEQ ID NO: 63      moltype = AA length = 726
FEATURE           Location/Qualifiers
source            1..726
                  mol_type = protein
                  note = AAV5/AAVhu32 VP1 aa
                  organism = Synthetic construct

SEQUENCE: 63
MSFVDHPPDW LEEVGEGLRE FLGLEAGPPK PKPNQQHQDQ ARGVLVPGYN YLGPNGNGLDR 60
GEPVNRADRV AREHDI SYNE QLEAGDNPYL KYNHADAEFQ EKLADDTSFQ GNLGKAVFQA 120
KKRVLEPFG LVEEGAKTAP GKRIIDHFPK RKKARTEEDS KPSTSSDAEA GPSGSQQLQI 180
PAQPASSLGA DTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI TTSTRTWALP 240
TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR LINNNWGFPR 300

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KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	EGCLPPFPAD	360
VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	QFQSYEFENV	PFHSSSYAHSQ	420
SLDRLMNPIL	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	GPSYRQQRVS	480
TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	LIFGKQGTGR	540
DNVDADKVM	TNEEEIKTTN	PVATESYGVV	ATNHQSAQAQ	AQTGWVQNGG	ILPGMVWQDR	600
DVYLQGPWA	KIPHTDGNFH	PSPLMGGFGM	KHPPPIILIK	NTPVPADPPT	AFNKDKLNSF	660
ITQYSTGQVS	VEIEWELQKE	NSKRWNPEIQ	YTSNYYKSNN	VEFAVNTBEGV	YSEPRPIGTR	720
YLTRNL						726

SEQ ID NO: 64 moltype = DNA length = 2181
 FEATURE Location/Qualifiers
 source 1..2181
 mol_type = other DNA
 note = AAV5/AAVhu32 VP1 dna
 organism = Synthetic construct

SEQUENCE: 64

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gcccgtggt	ttgtgtgct	tggttataac	tatctcggac	ccggaacggg	tctcgatcga	180
ggagagcct	tcaacagggc	agacgaggtc	gcgcgagagc	acgacatctc	gtacaacgag	240
cagcttgag	cgggagacaa	cccctacctc	aagtacaacc	acgcggacgc	cgagtttcag	300
gagaagctc	cgacgacac	atccttcggg	ggaacacctc	gaaaggcagt	cttcagggcc	360
aagaaaagg	ttctcgaa	ttttggcctg	gttgaagagg	gtgctaagac	ggccccctac	420
ggaaagcga	tagacgacca	ctttccaaaa	agaaagaagg	ctcggaccga	agaggactcc	480
aagccttcca	cctcgtcaga	gcgcgaagct	ggaccacgag	gatcccgaga	gctgcaaatc	540
ccagcccaac	cagcctcaag	tttgggagct	gatacaatgg	cttcaggtag	tgccgcacca	600
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acctacaaca	atcacctcta	caagcaaatc	tccaacagca	catctggagg	atcttcaaat	780
gacaaagcct	acttcggcta	cagcaccccc	tggggggtatt	ttgacttcaa	cagattccac	840
tgccacttct	caccacgtga	ctggcagcga	ctcatcaaca	acaactgggg	attccggcct	900
aagcgactca	acttcaagct	cttcaacatt	caggtcaaa	aggttacgga	caacaatgga	960
gtcaagacca	tcgccaataa	ccttaccagc	acggtcacag	tcttcacgga	ctcagactat	1020
cagctcccg	acgtgctcgg	gtcggctcac	gagggctgcc	tcccgccgtt	cccagcggac	1080
gttttcacga	ttcctcagta	cggggtatct	acgcttaatg	atggggagcca	ggccgtgggt	1140
cgttcgtcct	tttactgcct	ggaatatttc	ccgtcgcgaa	tgctaagaac	gggtaacaac	1200
ttccagttca	gctacgagtt	tgagaacgta	cctttccata	gcagctacgc	tcacagccaa	1260
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atggctgtcc	agggagaata	ctacatacct	ggaccacgct	accgacaaca	acgtgtctca	1440
accactgtga	ctcaaaaaca	caacagcgaa	tttgcttggc	ctggagcttc	ttcttgggct	1500
ctcaatggac	gtaatagctt	gatgaatcct	ggacgtgcta	tgccacgcca	caaagaagga	1560
gaggacccgt	tctttccttt	gtctggatct	ttaatttttg	gcaaacaaag	aactggaaga	1620
gacaacgtgg	atgcggacaa	gtcatgata	accacgaag	aagaaattaa	aactactaac	1680
ccggtagcaa	cggagtccta	tgacaaagtg	gccacaaacc	accagagtgc	ccaagcacag	1740
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ccttctccgc	taattggagg	gtttggaatg	aagcaccgcg	ctcctcagat	cctcatcaaa	1920
aaacacacct	tacctccgga	tcctccaacg	gctttcaata	aggacaagct	gaactctttc	1980
atcacccagt	attctactgg	ccaagtcagc	gtggagattg	agtgggagct	gcagaaggaa	2040
aacagcaagc	gctggaaccc	ggagatccag	tacacttcca	actattacaa	gtctaataat	2100
gttgaatttg	ctgttaatac	tgaagtgta	tatagtgaac	cccgcgccat	tggcaccaga	2160
tacctgactc	gtaatctgta	a				2181

SEQ ID NO: 65 moltype = AA length = 732
 FEATURE Location/Qualifiers
 source 1..732
 mol_type = protein
 note = AAVhu32/AAVrh15 VP1 aa
 organism = Synthetic construct

SEQUENCE: 65

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFQQTGDTE	180
SVPDPQPIGE	PNAAPSGVGS	LTMAGGGAP	MADNNEGADG	VGNASGNWHC	DSTWLGRDVI	240
TTSTRTWALP	TYNHNLKYQI	SSQSGATNDN	HFFGYSTPWG	YFDFNRFHCH	FSPRDWQRLI	300
NNNWGFPRPK	LRFKLFNIQV	KEVTNDGVT	TIANNLTSTI	QVFSDESYQL	PYVLGSAHQG	360
CLPPFPADVF	MIPQYGYLTL	NNGSQSVGRS	SFYCLEYFPS	QMLRTGNIFE	FSYTFEEVFP	420
HSSYAHSQSL	DRLMNPILID	YLYYLARTQS	TTGSTRELQF	HQAGPNTMAE	QSKNWLPGPC	480
YRQQRLSKNI	DSNNTSNFAW	TGATKYHLNG	RNSLTNPQVA	MATNKDDEDQ	FFPINGVLVF	540
GKTGAANKTT	LENVLMYSEE	EIKTTNPVAT	EQYGVVSNL	QSSTAGPOTQ	TVNSQGALPG	600
MVWQNRDVYL	QGPIWAKIPH	TDGNFHPSP	MGGFGLKHPP	PQILIKNTPV	PANPPEVFTP	660
AKFASFITQY	STGQVSVEIE	WELQKENSKR	WNPEIQYTSN	YAKSNNVEFA	VNNEGVTYEP	720
RPIGTRYLTR	NL					732

SEQ ID NO: 66 moltype = DNA length = 2199

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FEATURE
source Location/Qualifiers
1..2199
mol_type = other DNA
note = AAVhu32/AAVrh15 VP1 dna
organism = Synthetic construct

SEQUENCE: 66

atggctgccg	atggttatct	tccagattgg	ctcgaggaca	ctctctctga	aggaataaga	60
cagtggtgga	agctcaaac	tggcccacca	ccaccaaagc	ccgcagagcg	gcataaggac	120
gacagcaggg	gtcttgtgct	tccctgggtac	aagtacctcg	gaccgcgcaa	cggactcgac	180
aaggggggagc	cggccaacgc	agcagacgcg	gcggccctcg	agcagcagaa	ggcctacgac	240
cagcagctca	aggccggaga	caacccgtac	ctcaagtaca	accacgcgga	cgcgaggttc	300
caggagcggc	tcaagaaga	tacgtctttt	gggggcaacc	tcgggcgagc	agtcttcacg	360
gccaaaaaga	ggcttcttga	acctcttggg	ctgggttagg	aagcgggtaa	gacggctcct	420
ggaaaagaaga	ggcctgtaga	gcagtctcct	caggaaccgg	actcctcgcg	gggtattggc	480
aaatcgggtt	cacagcccg	taaaaagaaa	ctcaatttgc	gtcagactgg	cgacacagag	540
tcagtccccg	accctcaacc	aatcggagaa	cctcccgagc	ccccctcagg	tgtgggatct	600
cttacaatgg	ctgcaggcgg	tggcgctcca	atggcagaca	ataacgaagg	cgcgcagcga	660
gtgggtaatg	cctccggaag	tggcattgac	gattccacat	ggctgggcga	cagagtcac	720
accaccagca	cccgcacctg	ggcctgcccc	acctacaaca	accacctcta	caagcagata	780
tcaagttaga	gcggggctac	caacgacaa	cactctctcg	gctacagcac	ccccggggc	840
tattttgact	tcaacagatt	ccactgccac	ttctcaccac	gtgactggca	gcgactcatc	900
aacaacaact	ggggattccg	gcccagaaag	ctgcggttca	agttgttcaa	catccaggtc	960
aaggaggtca	gcagcaacga	cggcggttac	accatcgcta	ataaccttac	cagcacgatt	1020
caggtctctt	cggactcgga	gtaccaactg	ccgtacgtcc	tcggctctgc	gcaccagggc	1080
tgctccctc	cggtccctgc	ggacgtgttc	atgattcctc	agtacggata	tctgactcta	1140
aacaacggca	gtcagtctgt	gggacgttcc	tcctctact	gcctggagta	ctttccttct	1200
cagatgtctg	gaacgggcaa	taactttgaa	ttcagctaca	cctttgagga	agtgcctttc	1260
cacagcagct	atgcgcacag	ccagagcctg	gaccggctga	tgaatccctc	catcgaccag	1320
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accggggcca	ctaaataacca	tctgaatggt	agaaattcat	taaccaaccc	gggcgtagcc	1560
atggccacca	acaaggacga	cgaggaccag	ttctttccca	tcaacggagt	gctggttttt	1620
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gagatcaaaa	ccaccaatcc	cgtgggtaca	gaacagtagc	gtgtggtctc	cagcaacctg	1740
caatcgtcta	cggccggacc	ccagacacag	actgtcaaca	gccagggggc	tctgccccgc	1800
atggtctggc	agaaccggga	cgtgtacctg	cagggtccca	tctgggccaa	aattcctcac	1860
acggacggca	actttcaccc	gtctccccct	atgggcggat	ttggactcaa	acaccgcct	1920
cctcaaatcc	tcatcaaaaa	caccccggtg	cctgtcaatc	ctccagaggt	gtttactcct	1980
tgcaagtttg	cctcatttat	cacgcagtag	agcaccggcc	aggtcagcgt	ggagatcgag	2040
ggggaactgc	agaagaagaa	cagcaaacgc	tggaatccag	agattcagta	cacctcaaat	2100
tatgcccaag	ctaataatgt	ggaatttgct	gtcaacaacg	aagggggtta	tactgagcct	2160
cgccccattg	gcacccgtta	cctcaccctg	aacctgtaa			2199

SEQ ID NO: 67 moltype = AA length = 732
FEATURE Location/Qualifiers
source 1..732
mol_type = protein
note = AAV6/AAVrh15 VP1 aa
organism = Synthetic construct

SEQUENCE: 67

MAADGYLPDW	LEDNLSEGIR	EWWDLKPAGP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPFG	LVEEGAKTAP	GKKRPVEQSP	QEPDSSSGIG	KTGQQPAKKR	LNFGQTGDS	180
SVPDPQPLGE	PPATPAAVGP	TTMAAGGGAP	MADNNEGADG	VGNASGNWHC	DSTWLGDRVI	240
TTSTRTWALP	TYNNHLYKQI	SSQSGATNDN	HFFGYSTPWG	YFDNRFHCH	FSPRDWQRLI	300
NNNWGFRPRK	LRFLKFNIVQ	KEVTNDGVT	TIANNLTSTI	QVFSDEYQL	PYVLGSAHQG	360
CLPPFPADV	MIPQYGLTL	NNGSQSVGRS	SFYCLEYFPS	QMLRTGNNE	FSYTFEEVFP	420
HSSYAHSQSL	DRLMNLIDQ	YLYLARTQS	TTGSTRELQF	HQAGPNTMAE	QSKNWLPGPC	480
YRQRLSKNI	DSNNTSNFAW	TGATKYHLNG	RNSLTNPQVA	MATNKDDEDQ	FFPINGVLVF	540
GKTGAANKTT	LENVLMTSEE	EIKTTNPVAT	EQYGVSSNL	QSSTAGPQTQ	TVNSQGALPG	600
MVWQNRDVYL	QGPIWAKIPH	TDGNFHPSP	MGGFGLKHPP	PQILIKNTPV	PANPPEVFT	660
AKFASFITQY	STGQVSVEIE	WELQKENSKR	WNPEIQYTSN	YAKSNNVEFA	VNNEGVYTEP	720
RPIGTRYLTR	NL					732

SEQ ID NO: 68 moltype = DNA length = 2199
FEATURE Location/Qualifiers
source 1..2199
mol_type = other DNA
note = AAV6/AAVrh15 VP1 dna
organism = Synthetic construct

SEQUENCE: 68

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gagtggtggg	acttgaaac	tggagccccg	aaacccaag	ccaaccagca	aaagcaggac	120
gacggccggg	ctctgggtct	tcttggtctc	aagtacctcg	gaccttcaa	cggactcgac	180
aaggggggagc	ccgtcaacgc	ggcggatgca	gcggccctcg	agcagcagaa	ggcctacgac	240

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cagcagctca aagcgggtga caatccgtac ctgcggtata accacgccga cgccgagttt 300
caggagcgctc tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcacag 360
gccaagaaga ggggtctcga accttttgggt ctggttgagg aaggtgctaa gacggctcct 420
ggaaagaaac gtccggtaga gcagtcgcca caagagccag actcctctc gggcattggc 480
aagacagggc agcagccgcg taaaaagaga ctcaattttg gtcagactgg cgactcagag 540
tcagtcgccg acccacaaac tctcggagaa cctccagcaa ccccgctgc tgtgggacct 600
actacaatgg ctgcaggcgg tgccgctcca atggcagaca ataacgaagg cgccgacgga 660
gtgggtaatg cctcgggaaa ttggcattgc gattccacat ggctgggcga cagagtcac 720
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tcaagtcaga gcggggctac caacgacaa cacttcttcg gctacagcac cccctggggc 840
tattttgact tcaacagatt ccactgccac ttctcaccac gtgactggca gcgactcac 900
aacaacaact ggggattccg gcccagaaag ctgcgggtta agttgttcaa catccaggtc 960
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cagggtctctt cggactcgga gtaccactg cegtacgtcc tcggctctgc gcaccagggc 1080
tgccctccctc cgttccctgc ggaactgttc atgattcctc agtacggata tctgactcta 1140
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cagatgctga gaacgggcaa taactttgaa ttcagctaca cctttgagga agtgcctttc 1260
cacagcagct atgcgcacag ccagagcctg gaccggctga tgaatccctt catcgaccag 1320
taactgtact acctggcccg caaccagagc actacggggg ccacaaggga gctgcagttc 1380
catcaggctg ggcaccaaac catggccgag caatcaaaaga actggctgcc cggacctgt 1440
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acgggggcca ctaaatacca tctgaatggg agaaattcat taaccaaccc gggcgtagcc 1560
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gagatcaaaa ccaccaatcc cgtggctaca gaacagtacg gtgtggtctc cagcaacctg 1740
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atggtctggc agaaccggga cgtgtacctg cagggtccca tctgggcca aattcctcac 1860
acggacggca actttcacc gtcctccctg atgggaggat ttgactcaa acaccgcct 1920
cctcaaatte tcatcaaaaa caccgccgta cctgctaate ctccagaggt gtttactcct 1980
gccaagtttg cctcatttat cacgcagtag agcacgggcc aggtcagcgt ggagatcgag 2040
tggaactgc agaagaaaaa cagcaaacgc tggaatccag agattcagta cacctcaaat 2100
tatgccaaat taataatgt ggaatttgc gtcaacaacg aaggggttta tactgagcct 2160
cgccccattg gcaaccgtta cctcaccctg aacctgttaa 2199

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SEQ ID NO: 69      moltype = AA  length = 722
FEATURE           Location/Qualifiers
source            1..722
                  mol_type = protein
                  note = AAV5/AAVrh15 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 69
MSFVDHPPDW LEEVEGLRE FLGLEAGPPK PKPNQQHQDQ ARGVLVLPYN YLGPNGGLDR 60
GEPVNRADVE AREHDI SYNE QLEAGDNPYL KYNHADAEPQ EKLADDSFG GNLGKAVFQA 120
KKRVLEPFG LVEGAKTAPT GKRIDDHFPK RKKARTEEDS KPSTSSDAEA GPSGSQQLQI 180
PAQPASSLGA DTMAAGGGAP MADNNEGADG VGNASGNWHC DSTWLGDRI TTSRTWALP 240
TYNNHLYKQI SSQSGATNDN HFFGYSTPWG YFDNFRFHCH FSPRDWQRLI NNNWGFPRPK 300
LRFKLFNIQV KEVTTNDGVT TIANNLTSTI QVFSDEYQL PYVLGSAHQG CLPPFPADVF 360
MTPQYGYLTL NNGSQSVGRS SFYCLEYFPS QMLRTGNNEF FSYTFEEVFP HSSYAHSQL 420
DRLMNPIDQ YLYYLARTQS TTGSTRELQF HQAGPNTMAE QSKNWLPGPC YRQRLSKNI 480
DSNNTSNFAW TGATKYHLNG RNSLTNPGVA MATNKDDEDQ FFPINGVLVF GKTGAANKTT 540
LENVLMTSEE EIKTTNPVAT EQYGVVSSNL QSSTAGPQTQ TVNSQGLPG MVWQNRDVYL 600
QGIWAKIPH TDGNFHPPL MGGFGLKHP PQLIKNTPV PANPPEVFTP AKFASFTQY 660
STGQVSVEIE WELQKNSKR WNPEIQYTSN YAKSNNVEFA VNNEGVYTEP RPIGTRYLTR 720
NL 722

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SEQ ID NO: 70      moltype = DNA  length = 2169
FEATURE           Location/Qualifiers
source            1..2169
                  mol_type = other DNA
                  note = AAV5/AAVrh15 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 70
atgtcttttg ttgatcccc tccagattgg ttggaagaag ttggtgaagg ttttcgcgag 60
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gcccggtggtc ttgtgctgcc tggttataac tatctcgac ccggaacagg tctcgatcga 180
ggagagcctg tcaacagggc agacgaggtc gcgcgagagc acgacatctc gtacaacgag 240
cagcttgagg cgggagacaa cccctacctc aagtacaacc acgcgagcgc cgagtttcag 300
gagaagctcg ccgacgacac atccttcggg ggaacacctg gaaaggcagt ctttcaggcc 360
aagaaaaggg ttctcgaaac ttttggcctg gttgaagagg gtgctaagac ggccctacc 420
ggaaagcggg tagacgacca ctttcaaaaa agaaagaagg ctcgggacga agaggactcc 480
aagccttcca accctgcaga gcgcgaagct ggacccagcg gatcgcaatc gctgcaaatc 540
ccagcccaac cagcctcaag tttgggagct gatacaatgg ctgcaggcgg tggcgctcca 600
atggcagaca ataacgaag cgcgcagcga gtgggtaatg cctccggaaa ttggcattgc 660
gattccacat ggctggggcg cagagtcac accaccagca cccgcacctg ggccctgccc 720
acctacaaca accacctcta caagcagata tcaagtcaga gcggggctac caacgacaac 780
cacttcttcg gctacagcac cccctggggc tattttgact tcaacagatt ccactgccac 840

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ttctcaccac gtgactggca gcgactcacc aacaacaact ggggattccg gccagaaaag 900
ctgcgggttca agttgttcaa catccagggtc aaggagggtca cgacgaacga cggcgttacg 960
accatcgcta ataaccttac cagcacgatt caggtcttct cggactcgga gtaccaactg 1020
ccgtacgtcc tcggctctgc gcaccagggc tgctccctc cgttccctgc ggacgtgttc 1080
atgattcctc agtacggata tctgactcta aacaacggca gtcagtctgt gggacgttcc 1140
tccttctact gcctggagta ctttcttct cagatgctga gaacgggcaa taactttgaa 1200
ttcagctaca ctttgagga agtgcccttc cacagcagct atgcgcacag ccagagcctg 1260
gaccggctga tgaatccct catcgaccag tacctgtact acctggcccg gaccagagc 1320
actacggggt ccacaaggga gctgcagttc catcaggctg ggcccaacac catggccgag 1380
caatcaaaga actggctgcc cggaccctgt tatcggcagc agagactgtc aaaaaacata 1440
gacagcaaca acaccagtaa ctttgcctgg accggggcca ctaaatacca tctgaatggt 1500
agaaattcat taaccaaccc gggcgtagcc atggccacca acaaggacga caggagccag 1560
ttctttccca tcaacggagt gctggttttt ggcaaaacgg gggtgccaa caagacaacg 1620
ctggaaaacg tgcacggag gagatcaaaa ccaccaatcc cgtggctaca 1680
gaacagtacg gtgtggtctc cagcaacctg caatcgtcta cggccggacc ccagacacag 1740
actgtcaaca gccagggggc tctgccgggc atggtctggc agaaccggga cgtgtacctg 1800
cagggtccca tctgggcca aattcctcac acggacggca accttcaacc gtctccctg 1860
atggggcggt ttggactcaa acaccgcct cctcaaatc tcatcaaaaa caccocggta 1920
cctgctaact ctcagaggt gttactcct gccagtttg cctcattat cagcagtag 1980
agcaccggcc gtgcacagc ggagactgc agaagaaaa cagcaaacgc 2040
tggaatccag agattcagta cacctcaaat tatgccaaat ctaataatgt ggaatttgc 2100
gtcaacaacg aagggttata tactgagcct cgccccattg gcaccggtta cctcaccgct 2160
aacctgtaa

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SEQ ID NO: 71      moltype = AA length = 732
FEATURE           Location/Qualifiers
source            1..732
                  mol_type = protein
                  note = AAV6/AAVrh13 VP1 aa
                  organism = Synthetic construct

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SEQUENCE: 71
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD DGRGLVLPGY KYLGPFGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFO 120
AKKRVLPEPG LVEEGAKTAP GKRPVEQSP QEPDSSSGIG KTGQPAKKR LNFQQTGDSE 180
SVPDPQPLGE PPATPAAVGP TTMAAGGGAP MADNNEGADG VGNASGNWHC DSTWLGDRVI 240
TTSTRTWALP TYNHLYKQI SSQSGATNDN HFFGYSTPWG YFDNFRFHC FSPRDWQRLI 300
NNNWGFRPRK LRFKLFIQV KEVTTNDGVT TIANNLTSTI QVFSDSYQL PYVLGSAHQG 360
CLPPFPADV MIPQGYLTL NNGSQSVGRS SFYCLEYFPS QMLRTGNNFE FSYTFEEVFP 420
HSSYAHQSLS DRLEMLIDQ YLYLARTQS TTGSTRELQF HQAGPNTMAE QSKNWLPGPC 480
YRQQLSLKNI DRNNNSNFAW TGATKYHLNG RNSLTNPQVA MATNKDDEQ FFPINGVLVF 540
GETGAANKTT LENVMTSEE EIKTTNPVAT EEYGVVSSNL QSSTAGPQTQ TVNSQGALPG 600
MVWQNRDVL QGPIWAKIPH TDGNFHPSPL MGGFGLKHPP PQILIKNTPV PANPPEVFTP 660
AKFASFITQY STGQVSVEIE WELQKENSKR WNPEIQYTSN YAKSNNVEFA VNNEGVYTEP 720
RPIGTRYLTR NL

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SEQ ID NO: 72      moltype = DNA length = 2199
FEATURE           Location/Qualifiers
source            1..2199
                  mol_type = other DNA
                  note = AAV6/AAVrh13 VP1 dna
                  organism = Synthetic construct

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SEQUENCE: 72
atggctgccc atggttatct tccagattgg ctgcaggaca acctctctga gggcattcgc 60
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gacggccggg gtctggtctc tcttggtac aagtacctcg gaccttcaa cggactcgac 180
aagggggagc ccgtcaacgc ggcggatgca gcggccctcg agcacgacaa ggctacgac 240
cagcagctca aagcgggtga caatccgtac ctgcgggata accacgcga cgcagagttt 300
caggagcgtc tgcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
gccaagaaga gggttctcga accttttggg ctgggtgagg aaggtgctaa gacggctcct 420
ggaaagaaac gtccggtaga gcagtcgcca caagagccag actcctcctc gggcattggc 480
aagacagccc agcagcccg ctcaaatctg gtcagactgg cgactcagag 540
tcagtcctcg acccacaacc tctcggagaa cctccagcaa ccccgctgc tgtgggacct 600
actacaatgg ctgcaggcgg tggcgacca atggcagaca ataacgaagg cgcgacgga 660
gtgggtaatg cctccgaaa ttggcattgc gattccacat ggctgggcca cagagtcac 720
accaccagca ccgcacctg ggcctgccc acctacaaca accactcta caagcagata 780
tcaagtccga gcggggtac caacgacaac cacttcttcg gctacagcac cccctggggc 840
tattttgact tcaacagatt ccactgccac ttctcaccac gtgactggca gcgactcacc 900
aacaacaact ggggattccg gccagaaaag ctgcgggtca agttgttcaa catccaggtc 960
aaggagggtca cgacgaacga cggcgttac accatcgcta ataaccttac cagcacgatt 1020
caggtcttct cggactcgga gtaccaactg ccgtacgtcc tcggctctgc gcaccagggc 1080
tgctccctc cgttccctgc ggacgtgttc atgattcctc agtacggata tctgactcta 1140
aacaacggca gtcagtctgt gggacgttcc tccttctact gcctggagta ctttcttct 1200
cagatgctga gaacgggcaa taactttgaa ttcagctaca ctttgagga agtgcccttc 1260
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tacctgtact acctggcccg gaccagagc actacggggt ccacaaggga gctgcagttc 1380
catcaggctg ggcccaacac catggccgag caatcaaaga actggctgcc cggaccctgt 1440

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tatcggcagc	agagactgtc	aaaaaacata	gacagcaaca	acaacagtaa	ctttgcctgg	1500
acgggggcca	ctaaatacca	tctgaatggt	agaaattcat	taaccaaccc	ggcgctagcc	1560
atggccacca	acaaggacga	cgaggaccag	ttctttccca	tcaacggagt	gctggttttt	1620
ggcgaaacgg	gggctgccaa	caagacaacg	ctggaaaacg	tgctaatac	cagcgaggag	1680
gagatcaaaa	ccaccaatcc	cgtggctaca	gaagaatacg	gtgtggtctc	cagcaacctg	1740
caatcgtcta	cgcccgagcc	ccagacacag	actgtcaaca	gccagggggc	tctgcccgcg	1800
atggtctggc	agaaccggga	cgtgtacctg	cagggtccca	tctgggcccc	aattcctcac	1860
acggacggca	actttcacc	gtctccctg	atggcgcgat	ttggactcaa	acaccgcct	1920
cctcaaatto	tcatacaaaa	caccccggt	cctgctaata	ctccagaggt	gtttactcct	1980
gccaagtgtg	cctcatttat	cacgcagtac	agcaccggcc	aggtcagcgt	ggagatcgag	2040
tggaactg	agaaagaaaa	cagcaaacgc	tggaatccag	agattcagta	cacctcaaat	2100
tatgccaagt	ctaataatgt	ggaatttgc	gtcaacaacg	aaggggttta	tactgagcct	2160
cgccccattg	gcacccggt	cctcaccg	aacctgtaa			2199

SEQ ID NO: 73 moltype = AA length = 1337
 FEATURE Location/Qualifiers
 source 1..1337
 mol_type = protein
 note = DYS1
 organism = Synthetic construct

SEQUENCE: 73

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KLPKEKGSTR	VHALNNVNKA	LRVLQNNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNILHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
FDWNSVVCQQ	SATORLEHAF	NIARYQLGIE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVLP	240
QQVSIEAIQE	VEMLPRPPKV	TKEEHFQLHH	QMHSYQQITV	SLAQGYERTS	SPKPRFKSYA	300
YTQAAYVTTT	DPTRSPFPFSQ	HLEAPEDKSF	GSSLMSEEVN	LDRYQTALAE	VLSWLLSAED	360
TLQAQGEISN	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QLGSKLIGTG	KLSEDEETEV	420
QEQMNLNLSR	WECLRVASME	KQSNLHRVLM	DLQNQKLKEL	NDWLTKTEER	TRKMEEEPGL	480
PDLEDLKRQV	QQHKVLQEDL	EQEQVRVNSL	THMVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLLST	LKNKSVTQKT	EAWLDNFARC	WDNLVQKLEK	660
STAQISQQPD	LAPGLTTIGA	SPTQVTTLVT	QPVTTKETAI	SKLEMPSSLM	LEVPTLERLQ	720
ELQEATDELD	LKLRAEVIK	GSWQPVGDLL	IDSLLQDHLEK	VKALRGEIAP	LKENVSHVND	780
LARQLTTLGI	QLSPYNLSTL	EDLNRWKLL	QVAVEDRVRO	LHEAHRDFGP	ASQHFLSTSV	840
QGPWERAIAP	NKVPYYINHE	TQTTCWDHPK	MTELYQSLAD	LNNVRFSAIR	TAMKLRLRLQ	900
ALCLDLLSL	AACDALDQHN	LKQNDQPMDI	LQIINCLTTI	YDRLEQEHNN	LVNVPLCVDN	960
CLNWLNLVND	TGRTGRIRVL	SFKTGIISLC	KAHLEDKRYR	LFKQVASSTG	FCDQRRLLGL	1020
LHDSIQIPRQ	LGEVASFGGS	NIEPSVRSCF	QFANNKPEIE	AALFLDWML	EPQSMVWLVP	1080
LHRVAAAETA	KHQAKCNICK	ECPIIGFRYR	SLKHFNIDIC	QSCFFSGRVA	KGHKMHYPMV	1140
EYCTPTTSGE	DVRDFAKVLK	NKPRTKRYFA	KHPRMGYLPV	QTVLEGDNME	TPVTLINFWP	1200
VDSAPASSPQ	LSHDDTHSRI	EHYASRLAEM	ENSNGSYLND	SISPNEISDD	EHLILQHYCQ	1260
SLNQDSPLSQ	PRSPAQLLIS	LESEERGELE	RILADLEEN	RNLQAEYDRL	KQQHEHKGLS	1320
PLPSPPEMMP	TSPQSPR					1337

SEQ ID NO: 74 moltype = AA length = 1191
 FEATURE Location/Qualifiers
 source 1..1191
 mol_type = protein
 note = DYS3
 organism = Synthetic construct

SEQUENCE: 74

MLWWEVEEDC	YEREDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPKEKGSTR	VHALNNVNKA	LRVLQNNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNILHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
FDWNSVVCQQ	SATORLEHAF	NIARYQLGIE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVLP	240
QQVSIEAIQE	VEMLPRPPKV	TKEEHFQLHH	QMHSYQQITV	SLAQGYERTS	SPKPRFKSYA	300
YTQAAYVTTT	DPTRSPFPFSQ	HLEAPEDKSF	GSSLMSEEVN	LDRYQTALAE	VLSWLLSAED	360
TLQAQGEISN	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QLGSKLIGTG	KLSEDEETEV	420
QEQMNLNLSR	WECLRVASME	KQSNLHRVLM	DLQNQKLKEL	NDWLTKTEER	TRKMEEEPGL	480
PDLEDLKRQV	QQHKVLQEDL	EQEQVRVNSL	THMVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLLST	LKNKSVTQKT	EAWLDNFARC	WDNLVQKLEK	660
STAQISQQPD	LAPGLTTIGA	SPTQVTTLVT	QPVTTKETAI	SKLEMPSSLM	LEVPTLERLQ	720
ELQEATDELD	LKLRAEVIK	GSWQPVGDLL	IDSLLQDHLEK	VKALRGEIAP	LKENVSHVND	780
LARQLTTLGI	QLSPYNLSTL	EDLNRWKLL	QVAVEDRVRO	LHEAHRDFGP	ASQHFLSTSV	840
QGPWERAIAP	NKVPYYINHE	TQTTCWDHPK	MTELYQSLAD	LNNVRFSAIR	TAMKLRLRLQ	900
ALCLDLLSL	AACDALDQHN	LKQNDQPMDI	LQIINCLTTI	YDRLEQEHNN	LVNVPLCVDN	960
CLNWLNLVND	TGRTGRIRVL	SFKTGIISLC	KAHLEDKRYR	LFKQVASSTG	FCDQRRLLGL	1020
LHDSIQIPRQ	LGEVASFGGS	NIEPSVRSCF	QFANNKPEIE	AALFLDWML	EPQSMVWLVP	1080
LHRVAAAETA	KHQAKCNICK	ECPIIGFRYR	SLKHFNIDIC	QSCFFSGRVA	KGHKMHYPMV	1140
EYCTPTTSGE	DVRDFAKVLK	NKPRTKRYFA	KHPRMGYLPV	QTVLEGDNME	T	1191

SEQ ID NO: 75 moltype = AA length = 1283
 FEATURE Location/Qualifiers
 source 1..1283

-continued

mol_type = protein
note = DYS5
organism = Synthetic construct

SEQUENCE: 75

MLWWEVEEDC	YEREDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPKEKGSTR	VHALNNVNKA	LRVLQNNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNIIILHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
FDWNSVVCQQ	SATQRLHAF	NIARYQLGIE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVLP	240
QQVSIEAIQE	VEMLPRPPKV	TKEEHFQLHH	QMHYSQQITV	SLAQGYERTS	SPKPRFKSYA	300
YTQAAYVTTT	DPTRSPPFSQ	HLEAPEDKSF	GSSLMESVFN	LDRYQTALEE	VLSWLLSAED	360
TLQAQGEISN	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QLGSKLIGTG	KLSEDEETEV	420
QEQMNLNLSR	WECLRVASME	KQSNLHRVLM	DLQNQKLKEL	NDWLTKTEER	TRKMEEEPGL	480
PDLEDLKRQV	QQHKVLEQDL	EQEQRVNSL	THMVVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLLST	LKNKSVTQKT	EAWLNDNFARC	WDNLVQKLEK	660
STAQISQQPD	LAPGLTTIGA	SPTQTVTLVT	QPVVTKETAI	SKLEMPSSLM	LEVPTLERLQ	720
ELQEATDELD	LKLQAEVIK	GSWQPVGDLL	IDSLQDHLEK	VKALRGEIAP	LKENVSHVND	780
LARQLTTLGI	QLSPYNLSTL	EDLNTRWKLL	QVAVEDRVQ	LHEAHRDFGP	ASQHFLSTSV	840
QGFWEAIAIS	NKVPYYINHE	TQTTCWDHPK	MTELYQSLAD	LNNVRFSAYR	TAMKLRRLQK	900
ALCLDLLSL	AACDALDQHN	LKQNDQPMDI	LQIINCLTTI	YDRLEQEHNN	LVNVPLCVDN	960
CLNWLNLNVD	TGRTGRIRVL	SFKTGIIISL	KAHLEDKYRY	LFKQVASSTG	FCDQRRLLGL	1020
LHDSIQIPRQ	LGEVASFVSG	NIEPSVRSCF	QFANNKPEIE	AALFLDWMRL	EPQSMVWLPV	1080
LHRVAAAETA	KHQAACNICK	ECPIIGFRYR	SLKHFNIDIC	QSCFFSGRVA	KGHKMHYPMV	1140
EYCTPTTSGE	DVRDFAKVLK	NKFRTKRYFA	KHPRMGYLPV	QTVLEGDNME	TPVTLLINFWP	1200
VDSAPASSPQ	LSHDDTHSRI	EHYASRLAEM	ENSGSYLND	SISPNESIDD	EHLIIQHYCQ	1260
SLNQDSPLSQ	PRSPAQILIS	LES				1283

SEQ ID NO: 76 moltype = AA length = 1197
FEATURE Location/Qualifiers
source 1..1197
 mol_type = protein
 note = human MD1 (R4-R23/deltaCT)
 organism = Synthetic construct

SEQUENCE: 76

MLWWEVEEDC	YEREDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPKEKGSTR	VHALNNVNKA	LRVLQNNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNIIILHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
FDWNSVVCQQ	SATQRLHAF	NIARYQLGIE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVLP	240
QQVSIEAIQE	VEMLPRPPKV	TKEEHFQLHH	QMHYSQQITV	SLAQGYERTS	SPKPRFKSYA	300
YTQAAYVTTT	DPTRSPPFSQ	HLEAPEDKSF	GSSLMESVFN	LDRYQTALEE	VLSWLLSAED	360
TLQAQGEISN	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QLGSKLIGTG	KLSEDEETEV	420
QEQMNLNLSR	WECLRVASME	KQSNLHRVLM	DLQNQKLKEL	NDWLTKTEER	TRKMEEEPGL	480
PDLEDLKRQV	QQHKVLEQDL	EQEQRVNSL	THMVVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLLST	LKNKSVTQKT	EAWLNDNFARC	WDNLVQKLEK	660
STAQISQAVT	TTQPSLTQTT	VMETVTTVTT	REQILVKHAQ	EELPPPPPPQK	KRQITVDTLE	720
RLQELQEATD	ELDLKLRAE	VIKGSWQPVG	DLIDSLQDH	LEKVKALRGE	IAPLKENVSH	780
VNDLARQLTT	LGIQLSPYNL	STLEDLNTRW	KLLQVAVEDR	VRQLHEAHRD	FGPASQHFSL	840
TSVQGFWEA	NRQPSLTQTT	NHETQTTTCD	HPKMTLEYQS	LADLNNVRFS	AYRTAMKLRR	900
LQKALCLDLL	SLSAACDALD	QHNKQNDQD	MDILOIINCL	TTIYDRLEQE	HNNLVNVPLC	960
VDMCLNWLIN	VDYTGRTGRI	RVLSFKTGII	SLCKAHLEDK	YRYLFKQVAS	STGFCDQRR	1020
GLLLHDSIQI	PRQAGEVASF	GGSNIEPSVR	SCFQFANNKP	EIEAALFLDW	MRLEPQSMVW	1080
LPVLHRVAAA	ETAKHQACN	ICKECPIIGF	RYRSLKHFN	DICQSCFFSG	RVAKGHKMHY	1140
PMVEYCTPTT	SGEDVRDFAK	VLNKFKRTRK	YFAKHPRMGY	LPVQTVLEGD	NMETDTM	1197

SEQ ID NO: 77 moltype = AA length = 1192
FEATURE Location/Qualifiers
source 1..1192
 mol_type = protein
 note = Human microdystrophin
 organism = Synthetic construct

SEQUENCE: 77

MLWWEVEEDC	YEREDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPKEKGSTR	VHALNNVNKA	LRVLQNNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNIIILHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
FDWNSVVCQQ	SATQRLHAF	NIARYQLGIE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVLP	240
QQVSIEAIQE	VEMLPRPPKV	TKEEHFQLHH	QMHYSQQITV	SLAQGYERTS	SPKPRFKSYA	300
YTQAAYVTTT	DPTRSPPFSQ	HLEAPEDKSF	GSSLMESVFN	LDRYQTALEE	VLSWLLSAED	360
TLQAQGEISN	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QLGSKLIGTG	KLSEDEETEV	420
QEQMNLNLSR	WECLRVASME	KQSNLHRVLM	DLQNQKLKEL	NDWLTKTEER	TRKMEEEPGL	480
PDLEDLKRQV	QQHKVLEQDL	EQEQRVNSL	THMVVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLLST	LKNKSVTQKT	EAWLNDNFARC	WDNLVQKLEK	660
STAQISQAVT	TTQPSLTQTT	VMETVTTVTT	REQILVKHAQ	EELPPPPPPQK	KRTLERLQEL	720
QEATDELDLK	LQAEVIKGS	WQPVGDLLID	SLQDHLEKVK	ALRGEIAPLK	ENVSHVNDLA	780
RQLTTLGIQL	SPYNLSTLED	LNTRWKLLQV	AVEDRVRLQH	EAHRDFGPAS	QHFLSTSVQG	840

SEQ ID NO: 78	moltype = AA length = 1325					
FEATURE	Location/Qualifiers					
source	1..1325					
	mol_type = protein					
	note = Dys3978					
	organism = Synthetic construct					
SEQUENCE: 78						
MLWWEVEVDC	YEREDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPEKEGSTS	VHALNVNKA	LRVLQNNVD	LVNIGSTDIV	DGNHKLTLGL	IWNIIHLHWQV	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
QDWSNVSCQQ	SATQRLHFAF	NIARYQLGHE	KLLDPEDVDT	TPYDKKSILM	YITSLFQVPL	240
QQVSIIEAIQE	VEMLPRPPKV	TKEEHQLHGE	QMHSYQQITV	SLAQGYRTS	SPKPRFKSAV	300
YTQAAYVVTTS	DPTRSPFPSPQ	HLEAPEDKSF	GSSLMSEEVN	LDRYQTAAEE	VLSWLLSAED	360
TLQAQGEIIS	DVEVVKDQFH	THEGYMMDLT	AHQGRVGNIL	QSGKSLITGE	KLSEDEETE	420
QGMNLLNSR	WBCLRVASME	KQSNLHRVLM	DLQNKKLKEL	NDLWLTKEER	TRKMEEPLEG	480
PDLEDLKRQV	QQHKVLQEDL	EQEQVRVNSL	THMVVVVDES	SGDHATAALE	EQKVLVGLDRW	540
ANICRWTEDR	WVLLQDQPD	APGLTTIGAS	PTQTVTLVTQ	PVVTKETAI	KLEMPSSML	600
EVPTHRLLQQ	FLPLDLKFLA	WLTEAETIAN	VLQDATRKER	LLEDSKGVKE	LMKQSQDLQG	660
EIEAHTDVYH	NLDENSQKIL	RSLEGSDDAV	LLQRRLDNMN	FKWSELRKKS	LNIRSHLEAS	720
SDQWKRLLHS	LQELLVWLQL	KDDELRSQAP	IGGDFPAVQF	LRQVHRAFFT	ELKTKEPVM	780
SLTETVRIFL	TEQPLEGLEK	LYQEPRELPE	BERAQNVTRL	KNDQAEEVNT	EWEKLNLSHA	840
DWQRKIDETL	ERLQELQEAT	DELDDLKLRA	EVIKGSWQPV	GDLLIDSLOD	HLEKVKALRG	900
EIAPLKENVS	HWNDLARLOL	TLGIQLSPYN	LSLTEDLNTN	WKLQLVAVED	VRVLQHEAHR	960
DFAGSCQHL	STSVQGPWER	AISPKNKVPY	INHETQCTCW	DHKPMTELYS	SADLNLNVRF	1020
SAYRTAMKLR	RLQKALCLDL	LSLSAACDAL	DQHNLKQNDQ	PMDILQIINC	LTTIYDRLEQ	1080
EHNHNLNVPL	CVDMCLNWL	NVYDTRGTAR	IRVLSFKTGI	ISLSCAHANL	KYRYLKFQVA	1140
STSGFCDDRP	KGLLLHDSIQ	IPRLQELGVS	FGGSNIBPSV	RSCEFQAKKK	PEYEAALFQD	1200
WMRLEPQSMV	WLPVLHVRVA	AETAKHQAKC	NICKECPIIG	FRYRSCLKHN	YDICQSCFFS	1260
GRVAKGHKMH	YPMVEYCTPT	TSGEDVRDFA	KVLNKNFRFK	RYFAKHPRMG	YLPVQTVLLE	1320
DNMET						1325

SEQ ID NO: 79	moltype = AA length = 1392					
FEATURE	Location/Qualifiers					
source	1..1392					
	mol_type = protein					
	note = Human MD3					
	organism = Synthetic construct					
SEQUENCE: 79						
MLNWEVEEDC	YERLEDVQKKT	FTKWVNAQFS	KFGKQHIENL	FSDLQDGRRL	LDLLEGLTGQ	60
KLPEKEGSTR	YHALNVNVKA	LRVLQNNVVD	LVNIGSTDIV	DGNHKLTLGL	IWNIIILHWVQ	120
KNVMKNIMAG	LQQTNSEKIL	LSWVRQSTRN	YPQVNVINFT	TSWSDGLALN	ALIHSHRPDL	180
QDWSVVCQQ	SATORLEHAF	NIARYQLGHE	KLLDPEDVDT	TYPDKKSILM	YITSLFQVPL	240
FDVNSIEAQE	VEMLRPPEVK	TKEEHFQLHG	QMHSYQQITV	SLAQSQYRTS	SPKPRFKSYA	300
YTQAAVYVTS	DPTRSPFPQS	HLEAPEDKSF	GSSLMSEEVN	LDRYQTAAEE	VLSWLLSAED	360
TLQAGBEISN	WVEVLWDQFH	THEGYMMDLT	AHQGRVGNIL	QSGKSLIGTG	KLSEDEETEV	420
QEQMNNLSNR	DEVCRLVASME	KQSNLRHVLIM	DLQNKGLKEL	NWGLTKTEER	TRKMEEPLEQ	480
PDLEDLKRQV	QQHKVLQEDL	EEQEQVRVNSL	THMVVVVDES	SGDHATAALE	EQLKVLGDRW	540
ANICRWTEDR	WVLLQDILLK	WQRLTEEQCL	FSAWLSEKED	AVNKIHTTGF	KDQNEMLSSL	600
QKLAVLKADL	EKKKQSMGKL	YSLKQDLBST	LNNKSVTQKT	EAWLDNFARC	WDNLVQKLEK	660
STAQISQAVT	TTQPSLTQTT	VMETVTVTVT	REQILVKHAQ	EELPPPPPPQ	KRQITVDTLE	720
RLQELQEATD	ELDLKLKQAE	VIKGSSQWPV	LDLIDSQDHD	LEKVLKALRE	IAPLKNVVSH	780
VNDLARQLTL	LGIQLSPYNL	STLEDNPTWG	KLLQVAEVRD	LQKVLAHEARD	FGPASQHFLS	840
TSVQGPWERA	ISPKNVPYYI	NHETQTTCWD	HPKMTELYQS	LADLNNVRFS	AYRTAMKLRR	900
LQKALCLDLL	LSLSAACDALD	QHNLKQNDQP	MDILQIINCL	TYTIDRLBQE	HNNLVNVPLC	960
VDMLCNLLDL	VSDATGTGRI	RVLSTQGTGI	SLCKAHLEDD	RRYDFKVLWS	STGFCNQDRL	1020
GLLLHDSIQI	PRQLGEVASF	GGSNIEPSVR	SCQFQANNKP	EIEAALFDWV	MRLEPQSMVW	1080
LPVLHVRVAA	ETAKHQAKCN	ICEKCPGIFG	RYRSLKHFNY	DICQSCFSEG	RVAKGHKMHY	1140
PMVEYCTPTT	SGEDVRDFAK	VLKNKFRTRK	YFAKHPRMGY	LPVQTVLEFG	NMETPVTLIN	1200
FWPVDAPAS	SPQLSHDDTH	SRIEHYASRL	AEEMNSNGSY	LNDSTSPNES	IDDEHLLIQH	1260
YCQSLNQDSP	LSQPRSPAQI	LISLESEERG	ELERILADLE	EENRNLAQAY	DRLKQOQHEK	1320
GLSPPLSPPE	NMPTSPQSPR	DAELIAEAKL	LRQHKGRLEA	RMQILEDHNK	QLESQHLHLR	1380
OLLEPOAED	TM					1392

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SEQ ID NO: 80      moltype = AA length = 1471
FEATURE            Location/Qualifiers
source             1..1471
                  mol_type = protein
                  note = Human MD4
                  organism = Synthetic construct
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-continued

SEQUENCE: 80

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MLWWEVEEDC YEREDVQKKT FTKWVNAQFS KFGKQHIENL FSDLQDGRRL LDLLLEGLTGQ 60
KLPKEKGSTR VHALNNVNKA LRVLQNNNVD LVNIGSTDIV DGNHKLTLGL IWNIIILHWQV 120
KNVMKNIMAG LQQTNSEKIL LSWVRQSTRN YPQVNVINFT TSWSDGLALN ALIHSRHPDL 180
FDWNSVVCQQ SATQRLEHAF NIARYQLGIE KLLDPEDVDT TYPDKKSILM YITSLFQVLP 240
QQVSIETAIQE VEMLP RPFPKV TKEEHFQLHH QMHYSQQITV SLAQGYERTS SPKPRFKSYA 300
YTQAAYVTTT DPTRSPFPQS NLEAPEDKSF GSSLMESVFN LDRYQTAL EE VLSWLLSAED 360
TLQAQGEISN DVEVVKDQFH THEGYMMDLT AHQGRVGNIL QLGSKLIGTG KLSDEETEVE 420
QEQMNLNLSR WECLRVASME KQSNLHRVLM DLQNKQLKEL NDWLTKEER TRKMEEEPGL 480
PDLEDLKRQV QQHKVLQEDL EQEQVRVNSL THMVVVDES SGDHALALE EQLKVLGDRW 540
ANICRWTEDR WVLLQDILLK WQRLTEEQCL FSAWLSEKED AVNKIHTTGF KDQNEMLSSL 600
QKLAVLKADL EKKKQSMGKL YSLKQDLLST LKNKSVTQKT EAWLDNFARC WDNLVQKLEK 660
STAQISQAVT TTQPSLTQTT VMETVTVTVT REQILVKHAQ EELPPPPQK KRQITVDTLE 720
RLQELQEATD ELDLKLQRAE VIKGSWQPVG DLLIDSLQDH LEKVKALRGE IAPLKENVSH 780
VNDLARQLTT LGIQLSPYNL STLEDLNRW KLLQVAVERD VRQLHEAHRD FGPASQHFSL 840
TSVQGWERA PRQLGEVASF GSGNIEPSVR SCFQFANNKP EIEAALFLDW MRLEPQSMVW 900
LQKALCLDLL SLSAACDALD QHNLKQNDQP MDILQIINCL TTIYDRLEQE HNNLVNVPLC 960
VDMCLNWLIN VYDTGRTRGRI RVLSFKTGII SLCKAHLEDK YRYLFKQVAS STGFCQRRRL 1020
GLLLHDSIQV QRLGEVASF GSGNIEPSVR SCFQFANNKP EIEAALFLDW MRLEPQSMVW 1080
LPVLHRVAAA ETAKHQAKCN ICKECPPIGF RYRSLKHPNY DICQSCFFSG RYAKGHKMHY 1140
PMVEYCTPTT SGEVDVRFAK VLKPKFRTKR YFAKHPRMGY LPVQTVLEGD NMETPVTLIN 1200
FWPVSAPAS SPQLSHDFTI NHEHYASRL AEMENSNGSY LNDISIPNES IDDEHLLIQH 1260
YCQSLNQDSP LSQPRSPAQI LISLESEERG ELERILADLE EENRNLAQAY DRKQQQHEHK 1320
GLSPLSPSPE MMPTSPQSPR DAELIAEAKL LRQHKGRLEA RMQILEDHNC QLESQHLRLR 1380
QLLEQPAQAA KVNQTTVSSP STSLQSDSS QPMLLRVVGVS QTSDSMGEDD LLSPPQDTST 1440
GLEEVMEQLN NSPSSSRGRN TPGKPMREDT M 1471

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SEQ ID NO: 81          moltype = AA length = 743
FEATURE
source                Location/Qualifiers
                      1..743
                      mol_type = protein
                      note = Hu32 Variant 1
                      organism = Synthetic construct

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SEQUENCE: 81

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MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFO 120
AKKRLEPLIG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGRDVI 240
TTSTRTWALP TYNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNFRFH CHFSPRDWQR 300
LNNNWGFRP KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTSDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMPLI DQYLYYLSKT INSGQGNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVADKVM I TNEEEIKTTN PVATESYGVQV ATNHQSAQST TLYSPAQAQT 600
GWVQNGQILP GMVWQDRDVI LQGPWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNTP 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYKSNNVFV 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 82          moltype = DNA length = 2232
FEATURE
source                Location/Qualifiers
                      1..2232
                      mol_type = other DNA
                      note = Hu32 Variant 1
                      organism = Synthetic construct

```

SEQUENCE: 82

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atggctgcgc atgggttatct tccagattgg ctcgaggaca ctctctctga aggaataaga 60
cagtgggtgga agctcaaaacc tggcccacca ccaccaaaagc ccgcagagcg gcataaggac 120
gacagcagggt gtcttgtgct tctctgggtac aagtaacctcg gaccgggcaa cggactcgac 180
aagggggagc cgggtcaacgc agcagacgcg gcggccctcg agcagcagaa ggctacgac 240
cagcagctca agcccgagaga caaccctgac ctcaagtaca accacgcga cgcagagttc 300
caggagcggc tcaagaaga taccgtctttt gggggcaacc tcgggcgagc agtcttcacg 360
gccccaaaaga ggcttcttga acctcttggt ctggttgagg aagcgggctaa gacggctcct 420
ggaaagaaga ggctgttaga gcagctctct caggaaaccgg actcctcgc gggtattggc 480
aaatcgggtt cacagccgc taaaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtccccc accctcaacc aatcggagaa cctccgcgag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg tggcgacca gtggcgagaca ataacgaagg tgcgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca ccggaacctg tggccctgcc acctacaaca atcacctcta caagcaaatc 780
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ctcatcaaca caactcggg attcgggcct aagcgactca actcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
gagggctgcc tcccgcgctt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
acgttaaatg atgggagcca ggcctgggt cgttcgctct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtacaac ttccagttca gctacgagtt tgagaacgta 1260

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ctaaaattca	gcgtggcccg	accagcaac	atggctgtcc	aggggaagaa	ctacatacct	1440
ggaccagct	accgacaaca	acgtgtctca	accactgtga	ctcaaaacaa	caacagcgaa	1500
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ggacctgcta	tggccagcca	caaagaagga	gaggaccgtt	tctttccttt	gtctggatct	1620
ttaatttttg	gcaaaacagg	aactggaaga	gacaacgtgg	atgctggacaa	agtcatgata	1680
accaacgaag	aagaaattaa	aactactaac	cggtagcaaa	cggagtctta	tggacaagtg	1740
gccacaaacc	accagagtgc	ccaatcaacg	acgtctctatt	ctccagcaca	ggcgagacc	1800
ggctgggttc	aaaaccaagg	aatacttcg	ggtatggttt	ggcaggacag	agatgtgtac	1860
ctgcaaggac	ccatttgggc	caaaattcct	cacacggacg	gcaactttca	cccttctccg	1920
ctaattgggag	ggtttggaat	gaagcaccgc	cctcctcaga	tcctcatcaa	aaacacacct	1980
gtacctgcgg	atcctccaac	ggctttcaat	aaggacaagc	tgaactcttt	catcaccag	2040
tattctactg	gccaaagctg	cgtggagatt	gagtgaggag	tgcagaagga	aaacagcaag	2100
cgctggaacc	cggagatcca	gtacacttcc	aactattaca	agtctaataa	tgttgaattt	2160
gctgttaata	ctgaaggtgt	atatagtga	ccccgcccc	ttggcaccag	atactgact	2220
cgtaatctgt	aa					2232

SEQ ID NO: 83 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 Variant 2
 organism = Synthetic construct

SEQUENCE: 83

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPNGGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFO	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRVI	240
TTSTRTWALP	TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNRFH	CHFSPRDWQR	300
LINNNWGRFP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSYAHSQ	SLDRLMNLFI	DQYLYLSKT	INGSGQNQOT	LKFSVAGPSN	MAVQGRNIYP	480
GPSYRQQRVS	TTVQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGOV	ATNHQSAQFV	VGQSYAQQT	600
GWVQNGQILP	GMVWQDRDVI	LQGPWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNTP	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSNNVF	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 84 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 Variant 2
 organism = Synthetic construct

SEQUENCE: 84

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gacagcaggg	gtcttctgtc	tctctgggtac	aagtaacctc	gaccggcgaa	cggactcgac	180
aaggggggag	cggctcaacg	agcagacgcg	ggggccctcg	agcagcgaca	ggcctacgac	240
cagcagctca	agcccgagga	caaccgcgtac	ctcaagtaca	accagccgga	cgcagagttc	300
caggagcggc	tcaaaagaaga	tacgtctttt	gggggcaacc	tggggcgagc	agtcttcacg	360
gcaaaaaaga	ggcttcttga	acctcttggt	ctgggttgag	aagcgggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctcct	caggaaccgg	actcctccgc	gggtattggc	480
aaatcgggtt	cacagcccg	taaaaagaaa	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcctcc	accctcaacc	aatcggagaa	cctcccgag	ccccctcagg	tgtgggatct	600
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accaccagca	cccgaacctg	ggccctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tccaacagca	catctggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccctc	840
tgggggtatt	ttgacttcaa	cagattccac	tgccacttct	caccacgtga	ctggcagcga	900
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caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tgcgaataa	ccttaccagc	1020
acggtccagg	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	cccgcgcgtt	ccagcggac	gttttcatga	ttcctcagta	cgggtatctg	1140
acgcttaaat	atgggagcca	ggccgtgggt	cgcttcgtct	tttactgcct	ggaatatttc	1200
cgctcgcaaa	tgctaagaac	gggtaacaac	ttccagttca	gctacagagt	tgagaacgta	1260
cctttccata	gcagctacgc	tcacagccaa	agcctggacc	gactaatgaa	tccactcatc	1320
gaccaatact	tgtactatct	ctcaaagact	attaacgggt	ctggacagaa	tcaacaaacg	1380
ctaaaattca	gcgtggcccg	accagcaac	atggctgtcc	aggggaagaa	ctacatacct	1440
ggaccagct	accgacaaca	acgtgtctca	accactgtga	ctcaaaacaa	caacagcgaa	1500
tttgcttggc	ctggagcttc	ttcttgggct	ctcaatggac	gtaatagctt	gatgaatcct	1560
ggacctgcta	tggccagcca	caaagaagga	gaggaccgtt	tctttccttt	gtctggatct	1620
ttaatttttg	gcaaaacagg	aactggaaga	gacaacgtgg	atgctggacaa	agtcatgata	1680
accaacgaag	aagaaattaa	aactactaac	cggtagcaaa	cggagtctta	tggacaagtg	1740
gccacaaacc	accagagtgc	ccaattcgtt	gtgggacagt	cctatgcaca	ggcgagacc	1800

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ggctgggttc aaaaccaagg aatacttcg ggtatggtt ggcaggacag agatgtgtac 1860
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ctaattgggag ggtttggaat gaagcaccgc cctcctcaga tctcatcaa aaacacacct 1980
gtacctgcgg atccccaac ggctttcaat aaggacaagc tgaactcttt catcaccag 2040
tattctactg cccaagtcag cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
gctgttaata ctgaaggtgt atatagtga ccccgcccca ttggcaccag atacctgact 2220
cgtaactctg aa

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SEQ ID NO: 85      moltype = AA length = 746
FEATURE           Location/Qualifiers
source            1..746
                  mol_type = protein
                  note = Hu32 Variant 3
                  organism = Synthetic construct

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SEQUENCE: 85
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFO 120
AKKRLLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFGQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNRFH CHFSRPDWR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSYAHSQ SLDRLMNP LI DQYLYLSKT INSGQSGRG DLGLGQNGQT LKFSVAGPSN 480
MAVQGRNYP GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG 540
EDRFFPLSGS LIPFGQGTGR DNVDADKVI TNEEEIKTTN PVATESYQV ATNHQSAQAQ 600
AQTGWVQNG ILPGMVWQDR DVYLQGP IWA KIPHTDGNFH PSPLMGGFGM KHPPPQILIK 660
NTPVPADPPT AFNKDKLNSF ITQYSTGQVS VEIEWELQKE NSKRWNPEIQ YTSNYKSN 720
VEFAVNTGEG YSEPRPIGTR YLTRNL

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SEQ ID NO: 86      moltype = DNA length = 2241
FEATURE           Location/Qualifiers
source            1..2241
                  mol_type = other DNA
                  note = Hu32 Variant 3
                  organism = Synthetic construct

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SEQUENCE: 86
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gacagcaggg gtcttgtgtc tcttgggtac aagtacctcg gaccggcaa cggactcgac 180
aagggggagc cggccaacgc agcagacgcg gcggccctcg agcacgaca ggccctacgac 240
cagcagctca agcccgagaga caaccgcgtac ctcaagtaca accacgcga cgcgaggttc 300
caggagcggc tcaaaagaag taccgtcttt gggggcaacc tcgggcgagc agtcttcag 360
gcaaaaaaga ggcttcttga acctcttggc ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggctgtgata gcagctcctc caggaaccgg actcctcgcg gggtattggc 480
aaatcgggtt cacagccgcg taaaaaagaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctcg accctcaacc aatcggagaa cctccgcgag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg ttgcgcacca gtggcagaca ataacgaagg tgcgagtgga 660
gtgggtagtt cctcgggaaa tggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcacctcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgaactcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acgggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
gagggctgcc tcccgccgtt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcgtgggt cgttcgtcct tttactgcct ggaatatctc 1200
ccgtcgcaaa tgctaagaag tggtaacaa ttccagttca gctacgagtt tgagaacgta 1260
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gaccaatact tgtactatct ctcaaagact attaacggtt ctggccagag tggccgcggc 1380
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accacttgta ctcaaaacaa caacagcgaa tttgctggc ctggagcttc tcttgggct 1560
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gaggaccgtt tcttctcttt gcttgatct ttaatttttg gcaaacgaag aactggaaga 1680
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gcgcagaccg gctgggttca aaaccaagga atactccgg gtatgggttg gcaggacaga 1860
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ccttctccgc taatgggagg gtttggaaat aagcaccgc ctcctcagat cctcatcaaa 1980
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aacagcaagc gctggaaccc ggagatccag tacacttcca actattacaa gtctaataat 2160
gttgaatttg ctgttaatac tgaaggtgta tatagtgaac cccgccccat tggcaccaga 2220
tacctgactc gtaactctga a

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SEQ ID NO: 87      moltype = AA  length = 745
FEATURE           Location/Qualifiers
source            1..745
                  mol_type = protein
                  note = Hu32 variant 4
                  organism = Synthetic construct

SEQUENCE: 87
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
AKKRLEPLG LVVEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGD RVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNRFH CHFSPRDWQR 300
LINNNWGFRR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSYAHSQ SLDRLMNLPI DQYLYYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVQNINNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVADAKVMI TNEEEIKTTN PVATESYGVQV ATNHQQQSGR GDLGLSAQAA 600
QTGWVQNQGI LPGMVWQDRD VYLQGGPIWAK IPHTDGNFHP SPLMGGFGMK HPPPIILIKN 660
TPVPADPPTA FNKDKLNSFI TQYSTGQVSV EIEWELQKEN SKRWNPEIQY TSNNYKSNNV 720
EFAVNTEGVY SEPRPIGTRY LTRNL 745

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SEQ ID NO: 88      moltype = DNA  length = 2238
FEATURE           Location/Qualifiers
source            1..2238
                  mol_type = other DNA
                  note = Hu32 variant 4
                  organism = Synthetic construct

SEQUENCE: 88
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gacagcaggg gtcttgtgtac tctctgggtac aagtaacctcg gaccggcaca cggactcgac 180
aaggggggagc cggctcaacgc agcagacgcg cgggcccctcg agcacgacaa ggctacgac 240
cagcagctca aggccggaga caaccgcgtac ctcaagtaca accacgcga cgcagagttc 300
caggagcggc tcaagaaga taccgtctttt gggggcaacc tcgggcgagc agtcttcag 360
gccaaaaaga ggctctctga acctcttggt ctgggtgagg aagcggctaa gacggctcct 420
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aaatcgggtt cacagccgcg taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctccg acctcaacc aatcggagaa cctccgcgag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg ttgcccacca gtggcagaca ataacgaagg tgcgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcacctcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tgccaactct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acgggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
gagggctgcc tcccgcggtt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
acgtttaatg atgggagcca ggcctgggtt cgctcgtcct tttactgctt ggaatatttc 1200
ccgtcgcaaa tgctaagaa gggtaacaa ttccagttca gctacgagt tgagaacgta 1260
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ctaaaattca gcgtggccgg acccagcaac atggctgttc agggaagaaa ctacatacct 1440
ggaccagct accgacaaca acgtgtctca accactgtga ctcaaaacaa caacagcgaa 1500
tttgcttggt ctggagcttc ttcttggtt ctcaatggac gtaatagctt gatgaatcct 1560
ggactcgcta tggccagcca caaagaagga gaggaaccgtt tcttctctt gtcggatct 1620
ttaattttt gcaaacaaag aactggaaga gacaacgtgg atgcggacaa agtcattgata 1680
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gccacaaacc accagggcca gagtggccgc ggcgatctgg gctgagcgc ccaggcgcc 1800
cagaccggct ggggtcaaaa ccaaggaata cttccgggta tgggtttgga ggacagagat 1860
gtgtacctgc aaggacctat ttgggcaaaa attcctcaca cggacggcaa ctttcacct 1920
tctccgctaa tgggaggggt ttggaatgaag caccgcctc ctacagatcct catcaaaaac 1980
acacctgtac ctgaggatcc tccaacggct ttcaataagg acaagctgaa ctctttcatc 2040
acccagttat ctactggcca agtcagcgtg gagattgagt gggagctgca gaaggaacac 2100
agcaagcgct ggaaccggga gatccagtag acttccaact attacaagtc taataatgtt 2160
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ctgactcgta atctgtaa 2238

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SEQ ID NO: 89      moltype = AA  length = 743
FEATURE           Location/Qualifiers
source            1..743
                  mol_type = protein
                  note = Hu32 variant 5
                  organism = Synthetic construct

SEQUENCE: 89
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120

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AKKRLLEPLG	LVVEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRVI	240
TTSTRTWALP	TYNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGOV	ATNHQSAQGR	GDLGLAQAT	600
GWVQNQGILP	GMVWQDRDVI	LQGPWIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNT	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSNNEVF	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 90 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 5
 organism = Synthetic construct

SEQUENCE: 90

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cagtggtgga	agctcaaacc	tggcccacca	ccaccaaaagc	ccgcagagcg	gcataaggac	120
gacagcaggg	gtcttctgct	tcttggttac	aagtacctcg	gaccggcaca	cggactcgac	180
aagggggagc	cggtcaacgc	agcagacgcg	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctca	agggccgaga	caaccgcgtac	ctcaagtaca	accacgcgca	cgcagagttc	300
caggagcggc	tcaaagaaga	tacgtctttt	gggggcaacc	tcgggcgagc	agtcttcacg	360
gccccaaaaga	ggcttcttga	acctcttggt	ctgggtgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaaccg	actcctccgc	gggtatttgc	480
aaatcgggtt	cacagcccg	taaaaagaaa	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcctcc	acctctaac	aatcggagaa	cctcccgacg	ccccctcagg	tgtgggtatc	600
cttacaatgg	cttcaggtgg	tggcgaccca	gtggcagaca	ataacgaagg	tgccgatgga	660
gtgggtagtt	cctcgggaaa	tggcattg	gattcccaat	ggctggggga	cagagtcac	720
accaccagca	cccgaacctg	ggccctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tccaacagca	catctggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
tgggggtatt	ttgacttcaa	cagattccac	tgccacttct	caccacgtga	ctggcagcga	900
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caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acgggtccagg	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tcccgcggtt	cccagcggac	gttttcatga	ttcctcagta	cgggtatctg	1140
acgcttaatg	atgggagcca	ggcctggggt	cggttcgtct	tttactgcct	ggaatatttc	1200
ccgtcgcaaa	tgctaagaac	gggtaacaa	ttccagttca	gctacgagtt	tgagaacgta	1260
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gaccaatact	tgtactatct	ctcaaagact	attaacggtt	ctggacagaa	tcaacaaacg	1380
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accaaagca	aagaaattaa	aactactaac	ccggttagcaa	cggagtccta	tggaacaagt	1740
gccacaaccc	ccaagggccg	ggcgatctgg	gcctggcaca	ggcgagacac		1800
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gtacctcgcg	atctctcaac	ggctttcaat	aaggacaagc	tgaactcttt	catcacccag	2040
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gctgttaata	ctgaagggtg	atatagttaa	ccccgcccc	ttggcaccag	atacctgact	2220
cgtaatctgt	aa					2232

SEQ ID NO: 91 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 6
 organism = Synthetic construct

SEQUENCE: 91

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PKPAERHKD	DSRGLVLPY	KYLGPNGNLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LKYNHADAEP	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLEPLG	LVVEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTGDTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRVI	240
TTSTRTWALP	TYNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGOV	ATNHQSAQQR	GDYNSAQAT	600
GWVQNQGILP	GMVWQDRDVI	LQGPWIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNT	660

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VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYYKSNNVEF 720
 AVNTEGVYSE PRPIGTRYLT RNL 743

SEQ ID NO: 92 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 6
 organism = Synthetic construct

SEQUENCE: 92
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 gacagcagggg gtctttgtgct tctctgggtac aagtacctcg gaccgcggcaa cggactcgac 180
 aaggggggagc cgggtcaacgc agcagacgcg gcggccctcg agcagacaaa ggcttacgac 240
 cagcagctca aggcgcggaga caaccgcgtac ctcaagtaca accacgcgca cgcgcagttc 300
 caggagcggcg tcaaaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
 gccaaaaaga ggcttcttga acctcttggt ctgggtgagg aagcggctaa gacggctcct 420
 ggaaagaaga ggctgttaga gcagttctct caggaaccgg actcctccgc ggggtattgc 480
 aaatcggggt caccgcccgc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
 tcagtcctccg acctccaacc aatcggagaa cctccgcgag cccctcagg tgtgggatct 600
 cttacaatgg cttcaggtgg tggcgacca gtggcagaca ataacgaagg tgcgatgga 660
 gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
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 tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
 ctcatcaaca acaactgggg attccggcct aagcagctca acttcaagct cttcaacatt 960
 caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
 acggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
 gagggctgcc tccgcgcgtt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
 acgcttaatg atgggagcca ggcctggggt cgttcgtcct tttactgctt ggaatatttc 1200
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 gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg 1380
 ctaaaattca cgcgtggcgg acccagcaac atggctgtcc agggaagaaa ctacatacct 1440
 ggaccagctc accgacaaca acgtgtctca accactgtga ctcaaaacaa caacacggaa 1500
 tttgcttggt ctggagcttc ttcttggggt ctcaatggac gtaatatgct gatgaatcct 1560
 ggactcgcta tggcagccca caaagaagga gaggaccgtt tctttccttt gtcggatct 1620
 ttaatttttg gcaaaccaag aactggaaga gacaacgtgg atgcggacaa agtcatgata 1680
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 gccacaaacc accagagtgc ccaacagaga ggcgactaca actctgcaca ggcgcagacc 1800
 ggctgggttc aaaaccgaag aatacttccg ggtatgggtt ggcaggacag agatgtgtac 1860
 ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaactttca ccttctccg 1920
 ctaattggag ggtttggaat gaagcaccgc cctcctcaga tctcatcaa aaacacacct 1980
 gtactcggcg atctctcaac ggtttcaat aaggacaagc tgaactcttt catcaccag 2040
 tattctactg gccaaagtcag cgtggagatt gaggtaggagc tgcagaagga aaacagcaag 2100
 cgctggaaac ccgagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
 gctgttaata ctgaagggtg atatatgtga ccccgcccca ttggcaccag atacctgact 2220
 cgtaatctgt aa 2232

SEQ ID NO: 93 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 7
 organism = Synthetic construct

SEQUENCE: 93
 MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPGNGLD 60
 KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
 AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQGTGDTE 180
 SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNNEGADG VGSSSGNWHC DSQWLGDRI 240
 TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNFRFH CHFSPRDWQR 300
 LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
 EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
 PFHSSYASHQ SLDRMLNPLI DQLYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYP 480
 GPSYRQQRVS TTVQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
 LIFGQGTGR DNVDADKMI TNEEEIKTTN PVATESYGQV ATNHQSAQNS RGDYNAQAQT 600
 GWVQNQGILP GMVWQDRDVY LQGPWIWAKIP HTDGNFHPSP LMGFGMKHP PPQILIKNTP 660
 VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYYKSNNVEF 720
 AVNTEGVYSE PRPIGTRYLT RNL 743

SEQ ID NO: 94 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 7
 organism = Synthetic construct

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SEQUENCE: 94

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gacagcaggg gtcttgtgtc tctctgggtac aagtaacctcg gaccgggcaa cggactcgac 180
aagggggagc cgggtcaacgc agcagacgcg gcggccctcg agcacgacaa ggctacgac 240
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gccacaaccc accagatgac ccaaaattct agaggcgact acaacgcaca ggccgagacc 1800
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ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaactttca ccttctccg 1920
ctaattggag ggtttggaat gaagcaccgc cctcctcaga tctcatcaa aaacacacct 1980
gtacctgcgg atctctcaac ggctttcaat aaggacaagc tgaactcttt catcaccag 2040
tattctactg gccaaagtcg cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaaac cggagatcca gtacacttcc aactattaca agtctaataa tgttgaaatt 2160
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cgtaatctgt aa 2232

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SEQ ID NO: 95

moltype = AA length = 743

FEATURE

Location/Qualifiers

source

1..743

mol_type = protein

note = Hu32 variant 8

organism = Synthetic construct

SEQUENCE: 95

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MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPFY KYLGPGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFO 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDTE 180
SVDPDQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGLI TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMNLPI DQYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDADKMI TNEEEIKTTN PVATESYGQV ATNHQSAQEN RRGDFQAQAT 600
GWVQNGILP GMVWQDRDVI LQGPWIWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNT 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS RWNPEIQYTS NYYKSNNEVF 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 96

moltype = DNA length = 2232

FEATURE

Location/Qualifiers

source

1..2232

mol_type = other DNA

note = Hu32 variant 8

organism = Synthetic construct

SEQUENCE: 96

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gacagcaggg gtcttgtgtc tctctgggtac aagtaacctcg gaccgggcaa cggactcgac 180
aagggggagc cgggtcaacgc agcagacgcg gcggccctcg agcacgacaa ggctacgac 240
cagcagctca aggcgggaga caaccctgtac ctcaagtaca accacgcgga cgcgaggttc 300
caggagcggc tcaaaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
gccaaaaaga ggcttcttga acctcttggc ctgggtgagg aagcggctaa gacggctcct 420
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aaatcggggtt cacagcccgcc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
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tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
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gtactcgcg atcctccaac ggctttcaat aaggacaagc tgaactcttt catcaccag 2040
tattctactg gccaaagtc cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
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cgtaattctg aa 2232

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SEQ ID NO: 97      moltype = AA length = 743
FEATURE           Location/Qualifiers
source            1..743
                  mol_type = protein
                  note = Hu32 variant 9
                  organism = Synthetic construct

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SEQUENCE: 97
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPFY KYLGPNGNLD 60
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AKKRLLLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQGTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYDFNFRH CHFSPRDWQR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPOQYGL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRMLNPLI DQLYYLSKT INSGQONQOT LKFSVAGPSN MAVQGRNYP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDAKVM I TNEEEIKTTN PVATESYGVV ATNHQSAQRG DYVGLAQAT 600
GWVQNGILP GMVWQDRDVY LQGPWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNT 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS RWNPEIQYTS NYKSNNVF 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 98      moltype = DNA length = 2232
FEATURE           Location/Qualifiers
source            1..2232
                  mol_type = other DNA
                  note = Hu32 variant 9
                  organism = Synthetic construct

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SEQUENCE: 98
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aagggggagc cggccaacgc agcagacgcg gcggccctcg agcagcaca ggctacgac 240
cagcagctca agccgggaga caaccgtac ctcaagtaca accacgccga cgcaggttc 300
caggagcggc tcaagaaga taccgtcttt gggggcaacc tcgggcgagc agtcttcag 360
gcaaaaaaga ggctctctga acctcttggt ctgggttagg aagcggctaa gacggctcct 420
ggaaagaaga ggcctgtaga gcagctctct caggaaacgg actcctccg gggatttggc 480
aaatcgggtt cacagcccgcc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtccccc accctcaacc aatcggagaa cctcccgag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg tggcgaccca gtggcagaca ataacgaagg tgcgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcate 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcacctcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020

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acgggtccagg tcttcacgga ctcagactat cagctcccgt acgtgctcgg gtcgggtcac 1080
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acgcttaatg atgggagcca ggcctgggtt cgttcgtcct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagtcca gtaacaggtt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320
gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg 1380
ctaaaattca gcgtggcccg acccagcaac atggctgtcc agggaagaaa ctacatacct 1440
ggaccagctc accgacaaca acgtgtctca accactgtga ctcaaaacaa caacagcgaa 1500
tttgcttggc ctggagcttc ttcttgggct ctcaatggac gtaatagctt gatgaatcct 1560
ggactgcta tggccagcca caaagaagga gaggaccgtt tctttccttt gctggatct 1620
ttaatttttg gcaacaagg aactggaaga gacaacgtgg atgcggacaa agtcatgata 1680
accaacgaag aagaaattaa aactactaac ccggtagcaa cggagtccca tggacaagt 1740
gccacaaacc accagagtgc caaagaggc gactacgtgg gcctggcaca ggcgcagacc 1800
ggctgggttc aaaaccaagg aatacttccg ggtatggttt ggcaggacag agatgtgtac 1860
ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaactttca ccttctccg 1920
ctaatgggag ggtttggaat gaagcaccgc cctcctcaga tctcatcaa aaacacacct 1980
gtactgcggg acctccaac ggcttcaat aaggacaagc tgaactcttt catcacccag 2040
tattctactg gccaaagtcag cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
gctgttaata ctgaaggtgt atatagttaa ccccgcccca ttggcaccag atacctgact 2220
cgtaatctgt aa 2232

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SEQ ID NO: 99      moltype = AA length = 743
FEATURE           Location/Qualifiers
source            1..743
                  mol_type = protein
                  note = Hu32 variant 10
                  organism = Synthetic construct

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SEQUENCE: 99
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPFY KYLGPNGNGLD 60
KGEFVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYDFDNRFH CHFSPRDWQR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMNLPI DQYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVADKVM I TNEEEIKTTN PVATESYGVV ATNHQNSRSG DYNLAQAQT 600
GWVQNQGIPL GMVWQDRDVI LQGIWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNT 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS RWNPEIQYTS NYYKSNNV 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 100     moltype = DNA length = 2232
FEATURE           Location/Qualifiers
source            1..2232
                  mol_type = other DNA
                  note = Hu32 variant 10
                  organism = Synthetic construct

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SEQUENCE: 100
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gacagcaggg gtcttgtgct tcctgggtac aagtacctcg gaccocggca cggactcgac 180
aagggggagc cggtaaacgc agcagacgcg gcggccctcg agcacgacaa ggcctacgac 240
cagcagctca aggcgggaga caaccggtac ctcaagtaca accacgcgca cgcaggttc 300
caggagcgcc tcaagaaga tacgtctttt gggggcaacc tcggggcagc agtctccag 360
gccaaaaaga ggcttcttga acctcttggt ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggcctgttga gcagtctcct caggaaacgg actcctccg cggatttgcc 480
aaatcggttt caccagccgc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctcc accctcaacc aatcggagaa cctcccgag cccctcagg tgtgggatct 600
cttacaatgg ctccaggtgg tggcgacaca gtggcagaca ataacgaag tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcactcta caagcaaat 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaa ggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acggtccagg tcttcacgga ctccagctat cagctcccgt acgtgctcgg gtcgggtcac 1080
gaggggtgcc tcccgcggtt cccagcggac gttttcatga ttectcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcctgggtt cgttcgtcct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagtcca gtaacaggtt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320
gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg 1380
ctaaaattca gcgtggcccg acccagcaac atggctgtcc agggaagaaa ctacatacct 1440
ggaccagctc accgacaaca acgtgtctca accactgtga ctcaaaacaa caacagcgaa 1500
tttgcttggc ctggagcttc ttcttgggct ctcaatggac gtaatagctt gatgaatcct 1560

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ggacctgcta tggccagcca caaagaagga gaggaccgtt tctttccttt gtctggatct 1620
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accaacgaag aagaaattaa aactactaac ccggttagcaa cggagtcccta tggacaagtg 1740
gccacaaacc accagagcaa ctctagaggg gactacaaca gcctggcaca ggcgcagacc 1800
ggctgggttc aaaaccaagg aatacttccg ggtatgggtt ggcaggacag agatgtgtac 1860
ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaactttca cccttctccg 1920
ctaattgggag ggtttggaat gaagcaccgg cctcctcaga tcctcatcaa aaacacacct 1980
gtacctgcgg atcctccaac ggctttcaat aaggacaagc tgaactcttt catcacccag 2040
tattctactg gccaaagtca cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
gctgttaata ctgaaggtgt atatatgtga ccccgcccca ttggcaccag atacctgact 2220
cgtaatctgt aa 2232

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SEQ ID NO: 101      moltype = AA  length = 743
FEATURE             Location/Qualifiers
source              1..743
                    mol_type = protein
                    note = Hu32 variant 11
                    organism = Synthetic construct

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SEQUENCE: 101
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPFY KYLGPNGNLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDTE 180
SVDPDQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWALP TYNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYDFDNRFH CHFSPRDWQR 300
LINNNWGFPR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRMLNPLI DQYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVADKVM I TNEEEIKTTN PVATESYGVV ATNHQENRRG DFNNTAQAT 600
GWVQNGILP GMVWQDRDVI LQGPIWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNT 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYYKSNNV 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 102      moltype = DNA  length = 2232
FEATURE             Location/Qualifiers
source              1..2232
                    mol_type = other DNA
                    note = Hu32 variant 11
                    organism = Synthetic construct

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SEQUENCE: 102
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gacagcaggg gtcttgtgct tccctgggtac aagtacctcg gaccgcgcaa cggactcgac 180
aagggggagc cggccaacgc agcagacgcg gcggccctcg agcacgacaa ggcctacgac 240
cagcagctca cgcacggaga caaccggtac ctcaagtaca accacgcgca cgcagagttc 300
caggagcggc tcaaaagaaga tacgtctttt gggggcaacc tcggggcgagc agtcttccag 360
gccaaaaaga ggcttcttga acctcttggt ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggcctgtaga gcagtctcct caggaaacgg actcctccgc gggatttggc 480
aaatcgggtt cacagcccg taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctcg acctcaacc aatcggagaa cctcccgag cccctcagg tgtgggactc 600
cttacaatgg cttcaggtgg tggcgacaca gtggcagaca ataacgaag tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca ccggaacctg ggccctgccc acctacaaca atcacctcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tggggggtatt ttgacttcaa cagattccac tggcacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attcgggcct aagcgactca actcaagct ctccaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acggteccag tcttcacgga ctacagactat cagctcccgt acgtgctcgg gtccggtcac 1080
gagggctgct tcccgcggt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcgtgggt cgttcgtcct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagtt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320
gaccaatact tgtactatct ctcaaagact attaacgggt ctggacagaa tcaacaaacg 1380
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accaacgaag aagaaattaa aactactaac ccggttagcaa cggagtcccta tggacaagtg 1740
gccacaaacc accaggagaa tagacggggc gacttcaaca acacgcgaca ggcgcagacc 1800
ggctgggttc aaaaccaagg aatacttccg ggtatgggtt ggcaggacag agatgtgtac 1860
ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaactttca cccttctccg 1920
ctaattgggag ggtttggaat gaagcaccgg cctcctcaga tcctcatcaa aaacacacct 1980
gtacctgcgg atcctccaac ggctttcaat aaggacaagc tgaactcttt catcacccag 2040
tattctactg gccaaagtca cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100

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cgctggaacc	cggagatcca	gtacacttcc	aactattaca	agtctaataa	tgttgaattt	2160
gctgttaata	ctgaaggtgt	atatagttaa	ccccgcccc	ttggcaccag	atacctgact	2220
cgtaatctgt	aa					2232

SEQ ID NO: 103 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 12
 organism = Synthetic construct

SEQUENCE: 103

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPFY	KYLGPGNGLD	60
KGEFVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTDGTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYPDFNRFH	CHFSRPRDQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPFPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGGV	ATNHQSAQRG	DYVGVAQAQT	600
GWVQNGQILP	GMVWQDRDVI	LQGPIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNT	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSNNVF	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 104 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 12
 organism = Synthetic construct

SEQUENCE: 104

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gacagcaggg	gtcttgtgct	tccctgggtac	aagtacctcg	gacccggcaa	cggactcgac	180
aagggggagc	cggccaacgc	agcagacgcg	cgcccccctcg	agcagcagaa	ggcctacgac	240
cagcagctca	agcccgagga	caaccgctac	ctcaagtaca	accacgccga	cgcagagttc	300
caggagcggc	tcaaaagaaga	tacgtctttt	gggggcaacc	tcggggcagc	agtcctccag	360
gccccaaaaga	ggcttcttga	acctcttgg	ctgggttgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaaccg	actcctccgc	gggtattggc	480
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tcacaacagca	catctggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
tgggggtatt	ttgacttcaa	cagattccac	tggcacttct	caccacgtga	ctggcagcga	900
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caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acggctccag	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tcccgcggtt	cccagcggac	gtttctcatga	ttcctcagta	cgggtatctg	1140
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ccgtcgcaaa	tgctaagaac	gggttaacaac	ttccagttca	gctacagagt	tgagaacgta	1260
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ttaatttttg	gcaacaaggg	aactggaaga	gacaacgtgg	atgctggaca	agtcacgata	1680
accacgaag	aagaaattaa	aactactaac	ccggtagcaa	cggagtcccta	tggaacagtg	1740
gccacaaacc	accagagtgc	ccaagagggc	gactacgtgg	gcgtggcaca	ggcgcagacc	1800
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cgctgggaacc	cggagatcca	gtacacttcc	aactattaca	agtctaataa	tgttgaattt	2160
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SEQ ID NO: 105 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 13

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organism = Synthetic construct
SEQUENCE: 105
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFGQTDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LINNNWGFRR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMNLPI DQYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDAKVM I TNEEEIKTTN PVATESYGVV ATNHQSAQRG DVLGSAQAQT 600
GWVQNQGILP GMVWQDRDVI LQGPIWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNT 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS K RWNPEIQYTS NYYKSNNVF 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 106      moltype = DNA length = 2232
FEATURE            Location/Qualifiers
source              1..2232
                    mol_type = other DNA
                    note = Hu32 variant 13
                    organism = Synthetic construct

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SEQUENCE: 106
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gacagcaggg cgctctgtgct tgcctgggtac aagtacctcg gaccggcaca cggaactcgac 180
aagggggagc cggtcaacgc agcagacgcg cgggccctcg agcagcagaa ggccctacgac 240
cagcagctca agcccgagga caaccgcgtac ctcaagtaca accacgcgca cgccgagttc 300
caggagcggc tcaagaaga taccgtctttt gggggcgaacc tcggggcgagc agtctctccag 360
gccccaaaag ggcttcttga acctcttggt ctggttgagg aagcggctaa gacggctcct 420
ggaaaagaag ggcctgtaga gcaagtctct caggaaacgc actctccgc gggattattgc 480
aaatcggtt cagacccgc taaaaaagaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcgccg acctcaacc aatcggagaa cctcccgag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg ttggcgacca gtggcagaca ataacgaagg tgcgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcactcta caagcaaat 780
tccaaacgca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tggcacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca actcaagct ctccaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
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gagggctgcc tcccgccggt cccagcggac gtttctcatg ttctcagta cgggatctcg 1140
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ccgtcgcaaa tgctaagaac gggtaacaac ttccagtcca gctacgagt tgagaacgta 1260
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tttgcttggc ctggagcttc ttcttgggct ctcaatggac gtaatagct gatgaatcct 1560
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gccacaaacc accagagtgc ccaagagggc gacgtgctgg gctctgcaca ggcgcagacc 1800
ggctgggttc aaaaccaagg aatacttccg ggtatgggtt ggcaggacag agatgtgtac 1860
ctgcaaggac caatttgggc caaaattcct cacacggacg gcaactttca cccttctccg 1920
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gtacctcgcg atctccaac ggctttcaat aaggacaagc tgaactctt catcaccag 2040
tattctactg gccaagtcat cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtetaataa tgttgattt 2160
gctgttaata ctgaaggtgt atatatgtaa ccccgcccca ttggcaccag atacctgact 2220
cgtaatctgt aa 2232

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SEQ ID NO: 107      moltype = AA length = 743
FEATURE            Location/Qualifiers
source              1..743
                    mol_type = protein
                    note = Hu32 variant 14
                    organism = Synthetic construct

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SEQUENCE: 107
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFGQTDTE 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LINNNWGFRR KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420

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PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGVQ	ATNHQSAQRG	DFVGLAQAAQT	600
GWVQNQGILP	GMVWQDRDVY	LQGPWIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNTN	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYYSNNNVEF	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 108 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 14
 organism = Synthetic construct

SEQUENCE: 108

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cagtggtgga	agctcaaac	tggcccacca	ccaccaaac	ccgcagagcg	gcataaggac	120
gacagcaggg	gtcttgtgct	tccctgggtac	aagtacctcg	gacccggcaa	cggactcgac	180
aagggggagc	cgggtcaacgc	agcagacgcg	gcggccctcg	agcagcacia	ggcctacgac	240
cagcagctca	agggcggaga	caaccgcgtac	ctcaagtaca	accacgccga	cgcgcagttc	300
caggagcggc	tcaagaaga	tacgtctttt	gggggcaacc	tcggggcagc	agtcctccag	360
gccccaaaaga	ggcttcttga	acctcttgg	ctgggttgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagtcctct	caggaaccgg	actcctccgc	gggtattggc	480
aaatcgggtt	cacagccgcg	taaaaagaaa	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtcctccg	acctcaacc	aatcggagaa	cctcccgacg	ccccctcagg	tgtgggatct	600
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gtgggtatgt	cctcgggaaa	tggcattgc	gattcccaat	ggctggggga	cagagtcac	720
accaccagca	cccgaacctg	ggccctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tccaacagca	cactggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
tgggggtatt	ttagcttcaa	cagattccac	tgccacttct	caccacgtga	ctggcagcga	900
ctcatcaaca	acaactgggg	attccggcct	aagcgactca	actcaagct	cttcaacatt	960
caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acggctccag	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tcccgccggt	cccagcggac	gttttcacga	ttcctcagta	cgggtatctg	1140
acgcttaatg	atgggagcca	ggcgtgggt	cgcttcgtct	tttactgcct	ggaatatttc	1200
ccgtcgcaaa	tgctaagaac	gggtaacaac	ttccagttca	gctacgagtt	tgagaacgta	1260
cctttccata	gcagctacgc	tcacagccaa	agcctggacc	gactaatgaa	tccactcatc	1320
gaccaatact	tgtctatct	ctcaagact	attaacgggt	ctggacagaa	tcaacaaacg	1380
ctaaaattca	cgctggccgg	accagcaaac	atggctgtcc	agggagaagaa	ctacatacct	1440
ggaccagct	accgacaaca	acgtgtctca	accactgtga	ctcaaaaaca	caacagcgaa	1500
tttgcttggc	ctggagcttc	ttcttgggct	ctcaatggac	gtaatagctt	gatgaatcct	1560
ggacctgcta	tggccagcca	caaagaagga	gaggaccgtt	tccttctctt	gtctggatct	1620
ttaatttttg	gcaaacagg	aactggaaga	gacaacgtgg	atgctggcaa	agtcacgata	1680
accaacgaag	agvaaaattaa	aactactaac	ccggtagcaa	cggagtcccta	tggacaagt	1740
gccacaaacc	accagagtgc	ccaagaggc	gacttcgtgg	gcctggcaca	ggcgcagacc	1800
ggctgggttc	aaaaccaagg	aatacttccg	ggtatggttt	ggcaggacag	agatgtgtac	1860
ctgcaaggac	ccatttgggc	caaaaattcct	cacacggacg	gcaactttca	cccttctccg	1920
ctaattggag	gggttgggaat	gaagcaccgc	cctcctcaga	tcctcatcaa	aaacacacct	1980
gtacctcgcg	atctctcaac	ggctttcaat	aaggacaagc	tgaactcttt	catcaccacg	2040
tattctactg	cccgaagtca	cggtggagatt	gagtgaggag	tgacagaagg	aaacagcaag	2100
cgctggaacc	ggcagatcca	gtacacttcc	aactattaca	agtcataata	tgttgaaatt	2160
gctgttaata	ctgaagggtg	atatagttaa	ccccgcccc	ttggcaccag	atacctgact	2220
cgtaattctg	aa					2232

SEQ ID NO: 109 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 15
 organism = Synthetic construct

SEQUENCE: 109

MAADGYLPDW	LEDTLSEGR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPY	KYLGPGNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTDGTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRLV	240
TTSTRTWALP	TNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNFRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGVQ	ATNHQSAQRG	DYTLQAQAQT	600
GWVQNQGILP	GMVWQDRDVY	LQGPWIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNTN	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYYSNNNVEF	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 110 moltype = DNA length = 2232
 FEATURE Location/Qualifiers

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source          1..2232
                 mol_type = other DNA
                 note = Hu32 variant 15
                 organism = Synthetic construct

SEQUENCE: 110
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gacagcaggg gtcttgtgct tcttgggtac aagtacctcg gaccggcga cggactcgac 180
aaggggggagc cggtaaacgc agcagacgcg gcggccctcg agcagcaca ggccacgac 240
cagcagctca aggcgggaga caaccgcgtac ctcaagtaca accacgcccga cggcaggttc 300
caggagcggc tcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcacg 360
gccaaaaaga ggcttcttga acctcttggg ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggctgttaga gcagtcctct caggaaccgg actcctccgc gggatttggc 480
aaatcggggt cacagccgcg taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctccg acctcaacc aatcggagaa cctcccgagc cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg ttggcgacca gtggcagaca ataacgaagg tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcactcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccgcc 840
tggtgggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acggtccagg tcttcacgga ctccagactat cagctcccg acgtgctcgg gtcggctcac 1080
gaggggtgct tcccgccggt cccagcggac gttttcatga ttctcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcggtgggt cgttcgtcct tttactgctt ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagt tgagaacgta 1260
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tattctactg gccaaagtca cgtggagatt gagtgggagc tgacgaagaa aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaattt 2160
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cgtaatctgt aa 2232

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SEQ ID NO: 111      moltype = AA length = 743
FEATURE            Location/Qualifiers
source              1..743
                   mol_type = protein
                   note = Hu32 variant 16
                   organism = Synthetic construct

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SEQUENCE: 111
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEA AKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDT 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNFNRFH CHFSPRDWQR 300
LINNNWGFRP KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTSDSY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHQ SLDRLMNPLI DQYLYLSKT INSGQNQQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDA DKVMI TNEEEIKTTN PVATESYGQV ATNHQSAQSR GDFNLAQAQT 600
GWVQNQGILP GMVWQDRDVI LQGPWIWAKIP HTDGNFHPSP LMGGFGMKHP PPQILIKNTP 660
VPADPPTAFN KDKLNSFITQ YSTGQVSVEI EWELQKENS RWNPEIQYTS NYYKSNVVEF 720
AVNTEGVYSE PRPIGTRYLT RNL 743

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SEQ ID NO: 112      moltype = DNA length = 2232
FEATURE            Location/Qualifiers
source              1..2232
                   mol_type = other DNA
                   note = Hu32 variant 16
                   organism = Synthetic construct

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SEQUENCE: 112
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gacagcaggg gtcttgtgct tcttgggtac aagtacctcg gaccggcga cggactcgac 180
aaggggggagc cggtaaacgc agcagacgcg gcggccctcg agcagcaca ggccacgac 240

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cagcagctca aggccggaga caaccgctac ctcaagtaca accacgccga cgccgagttc 300
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gccccaaaga ggctctctga acctcttggt ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggctctgtag gacgtctcct caggaaccgg actcctccgc gggatttggc 480
aaatcggtt caccgcccgc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
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cttacaatgg cttcaggtgg tggcgcacca gtggcagaca ataacgaagg tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
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tgggggtagt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
caggtcaaa aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acggtccagg tcttcacgga ctacagactat cagctcccg acgtgctcg gtcggctcac 1080
gagggctgct tcccgccgtt cccagcggac gttttcatga ttcctcagta cgggtatctg 1140
acgtttaatg atgggagcca ggcgtgggt cgttcgtcct tttactgctt ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320
gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg 1380
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ggactgcta tggccagcca caaagaagga gaggaaccgt tcttctctt gctggatct 1620
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accaacgaag aagaaattaa aactactaac ccggtagcaa cggagtccta tggacaagt 1740
gccacaacc accagagtgc ccaactaga ggcgacttca acctggcaca ggcgagacc 1800
ggctgggttc aaaaccaagg aatacttccg ggtatggtt ggcaggacag agatgtgtac 1860
ctgcaaggac ccatttgggc caaaattcct cacacggacg gcaacttca ccttctccg 1920
ctaattggag gatttggaaat gaagcaccg cctcctcaga tctcctcaa aaacacacct 1980
gtactcgcg atcttcaac ggtttcaat aaggacaagc tgaactctt catcaccag 2040
tattctactg gccaaagtca cgtggagatt gagtgggagc tgcagaagga aaacagcaag 2100
cgctggaacc cggagatcca gtacacttcc aactattaca agtctaataa tgttgaatt 2160
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cgtaatctgt aa 2232

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SEQ ID NO: 113      moltype = AA length = 741
FEATURE
source             Location/Qualifiers
                   1..741
                   mol_type = protein
                   note = Hu32 variant 17
                   organism = Synthetic construct

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SEQUENCE: 113
MAADGYLPDW LEDTLSEGR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPNGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQGTGDT 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LNNNWGFRRP KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMLPQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRLMNPLI DQYLYLSKT INSGGQNGQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVQNNNS FAWPGASSWA LNRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVQADKVM I TNEEEIKTTN PVATESYGOV ATNHQSAORG DYVAQAQTGW 600
VQNQGILPGM VWQDRDYLQ GPIWAKIPHT DGNFHPSPLM GGFGMKHPPP QILIKNTPVP 660
ADPPTAFNKD KLNSFITQYS TGQVSVEIEW ELQKENS KRW NPEIQYTSNY YKSNNVEFAV 720
NTEGVYSEPR PIGTRYLTRN L 741

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SEQ ID NO: 114      moltype = DNA length = 2226
FEATURE
source             Location/Qualifiers
                   1..2226
                   mol_type = other DNA
                   note = Hu32 variant 17
                   organism = Synthetic construct

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SEQUENCE: 114
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gacagcaggg gtcttgtgct tcttgggtac aagtacctcg gaccggcaa cggactcgac 180
aagggggagc cggccaacgc agcagacgcg gggccctcg agcagcaca ggcctacgac 240
cagcagctca agcccgtag caaccgctac ctcaagtaca accacgccga cgccgagttc 300
caggagcggc tcaaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttccag 360
gccccaaaga ggctctctga acctcttggt ctggttgagg aagcggctaa gacggctcct 420
ggaaagaaga ggctctgtag gacgtctcct caggaaccgg actcctccgc gggatttggc 480
aaatcggtt caccgcccgc taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcctccg acctcaacc aatcggagaa cctcccgtag cccctcagg tgtgggtagct 600
cttacaatgg cttcaggtgg tggcgcacca gtggcagaca ataacgaagg tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcactcta caagcaaatc 780

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tccaacagca catctggagg atettcaa atgacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
cagggtcaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acgggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
gagggctgcc tcccgcggtt ccagcgggac gttttcatga ttctcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcgctgggt cgttcgtcct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacaggt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320
gaccaatact tgtactatct ctcaaagact attaacgggt ctggacagaa tcaacaaacg 1380
ctaaaattca gcgtggcggg acccagcaac atggctgtcc agggaaagaaa ctacatacct 1440
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tttgcctggc ttggagcttc ttcttgggct ctcaatggac gtaatagctt gatgaatcct 1560
ggacctgcta tggccagcca caaagaagga gaggaaccgtt tctttccttt gctgggatct 1620
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gccacaaacc accagagtg ccaaaagagg gactacgtgg cacaggcgca gaccggctgg 1800
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gcggatcctc caacggcttt caataaggac aagctgaact ctttcatcac ccagtattct 2040
actggccaag tgcagtgga gatagagtg gagctgcaga aggaaacacg caagcgtctg 2100
aaccgggaga tccagtacac ttccaactat tacaagtcta ataagtgtga atttgcgtt 2160
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ctgtaa 2226

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SEQ ID NO: 115      moltype = AA length = 736
FEATURE
source              Location/Qualifiers
                    1..736
                    mol_type = protein
                    note = Hu32 variant 18
                    organism = Synthetic construct

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SEQUENCE: 115
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KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
AKRRLLEPLG LVEEAAKTAP GKRRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDT 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI 240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR 300
LNNNNWGFPP KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH 360
EGCLPPFPAD VFMI PQYGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN QFQSYEFENV 420
PFHSSYAHSQ SLDRLMNPLI DQYLYYLSKT INSGGQNNQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVTQNNNSE FARPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVDAKVM I TNEEEIKTTN PVATESYGOV ATNHQSAQAQ AQTGWVQNGQ 600
ILPGMVWQDR DVYLQGP IWA KIPHTDGNFH PSPLMGGFGM KHPPPQILIK NTPVPADPPT 660
AFNKDKLNSF ITQYSTGQVS VEIEWELQKE NSKRWNPEIQ YTSNYYKSNN VEFVNTGTV 720
YSEPRPIGTR YLTRNL 736

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SEQ ID NO: 116      moltype = DNA length = 2211
FEATURE
source              Location/Qualifiers
                    1..2211
                    mol_type = other DNA
                    note = Hu32 variant 18
                    organism = Synthetic construct

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SEQUENCE: 116
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gacagcaggg gtcttgtgct tcttgggtac aagtacctcg gacccgcaa cggactcgac 180
aagggggagc cggccaacgc agcagacgcg gcggccctcg agcacgaca ggctacagc 240
cagcagctca agggcggaga caaccgctac ctcaagtaca accacgccga cggcgagttc 300
caggagcgcg tcaaaaga taactctttt gggggcaacc tcggggcagc agtcttccag 360
gccaaaaga ggcctcttga acctcttgg ctgggtgagg aagcggctaa gacggctcct 420
ggaaagaaga ggctgtaga gcagctcctc caggaaaccg actcctccgc ggggtattggc 480
aatcggtt cagagccgcg taaaaagaaa ctcaatttcg gtcagactgg cgacacagag 540
tcagtcgccg acctcaacc aatcgagaga cctccgcag cccctcagg tgtgggatct 600
cttacaatgg cttcaggtgg tggcgagaca ataacgaagg tgccgatgga 660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac 720
accaccagca cccgaacctg ggcctgccc acctacaaca atcactcta caagcaaatc 780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc 840
tgggggtatt ttgacttcaa cagattccac tgccacttct caccacgtga ctggcagcga 900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt 960
cagggtcaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc 1020
acgggtccagg tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcggctcac 1080
gagggctgcc tcccgcggtt ccagcgggac gttttcatga ttctcagta cgggtatctg 1140
acgcttaatg atgggagcca ggcgctgggt cgttcgtcct tttactgcct ggaatatttc 1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacaggt tgagaacgta 1260
cctttccata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcatc 1320

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gaccaatact	tgtactatct	ctcaaagact	attaacgggt	ctggacagaa	tcaacaaacg	1380
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ggaccagct	accgacaaca	acgtgtctca	accactgtga	ctcaaaacaa	caacagcgaa	1500
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gccacaaacc	accagagtgc	ccaagcacag	gcgcagaccg	gctgggttca	aaaccaagga	1800
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aaaattcctc	acacggacgg	caactttcac	ccttctccgc	taatgggagg	gtttggaatg	1920
aagcacccgc	ctcctcagat	cctcatcaaa	aacacacctg	tacctgcgga	tcctccaacg	1980
gctttcaata	aggacaagct	gaactctttc	atcacccagt	attctactgg	ccaagtcagc	2040
gtggagattg	agtgggagct	gcagaaggaa	aacagcaagc	gctggaaccc	ggagatccag	2100
tacactttca	actattacaa	gtctaataat	gttgaatttg	ctgttaatac	tgaaggtgta	2160
tatagtgaac	cccgcctcat	tggcaccaga	tacctgactc	gtaatctgta	a	2211

SEQ ID NO: 117 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 19
 organism = Synthetic construct

SEQUENCE: 117

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPFY	KYLGPGNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTDGTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYPDFNRFH	CHFSPRDWQR	300
LINNWNWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRMLNPLI	DQYLYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYP	480
GPSYRQQRVS	TTVTQNNNSE	FARPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKMI	TNEEEIKTTN	PVATESYGVQ	ATNHQSAQST	TLYSPAQAQT	600
GWVQNGILP	GMVWQDRDVI	LQGIWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNTP	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSNNVFEP	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 118 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 19
 organism = Synthetic construct

SEQUENCE: 118

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gacagcaggg	gtcttgtgct	tcctgggtac	aagtacctcg	gaccgcgcaa	cggactcgac	180
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cagcagctca	agcccgga	caaccgtac	ctcaagtaca	accacgccga	cgcagagttc	300
caggagcgcc	tcaagaaga	tacgtctttt	gggggcaacc	tcgggagcgc	agtcctccag	360
gccaaaaaga	ggcttcttga	acctcttggt	ctgggtgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaacgc	actcctccgc	gggtattggc	480
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accaccagca	cccgaacctg	ggccctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tcacaacgca	catctggagg	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
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caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acggtccagg	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tccgcgcgtt	cccagcggac	gtttctcatga	ttcctcagta	cgggtatctg	1140
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gccacaaacc	accagagtgc	ccaatcaacg	acgctctatt	ctccagcaca	ggcgcagacc	1800
ggctgggttc	aaaaccagag	aatactccg	ggtatgggtt	ggcaggacag	agatgtgtac	1860
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ctaattgggag	ggtttgggaat	gaagcaccg	cctcctcaga	tectcatcaa	aaacacacct	1980
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tattctactg	gccaaagtca	cgtggagatt	gagtgggagc	tgacagaaga	aaacagcaag	2100
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gctgttaata	ctgaaggtgt	atatagtgaa	ccccgcccc	ttggcaccag	atacctgact	2220
cgtaatctgt	aa					2232

SEQ ID NO: 119 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 note = Hu32 variant 20
 organism = Synthetic construct

SEQUENCE: 119

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPFY	KYLGPGNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTDGTE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TYNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYPDFNRFH	CHFSRPDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNLFI	DQYLYLSKT	INGSGNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FARPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKMI	TNEEEIKTTN	PVATESYGVV	ATNHQSAQFV	VGQSYAQAT	600
GWVQNGILP	GMVWQDRDVI	LQGPWAKIP	HTDGNFHPSP	LMGGFGMKHP	PPQILIKNTP	660
VPADPPTAFN	KDKLNSFITQ	YSTGQVSVEI	EWELQKENS	RWNPEIQYTS	NYKSNNVFV	720
AVNTEGVYSE	PRPIGTRYLT	RNL				743

SEQ ID NO: 120 moltype = DNA length = 2232
 FEATURE Location/Qualifiers
 source 1..2232
 mol_type = other DNA
 note = Hu32 variant 20
 organism = Synthetic construct

SEQUENCE: 120

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caggagcggc	tcaagaaga	tacgtctttt	gggggcaacc	tcgggagcgc	agtcctccag	360
gcaaaaaaga	ggcttcttga	acctcttgg	ctggttgagg	aagcggctaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaacgc	actcctccgc	gggtattggc	480
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ggctgggttc	aaaaccaagg	aatacttccg	ggtatggttt	ggcaggacag	agatgtgtac	1860
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gtacctgcgg	atcctccaac	ggctttcaat	aaggacaagc	tgaactcttt	catcacccag	2040
tattctactg	gccaaagtca	cgtggagatt	gagtgggagc	tgacagaaga	aaacagcaag	2100
cgttggaacc	cggagatcca	gtacacttcc	aactattaca	agtctaataa	tgttgaattt	2160
gctgttaata	ctgaaggtgt	atatagtgaa	ccccgcccc	ttggcaccag	atacctgact	2220
cgtaatctgt	aa					2232

SEQ ID NO: 121 moltype = AA length = 58
 FEATURE Location/Qualifiers

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source          1..58
                 mol_type = protein
                 note = AAV9/hu32 consensus N-terminal sequence
                 organism = Synthetic construct
VARIANT         14
                 note = X = N or T
VARIANT         21
                 note = X = E or Q
VARIANT         24
                 note = X = A or K
VARIANT         29
                 note = X = A or P
VARIANT         31
                 note = X = Q or P
VARIANT         33
                 note = X = A or P
VARIANT         35
                 note = X = N or A
VARIANT         36
                 note = X = Q or E
VARIANT         37
                 note = X = Q or R
VARIANT         39
                 note = X = Q or K
VARIANT         41
                 note = X = N or D
VARIANT         42
                 note = X = A or S
VARIANT         34
                 note = X = A or P

SEQUENCE: 121
MAADGYLPDW LEDXLSEGIR XWWXLKPGXP XPKXXXXHXD XXRGLVLPGY KYLGPGNG    58

SEQ ID NO: 122      moltype = AA length = 58
FEATURE            Location/Qualifiers
source             1..58
                  mol_type = protein
                  note = AAV9 N-terminal sequeunce
                  organism = Synthetic construct

SEQUENCE: 122
MAADGYLPDW LEDNLSEGIR EWWALKPGAP QPKANQQHQD NARGLVLPGY KYLGPGNG    58

SEQ ID NO: 123      moltype = AA length = 58
FEATURE            Location/Qualifiers
source             1..58
                  mol_type = protein
                  note = Hu32 N-terminal sequeunce
                  organism = Synthetic construct

SEQUENCE: 123
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPGNG    58

SEQ ID NO: 124      moltype = AA length = 736
FEATURE            Location/Qualifiers
source             1..736
                  mol_type = protein
                  note = AAV9
                  organism = Adeno-associated virus

SEQUENCE: 124
MAADGYLPDW LEDNLSEGIR EWWALKPGAP QPKANQQHQD NARGLVLPGY KYLGPGNGLD    60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ    120
AKKRLLLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGAQPAKKR LNFGQTGDTE    180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRVI    240
TTSTRTWALP TYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDFNRFH CHFSPRDWQR    300
LINNNWGFRP KRLNFKLFNI QVKEVTDNNG VKTIANNLTS TVQVFTDSY QLPYVLGSAH    360
EGCLPPFPAD VFMIPQGYL TLNDGSQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV    420
PFHSSYAHSQ SLDRLMNPLI DQYLYLSKT INSGGQNOQT LKFSVAGPSN MAVQGRNYIP    480
GPSYRQQRVS TTVTQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS    540
LIFGKQGTGR DNVDADKVI TNEEEIKTTN PVATESYGVV ATNHQSAQAQ AQTGWVQNQG    600
ILPGMVWQDR DVYLGQPIWA KIPHTDGNFH PSPLMGGFGM KHPPPQILIK NTPVPADPPT    660
AFNKDKLNSF ITQYSTGQVS VEIEWELQKE NSKRWNPEIQ YTSNYYKSNN VEFVNTGTV    720
YSEPRPIGTR YLTRNL                                     736

SEQ ID NO: 125      moltype = DNA length = 2211
FEATURE            Location/Qualifiers
source             1..2211
                  mol_type = genomic DNA

```

-continued

note = AAV9
organism = Adeno-associated virus

SEQUENCE: 125

atggtgtgccc	atgtgttatct	tccagattgg	ctcgaggaca	accttagtga	aggaattcgc	60
gagtggtggg	ctttgaaacc	tggagccct	caacccaagg	caaatcaaca	acatcaagac	120
aacgctcgag	gtctctgtgct	tccgggttac	aaataccttg	gaccgcgcaa	cggactcgac	180
aagggggagc	cgggtcaacgc	agcagacgcg	gcggccctcg	agcacgacaa	ggcctacgac	240
cagcagctca	aggcgggaga	caacccgtag	ctcaagtaca	accacgcoga	cgccgagttc	300
caggagcggc	tcaagaaga	tacgtctttt	gggggcaacc	tggggcgagc	agtcttcacg	360
gccaaaaaga	ggcttcttga	acctcttgg	ctggttgagg	aagcgggtaa	gacggctcct	420
ggaaagaaga	ggcctgtaga	gcagctctct	caggaaccgg	actcctcgcg	gggtattggc	480
aatcggtgtg	cacagcccg	taaaaagaga	ctcaatttcg	gtcagactgg	cgacacagag	540
tcagtccag	accctcaacc	aatcggagaa	cctcccgag	ccccctcagg	tgtgggatct	600
cttacaatgg	cttcaggtgg	tggcgaccca	gtggcagaca	ataacgaagg	tgccgatgga	660
gtgggtagtt	cctcgggaaa	tggcattgc	gattcccaat	ggctggggga	cagagtcac	720
accaccagca	cccgaacctg	ggcctgccc	acctacaaca	atcacctcta	caagcaaatc	780
tccaacagca	cactctggag	atcttcaaat	gacaacgcct	acttcggcta	cagcaccccc	840
tgggggtatt	ttgacttcaa	cagattccac	tgccacttct	caccacgtga	ctggcagcga	900
ctcatcaaca	acaactgggg	attccggcct	aagcgactca	acttcaagct	cttcaacatt	960
caggtcaaa	aggttacgga	caacaatgga	gtcaagacca	tcgccaataa	ccttaccagc	1020
acggtccagg	tcttcacgga	ctcagactat	cagctcccg	acgtgctcgg	gtcggctcac	1080
gagggctgcc	tcccgccgtt	cccagcggac	gttttcatga	tctctcagta	cgggtatctg	1140
acgttaaatg	atggaagcca	ggccgtgggt	cgctctgct	tttactgct	ggaatatctc	1200
ccgtcgcaaa	tgctaagaac	gggtaacaac	ttccagtcca	gctacgagtt	tgagaacgta	1260
cctttccata	gcagctacgc	tcacagccaa	agcctggacc	gactaatgaa	tccactcatc	1320
gaccaatact	tgactatct	ctcaaaagact	attnaacggt	ctggacagaa	tcaacaaacg	1380
ctaaaattca	gtgtggccgg	accagcaaac	atggctgtcc	agggaaagaa	ctacatacct	1440
ggaccagct	accgacaaca	accgtgtcta	accactgtga	ctcaaaacaa	caacagcgaa	1500
tttcttggc	ctggagcttc	ttcttgggct	ctcaatggac	gtaaatagct	gatgaatcct	1560
ggacctgcta	tggccagcca	caaagaagga	gaggaccgtt	tcttctcttt	gtctggatct	1620
ttaatttttg	gcaacaagg	actggaaga	gacaacgtgg	atgcccagaa	agtcatgata	1680
accaacgaag	agaagaattaa	aactactaac	ccggtagcaa	cggagtccta	tggacaagtg	1740
gccacaaacc	accagagtgc	ccaagcacag	gcgcagaccg	gctgggtcca	aaaccaagga	1800
atacttcgg	gtatgggttg	gcaggacaga	gatgtgtacc	tgcaaggacc	catttgggcc	1860
aaaattcctc	acacggacgg	caactttcac	ccttctccgc	tgatgggagg	gtttgggaatg	1920
aagcaccgcg	ctcctcagat	cctcatcaaa	aacacacctg	tacctgcgga	tcctccaacg	1980
gccttcaaca	aggacaagct	gaactctttc	atcacccagt	attctactgg	ccaagtcagc	2040
gtggagatcg	agtgaggagct	gcagaaggaa	aacagcaagc	gctggaaccc	ggagatccag	2100
tacacttcca	actattacaa	gtctaataat	gttgaatttg	ctgttaatac	tgaagggtga	2160
tatagtgaac	cccgcgccat	tggcaccaga	tacctgactc	gtaatctgta	a	2211

SEQ ID NO: 126 moltype = AA length = 736
FEATURE Location/Qualifiers
source 1..736
 mol_type = protein
 note = AAV1
 organism = Adeno-associated virus

SEQUENCE: 126

MAADGYLPDW	LEDNLSEGR	EWWDLPKGP	KPKANQQKQ	DGRGLVLPY	KYLGPFGNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPVEQSP	QEPDSSSGIG	KTGQQPAKKR	LNFGQTDGSE	180
SVPDPQPLGE	PPATPAAVGP	TTMASGGGAP	MADNNEGADG	VGNASGNWHC	DSTWLGDRI	240
TTSTRTWALP	TYNNHLYKQI	SSASTGASND	NHYFGYSTPW	GYFDNRFHC	HFSPRDWQRL	300
INNNGFRPRK	RLNFKLFNIQ	VKEVTTNDGV	TTIANNLTS	VQVFSDSYQL	LPYVLGSAHQ	360
GCLPPFPADV	FMIPQGYLT	LNNGSQAVGR	SSFYCLEYFP	SQMLRTGNF	TFSYTFEEVP	420
FHSSYAHSQS	LDRLMNLID	QYLYLNRTO	NQSGSAQNKD	LLFSRGSPAG	MSVQPKNWLP	480
GPCYRQQRVS	KTKTDNNNSN	FTWTGASKYN	LNGRESIINP	GTAMASHKDD	EDKFFPMSPV	540
MIFGKESAGA	SNTALDNNMI	TDEEEIKATN	PVATERFRTV	AVNFQSSSTD	PATGDVHAMG	600
ALPGMVWQDR	DVYLQGPWA	KIPHTDGHFH	PSPLMGGFGL	KNPPPPQILIK	NTPVPANPPA	660
EFSA TKFASF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEVQ	YTSNYAKSAN	VDFTVDNNGL	720
YTEPRPIGTR	YLTRPL					736

SEQ ID NO: 127 moltype = AA length = 735
FEATURE Location/Qualifiers
source 1..735
 mol_type = protein
 note = AAV2
 organism = Adeno-associated virus

SEQUENCE: 127

MAADGYLPDW	LEDTLSEGR	QWWKLKPGPP	PKPAERHKD	DSRGLVLPY	KYLGPFGNGLD	60
KGEPVNEADA	AALHDKAYD	RQLDSGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEPVKTAP	GKKRPVEHSP	VEPDSSSGTG	KAGQQPARKR	LNFGQTDGAD	180
SVPDPQPLGQ	PPAAPSLG	NTMATGSGAP	MADNNEGADG	VGNSSGNWHC	DSTWMDRI	240
TTSTRTWALP	TYNNHLYKQI	SSQSGASNDN	HYFGYSTPWG	YFDNRFHCH	FSPRDWQRLI	300
NNNGFRPRK	RLNFKLFNIQ	KEVTQNDGTT	TIANNLTSTV	QVFTDSYQL	PYVLGSAHQ	360
CLPPFPADV	MVPQGYLT	LNNGSQAVGRS	SPYCLEYFPS	QMLRTGNFT	FSYTFEDVFP	420

```
SEQ ID NO: 128      moltype = AA  length = 736
FEATURE             Location/Qualifiers
source              1..736
                   mol_type = protein
                   note = AAV3
                   organism = Adeno-associated virus
```

```
SEQ ID NO: 129      moltype = AA  length = 734
FEATURE             Location/Qualifiers
source              1..734
                   mol_type = protein
                   note = AAV4
                   organism = Adeno-associated virus
```

```
SEQ ID NO: 130      moltype = AA  length = 736
FEATURE             Location/Qualifiers
source              1..736
                   mol_type = protein
                   note = AAV6
                   organism = Adeno-associated virus
```

```
SEQ ID NO: 131      moltype = AA  length = 737
FEATURE             Location/Qualifiers
source              1..737
                   mol_type = protein
                   note = AAV7
                   organism = Adeno-associated virus
```

-continued

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SEQUENCE: 131
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD NGRGLVLPGY KYLGPFNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEGAKTAP AKKRPVEPSP QRSPTSSTGI GKKGQQPARK RLNFGQTGDS 180
ESVPDPQPLG EPPAAPSSVG SGTVAAGGGA PMADNNEGAD GVGNASGNWH CDSTWLGDRV 240
ITTSTRTWAL PTYNNHLYKQ ISSSETAGSTN DNTYFGYSTP WGYFDNRFH CHFSPRDWQR 300
LINNNWGFRR KKLRFKLFNI QVKEVTINDG VTTIANNLTS TIQVFSSEY QLPYVLGSAH 360
QGCLPPFPAD VFMIPQYGYL TLNNGSQSVG RSSFYCLEYF PSQMLRTGNN FEFSYSFEDV 420
PFHSSYAHSQ SLDRMLNPLI DQYLYYLART QSNPGGTAGN RELQFYQGGP STMAEQAKNW 480
LPGPCFRQQR VSKTLDQNNN SNFAWTGATK YHLNGRNSLV NPGVAMATHK DDEDRFFPSS 540
GVLIFGKTGA TNKTTLENVL MTNEEEIRPT NPVATEEYGI VSSNLQAANT AAQTQVVNNQ 600
GALPGMVWQN RDVYLQGPW AKIPHDTGNF HPSPLMGGFG LKHPPQIL I KNTVPVNP 660
EVFTPAKFAS FITQYSTGQV SVEIEWELQK ENSKRWNPEI QYTSNFEKQT GVDFAVDSQG 720
VYSEPRPIGT RYLTRNL 737

```

```

SEQ ID NO: 132      moltype = AA length = 736
FEATURE            Location/Qualifiers
source              1..736
                    mol_type = protein
                    note = AAVhu31
                    organism = Adeno-associated virus

```

```

SEQUENCE: 132
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADAEF QERLKEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEAAKTAP GKRPVEQSP QEPDSSAGIG KSGSQPAKKK LNFQQTGDT 180
SVPDPQPIGE PPAAPSGVGS LTMASGGGAP VADNNEGADG VGSSSGNWHC DSQWLGDRI 240
TTSTRTWAL PTYNNHLYKQI SNSTSGGSSN DNAYFGYSTP WGYFDNRFH CHFSPRDWQR 300
LINNNWGFRR KRLNFKLFNI QVKEVTINDG VTKIANNLTS TVQVFTSDY QLPYVLGSAH 360
EGCLPPFPAD VFMIPQYGYL TLNDGGQAVG RSSFYCLEYF PSQMLRTGNN FQFSYEFENV 420
PFHSSYAHSQ SLDRMLNPLI DQYLYYLSKT INSGGQNGQT LKFSVAGPSN MAVQGRNYIP 480
GPSYRQQRVS TTVQNNNSE FAWPGASSWA LNGRNSLMNP GPAMASHKEG EDRFFPLSGS 540
LIFGKQGTGR DNVADKVM I TNEEEIKTTN PVATESYQGV ATNHQSAQAQ AQTGWVQNGQ 600
ILPGMVWQDR DVYLQGPW AKIPHDTGNF HPSPLMGGFG KHPPPQIL I KNTVPVNP 660
AFNKDKLNSF ITQYSTGQVS VEIEWELQKE NSKRWNPEIQ YTSNYYKSNN VFAVSTEGV 720
YSEPRPIGTR YLTRNL 736

```

```

SEQ ID NO: 133      moltype = AA length = 738
FEATURE            Location/Qualifiers
source              1..738
                    mol_type = protein
                    note = AAVrh10
                    organism = Adeno-associated virus

```

```

SEQUENCE: 133
MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD DGRGLVLPGY KYLGPFNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFQ 120
AKKRVLEPLG LVEEGAKTAP AKKRPVEPSP QRSPTSSTGI GKKGQQPAKK RLNFGQTGDS 180
ESVPDPQPIG EPPAGPSGLG SGTMAAGGGA PMADNNEGAD GVGSSSGNWH CDSTWLGDRV 240
ITTSTRTWAL PTYNNHLYKQI ISNGTSGGST NDNTYFGYST PWGYFDNRFH CHFSPRDWQ 300
RLINNNWGFRR PKRLNFKLFNI IQVKEVTQNE GTKTIANNLT STIQVFTDSE YQLPYVLGSA 360
HQGCLPPFPA DVFMIPQYGYL LTLNNGSQAV GRSSFYCLEY FPSQMLRTGN NFEFSYQFED 420
VPFHSSYAHS QSLDRMLNPLI IDQYLYLSR TQSTGGTAGT QQLLFSQAGP NNMSAQAKNW 480
LPGPCYRQQR VSTTSLQNNN SNFAWTGATK YHLNGRDSL NPGVAMATHK DDEERFFPSS 540
GVLMPGKQGA GKDNVDYSSV MLTSEEEIKT TNPVATEQY VVADNLQQQN AAPIVGAVNS 600
QGALPGMVWQ NRDVYLQGP WAKIPHDTGN FHPSPLMGGF GLKHPPQIL I KNTVPVNP 660
PTTFSQAKLA SFITQYSTGQ VSVEIEWELQ KENSKRWNP I QYTSNYYKS TNVDFAVNTD 720
GTYSERPIG TRYLTRNL 738

```

```

SEQ ID NO: 134      moltype = AA length = 12
FEATURE            Location/Qualifiers
source              1..12
                    mol_type = protein
                    note = Muscle-homing peptide M12
                    organism = Synthetic construct

```

```

SEQUENCE: 134
RRQPPRSISS HP 12

```

```

SEQ ID NO: 135      moltype = AA length = 7
FEATURE            Location/Qualifiers
source              1..7
                    mol_type = protein
                    note = Muscle-specific peptide MSP
                    organism = Synthetic construct

```

```

SEQUENCE: 135
ASSLNIA 7

```

```

SEQ ID NO: 136      moltype = AA length = 27

```

-continued

FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = protein	
	note = Muscle-homing peptide	
	organism = Synthetic construct	
SEQUENCE: 136		
AGSTSAGSAA GSSGDRRQPP RSISSHP		27
SEQ ID NO: 137	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	note = Muscle-homing peptide	
	organism = Synthetic construct	
SEQUENCE: 137		
CYAIGSFDC		9
SEQ ID NO: 138	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	note = Muscle-homing peptide	
	organism = Synthetic construct	
SEQUENCE: 138		
SGASAV		6
SEQ ID NO: 139	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	note = Muscle-homing peptide	
	organism = Synthetic construct	
VARIANT	5	
	note = X = any amino acid	
SEQUENCE: 139		
GRSGXR		6
SEQ ID NO: 140	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = RGD peptide	
	organism = Synthetic construct	
SEQUENCE: 140		
SLRSPPS		7
SEQ ID NO: 141	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = RGD peptide	
	organism = Synthetic construct	
SEQUENCE: 141		
RGDLGLS		7
SEQ ID NO: 142	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	note = 4C-RGD peptide	
	organism = Synthetic construct	
SEQUENCE: 142		
CDCRGDCFC		9
SEQ ID NO: 143	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	note = 4C-RGD peptide	
	organism = Synthetic construct	
SEQUENCE: 143		
CDCRGDCFCG LS		12
SEQ ID NO: 144	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	

-continued

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mol_type = protein
note = RGD peptide
organism = Synthetic construct

SEQUENCE: 144
RGDLTTP 7

SEQ ID NO: 145
FEATURE
source
mol_type = AA length = 7
Location/Qualifiers
1..7
mol_type = protein
note = RGD peptide
organism = Synthetic construct

SEQUENCE: 145
RGDLSTP 7

SEQ ID NO: 146
FEATURE
source
mol_type = AA length = 137
Location/Qualifiers
1..137
mol_type = protein
organism = Adeno-associated virus

SEQUENCE: 146
MAADGYLPDW LEDTLSEGIR QWWKLKPGPP PPKPAERHKD DSRGLVLPGY KYLGPGNGLD 60
KGEPVNAADA AALEHDKAYD QQLKAGDNPY LKYNHADA EF QERLKEDTSF GGNLGRAVFQ 120
AKKRLLEPLG LVEEAAK 137

SEQ ID NO: 147
FEATURE
source
mol_type = DNA length = 411
Location/Qualifiers
1..411
mol_type = genomic DNA
note = AAVhu32 VP1u
organism = Adeno-associated virus

SEQUENCE: 147
atggctgccg atggttatct tccagattgg ctcgaggaca ctctctctga aggaataaga 60
cagtgggtgga agctcaaacc tggcccacca ccaccaaagc ccgcagagcg gcataaggac 120
gacagcaggg gtcttgtgct tcttgggtac aagtacctcg gaccgggcaa cggactcgac 180
aagggggagc cggccaacgc agcagacgcg gcggccctcg agcacgacaa ggcctacgac 240
cagcagctca aggcgggaga caaccctgac ctcaagtaca accacgccga cgccgagttc 300
caggagcgcg tcaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcag 360
gccaaaaaga ggcttcttga acctcttggt ctggttgagg aagcggetaa g 411

SEQ ID NO: 148
FEATURE
source
mol_type = AA length = 65
Location/Qualifiers
1..65
mol_type = protein
note = AAVhu32 VP1/2s (amino acids 138-202)
organism = Adeno-associated virus

SEQUENCE: 148
TAPGKKRPVE QSPQEPDSSA GIGKSGSQPA KKKLNFGQTG DTESVPDPQP IGEPPAAPSG 60
VGSLT 65

SEQ ID NO: 149
FEATURE
source
mol_type = DNA length = 195
Location/Qualifiers
1..195
mol_type = genomic DNA
note = AAVhu32 VP1/2s
organism = Adeno-associated virus

SEQUENCE: 149
acggctcctg gaaagaagag gctgttagag cagtctcttc aggaaccgga ctctccgcg 60
ggtattggca aatcggttc acagcccgct aaaaagaaac tcaatttcgg tcagactggc 120
gacacagagt cagtcccga ccctcaacca atcggagaac ctcccgcagc cccctcaggt 180
gtgggatctc ttaca 195

SEQ ID NO: 150
FEATURE
source
mol_type = AA length = 7
Location/Qualifiers
1..7
mol_type = protein
note = Muscle-homing peptide
organism = Synthetic construct

VARIANT
7
note = X = any amino acid

SEQUENCE: 150
DFSGIAX 7

```

-continued

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SEQ ID NO: 151      moltype = AA  length = 7
FEATURE            Location/Qualifiers
source              1..7
                    mol_type = protein
                    note = RGD peptide
                    organism = Synthetic construct

SEQUENCE: 151
RGDLRV

```

7

What is claimed is:

1. A polypeptide comprising:
 - a. the VP1/2s region of AAV5; and
 - b. the VP3 region of AAV8.
2. The polypeptide of claim 1, further comprising the VP1u region of AAV5 or AAV8.
3. The polypeptide of claim 2, wherein the regions are present in the following order from amino- to carboxy-terminus of the polypeptide:
 - a. the VP1u region of AAV5 or AAV8;
 - b. the VP1/2s region of AAV5; and
 - c. the VP3 region of AAV8.
4. A polypeptide comprising:
 - a. the VP1u region of AAV Hu32;
 - b. the VP1/2s region of AAV Hu32; and
 - c. the VP3 region of AAV Rh13.
5. The polypeptide of any one of claims 1-3, wherein the VP1/2s region of AAV5 comprises SEQ ID NO: 3.
6. The polypeptide of any one of claims 2, 3, and 5, wherein the VP1u region of AAV5 or AAV8 comprises SEQ ID NO: 7 or 17, respectively.
7. The polypeptide of any one of claims 1-3 and 5-6, wherein the VP3 region of AAV8 comprises SEQ ID NO: 11.
8. The polypeptide of any one of claims 1-3 and 6-7, wherein the VP1/2s region of AAV5 comprises an amino acid sequence consisting of SEQ ID NO: 3.
9. The polypeptide of any one of claims 2, 3, 5, and 7-8, wherein the VP1u region of AAV5 or AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 7 or 17, respectively.
10. The polypeptide of any one of claims 1-3, 5-6, and 8-9, wherein the VP3 region of AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 11.
11. The polypeptide of any one of claims 1-3, 6-7, and 8-9, wherein the VP1/2s region of AAV5 consists of SEQ ID NO: 3.
12. The polypeptide of any one of claims 2, 3, 5, 7, 8, 10, and 11, wherein the VP1u region of AAV5 or AAV8 consists of SEQ ID NO: 7 or 17, respectively.
13. The polypeptide of any one of claims 1-3, 5, 6, 8, 9, 11, and 12, wherein the VP3 region of AAV8 consists of SEQ ID NO: 11.
14. The polypeptide of any one of claims 1-3, wherein the polypeptide comprises SEQ ID NO: 23 or 25.
15. The polypeptide of any one of claims 1-3, wherein the polypeptide consists of SEQ ID NO: 23 or 25.
16. The polypeptide of any one of claims 1-3, wherein the polypeptide comprises an amino acid sequence consisting of SEQ ID NO: 23 or 25.
17. The polypeptide of claim 4, wherein the VP1u region of AAVhu32 comprises the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146.
18. The polypeptide of claim 4, wherein the VP1/2s region of AAVhu32 comprises the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148.
19. The polypeptide of claim 4, wherein the VP3 region of AAVrh13 comprises SEQ ID NO: 27.
20. The polypeptide of any one of claims 4 and 18-19, wherein the VP1u region of AAVhu32 comprises the VP1u portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 146.
21. The polypeptide of any one of claims 4, 17, and 19-20, wherein the VP1/2s region of AAVhu32 comprises the VP1/2s portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 148.
22. The polypeptide of any one of claims 4, 17, 18, and 20-21, wherein the VP3 region of AAVrh13 comprises an amino acid sequence consisting of SEQ ID NO: 27.
23. The polypeptide of any one of claims 4, 18-19, and 21-22, wherein the VP1u region of AAVhu32 consists of the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146.
24. The polypeptide of any one of claims 4, 17, 19-20, and 22-23, wherein the VP1/2s region of AAVhu32 consists of the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148.
25. The polypeptide of any one of claims 4, 17, 18, 20-21, and 23-24, wherein the VP3 region of AAVrh13 consists of SEQ ID NO: 27.
26. The polypeptide of claim 4, wherein the polypeptide comprises an amino acid sequence of SEQ ID NO: 29.
27. The polypeptide of claim 4, wherein the polypeptide consists of an amino acid sequence of SEQ ID NO: 29.
28. The polypeptide of claim 4, wherein the polypeptide comprises an amino acid sequence consisting of SEQ ID NO: 29.
29. A nucleotide sequence encoding the amino acid sequence of the polypeptide of any one of claims 1-28.
30. An expression vector comprising the nucleotide sequence of claim 29.
31. A host cell comprising the nucleotide sequence of claim 29 or the expression vector of claim 30.
32. A recombinant AAV comprising the polypeptide of any one of claims 1-28.
33. A recombinant adeno-associated virus (AAV) comprising a capsid protein, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAV8, and wherein the rAAV comprises an expression cassette.
34. The rAAV of claim 33, wherein the expression cassette comprises an open reading frame (ORF) encoding a transgene.
35. The rAAV of claim 33 or 34, wherein the expression cassette further comprises a promoter.

36. The rAAV of any one of claims 33-35, wherein the capsid protein further comprises an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAV5 or AAV8.

37. The rAAV of any one of claims 33-35, wherein the capsid protein further comprises an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAV5 or AAV8.

38. The rAAV of any one of claims 33-35, wherein the capsid protein further comprises an amino acid sequence of the VP1u region of AAV5 or AAV8.

39. The rAAV of any one of claims 33-38, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAV8.

40. The rAAV of any one of claims 33-38, wherein the capsid protein comprises: (i) an amino acid sequence of the VP1/2s region of AAV5, and (ii) an amino acid sequence of the VP3 region of AAV8.

41. The rAAV of any one of claims 33-40, wherein the VP1/2s region of AAV5 comprises SEQ ID NO: 3.

42. The rAAV of any one of claims 36-41, wherein the VP1u region of AAV5 or AAV8 comprises SEQ ID NO: 7 or 17, respectively.

43. The rAAV of any one of claims 33-42, wherein the VP3 region of AAV8 comprises SEQ ID NO: 11.

44. The rAAV of any one of claims 33-40 and 42-43, wherein the VP1/2s region of AAV5 comprises an amino acid sequence consisting of SEQ ID NO: 3.

45. The rAAV of any one of claims 36-41 and 43-44, wherein the VP1u region of AAV5 or AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 7 or 17, respectively.

46. The rAAV of any one of claims 33-42 and 44-45, wherein the VP3 region of AAV8 comprises an amino acid sequence consisting of SEQ ID NO: 11.

47. The rAAV of any one of claims 33-40, 42-43, and 45-46, wherein the VP1/2s region of AAV5 consists of SEQ ID NO: 3.

48. The rAAV of any one of claims 36-41, 43-44, and 46-47, wherein the VP1u region of AAV5 or AAV8 consists of SEQ ID NO: 7 or 17, respectively.

49. The rAAV of any one of claims 33-42, 44-45, and 47-48, wherein the VP3 region of AAV8 consists of SEQ ID NO: 11.

50. A recombinant adeno-associated virus (AAV) comprising a capsid protein, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAVrh13, and an expression cassette.

51. The rAAV of claim 50, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAVrh13.

52. The rAAV of any one of claims 50-51, wherein the capsid protein comprises: (i) an amino acid sequence of the VP1u region of AAVhu32, (ii) an amino acid sequence of the

VP1/2s region of AAVhu32, and (iii) an amino acid sequence of the VP3 region of AAVrh13.

53. The rAAV of any one of claims 50-52, wherein the expression cassette comprises an open reading frame (ORF) encoding a transgene.

54. The rAAV of any one of claims 50-53, wherein the expression cassette further comprises a promoter.

55. The rAAV of any one of claims 50-54, wherein the VP1u region of AAVhu32 comprises the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146.

56. The rAAV of any one of claims 50-54, wherein the VP1u region of AAVhu32 consists of the VP1u portion of SEQ ID NO: 29, or SEQ ID NO: 146.

57. The rAAV of any one of claims 50-54, wherein the VP1u region of AAVhu32 comprises the VP1u portion of an amino acid sequence consisting of SEQ ID NO: 29, or SEQ ID NO: 146.

58. The rAAV of any one of claims 50-57, wherein the VP1/2s region of AAVhu32 comprises the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148.

59. The rAAV of any one of claims 50-57, wherein the VP1/2s region of AAVhu32 consists the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148.

60. The rAAV of any one of claims 50-57, wherein the VP1/2s region of AAVhu32 comprises an amino acid sequence consisting of the VP1/2s portion of SEQ ID NO: 29, or SEQ ID NO: 148.

61. The rAAV of any one of claims 50-60, wherein the VP3 region of AAVrh13 comprises an amino acid sequence of SEQ ID NO: 27.

62. The rAAV of any one of claims 50-60, wherein the VP3 region of AAVrh13 consists of an amino acid sequence of SEQ ID NO: 27.

63. The rAAV of any one of claims 50-60, wherein the VP3 region of AAVrh13 comprises an amino acid sequence consisting of SEQ ID NO: 27.

64. The rAAV of any one of claims 33-49, wherein the capsid protein comprises an amino acid sequence of SEQ ID NO: 21 or 23.

65. The rAAV of any one of claims 33-49, wherein the capsid protein consists of an amino acid sequence of SEQ ID NO: 21 or 23.

66. The rAAV of any one of claims 33-49, wherein the capsid protein comprises an amino acid sequence consisting of SEQ ID NO: 21 or 23.

67. The rAAV of any one of claims 50-63, wherein the capsid protein comprises an amino acid sequence of SEQ ID NO: 29.

68. The rAAV of any one of claims 50-63, wherein the capsid protein consists of an amino acid sequence of SEQ ID NO: 29.

69. The rAAV of any one of claims 50-63, wherein the capsid protein comprises an amino acid sequence consisting of SEQ ID NO: 29.

70. The rAAV of any one of claims 33-69, further comprising an AAV inverted terminal repeat.

71. A composition comprising the rAAV of any one of claims 33-70, and a physiologically acceptable carrier.

72. The rAAV of claim 32, wherein the rAAV further comprises a transgene.

73. A method of delivering the transgene to a cell, comprising contacting the cell with the rAAV of any one of claims 33-70 and 72.

74. The method of claim **73**, wherein the cell is a muscle cell.

75. The rAAV of any one of claims **33-69** and **72**, or the method of claim **73** or **74**, wherein the transgene comprises a heterologous gene associated with a muscle related disease or disorder.

76. The rAAV or the method of claim **75**, wherein the heterologous gene is operably linked to a regulatory sequence that controls expression of the heterologous gene in a host cell.

77. The host cell of claim **31**, or the rAAV of claim **76**, or the method of claim **76**, wherein the host cell is a muscle cell.

78. The rAAV of any one of claims **33-69** and **72**, or the method of claim **73** or **74**, wherein the transgene encodes a therapeutic protein.

79. The rAAV or the method of claim **78**, wherein the therapeutic protein is associated with a muscle related disease or disorder.

80. The rAAV or the method of claim **79**, wherein the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedrich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy.

81. The rAAV or the method of claim **80**, wherein the muscle related disease or disorder is Duchenne muscular dystrophy.

82. The rAAV or the method of any one of claims **78-81**, wherein the transgene encodes a microdystrophin protein.

83. A vector comprising a nucleic acid sequence encoding the capsid protein as defined in any one of claims **33-70**.

84. The vector of claim **83**, wherein the nucleic acid sequence is operably linked to a heterologous regulatory element that controls expression of the capsid protein in a host cell.

85. An in vitro cell comprising the vector of claim **83** or **84**.

86. A method of producing a recombinant adeno-associated virus (rAAV), the method comprising the steps of:

- i. culturing a cell in a cell culture to produce the rAAV, the cell comprising a nucleotide sequence encoding a capsid protein, and a nucleotide sequence comprising a transgene, and wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAV8; and
- ii. collecting the rAAV from the cell culture.

87. The method of claim **86**, wherein the capsid protein further comprises an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAV5 or AAV8.

88. The method of claim **86**, wherein the capsid protein further comprises an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAV5 or AAV8.

89. The method of claim **86**, wherein the capsid protein further comprises an amino acid sequence of VP1u region of AAV5 or AAV8.

90. The method of any one of claims **86-89**, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAV5, and (ii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAV8.

91. The method of any one of claims **86-89**, wherein the capsid protein comprises: (i) an amino acid sequence of VP1/2s region of AAV5, and (ii) an amino acid sequence of VP3 region of AAV8.

92. A method of producing a recombinant adeno-associated virus (rAAV), the method comprising the steps of:

- i. culturing a cell in a cell culture to produce the rAAV, the cell comprising a nucleotide sequence encoding a capsid protein, and a nucleotide sequence comprising a transgene, and wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 95% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 95% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 95% sequence identical to VP3 region of AAVrh13; and
- ii. collecting the rAAV from the cell culture.

93. The method of claim **92**, wherein the capsid protein comprises: (i) an amino acid sequence that is at least about 98% sequence identical to VP1u region of AAVhu32, (ii) an amino acid sequence that is at least about 98% sequence identical to VP1/2s region of AAVhu32, and (iii) an amino acid sequence that is at least about 98% sequence identical to VP3 region of AAVrh13.

94. The method of claim **92** or **93**, wherein the capsid protein comprises: (i) an amino acid sequence of VP1u region of AAVhu32, (ii) an amino acid sequence of VP1/2s region of AAVhu32, and (iii) an amino acid sequence of VP3 region of AAVrh13.

95. The method of any one of claims **86-94**, wherein the cell comprises a nucleotide sequence encoding an AAV rep protein.

96. The method of any one of claims **86-95**, wherein the transgene is flanked by AAV inverted terminal repeats.

97. The method of any one of claims **86-96**, wherein the transgene comprises a heterologous gene associated with a muscle related disease or disorder.

98. The method of claim **97**, wherein the heterologous gene is operably linked to a regulatory sequence that controls expression of the heterologous gene in a host cell.

99. The method of any one of claims **86-96**, wherein the transgene encodes a therapeutic protein.

100. The method of claim **99**, wherein the therapeutic protein is associated with a muscle related disease or disorder.

101. The method of claim **99** or **100**, wherein the therapeutic protein encodes microdystrophin protein.

102. The method of any one of claims **86-96**, wherein the transgene encodes a functional gene product.

103. The method of any one of claims of any one of claims **86-102**, wherein the cell is selected from an invertebrate cell, an insect cell, or a mammalian cell.

104. The method of claim **103**, wherein the mammalian cell is selected from HEK293, HeLa, CHO, NS0, SP2/0, PER.C6, Vero, RD, BHK, HT-1080, A549, Cos-7, ARPE-19, MRC-5, or any combination thereof.

105. The method of claim **103**, wherein the insect cell is selected from High Five, Sf9, Se301, SeIZD2109, SeUCR1, Sf900+, Sf21, Bti-tn-5b1-4, MG-1, Tn368, HzAm1, BM-N, Ha2302, Hz2E5, Ao38, or any combination thereof.

106. A method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering the rAAV of any one of claims **33-70**, **72**, and **78-82**, or the composition of claim **71**, to the subject.

107. The method of claim **106**, wherein the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy.

108. The method of claim **106**, wherein the muscle related disease or disorder is muscular dystrophy.

109. The method of claim **108**, wherein the muscular dystrophy is Duchenne muscular dystrophy.

110. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV provides at least one improvement of

packaging efficiency, yield, titer, infectivity, transduction efficiency, and transfection efficiency, as compared to a reference AAV.

111. The rAAV of claim **110**, wherein the reference AAV is AAV8.

112. The rAAV of claim **110**, wherein the reference AAV comprises VP1u of AAV8.

113. The rAAV of claim **110**, wherein the reference AAV is AAVrh13.

114. The rAAV of any one of claims **110-113**, wherein the improvement is by about or at least about 1-fold, 1.5-fold, 2-fold, 2.5-fold, 3-fold, 3.5-fold, 4-fold, 4.5-fold, 5 fold, 5.5-fold, 6-fold, 6.5-fold, 7-fold, 7.5-fold, 8-fold, 8.5-fold, 9-fold, 9.5-fold, 10-fold, or higher than 10-fold.

115. The rAAV of claim **114**, wherein the improvement is by about or at least about 2.5-fold.

116. The rAAV of any one of claims **110-115**, wherein the at least one improvement is an improvement in packaging efficiency.

117. The rAAV of any one of claims **110-115**, wherein the at least one improvement is an improvement in titer.

118. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV transduces liver tissues at lower levels as compared to a reference AAV.

119. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV results in lower RNA expression in liver as compared to a reference AAV.

120. The rAAV of claim **119**, wherein the RNA expression is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the RNA expression from a reference AAV.

121. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV results in higher RNA/DNA ratio in muscle tissue as compared to a reference AAV.

122. The rAAV of claim **121**, wherein the RNA/DNA ratio is higher by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100%.

123. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV results in lower DNA levels in muscle tissue as compared to a reference AAV.

124. The rAAV of claim **123**, wherein the DNA levels is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the DNA levels from a reference AAV.

125. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV results in a lower level of liver toxicity as compared to the level of liver toxicity from a reference AAV.

126. The rAAV of claim **125**, wherein the level of liver toxicity is lower by about or at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more than 100% as compared to the level of liver toxicity from a reference AAV.

127. The rAAV of any one of claims **33-70**, **72**, and **78-82**, wherein the rAAV results in transcription-specific liver de-targeting as compared to a reference AAV.

128. The method of claim **106**, wherein the administering comprises administering a lower amount of vector genome of the rAAV as compared to the amount of vector genome

of a reference AAV necessary to obtain the same therapeutic effect after the administering.

129. The rAAV of any one of claims **118-127** or the method of claim **128**, wherein the reference AAV is AAV8.

130. The rAAV of any one of claims **118-127** or the method of claim **128**, wherein the reference AAV comprises VP1u of AAV8.

131. The rAAV of any one of claims **118-127** or the method of claim **128**, wherein the reference AAV is AAVrh13.

132. A polypeptide comprising one or more of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71.

133. A polypeptide consisting of one or more of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71.

134. A polypeptide comprising one or more of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

135. A polypeptide consisting of one or more of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

136. A polynucleotide encoding an AAV capsid protein comprising the amino acid sequence of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71.

137. A polynucleotide encoding an AAV capsid protein consisting of any one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, and 71.

138. A polynucleotide comprising any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and 72, wherein the polynucleotide encodes an AAV capsid protein.

139. A polynucleotide encoding an AAV capsid protein comprising the amino acid sequence of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

140. A polynucleotide encoding an AAV capsid protein consisting of any one of SEQ ID NOs: 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

141. A polynucleotide comprising any one of SEQ ID NOs: 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120, wherein the polynucleotide encodes an AAV capsid protein.

142. A polynucleotide consisting of any one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, and 72, wherein the polynucleotide encodes an AAV capsid protein.

143. A polynucleotide consisting of any one of SEQ ID NOs: 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120, wherein the polynucleotide encodes an AAV capsid protein.

144. A method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering an rAAV comprising a polypeptide comprising an amino acid sequence of SEQ ID NO:31 or an rAAV encoded by SEQ ID NO:32.

145. The method of claim **144**, wherein the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy (SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy.

146. The method of claim **144**, wherein the muscle related disease or disorder is muscular dystrophy.

147. The method of claim **146**, wherein the muscular dystrophy is Duchenne muscular dystrophy.

148. The method of any one of claims **146-147**, wherein the rAAV further comprises a transgene encoding a functional dystrophin, a minidystrophin, a microdystrophin, or a dystrophin exon-skipping snRNA.

149. The method of claim **148**, wherein the transgene comprises a polynucleotide sequence encoding any one of SEQ ID NOs: 73, 74, 75, 76, 77, 78, 79, and 80.

150. The method of any one of claims **144-149**, wherein the rAAV provides at least one improvement of packaging efficiency, yield, titer, infectivity, transduction efficiency, and transfection efficiency, as compared to a reference AAV, wherein the reference AAV is AAV9.

151. The method of any one of claims **106-109, 128-131**, and **144-150**, wherein the rAAV provides a higher vector yield after the rAAV is administered to the subject, as compared to a reference AAV.

152. The method of claim **151**, wherein the vector yield is higher by about or at least about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times.

153. The method of claim **151**, wherein the vector yield is higher by between about 1.5 to about 3 times.

154. The method of any one of claims **106-109, 128-131**, and **144-153**, wherein the RNA level is increased after the rAAV is administered to the subject, as compared to a reference AAV.

155. The method of claim **154**, wherein the RNA level is increased by about or at least about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times.

156. The method of claim **154**, wherein the RNA level is increased by between about 2.5 to about 3.5 times.

157. The method of any one of claims **106-109**, **128-131**, and **144-156**, wherein the ratio of RNA to DNA is increased after the rAAV is administered to the subject.

158. The method of claim **157**, wherein the ratio of RNA to DNA is increased by about or at least about 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10 times, or more than 10 times.

159. The method of claim **157**, wherein the ratio of RNA to DNA is increased by between about 2.5 to about 3.5 times.

160. The method of any one of claims **106-109**, **128-131**, and **144-159**, wherein the reference AAV is AAV5.

161. The method of any one of claims **106-109**, **128-131**, and **144-159**, wherein the reference AAV is AAV8.

162. The method of any one of claims **106-109**, **128-131**, and **144-161**, where the reference AAV comprises the expression cassette.

163. A recombinant adeno-associated virus (rAAV) hu32 capsid protein comprising SEQ ID NO: 31, or an rAAV capsid protein comprising an amino acid sequence that is about or at least about 90% identical to SEQ ID NO: 31, comprising a peptide insertion of at least 4 and up to 12 contiguous amino acids from a heterologous protein that is not an AAV protein, wherein the peptide insertion is inserted immediately after an amino acid residue corresponding to one of any one of amino acids 451 to 461 of AAVhu32 capsid protein of SEQ ID NO: 31.

164. The method of claim **163**, wherein the peptide insertion is between amino acid 454 and amino acid 455 of SEQ ID NO: 31.

165. A recombinant adeno-associated virus (rAAV) hu32 capsid protein comprising SEQ ID NO: 31, or an rAAV capsid protein comprising an amino acid sequence that is about or at least about 90% identical to SEQ ID NO: 31, comprising a peptide insertion of at least 4 and up to 12 contiguous amino acids from a heterologous protein that is not an AAV protein, wherein the peptide insertion is inserted immediately after an amino acid residue corresponding to one of amino acids 570 to 595 of AAVhu32 capsid protein of SEQ ID NO: 31.

166. The method of claim **165**, wherein the peptide insertion is between amino acid 589 and amino acid 590 of SEQ ID NO: 31.

167. The method of any one of claims **163** to **166**, wherein the peptide insertion is a muscle-homing peptide.

168. The method of claim **167**, wherein the muscle-homing peptide comprises an integrin receptor-binding domain or an integrin-binding domain.

169. A recombinant adeno-associated virus (rAAV) capsid protein comprising an amino acid sequence that is about or at least about 90% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

170. The rAAV capsid protein of claim **169**, wherein the rAAV capsid protein comprises an amino acid sequence that is about or at least about 98% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

171. The rAAV capsid protein of claim **169** or **170**, wherein the rAAV capsid protein comprises an amino acid

sequence of any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

172. A recombinant adeno-associated virus (rAAV) vector comprising a nucleic acid sequence encoding the rAAV capsid protein of any one of claims **169** to **171**.

173. A recombinant adeno-associated virus (rAAV) vector comprising a nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120.

174. A cell comprising the rAAV vector of claim **172** or **173**.

175. A cell expressing the rAAV capsid protein of any one of claims **169** to **171**.

176. A recombinant AAV viral particle comprising the rAAV capsid protein of any one of claims **169** to **171**.

177. The rAAV capsid protein of any one of claims **169** to **171**, wherein a first AAV viral particle comprising the rAAV capsid protein results in increased transduction of muscle cells or heart cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 19, 31, and 124.

178. The rAAV capsid protein of claim **177**, wherein the first AAV viral particle has at least about 1.5 fold increase in transduction of muscle cells or heart cells relative to the second AAV viral particle.

179. The rAAV capsid protein of any one of claims **169** to **171**, wherein a first AAV viral particle comprising the rAAV capsid protein has reduced transduction of liver cells relative to a second AAV viral particle comprising a capsid protein that comprises any one of SEQ ID NOs: 11, 15, 19, 31, and 124.

180. The rAAV capsid protein of claim **179**, wherein the first AAV viral particle has about or at least about 1 fold decrease in transduction of liver cells relative to the second AAV viral particle.

181. A pharmaceutical composition comprising a recombinant adeno-associated virus (rAAV) vector comprising a recombinant AAV capsid protein, wherein the recombinant AAV capsid protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 21, 23, 25, 29, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 115, 117, and 119.

182. A method of treating a muscle related disease or disorder in a subject in need thereof, the method comprising administering the rAAV viral particle of claim **176**, the cell of claim **174** or **175**, the rAAV vector of claim **172** or **173**, or the pharmaceutical composition of claim **181** to the subject.

183. The method of claim **182**, wherein the muscle related disease or disorder is at least one of Duchenne muscular dystrophy, Becker muscular dystrophy, Bethlem congenital muscular dystrophy (CMD), Fukuyama CMD, muscle-eye-brain disease, rigid spine syndrome, Ullrich CMD, walker-warburg syndrome, Emery-Dreifuss muscular dystrophy (EDMD), Facioscapulohumeral muscular dystrophy (FSHD), Limb-girdle muscular dystrophies (LGMD), Myotonic dystrophy (DM), Oculopharyngeal muscular dystrophy (OPMD), amyotrophic lateral sclerosis (ALS), Spinal-bulbar muscular atrophy (SBMA), Spinal muscular atrophy

(SMA), Andersen-Tawil syndrome, Hyperkalemic periodic paralysis, Hypokalemic periodic paralysis, Myotonia congenita, Becker myotonia, Thomsen myotonia, Paramyotonia congenita, Potassium-aggravated myotonia, Friedreich's ataxia (FA), Kearns-Sayre syndrome (KSS), Leigh syndrome (subacute necrotizing encephalomyopathy), Mitochondrial DNA depletion syndromes, Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes (MELAS), Mitochondrial neurogastrointestinal encephalomyopathy (MNGIE), Myoclonus epilepsy with ragged red fibers (MERRF), Neuropathy, ataxia and retinitis pigmentosa (NARP), Pearson syndrome, Progressive external ophthalmoplegia (PEO), Congenital myasthenic syndromes (CMS), Lambert-Eaton myasthenic syndrome (LEMS), Myasthenia gravis (MG), Charcot-Marie-Tooth disease (CMT), Giant axonal neuropathy (GAN), Myofibrillar myopathy (MFM), Scapuloperoneal myopathy, Metabolic myopathy, inflammatory myopathy, endocrine myopathy, distal myopathy, and congenital myopathy.

183. The method of claim **182** or **183**, wherein the muscle related disease or disorder is muscular dystrophy.

184. The method of claim **183**, wherein the muscular dystrophy is Duchenne muscular dystrophy.

185. The method of any one of claims **182** to **184**, wherein the rAAV further comprises a transgene encoding a functional dystrophin, a minidystrophin, a microdystrophin, or a dystrophin exon-skipping snRNA.

186. An isolated nucleic acid molecule comprising a nucleic acid sequence encoding the rAAV capsid protein of any one of claims **169** to **171**.

187. An isolated nucleic acid molecule comprising the nucleic acid sequence of any one of SEQ ID NOs: 22, 24, 26, 30, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 116, 118, and 120.

188. A cultured cell comprising the nucleic acid molecule of claim **186** or **187**.

189. The rAAV capsid protein of any one of claims **177** to **180**, wherein the second AAV viral particle comprises an AAVhu32, AAV8, and/or AAV9 capsid protein.

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